Recipes for Rare Metal Ores: Magmas, Metasomatism, and Meteoric Mobilization via Sulfide/Goethite (U-Th)/He Geochronology and ML-Accelerated Thermodynamic Modeling

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ABSTRACT

Rising demand for critical metals, driven by computing infrastructure and the transition to renewable energy, foreground the need to understand how ore deposits form, change, and survive. This thesis develops geochemical tools for three families of processes that concentrate and rework rare metals: magmatic, metasomatic, and meteoric. The first three chapters extend U-Th/He geochronology to minerals that decompose under the vacuum of conventional helium step-heating experiments. Chapter 1 presents a method that seals samples in helium-permeable silica ampoules holding water- or sulfur-saturated conditions, with a framework for disentangling the proceeses of helium permeation through the glass from release by the sample. Chapter 2 applies this method to goethite, a direct chronometer of weathering and laterite formation. Under water-saturated conditions helium release is diffusive, with an activation energy of 157 ± 5 kJ/mol and near-quantitative retention at surface temperatures over billion-year timescales for the sample studied, whereas earlier vacuum experiments recorded conversion to hematite. Chapter 3 shows that pyrite and arsenopyrite are highly helium retentive, with pyrite in particular being among the most helium-retentive minerals studied, comparable to or exceeding monazite. U-Th/He ages from the Gympie gold field and the Kidd Creek massive sulfide deposit depend on uranium- and thorium-bearing inclusions and must be interpreted locality by locality. Chapter 4 presents *nGibbs*, a software package of compositionally general deep-learning emulators of the MELTS and HeFESTo thermodynamic models that conserve mass and run two to four orders of magnitude faster, making composition a practical free variable in models of magmatic and mantle systems. Chapter 5 turns to the Bushveld Complex, the world's dominant source of platinum-group metals. Field mapping at Draaikraal, Mpumalanga shows that local deformation at the intrusion's floor is consistent with a synmagmatic origin, and that the area's recessive, laterite-capped topography is the result of contamination by fluids from devolatilizing floor sediments. The Bushveld, with its sulfide-hosted platinum-group metals, goethite laterites, and unresolved petrological puzzles, is a natural place to apply these tools together.

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PUBLISHED CONTENT AND CONTRIBUTIONS

None of the chapters of this thesis has yet been published in a peer-reviewed journal.

- Chapter 1: B. J. Thyer, J. Treffkorn, and K. A. Farley. In preparation.
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- Chapter 4: B. J. Thyer and P. D. Asimow. In preparation.
- Chapter 5: B. J. Thyer and J. Roberts. In preparation.

B. J. Thyer is the lead author of each manuscript. The contributions of each author are stated at the start of each chapter.

Software and data. The following software and data, described in this thesis, have been published by B. J. Thyer, who is their sole author:

- Thyer, B. J. (2026). *nGibbs* (Version v0.1.0b4) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.21345830. The software described in Chapter 4.
- Thyer, B. (2026). Datasets for "Hydrothermal Helium Step Heats Reveal High Helium Retentivity and Multiple Diffusion Domain Systematics in Goethite" [Data set]. Zenodo. https://doi.org/10.5281/zenodo.22113166. The step heat data and ampoule-cooling simulation code for Chapter 2.

Related software. The following software was developed during the author's doctoral work but is not described in this thesis:

- Thyer, B. (2026). *DustMAP*: an Automated Mass Calibration and Categorization Pipeline for TOF Mass Spectra (Version 0.0.1) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.18895490.

Use of Generative AI

Anthropic's Claude generative AI was used in preparing this thesis to (1) write and debug data-analysis code, figure-generation scripts, and parts of the nGibbs software; (2) typeset the document in LaTeX, including tables, cross-references, and bibliography management; (3) edit text written by the author for clarity, grammar, and length; and (4) produce first drafts of some passages of the introduction, abstract, and chapters 2-4, which the author then revised. The research design, experiments, field work, and scientific interpretations are the work of the author and the co-authors named in each chapter. The author reviewed and verified all AI-assisted code, text, and citations, and takes full responsibility for the content of this thesis.

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INTRODUCTION

Modern energy and computing infrastructure is built from metals whose supply is concentrated in a select few places and depends on deposits that are difficult to find. The International Energy Agency projects that, under stated policies, global demand between 2024 and 2040 will grow by more than a factor of three for lithium, a factor of two for graphite, 65% for nickel, about 50% for magnet rare earth elements, and more than 25% for copper, which alone adds roughly 7 million tons (IEA, 2026). Two critical innovations are driving this demand: computational infrastructure in "the age of AI", as well as the transition to renewable energy sources, both of which depend on copper, silver, gold and platinum-group metals, as well as other rare metals such as gallium, germanium and indium. Technological solutions to the anthropogenic climate crisis (electric vehicles, battery storage, solar and wind generation, power-grid modernization) in effect shift much of the environmental burden from fossil fuel consumption to other, similarly non-renewable mined materials. Gracefully navigating this transition will require interdisciplinary innovations within the fields of geology, chemistry, and biology. Improved techniques in metal extraction from ores through bioengineered fermentation reactors, for example, could reduce the environmental footprint of mining and processing while increasing the supply of critical metals (Fernando et al., 2024).

Policy plays an essential role in ensuring that the transition to a low-carbon economy is sufficiently rapid and equitable (Sovacool et al., 2020). When I was conducting geological field work in South Africa in 2023, I was struck by how local communities viewed large-scale mining efforts. In Draaikraal, Mpumalanga, where economic-grade Platinum Group Element (PGE) ores lie underfoot at shallow depths (van der Merwe, 2007), locals overwhelmingly felt positive about the expansion of mining into their village. To them, the mines represented a source of employment and economic growth for their community, wherein livelihoods are currently made from small-scale livestock operations. But they also expressed disappointment in how few of their neighbors were hired when the nearest PGE mine (Booysendal, 30 km away) had opened. In much of Africa, informal small-scale mining (framed either as "illegal" mining or "artisanal" mining, depending on context) can generate revenue for local communities who often benefit little from large-scale mining operations

(Sovacool et al., 2020), although it also is known to fund crime syndicates and spur illegal immigration (Martin, 2019). These small-scale operations are often unregulated and unsafe: a few months before this was written, more than a dozen of these independent, informal miners were killed while mining PGEs in an abandoned mine in the western limb of the Bushveld Complex in a single incident (Ngcobo, 2026). Such accidents are common. Nonetheless, dispersed and decentralized artisanal mining operations contribute to the global supply of critical metals from sites that cannot be utilized (and are often abandoned by) larger operations, thereby producing needed resources with minimal additional environmental impacts and playing an important role in meeting global demand (Sovacool et al., 2020). Strong, thoughtful, humanist policies may resolve such challenges.

I often argue that policy is the most important field of work to direct humanity to a sustainable future, but I also find policy intolerably uninteresting and am often discouraged by sluggish political action and the recent regressions within environmental policy in the United States and elsewhere. I much prefer to study the natural world. Indeed, my interest in critical metals is rooted in the desire to better understand the natural processes that operate within the planet's interior. The application of these studies to mineral prospecting to fuel the transition to a low-carbon economy is serendipitous.

The role of geoscientists to this end is to locate accessible, high-grade deposits of critical metals. Intuitively, mining higher grade deposits damages less land and produces less waste to produce a given amount of critical metal. The foundation of this work is to understand the processes by which these deposits form, change, and survive.

In this thesis I develop geochemical tools for three families of processes wherein natural fluids concentrate and rework rare and valuable metals: magmas, metasomatism, and meteoric mobilization. Magmas deliver critical metals from the mantle to the crust, where they can become concentrated into cumulates and immiscible sulfide or oxide liquids as they crystallize. Metasomatic and hydrothermal fluids then transport and redeposit them, commonly as sulfide minerals. Meteoric (surface) water weathers the resulting deposits, upgrading some into supergene and lateritic ores and dispersing others. Each stage leaves a record in minerals, which can be read and interpreted through geochronometry, which bears on the rates and orders of these processes, and thermodynamic

models, which disentangle the chemical behaviors of these systems and the effects of the processes that operate on them. Together, these tools allow us to follow the journey of critical metals within magma through the mantle and hydrothermal reworking into the crust where they can be observed and collected.

The Olympic Dam Iron Oxide Copper Gold (IOCG) deposit in South Australia offers a striking example of why understanding process matters for discovery. The deposit lies 520 km north-northwest of Adelaide beneath about 260 m of flat-lying sedimentary rock and has no surface expression. It was found in 1975 by drilling coincident gravity and magnetic anomalies that had been selected using a conceptual model of copper source, transport and deposition, together with continental tectonic and lineament analysis. It holds a resource of 2660 Mt at 1.2% Cu, 1.4 kg/t U_3O_8 and 0.5 g/t Au, which made it the world's largest uranium resource and fifth largest copper resource at the end of 2001 (Esdale et al., 2003). As scientists have learned more about the Olympic Dam, more IOCG deposits have been discovered, and the group's growing diversity challenges the unifying models that explain the origin of IOCG deposits. Nature is complex, and demands geochemical tools that can be confidently applied and interpreted to disentangle the processes that produce these resources.

U-Th/He dating is a powerful tool for understanding the timing and duration of geological processes, and its application to different minerals is sensitive to processes occurring at a variety of temperatures and depths, including shallow crustal processes that are difficult to study with other geochronometers (Farley, 2002), making it uniquely well-suited for studying the full history of an ore deposit, including hydrothermal processes (McInnes et al., 2005). The key to applying U-Th/He dating to a given mineral is to understand how helium behaves within it; this is the basis on which scientists relate these observations to natural phenomena. For apatite, titanite, and zircon, helium's behavior at different temperatures is well understood (Farley, 2000; Reiners and Farley, 1999; Guenthner et al., 2013). More recently, helium in hematite has become well understood, expanding the domain of processes accessible to U-Th/He dating (Farley and Flowers, 2012; Farley, 2018). Another important target mineral for U-Th/He dating is goethite, a hydrous iron oxide that is a direct chronometer of weathering and laterite formation (Shuster et al., 2005; Vasconcelos et al., 2013; H. S. Monteiro et al., 2014). However, these data

are difficult to interpret because goethite decomposes during conventional helium step-heating experiments, producing data that reflect the kinetics of its decomposition and not helium's behavior within it in nature (Farley et al., 2024). Sulfide minerals, too, are promising targets for U-Th/He dating, because they may directly date primary hydrothermal mineralization itself (O. Yakubovich et al., 2020), but they too decompose during conventional helium step-heating experiments. In Chapter 1, I address this problem by developing a method to stabilize these minerals during helium step heat experiments by sealing samples in silica glass ampoules that maintain water- or sulfur-saturated conditions internally while remaining permeable to helium. Therein I develop the mathematical and experimental framework to separate the process of helium permeation through the glass from release by the sample. Chapter 2 applies this method to goethite to measure helium diffusion, demonstrating that it has remarkably high helium retentivity and relatively simple helium systematics. Chapter 3 applies the method to sulfides, the primary hosts of many critical metals. Pyrite and arsenopyrite prove highly retentive, so much so that pyrite is amongst the most helium-retentive minerals studied, comparable to or exceeding monazite (Farley, 2007). I then apply the U-Th/He method to several sulfide samples to date Mesozoic mineralization at the Gympie gold field in Queensland, Australia, and Archean mineralization at the Kidd Creek volcanogenic massive sulfide deposit in Ontario, Canada. These results are promising, but expose the complication that the uranium and thorium in sulfides are held by inclusions of other minerals, requiring that the interpretation of such data be bespoke by locality.

Chapter 4 turns from clocks to models. Thermodynamic models of equilibrium phase assemblages, such as MELTS (Ghiorso and Sack, 1995; Gualda et al., 2012) and HeFESTo (Stixrude and Lithgow-Bertelloni, 2011; Stixrude and Lithgow-Bertelloni, 2024), are the standard way to follow magmas and the mantle through pressure, temperature and composition, but they are too slow and unstable to be applied to detailed models that more accurately represent the complexity of natural systems in which composition varies in space and time. *nGibbs* is a set of general deep-learning emulators of these models that run orders of magnitude faster, conserve mass, and can stand in for the parent models in existing workflows (M. L. M. Gleeson et al., 2025).

Chapter 5 concludes the thesis with a field study of what I would consider

to be the most important ore deposit in the world: the Bushveld Complex in South Africa. Not only does the Bushveld provide the majority of the world's platinum-group metals, as well as vast reserves of chromium, vanadium, iron, and titanium, but it is also the type locality for layered mafic intrusions: a remarkably well-preserved natural laboratory for studying the fundamental processes that operate within the magmatic plumbing systems that supply volcanoes on the surface. At the Bushveld Complex, I studied the platiniferous intersection of the Main and Critical Zones with the sedimentary country rock that the Bushveld intrudes, transforms, and consumes. With detailed field work and structural mapping in the vicinity of the village of Draaikraal in the Mpumalanga Province, I show that the local deformation of the country rock was synmagmatic, whereas previous researchers had suggested either that this deformation occurred beforehand (van der Merwe, 2007), or afterwards as the intrusion cooled and sank (Sharpe and Chadwick, 1982). I show that this deformation is structurally related to a prominent topographic lineament in the area, and I correlate distinctive marker beds across it to show that this feature did not significantly displace rocks on either side of it, arguing instead that this feature coincides with the edge of the underlying mid-lower Critical Zone, which had buffered the overlying Main Zone to the north from crustal contamination, leaving the south to be contaminated by iron-rich fluids that facilitated its oxidation and weathering.

The Bushveld is a fitting place to end. Indeed, the valley in which Draaikraal resides is blanketed by goethite laterites, lying just above the mineralized upper Critical Zone which hosts its platinum group elements within exotic sulfides. Furthermore, the Bushveld Complex hosts countless classical problems in igneous petrology that a reliable, fast surrogate thermodynamic model may be applied to, exploring the chemical consequences of many scenarios of magma evolution and emplacement to explain the origin of the Bushveld and other PGE-bearing ore deposits.

Chapter 1

A SIMPLE METHOD FOR HELIUM STEP HEAT EXPERIMENTS AT HIGH WATER OR SULFUR FUGACITY

Helium isotope studies and U-Th/He geochronology of sulfur-bearing and hydrous minerals offer direct windows into the timing and context of hydrothermal mineralization, provided the conditions governing radiogenic helium retention and loss in nature are known. However, assessing helium retentivity at elevated temperatures is currently hampered by these minerals' instability under the high-temperature, high-vacuum conditions of conventional helium step-heating experiments. This paper introduces a novel method for conducting helium step-heating experiments under hydrothermal and sulfur-saturated conditions, designed to extend our knowledge of helium diffusion kinetics in hydrous and sulfide mineral phases. In this approach, silica glass ampoules encapsulate the mineral of interest, maintaining internal sulfur vapor or water vapor saturated conditions within the ultra-high vacuum of the surrounding extraction line, while the high helium permeability of fused silica permits quantification of evolved helium using standard temperature control and mass spectrometry procedures. This work presents the experimental framework and key equations necessary to deconvolve the secondary processes of helium permeation through the silica ampoule (referred to here as permeation lag*) from the thermally activated degassing of helium from the encapsulated mineral. Permeation lag is quantified with a first-order permeation model and is accounted for by one of two methods: the* permeation step method*, which measures the ampoule's residual helium directly with low-temperature permeation steps, or the* bounding method*, which constrains the residual helium from above by assuming steady-state permeation during long steps. Both approaches rest on the ampoule's permeation rate constant, which is experimentally determined. Validation experiments measuring helium permeation through empty ampoules and ampoules containing only water, sulfur, or Durango*

apatite demonstrate that permeation lag is readily correctable in well-designed experiments. While sulfur vapor saturated experiments are straightforward to carry out with this method, the helium permeation rate for water-vapor saturated experiments is observed to decrease over time, requiring continuous monitoring during the experiment. This method expands standard experimental capabilities to encompass mineral phases that decompose under high vacuum, including sulfides and iron oxyhydroxides, opening new avenues for thermochronometric investigation of hydrothermal ore deposits.

Author contributions. This chapter is being prepared for publication with co-authors J. Treffkorn and K. A. Farley. **B.J.T.:** methodology, investigation, formal analysis, writing – original draft. **J.T.:** methodology, investigation. **K.A.F.:** conceptualization, funding acquisition, writing – review and editing. Generative AI was used as described in the front matter.

1.1 Introduction

Noble gas geochronology and thermochronometry ($^{40}Ar/^{39}Ar$ and U-Th/He) on hydrous and sulfur-bearing mineral phases such as hornblende, clay minerals, biotite, goethite, and pyrite have already been leveraged as a powerful tool for understanding regional thermal histories as well as the timing of iron or rare metal mineralization (McInnes et al., 2005; Heim et al., 2006; Vasconcelos et al., 2013; Nomade et al., 2004; O. Yakubovich et al., 2020; O. Yakubovich et al., 2021). The interpretation of these data, however, is often complicated by the fact that helium (or argon) within hydrous or sulfur-bearing phases often degasses differently in the laboratory and in nature (Hanson et al., 1975; J. K. W. Lee, 1993; Lo et al., 2000; Farley et al., 2024).

Experiments aiming to characterize the migration mechanisms and rates of noble gases through hydrous phases yield different results when conducted under hydrothermal conditions as opposed to traditional experimental methods in vacuo. Hydrothermal step heat experiments for argon in biotite and hornblende showed higher retentivity and comparatively favored diffusive loss mechanisms relative to experiments conducted in vacuo, and these results are more likely to accurately reflect loss mechanisms in nature (Harrison, 1982; Harrison et al., 1985); however, such experiments are labor-intensive and comparatively inflexible relative to modern helium step heat methods (Farley et al., 1999).

Figure 1.1: A typical silica glass ampoule used for helium step heat experiments. The ampoule shown here contains Durango Apatite crushate sieved to 300 µm (radii ≈ 150 µm). The ampoule is roughly one cm long and one mm wide. The bars visible in the background are each separated by 1 mm.

Recently, some workers have taken advantage of the high helium permeability and selectivity of pure silica glass when designing novel experimental methods for U-Th/He geochronology (O. Yakubovich et al., 2020; O. Yakubovich et al., 2021), but these ideas have only been applied in a limited way to characterize helium migration in sulfur-bearing or hydrous phases (e.g., monotonic ramped heating experiments: O. V. Yakubovich et al., 2019). This work aims to apply decades of research on helium permeability, solubility, and diffusion in fused silica glass (e.g., Jones, 1953; Norton, 1953; Rogers et al., 1954; Altemose, 1961; Swets et al., 1961; Doremus, 1966; Masaryk, 1968; Shelby, 1971; Shelby, 1972; Shelby, 1976; Shackelford et al., 1972) to adapt existing agile step-heat methods (Farley et al., 1999) to investigate helium loss rates and mechanisms in sulfur-bearing and hydrous mineral phases.

Encapsulating minerals in silica glass ampoules allows experimentalists to maintain hydrothermal or sulfur-saturated conditions within the ampoule without otherwise altering the experimental design. Depending on the thermodynamics of the phase of interest, this method may stabilize the phase at elevated experimental temperatures, allowing for detailed characterization of helium migration in sulfide and hydrous minerals that hitherto has remained poorly understood.

Table 1.1: Silica Glass tubes that were the subject of helium step heat experiments in this work. Ampoules labeled "Quad" had evolved 4He quantified by isotope dilution using a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer, whereas the ampoule labeled "MC" had both evolved 3He and 4He quantified using a Helix-MC-Plus^{10K} multi-collector mass spectrometer. Where 3He was measured, the ampoule was soaked in tank helium with an isotope ratio of ~0.15 Ra.

Ampoule Contents	Mass Spec.	Inner Diameter µm	Outer Diameter µm	Geometric factor G cm^{-2}	Volume Ratio $\left(\frac{V_{glass}}{V_{gas}}\right)$	Content Mass	Helium Source
Empty	Quad	630	960	4785	1.32	—	Atmosphere
Empty	MC	747	1159	3264	1.41	—	Tank He
Sulfur	MC	529–550	835–872	5992	1.5	~0.2 mg	Tank He
Water	MC	487–515	734–776	7789	1.27	~0.5 mg	Tank He
Durango Apatite	Quad	850	1350	2394	1.52	3.18 mg	Radiogenic

1.2 Experimental Methods

Five helium step heat experiments of sealed silica glass ampoules were undertaken at Caltech to demonstrate the applicability of equations derived here to real helium step heat experiments of geologic materials using a projector-lamp heating apparatus, described in detail in Farley et al. (1999). Four of these experiments involved introducing helium into empty, water-filled, and sulfur-filled ampoules, then measuring helium permeation to assess the applicability of theory outlined in this chapter to helium step heat experiments at Caltech, and to reproduce measurements of permeability rates from the literature (Norton, 1953; Jones, 1953; Swets et al., 1961; Altemose, 1961). The fifth experiment was a standard helium step heat experiment on encapsulated Durango apatite, a widely used standard for helium diffusion experiments and U-Th/He thermochronometry (Zeitler et al., 1987; Wolf et al., 1996; Farley, 2000; Shuster et al., 2004; Shuster et al., 2006; Flowers et al., 2007). The aim of that experiment was to be a practical test in deconvolving the process of helium permeation through silica glass ampoules from thermal helium loss of a well-studied geologic material. Table 1.1 summarizes the dimensions, geometric factors, and encapsulated materials of this work.

To prepare the silica glass ampoules, ~10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in

sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thin pipettes with a jewelry welding torch and their thin ends were fused closed.

After the glass had cooled, degassed lab-grade sulfur, distilled water, or Durango apatite were added to some of the tubes. Two tubes were left empty. To seal, the top of each pipette was interfaced with a vacuum line and was pumped to a pressure of <0.1 torr. The encapsulated water tube was flash-frozen in liquid nitrogen and then pumped and sealed in a saltwater ice bath to selectively pump away atmospheric gases and not water. Sublimation of sulfur and water ice during pumping was minor. The primary motivation for pumping the tube to rough vacuum was to aid the sealing process. After rough vacuum was achieved, a fine-tipped welding torch was used to seal the top of each ampoule. To determine the geometric factors for permeation of the ampoules, the inner and outer diameters of each ampoule were determined using the measurement function of the Amscope stereo microscope software, which was calibrated and tested during early development of this method.

Except for the encapsulated Durango apatite ampoule, helium gas needed to be introduced into the sealed ampoules before they were loaded into the projector lamp heating cells. About 15 nccSTP of atmospheric helium was introduced into the empty ampoule used for the quadrupole mass spectrometer experiment by heating the ampoule in a muffle furnace to >400°C in atmosphere overnight, which was sufficient to support a short permeation experiment of 4He. However, to conduct step heat permeation experiments on 3He, much larger quantities of helium were necessary. The encapsulated sulfur and water ampoules and the other empty ampoule were soaked in 2–20 torr of high purity tank helium with an isotope ratio of ~0.15 Ra at 300°C for about 24 hours. This procedure introduced 0.6–10 pccSTP 3He and up to 52,000 nccSTP 4He into each tube, allowing for measurements of both 3He and 4He permeation simultaneously.

The sealed ampoules were placed in copper foil packets and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley et al. (1999). Before each experiment began, each ampoule was washed with ethanol, air-dried, loaded in the lamp cell, pumped to high vacuum, and was held at 100°C for 30 minutes under dynamic vacuum to desorb ethanol, water, and other atmospheric species on the ampoule, copper packet, and lamp cell. Less than 0.5 ‰ of the radiogenic

helium in 150µm radius Durango apatite would be expected to degas in 30 minutes at 100°C, which would amount to ~0.01 nccSTP for this experiment, using the diffusion parameters of Farley (2000) and the forward equations of Fechtig and Kalbitzer (1966). The atmospheric helium component within the encapsulated Durango Apatite ampoule constrained by permeation steps at the beginning of the experiment was much larger: 1.47 ± 0.18 nccSTP 4He.

We found that the fused silica ampoules were capable of maintaining surprisingly high internal pressures, but ampoules filled with water would often explode at temperatures above 315°C. It is recommended that experimentalists limit the temperature of water-filled ampoules to below 300°C. We have not observed any explosions of sulfur-filled ampoules at any temperature.

Heating steps and isotopic measurements for all samples were automated using custom LabView routines with the exception of the tank helium-soaked empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer after the 7th measurement. For all other ampoules, a Watlow temperature controller held the ampoule at the temperature set point for a pre-programmed duration, after which the ampoule was allowed to cool while the evolved gas was prepared for measurement for about 25 minutes. After the first 7 measurements, the empty tube experiment on the Helix-MC-Plus^{10K} was instead set to an elevated temperature manually while the evolved gas was allowed to accumulate continuously in an attempt to mitigate uncertainties related to frequently changing temperatures (supplementary Eq. (1.48); Fig. 1.24) and continued permeation at room temperature during gas cleanup and measurement. The accumulated gas was measured every 40-120 minutes. For this experiment, the temperature was changed either during the 1 minute pump-out of the lamp cell after inletting gas into the mass spectrometer, or during a period of helium accumulation for measurement. When the temperature was changed during an accumulation period, no permeability was calculated for that step, and the measured abundances of those steps were used only to track the helium contents of the tube.

Helium isotope measurements of evolved gases from step heats of one empty silica glass ampoule and glass-encapsulated sulfur and water ampoules equilibrated with 2-20 torr of high purity tank helium (R ≈ 0.15 R_a) were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from November 19th, 2023 to February 15th, 2024. Helium was

purified by exposing evolved gases to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray. 4He abundances were very large during these experiments, resulting in a small but measurable 4He overspray of up to 0.68 cps at mass 3.5. These overspray signals at mass 3.5 were always less than 1% of the 3He signal at mass 3, suggesting that 4He overspray did not significantly affect 3He signals. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility. These standards show 3He was internally reproducible to within 5% (2σ), whereas 4He was internally reproducible to within 2% (2σ), where reproducibility is estimated as the standard deviation divided by the mean.

Helium-4 evolved during step heats of the empty silica glass ampoule equilibrated with atmosphere and the silica glass-encapsulated Durango apatite was quantified using isotope dilution with a high purity 3He spike on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer from October 30th to November 10th, 2023. Evolved gas from the ampoule in the lamp heating cell was allowed to mix with the 3He spike and the mixture was frozen on charcoal held at 14 K within a cryostat, then helium was selectively released to the mass spectrometer by raising the temperature to 36 K.

A standard gas with a known amount of 4He is routinely measured, at least one time per experiment, to monitor drift in the relative sensitivity of 3He and 4He on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer. Relative sensitivities are determined for each standard measurement and are linearly interpolated in time for sample measurements. Blanks are approximately 5 pccSTP ^{4}He, which is very small compared to all interpreted ^{4}He measurements.

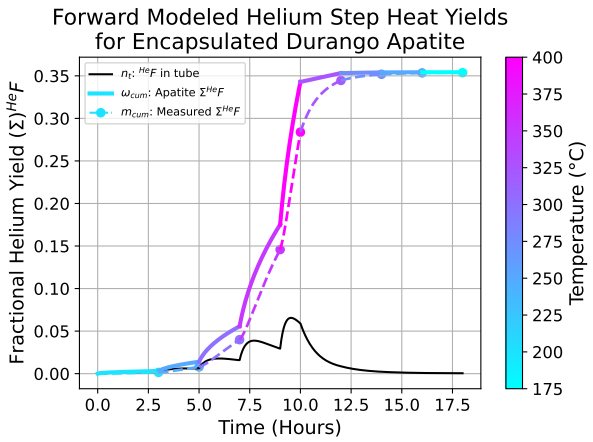

Figure 1.2: An illustrative simulation of the effects of permeation lag on step heat measurements of 150 µm radii spheres of Durango apatite encapsulated in a "standard ampoule" with an outer diameter of 1 mm and inner diameter of 0.6 mm, corresponding to a geometric factor G of 4350.26 cm^{-2}. Some amount of helium degassed from the apatite during an experimental step remains in the tube during measurement and permeates out in later steps. The heating schedule of this model system is given in Table 1.2. This simple model assumes new temperatures and their corresponding steady-state diffusive concentration profile within the glass ampoule are reached instantaneously. Effects of diffusive disequilibrium, if considered, would change the permeated amount for each step by less than 5%.

1.3 Theoretical Framework of Permeation Lag

The process of helium permeation through thin walls of an ampoule is rapid but not instantaneous on experimental timescales. Helium evolved from encapsulated minerals is only available for measurement following permeation through the glass ampoule (Fig. 1.2), producing a predictable "permeation lag" effect in helium measurements that commonly manifests with undermeasurement of helium during prograde heating sequences and corresponding overmeasurement of helium produced during retrograde sequences (Fig. 1.4). This is not merely a theoretical concern: Fig. 1.5 shows the same artifact in real, uncorrected step heat data for the encapsulated Durango apatite experiment described above (§1.2); the correction developed below is applied to this dataset in full in §1.6 (Fig. 1.5).

This effect can be modeled (and corrected for) for a system obeying the simple equation:

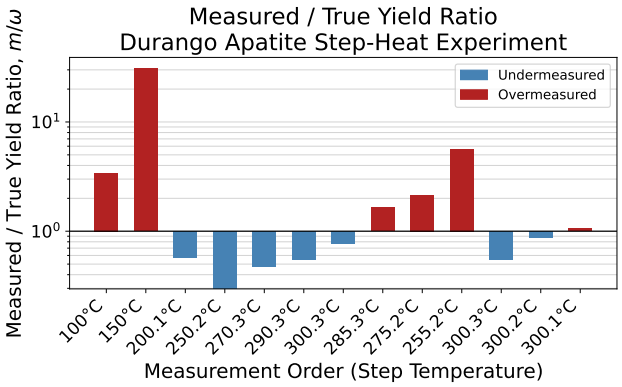

Figure 1.3: Measured-to-true yield ratio, m/ω, for the first 13 experimental steps (through the isothermal 300°C triplicate) of the real encapsulated Durango apatite step heat experiment described above (§1.2) and tabulated in full in Table 1.3, plotted in measurement order and labeled by nominal step temperature. Here m is the raw measured helium yield for a step and ω is its permeation lag-corrected yield; bars are anchored at a ratio of 1 (no anomaly) on a logarithmic axis, so a doubling and a halving of the ratio are given equal visual weight. Steps taken while ramping to higher temperatures are generally undermeasured (blue), as some of the sample's evolved helium has not yet fully permeated out of the tube at the time of measurement, while steps taken while ramping to lower temperatures are generally overmeasured (red), as helium retained in the tube from earlier, hotter steps continues to permeate out. The first two steps (100-150°C) are strongly overmeasured despite being on the heating ramp because little helium had yet evolved from the apatite, so measured yield there is dominated by pre-existing atmospheric helium draining from the tube rather than fresh sample degassing.

$$n_t = n_0 + \omega - m \tag{1.1}$$

Where n_t is the amount of helium within the ampoule after time t, n_0 is the initial amount of helium within the ampoule (which is inherited from previous heating steps), ω is the amount of helium evolved by the encapsulated sample during time t, and m is the amount of measurable helium that has permeated out of the ampoule into the high-vacuum chamber during time t.

Heating steps are consecutive, so the helium inventory quantified between two steps applies to both the previous and the subsequent step:

$$n_t^i = n_0^{i+1}, \qquad n_0^i = n_t^{i-1} \tag{1.2}$$

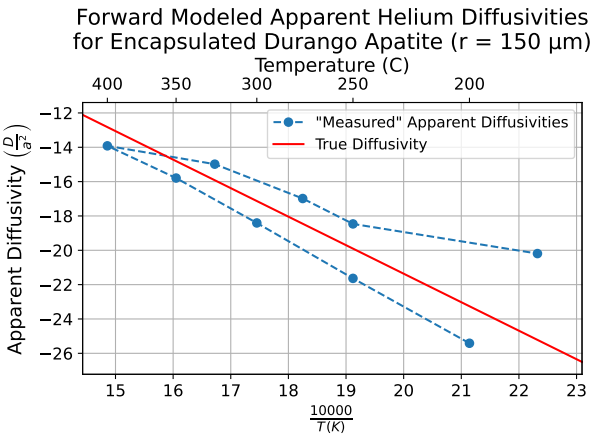

Figure 1.4: An Arrhenius plot of apparent diffusivities inverted from modeled hypothetical measurements of an encapsulated sample of Durango apatite. Permeation lag produces a characteristic effect where measurements of prograde sequences of increasing temperature produce erroneously low apparent diffusivities when inverted using the equations of Fechtig and Kalbitzer (1966), whereas retrograde (decreasing) temperature sequences exhibit the opposite effect. Diffusion parameters are taken from Farley (2000).

Table 1.2: Heating schedule for numerically modeled step heat experiment of silica glass encapsulated Durango apatite.

Temperature (°C)	Time (min)
200	180
250	120
300	120
350	120
400	60
325	120
275	120
250	120
175	120

Recovering ω for a step therefore requires knowledge of the ampoule's helium content immediately before and immediately after it. Sections 1.4 and 1.4 give the two routes used in this work to obtain those quantities.

Helium permeation through fused silica in diffusive steady state is a first order process that can be described with the simple differential equation:

$$\frac{dn}{dt} = -GKnRT \tag{1.3}$$

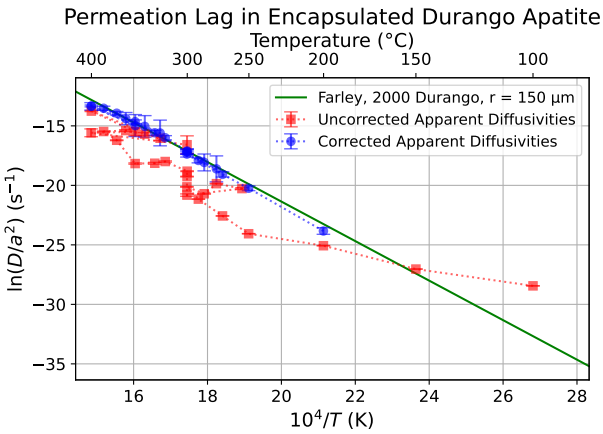

Figure 1.5: An Arrhenius plot of a helium step heat experiment of silica-glass encapsulated Durango Apatite with 2σ error bars. Without permeation lag corrections, the apparent diffusivities display highly irregular behavior that would be unrecognizable to this well-behaved sample – exactly the artifact anticipated by the simulation of Fig. 1.2 above. With permeation lag corrections, the sample agrees well with the reported diffusive parameters of Farley (2000) for the appropriate sieve size ($r = 150\mu m$). As the experiment proceeds beyond temperatures of 300°C, the activation energy of diffusion decreases to reveal a slight downward curl, a phenomenon well-documented for Durango apatite likely due to annealing of crystal defects and radiation damage (Wolf et al., 1996; Farley, 2000). Apparent diffusivities are calculated with the equations of Fechtig and Kalbitzer (1966) with and without permeation lag corrections. Four measurements of the permeation lag corrected apparent diffusivities are within error of 0 at the 2σ level and thus are excluded from the plot.

Where n is the number of helium atoms, t is the time, G is the geometric parameter in units of cm^{-2}, which is a function only of the geometry of the glass membrane, K is the permeation constant for helium in fused silica, which follows the Arrhenius law: $\left(K(T) = K_0 e^{-\frac{E_a}{RT}}\right)$ and has been repeatedly measured for fused silica (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), R is the gas constant, and T is the temperature in Kelvin. Equation (1.3) is identical to the permeation equation used in early studies (Norton, 1953; Altemose, 1961; Swets et al., 1961; Studt, 1962; Masaryk, 1968), except that it is formulated in terms of absolute atomic abundances instead of helium gas pressure by the ideal gas law. It is easily integrated by separation of variables to yield $n_t = n_0 e^{-GKRTt}$ (see supplement, §1.1s, for details). Relating that result to the measured quantity through Eq. (1.1) in the absence of any release from the sample $(m(\omega = 0) = n_0 - n_t)$ gives an expression most intuitively

stated in terms of the fraction of the ampoule's initial helium reservoir that permeates during time t:

$$^{He}F = \frac{m}{n_0} = 1 - e^{-\lambda t}, \qquad \lambda = KGRT \tag{1.4}$$

Where ^{He}F is the fractional permeation yield: the fraction of the ampoule's initial helium reservoir (n_0) that permeates as m (Eq. (1.1)) during a time t of steady state permeation, and λ is the ampoule's permeation rate constant. Collecting K, G, R, and T into λ is useful because λ is the quantity that can be measured directly during an experiment (§1.4), whereas K and G cannot be separated without independent knowledge of the ampoule geometry.

The geometric factor G for a cylindrical membrane with negligible permeation through the top or bottom is determined using the effective area of permeation given by Masaryk (1968):

$$G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} \tag{1.5}$$

where a is the cylinder's internal radius, and b is the external radius. There is no dependence on the cylinder length when permeation through the top and bottom is negligible. This geometric constant will vary according to the experimental design and therefore must be measured by the experimentalist.

Equation (1.4) is only strictly valid with two assumptions:

1. The helium partial pressure in the "measurable" reservoir outside of the ampoule is negligible compared to the partial pressure within the ampoule. Because the ratio of typical ampoule to diffusion cell volumes is ~1/10,000 (Farley et al., 1999), this assumption is essentially always reasonable.

2. The helium concentration profile within the ampoule's glass walls is in diffusive steady state between the inner and outer walls. The real steady state helium concentration profile is well-approximated by the ideal steady state profile for ampoules with a glass/gas volume ratio less than 2 (Figs. 1.16, 1.20). For these ampoules, (1.4) is accurate to within 1% after the establishment of a diffusive steady state. Additionally, the diffusive

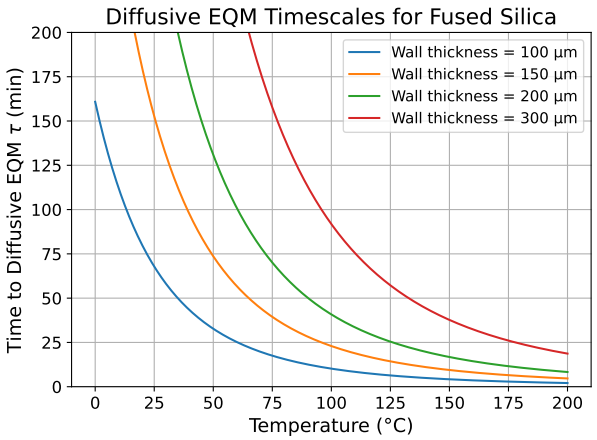

Figure 1.6: Timescales as a function of temperature to reestablish a steady state diffusive concentration profile for different glass thicknesses. These curves correspond to the timescales at which the Fourier number equals 1. 4He diffusivities are taken from Swets et al. (1961).

steady state assumption is ephemerally violated when the temperature is changed due to the temperature dependence of the glass/gas partitioning of helium. The ampoule should therefore be held at the permeation temperature for at least one diffusive timescale $\tau = L^2/D$ after every temperature change before measuring, where L is the wall thickness and D the diffusivity of helium in silica glass (Swets et al., 1961). In this work, low-temperature heating steps are repeated to allow for the new diffusive steady state concentration profile to become established. The ampoules of this work reach diffusive equilibrium in 30–70 minutes at 80°C (Fig. 1.6; supplementary Figs. 1.25 and 1.26). Comparisons between the abundance of helium within glass as a function of temperature limit the error introduced by thermal perturbation to less than 5% (supplementary Fig. 1.24).

1.4 Experimental Approaches Addressing Permeation Lag

A combination of analytical and experimental methods may be used to correct for permeation lag and properly account for all helium evolved from the sample. Some common practices for noble gas measurements may introduce error when using this method. For example, it is common practice to purify and collect emitted helium with the diffusion cell open to the vacuum line at room temperature for extended times, often 15-30 minutes, and it is assumed

that no additional helium evolution occurs. This assumption is reasonable for commonly measured geological materials, but it can be problematic when unexamined during permeation steps of silica glass encapsulated materials because helium permeation in silica glass is significant even at room temperature. In this work, permeation during these waiting times is often mathematically corrected for using Eq. (1.4), assuming permeation continues at room temperature. Alternatively, there are experimental approaches to address ongoing permeation during these waiting periods: they may be shortened to minimize their impact on the measurement, or the ampoule may be held at the warm permeation temperature during the gas purification and cryofocusing processes, so that permeation continues at the same temperature during this time without introducing complicated thermal histories.

From Eq. (1.1), it is immediately clear that when the evolved helium ω is large with respect to the amount of helium within the ampoule, n_t or n_0, then $\omega \approx m$ and permeation lag can be neglected. Unfortunately, this is not the case for most phases commonly subjected to helium step heat experiments with routine heating schedules (e.g. hour-timescale steps at 100-400°C for apatite or zircon), but it is sometimes possible to design an experiment with long time-steps where the total evolved helium is large compared to the amount of helium in-residence within the tube. Nonetheless, permeation lag can be quantified and corrected for in *any* experiment where the encapsulated material is effectively closed to helium loss at room temperature. Two approaches are described here to recover the helium evolved from an encapsulated sample (ω) from measured amounts (m): the *permeation step method* (§1.4), which measures the ampoule's helium inventory directly between experimental steps, and the *bounding method* (§1.4), which constrains that inventory from above when it cannot be measured. Both require the ampoule's permeation rate constant, which should be measured during the experiment itself rather than assumed from the literature (§1.4).

The Permeation Step Method

In typical experimental heating steps, there are two fluxes of helium controlling the ampoule's helium abundance: an influx of helium evolved from the sample and an outflux of helium due to permeation. For lower temperature steps, for which helium release from the sample is negligible ($\omega = 0$), the helium abundance within the ampoule is controlled only by the process of helium permeation, simplifying Eq. (1.1) and allowing for Eqs. (1.1) and (1.4) to form

a system of equations that can be solved for n_t and n_0 given a measurement m for an isothermal heating step of known duration.

$$n_0 = \frac{m}{{}^{He}F} \tag{1.6}$$

Where ${}^{He}F$ is the fractional permeation yield defined in Eq. (1.4). Such a low temperature heating step is hereby referred to as a *permeation step* in this work. A permeation step does double duty: through the step-to-step continuity of the ampoule's inventory, it constrains both the helium remaining after the preceding high temperature step and the helium inherited by the following one. Placing a permeation step after every high temperature *experimental step* ($\omega > 0$) therefore supplies n_t and n_0 throughout the experiment, and Eq. (1.1) returns the helium evolved from the natural sample, ω (Fig. 1.7).

Equation (1.6) requires the rate constant of permeation for the ampoule in use, $\lambda = KGRT$. This may be assembled from the ampoule's measured dimensions through the geometric factor G (Eq. (1.5)) together with a literature permeability coefficient K for helium in fused silica, which has been measured repeatedly as a function of temperature (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968). Published values should not be relied upon, however, especially for hydrothermal experiments: water entering the glass changes its permeability (Shelby, 1972; Tsuru et al., 2011), and water-bearing ampoules show anomalously high rate constants (§1.7). The rate constant should instead be measured on the ampoule actually in use, during the experiment itself (§1.4). That measurement is best understood as a prerequisite and extension of the permeation step method rather than as an alternative to it: the consecutive isothermal permeation steps it requires are already part of the method, so the only additional cost is the discipline of running enough of them in sequence.

Measuring the Ampoule's Rate Constant In Situ

The permeability coefficient K of helium through nominally anhydrous silica glass is well known at relevant experimental temperatures for water-free glass (Norton, 1953; Altemose, 1961; Swets et al., 1961; Rogers et al., 1954; Masaryk, 1968). Literature values need not describe the ampoule actually in use. Water diffuses into glass and changes its helium permeability (Shelby, 1972; Tsuru et al., 2011), so K drifts during hydrothermal experiments; and the volume occupied by a sample or buffering phase raises the effective rate constant above

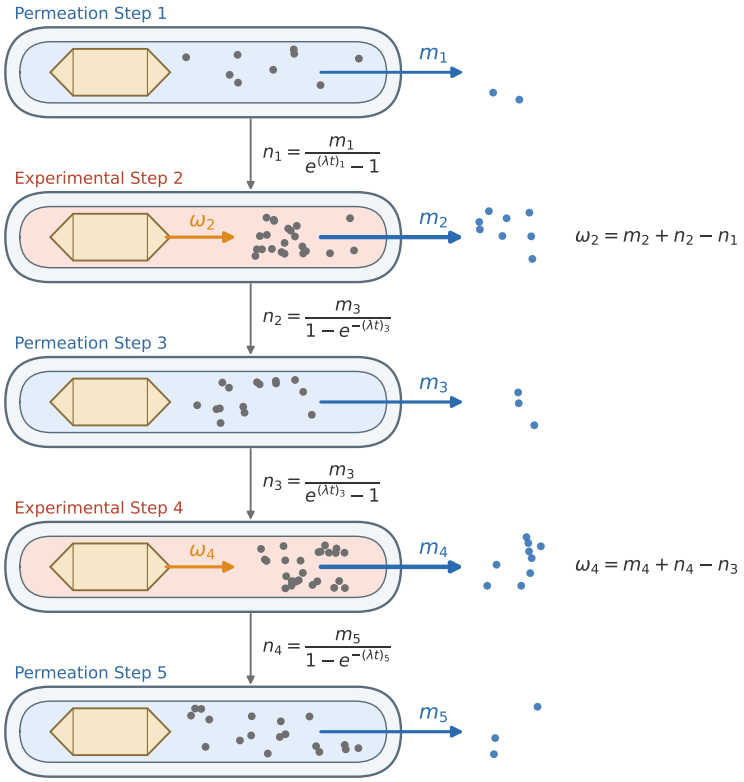

Figure 1.7: A schematic illustration of the permeation step method. Each row is one step of an experiment in chronological order. The ampoule interior is tinted by step temperature. During high-temperature experimental steps the sample releases helium (ω) into the ampoule, and during every step helium permeates out of the ampoule and is measured (m). n_i is the helium in the ampoule at the end of step i. For each low-temperature permeation step, the sample releases no helium, so the measured helium determines the ampoule's helium content (Eq. (1.4)). These determinations of the ampoule's helium inventory are used to return ω for each experimental step (Eq. (1.1)).

the value implied by the ampoule's nominal internal geometry (§1.7). Neither effect is captured by G and a tabulated K.

Fortunately, the rate constant can be measured without knowing either quantity separately. Regardless of the values of K and G, helium permeation through glass remains a first order process obeying Eq. (1.4), which is the recognizable decay equation written with the rate constant $\lambda = KGRT$ defined above:

$$n_t = n_0 e^{-\lambda t} \tag{1.7}$$

λ can be experimentally determined alongside n_0 by using a non-linear least

squares optimization to fit a consecutive series of at least three isothermal permeation step measurements to the following equation, expressed in terms of cumulative measured helium over cumulative time:

$$m_{cum} = n_0 \left(1 - e^{-\lambda t_{cum}}\right) \tag{1.8}$$

We fit individual step yields rather than cumulative yields, because cumulative fitting propagates the errors of early measurements into later measurements, unduly biasing the solver to fit the earlier measurements.

$$m_{i+1} = n_0 \left(1 - e^{-\lambda t_{i+1}}\right) - n_0 \left(1 - e^{-\lambda t_i}\right) \tag{1.9}$$

We used the Levenberg-Marquardt implementation in SciPy (Virtanen et al., 2020), weighted by each measurement's own uncertainty, with initial guesses $\lambda \approx 6 \times 10^{-6}\,\mathrm{s}^{-1}$ and $n_0 \approx m_i / \left(1 - e^{-\lambda t_i}\right)$ evaluated at the first measurement of each series. Uncertainties come from the fit covariance matrices. The same optimization can be accomplished with the "Solver" function in Microsoft Excel.

This fit requires that enough cumulative time has passed to reveal the curvature of cumulative helium yield against time, or else the problem will be poorly conditioned. This can be assured in practice by waiting until the rate of helium permeation has slowed markedly after three or more isothermal permeation measurements before advancing to the next high-temperature experimental step. The exercise can technically be done with as few as two consecutive measurements, although the solver would then be unable to estimate uncertainties on the fit parameters.

In addition to the initial helium contents of the ampoule n_0, this procedure simultaneously determines the product of the permeability and geometric factor for the silica glass ampoule. Once $\lambda = KGRT$ has been satisfactorily constrained over a range of temperatures, the activation energy E_a of permeation and the product of the geometric factor and permeability at infinite temperature, $K_0 G$, can be fit using linear regression. Determining these values allows for extrapolation of the permeability to the high temperatures of the experimental steps, which can be useful to an experimentalist who wishes to design a heating schedule wherein the permeation lag is negligible for some experimental steps

but does not know the permeability coefficient K and/or the geometric factor G.

The rate constant costs little to measure. A series of consecutive isothermal permeation steps is already required by the permeation step method (§1.4), so the additional expense is the discipline of running enough of them in sequence to resolve curvature rather than a single step in isolation. We therefore treat in-experiment measurement of λ as a routine part of the permeation step method rather than as an optional refinement, and as indispensable for hydrothermal experiments, where the glass takes up water during the run and no tabulated value can be expected to hold. Where λ is measured repeatedly through an experiment, its drift can be tracked and interpolated; where it is measured once, it at least applies to the ampoule at hand rather than to an idealized dry one.

The Bounding Method

Permeation steps require a measurable signal at low temperature. When the encapsulated sample is retentive and its yields are small, the low-temperature signal can fall to blank level and Eq. (1.6) cannot be applied. An alternative is available when steps are long compared with the permeation timescale of the ampoule, which is common for retentive phases: rather than measuring the ampoule's residual helium, it can be bounded from above. Permeation lag is a significant effect for typical heating schedules operating on timescales of a few hours (e.g., Farley, 2000; Shuster et al., 2004; Figs. 1.2 and 1.4), but its relative significance decreases with step duration and increased temperature (Fig. 1.8). Consider an arbitrarily long experimental step: after some time, a steady state $\left(\frac{dn}{dt} = 0\right)$ will be reached where the helium permeation rate out of the ampoule is balanced by the production rate of helium within the ampoule from the degassing of the natural sample. The upper bound for this steady state concentration as a function of measured helium abundance (Eq. (1.10)) is derived in the supplement, §1.3s (Eq. (1.58)).

$$n^{ss} \leq \frac{m}{\lambda t - 1} \tag{1.10}$$

Equation (1.10) only applies when step heats are long enough (roughly 5-24 hours depending on temperature and ampoule geometry, Fig. 1.27) to assume steady state permeation ($\lambda t >\sim 5$), which requires that G and K are known or

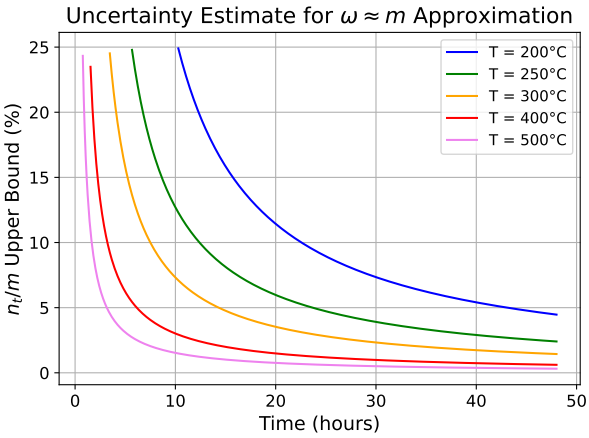

Figure 1.8: For long steps and high temperatures, the abundance of helium within the ampoule will reach a steady state and will be constant as the measurable helium outside the ampoule in the lamp cell increases. As the step gets longer, this amount becomes small relative to the amount of measurable helium, and the measurable helium can be used to approximate the total amount of degassed helium with only minor inaccuracy. Eq. (1.10) is plotted here to show the upper bound relative amount of helium trapped within the ampoule normalized to the measurable amount. The geometric factor for these calculations is 4350 cm^{-2}, corresponding to an ampoule with an inner and outer diameter of 0.6 and 1 mm respectively. Permeability values (K) are for 4He in silica glass and are taken from Swets et al. (1961).

can be conservatively estimated. It also assumes that the production rate of helium within the ampoule due to degassing of the encapsulated natural sample is constant over time at a given temperature. Typically, the helium production rate Ω decreases over time in most experiments, which would produce a smaller steady state He concentration than what is predicted by Eq. (1.10) (Fig. 1.9). Ω decreases over time for helium loss governed by volume diffusion (Wolf et al., 1996; Reiners and Farley, 1999; Farley and McKeon, 2015; Farley, 2018; Farley et al., 2024), or first order processes (O. V. Yakubovich et al., 2010; W. D. Wilson and Bisson, 1973), except for the case of initially inhomogeneous distributions at low helium loss fractions (Goldsmith et al., 2020; Ketcham et al., 2011; Shuster et al., 2004; Farley et al., 1996) or radiation damage annealing (Cherniak, 2019; Guenthner et al., 2013; Flowers et al., 2007; Flowers et al., 2009; Shuster et al., 2006). For experiments where these factors are of interest, the permeation step method (§1.4), which handles permeation lag explicitly, is superior. Additionally, Eq. (1.10) is conservative whenever the value of λ used does not exceed the true one, because a lower bound on

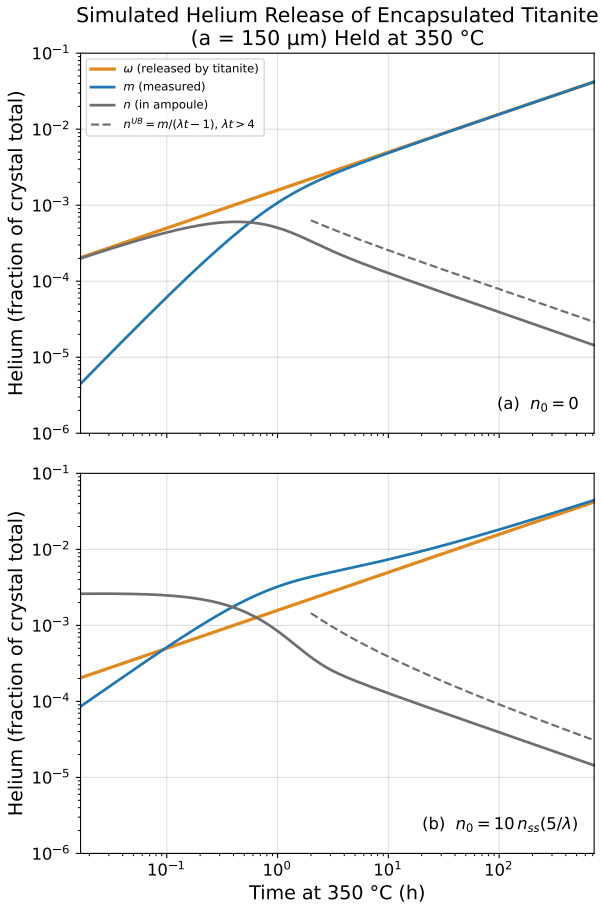

Figure 1.9: Simulated isothermal hold at 350°C of titanite (spherical diffusion domain of radius 150 µm; $D_0 = 60\,\mathrm{cm^2\,s^{-1}}$, $E_a = 44.6\,\mathrm{kcal\,mol^{-1}}$; Reiners and Farley, 1999) sealed in an ampoule with outer and inner diameters of 1.0 and 0.8 mm ($1/\lambda = 30$ min), showing the helium released by the crystal (ω), measured (m), and held in the ampoule (n), together with the upper bound on the ampoule's helium from the measurement alone, $n^{UB} = m/(\lambda t - 1)$ (Eq. (1.10)), drawn for $\lambda t > 4$. The ampoule starts with inherited helium equal to 0.0 (a) or 10 (b) times the idealized steady inventory $n_{ss} = \dot{\omega}/\lambda$ at $\lambda t = 5$. Because release by volume diffusion slows with time, n stays below about half of n^{UB}. As λt grows, m converges on ω: permeation lag becomes negligible.

λ yields an upper bound on the residual. Any λ that can be independently shown to underestimate the ampoule's true rate constant is therefore admissible. The bounding method is consequently an excellent fallback when the small permeation step yields of retentive phases cannot be well resolved.

Using the steady state concentrations calculated with Eq. (1.10) from a long duration step (1) and the previous step (0) as upper bound estimates in Eq. (1.1)

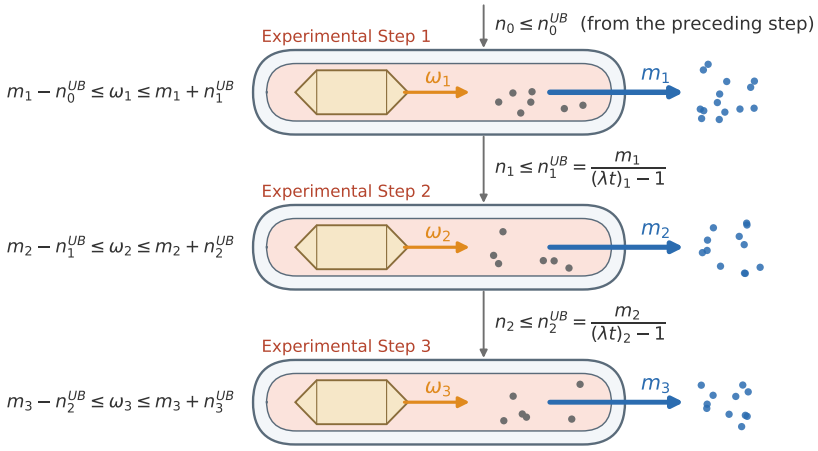

Figure 1.10: The bounding method, for a sequence of long experimental steps ($\lambda t \gtrsim 5$). The helium left in the ampoule at the end of each step is bounded from above by that step's own measurement (Eq. (1.10)), and the bounds at the two ends of a step bound ω (Eq. (1.11)). Symbols as in Fig. 1.7.

yields the inequality that can be used to bound the amount of helium evolved from the sample during step 1 without permeation steps (Fig. 1.10):

$$m_1 - \frac{m_0}{(\lambda t)_0 - 1} \le \omega_1 \le m_1 \left(1 + \frac{1}{(\lambda t)_1 - 1}\right) \tag{1.11}$$

The relative uncertainty $\left(\frac{m}{n^{ss}} = \frac{1}{\lambda t - 1}\right)$ is plotted in Fig. 1.8 as a function of time and temperature for a standard ampoule with geometric factor G of 4350.26 cm^{-2}. To achieve relative uncertainties below $\sim 5\%$, steps longer than 24 hours are typically required at temperatures below 300°C.

1.5 Analytical Methods

Permeabilities were calculated for the empty ampoule experiments by solving Eq. (1.4) for permeability K.

For the empty ampoule experiment, the rate constant λ and the ampoule's initial helium content n_0 were fit independently for ^{3}He and ^{4}He to each run of consecutive isothermal permeation steps using Eq. (1.9) (§1.4). The first steps of the 125 and 200°C runs were excluded because the glass had not yet reached a diffusive steady state (assumption 2 of §1.3); excluding them raises the R^2 of the Arrhenius relationship in Fig. 1.12a from 0.79 to 0.99 (^{4}He) and 0.98 (^{3}He). The last two 250°C accumulations were also excluded: doing so decreases the χ^2 statistic of the fit for ^{4}He by three orders of magnitude.

Besides the encapsulated Durango experiment, which contained a helium source, the helium contents (n_0, n_t) of the other ampoules changed only by decreasing due to helium permeation through silica glass throughout the experiments. These experiments were concluded when measured helium permeate of a high temperature step was indistinguishable from blank. The values of n_0 and n_t for each step were directly quantified after the experiments concluded using the total yield. Assuming that the helium permeated during the short pump-out time of the lamp cell is negligible, the total yield is simply the sum of all the measured abundances of helium for each step. The atmosphere-equilibrated empty tube experiment on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer was pumped for 15 minutes between steps, the encapsulated apatite experiment was pumped for 5 minutes between steps, and the Helix-MC-Plus^{10K} experiments on silica glass ampoules equilibrated in 2-20 torr tank helium were pumped for only 1 minute between steps. This pump-out time was short enough to neglect for all but the atmosphere-equilibrated empty tube, whose longer 15-minute pump-out and short (45-minute) heated steps required small explicit pump-out and gas-preparation corrections using Eq. (1.4); see the supplement, §1.5s, for the full derivation.

One high temperature measurement – the last step of the empty ampoule experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer – showed unusual, non-linear signal evolution and required a bespoke recalculation to recover its 4He and 3He yields; see the supplement, §1.5s, for details.

The encapsulated apatite experiment included high temperature steps ranging from 100-400°C, each preceded and followed by a sequence of two or three permeation steps. Equation (1.6) was applied to solve for the helium contents of the ampoule (n_0) between steps for permeability corrections. Only the last of each sequence of room temperature measurements (shown in **bold** in Table 1.3) was used to solve for n_0 using Eq. (1.6), because the temperature records showed that the encapsulated apatite had not yet cooled to room temperature by the time the first permeation step at room temperature began, although each pair of permeation steps always yielded values of n_0 within uncertainty (2σ) of one another. Measured helium abundances of the other permeation steps taken before reestablishing an equilibrium steady state concentration profile (shown in *italics* in Table 1.3) were added to or subtracted from the n_0 value to propagate the determined helium contents of the ampoule back or

forward in time to the high temperature experimental steps for permeation lag corrections using Eq. (1.1). Measurement uncertainties are estimated using standard deviations of sensitivities and blanks determined by standards. These measurement uncertainties are propagated through Eqs. (1.4) and (1.1) in quadrature, in addition to a general 10% uncertainty for the use of Eq. (1.4) following the recommendations of this work (see discussion). Four of the 25 high temperature experimental steps of the encapsulated Durango apatite produced corrected yields within error (2σ) of 0 and were excluded from Fig. 1.5 (see figure caption). Uncertainty for a step's fractional cumulative yield was estimated as the sum of the uncertainty of its fractional yield plus the uncertainties of the measured fractional yields of all previous measurements, added in quadrature. The total yield was estimated based on the precisely-measured mass of the encapsulated Durango apatite (3.18 mg) and its helium concentration of 194.2 nccSTP/mg (Farley, 2000), a value independently verified in-lab (supplement, §1.5s). No uncertainty in the total yield estimation is propagated through to the diffusivities reported here.

Numerical Diffusive Permeation Model for Arbitrary Thermal History and Diffusive Disequilibrium

Most of the permeation measurements performed on the Helix-MC-Plus^{10K} mass spectrometer were complicated by additional permeation occurring at variable temperature during long cooling times following high temperature experimental steps. Such a complication can be effectively avoided by limiting the duration of time for which the diffusion cell is opened to the rest of the vacuum line, or by keeping the sample at-temperature during this equilibration time. To address this, a numerical diffusive permeation model is used to calculate the ampoule's expected permeation fraction (n_t/n_0) for each step with a complex thermal history (see supplement, §1.2s, for details).

1.6 Results

Predicted and measured fractional yields for all permeation steps agree well (within 1σ; Fig. 1.11, supplementary Figs. 1.28–1.29) with the predictive equations presented in this work and with the permeabilities reported by Swets et al. (1961) and Jones (1953) for the empty and sulfur-filled ampoules across all studied temperatures (24-300°C) and step fractional yields (up to ~80%), but not for the water-filled ampoule.

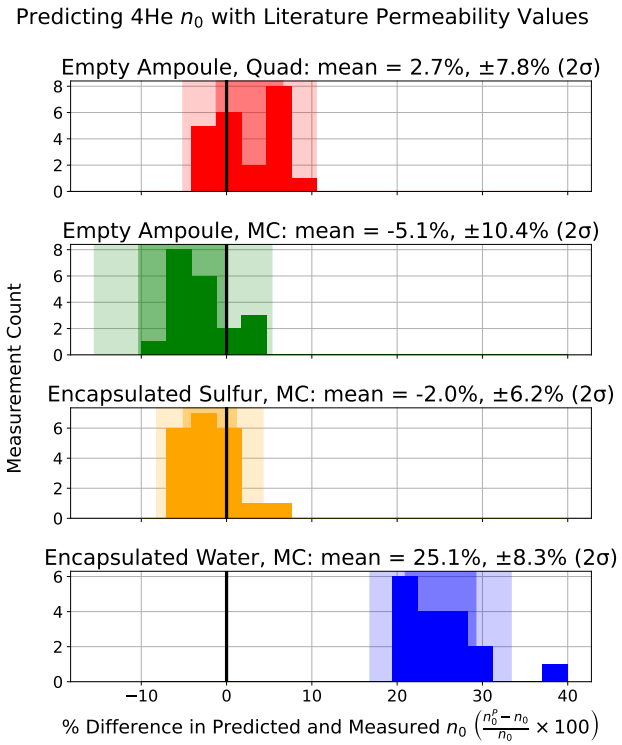

Figure 1.11: Histograms showing the accuracy of 4He n_0 predictions using permeabilities of Swets et al. (1961) from either Eq. (1.22) or a numerical diffusion model from the measured value, with shaded regions behind the histograms corresponding to 1 and 2 sigma (σ) from the average relative difference between measured and predicted values. The measured n_0 values of the empty and sulfur-filled ampoules are within 1σ of the predicted values, suggesting that the permeation model and literature values apply well to those systems, and such models can be used to reliably recover helium contents of permeating ampoules for use in permeation lag corrections. On the other hand, measured and predicted 4He n_0 for water-filled ampoules are well resolved beyond 2σ, suggesting that their helium permeability differs strongly from the literature values. Internal reproducibilities (1σ) of a gas standard repeatedly analyzed during these experiments were better than 1% for 4He, requiring that most of the scatter in these data be related to factors outside of analytical precision, such as long/inconsistent ramp times during heating or uncertainties or rounding of heating times.

Conversely, helium permeation for the ampoule containing water was inconsistent with the permeabilities reported by Swets et al. (1961) and Jones (1953). For these hydrothermal step heat experiments, measured helium partial yields are systematically high by about 25% relative to the expected values (supplementary Figs. 1.11, 1.29), meaning that permeation was systematically faster than expected for the ampoule filled with water. An analysis of this discrepancy

and a hypothesis for its origin are presented in the discussion.

Permeation ratios for 3He and 4He measured in the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer are plotted in supplementary Fig. 1.30. These data present significant scatter, but generally agree with the reported results of Jones (1953), except for measurements at room temperature, which cluster tightly at a permeation ratio of 1:1 for the two isotopes. Additionally, fields corresponding to the ratios of apparent permeabilities for the water-filled and sulfur-filled experiments are plotted for the temperature ranges of those experiments. Results for those experiments are plotted as fields because each step integrated permeation behavior over a complex (non-isothermal) cooling history. These experiments also agree with the results of Jones (1953) and the empty ampoule experiment. The qualitative result of Jones (1953) that silica glass is only slightly more permeable for 3He than 4He is supported by these data, which are in marked contrast to the reported extreme differences in 3He and 4He diffusivity in silica glass of McMurtry et al. (2019), who report instantaneous isotopic fractionations up to a factor of 1000.

Rate constants fit in situ to consecutive isothermal permeation steps of the empty ampoule are shown in Fig. 1.12. Examples of the fit data are shown in Fig. 1.13 and listed in Table 1.4. For the runs fit with three or more steps (150, 175 and 250°C), the fits recover the ampoule's helium content, which is known independently from the total yield but was not used in the fits, to within 3.3 % for ^{4}He and 1.9 % for ^{3}He. The fitted rate constants follow an Arrhenius relationship in λ/T with an activation energy of 20.4 kJ mol^{-1} for ^{4}He, and the fits with three or more steps fall within 7 % of the rate constants predicted from the measured ampoule geometry and permeability from Swets et al. (1961), and Jones (1953) (Fig. 1.12a). These temperatures (125–250°C) are too high to be used for permeation steps of apatite or similarly retentive phases, which must be effectively closed to helium loss during permeation steps (generally below ~100°C, and lower for less retentive phases), but the procedure is identical at lower temperatures and only requires longer cumulative times because λ is smaller.

The same fits provide an isotopic fractionation check (Fig. 1.12b): the fits with three or more steps give $\lambda(^3\text{He})/\lambda(^4\text{He}) = 1.13$–$1.17$, close to the ratio reported by Jones (1953). We do not observe the extreme isotope effects reported by

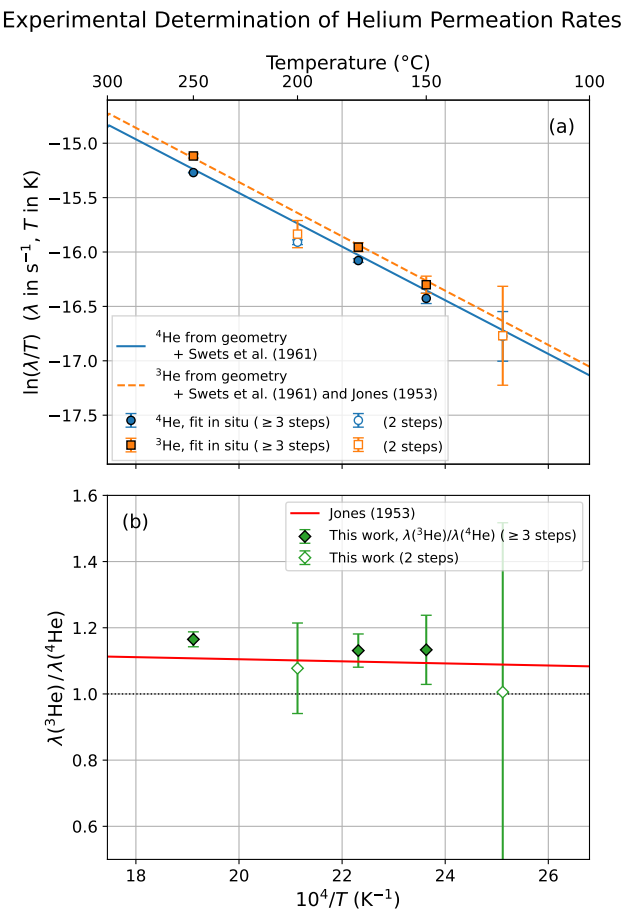

Figure 1.12: (a) Permeation rate constants λ of the empty ampoule fit in situ to consecutive isothermal permeation steps (Eq. (1.9)) for ^{4}He (blue circles) and ^{3}He (orange squares), plotted as $\ln(\lambda/T)$ against inverse temperature. Filled symbols are fits of three or more steps; open symbols (125 and 200°C, first step excluded; §1.5) are exactly determined two-step fits. Lines are predictions from the measured ampoule geometry and permeabilities from (Swets et al., 1961; Jones, 1953), shown as a consistency check. (b) Ratio of the fitted ^{3}He and ^{4}He rate constants compared with Jones (1953). Error bars are 2σ.

McMurtry et al. (2019) for fused silica glass.

The permeation lag-corrected helium diffusivities for the encapsulated Durango apatite experiment are within uncertainty of the Arrhenius relationship for 150µm radii Durango apatite reported by Farley (2000), except for the first two plotted points and the 400°C step (Fig. 1.5). Farley (2000) also observed some low apparent diffusivities for helium in Durango apatite in early steps, although rounding of the concentration profile by diffusive loss or alpha ejection is not expected for interior fragments of large gemmy apatites. The high temperature

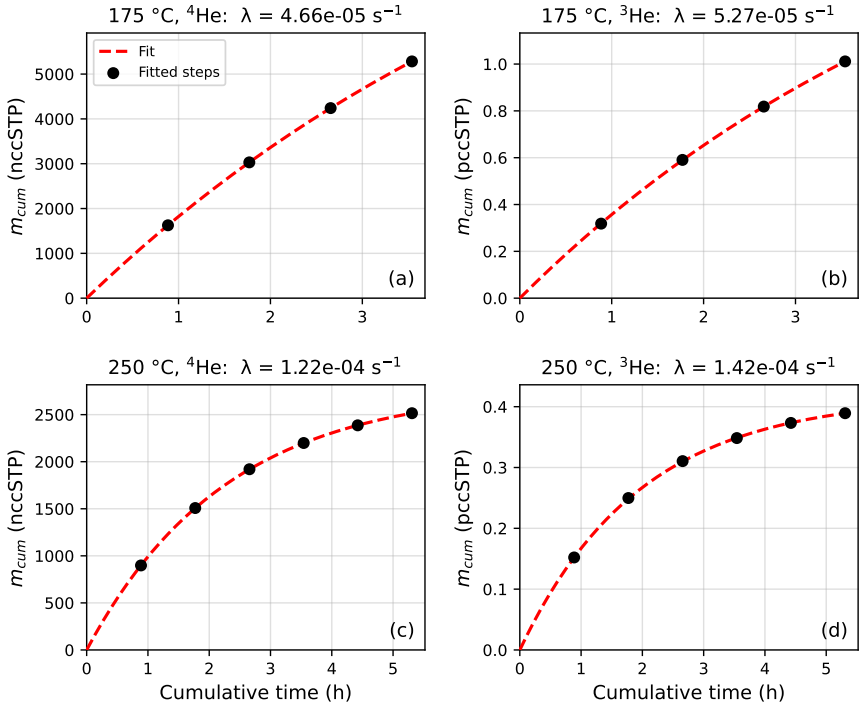

Figure 1.13: Examples of the in situ rate-constant fits for the empty ampoule: cumulative measured helium against cumulative time for the 175°C (a, b) and 250°C (c, d) runs of consecutive isothermal permeation steps for ^{4}He (a, c) and ^{3}He (b, d). Red dashed curves are fits of Eq. (1.9). All fits are shown in supplementary Fig. 1.31.

downward curvature of the apparent diffusivities in Fig. 1.5, however, is well-understood. Wolf et al. (1996) and Farley (2000) discuss similar downward curvature of the diffusion array of Durango apatite at high temperatures. The annealing of radiation damage 'traps' and crystalline defects is the favored explanation for this irreversible change in diffusivity at high temperatures for Durango apatite (Farley, 2000; Shuster et al., 2006). Overall, the permeation lag-corrected apparent diffusivities for encapsulated Durango Apatite reproduce the same results of unencapsulated Durango apatite (Fig. 1.5), suggesting that permeation lag can be reasonably deconvolved from thermal helium emission from minerals or other solid materials of interest when performing step heat experiments to characterize helium retentivity and loss mechanisms in solids.

The encapsulated Durango apatite data also implicitly measure the rate constant of helium permeation through fused silica glass. This experiment uses the permeation step approach to correct for permeation lag, but only uses room

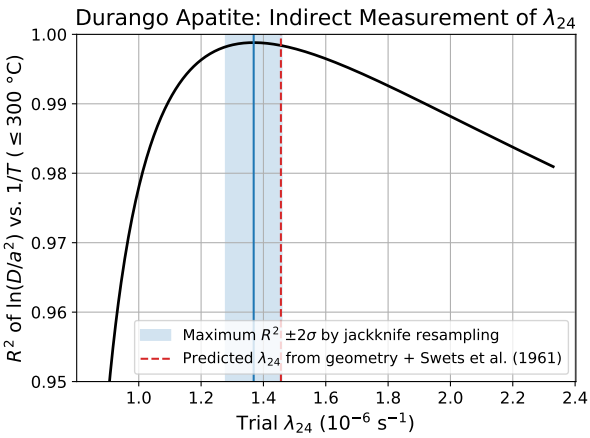

Figure 1.14: Coefficient of determination (R^2) of a linear regression of permeation lag-corrected $\ln(D/a^2)$ on $1/T$ for the 10 encapsulated Durango apatite steps below the high-temperature curl, as a function of the trial rate constant at 24°C (λ_{24}) used for the permeation lag corrections by Eq. (1.6). The prediction from the measured ampoule geometry and permeability from Swets et al. (1961) (red dashed line) is shown for comparison.

temperature ($\sim$ 24°C) permeation steps, so the only free parameter is the rate constant of helium permeation at $\sim$ 24°C: λ_{24}. Assuming that helium diffusivity measured for Durango Apatite should yield linear arrays on an Arrhenius plot, the R^2 statistic can be used as an objective function to be maximized by solving λ_{24}. This approach determines $\lambda_{24} = 1.37 \pm 0.09 \times 10^{-6}\,s^{-1}$ (2σ uncertainty estimated by Jackknife resampling; $R^2 = 0.9988$; Fig. 1.14). The rate constant predicted from the ampoule's measured geometry and permeability from Swets et al. (1961), which was used for the correction shown in Fig. 1.5 ($R^2 = 0.9984$), is $1.46 \times 10^{-6}\,s^{-1}$, agrees with this implied rate constant for helium permeation at room temperature (λ_{24}). Because this self-calibration uses only the measured yields, the known total yield, and the expectation that helium diffusion in Durango apatite is Arrhenian below 300°C, the agreement of the corrected diffusivities with Farley (2000) remains an independent test of the correction.

Table 1.3: Step heat data for the silica glass encapsulated Durango apatite experiment. ${}^{He}F_{Apatite}$ is the permeation lag-corrected cumulative fractional yield of helium released from the apatite. n_0 values (initial ^{4}He content of the ampoule, Eq. (1.6)) are shown in **bold** for permeation steps used directly to solve for n_0, and in *italics* for all other steps, where n_0 is instead propagated forward or backward in time from the nearest bolded permeation step using Eq. (1.1). Blank "Type" entries denote room-temperature measurements preceding the permeation step used to solve for n_0, taken before the diffusive steady state concentration profile was reestablished (see text). Uncertainties are 2σ. Corrected $\ln(D/a^2)$ values shown as "–" correspond to steps where the corrected apatite yield (ω) was within 2σ of 0 and were excluded from the Arrhenius plot (Fig. 1.5).

Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	${}^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	24.3	0.00704 ± 0.00046	**1.48 ± 0.18**				
Experimental	240	100.0	0.1700 ± 0.0011	*1.47 ± 0.18*	0.05 ± 0.24	0.00803%	–	-28.39
	15	28.0	0.00721 ± 0.00039	*1.35 ± 0.16*				
	15	24.0	0.00639 ± 0.00039	*1.34 ± 0.16*				
Permeation	15	24.0	0.00630 ± 0.00039	**1.34 ± 0.16**				
Experimental	240	150.0	0.3090 ± 0.0023	*1.33 ± 0.16*	0.01 ± 0.21	0.00983%	–	-26.98
	15	28.1	0.00542 ± 0.00039	*1.03 ± 0.13*				
	15	24.3	0.00484 ± 0.00039	*1.03 ± 0.13*				
Permeation	15	24.2	0.00485 ± 0.00039	**1.02 ± 0.13**				
Experimental	240	200.1	0.9416 ± 0.0085	*1.02 ± 0.13*	1.64 ± 0.23	0.276%	-23.80 ± 0.28	-25.02
	15	29.9	0.00947 ± 0.00041	*1.72 ± 0.19*				
	15	23.8	0.00786 ± 0.00040	*1.71 ± 0.19*				
Permeation	15	24.0	0.00804 ± 0.00040	**1.70 ± 0.19**				
Experimental	60	250.2	1.097 ± 0.011	*1.69 ± 0.19*	3.70 ± 0.48	0.874%	-20.21 ± 0.20	-24.01
	15	33.0	0.02653 ± 0.00050	*4.30 ± 0.44*				
	15	23.5	0.02016 ± 0.00046	*4.27 ± 0.44*				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	24.0	0.02003 ± 0.00047	**4.25 ± 0.44**				
Experimental	60	270.3	2.508 ± 0.027	*4.23 ± 0.44*	5.26 ± 0.83	1.73%	-19.04 ± 0.22	-22.51
	15	33.9	0.04414 ± 0.00067	*6.98 ± 0.70*				
Permeation	15	23.6	0.03246 ± 0.00057	**6.93 ± 0.70**				
Experimental	60	290.3	4.979 ± 0.058	*6.90 ± 0.70*	9.0 ± 1.3	3.18%	-17.87 ± 0.20	-21.10
	15	33.6	0.06816 ± 0.00095	*10.9 ± 1.1*				
Permeation	15	23.6	0.05050 ± 0.00078	**10.8 ± 1.1**				
Experimental	60	300.3	7.243 ± 0.091	*10.8 ± 1.1*	9.5 ± 1.7	4.72%	-17.34 ± 0.25	-20.07
	15	34.2	0.0843 ± 0.0012	*13.1 ± 1.3*				
	15	23.5	0.06160 ± 0.00094	*13.0 ± 1.3*				
Permeation	15	24.1	0.06097 ± 0.00095	**12.9 ± 1.3**				
Experimental	60	285.3	5.919 ± 0.080	*12.9 ± 1.3*	3.6 ± 1.7	5.30%	-18.04 ± 0.71	-20.65
	15	36.8	0.0729 ± 0.0011	*10.5 ± 1.1*				
Permeation	15	23.6	0.04846 ± 0.00084	**10.4 ± 1.1**				
Experimental	80	275.2	5.382 ± 0.077	*10.4 ± 1.1*	2.5 ± 1.3	5.71%	-18.6 ± 1.1	-19.80
	15	37.0	0.05243 ± 0.00092	*7.53 ± 0.76*				
Permeation	15	23.6	0.03487 ± 0.00070	**7.48 ± 0.76**				
Experimental	100	255.2	3.382 ± 0.052	*7.44 ± 0.76*	0.60 ± 0.90	5.81%	–	-20.22
	15	37.1	0.03206 ± 0.00069	*4.66 ± 0.48*				
Permeation	15	23.6	0.02147 ± 0.00056	**4.63 ± 0.48**				
Experimental	60	300.3	3.684 ± 0.059	*4.61 ± 0.48*	6.68 ± 0.91	6.89%	-17.17 ± 0.20	-20.76
	15	36.3	0.0518 ± 0.0010	*7.60 ± 0.77*				
Permeation	15	23.6	0.03513 ± 0.00077	**7.55 ± 0.77**				
Experimental	120	300.2	9.78 ± 0.17	*7.51 ± 0.77*	11.1 ± 1.2	8.69%	-17.13 ± 0.22	-19.21
	15	35.7	0.0595 ± 0.0012	*8.85 ± 0.90*				
Permeation	15	23.6	0.04087 ± 0.00089	**8.79 ± 0.90**				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	$^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Experimental	180	300.1	14.58 ± 0.27	*8.75 ± 0.90*	13.8 ± 1.2	10.9%	-17.08 ± 0.18	-18.75
	15	35.9	0.0547 ± 0.0012	*7.96 ± 0.81*				
Permeation	15	23.6	0.03782 ± 0.00089	**7.91 ± 0.81**				
Experimental	60	320.3	7.49 ± 0.14	*7.87 ± 0.81*	11.4 ± 1.5	12.8%	-15.96 ± 0.18	-17.94
	15	36.9	0.0813 ± 0.0017	*11.7 ± 1.2*				
Permeation	15	23.5	0.0541 ± 0.0012	**11.7 ± 1.2**				
Experimental	60	330.4	11.18 ± 0.22	*11.6 ± 1.2*	14.5 ± 1.9	15.1%	-15.54 ± 0.21	-18.08
	15	37.4	0.1076 ± 0.0023	*14.9 ± 1.5*				
Permeation	15	23.7	0.0691 ± 0.0016	**14.8 ± 1.5**				
Experimental	60	350.4	19.78 ± 0.41	*14.7 ± 1.5*	28.3 ± 2.8	19.7%	-14.61 ± 0.15	-18.11
	15	37.0	0.1615 ± 0.0035	*23.3 ± 2.4*				
Permeation	15	23.5	0.1071 ± 0.0024	**23.1 ± 2.4**				
Experimental	60	370.4	32.97 ± 0.71	*23.0 ± 2.4*	41.3 ± 4.0	26.4%	-13.90 ± 0.15	-16.15
	15	37.3	0.2207 ± 0.0049	*31.4 ± 3.2*				
Permeation	15	23.5	0.1447 ± 0.0033	**31.1 ± 3.2**				
Experimental	60	385.4	42.32 ± 0.94	*31.0 ± 3.2*	44.8 ± 4.8	33.6%	-13.48 ± 0.17	-15.41
	15	37.8	0.2393 ± 0.0055	*33.5 ± 3.4*				
Permeation	15	23.5	0.1548 ± 0.0037	**33.2 ± 3.4**				
Experimental	60	360.4	27.53 ± 0.64	*33.1 ± 3.4*	15.3 ± 4.1	36.1%	-14.35 ± 0.51	-15.33
	15	39.6	0.1620 ± 0.0039	*20.9 ± 2.1*				
Permeation	15	23.6	0.0963 ± 0.0024	**20.7 ± 2.1**				
Experimental	60	340.4	14.78 ± 0.35	*20.6 ± 2.1*	7.4 ± 2.6	37.3%	-15.00 ± 0.90	-15.63
	15	40.0	0.1029 ± 0.0026	*13.2 ± 1.4*				
Permeation	15	23.5	0.0610 ± 0.0017	**13.1 ± 1.4**				
Experimental	100	325.2	12.37 ± 0.31	*13.1 ± 1.4*	6.8 ± 1.6	38.4%	-15.6 ± 1.1	-13.69
	15	40.1	0.0589 ± 0.0017	*7.50 ± 0.78*				

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Type	Time (min)	Temp. (°C)	m (nccSTP ^{4}He)	n_0 (nccSTP ^{4}He)	ω (nccSTP ^{4}He)	${}^{He}F_{Apatite}$	Corrected $\ln(D/a^2)$	Uncorrected $\ln(D/a^2)$
Permeation	15	24.0	0.0351 ± 0.0011	**7.44 ± 0.78**				
Experimental	240	400.1	101.1 ± 2.7	*7.40 ± 0.78*	110.0 ± 3.2	56.2%	-13.281 ± 0.076	-13.70
	15	34.9	0.1096 ± 0.0031	*16.3 ± 1.7*				
Permeation	15	23.8	0.0757 ± 0.0022	**16.2 ± 1.7**				
Experimental	90	350.3	14.88 ± 0.41	*16.1 ± 1.7*	5.8 ± 1.9	57.2%	-14.92 ± 0.92	-15.42
	15	39.5	0.0567 ± 0.0017	*7.06 ± 0.74*				
Permeation	15	24.0	0.0330 ± 0.0011	**7.00 ± 0.74**				
Experimental	60	400.4	14.21 ± 0.40	*6.97 ± 0.74*	18.1 ± 1.4	60.1%	-13.31 ± 0.31	-15.50
	15	40.4	0.0866 ± 0.0026	*10.9 ± 1.1*				
Permeation	15	24.0	0.0508 ± 0.0016	**10.8 ± 1.1**				
Experimental	120	300.2	7.65 ± 0.22	*10.7 ± 1.1*	0.7 ± 1.2	60.2%	–	-16.50
	15	39.4	0.0291 ± 0.0011	*3.79 ± 0.41*				
Permeation	15	23.7	0.01758 ± 0.00077	**3.76 ± 0.41**				

1.7 Discussion

Except for ampoules with encapsulated water, the experimental results of this work are predicted within 10% when assuming the literature values for permeabilities for 4He from Swets et al. (1961) and the 3He/4He permeability ratio of Jones (1953). This suggests that permeation lag is a tractable complexity, enabling the use of fused silica ampoules to control the sulfur and water fugacity of helium step heat experiments, but that literature values for helium permeability do not apply to hydrothermal experiments, which instead require the ampoule's rate constant to be measured in situ (§1.4).

Resolving the Paradoxically High Apparent Helium Permeability in Water-Filled Tubes

The high measured permeability of helium for water-filled silica glass ampoules (supplementary Figs. 1.11, 1.29) is initially surprising and warrants further explanation, as the dissolved water and/or hydroxyl ion content of fused silica glass is only known to decrease helium permeability (Tsuru et al., 2011; Shelby, 1972; Masaryk, 1968). It is hypothesized here to result from an increase in the partial pressure of helium within the gas phase of the ampoule due to volume displacement of the gas phase by liquid water. This results from the low solubility and high diffusivity of helium in water, which allow for helium to be concentrated within the gas phase while remaining in equilibrium with the glass throughout the entire ampoule, even where obstructed by water.

The diffusivity of helium in water is about 7×10^{-5} cm^2/s at room temperature (Verhallen et al., 1984), whereas the diffusivity of helium in glass is three orders of magnitude slower: about 2×10^{-8} cm^2/s (Swets et al., 1961). Helium diffusion in water can be considered instantaneous relative to the timescales of helium diffusion through glass, so the helium concentration in the gas phase is expected to be in equilibrium with the glass throughout the entire tube even where mediated by liquid water. Therefore, adding water to an ampoule changes the volume available for the gas phase while not affecting the effective area of permeation.

Helium partial pressures in the gas phase can be increased by volume displacement assuming that helium strongly partitions into the gas phase over glass and water. Helium solubilities in water are similar to but slightly lower than in glass. At room temperature, the solubility of helium in water is ~

$4 \times 10^{-6} \frac{moles}{m^3 * Pa}$ (Sander, 2015) or $\sim 2 \times 10^{17} \frac{atoms}{cm^3 * atm}$. The solubility of helium in silica glass is $\sim 6 \times 10^{17} \frac{atoms}{cm^3 * atm}$ (Swets et al., 1961), a factor of 3 higher. At room temperature, the concentration of helium atoms in glass and in water is about 3% and 1% respectively of the concentration of helium atoms in the gas phase (supplementary Eq. (1.41)). The net result of this differential partitioning is that the effective permeability of helium roughly scales with the inverse volume fraction of the gas phase, following the equation:

$$K_{eff} = K_{dry} \frac{V_{water} + V_{gas}}{V_{gas}} \tag{1.12}$$

Where K_{eff} is the effective helium permeability, K_{dry} is the nominal helium permeability at anhydrous conditions, and V_{water} and V_{gas} are the relative volumes of water and gas respectively. This effect is potentially larger than the decrease in helium permeability of fused silica glass due to elevated water contents; therefore an ampoule that is filled with significant water may paradoxically permeate helium faster than the same ampoule without water, despite the lower helium permeability of the glass itself.

Evolving Helium Permeability in Water-Filled Tubes

Helium permeability measurements of water-filled ampoules show decreasing fractional yields for identical steps (Fig. 1.15). This observation is consistent with previous studies that show decreased helium permeability in silica glass following hydrothermal treatment (Tsuru et al., 2011; Masaryk, 1968). Other studies have shown that diffusion of water in silica glass is complex, but it is always significantly slower than diffusion of helium in silica glass. At 250°C, the effective diffusivity of both hydroxyl groups and molecular water in silica glass is initially ~10^{-11} cm^2/s and further slows by a factor of 4 over the first 500 hours of observation (Oehler and Tomozawa, 2004). Helium, on the other hand, has a diffusivity of ~10^{-6} cm^2/s at 250°C (Swets et al., 1961). For a 200 µm thick silica ampoule, the diffusive timescale ($\tau = \frac{L^2}{D}$) of water is about 76 years, whereas the diffusive timescale of helium in the same ampoule is less than 7 minutes. Even so, we observe a gradual decrease in helium permeability of about 7% over the course of a few days at 100°C in this work (Fig. 1.15). The implication of this observation is that the helium permeability in water-filled tubes must be continuously monitored by measuring the rate constant in situ (§1.4; Eqs. (1.8), (1.9)) during a hydrothermal step heat experiment to track

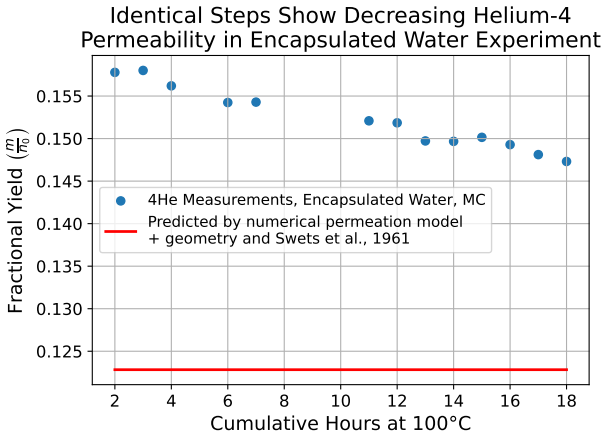

Figure 1.15: Identical permeation steps at 100°C with identical cooling times of the encapsulated water tube show higher 4He fractional yields than predicted with a numerical diffusion model and literature values of permeability from Swets et al. (1961). These identical steps also show a striking systematic decrease in fractional yields as the ampoule is held at 100°C, suggesting that the permeability of the ampoule is slowing during the experiment, possibly due to diffusion of water and hydroxyl groups into the silica glass (Tsuru et al., 2011; Oehler and Tomozawa, 2004). This trend is not visible in the 3He data due to higher scatter in the data.

(or exclude) significant temporal evolution of helium permeability throughout the experiment.

Previous studies have shown that the diffusion of water-derived species (OH^-, H_2O, H^+) into silica glass slows significantly as the water concentration increases within the glass (Oehler and Tomozawa, 2004; Davis and Tomozawa, 1995). Thus, it may be possible that a hydrothermal pretreatment (or early high-temperature heating steps) could sufficiently increase the trace water content of the silica glass for it to enter a regime of slower solubilization of water, thereby stabilizing the trace water content and helium permeability of silica glass for hydrothermal experiments. This hypothesis would be a fruitful direction for future work related to hydrothermal helium step heat experiments in silica glass ampoules.

1.8 Conclusions

Encapsulating minerals in pure silica glass allows helium step heat experiments to be conducted within controlled microenvironments of high water or sulfur partial pressure. Helium permeation through fused silica glass is a rapid but

non-instantaneous first-order process that complicates the interpretation of step heat data; however, its effects can be reliably deconvolved from the sample's true helium release using the equations and procedures described above. A general workflow combining the two approaches described above is suggested below:

1. Pull a fused silica tube (e.g., 3 mm inner diameter, 4 mm outer diameter stock) into a pipette with the desired inner diameter and wall thickness for the experiment.

2. Load the mineral sample and the desired buffering phase (e.g., water or sulfur) into the ampoule and seal it, taking care not to heat the sample itself during sealing.

3. If possible, measure the ampoule's inner and outer diameters with a calibrated microscope. This is not strictly necessary, but is helpful for planning the experiment: these measurements give the geometric factor G (Eq. (1.5)), which, combined with literature permeability values, allows the rate constant of permeation λ to be estimated in advance.

4. Before beginning the experiment, and after each high-temperature step thereafter, hold the ampoule at a permeation temperature low enough for the sample to be effectively closed to helium loss for at least the diffusive timescale $\tau = \frac{L^2}{D}$, where L is the ampoule's wall thickness and D is the diffusivity of helium in fused silica glass (e.g., from Swets et al., 1961). This resting period allows a steady state helium concentration profile to establish within the glass, which is required for accurate permeation corrections (Eqs. (1.1), (1.6)).

5. Perform the permeation step measurement after establishing the steady state concentration profile. Use this measurement to solve Eq. (1.6) to determine the residual helium content in the ampoule to recover the sample's helium yield using Eq. (1.1). Accounting for the helium lost during the permeation steps, this measurement applies for both the previous and subsequent high temperature experimental steps through the relation of Eq. (1.2).

6. At several points throughout the experiment, run a series of at least three consecutive permeation steps to determine the permeation rate

constant λ in situ (§1.4; Eq. (1.9)). Do this in every experiment rather than relying on literature permeabilities, and repeat it through the run for hydrothermal experiments, where λ may drift as the glass takes up water.

For high temperature experiments of retentive phases, where low temperature permeation steps may yield signals indistinguishable from blank, experimentalists may instead use the bounding method (§1.4) to constrain the ampoule's residual helium from above rather than measuring it. This requires only a lower bound on the rate constant of permeation at the experimental temperature, and can only be done for steps where $\lambda t > 5$. No permeation steps are necessary for this approach, although a series of them remains the most reliable way to establish the rate constant it depends on.

Supplementary Information

for Chapter 1: *A Simple Method for Helium Step Heat Experiments at High Water or Sulfur Fugacity*

1.1s A Fick's Law Formulation Describing Helium Migration Across Semi-Permeable Membranes

We derive the relevant equations describing helium permeation through glass by beginning with Fick's Law, similarly to previous works (Rogers et al., 1954; Altemose, 1961).

$$J = -D\frac{d\varphi}{dx} \tag{1.13}$$

Where J is the area-normalized Fickian diffusive flux in units of $\frac{atoms}{s \times cm^2}$, D is the diffusivity (of helium in glass) in units of $\frac{cm^2}{s}$, and $\frac{d\varphi}{dx}$ is the concentration gradient, in units of $\frac{\left(\frac{atoms}{cm^3}\right)}{cm}$ or $\frac{atoms}{cm^4}$.

For steady state permeation across a planar membrane, the concentration gradient is constant. If we take l to be the glass thickness and we rewrite the Fickian flux J as $\frac{F}{A}$, where F is the absolute atom flux in units of $\frac{atoms}{s}$, and A is simply the membrane area, we can rewrite Fick's law as:

$$F = -DA\frac{\Delta\varphi}{l} \tag{1.14}$$

Finally, we assume that the concentration of helium within the glass membrane is Henrian, i.e. proportional to the gas pressure in contact with the membrane. Experimental work has shown that helium permeation through silica glass is Henrian at pressures of several hundred bars and lower (Shelby, 1976), well above the experimental helium partial pressures for this work. We define this proportionality factor as the solubility, S, in units of $\frac{atoms}{atm \times cm^3}$. Now we may write the expression in terms of the gas pressure gradient, or the difference of helium partial pressures on each side of the membrane.

$$F = -DA\frac{S\Delta P}{l} \tag{1.15}$$

Diffusivity and solubility are material properties that are functions of temperature following the Arrhenius equation, and therefore their mathematical product (permeability, K) also follows an Arrhenius relation with temperature (Rogers et al., 1954; Swets et al., 1961). Now, recognizing that K = DS, we may express the equation in terms of permeability $\frac{atoms}{s \times cm \times atm}$.

$$F = -\frac{KA\Delta P}{l} \tag{1.16}$$

Studt (1962) and Masaryk (1968) also use this equation to describe helium permeation through silica glass, taken from Darcy's Law for gas flow through semi-permeable membranes (Carman, 1956). This ordinary, first-order differential equation is well suited for describing permeation through a silica membrane when the permeate pressures on each side of the membrane are constant (Norton, 1953; Altemose, 1961; Swets et al., 1961). Cases where permeate partial pressure changes during the experiment (such as this work) require solving the first order ODE above to relate the abundance of an analyzed permeate to the abundance of unanalyzed permeate remaining in the depleting reservoir.

Recalling that F is an atomic flux in units of $\frac{atoms}{s}$ that could be notated as $\frac{dn}{dt}$:

$$\frac{dn}{dt} = -\frac{KA\Delta P}{l} \tag{1.17}$$

We assume that the helium partial pressure in the ultra-high vacuum (UHV) heating cell surrounding the ampoule is 0, so that the pressure gradient equals the partial pressure within the ampoule ($\Delta P = P$). This is a valid assumption for this work because the heating cell is 4-5 orders of magnitude more voluminous than the glass ampoule. The assumption therefore is only appreciably violated for the trivial case in which effectively all the helium present within the tube is permeated out and the helium partial pressures within and outside of the ampoule are equal. In this case, net helium flux goes to zero even though the ampoule is not entirely evacuated. Relative to the amount of helium within the ampoule at time 0 (n_0), the amount of helium remaining in the ampoule after infinite time *(n_{eqm})* will be set by the volume ratio of the ampoule and the lamp cell.

$$\frac{n_{eqm}}{n_0 - n_{eqm}} \approx \frac{n_{eqm}}{n_0} = \frac{V_{ampoule}}{V_{lamp\ cell}} \approx 10^{-4}$$

For this experimental setup, only 1 part in 10000 helium atoms will remain in the ampoule after infinite time in the closed heating cell.

Before integrating Eq. (1.17), it is useful to experimentalists working with mass spectrometry to express the pressure term P in terms of atoms n, the atomic abundance of helium within the ampoule. Helium is well-approximated as an ideal gas at experimental conditions, so the ideal gas law (PV = nRT) relates the pressure, P, to the atomic abundance, n, yielding Eq. (1.18):

$$\frac{dn}{dt} = -\frac{AKnRT}{lV} \tag{1.18}$$

Where R is the gas constant $\left(1.36 \times 10^{-22} \frac{cm^3 atm}{K\ atoms}\right)$, T is the temperature in Kelvin, and V is the volume in cm^3. It is now obvious that the helium atoms in the gas phase (n) decrease over time. Because Eq. (1.18) assumes a steady state helium concentration profile in the glass walls of the ampoule in equilibrium with the helium partial pressure inside the ampoule, one must make another essential assumption to continue: that the permeation timescale be small with respect to the diffusive timescale so that the helium concentration profile of the glass effectively remains in an equilibrium with the helium atoms in the gas phase (n). If the inner gas volume is small relative to the volume of the ampoule glass, then $\frac{dn}{dt}$ will be relatively large and the helium atoms in the gas phase (n) will decrease faster than diffusion's ability to maintain an equilibrium concentration profile in the glass, violating the assumption of a steady state concentration profile in equilibrium with the inner gas phase. Then, Eq. (1.18) and all equations derived from it will be invalid. In other words, the timescale of diffusion must be much smaller than the timescale of permeation ($\tau_D \ll \tau_P$) for Eq. (1.18) and all derived equations to apply. A more thorough analysis of this assumption is presented in §1.2s.

Eq. (1.18) can be simplified by defining a geometric factor (G) in units of cm^{-2} that incorporates the spatial parameters for area, volume, and length of the experimental glass ampoule:

$$\frac{dn}{dt} = -GKnRT \tag{1.19}$$

Eq. (1.19) can be integrated using a straightforward separation of variables to get a useful general equation describing helium reservoirs depleting via permeation.

$$n_t = n_0 e^{-KGRTt} \tag{1.20}$$

Where n_t is the amount of helium atoms within the reservoir after time t in seconds, and n_0 is the amount of helium atoms within the reservoir at time t = 0.

Eq. (1.20) can equivalently be expressed in terms of the fraction of the initial helium reservoir that has permeated during time t, which is often a more intuitive quantity when relating this equation to a measured amount of helium. Defining the fractional permeation yield ${}^{He}F \equiv \frac{m}{n_0}$, where $m \equiv n_0 - n_t$ is the amount of helium permeated and available to be measured during time t, Eq. (1.20) can be recast as:

$$ {}^{He}F = \frac{m}{n_0} = 1 - e^{-KGRTt} \tag{1.21}$$

Eq. (1.20) can be solved for permeability (K) for use in experimentally determining permeability for experiments in which the permeating helium source is significantly depleted during the experiment, yielding Eq. (1.22):

$$ K = -\frac{\ln(n_t) - \ln(n_0)}{GRTt} \tag{1.22}$$

In typical experimental measurements of helium permeability through glass, permeation rates are measured in real time, and the pressure gradient is constant throughout the measurement, so an equation like Eq. (1.16) is used to solve for permeability instead (Norton, 1953; Swets et al., 1961; Altemose, 1961).

Permeation Through Cylindrical Membranes

For non-planar membranes, the effective area of permeation, A, is not straightforward to evaluate because the surface area on each side of the membrane differs. Masaryk (1968) gives the effective area of permeation for a cylinder (such as for this experiment) as:

$$ A = \frac{2\pi hl}{\ln\left(\frac{b}{a}\right)} \tag{1.23}$$

Where a is the cylinder's internal radius, b is the cylinder's external radius, and h is its length. This effective area of permeation (Eq. (1.23)) can be used with the equation for the internal volume of the cylinder: $V = h\pi a^2$ to evaluate the geometric factor G (Eq. (1.19)) for a cylinder in terms of the ampoule geometry. Reducing yields Eq. (1.24):

$$G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} \tag{1.24}$$

Interestingly, h cancels so that the permeation rate has no dependence on the ampoule length when permeation through the top and bottom is negligible. Eq. (1.24) displays the intuitive behavior that thinner walls and narrower tubes lead to more efficient permeation. Where G is used elsewhere in this work, it refers to the cylindrical geometric factor given in Eq. (1.24). Most silica ampoules sealed at Caltech for helium step heat experiments were 0.6-1.2 mm wide with 150-300 µm thick glass, translating to values of G between 2000-7000 cm^{-2}. In this work, if no value of G is specified, then a typical 200 µm thick ampoule with an outer diameter of 1 mm is assumed, corresponding to $a = 0.03$ cm, $b = 0.05$ cm and $G = 4350.26$ cm^{-2}.

1.2s Diffusive Steady State Concentration Profiles in Cylindrical Membranes

To establish a steady state concentration profile, the diffusive timescale must be much faster than the permeation timescale, otherwise the helium concentration profile within the glass will respond too slowly to maintain equilibrium with the rapidly depleting internal gas phase (Fig. 1.16). The ratio of the permeation timescale and diffusive timescale is a strong function of the ratio of the ampoule's glass and internal gas volumes with a very minor dependence on temperature (Fig. 1.17 below). An ampoule with a glass/gas volume ratio of 2 has a diffusive timescale that is ~5.6% of the permeation timescale (Eq. (1.28) below). Numerical diffusion-permeation simulations show that Eq. (1.22) accrues less than 1% error due to deviation from the idealized helium concentration profile (Figs. 1.18 and 1.19 below). Smaller glass/gas volume ratios are better: the ampoules in the main text have glass/gas volume ratios ~1.25-1.5. Numerical simulations show that this error is determined nearly entirely by the volume ratio of the ampoule glass and gas and is otherwise insensitive to ampoule geometry (Figs. 1.20 and 1.21 below).

The derivation for Eq. (1.22) always assumes an equilibrium steady state helium concentration profile across the ampoule. The most important corollary of this assumption is that the timescale of permeation must be much larger than the diffusive timescale ($\tau_D \ll \tau_P$) for Eq. (1.22) and all derived equations to apply.

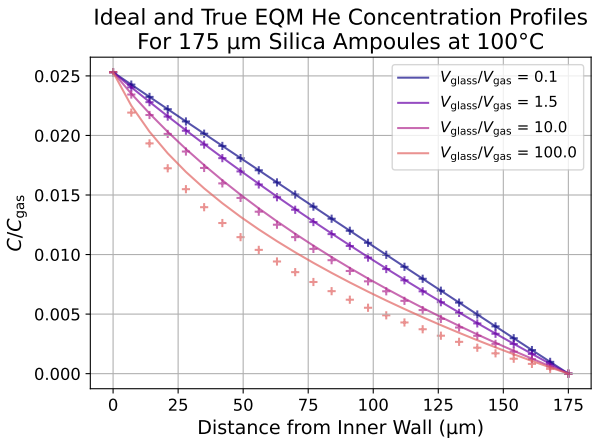

Figure 1.16: Ideal diffusive steady state concentration profiles (demarcated by the '+' symbols) are contrasted with the numerically calculated "effective" steady state concentration profile, represented by the solid lines. These effective concentration profiles are not strictly in steady state, but they are effectively in equilibrium with a rapidly decreasing internal helium reservoir. These profiles are retrieved by numerically simulating a model ampoule permeating helium for at least one diffusive timescale. At low glass/gas volume ratios, the idealized and true concentration profiles are indistinguishable. Ideal steady state concentration profiles for cylindrical membranes are strongly curved at high glass/gas volume ratios and become linear as the glass/gas volume ratio approaches 0. For closed ampoules where internal pressure is not constant, but instead decreases due to permeation, the internal pressure decreases faster than the glass concentration profile can respond to the new internal helium pressure, so the concentrations are higher (and less curved) than the ideal profile, because of memory of higher internal helium pressures. 4He diffusivities and solubilities are taken from Swets et al. (1961).

An analysis is presented here exploring the ampoule geometries that permit the use of the analytical solutions for helium permeation through silica glass.

The diffusive timescale is the timescale at which diffusion is generally expected to establish a steady state concentration profile between two (effectively) constant reservoirs. For a diffusive domain size L, the timescale is given as:

$$\tau_D = \frac{L^2}{D} \tag{1.25}$$

The permeation timescale is analogous to an "e-folding time" or "mean life" in radioactive decay, the time after which point the original abundance of something decaying by a first order process following the form $\frac{dn}{dt} = -\lambda n$ would have decreased by a factor of Euler's number $(1/e)$. After one permeation

timescale has passed, more than half of the original helium has permeated through the ampoule walls, leading to the intuition that the dimensionless timescale ratio must satisfy $\frac{\tau_D}{\tau_P} \ll 1$ in order for Eq. (1.22) and Eq. (1.20) to apply, because only vanishingly small changes in the total helium are allowed within one diffusive timescale in order for an equilibrium helium concentration profile to be stabilized and maintained.

$$\tau_P = \frac{1}{\lambda} = \frac{1}{GRTK} \tag{1.26}$$

The ratio of these two timescales is a weak function of temperature and a strong function of the ratio of ampoule glass and internal gas volumes $\left(\frac{V_{glass}}{V_{gas}}\right)$ (Fig. 1.17). By the equation for the volume of cylinders:

$$V_{\frac{glass}{gas}} = \frac{h\pi\left(b^2 - a^2\right)}{h\pi a^2} = \frac{\left(b^2 - a^2\right)}{a^2} \tag{1.27}$$

$\frac{\tau_D}{\tau_P}$ can be expanded and recast to be a function of only temperature and $V_{\frac{glass}{gas}}$. Recall that L = b-a and K = SD, and the identity of G for cylinders from Eq. (1.24):

$$\frac{\tau_D}{\tau_P} = \frac{L^2}{D}GRTK = \frac{(b-a)^2}{D}\frac{2}{a^2 \ln\left(\frac{b}{a}\right)}RTSD$$

Factoring $(b^2 - a^2) = (b - a)(b + a)$, substituting, and simplifying:

$$\frac{\tau_D}{\tau_P} = V_{\frac{glass}{gas}} \bullet \frac{(b-a)}{(b+a)}\frac{2}{\ln\left(\frac{b}{a}\right)}RTS$$

Now using Eq. (1.27) to relate the terms containing a and b to $V_{glass/gas}$, all spatial terms can be expressed in terms of this ratio.

$$\left(\frac{b}{a}\right) = \sqrt{V_{\frac{glass}{gas}} + 1}$$

For example:

$$b = a\sqrt{V_{\frac{glass}{gas}} + 1}$$

$$(b-a) = a\left(-1+\sqrt{V_{\frac{glass}{gas}}+1}\right)$$

$$(b+a) = a\left(1+\sqrt{V_{\frac{glass}{gas}}+1}\right)$$

$$\frac{\tau_D}{\tau_P} = S(T)\bullet\, RT\, \bullet V_{\frac{glass}{gas}} \bullet \frac{-1+\sqrt{V_{\frac{glass}{gas}}+1}}{1+\sqrt{V_{\frac{glass}{gas}}+1}} \bullet \frac{2}{\ln\left(\sqrt{V_{\frac{glass}{gas}}+1}\right)} \quad (1.28)$$

Eq. (1.28) is plotted for three temperatures as a function of glass/gas volume ratio in Fig. 1.17. $\frac{\tau_D}{\tau_P}$ increases monotonically and nearly linearly in log-log space with the glass/gas volume ratio of ampoules, whereas temperature plays only a minor role. Ampoules with high glass/gas volume ratios and $\frac{\tau_D}{\tau_P}$ ratios can be considered "glass dominated". For these glass-dominated ampoules, the relative difference in surface area between the inner and outer surfaces is large, resulting in a more profoundly curved ideal concentration profile (Fig. 1.16 above, Eq. (1.33)). For glass dominated ampoules, this ideal curved concentration profile is only realized if the internal pressure is held constant, otherwise, this internal helium partial pressure decays rapidly due to permeation relative to the rate of diffusion within the glass, thereby stabilizing a helium concentration profile in the glass that will be less curved and higher than the idealized steady state of Eq. (1.33) (Fig. 1.16 above). When normalized to the concentration, quantity, or pressure of helium within the gas phase of the ampoule, this quasi-steady state concentration profile is reached within one diffusive timescale and remains constant as diffusion and permeation proceed, regardless of the ratio of glass and gas volumes (this property was tested by varying the permeation time between one and ten diffusive timescales, which had no effect on the concentration profile normalized to the concentration, abundance, or partial pressure of helium in the gas phase). Indeed, all model ampoule calculations from Figs. 1.18–1.20 were initialized to have this effective steady state concentration profile, which is done by allowing the model ampoule to permeate and diffuse for one diffusive timescale, after which the concentration profile is multiplied by the simulated $\frac{n_0}{n_t}$ to renormalize it to the original helium contents of the ampoule. The helium content of the gas is then recalculated using the relation $n_{gas} = n_0 - n_{glass}$, and the permeated helium outside of the

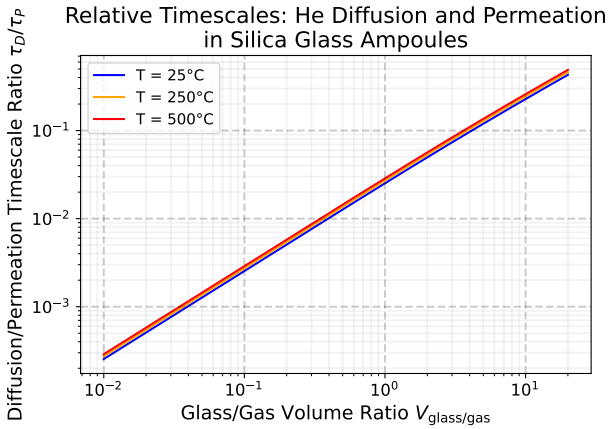

Figure 1.17: The accuracy of Eq. (1.20) and equations derived from it is dependent on a steady state concentration profile in equilibrium with the helium gas concentration inside the ampoule. If the timescale of permeation is comparable to or lower than the timescale of diffusion for the ampoule, then the gas concentration inside the ampoule will change faster than the response of the concentration profile within the glass, and these equations will become inaccurate. In other words, these equations are only accurate when $\tau_D/\tau_P \ll 1$. For helium permeation in silica glass ampoules, the ratio of the diffusive and permeation timescales is a strong function of the ratio of glass and gas volume, and a weak function of temperature. τ_D/τ_P is nearly linear with V_{glass}/V_{gas} in log-log space, forming subtly-offset parallel arrays at different temperatures. 4He solubilities are taken from Swets et al. (1961).

ampoule is reset to 0. Numerical simulations of model ampoules with glass/gas volume ratios above ~ 0.1 that were initialized to the ideal steady state of Eq. (1.33) recorded an early spike in relative error with Eq. (1.22) that was maximized before one diffusive timescale because of an incorrect initial steady state assumption. This artifact vanished completely when the true steady state profile was used.

Numerical simulations of model ampoules of varying dimensions, thicknesses, and geometric factors show that the only geometric parameter that directly affects the accuracy of the analytical solutions, e.g. Eq. (1.22), is the glass/gas volume ratio of the ampoules (Figs. 1.18–1.20). This geometric dependence is much more important than subtle effects of temperature (Figs. 1.17; 1.20). Ampoules from this work have glass/gas volume ratios between 1.2 and 1.6, for which Eq. (1.22) is accurate to within 1% for temperatures less than 500°C (Fig. 1.19). For more precise handling, odd thermal schedules, or for ampoules with higher glass/gas volume ratios, permeation fractions would need

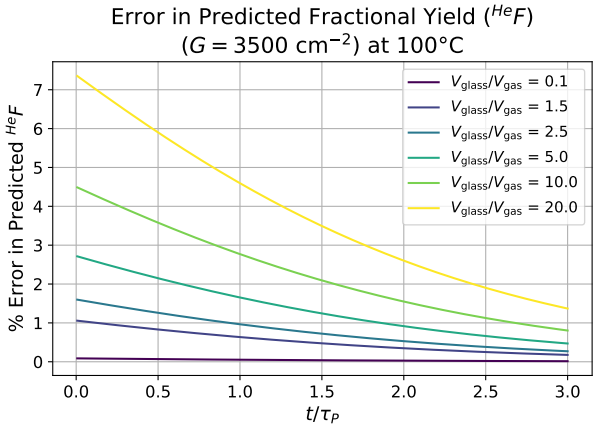

Figure 1.18: Numerical simulations confirm that Eq. (1.22) is accurate for ampoules with low glass/gas volume ratios. For ampoules with higher glass/gas volume ratios, the relative error is not constant but depends on the duration of the step. The relative error reaches a maximum at a step shorter than the permeation timescale (τ_P) for ampoules with glass/gas volume ratios higher than 1, then decreases as steps continue and most of the original gas within the ampoule has been permeated. 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

to be numerically solved to achieve sufficient accuracy to use a permeation measurement to constrain the helium contents of an ampoule.

To understand the steady state helium concentration profile in the glass, it is necessary to model the radial diffusion through a symmetrical, hollow cylinder. Carslaw and Jaeger (1959) give the equation for the change in concentration by diffusion in a cylinder (eqn. 5, page 17, second edition), expressed here in terms of concentration and diffusivity.

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r} + \frac{1}{r^2}\frac{\partial^2 C}{\partial \theta^2} + \frac{\partial^2 C}{\partial z^2}\right] \tag{1.29}$$

For typical ampoules used in this work, the surface area of the top and bottom together is about 5-10% of the surface area of the walls, but they are neglected here for simplicity. Assuming that the concentration profiles are radially symmetrical and uniform with height (z), Eq. (1.29) is reduced to:

$$\frac{dC}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right] \tag{1.30}$$

At steady state, $\frac{dC}{dt} = 0$, so

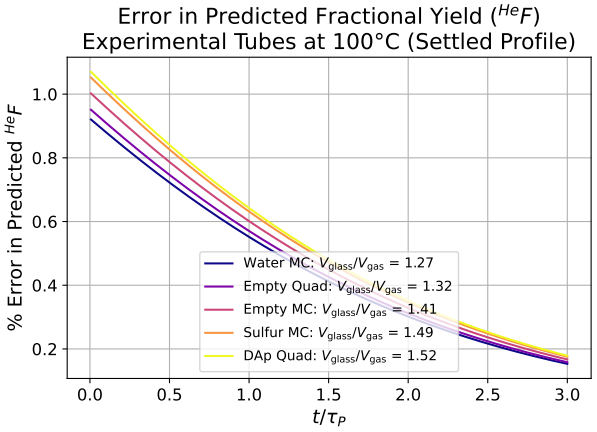

Figure 1.19: Numerical experiments of this work show that these equations will be accurate to within 1% for ampoule glass/gas volume ratios less than 2 for temperatures less than 500°C, which corresponds to $\tau_D/\tau_P < 0.056$. Experimental ampoules at Caltech have glass/gas volume ratios generally between 1.25 and 1.5, and thus Eq. (1.22) is accurate to within 1% for ampoules with concentration profiles in steady state. 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

$$\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r} = 0$$

Which, by the product rule, can be expressed as:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C}{\partial r}\right) = 0$$

Now, the $\frac{1}{r}$ term can be dropped, and both sides can be integrated with respect to r to yield:

$$r\frac{\partial C}{\partial r} = A$$

Where A is a constant of integration. Now both sides are integrated again with respect to r, leaving the equation describing the geometry of the diffusive steady state concentration profile of a cylinder (Eq. (1.31)):

$$\frac{\partial C}{\partial r} = \frac{A}{r}$$

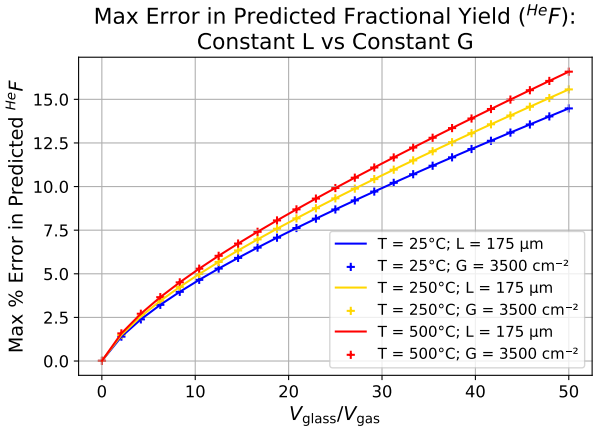

Figure 1.20: Numerical simulations of two suites of model ampoules with varying glass/gas volume ratios and a constant thickness (L) or permeation geometric factor (G) show that the only factors that determine the applicability of Eq. (1.22) for helium permeation of closed ampoules are glass/gas volume ratio (primarily) and temperature (secondarily). These two suites have different geometries and permeation rates, yet the deviation of the numerical model from Eq. (1.22) is the same for different ampoules with the same glass/gas volume ratio at the same temperature. This is clear evidence that the ratio of diffusive and permeation timescales is the primary control on the accuracy of Eq. (1.22). 4He diffusivities, solubilities, and permeabilities are taken from Swets et al. (1961).

$$C(r) = \int \frac{A}{r} dr$$

$$C(r) = Aln(r) + B \tag{1.31}$$

Where B is another constant of integration. Constants A and B can be solved for once the boundary conditions are defined, that is, the helium concentrations at the inner and outer walls of the glass ampoule. The concentration of helium at the outer wall is effectively 0 because the high vacuum heating cell is essentially an infinite volume compared to the glass ampoule. At the inner wall, the concentration is governed by the solubility and helium partial pressure of the encapsulated volume. Letting a be the radius of the ampoule inner wall, b be the radius of the outer wall, and P_{He}^{gas} be the partial pressure of helium in the encapsulated gas of the ampoule, the following boundary conditions are enforced:

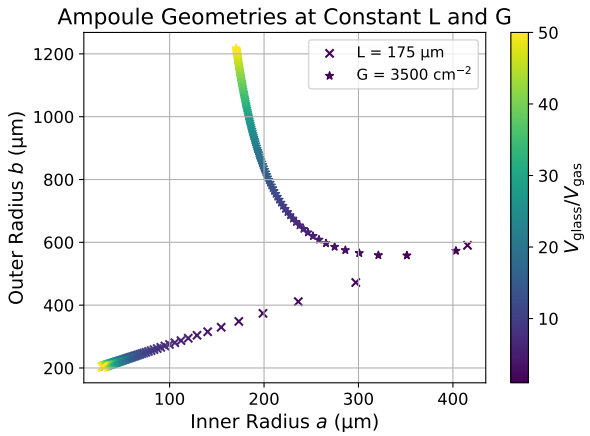

Figure 1.21: Model ampoules with uniformly distributed glass/gas volume ratios between 0.01 and 50 and one other constraint (either constant thickness or constant geometric permeation factor G, Eq. (1.24)) have defined inner and outer radii. These are the same suites of ampoules for which helium permeation was numerically simulated in Fig. 1.20.

$$C(r = b) = 0$$

$$C(r = a) = S(T) \bullet P_{He}^{gas}$$

Now a system of equations can be formulated so that A can be solved first, then B.

$$0 = Aln(b) + B$$

$$S(T) \bullet P_{He}^{gas} = Aln(a) + B$$

$$S(T) \bullet P_{He}^{gas} = Aln(a) - Aln(b)$$

$$S(T) \bullet P_{He}^{gas} = A\ ln\left(\frac{a}{b}\right)$$

$$A = \frac{S(T) \bullet P_{He}^{gas}}{ln\left(\frac{a}{b}\right)}$$

$$B = -\frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} ln(b)$$

Substituting A and B back into Eq. (1.31):

$$C(r) = \frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} \ln(r) - \frac{S(T) \bullet P_{He}^{gas}}{\ln\left(\frac{a}{b}\right)} \ln(b)$$

$$C(r) = S(T) \bullet P_{He}^{gas} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)} \tag{1.32}$$

Using the ideal gas law and the volume for the cylinder, the partial pressure of helium within the ampoule can be related to the geometry of the ampoule and the abundance of helium atoms within the ampoule:

$$P_{He}^{gas} = \frac{n_{He}^{gas} RT}{V_{gas}}$$

$$V_{gas} = \pi a^2 h$$

$$P_{He}^{gas} = \frac{n_{He}^{gas} RT}{\pi a^2 h}$$

$$C(r) = S(T) \bullet \frac{n_{He}^{gas} RT}{\pi a^2 h} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)} \tag{1.33}$$

Where h is the length of the ampoule.

Because the concentration profile of the ampoule is radially symmetrical, Eq. (1.33) can be expressed as a radial density of helium atoms instead to eliminate the dependence on h. The concentration ($C(r)\frac{atoms}{cm^3}$) can be converted into a radial density ($\epsilon(r)\frac{atoms}{cm}$) by multiplying by the circumference of a circle with radius r and the cylinder height h.

$$\epsilon(r) = C(r) * 2\pi r h \tag{1.34}$$

Substituting in $C(r)$ from Eq. (1.33) and the geometric constant of permeation G from Eq. (1.24) $\left(G = \frac{2}{a^2 \ln\left(\frac{b}{a}\right)} = -\frac{2}{a^2 \ln\left(\frac{a}{b}\right)}\right)$:

$$\epsilon(r) = \ S(T) \bullet \frac{2 n_{He}^{gas} r RT}{a^2} \bullet \frac{\ln\left(\frac{r}{b}\right)}{\ln\left(\frac{a}{b}\right)}$$

$$\epsilon(r) = \ -S(T) \bullet n_{He}^{gas} GRTr \ln\left(\frac{r}{b}\right) \tag{1.35}$$

To find the proportion of helium within the glass at steady state, the radial density is integrated from $r = a$ to $r = b$.

$$n_{He}^{glass} = \int_a^b \epsilon(r) dr = \int_a^b -S(T) \bullet n_{He}^{gas} GRTr \ln\left(\frac{r}{b}\right) dr$$

$$n_{He}^{glass} = -S(T) \bullet n_{He}^{gas} GRT \bullet \int_a^b \left[rln(r) - rln(b)\right] dr$$

Applying the sum rule, then integration by parts to one term yields the indefinite integral:

$$\int \left[rln(r) - rln(b)\right] dr = \frac{1}{2} r^2 \ln(r) - \frac{r^2}{4} - \frac{1}{2} r^2 \ln(b) + C$$

Evaluating the definite integral from the bounds of $r = a$ to $r = b$ yields:

$$\int_a^b \left[rln(r) - rln(b)\right] dr = \frac{1}{2} b^2 \ln(b) - \frac{b^2}{4} - \frac{1}{2} b^2 \ln(b) - \frac{1}{2} a^2 \ln(a) + \frac{a^2}{4} + \frac{1}{2} a^2 \ln(b)$$

Reducing and simplifying:

$$\int_a^b \left[rln(r) - rln(b)\right] dr = \frac{1}{2} a^2 \left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}$$

$$n_{He}^{glass} = -S(T) \bullet n_{He}^{gas} GRT \bullet \left(\frac{1}{2} a^2 \left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right) \tag{1.36}$$

Eq. (1.36) can be easily expressed in terms of total helium in the ampoule by substituting for the relationship $n_{He}^{gas} = n_0 - n_{He}^{glass}$

$$n_{He}^{glass} = -S(T) \bullet \left(n_0 - n_{He}^{glass}\right) GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)$$

$$\frac{\left(n_0 - n_{He}^{glass}\right)}{n_{He}^{glass}} = \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1}$$

$$\frac{n_0}{n_{He}^{glass}} = 1 + \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1} \tag{1.37}$$

A Numerical Model for Diffusive Permeation of Helium Through Silica Glass Ampoules

A numerical model to estimate diffusive permeation is invaluable, as it provides a flexible way to estimate the permeation fraction $\left(\frac{m}{n_0}\right)$ even when the required assumptions for the analytical solutions are not met. For example, a numerical approach can calculate the expected permeation fraction for a non-isothermal time-temperature history, and it needn't obey the assumption that the helium concentration profile in the glass is in diffusive equilibrium with the partial pressure of helium in the encapsulated gas phase. As will be investigated below, the equilibrium concentration profile assumption is ephemerally broken when temperatures are changed, and for ampoules with high glass/gas ratios it is never true.

We numerically solve the radial diffusion equation for cylinders (Eq. (1.30)) using the Forward-Time Central-Space (FTCS) Finite Difference Method (FDM) along a radial transect of a model ampoule of arbitrary geometry for small, discrete time intervals. The system is spatially discretized into 100 cells along a radial transect, and numerical stability was ensured by solving for the time interval of a single step, Δt, such that $\frac{D\Delta t}{\Delta x^2} = 0.5$, satisfying the Courant-Friedrichs-Lewy (CFL) condition.

Helium concentrations along the transect (except for the boundaries, where r = a and r = b) are updated according to the following equation, an FTCS discretization of the radially symmetric diffusion equation for a cylinder (Eq. (1.30)):

$$\frac{dC_r}{dt} = D\left[\frac{\partial^2 C}{\partial r^2} + \frac{1}{r}\frac{\partial C}{\partial r}\right]$$

$$\Delta C_r \approx \Delta t \ \bullet D\left[\frac{C_{r-\Delta r} - 2C_r + C_{r+\Delta r}}{\Delta r^2} + \frac{1}{r}\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r}\right] \tag{1.38}$$

Dirichlet boundary conditions were enforced at the inner and outer boundaries. The concentration at the model ampoule's outer boundary (where r = b) was held at 0, while the concentration at the inner boundary (where r = a) was calculated at each discrete step by multiplying the solubility (in units of $\frac{atoms}{atm \times cm^3}$) by the helium partial pressure within the ampoule, according to Eq. (1.39) or (1.40).

$$C_{He}^{glass} = S(T) \bullet P_{He}^{gas} \tag{1.39}$$

Eq. (1.39) can be related to the atomic abundance of helium in the gas phase using the ideal gas law and the formula for the volume of a cylinder:

$$P_{He}^{gas} = \frac{n_{gas} RT}{V_{gas}}$$

$$V_{gas} = \pi a^2 h$$

$$C_{He}^{glass} = S(T) \bullet \frac{n_{gas} RT}{\pi a^2 h} \tag{1.40}$$

For simulations where the internal pressure is not constant, n_{gas} changes after every step and Eq. (1.39) is reapplied to the inner boundary cell where r = a. While the helium concentrations in the glass and in the encapsulated gas phase are both dependent on the length of the ampoule (h), they compensate for one another such that the absolute atom flux into and out of the ampoule is not a function of h. Within the numerical model, h was varied by orders of magnitude and the results of the simulations did not change, confirming this property.

It is sometimes useful to normalize the helium concentration of the glass to the concentration of helium in the gas. To do this, the internal helium partial pressure is related to an atomic or molar concentration using the ideal gas law:

$$C_{He}^{gas} = \frac{n}{V} = \frac{P_{He}^{gas}}{RT}$$

The pressure term vanishes in the ratio of the two concentrations and depends only on temperature:

$$C_{He}^{glass/gas} = \frac{C_{He}^{glass}}{C_{He}^{gas}} = S(T) \bullet RT \tag{1.41}$$

Eq. (1.41) is plotted in Fig. 1.22. It shows that this partitioning ratio strongly favors the gas and is only a weak function of temperature, varying between 2.5 and 2.8% between 0 and 300°C. This implies that departures from the equilibrium steady state assumption due to temperature changes will be small and perhaps negligible, which is confirmed in Fig. 1.24.

In order to simulate permeation, the atomic abundances of helium within the gas phases both outside the ampoule (at high vacuum) and inside the ampoule must be tracked, as must the fluxes between the glass and these two gas phases. These fluxes were calculated simply as the diffusive flux in $\frac{atoms}{s \bullet cm^2}$ as is given by Fick's law (Eq. (1.13)). For spatially discrete cells throughout the ampoule at any radius except for the boundaries, the diffusive flux can be approximated with the finite difference:

$$J_r = -D\frac{\partial C_r}{\partial r} \approx -D\frac{C_{r+\Delta r} - C_{r-\Delta r}}{2\Delta r} \tag{1.42}$$

However, Eq. (1.42) cannot be applied at the ampoule's inner wall when r = a or outer wall when r = b because there is no remaining glass on one side of the cell. In that case, second order terms from Taylor expansions centered at $a + \Delta r$ and $b - \Delta r$ can be used to better approximate the $\frac{\partial C_a}{\partial r}$ and $\frac{\partial C_b}{\partial r}$ terms. Letting $f(r) = \frac{\partial C_r}{\partial r}$ and $f'(r) = \frac{\partial^2 C_r}{\partial r^2}$:

$$J_a = -D\frac{\partial C_a}{\partial r} \approx -D\left(\frac{\partial C_{a+\Delta r}}{\partial r} + \frac{\partial^2 C_{a+\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_{a+2\Delta r} - C_a}{2\Delta r} + \frac{C_a - 2C_{a+\Delta r} + C_{a+2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{1.43}$$

$$J_b = -D\frac{\partial C_b}{\partial r} \approx -D\left(\frac{\partial C_{b-\Delta r}}{\partial r} - \frac{\partial^2 C_{b-\Delta r}}{\partial r^2}\Delta r\right) \approx -D\left(\frac{C_b - C_{b-2\Delta r}}{2\Delta r} - \frac{C_b - 2C_{b-\Delta r} + C_{b-2\Delta r}}{2\Delta r^2}\Delta r\right) \tag{1.44}$$

The helium contents in the measurable (external, high vacuum) reservoir and the inner encapsulated gas phase are updated every step using fluxes determined from Eq. (1.44) and (1.43) respectively. To convert these area-normalized fluxes into absolute atomic abundances, these fluxes are multiplied

by the time increment (Δt) and the surface area of the cylinder at the outer and inner walls respectively.

Modeling permeation for ampoules that experienced complex thermal histories

In order to compare the experimental results for ampoules that experienced complex cooling histories with the expectations of the first order permeation model developed in this work, the numerical model developed above was used. The model predicts the relative helium permeation for the three experiments on the Helix-MC-Plus^{10K} multi-collector mass spectrometer for both 4He and 3He by simulating helium permeation of a model ampoule with the same dimensions and thermal history of the real experiment. For ampoules with multiple measurements of the geometric dimensions, the smaller set of measurements is used for the numerical model.

To simulate permeation during cooling of an ampoule, the temperature is decreased at every step following a parabolic rate law determined using the Watlow temperature records of the cooling ampoule. A parabolic function is fit to cooling rates evaluated as a function of temperature. Cooling rates were algorithmically calculated for discrete time intervals using the Watlow temperature files of the relevant ampoule in Python. Temperatures were taken to be the mid-point between the maximum and minimum temperatures of the interval over which the temperature change was evaluated. The quadratic functions fit to these data were forced to have a single root at T = 24°C so that the simulated ampoules would cool to a final temperature of 24°C, a common room temperature of the Noble Gas Laboratory at Caltech.

The model ampoules were initially assigned an arbitrary and unitless amount of helium, and the permeation predictions were output as step fractional yields $\left(\frac{n_t}{n_0}\right)$, where n_0 was the amount of helium in the ampoule at the beginning of the step. Before the first heating step was simulated, the ampoule was assumed to be in diffusive steady state at 24°C. Afterwards, the ampoule was not assumed to be in diffusive equilibrium except for isothermal steps at times longer than the diffusive timescale at that temperature, so that the concentration profile of the model ampoule was tracked throughout the entire heating history. For steps above room temperature, the ampoule was simulated to immediately transition to the target temperature, then the helium concentration profile,

influx, and outflux of the glass were tracked for the nominal (programmed) heating times for times smaller than the diffusive timescale at that temperature. After a heated step, an equilibrium concentration profile was reinitialized at that temperature only if the equilibrium equation (Eq. (1.22)) was used for permeation timescales longer than the diffusive timescale, then the temperature of the ampoule was cooled continuously according to the quadratic rate law determined from the experimental ampoule's thermal history while helium diffusion and permeation were continuously simulated.

Errors related to violation of the steady state concentration profile assumption due to temperature change

The equilibrium concentration profile is always a function of temperature because the partitioning of helium between the gas phase and the glass at the ampoule's inner boundary is temperature dependent (Shackelford et al., 1972). A new steady-state helium concentration profile must be established every time the temperature is changed. Because of this behavior, it is advantageous to characterize the timescales necessary to achieve a steady state diffusive profile at different temperatures, as well as the magnitude of the uncertainties that result from applying Eq. (1.22) immediately after a temperature change, when the diffusive steady state concentration profile is violated.

A new steady state concentration profile must first be established when the temperature is changed for Eq. (1.22) to strictly apply. It is possible to design experiments where permeation measurements are taken only after a diffusive steady state concentration profile is established within the ampoule, which is further discussed below.

It may unduly inconvenience the experimentalist to experimentally enforce the diffusive steady state condition when the uncertainties introduced by violating the assumption are small. In Fig. 1.22, Eq. (1.41) is plotted for relevant experimental temperatures using the solubilities reported by Swets et al. (1961). Fig. 1.22 shows that the helium volume density of glass at the ampoule's inner boundary will be between 2.5 and 3% of the volume density of equilibrated gaseous helium. The overall variation in helium partitioning is small ($\leq$12%) across temperatures relevant to step heat experiments on encapsulated goethite (0-300°C). There will be excess helium within the glass when changing from a higher temperature experimental step ($\geq$175°C) to a lower temperature

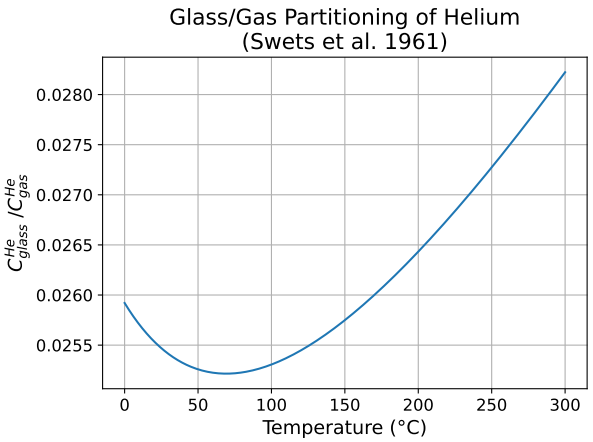

Figure 1.22: Helium preferentially partitions into the gas phase over silica glass. Eq. (1.41) is plotted here: $\frac{C^{glass}_{He}}{C^{gas}_{He}} = S \cdot RT$. It is the ratio of concentrations (in units of atoms/mL) for helium in silica glass over gas at equilibrium using solubility values of 4He from Swets et al. (1961). This partitioning is only a weak function of temperature, so the concentration ratio is between 2.5% and 2.9% for temperatures between 0 and 300°C.

permeation step (≤120°C) that would result in a slightly higher measured value than what would be predicted by Eq. (1.20). Other temperature shifts could also result in a deficit of dissolved helium instead of an excess (e.g. when shifting temperatures from 70°C to 25°C: Figs. 1.23, 1.24).

$$Disequilibrium\ Excess = \ n^{glass}_{He}\ (T_1) - n^{glass}_{He}\,(T_2) \tag{1.45}$$

The disequilibrium excess of Eq. (1.45) can be made into a relative error by dividing by the ideal amount of permeated helium measured in a permeation step from Eq. (1.20).

$$Relative\ Disequilibrium\ Error\ = \frac{n^{glass}_{He}\ (T_1) - n^{glass}_{He}\,(T_2)}{n_0(1 - e^{-\lambda(T_2)t})} \tag{1.46}$$

To evaluate Eq. (1.45) and (1.46), n^{glass}_{He} must be calculated. An upper bound can be obtained by assuming a steady state diffusion profile and an ideal open tube geometry (with small tops and bottoms neglected for simplicity), assuming that the helium partial pressure outside of the ampoule is 0 and defining the abundance of helium within the gas phase as n^{gas}_{He} . The radial concentration profile at diffusive steady state as a function of radius and temperature is

given in Eq. (1.35). Radially integrated, the total amount of helium dissolved within the glass is given in Eq. (1.37), which can be solved for n_{He}^{glass} , yielding Eq. (1.47).

$$n_{He}^{glass} = \frac{n_0}{1 + \left[-S(T) \bullet GRT \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)\right]^{-1}} \tag{1.47}$$

Where a is the radius of the ampoule's inner wall and b is the radius of the ampoule's outer wall. For legibility, the terms that are not a function of temperature are grouped as the parameter H.

$$H = -GR \bullet \left(\frac{1}{2}a^2\left(\ln\left(\frac{b}{a}\right) + \frac{1}{2}\right) - \frac{b^2}{4}\right)$$

Now Eq. (1.47) can be expressed as:

$$n_{He}^{glass} = \frac{n_0}{\left(\frac{1}{HTS} + 1\right)}$$

$\frac{n_{He}^{glass}}{n_0}$, the fraction of helium atoms within the ampoule dissolved within glass, is a function only of temperature for a given ampoule. This quantity is evaluated using Eq. (1.47) for several tubes of varying glass/gas volume ratios that share a geometric factor for permeation of $G = 4350$ cm^{-2} and plotted in Fig. 1.23. Although these model tubes would all permeate helium at the same rate in diffusive steady state, they have different sizes, thicknesses, and relative internal (gas) and glass volumes, which affect how long they would need to achieve a steady state diffusive concentration profile after thermal perturbation and how severely diffusive disequilibrium would cause the amount of permeated helium to vary from the predicted amount from Eq. (1.19). Fig. 1.23 gives the intuitive result that ampoules containing more glass relative to gas will produce larger helium excesses/deficits when permeating immediately after thermal perturbation.

Now that the amount of helium atoms dissolved in the glass is defined as a function of temperature, Eq. (1.47) evaluated at the old and new temperatures (T_1 and T_2 respectively) can be substituted into Eq. (1.46) to estimate the relative measured helium excess caused by thermal perturbation as a function of time. Considering that the predicted value of n_0 is proportional to the

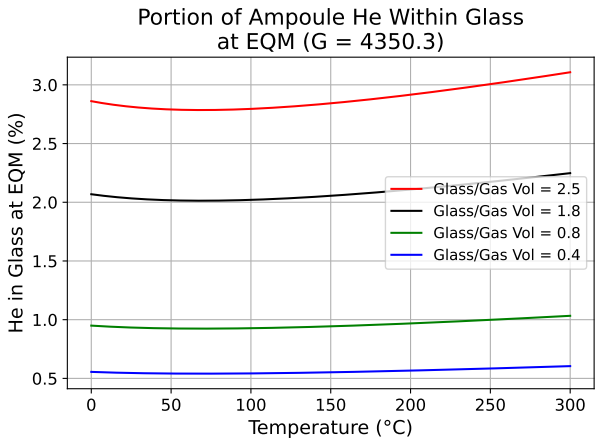

Figure 1.23: Most helium trapped within the ampoules of this work resides in the gas phase. Permeation from ampoules with more glass relative to gas will deviate more strongly from the expected permeation at diffusive steady state (Eq. (1.20)) after thermal perturbation. At diffusive equilibrium, the concentration of helium at the ampoule's inner wall is defined by the equilibrium concentration ratio with the gas phase plotted in Fig. 1.22. The concentration at the outer wall is effectively 0. The concentration profile at any point along a transect through the tube is defined by Eq. (1.35). Eq. (1.36) integrates this concentration profile along the radius to calculate the total amount of helium within the glass, and this value is divided by the total amount of helium within the gas to show the relative distribution of helium between glass and gas for ampoules as a function of inner and outer radii and temperature. It is solved here using solubilities of 4He from Swets et al. (1961) for several tubes with varying glass/gas volume ratios that share the same geometric constant of permeation G. Red, black, green, and blue lines correspond to inner radii a of 270, 300, 400, and 500 microns and outer radii of 507, 500, 533, and 601 microns respectively. The black line follows the "standard ampoule" of this work.

measured amount of helium in a permeation step according to Eq. (1.19), the *Relative Disequilibrium Excess* defined below applies to both the measured value of helium atoms m and to the calculated value of helium within the ampoule at the beginning of the permeation step n_0.

$$Relative\ Disequilibrium\ Excess = \frac{\left(\frac{n_0}{\left(\frac{1}{HT_1S_1}+1\right)} - \frac{n_0}{\left(\frac{1}{HT_2S_2}+1\right)}\right)}{n_0(1-e^{-KGRT_2t})}$$

Upon factoring and simplifying, n_0 cancels out, removing the dependence of this error estimation on knowledge of the true initial helium content of the

ampoule.

$$\textit{Relative Disequilibrium Excess} = \frac{\left(\frac{1}{\left(\frac{1}{HT_1S_1}+1\right)} - \frac{1}{\left(\frac{1}{HT_2S_2}+1\right)}\right)}{1-e^{-KGRT_2t}} \tag{1.48}$$

The relative excess helium that would be measured in a permeation step following a high-temperature experimental step is plotted in percent in Fig. 1.24 as a function of permeation time according to Eq. (1.48) for several initial experimental temperatures and permeation temperatures. This relative error decreases as the new equilibrium concentration profile is established, approaching the permeation behavior expected from Eq. (1.19). These excesses assume an instantaneous transition from experimental step to permeation step and therefore are upper bounds. These excesses are dependent also on the ampoule dimension; in Fig. 1.24, a standard ampoule ($a = 0.03$ cm, $b = 0.05$ cm) is assumed. Thicker ampoules will produce higher magnitudes of excesses (or deficits) for the same heating schedules. Within the temperature range 0-300°C the maximum excess helium that could be introduced into a permeation step measurement due to diffusive disequilibrium would be for an 80°C permeation step following an experimental step at 300°C. Even for those circumstances, the maximum relative error would be $\leq$2% for a 3-hour permeation step (Fig. 1.24). It is important to recognize, however, that this error estimation does not account for non-instantaneous cooling, although fast thermal response for similar experiments and the exponential temperature dependence of diffusion and permeation typically permit experimentalists to neglect fast but non-instantaneous heating and cooling (Farley et al., 1999). These calculations show that, at least for ampoules of similar dimensions to those sealed at Caltech, the error in Eq. (1.20) resulting from violating the diffusive steady state assumption of Eq. (1.22) is small.

For cases where more precise determinations of n_0 and n_t from the application of Eq. (1.20) on a permeation step are desired, it is possible to experimentally ensure compliance with the diffusive steady state concentration assumption. For example, the ampoule could be programmed to be held at the target permeation temperature (for example, 80°C) immediately after a high temperature experimental step instead of completely cooling (to ~25°C) while evolved gases are processed and helium is purified from the high temperature

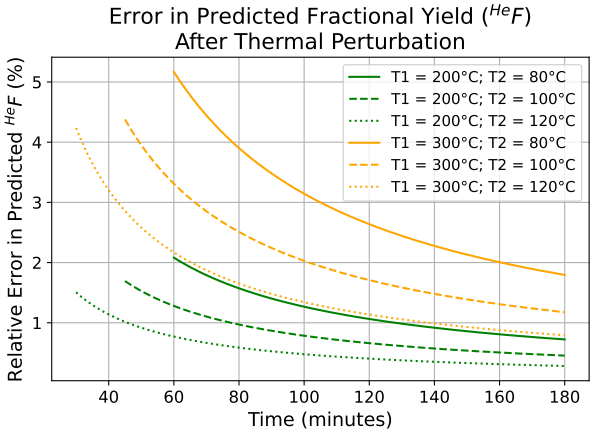

Figure 1.24: When changing temperatures, the inner boundary condition and diffusive steady state concentration profile of helium within a silica glass ampoule change, so the amount permeated will be different from the expected permeate amount of Eq. (1.20) that assumes a diffusive steady state profile has already been established at the new temperature. This difference is calculated here as the difference of helium dissolved within the tube at the starting and final temperature (neglecting the change in helium abundance in the gas phase) and is normalized to the expected permeation amount at the final temperature as a function of time for several temperature jumps, using Eq. (1.36) and solubilities of 4He from Swets et al. (1961) for a standard ampoule (with inner and outer radii of $a = 300$ microns and $b = 500$ microns respectively). Generally, larger temperature jumps produce larger relative errors. For a standard ampoule, the error of Eq. (1.20) will be less than 2% for any permeation step 3 hours or longer following a 300°C experimental step.

experimental step. To do this, one must be aware of the timescale necessary to reach diffusive equilibrium at a given temperature. At Caltech, helium purification after heating steps typically takes 30 minutes, which is only enough time to establish steady state concentration profiles for ampoules <175 µm thick at 100°C (main text Fig. 1.6). For larger ampoules or cooler permeation step temperatures, an additional delay may need to be programmed in order to experimentally enforce the diffusive steady state assumption. Alternatively, expected permeation fractions for any arbitrary thermal schedule can be numerically solved, provided that the solubility and diffusivity are each known for the experimental temperature range.

The Fourier number can be used to determine the time necessary to establish diffusive equilibrium of the helium concentration profile within the ampoule. The Fourier number is a dimensionless number that describes the extent of

diffusive equilibration or homogenization of a system by heat or mass transfer given a characteristic length scale and time (Ostrogorsky and Glicksman, 2015). It is the time divided by the diffusive timescale:

$$Fo = \frac{Dt}{L^2} = \frac{t}{\tau} \tag{1.49}$$

Where t is time and L is the diffusive length scale. For the purposes of helium permeation through silica glass ampoules, L is the glass thickness of the ampoule. When $Fo > 1$, a system is generally said to be effectively homogenized or at equilibrium with respect to diffusion, but it is somewhat arbitrary to define a specific Fourier number at which one can confidently say that diffusion has effectively re-established an equilibrium concentration profile (Ostrogorsky and Glicksman, 2015). To test the value of $Fo = 1$ as the relevant minimum value, we use the numerical diffusive permeation model described above for a characteristic ampoule where a = 300 µm and b = 500 µm. Dirichlet boundary conditions are imposed at the ampoule's inner and outer boundaries, where r = a and r = b, holding $C^{Glass}_{r=b} = 0$ and $C^{Glass}_{r=a} = C^{Gas}_{He} \bullet \left(\frac{C^{Glass}_{He}}{C^{Gas}_{He}}\right)_{EQM}$ as calculated in Eq. (1.41) and plotted in Fig. 1.22. For the analysis of the Fourier number during the approach to the diffusive equilibrium concentration profile in Figs. 1.25 and 1.26, C^{Gas}_{He} is arbitrary and held constant, so plotted concentrations are normalized to C^{Gas}_{He}. A steady state concentration profile (Eq. (1.33)) is taken as the initial condition. The absolute concentration is technically dependent on the ampoule height, but this dependence on h vanishes when normalizing the glass concentration to the gas concentration. The results of these calculations are plotted in Figs. 1.25 and 1.26, where Fig. 1.25 shows the transition from an equilibrium concentration profile at 300°C to 100°C, and Fig. 1.26 shows the more dramatic establishment of an equilibrium concentration profile at 100°C from an initial helium concentration of 0 throughout, such as could be expected after suddenly introducing helium into an ampoule that previously contained no helium. Both figures show that an equilibrium concentration profile is well-established for Fourier number $Fo = 1$, and Fig. 1.25 in particular shows that the difference in concentration profiles is small and nearly reestablished even when the Fourier number $Fo = 0.1$. We conclude that the timescale associated with $Fo = 1$ is the relevant time needed to hold the ampoule to ensure that Eq. (1.20) strictly applies without errors due to thermally-forced disequilibrium in the helium concentration profile.

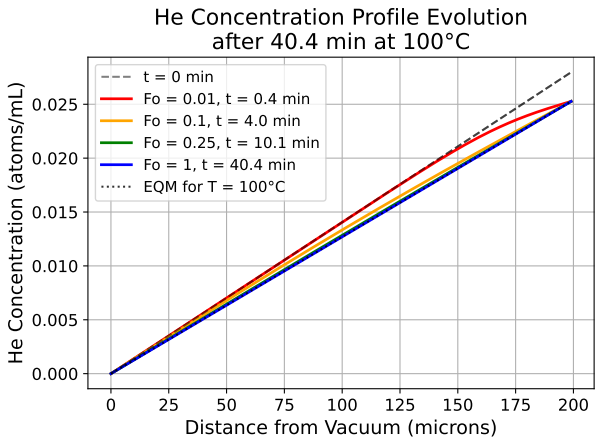

Figure 1.25: A numerical simulation of the evolution of 4He concentration profiles in silica glass ampoules from steady state at 300°C to steady state at 100°C. The steady state concentration profile is only subtly different for different temperatures. The diffusive timescale $\tau = L^2/D$ corresponding to a Fourier number of 1 is effectively sufficient to reestablish a steady state concentration profile at the new temperature. The concentration is arbitrary and is normalized to the concentration of helium in the gas phase of the ampoule, which is held constant for this simulation (therefore the ideal steady state of Eq. (1.35) applies regardless of the glass/gas volume ratio). The inner boundary condition is determined as a function of temperature using the values of Eq. (1.41), plotted in Fig. 1.22. The ampoule simulated here is a standard ampoule with an outer radius of 500 microns and inner radius of 300 microns.

The diffusive timescales associated with $Fo = 1$ for permeability values taken from Swets et al. (1961) are plotted in main text Fig. 1.6 for a few different thickness values (L) as a function of temperature for typical permeation step temperatures used in this study.

1.3s Estimating permeation lag uncertainty for uncorrected, long heating steps

The uncertainty of the helium evolution of the encapsulated natural material during a heating step (ω) due to permeation lag correction can be estimated by calculating the maximum possible helium residing in the tube between steps, i.e. for the values for n_0 and n_t: n_0^{UB} and n_t^{UB}. From the permeation lag equation (main text Eq. (1.1)), upper bound estimates for n_0 and n_t bound the encapsulated material contribution ω within:

$$m - n_0^{UB} \leq \omega \leq m + n_t^{UB} \tag{1.50}$$

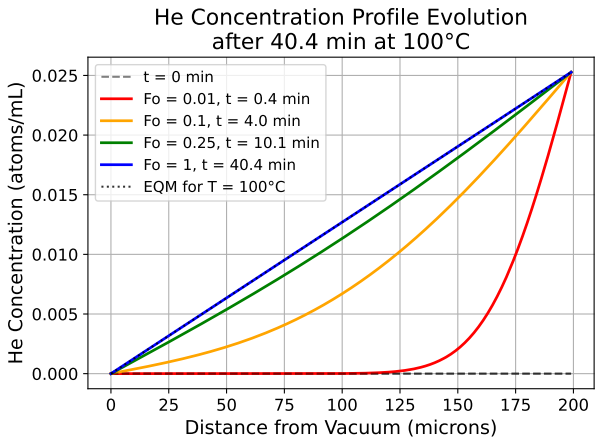

Figure 1.26: Similar to Fig. 1.25, but starting from a helium concentration of 0 throughout the ampoule. This is most useful to visualize the incomplete establishment of a steady state concentration profile for timescales corresponding to Fourier numbers up to 0.25. When the Fourier number is 1, the concentration profile is in diffusive steady state.

Upper bounds for n_t for a given high temperature experimental step can be calculated if step heats are long enough to assume steady state permeation, assuming G and K are known or can be conservatively estimated. To proceed, one must make the generally conservative assumption that the production rate of helium within the ampoule due to degassing of the encapsulated natural sample ($\Omega = \frac{\omega}{t}$) is constant over time at a given temperature. In reality, Ω will decrease over time for helium loss governed by first order processes (O. V. Yakubovich et al., 2010; W. D. Wilson and Bisson, 1973) or diffusion (except for the case of initially inhomogeneous distributions, the memory of which is often erased early in any diffusion experiment: Shuster et al., 2004). Eq. (1.52), given below, shows that the amount of helium trapped in the tube (n_t) is proportional to the rate of helium outgassing from the natural encapsulated sample (Ω), so the approximation that this rate is constant will overestimate the amount of unanalyzed helium remaining in the ampoule after a high temperature experimental step. The rate of helium degassing over time will decrease for degassing controlled by second-order volume diffusion *except* for the minor case of the alpha-ejection effect at low He-release fractions (Farley et al., 1996; Ketcham et al., 2011) or for the more serious effects of radiation damage annealing (Goldsmith et al., 2020) and mineral decomposition (Farley et al., 2024). In any case, sequential isothermal steps immediately reveal when $\frac{\omega}{t}$ increases over time.

A new first order differential equation like Eq. (1.19) is formulated, now incorporating a constant source term (Ω) to conservatively approximate the helium influx into the ampoule from the degassing natural sample.

$$\frac{dn}{dt} = \Omega - \lambda n \tag{1.51}$$

Where $\Omega = \frac{\omega}{t}$ and $\lambda = KGRT$. Immediately, the steady state quantity of helium atoms (n_{ss}) present in the ampoule can be found by solving Eq. (1.51) for the steady state condition where $\frac{dn}{dt} = 0$.

$$n_{ss} = \frac{\Omega}{\lambda} = \frac{\omega}{\lambda t} \tag{1.52}$$

Additionally, the helium concentration within the ampoule during the approach to the steady state concentration can be solved for by integrating Eq. (1.51) with respect to t. First, the integrating factor for Eq. (1.51) is identified: $I = e^{\lambda t}$, which is multiplied to both sides to yield Eq. (1.53).

$$e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n = e^{\lambda t}\Omega \tag{1.53}$$

Then both sides are integrated with respect to *t:*

$$\int \left(e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n \right) dt = \int e^{\lambda t}\Omega dt$$

Because $\frac{d}{dt}(ne^{\lambda t}) = e^{\lambda t}\frac{dn}{dt} + e^{\lambda t}\lambda n$ by the product rule Eq. (1.53) reduces to:

$$n_t e^{\lambda t} = \frac{\Omega}{\lambda}e^{\lambda t} + C \tag{1.54}$$

Where C is an unknown constant introduced by indefinite integration. This expression becomes more intuitive when both sides are divided by the integrating factor and by substituting the steady state concentration from Eq. (1.52).

$$n_t = n_{ss} + Ce^{-\lambda t} \tag{1.55}$$

The integration constant C can be solved by enforcing the condition that $n_t(t = 0) = n_0$, leading to:

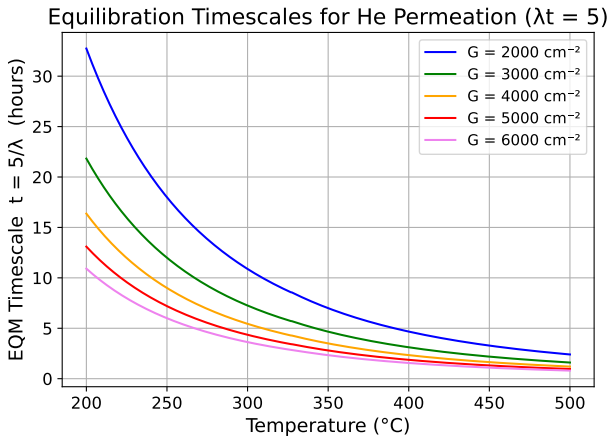

Figure 1.27: The upper limit of helium contents within the ampoule between high temperature experimental steps can be reasonably estimated without further measurement, provided permeative equilibrium has been reached. For typical ampoules with geometric factors between 3000 and 4000 cm^{-2}, this criterion is generally met for steps longer than 24 hours, or within 5 hours at 350°C or higher.

$$n_0 = n_{ss} + C \tag{1.56}$$

Solving Eq. (1.56) for C and substituting into Eq. (1.55) yields Eq. (1.57). Now an inequality is assigned to the equation to reflect the initial assumption that the isothermal helium outgassing rate (Ω) is constant *or decreasing*.

$$n_t \leq \ n_{ss} + (n_0 - n_{ss})\, e^{-\lambda t} \tag{1.57}$$

For long timescales (i.e. when $e^{-\lambda t} \approx 0$), Eq. (1.57) reduces to $n_t \leq \ n_{ss}$. This is effectively true for steps long enough such that $t \geq \frac{5}{\lambda}$, for which $e^{-\lambda t} \leq 0.007$. It is advisable to only consider the bounding method for steps of sufficiently long duration such that $t \geq \frac{5}{\lambda}$ so that the estimation for n_t can be reduced to simply the steady state value and not be dependent upon the estimation for n_0 or its uncertainties. The timescales required to reach permeation equilibrium are plotted as a function of temperature in Fig. 1.27 below for some select values of G, the geometric factor.

To estimate the upper bound on the permeation lag (n_t^{UB}), one must consider the circumstances that would produce the highest possible value for n_t. In addition, the constraint that $t \geq \frac{5}{\lambda}$ is enforced so that previous heating steps

are not significant, simplifying the problem significantly. Because n_t scales with ω by Eq. (1.52), the upper bound estimate n_t^{UB} must assume the upper bound estimate for ω, which is given in Eq. (1.50) as $m + n_t^{UB}$. Substituting $\omega^{UB} = m + n_t^{UB}$ into Eq. (1.52) to express n_t^{UB} in terms of the measured value m:

$$n_t^{UB} = n_{ss}(\omega^{UB} = m + n_t^{UB}) = \frac{m + n_t^{UB}}{\lambda t}$$

Solving for n_t^{UB} yields Eq. (1.58):

$$n_t^{UB} = \frac{m}{\lambda t - 1} \tag{1.58}$$

The upper bound amount of helium-4 within the ampoule normalized to the measured value in percent at steady state permeation is plotted in main text Fig. 1.8 using Eq. (1.58) $\left(\frac{n_t^{UB}}{m} = \frac{1}{\lambda t - 1}\right)$, assuming a standard tube where G = 4350.26 cm^{-2} and the permeability values given in Swets et al. (1961). Main text Fig. 1.8 shows that, in order to push this uncertainty to be less than 5%, even very hot steps require long step durations. For a 5% uncertainty using this method, a 10-hour step duration is required even at 400°C. Much longer durations are required for the same precision at lower temperatures: at 250°C, a step duration of 35 hours is required to push the uncertainty beneath 5%.

1.4s Comparison to Literature Permeability Values

Predicted and measured fractional yields for all permeation steps in ampoules besides the encapsulated water experiment agree well (within 1σ: Figs. 1.11, 1.29) with the predictive equations presented in this work and with the permeabilities reported by Swets et al. (1961) and Jones (1953) for all studied temperatures (24-300°C; Fig. 1.28) and step fractional yields (up to ~80%). One outlier measurement on the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer was excluded from Figs. 1.11 and 1.29.

Permeation ratios for 3He and 4He measured in the empty tube experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer are plotted in Fig. 1.30. These data present significant scatter, but generally agree with the reported results of Jones (1953), except for measurements at room temperature, which cluster tightly at a permeation ratio of 1:1 for the two isotopes. Additionally, fields corresponding to the ratios of apparent permeabilities for the water-filled

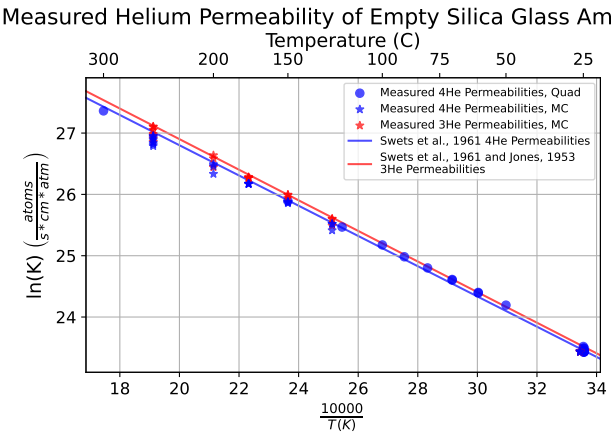

Figure 1.28: Helium permeabilities measured in silica glass ampoules at Caltech on a quadrupole mass spectrometer (4He only) and on a Helix-MC-Plus^{10K} (both 3He and 4He) multi-collector mass spectrometer are compared to literature values. Permeabilities of 3He and 4He are very similar, but 3He permeates slightly faster. The final measurement on the Helix-MC-Plus^{10K} mass spectrometer is an outlier and is excluded from the plot, as it is most sensitive to the value of the total yield remaining in the ampoule. These experiments on empty tubes at Caltech show excellent agreement with the literature values with allowance for some scatter. This result shows that existing understanding of helium permeability in glass can be reasonably leveraged for use in helium step heats of silica glass encapsulated geological samples. Literature 4He permeabilities are taken from Swets et al. (1961); whereas 3He permeabilities from the literature are calculated by multiplying the 3He/4He permeability ratios of Jones (1953) with the 4He permeabilities of Swets et al. (1961).

and sulfur-filled experiments are plotted for the temperature ranges of those experiments. Results for those experiments are plotted as fields because each step integrated permeation behavior over a complex (non-isothermal) cooling history. These experiments also agree with the results of Jones (1953) and the empty ampoule experiment. The qualitative result of Jones (1953) that silica glass is only slightly more permeable for 3He than 4He is supported by these data, which are in marked contrast to the reported extreme differences in 3He and 4He diffusivity in silica glass of McMurtry et al. (2019), who report instantaneous isotopic fractionations up to a factor of 1000.

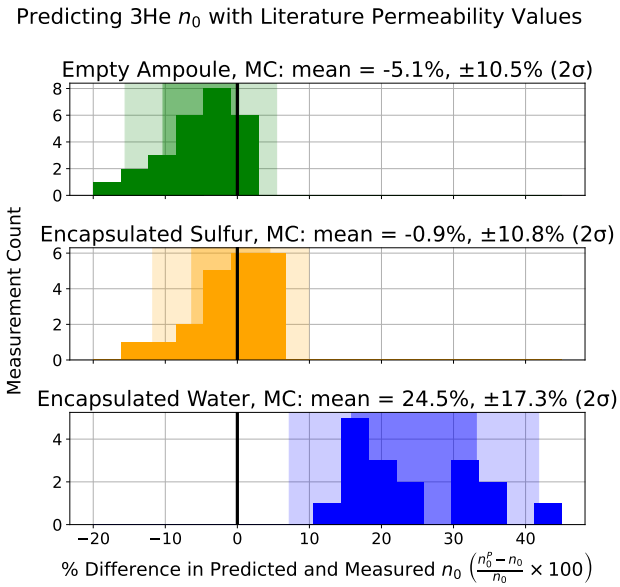

Figure 1.29: Similar to Fig. 1.11, but with 3He permeabilities taken from Jones (1953) and Swets et al. (1961). Shaded regions behind the histograms correspond to 1 and 2 sigma (σ) from the average relative difference between the predicted and measured values. The 3He permeation model and literature permeability values apply well to empty and sulfur-filled ampoules, and such models can be used to reliably recover 3He contents of permeating ampoules for use in permeation lag corrections. On the other hand, measured and predicted values of 3He n_0 for water-filled ampoules are well resolved beyond 2σ, suggesting that their permeability differs strongly from the literature values. Internal reproducibilities (1σ) of a gas standard repeatedly analyzed during these experiments were better than 3% for 3He, requiring that much of the scatter in these data be related to factors outside of analytical precision such as long/inconsistent ramp times during heating or uncertainties or rounding of heating times.

1.5s Additional Analytical Corrections for the Empty Ampoule Calibration Experiments

Pump-Out and Gas-Preparation Corrections for the Atmosphere-Equilibrated Empty Ampoule

Using the helium permeability values of Swets et al. (1961) for silica glass, an ampoule with a geometric factor (G) of 6000 cm^{-2} would permeate 0.2‰ of its helium in 1 minute at 26°C (Eq. (1.4)), which is negligible for the scope of this work. The atmosphere-equilibrated empty ampoule with a geometric factor (G) of ~4785 cm^{-2} would permeate 2.8‰ of its helium in its 15-minute pumping time between steps. For that experiment only, helium loss during pumping is accounted for by applying the permeation relationship of Eq. (1.4)

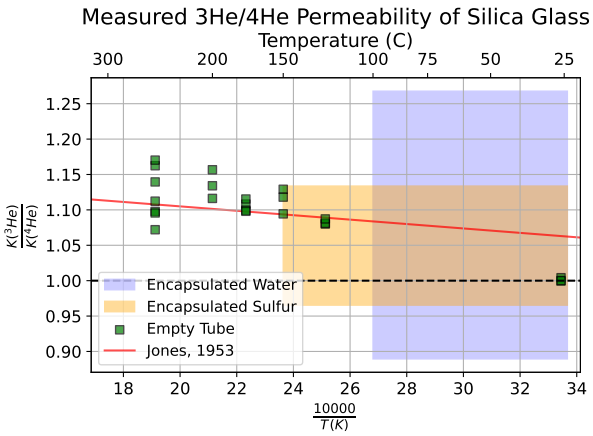

Figure 1.30: Ratios of calculated permeabilities from step heat experiments on silica glass ampoules on the Helix-MC-Plus^{10K} mass spectrometer at Caltech. The last measurement on the empty tube experiment is an outlier and was excluded from the plot. Experiments on encapsulated water and encapsulated sulfur were allowed to cool before isotopic measurements. Because of slow cooling rates, simple corrections could not be applied to these two experiments, which effectively queried permeabilities at a range of temperatures within a single step. Therefore, the ratios of their apparent permeabilities are plotted as a field with 2σ standard deviation across the experimental temperature range. These three experiments show general, albeit imprecise, agreement with the results of Jones (1953), except for the measurements at room temperature. Jones (1953) was likely able to measure the relative permeabilities of 4He and 3He with much higher precision because his experimental vessel was charged with 3He at a partial pressure of around 0.5 atmospheres, over one billion times higher than the 3He partial pressure within the fused silica ampoule used in this work. Interestingly, three room-temperature measurements of helium permeation of an empty tube at Caltech yielded essentially identical permeability of 3He and 4He.

($n_0 = n_t/(1 - {}^{He}F)$) to solve for n_0, the amount of helium in the ampoule before pump-out, from n_t, the amount remaining after pumping with T = 25°C and t = 15 minutes. A permeability coefficient (K) for helium in silica glass had to be assumed in order to apply this correction, so the value of 1.54×10^{10} $\frac{atoms}{s \times cm \times atm}$ at 25°C given by the Arrhenius law reported by Swets et al. (1961) was used. The validity of this choice is supported by the data; the average calculated permeability at room temperature for this experiment after correcting for helium lost during the 15-minute pump out is also 1.54×10^{10} $\frac{atoms}{s \times cm \times atm}$.

For that experiment only, a correction was also applied to the high temperature

steps because they were heated for short durations (45 minutes for all but one heated step), so that the permeation of the ampoule at room temperature during gas cryofocusing represented a significant portion of the step's yield. This was similarly corrected by calculating the expected fractional yield at room temperature during the gas preparation time, which is the step's total time minus the pump out time and the heated time (29–31 minutes for all heated steps). For that correction, the permeation relationship of Eq. (1.4) ($n_0 = n_t/(1 - {}^{He}F)$) is applied again to solve for the unknown n_0, the amount of helium within the ampoule immediately after the completion of the heated step, from n_t, the amount of helium within the ampoule when the gas is inlet into the mass spectrometer about 30 minutes later after spiking and helium purification. This cooling and gas preparation correction is not applied to the room temperature measurements for which the temperature remained constant during gas preparation.

Anomalous Signal Recalculation for the Tank Helium-Soaked Empty Ampoule

The last high temperature step of the empty ampoule experiment on the Helix-MC-Plus^{10K} multi-collector mass spectrometer had an unusual and non-linear signal evolution at the beginning of the measurement for both 3He and 4He. For this measurement, 4He was recalculated using a linear regression on the last three of six measurement cycles with an R^2 of 0.985 to regress to the 4He content at time 0. Typically, the signal evolution for a measurement like this would have a negative slope because the helium in the mass spectrometer is consumed as it is ionized and measured, but the slope for this measurement was positive. 3He from this measurement was calculated based on the recalculated 4He and an assumed isotope ratio for the measurement of 0.07 Ra. Three more measurements were taken over the next few days to ensure complete degassing of the ampoule. Two of these measurements yielded helium above blank, but were small compared to the final high temperature step. The yields of these steps were all considered in the calculation of the experiment's total yield.

Independent Verification of the Encapsulated Durango Apatite Mass

The total yield for the encapsulated Durango apatite experiment (main text, §1.5) is estimated from the precisely-measured mass of the encapsulated apatite (3.18 mg) and a helium concentration of 194.2 nccSTP/mg (Farley, 2000). This

Table 1.4: In-situ permeation rate constants of the empty ampoule, fit to consecutive isothermal permeation steps with Eq. (1.9). The first steps of the 125 and 200 °C runs are excluded (see main text), leaving exactly determined two-step fits. The last two 250 °C accumulations (steps 28–29) are also excluded. n_0 (true) is known independently from the experiment's total yield but is not used in the fits. λ_{pred} is predicted from the measured ampoule geometry and literature permeabilities (Swets et al., 1961; Jones, 1953) and is shown only as a check. Uncertainties are 1σ from the fit covariance.

Isotope	T (°C)	Steps	λ ($10^{-5}\,\mathrm{s}^{-1}$)	$\lambda/\lambda_{\mathrm{pred}}$	n_0 fit / true	λt_{cum}	$\lambda(^3\mathrm{He})/\lambda(^4\mathrm{He})$
^{4}He	125	2 (8–9)	2.063 ± 0.235	0.945	1.036	0.13	
^{3}He	125	2 (8–9)	2.075 ± 0.472	0.873	1.110	0.13	1.006 ± 0.256
^{4}He	150	3 (11–13)	3.110 ± 0.074	0.930	1.033	0.40	
^{3}He	150	3 (11–13)	3.525 ± 0.139	0.964	1.014	0.45	1.133 ± 0.052
^{4}He	175	4 (14–17)	4.662 ± 0.040	0.951	0.999	0.59	
^{3}He	175	4 (14–17)	5.273 ± 0.108	0.980	0.981	0.67	1.131 ± 0.025
^{4}He	200	2 (20–21)	5.825 ± 0.063	0.841	1.122	0.44	
^{3}He	200	2 (20–21)	6.277 ± 0.392	0.823	1.167	0.48	1.078 ± 0.068
^{4}He	250	6 (22–27)	12.216 ± 0.013	0.969	0.996	2.33	
^{3}He	250	6 (22–27)	14.233 ± 0.137	1.020	0.986	2.72	1.165 ± 0.011

concentration is independently supported by the measurement of 8 different >0.5 mg aliquots of the same Durango apatite sample in-lab, which yield a concentration of 194 ± 9 nccSTP/mg.

1.6s In-situ Measurements of the Rate Constant λ of Helium Permeation

1.7s Supplementary Data Tables

The following tables present the reduced step-by-step helium permeation data underlying the empty-tube, encapsulated-water, and encapsulated-sulfur calibration experiments summarized in the ampoule dimensions table of the main text, Table 1.1 and the permeability/isotope-ratio comparison figures above. m is the measured helium abundance of the step; n_0 and n_t are the helium contents of the ampoule at the beginning and end of the step (Eq. (1.20)); n_t/n_0 predicted and measured are the fractional yields expected from the permeability of Swets et al. (1961) (and, for 3He, the isotope ratios of Jones (1953)) and as measured, respectively.

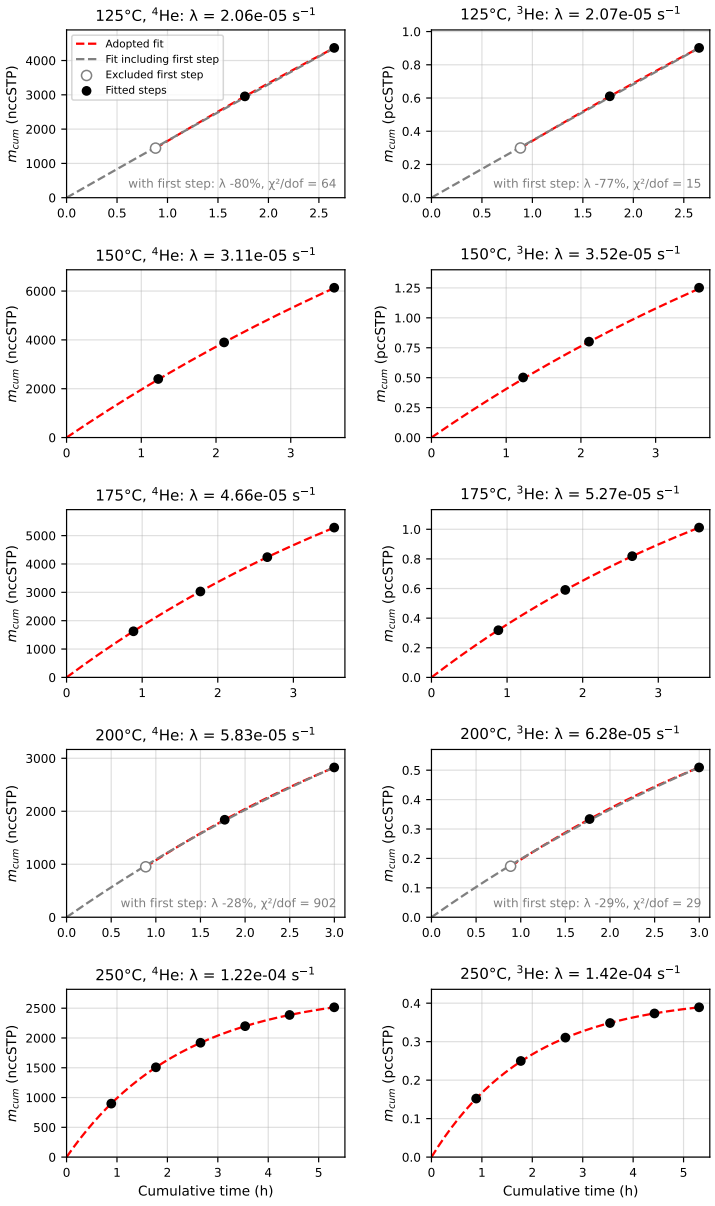

Figure 1.31: Cumulative yield against cumulative time for each isothermal run of permeation steps of the empty ampoule (rows: temperature; columns: isotope), with the rate constants fit in situ using Eq. (1.9) (red dashed curves). Hollow grey points are the first steps of the 125 and 200°C runs, which are excluded because the glass had not yet reached a diffusive steady state; the red curves for those runs are fit to the remaining two steps, and the grey dashed curves are the fits that include the first step (annotated with the change in λ and the resulting χ^2/dof).

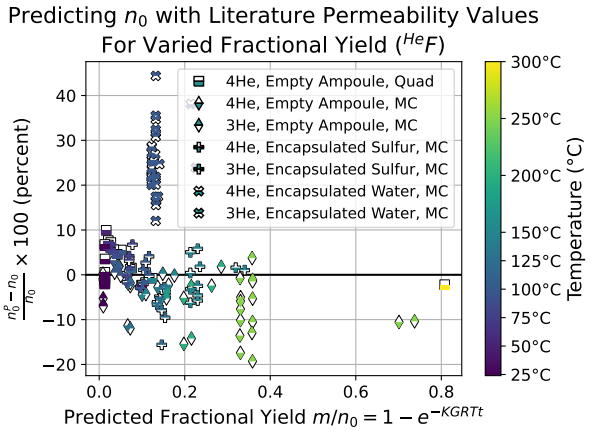

Figure 1.32: Helium measurements (m) of a permeating ampoule are related to the initial helium contents of the ampoule (n_0) by Eq. (1.22), using permeability values taken from Swets et al. (1961) and Jones (1953) (for 3He only). The n_0^P variable is the predicted value of n_0, the true value. For the empty ampoules, except for 3 measurements on the Helix-MC-Plus^{10K}, predicted values (n_0^P) are calculated using Eq. (1.20) (with some corrections applied to the quadrupole experiment: see main text for details). For the encapsulated sulfur and water ampoules, as well as 3 measurements of the blank ampoule on the Helix-MC-Plus^{10K}, predicted values (n_0^P) were determined by forward modeling diffusion and permeation on ampoules as they were held at high temperature and subsequently allowed to cool (see main text for details). This plot shows that the predicted values are correct to within ~10% for empty and sulfur-filled ampoules regardless of temperature or extent of permeation, whereas n_0^P values for water-filled ampoules are systematically high. The final Helix-MC-Plus^{10K} measurement on the empty ampoule yields an outlier n_0^P that was ~80% high — this measurement is not plotted here.

Table 1.5: Distilled helium permeation data for the empty silica glass ampoule equilibrated with atmosphere, measured by isotope dilution on the Pfeiffer Prisma QMS 200 quadrupole mass spectrometer ("Empty Tube, Pfeiffer Quad" in the ampoule dimensions table of the main text, Table 1.1). n_0 and n_t are the initial and final helium contents of the ampoule for each step (Eq. (1.20)); n_t "before cooling" is only populated for high-temperature experimental steps, for which the room-temperature pump-out/cool-down correction discussed in the main text is applied to obtain the "after cooling" value used to calculate permeability. $\ln(K)$ values are the natural log of the permeability coefficient in $\frac{atoms}{s \times cm \times atm}$.

Temp (°C)	Time (min)	m 4He (nccSTP)	n_0 4He (nccSTP)	n_t 4He (before cooling) (nccSTP)	n_t 4He (after cooling) (nccSTP)	n_t/n_0 measured	$10000/T$ (K^{-1})	$\ln(K)$ measured
25	77	0.1877	12.87		12.68	0.9854	33.56	23.52
25	77	0.1793	12.64		12.47	0.9858	33.56	23.49
50	45	0.2902	12.43	12.21	12.14	0.9819	30.96	24.19
24.8	74	0.1593	12.11		11.95	0.9868	33.58	23.46
100	45	0.7129	11.92	11.26	11.20	0.9452	26.81	25.18
24.7	74	0.1450	11.17		11.03	0.9870	33.59	23.44
60	45	0.3094	11.00	10.75	10.69	0.9771	30.03	24.40
70	45	0.3619	10.66	10.35	10.30	0.9712	29.15	24.60
80	45	0.4225	10.27	9.902	9.849	0.9640	28.33	24.80
24.7	74	0.1272	9.823		9.696	0.9870	33.59	23.44
24.8	74	0.1252	9.670		9.545	0.9871	33.58	23.44
24.8	75	0.1238	9.519		9.395	0.9870	33.58	23.43
70	45	0.3209	9.370	9.098	9.049	0.9709	29.15	24.61
60	45	0.2534	9.025	8.820	8.771	0.9773	30.03	24.39
24.8	74	0.1130	8.748		8.635	0.9871	33.58	23.44
90	45	0.4231	8.611	8.231	8.188	0.9558	27.55	24.98
119.8	45	0.6625	8.166	7.543	7.504	0.9236	25.46	25.47
24.8	74	0.09578	7.484		7.388	0.9872	33.58	23.43
150	45	0.9371	7.368	6.465	6.431	0.8775	23.64	25.89
25	581	0.6179	6.414		5.796	0.9037	33.56	23.43
300	90	4.549	5.780	1.238	1.231	0.2141	17.45	27.36
24.7	74	0.01673	1.228		1.211	0.9864	33.59	23.49
400	90	1.178	1.208		0.03073			
24.7	74	0.001170	0.03065					

Table 1.6: Distilled 3He permeation data for the empty silica glass ampoule equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Empty Tube, Helix MC" in the ampoule dimensions table of the main text, Table 1.1). Rows marked '–' in the predicted and measured n_t/n_0 and permeability columns are cooling-only intervals for which no isothermal permeability was calculated (see main text).

Temp (°C)	Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He K measured ($\frac{atoms}{s\ cm\ atm}$)
26	73.6	0.05159	0.00062	5.787	5.736	0.9902	0.9911	1.53×10^{10}
–	113.3	0.2689	0.0039	5.736	5.467			
26	73.7	0.04801	0.00085	5.467	5.419	0.9902	0.9912	1.50×10^{10}
26	75.3	0.04855	0.00099	5.419	5.370	0.9900	0.9910	1.50×10^{10}
–	112.9	0.2709	0.0057	5.370	5.099			
–	126.6	0.4175	0.0088	5.099	4.682			
125	52.9	0.2993	0.0063	4.682	4.382	0.9275	0.9361	1.18×10^{11}
125	53.1	0.3111	0.0066	4.382	4.071	0.9273	0.9290	1.31×10^{11}
125	53.1	0.2912	0.0062	4.071	3.780	0.9273	0.9285	1.32×10^{11}
–	73.6	0.4224	0.0087	3.780	3.358			
150	73.6	0.5017	0.0086	3.358	2.856	0.8512	0.8506	1.95×10^{11}
150	52.9	0.2993	0.0045	2.856	2.557	0.8907	0.8952	1.86×10^{11}
150	88.4	0.4502	0.0063	2.557	2.107	0.8240	0.8240	1.94×10^{11}
175	53.1	0.3184	0.0046	2.107	1.788	0.8427	0.8489	2.59×10^{11}
175	53.1	0.2722	0.0040	1.788	1.516	0.8427	0.8478	2.61×10^{11}
175	53.1	0.2278	0.0035	1.516	1.288	0.8427	0.8498	2.57×10^{11}
175	53.1	0.1929	0.0032	1.288	1.095	0.8427	0.8504	2.56×10^{11}
–	53.1	0.1635	0.0028	1.095	0.9317			
200	53.2	0.1735	0.0032	0.9317	0.7582	0.7843	0.8139	3.07×10^{11}
200	53.1	0.1604	0.0032	0.7582	0.5978	0.7847	0.7887	3.55×10^{11}
200	73.6	0.1754	0.0038	0.5978	0.4224	0.7146	0.7070	3.75×10^{11}
250	53.2	0.1521	0.0039	0.4224	0.2704	0.6410	0.6406	6.02×10^{11}
250	53.1	0.0977	0.0030	0.2704	0.1727	0.6413	0.6396	6.06×10^{11}
250	53.1	0.0607	0.0021	0.1727	0.1119	0.6413	0.6499	5.85×10^{11}
250	53.1	0.0379	0.0016	0.1119	0.07407	0.6414	0.6639	5.58×10^{11}
250	52.9	0.0249	0.0011	0.07407	0.04914	0.6427	0.6634	5.57×10^{11}
250	53.1	0.01599	0.00077	0.04914	0.03315	0.6415	0.6746	5.32×10^{11}
250	159.9	0.0260	0.0013	0.03315	0.007191	0.2625	0.2169	6.86×10^{11}
250	52.9	0.00494	0.00025	0.007191	0.002250	0.6424	0.3129	1.58×10^{12}
High T	52.2	0.00225		0.002250	0			

Table 1.7: Distilled 4He permeation data for the same empty silica glass ampoule experiment as Table 1.6. K_3/K_4 is the ratio of the measured 3He to 4He permeability coefficients for each step; 3He/4He measured (Ra) is the measured 3He/4He ratio of the step, normalized to the atmospheric ratio.

Temp	Time	m 4He	$\pm 2\sigma$	n_0 4He	n_t 4He	n_t/n_0 predicted	n_t/n_0 measured	4He K measured	K_3/K_4	3He/4He measured (Ra)
(°C)	(min)	(nccSTP)	(nccSTP)	(nccSTP)	(nccSTP)			$(\frac{atoms}{s\,cm\,atm})$		
26	73.6	263.7	1.3	29670	29410	0.9908	0.9911	1.52×10^{10}	1.0031	0.1408
–	113.3	1282.4	8.0	29410	28130					0.1509
26	73.7	247.4	2.0	28130	27880	0.9908	0.9912	1.51×10^{10}	0.9984	0.1396
26	75.3	250.0	2.3	27880	27630	0.9905	0.9910	1.50×10^{10}	0.9990	0.1397
–	112.9	1322	13	27630	26310					0.1475
–	126.6	1971	20	26310	24340					0.1524
125	52.9	1446	15	24340	22890	0.9332	0.9406	1.09×10^{11}	1.0786	0.1489
125	53.1	1509	16	22890	21380	0.9330	0.9341	1.21×10^{11}	1.0795	0.1483
125	53.1	1413	15	21380	19970	0.9330	0.9339	1.21×10^{11}	1.0856	0.1483
–	73.6	2032	21	19970	17940					0.1495
150	73.6	2399	22	17940	15540	0.8630	0.8663	1.73×10^{11}	1.1271	0.1505
150	52.9	1498	13	15540	14040	0.8996	0.9036	1.70×10^{11}	1.0922	0.1438
150	88.4	2238	17	14040	11800	0.8377	0.8406	1.74×10^{11}	1.1155	0.1447
175	53.1	1626	11	11800	10170	0.8556	0.8622	2.34×10^{11}	1.1053	0.1409
175	53.1	1404.1	8.5	10170	8771	0.8556	0.8620	2.34×10^{11}	1.1121	0.1395
175	53.1	1210.8	6.3	8771	7560	0.8556	0.8619	2.34×10^{11}	1.0959	0.1353
175	53.1	1041.8	4.8	7560	6518	0.8556	0.8622	2.34×10^{11}	1.0939	0.1332
–	53.1	895.4	3.7	6518	5623					0.1314
200	53.2	952.2	3.4	5623	4670	0.8020	0.8306	2.77×10^{11}	1.1107	0.1311
200	53.1	887.5	3.0	4670	3783	0.8023	0.8100	3.15×10^{11}	1.1275	0.1300
200	73.6	987.1	3.3	3783	2796	0.7370	0.7391	3.26×10^{11}	1.1485	0.1278

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(continued)

Temp (°C)	Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	4He K measured ($\frac{atoms}{s\,cm\,atm}$)	K_3/K_4	3He/4He measured (Ra)
250	53.2	898.0	3.0	2796	1898	0.6693	0.6788	5.23×10^{11}	1.1520	0.1218
250	53.1	609.8	2.1	1898	1288	0.6696	0.6787	5.24×10^{11}	1.1569	0.1153
250	53.1	412.2	1.4	1288	875.7	0.6696	0.6799	5.21×10^{11}	1.1231	0.1059
250	53.1	278.75	1.00	875.7	596.9	0.6697	0.6817	5.18×10^{11}	1.0778	0.0978
250	52.9	187.74	0.69	596.9	409.2	0.6709	0.6855	5.13×10^{11}	1.0868	0.0955
250	53.1	128.81	0.49	409.2	280.4	0.6697	0.6852	5.11×10^{11}	1.0411	0.0893
250	159.9	215.41	0.95	280.4	64.98	0.2990	0.2318	6.56×10^{11}	1.0452	0.0867
250	52.9	41.86	0.20	64.98	23.12	0.6706	0.3558	1.40×10^{12}	1.1246	0.0849
High T	52.2	23.12		23.12	0					0.0700

Table 1.8: Distilled 3He isothermal permeation data for the silica glass ampoule containing distilled water, equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Encapsulated Water, Helix MC" in the ampoule dimensions table of the main text, Table 1.1). Predicted n_t/n_0 values marked '–' correspond to steps for which the ampoule had not yet reached a diffusive steady state concentration profile (see main text).

Temp (°C)	Hot Time (min)	Cold Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured
100	60	26	0.1158	0.0012	0.6447	0.5289	–	0.8204
100	60	25.1	0.0942	0.0010	0.5289	0.4347	0.8677	0.8219
100	60	25	0.07568	0.00081	0.4347	0.3590	0.8678	0.8259
100	60	25.1	0.05784	0.00062	0.3590	0.3012	0.8678	0.8389
100	60	53.5	0.05298	0.00057	0.3012	0.2482	0.8589	0.8241
100	60	25.2	0.03824	0.00043	0.2482	0.2100	0.8678	0.8459
100	60	25.1	0.03368	0.00041	0.2100	0.1763	0.8678	0.8396
100	60	795.6	0.04934	0.00025	0.1763	0.1270	0.7741	0.7201
100	60	28.1	0.019725	0.000094	0.1270	0.1072	0.8666	0.8446
100	60	43.7	0.017349	0.000084	0.1072	0.08988	0.8622	0.8382
100	60	24.9	0.013779	0.000070	0.08988	0.07610	0.8673	0.8467
100	60	24.9	0.013153	0.000076	0.07610	0.06295	0.8678	0.8272
100	60	25	0.009771	0.000063	0.06295	0.05318	0.8678	0.8448
100	60	25.1	0.009520	0.000064	0.05318	0.04366	0.8679	0.8210
100	60	25.1	0.007641	0.000052	0.04366	0.03602	0.8678	0.8250
100	60	25	0.005717	0.000039	0.03602	0.03030	0.8678	0.8413
100	60	25	0.005784	0.000039	0.03030	0.02451	0.8678	0.8091
100	60	25.1	0.003629	0.000024	0.02451	0.02089	0.8679	0.8520
REST			0.02089	0.00014	0.02089	0		

Table 1.9: Distilled 4He isothermal permeation data for the same encapsulated-water ampoule experiment as Table 1.8.

Temp (°C)	Hot Time (min)	Cold Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He/4He measured (Ra)
100	60	26	604.0	1.8	3871	3267	–	0.8440	0.1379
100	60	25.1	515.5	1.5	3267	2751	0.8772	0.8422	0.1315
100	60	25	434.8	1.3	2751	2317	0.8772	0.8420	0.1252
100	60	25.1	361.9	1.1	2317	1955	0.8772	0.8438	0.1150
100	60	53.5	325.43	0.96	1955	1629	0.8688	0.8335	0.1171
100	60	25.2	251.31	0.74	1629	1378	0.8772	0.8458	0.1095
100	60	25.1	212.62	0.63	1378	1165	0.8772	0.8457	0.1140
100	60	795.6	344.78	0.80	1165	820.7	0.7858	0.7042	0.1029
100	60	28.1	127.20	0.28	820.7	693.5	0.8761	0.8450	0.1116
100	60	43.7	111.27	0.22	693.5	582.2	0.8719	0.8395	0.1122
100	60	24.9	88.54	0.17	582.2	493.7	0.8768	0.8479	0.1120
100	60	24.9	74.96	0.12	493.7	418.7	0.8773	0.8481	0.1262
100	60	25	62.687	0.088	418.7	356.0	0.8772	0.8503	0.1121
100	60	25.1	53.284	0.072	356.0	302.7	0.8773	0.8503	0.1285
100	60	25.1	45.452	0.061	302.7	257.3	0.8772	0.8499	0.1209
100	60	25	38.407	0.053	257.3	218.9	0.8773	0.8507	0.1071
100	60	25	32.416	0.048	218.9	186.4	0.8772	0.8519	0.1284
100	60	25.1	27.465	0.044	186.4	159.0	0.8773	0.8527	0.0950
REST			158.98	0.26	159.0	0			0.0945

Table 1.10: Distilled 3He isothermal permeation data for the silica glass ampoule containing fused sulfur, equilibrated with high-purity tank helium, measured on the Helix-MC-Plus^{10K} multi-collector mass spectrometer ("Encapsulated Sulfur, Helix MC" in the ampoule dimensions table of the main text, Table 1.1).

Temp (°C)	Hot Time (min)	Cold Time (min)	m 3He (pccSTP)	$\pm 2\sigma$ (pccSTP)	n_0 3He (pccSTP)	n_t 3He (pccSTP)	n_t/n_0 predicted	n_t/n_0 measured
80	60	25.6	0.699	0.028	9.569	8.870	0.9299	0.9270
80	60	61.3	0.736	0.029	8.870	8.134	0.9222	0.9170
80	60	25.1	0.591	0.022	8.134	7.543	0.9300	0.9274
80	60	25	0.544	0.018	7.543	6.999	0.9301	0.9279
80	60	25	0.494	0.016	6.999	6.505	0.9301	0.9294
100	60	91.8	0.759	0.024	6.505	5.746	0.8831	0.8834
100	60	25.9	0.601	0.020	5.746	5.145	0.8966	0.8955
100	60	52.0	0.575	0.021	5.145	4.570	0.8912	0.8882
100	60	26	0.463	0.017	4.570	4.108	0.8966	0.8988
100	60	89.9	0.498	0.020	4.108	3.609	0.8834	0.8787
120	60	24.8	0.510	0.023	3.609	3.099	0.8537	0.8586
120	60	52.8	0.457	0.022	3.099	2.641	0.8481	0.8524
120	60	24.9	0.356	0.018	2.641	2.286	0.8537	0.8653
120	60	61.3	0.318	0.016	2.286	1.967	0.8460	0.8607
120	60	25.1	0.243	0.012	1.967	1.724	0.8537	0.8766
150	60	25.0	0.420	0.020	1.724	1.304	0.7695	0.7562
150	60	25.0	0.283	0.013	1.304	1.021	0.7696	0.7828
150	60	690.8	0.351	0.010	1.021	0.6699	0.6598	0.6563
150	60	25	0.1472	0.0040	0.6699	0.5226	0.7696	0.7802
150	60	53.3	0.125	0.003	0.5226	0.3972	0.7643	0.7601
150	60	25.0	0.0893	0.0024	0.3972	0.3080	0.7695	0.7752
remainder			0.3080	0.0085				

Table 1.11: Distilled 4He isothermal permeation data for the same encapsulated-sulfur ampoule experiment as Table 1.10.

Temp (°C)	Hot Time (min)	Cold Time (min)	m 4He (nccSTP)	$\pm 2\sigma$ (nccSTP)	n_0 4He (nccSTP)	n_t 4He (nccSTP)	n_t/n_0 predicted	n_t/n_0 measured	3He/4He measured (Ra)
80	60	25.6	3354	11	52330	48980	0.9348	0.9359	0.1499
80	60	61.3	3617	12	48980	45360	0.9276	0.9262	0.1464
80	60	25.1	2949	10	45360	42410	0.9349	0.9350	0.1441
80	60	25	2759.6	9.6	42410	39650	0.9350	0.9349	0.1419
80	60	25	2587.1	9.1	39650	37060	0.9350	0.9348	0.1373
100	60	91.8	3980	15	37060	33080	0.8913	0.8926	0.1371
100	60	25.9	3088	14	33080	30000	0.9041	0.9066	0.1399
100	60	52.0	2978	15	30000	27020	0.8990	0.9007	0.1389
100	60	26	2517	13	27020	24500	0.9041	0.9068	0.1322
100	60	89.9	2687	14	24500	21810	0.8916	0.8903	0.1334
120	60	24.8	2769	15	21810	19040	0.8646	0.8731	0.1327
120	60	52.8	2551	13	19040	16490	0.8592	0.8661	0.1290
120	60	24.9	2097	10	16490	14400	0.8646	0.8728	0.1221
120	60	61.3	1948.5	9.4	14400	12450	0.8572	0.8647	0.1176
120	60	25.1	1573.3	7.4	12450	10870	0.8645	0.8736	0.1110
150	60	25.0	2435	11	10870	8439	0.7868	0.7761	0.1242
150	60	25.0	1806.5	8.2	8439	6633	0.7868	0.7859	0.1128
150	60	690.8	2151.5	8.5	6633	4481	0.6807	0.6756	0.1173
150	60	25	924.6	3.4	4481	3557	0.7868	0.7937	0.1146
150	60	53.3	753.9	2.8	3557	2803	0.7818	0.7880	0.1197
150	60	25.0	560.4	2.1	2803	2242	0.7868	0.8001	0.1146
remainder			2242.4	8.7					0.0988

Chapter 2

HELIUM STEP HEATS ON GOETHITE AT WATER SATURATED CONDITIONS SHOW DIFFUSIVE HELIUM LOSS AND HIGH RETENTIVITY AT SURFACE TEMPERATURES

U-Th/He geochronology of goethite (FeOOH) is a powerful method to quantitatively date weathering and erosion processes on Earth's surface; however, precise interpretation of goethite U-Th/He dates is challenging because uncertainty persists regarding the mineralogical and environmental conditions under which radiogenic helium is retained and how it may be lost over time. Standard vacuum step heating experiments have been successfully used on a variety of mineral phases to characterize helium migration, but the results of such experiments on goethite predominantly record dehydroxylation-induced helium loss, raising questions about their relevance to helium retention and loss in nature (Farley et al., 2024). Observations that thermal decomposition of goethite is greatly suppressed in the presence of high water pressure led us to develop a new method to conduct helium step heat experiments under water-saturated conditions up to 300°C within standard experimental setups. Using this method, we report results from the first helium step heat experiment of proton-irradiated goethite under hydrothermal conditions. Irradiated goethite sample Win-06-01b of the Flinders Ranges, South Australia and distilled water were encapsulated in a pure silica glass ampoule and were subjected to multiple prograde and retrograde thermal cycles over a temperature range of 170°C to 250°C under water vapor saturated conditions with step durations ranging from 2 hours to over a week. The results show apparent helium diffusivities approximately seven log units lower than those observed in vacuum, converging to a reproducible linear array on an Arrhenius plot with an activation energy of 157.1± 5.3kJ/mol and a closure temperature of 98.1± 8.6°C at a cooling rate of 10°C/Ma. These results are consistent with a volume-diffusion helium loss mechanism and show

high helium retentivity for natural goethite that extrapolates to near-quantitative helium retention at surface conditions over billion-year timescales. A four-domain diffusion model fitted to the ^{3}He release places 97% of the ^{3}He in a single domain and attributes the natural loss of radiogenic helium from that domain to thermally activated diffusion. Steady cooling from crystallization at ∼76–80 °C in the Jurassic fits the ^{4}He/^{3}He spectrum best, favoring a hydrothermal origin, while burial of surface-formed goethite to ∼60 °C in the Late Cretaceous remains a viable alternative. Reanalysis of published vacuum step heats of the same sample shows that they record first-order conversion to hematite, whereas the hydrothermal release records diffusion. Thermodynamic and nucleation-kinetic modeling, using the vacuum step heat kinetics of the same sample, explains this difference and shows that conducting step heats with this method is expected to slow dehydroxylation of Win-06-01b by many orders of magnitude, effectively suppressing dehydroxylation on experimental timescales. This analysis applies only to Win-06-01b, and can only be extended to other goethite samples for which kinetic data or vacuum step heat data are available due to the diversity and complexity of goethite morphologies and grain sizes. The method presented in this work can be systematically applied to experimentally investigate effects of crystallite size, chemical composition, and radiation damage to refine these results and to assess the applicability of existing vacuum step heat data to helium retention and loss in goethite in nature.

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2.1 Introduction

Goethite (FeOOH) crystallization is datable only by U-Th/He geochronology (Shuster et al., 2005; Farley et al., 2024). Goethite is a powerful chronometer

for dating surface processes such as weathering and ferruginization directly: laterite and duricrust formation and recrystallization (H. S. Monteiro et al., 2014; Allard et al., 2018; Ansart et al., 2022), and the ferruginization of organic matter and paleosols (Heim et al., 2006; Yapp and Shuster, 2011; Vasconcelos et al., 2013; Danišík et al., 2013; Hofmann et al., 2017).

Interpreting a goethite U-Th/He date requires knowing the conditions under which the mineral retains radiogenic helium. As with any radiochronometer, meaningful geological interpretation of a measured age depends on knowing when the system is open to loss or gain of parent or daughter nuclides. For goethite, step heats of proton-irradiated, crystalline samples are used to reveal the degree of partial helium loss by the 4He/3He method (Shuster et al., 2005), and suggest that helium loss fractions in nature are always non-zero but modest, ranging from 2 to 30% (Shuster et al., 2005; Heim et al., 2006; Hofmann et al., 2017; Deng et al., 2017). Although one study attempts to explain the variable loss principally as a result of variable radiation damage (Bassal et al., 2022), the effects of temperature, crystallite size, and chemical composition have not been determined experimentally and remain unresolved, because goethite decomposes to hematite (Fe_2O_3) in the step heat experiments used to characterize it (Farley et al., 2024).

Laboratory experiments of helium migration under controlled conditions have proven extremely useful in understanding the mechanisms and rates of helium migration in geological materials and have produced sensible results that are predominantly explainable by thermally activated volume diffusion (Reiners and Farley, 1999; Farley, 2000; Farley, 2007; Cherniak and Watson, 2011; Farley, 2018), albeit variably complicated by the effects of radiation damage and annealing (Shuster et al., 2006; Guenthner et al., 2013). This approach has also been leveraged for goethite; however, some complications arise when applying this approach to goethite. Above surface temperatures, goethite decomposes to hematite and water, often along crystallite elongation directions (Goss, 1987). Farley et al. (2024) measured evolved water by capacitance manometry during vacuum step heats and found helium and water release to be tightly correlated even below 200°C, indicating that decomposition, rather than helium migration through an unchanging crystal, governs the release. Vacuum step-heat results for goethite therefore need not describe helium behavior in nature.

Still, existing helium step heat measurements of goethite in vacuum likely remain informative to an extent, as goethite dehydration kinetics are highly influenced by some of the same factors that could be responsible for variable helium retention in natural goethite, like variable grain size (Navrotsky et al., 2008; Guo and Barnard, 2011; Goss, 1987), crystallinity (Schwertmann et al., 1985) or substitution of aluminum or other cations within the goethite structure (Majzlan et al., 2003; Bassal et al., 2022). For example, Shuster et al. (2005) initially conducted vacuum step-heating experiments on two vitreous, polycrystalline goethite samples, in which uniformly distributed 3He was introduced by proton irradiation for comparison with radiogenic 4He that was partially lost in nature. The results of Shuster et al. (2005) could be well-fit with a model of two distinct diffusion domains or migration mechanisms with significant differences in helium retention under surface conditions, e.g. crystallite interiors and intergranular space. Although these results may have been impacted by dehydration in vacuum, certain aspects may remain applicable to helium loss in goethite in nature if helium loss in nature and the laboratory advance through the samples with similar geometries. One important point is that if laboratory experiments degas domains in order of their helium retentivity in nature, even if the rates of loss are affected by goethite dehydration, then a $^{4}He/^{3}He$ experiment in vacuum can still be used to measure helium loss (Shuster et al., 2005; Vasconcelos et al., 2013).

An additional complexity is that natural goethite almost always occurs as aggregates of minute crystallites with cross-sections smaller than 1 µm in nature. Neglecting to account for multiple diffusion domains can lead to inaccurate determinations of activation energy (E_a) and frequency factor (D_o/a^2) for helium diffusion (Farley, 2018). Detailed step-heating with temperature cycling, as has been done for hematite (Farley and Flowers, 2012; Farley, 2018), is needed to characterize domain size and activation energy for polycrystalline samples.

Molecular simulations indicate that helium diffusion in perfectly crystalline goethite should be highly anisotropic, essentially reducing to 1-dimensional diffusion along the b-axis (typically the long axis of the polycrystalline aggregate) that would amount to near-instantaneous helium escape from goethite (Bassal et al., 2022), which is clearly contradicted by observed (U–Th)/He dates as old as ~275 Ma (Waltenberg, 2012) and dates that are concordant with $^{40}Ar/^{39}Ar$ ages of other phases (Vasconcelos et al., 2013). However, the models presented

by Bassal et al. (2022) suggest that crystallographic defects caused by radiation damage could block these channels, thereby reducing He loss. While these simulations are informative, they alone are insufficient for drawing conclusions about the mineralogical and environmental conditions needed for accurately dating the crystallization of goethite with the (U–Th)/He method and require experimental verification.

Goethite dehydration is suppressed at high water pressure (D. Langmuir, 1971; Navrotsky et al., 2008), so a step heat conducted under hydrothermal conditions may interrogate helium loss rather than decomposition kinetics. Until now, no studies have reported helium step-heating results for goethite under hydrothermal (high pH_2O) conditions, which could help stabilize goethite or slow its decomposition to illuminate the processes relevant to helium loss in natural settings. If experimental conditions can be controlled to slow the decomposition of goethite such that the rate of natural helium loss in goethite greatly exceeds the rate of goethite decomposition into hematite and water, it would open up an experimental approach to systematically assess how factors like crystallite size, temperature, radiation damage, and chemical composition could affect helium mobility and retentivity in nature.

Early trials of the method were encouraging (§2.2). SEM imagery taken of thermally treated natural goethites in vacuum and open air showed the development of contraction cracks on the outside and throughout the interior of goethite grains heat treated in vacuum and in open air (Figs. 2.1, 2.3), while goethite heat-treated when encapsulated with water or gypsum did not form cracks (Fig. 2.2). Goethite heat treated in open air turned red, whereas goethite heat treated in vacuum remained black.

We test this premise with the first helium step heat experiment on goethite under water-saturated conditions, and compare it with published vacuum step heats of the same sample (§2.4–2.4). A thermodynamic and kinetic analysis of goethite dehydroxylation then explains why water has this effect (§2.4–2.4).

2.2 Experimental Methods

This study builds upon previous helium step heat studies of the dense colloform goethite sample Win-06-01b (occasionally referred to as Win0601b) collected within float at the Flinders Ranges in South Australia (Waltenberg, 2012; H. Monteiro, 2017). It yields one of the oldest known goethite U-Th/He ages:

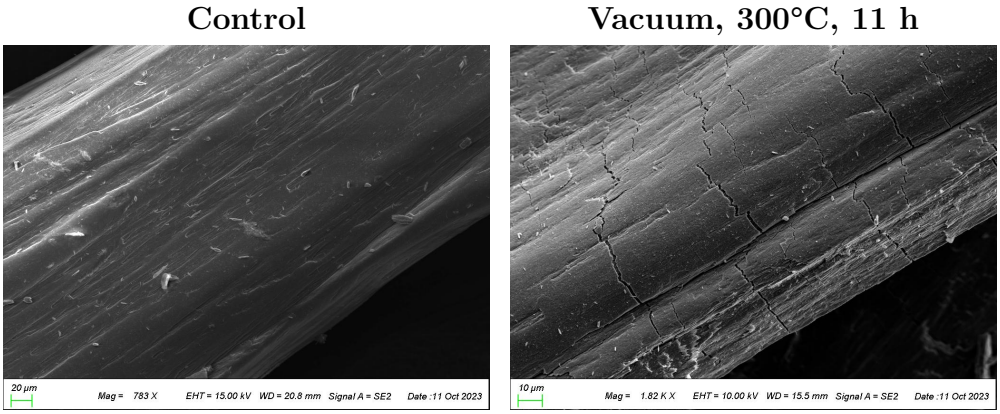

Figure 2.1: Secondary electron images of Bahia goethite aggregate before heat treatment (control) and after heat treatment at 300°C for 11 hours in vacuum. Cracks developed in the vacuum-heated sample both parallel and perpendicular to the long axis of the polycrystalline aggregate, representing volume contraction due to dehydration/decomposition of goethite into hematite. The buffered treatments at the same temperature are shown in Fig. 2.2.

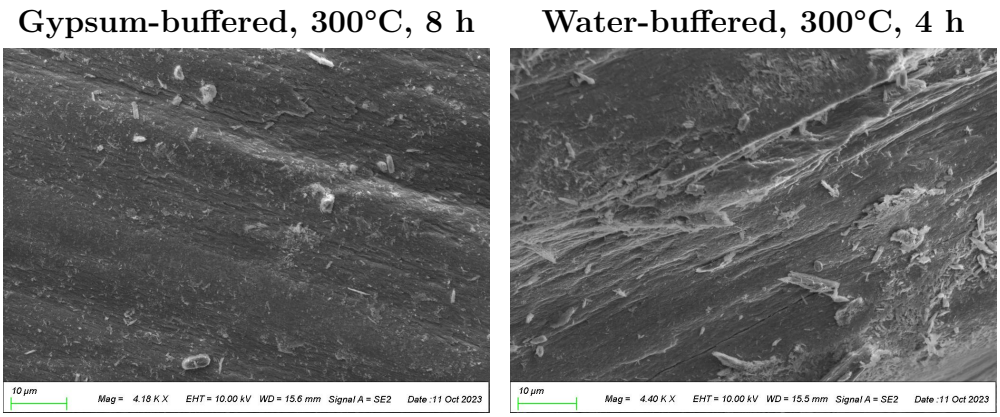

Figure 2.2: Secondary electron images of Bahia goethite aggregate heated at 300°C in a sealed glass tube with gypsum (8 hours) and with water (4 hours). Despite treatment at high heat, no cracks indicative of dehydration were visible on either goethite surface. In the gypsum experiment, the gypsum dehydrated first and produced enough water pressure to stop or slow dehydration of the goethite and prevent the development of microscopic cracks; at higher temperatures goethite and gypsum reacted, forming a surface layer of other minerals. For the water-buffered sample, no cracks were observed when investigating the goethite at scales similar to Fig. 2.1, and most of the aggregate's surface resembled the gypsum-buffered panel; the area pictured had the roughest texture on the grain. A very small crack parallel to the long axis may indicate some amount of dehydration had occurred, or it could have been inherited from before hydrothermal treatment, since even the untreated control grain (Fig. 2.1) had some rough surfaces and rare long-axis-parallel cracks.

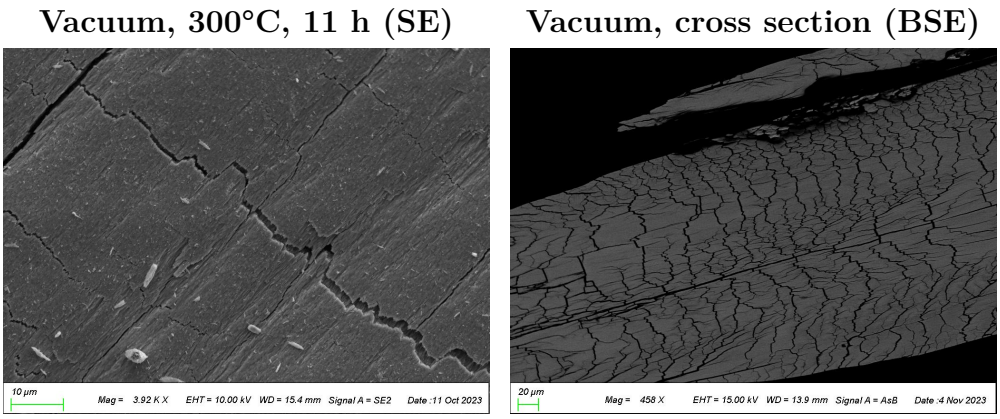

Figure 2.3: Vacuum-heated Bahia goethite at higher magnification and in cross section. Left: secondary electron image of goethite heated at 300°C for 11 hours in vacuum, at the same scale as the buffered samples of Fig. 2.2, showing rough cracks at various scales both parallel and orthogonal to the long axis of the goethite aggregate. These are the typical decomposition features that would be expected in Fig. 2.2 if significant decomposition had occurred there. Right: backscatter electron image of a polished cross section of epoxy-mounted, vacuum-heated Bahia goethite, from the same subsample as the vacuum-heated grain of Fig. 2.1, exposing the grain's interior. This image clearly shows that the cracks visible on the exterior of dehydrated goethite are pervasive throughout the interior as well.

180-135 Ma (Waltenberg, 2012). To produce a homogeneous distribution of 3He, Win-06-01b was irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster et al., 2004) and subsequently was subjected to helium step heat measurements in vacuum by H. Monteiro (2017) and Farley et al. (2024). In their step heat experiment of Win-06-01b, Farley et al. (2024) additionally measured evolved water using capacitance manometry to show that helium and water release were tightly correlated for goethite step heats in vacuum. Although they used subtly different heating schedules, the results of Farley et al. (2024) clearly agree well with H. Monteiro (2017).

In this study, an aliquot of the same irradiated Win-06-01b goethite used by H. Monteiro (2017) and Farley et al. (2024) was encapsulated with MilliQ water within a silica glass tube. Silica glass was chosen over other laboratory glasses such as borosilicate because it is the most permeable glass type to helium (Norton, 1953; Altemose, 1961). To prepare the silica glass tube, 10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an

inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thinner pipettes using a natural gas and oxygen glass-welding torch and the pipette ends were fused closed. After the glass had cooled, 1.18 mg of irradiated Win-06-01b goethite blades less than 2 mm in length were carefully transferred to the bottom of the tube in the presence of an ionizing fan. Less than a microliter of distilled water was transferred to the sealed pipette using a thin glass capillary, submerging the goethite, after which the tube was gently tapped to liberate the bubbles and ensure that the water and goethite were efficiently packed at the bottom of the pipette. To seal the pipette, the contents of the ampoule were flash-frozen in liquid nitrogen and transferred to a saltwater-ice bath at a temperature of -10°C. While the pipette end of the tube remained frozen in the saltwater-ice bath, the top was interfaced with a vacuum line and the tube was pumped to a pressure of <0.1 torr. The advantage of this approach was that water would remain solid within the tube, only slowly sublimating to vacuum while atmospheric gases would be readily pumped away. While it was desirable to remove atmospheric gases to reduce experimental complexity by removing potentially reactive gases like oxygen, the primary motivation for pumping the tube to rough vacuum was to aid the sealing process. When the tube contents were under vacuum, fused glass would deform inwards and self-seal, greatly improving the ease and quality of the seal while requiring far less heat input from the torch. After an acceptably low pressure was attained, the tube was closed to the vacuum line and the saltwater-ice bath was quickly replaced with a liquid nitrogen bath, ensuring that no ice was melted. The liquid nitrogen bath was preferable during sealing as it more effectively protected the goethite and ice from the high temperatures of the glass-welding torch. A fine-flamed welding torch was used to seal the top of the ampoule, which dropped into the liquid nitrogen bath and was retrieved by passing the liquid nitrogen through a sieve. Fig. 2.4 shows the sealed ampoule after the completion of the step heats with all of the original water remaining, a testament to the quality of the ampoule's seal, which had sustained a massive pressure gradient of over 11 orders of magnitude across the ampoule walls, with an inner vapor pressure of about 3×10^4 torr (~40 bars: Haar et al. (1984)) against the ~10^{-7} torr ultra-high vacuum of the lamp cell while the ampoule was held at 250°C for over a week.

Figure 2.4: A view of the entire experimental ampoule after the completion of the step heat experiment before it was broken for the final step in the resistance furnace. At room temperature, the tube is approximately half-filled with gas, one-third filled with liquid water, and one-sixth filled with <2 mm long shards of irradiated Win-06-01b goethite. A gel-like substance was observed lining the inner walls of the tube, which is likely to be amorphous silica dissolved from the tube and redeposited by the thermally-cycled, high pressure liquid water phase.

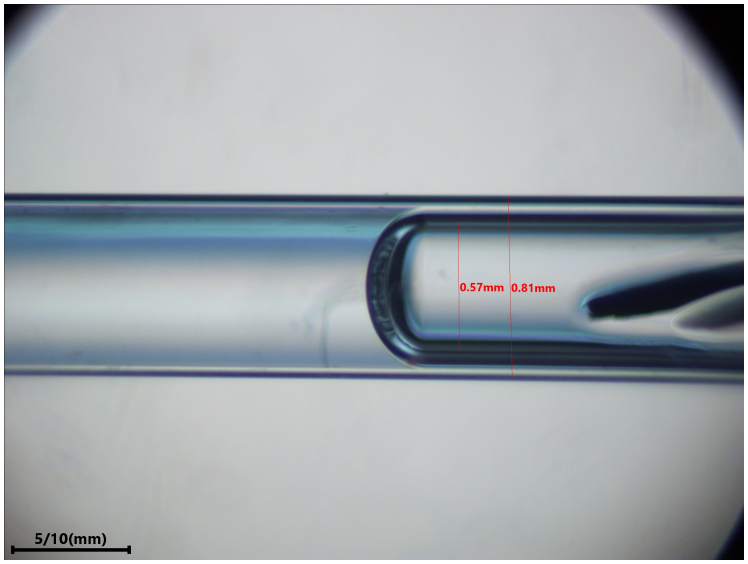

Figure 2.5: Measurements of the inner and outer diameter of the ampoule used in the hydrothermal step heats of Win-06-01b. The inner walls of the ampoule are only clearly visible within air bubbles.

The sealed ampoule, now referred to as HT-Win-06-01b, was placed in a copper foil packet and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in more detail by Farley et al. (1999). Before the experiment began, the ampoule was loaded in the lamp cell, pumped to high vacuum, and was held at 70°C for 4 days to degas atmospheric helium entrained within the ampoule.

HT-Win-06-01b was subjected to multiple prograde-retrograde heating cycles to investigate possible MDD behavior. Step durations varied from 2 hours to about a week; the total experiment was conducted over about 40 days and included experimental steps ranging from 170-250°C. We elected not to heat the sample hotter because high temperatures generated very high vapor pressures. In early trials, we found that similar ampoules most often withstood the ~86 bar vapor pressures associated with liquid water saturation at 300°C (Haar et al., 1984), but they would explode when further heated to temperatures beyond 315°C.

Experimental temperature records show ~1°C amplitude oscillations around the target temperature. While we cannot exclude the possibility that sustained temperature gradients were present throughout the sample due to the use of the comparatively large ~1 mm thick ampoule, we are confident that the conclusions of this work are robust. The ampoule was in direct contact with the thermocouple junction and was in front of the thermocouple junction with respect to the projector lamp heating source, suggesting that if there were a temperature discrepancy, the sample would have been hotter than the recorded temperature, which would not affect the conclusions of this work. Additionally, during early trials of this method for helium step heats, silica-glass encapsulated Durango apatite had been subjected to step heats at Caltech and gave corrected 4He yields indistinguishable from those of unencapsulated Durango apatite, suggesting that the temperature recorded by the thermocouple for our experimental setup is sufficiently accurate.

Helium isotope measurements of evolved gases from hydrothermal step heats of glass-encapsulated irradiated HT-Win-06-01b goethite were acquired at Caltech on a Thermo Fisher Helix-MC-Plus^{10K} mass spectrometer in peak hopping mode from January 31st, 2024 to March 13th, 2024. Helium was purified by exposing evolved gases to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding

into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray, which was never significant during the experiment. Measurements of high 3He MM gas standard were analyzed every 8 hours (totaling 116 measurements) to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility.

A cold blank was measured at the beginning of the experiment, showing a negligible 3He blank and a 4He blank signal corresponding to about 1.25 pccSTP, which was similar to 4He values attained during low temperature permeation step measurements. Permeation steps were thus assumed to measure shifting 4He blanks during the experiment, and the entire blank correction was added to all 4He uncertainties.

For the final step, the HT-Win-06-01b ampoule was broken and dried overnight in a 50°C oven, then completely degassed by heating to ~1100 °C in a resistance furnace. The final step measurement was conducted on a Thermo Fisher Helix SFT mass spectrometer. A hot furnace blank subjected to the same volume splits as the final HT-Win-06-01b measurement gave a 3He blank of 6.8 fccSTP and a 4He blank of 0.31 nccSTP. This blank correction was less than 1% of the measured 4He and negligible for 3He. A re-extract of HT-Win-06-01b immediately following the measurement yielded above-blank 3He and 4He signals, which were added to the total measurement, although both constituted less than 1% of the final step yield. Sensitivity was calculated by measuring a 3He-enriched air standard with known 3He and 4He contents immediately before the measurement.

Helium released by the sample reaches the mass spectrometer only after permeating the ampoule wall, so each measured step yield lags the true release. Step yields were corrected for this permeation lag with the bounding method of Chapter 1 (§1.4), which gives lower-bound, central and upper-bound values

for every step (Tables 2.1 and 2.2). The correction and the ampoule permeabilities measured during the experiment are described in the Supplementary Information (§2.1s), together with the diffusivity inversions, the isotope-ratio uncertainties and the kinetic analysis of the published vacuum step heats (§2.2s).

Kinetic and Diffusive Reductions of Step-Heat Data

We reanalyzed the published vacuum step heats of irradiated Win-06-01b by Farley et al. (2024) (hereafter *KAF*) and H. Monteiro (2017) (hereafter *HDM*), together with the water-saturated step heat of this study, by reducing every step both as a first-order conversion rate constant and as an apparent diffusivity. For the kinetic reduction, ^{3}He is assumed to be immediately released from the goethite lattice upon conversion to hematite, except for a fraction ϕ that is inherited by the hematite product. The fraction of helium released in each step is then $F_i = (1-\phi)\,\alpha_i$, where α_i is the fraction of goethite converted in step i.

We describe the kinetics of goethite dehydroxylation with a first-order rate law,

$$\frac{d\alpha}{dt} = k(T)\,(1-\alpha), \qquad k(T) = A\exp\!\left(-\frac{E_a}{RT}\right), \tag{2.1}$$

For a schedule of isothermal steps of duration Δt_j at T_j, Eq. (2.1) integrates to

$$\alpha_i = 1 - \exp\left[-A\sum_{j\le i},\Delta t_j \exp\!\left(-\frac{E_a}{RT_j}\right)\right], \tag{2.2}$$

Inverting Eq. (2.2) step by step gives an apparent rate constant for every step that is independent of any fitted parameter except ϕ:

$$k_i = -\frac{1}{\Delta t_i}\ln\frac{1-\alpha_i}{1-\alpha_{i-1}}, \qquad \alpha_i = \frac{F_i}{1-\phi}. \tag{2.3}$$

For comparison, the diffusion parameter of each step is obtained with the same Fechtig and Kalbitzer (1966) inversion for a sphere that is applied to the hydrothermal experiment (§2.2s).

Additionally, in both experiments release stalls at $F \approx 0.9$. The fitted $\phi = 0.101$ is the same for KAF and HDM. In HDM, the remaining ∼10% is released slowly between 340 and 500 °C and on fusion. We interpret this fraction as ^{3}He inherited by the hematite product and subsequently released by the much slower diffusion out of hematite. The conversion and the plateau correspond to

the "Stage 2" and "Stage 3" release of Farley et al. (2024), whose transition near $F \approx 0.9$ is independent of heating schedule (and tracked by a distinct shift in ^{4}He/^{3}He: Fig. 2.15). The kinetic inversions (Eq. (2.3)) for these latest steps are highly sensitive to the choice of ϕ and are excluded from the fits of the kinetic parameters. The KAF steps run with the trap isolated are also excluded from the Arrhenius regression.

Multiple Diffusion Domain and Thermal History Modeling

Together, the ^{3}He release and the bulk-normalized ^{4}He/^{3}He spectrum constrain the thermal history, if the activation energy for diffusion is known (Farley and Flowers, 2012). These data from HT-Win-06-01b are interpreted in two stages. First, the diffusion domain structure of the sample is inverted from the ^{3}He release alone. Second, natural thermal histories are fitted to the ^{4}He/^{3}He spectrum with that domain structure held fixed, so that every thermal history must work within the bounds set by the ^{3}He data. Derivations and numerical details are given in the SI (§2.6s).

Diffusion domain structure. The sample is represented by N non-interacting spherical domains that share the activation energy and frequency factor of the Stage 1 regression (§2.3). Domain j holds a fraction f_j of the uniformly distributed, proton-induced ^{3}He ($\sum_j f_j = 1$) and has radius r_j relative to the reference radius r_0 implied by the Stage 1 regression, so that

$$\left(\frac{D}{a^2}\right)_j = \left(\frac{r_j}{r_0}\right)^{-2} \frac{D_0}{a_0^2} \exp\!\left(-\frac{E_a}{RT}\right). \tag{2.4}$$

A model with N domains has $2N - 1$ free parameters. During the heating schedule, the reference domain accumulates the non-dimensional time $\tau_{0,i} = \sum_{m \le i} (D_0/a_0^2) \exp(-E_a/RT_m)\, \Delta t_m$ by the end of step i, and domain j reaches $\tau_{0,i}(r_j/r_0)^{-2}$. With $F(\tau)$ the cumulative fraction released by an initially uniform sphere (Crank, 1979), the model cumulative and fractional ^{3}He yields are

$$\Sigma F_i = \sum_{j=1}^{N} f_j\, F\!\left(\tau_{0,i}\,(r_j/r_0)^{-2}\right), \qquad \Delta F_i = \Sigma F_i - \Sigma F_{i-1}. \tag{2.5}$$

Following Farley and Flowers (2012), the fractional and cumulative yields are weighted equally, and $\{f_j, r_j\}$ minimize

$$S = \sum_i \left(\Delta F_i^{\mathrm{mod}} - \Delta F_i^{\mathrm{obs}}\right)^2 + \sum_i \left(\Sigma F_i^{\mathrm{mod}} - \Sigma F_i^{\mathrm{obs}}\right)^2, \tag{2.6}$$

summed over the 28 steps with more than 0.1 pccSTP of ^{3}He. The gas released in the six low-signal steps still counts toward the cumulative yields of later steps, and their durations toward the heating schedule. Domains are added one at a time, each search starting from the $(N-1)$-domain solution, until an added domain no longer changes the fit or the domain structure. The Arrhenius and delta plots are not fitted: they follow from applying the Fechtig and Kalbitzer (1966) inversion to the model cumulative release, and their misfit is reported as a reduced χ^2 over the same 28 steps.

Thermal histories. Any thermal history predicts a ^{4}He/^{3}He spectrum that is obtained following the procedure of Shuster and Farley (2004). A summary of this procedure is given in the SI (§2.6s). Thus, we evaluate how well each thermal history reproduces the measured ^{4}He/^{3}He spectrum. We compare three types of thermal histories, with two free parameters for steady cooling and the isothermal hold and three for burial and exhumation.

- **Steady cooling:** the temperature falls linearly from T_0 at t_{form} to 25 °C today. The goethite forms at an elevated temperature, for example from warm fluids at depth, and cools monotonically. Free parameters: t_{form} and T_0.
- **Burial and exhumation:** the path passes through (t_{form}, T_s), $(t_{\text{peak}}, T_{\text{max}})$, $(50\text{ Ma}, T_s)$ and $(0, T_s)$ with $T_s = 20\,°\text{C}$. For this model, we assume that the goethite forms at the surface, is buried and heated to a peak temperature, and is exhumed by the Paleogene, roughly following the thermal history (Gibson and Stüwe, 2000; M. M. Mitchell et al., 1998) interpreted for the region from apatite fission track thermochronometry. We fit t_{peak} and T_{max} at each of three fixed crystallization ages of 180, 195 and 210 Ma; because the crystallization age is also varied across the three fits, burial has three free parameters.
- **Isothermal hold:** $T = T_{\text{iso}}$ from t_{form} to the present. Free parameters: t_{form} and T_{iso}.

Each path is scored against

$$\chi^2 = \sum_i \left(\frac{R_i^{\text{mod}} - R_i^{\text{M}}}{\sigma_i} \right)^2, \qquad \sigma_i = \max\left[\tfrac{1}{2} \left(R_i^{\text{UB}} - R_i^{\text{LB}} \right), 0.02 \right], \tag{2.7}$$

over the 24 steps that have lag-correction bounds and more than 0.1 pccSTP of ^{3}He. Where M, LB and UB are the central, lower-bound and upper-bound bulk-normalized ratios of the bounding method (Table 2.2). This analysis fits 24 independent data points; subtracting the fit parameters gives $\nu = 22$ degrees of freedom for steady cooling and the isothermal hold and $\nu = 21$ for burial and exhumation, which are used to evaluate the reduced χ^2. χ^2 is evaluated for these free parameters along regularly spaced grids to produce a likelihood heatmap (Fig. 2.17). The measured He age (130–180 Ma; Waltenberg 2012) is treated as a hard constraint. The best fit of each family is the lowest-χ^2 path with a predicted age in that range, and parameter ranges are those of the age-consistent paths within the same $\Delta\chi^2$ of that best fit.

SEM Study of Thermally Treated Goethite

In early trials of the "glass tube method" for helium step heat experiments with controlled microenvironments presented here, sub-samples of goethite morphologically similar to Win-06-01b from Igarapé Bahia (Shuster et al., 2005; H. Monteiro, 2017) were heated in vacuum using the projector lamp heating apparatus of Farley et al. (1999) and in sealed glass ampoules with gypsum or water additionally added to provide moisture to slow/prevent dehydration of goethite. All samples were held at 300°C for 4-11 hours, and the glass ampoules were broken and the goethites were allowed to air dry before all samples were placed on carbon tape and carbon coated. These samples were imaged at Caltech in secondary electron mode by the ZEISS 1550VP Field Emission SEM. Synthetic goethite pigment, a commercial yellow iron-oxide pigment of acicular crystals ∼1 µm long, was treated in the same way but only with water, without vacuum or open-air runs: at least five larger silica glass ampoules, each loaded with pigment and an excess of water and fused shut at ambient pressure, were held at 200–250 °C for about a day in an oven whose voltage was adjusted by hand. These runs were a rapid, qualitative test; details of individual runs (temperature histories, durations, water-to-solid ratios) were not recorded, and reaction was assessed by color (yellow goethite, red hematite) before imaging.

2.3 Results

Water-Saturated Step Heat of HT-Win-06-01b

The complete corrected step-heat dataset underlying the diffusivities, delta plots, and Arrhenius regressions discussed below is reported in Tables 2.1

and 2.2, including the lower-bound, central, and upper-bound values from the bounding method's permeation-lag correction for ^{3}He and ^{4}He respectively.

Table 2.1: ^{3}He step-heat dataset for HT-Win-06-01b under the bounding method: measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), the lower- and upper-bound true step production (ω_{LB}, ω_{UB}), and the corresponding lower-bound, central, and upper-bound apparent diffusivities. "–" denotes early steps with $\lambda t < 4$, for which bounds could not be determined (see §2.1s). Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots (Fig. 2.15 and others derived from this dataset).

Temp	Time	^{3}He meas.	Cum. ^{3}He	$n_{t,\mathrm{UB}}$	ω (pccSTP)		$\ln(D/a^2)$		
(°C)	(min)	(pccSTP)	(pccSTP)	(pccSTP)	LB	UB	LB	m	UB
170	300	0.9630	0.0000	1.1660	–	–	–	–	–
180	463	0.5442	0.9630	0.2413	–	–	–	–	–
190	550	0.5292	1.6241	0.1538	–	–	–	–	–
200	500	0.5190	2.2112	0.1430	0.3651	0.6619	-22.16	-21.77	-21.50
210	475	0.5770	2.7646	0.1440	0.4341	0.7210	-21.71	-21.41	-21.16
220	450	0.6333	3.3799	0.1447	0.4894	0.7781	-21.34	-21.07	-20.84
230	425	0.7899	4.0551	0.1685	0.6452	0.9584	-20.82	-20.60	-20.39
225	908	0.7513	4.8885	0.0716	0.5827	0.8229	-21.52	-21.25	-21.15
215	450	0.1415	5.6842	0.0349	0.0699	0.1764	-22.84	-22.12	-21.90
205	520	0.0763	5.8529	0.0184*	0.0413*	0.0947*	-23.48*	-22.86*	-22.65*
195	1106	0.0670	5.9491	0.0080*	0.0487*	0.0750*	-24.05*	-23.73*	-23.62*
185	740	0.0227	6.0246	0.0049*	0.0147*	0.0276*	-24.84*	-24.40*	-24.21*

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(continued)

Temp	Time	^{3}He meas.	Cum. ^{3}He	$n_{t,\mathrm{UB}}$	ω (pccSTP)		$\ln(D/a^2)$		
(°C)	(min)	(pccSTP)	(pccSTP)	(pccSTP)	LB	UB	LB	m	UB
175	920	0.0180	6.0610	0.0036*	0.0131*	0.0216*	-25.17*	-24.85*	-24.66*
200	260	0.0224	6.0814	0.0163*	0.0189*	0.0388*	-23.53*	-23.36*	-22.81*
210	520	0.0812	6.1081	0.0181*	0.0649*	0.0993*	-22.98*	-22.76*	-22.55*
220	1167	0.5766	6.1914	0.0447	0.5585	0.6213	-21.59	-21.55	-21.48
230	370	0.3577	6.7796	0.0903	0.3130	0.4480	-20.95	-20.81	-20.58
240	320	0.6767	7.1563	0.1776	0.5864	0.8543	-20.06	-19.91	-19.67
245	250	0.5549	7.8582	0.1888	0.3773	0.7437	-20.18	-19.78	-19.47
250	210	0.6934	8.4743	0.2786	0.5047	0.9721	-19.63	-19.30	-18.94
245	270	0.4659	9.2086	0.1413	0.1872	0.6071	-20.80	-19.87	-19.60
240	320	0.2857	9.7206	0.0749	0.1445	0.3606	-21.18	-20.49	-20.25
235	370	0.2199	10.0357	0.0517	0.1450	0.2716	-21.28	-20.86	-20.65
230	420	0.1400	10.2736	0.0303	0.0883	0.1703	-21.89	-21.42	-21.22
225	652	0.1494	10.4279	0.0206	0.1191	0.1700	-22.01	-21.78	-21.65
230	500	0.1550	10.5908	0.0271	0.1344	0.1820	-21.61	-21.46	-21.30
235	450	0.1951	10.7582	0.0361	0.1681	0.2313	-21.26	-21.11	-20.94
240	400	0.2425	10.9638	0.0483	0.2063	0.2908	-20.91	-20.75	-20.57
245	350	0.3150	11.2328	0.0690	0.2667	0.3840	-20.50	-20.33	-20.12

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(continued)

Temp	Time	^{3}He meas.	Cum. ^{3}He	$n_{t,\mathrm{UB}}$	ω (pccSTP)		$\ln(D/a^2)$		
(°C)	(min)	(pccSTP)	(pccSTP)	(pccSTP)	LB	UB	LB	m	UB
250	300	0.3716	11.5689	0.0939	0.3026	0.4656	-20.18	-19.97	-19.74
250	300	0.3640	11.9627	0.0899	0.2700	0.4539	-20.26	-19.96	-19.73
250	300	0.3452	12.3490	0.0852	0.2553	0.4304	-20.29	-19.98	-19.75
250	10250	8.4829	12.7167	0.0494	8.3977	8.5324	-19.99	-19.98	-19.97
240	785	0.2261	21.2187	0.0209	0.1767	0.2470	-21.03	-20.78	-20.69
Total			146.1483						

Table 2.2: ^{4}He step-heat dataset and absolute ^{4}He/^{3}He ratios for HT-Win-06-01b under the bounding method, structured as in Table 2.1. "–" denotes steps for which bounds could not be determined. The Total row reports the bulk (total-yield) ^{4}He/^{3}He ratio, i.e. the normalizing value used to bulk-normalize these ratios for the plot in Fig. 2.15. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal from all plots, as in Table 2.1.

Temp	Time	^{4}He meas.	Cum. ^{4}He	$n_{t,\mathrm{UB}}$	ω (nccSTP)		$\ln(D/a^2)$			^{4}He/^{3}He		
(°C)	(min)	(nccSTP)	(nccSTP)	(nccSTP)	LB	UB	LB	m	UB	LB	m	UB
170	300	0.0192	0.0000	0.0280	–	–	–	–	–	–	–	–
180	463	0.0305	0.0192	0.0152	–	–	–	–	–	–	–	–
190	550	0.0558	0.0660	0.0179	–	–	–	–	–	–	–	–
200	500	0.1024	0.1375	0.0313	0.0845	0.1337	-26.21	-25.96	-25.62	–	197	–
210	475	0.1460	0.2500	0.0404	0.1147	0.1864	-25.31	-25.02	-24.72	248	253	269
220	450	0.2091	0.4055	0.0530	0.1686	0.2620	-24.41	-24.15	-23.88	326	330	349
230	425	0.3243	0.6319	0.0760	0.2714	0.4004	-23.43	-23.21	-22.96	404	411	432
225	908	0.3483	0.9799	0.0363	0.2722	0.3845	-23.81	-23.53	-23.41	443	464	474
215	450	0.0716	1.3487	0.0196	0.0353	0.0912	-24.95	-24.23	-23.98	429	506	537
205	520	0.0385	1.4359	0.0103*	0.0189*	0.0487*	-25.66*	-24.94*	-24.70*	392*	504*	535*
195	1106	0.0336	1.4912	0.0043*	0.0233*	0.0379*	-26.16*	-25.80*	-25.68*	447*	501*	515*
185	740	0.0079	1.5467	0.0019*	0.0037*	0.0098*	-27.59*	-26.81*	-26.59*	225*	350*	369*
175	920	0.0152	1.6020	0.0033*	0.0133*	0.0185*	-26.48*	-26.34*	-26.14*	843*	843*	982*

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Temp	Time	^{4}He meas.	Cum. ^{4}He	$n_{t,\mathrm{UB}}$	ω (nccSTP)		$\ln(D/a^2)$			^{4}He/^{3}He		
(°C)	(min)	(nccSTP)	(nccSTP)	(nccSTP)	LB	UB	LB	m	UB	LB	m	UB
200	260	0.0107	1.6351	0.0091*	0.0073*	0.0198*	-25.79*	-25.41*	-24.79*	376*	476*	555*
210	520	0.2166	1.6547	0.0535*	0.2075*	0.2701*	-23.07*	-23.02*	-22.79*	2667*	2667*	3104*
220	1167	0.3187	1.8714	0.0269	0.2652	0.3456	-23.50	-23.30	-23.21	472	553	563
230	370	0.2113	2.6766	0.0590	0.1844	0.2703	-22.39	-22.25	-21.99	575	591	628
240	320	0.4286	3.1616	0.1243	0.3697	0.5529	-21.36	-21.20	-20.93	612	633	675
245	250	0.3988	3.6430	0.1521	0.2746	0.5510	-21.29	-20.90	-20.56	666	719	778
250	210	0.4440	4.2359	0.2020	0.2918	0.6460	-20.91	-20.47	-20.07	542	640	703
245	270	0.3123	4.7263	0.1053	0.1103	0.4176	-22.04	-20.98	-20.68	469	670	721
240	320	0.1927	5.0846	0.0559	0.0874	0.2485	-22.38	-21.57	-21.32	506	674	718
235	370	0.1517	5.3031	0.0393	0.0958	0.1910	-22.39	-21.92	-21.69	600	690	730
230	420	0.0991	5.4714	0.0236	0.0598	0.1227	-22.96	-22.45	-22.23	608	708	746
225	652	0.0995	5.5835	0.0151	0.0760	0.1146	-23.14	-22.86	-22.72	608	666	689
230	500	0.1051	5.6969	0.0202	0.0900	0.1253	-22.68	-22.52	-22.34	652	678	708
235	450	0.1369	5.8251	0.0279	0.1167	0.1648	-22.29	-22.13	-21.94	675	702	734
240	400	0.1781	5.9762	0.0390	0.1502	0.2171	-21.89	-21.72	-21.52	705	735	771
245	350	0.2305	6.1807	0.0556	0.1915	0.2861	-21.48	-21.29	-21.07	694	732	772
250	300	0.2777	6.4385	0.0758	0.2221	0.3535	-21.13	-20.91	-20.66	703	747	793

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Temp	Time	^{4}He meas.	Cum. ^{4}He	$n_{t,\mathrm{UB}}$	ω (nccSTP)		$\ln(D/a^2)$			^{4}He/^{3}He		
(°C)	(min)	(nccSTP)	(nccSTP)	(nccSTP)	LB	UB	LB	m	UB	LB	m	UB
250	300	0.2654	6.7374	0.0725	0.1896	0.3379	-21.25	-20.91	-20.66	654	729	774
250	300	0.2489	7.0231	0.0680	0.1764	0.3169	-21.28	-20.93	-20.65	647	721	765
250	10250	6.8042	7.2901	0.0430	6.7362	6.8472	-20.70	-20.69	-20.68	801	802	803
240	785	0.1853	14.1156	0.0187	0.1423	0.2040	-21.69	-21.42	-21.32	770	819	838
Total			145.8145								998	

The diffusivities determined from the step heat 3He and 4He yields for HT-Win-06-01b are always significantly lower than the diffusivities determined from step heat experiments on the same sample in vacuum, and follow evolutions that are distinct from those determined from the vacuum experiments. The Arrhenius plot of the 3He diffusivities initially follows a declining concave-up trend until ~3.5% cumulative 3He yield, then slowly decreases in sub-parallel arrays during prograde and retrograde heating cycles from 3.5-7% cumulative 3He yields (Fig. 2.6, Fig. 2.7). The Arrhenius plot of the 4He diffusivities differs in shape: it is concave-down to about 5% cumulative 3He yield, after which it follows a roughly linear array about one log unit below the 3He array (Fig. 2.6). The 3He diffusivities settled into a linear array which was suitable for linear regression after ~7% of the cumulative 3He yield was lost (Fig. 2.8b). All remaining diffusivity measurements of the experiment were included in the linear regression, so that diffusivities from 3He loss fractions of about 7-15% were included. The linear regression yields an R^2 of 0.989, an activation energy of 157.1± 5.3kJ/mol, and a closure temperature of 98.1± 8.6°C assuming a cooling rate of 10°C/Ma and considering only the uncertainty in the regression intercept (D_0/a^2).

A delta plot was generated using the intercept and slope parameters from the 3He regression as a reference line, similarly to previous work (Farley, 2018). The 3He apparent diffusivity delta values monotonically decrease until a cumulative 3He loss of 7%, after which the delta values are within error of Delta = 0: the reference line (Fig. 2.7). A series of low-yield measurements on the first retrograde cycle produce high-error, scattered diffusivities which are omitted for clarity.

Kinetic Reanalysis of Vacuum Step Heats

All steps are plotted in Fig. 2.8, for both the kinetic and diffusion reductions, but the fits use only KAF steps 5–22 (185–280 °C) and HDM steps 8–26 (180–296 °C).

An unweighted least-squares fit of $\ln k_i$ against $1/T_i$ for the 35 steps in the fitting window gives

$$E_a = 137.0 \pm 2.7 \text{ kJ mol}^{-1}, \qquad \ln A = 21.91 \pm 0.66 \tag{2.8}$$

KAF alone gives 139.3±6.2 and HDM alone 135.4±2.5 kJ mol^{-1}), in agreement with the cumulative fit. The result does not depend on the fitting window:

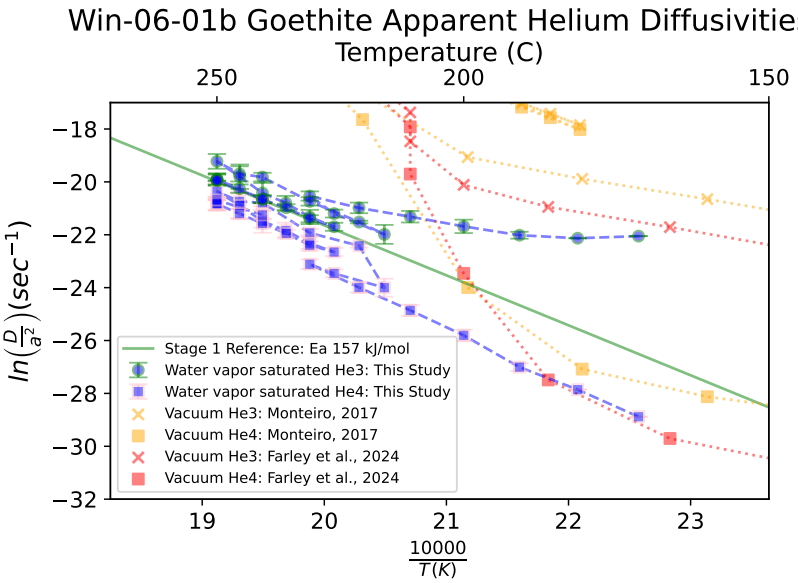


Figure 2.6: Linear regression of the last 12 measurements of 3He diffusivities in HT-Win-06-01b defining the "Stage 1" reference line. Apparent diffusivities under hydrothermal conditions are approximately seven natural log units lower than in vacuum. A diffusion domain corresponding to the Stage 1 reference line has a closure temperature of 98.1±8.6°C assuming a cooling rate of 10°C/Ma. Apparent diffusivities calculated from 4He step heats are sometimes within error of those calculated for 3He; however, they are occasionally resolved at the 2σ level and are about a single log unit lower than apparent diffusivities calculated for 3He after a cumulative fractional yield of 14% 3He.

fitting all steps instead gives $E_a = 122.7$ kJ mol^{-1} and the same ranking of models (§2.4s).

The kinetic rate parameters fitted to KAF alone predict the independent HDM schedule with an rms misfit in F of 0.018 over the fitting window. A single-domain spherical diffusion model fits both vacuum step heats worse, even when given the same hematite-inherited fraction ($\phi = 0.101$) as the first-order fit (Table 2.10).

KAF steps 1–4 and HDM steps 2–7 (below ∼180 °C) release ^{3}He 0.5–3 orders of magnitude faster than the extrapolated conversion rate (Fig. 2.8). This component amounts to less than 2% of the ^{3}He.

In two KAF steps the trap was isolated and evolved water accumulated in the line. At 230 °C the step rate constant fell to 0.33 times the mean of its neighbors at the same temperature, and at 240 °C to 0.22 times. The rate

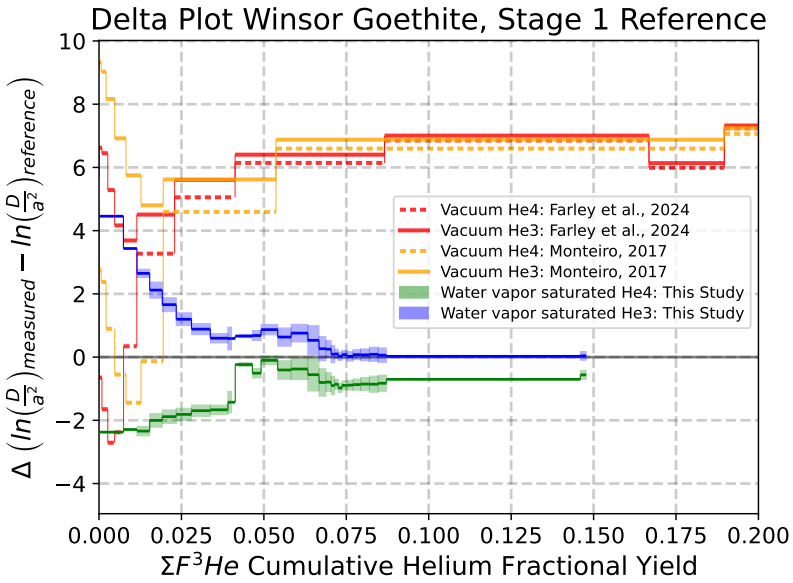


Figure 2.7: Delta plot similar to those presented in Farley (2018), or r/r0 plots in Lovera et al. (1991) and Farley and Flowers (2012). Delta plots are the difference between apparent log diffusivities and a reference line; they are designed to remove the temperature dependence of the dominant process of helium migration to reveal how apparent diffusivities evolve as an experiment proceeds. Six measurements from the water-saturated experiment are omitted for clarity due to low signals and high errors. For visual continuity, the gap left behind is bridged and no uncertainty is shown. 3He monotonically decreases until settling on the reference line from 7–15% 3He fractional yield. 4He diffusivities and delta values are generally highly uncertain but initially show an increase in delta values that settles to a slow increase or plateau about one log unit below the 3He reference line by a 3He fractional yield of 5%, which is consistent with diffusive loss of radiogenic 4He in variably retentive domains. Apparent diffusivities for helium in Win-06-01b in hydrothermal conditions are up to 7 log units lower than the same sample in vacuum.

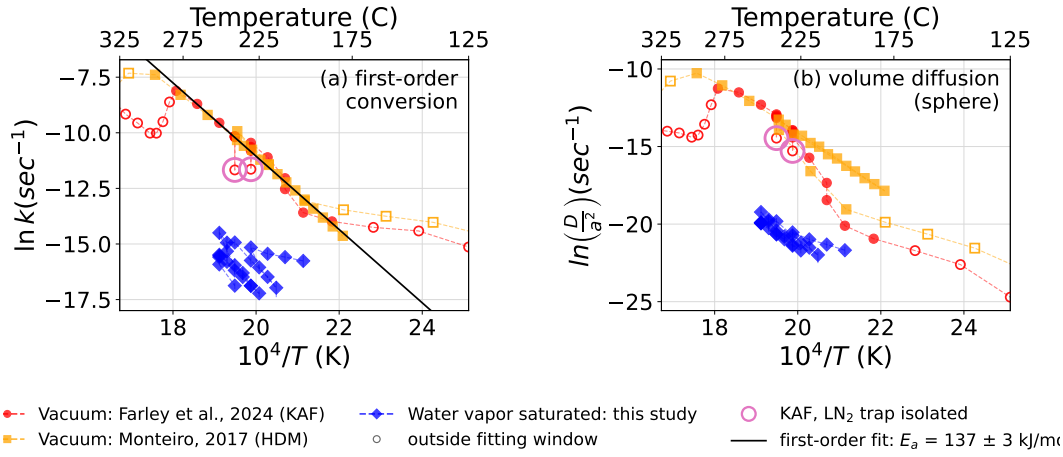


Figure 2.8: The ^{3}He release of every step of the two vacuum step heats of Win-06-01b (red circles: KAF, Farley et al. 2024; orange squares: HDM, H. Monteiro 2017) and of the hydrothermal step heat of this study (blue diamonds), reduced in two ways: (a) as first-order conversion rate constants k_i (Eq. (2.3), $\phi = 0.101$), and (b) as apparent diffusivities D/a^2 for a sphere (Fechtig and Kalbitzer, 1966). Only the filled points are used for the linear regressions. Large pink circles mark the two KAF steps with the LN_2 trap isolated. Six low-signal steps of the water-saturated experiment are excluded for clarity. The black line is a first-order Arrhenius fit to filled points of both vacuum experiments. These fitted parameters are given in Eq. (2.8). Below ~180 °C both vacuum experiments depart from the first-order line. Steps with $F > 1-\phi$ have no defined k and are absent from (a).

recovered fully when the trap was restored. Water and ^{3}He were suppressed together. We return to these two steps in §2.4.

Isothermal Discrimination Test at 250 °C

Sequences of isothermal steps can be particularly powerful in discriminating first and second order processes. Diffusion slows quickly as the diffusion front moves inward, depleting the grain rims of helium, whereas the rate of first-order conversion slows less for similar yields. We harnessed the principle to conduct a sensitive test by performing a series of 4 consecutive, long isothermal steps at 250 °C at the end of the water-saturated experiment: three consecutive 5-hour steps and one 170.8-hour step. Using the first three 5 h steps at 250 °C, we invert for both kinetic rate constants and apparent diffusivities, then use each model to project the yield for the 170.8 h step.

Both the first-order and diffusion models yield reproducible results for the three

5-hour calibration steps (Fig. 2.9), but their projections sharply diverge for the long step. Diffusion projects a final fractional cumulative yield of $F = 0.149$ against an observed $F = 0.146$, a much stronger agreement than the first-order model prediction of $F = 0.173$, which overpredicts the helium yield for that step by 46% (Fig. 2.9). The data's uncertainty is too small to affect this result. In the interest of thoroughness, we also tested the contracting-geometry and Avrami models of §2.4s, which fit the vacuum data less well. Calibrated the same way, they project cumulative fractional yields within 0.174–0.212, which diverge even more strongly from the data.

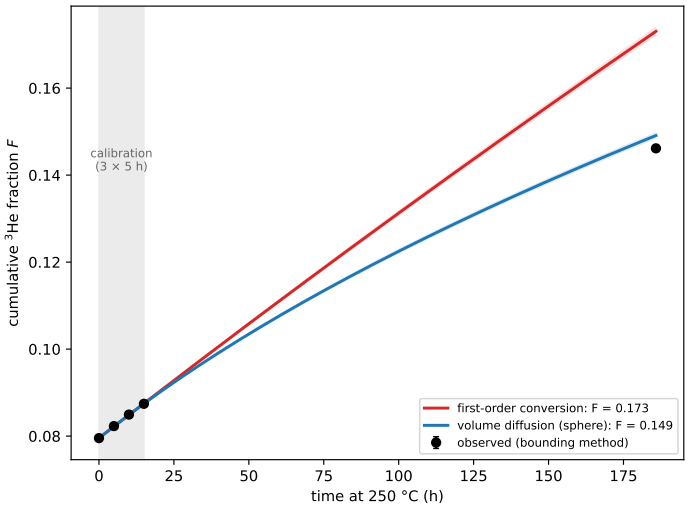


Figure 2.9: The isothermal 250 °C sequence of the water-saturated experiment. Black markers are measured cumulative ^{3}He fraction. The first-order conversion and spherical volume diffusion models are each calibrated on the three 5 h steps (grey band) and projected through the 170.8 h step.

2.4 Discussion

Dehydroxylation Kinetics in Vacuum Step Heats of Win-06-01b

Farley et al. (2024) showed that evolved helium in vacuum step heats of Win-06-01b (and many other goethite samples) is controlled by the conversion of goethite to hematite. Here, we present a reanalysis of the vacuum step heat data of Win-06-01b from KAF and HDM to show that the helium release of Win-06-01b is better described by a model of first-order reaction kinetics for goethite dehydroxylation than by a second order diffusion model.

We postulate that the early ^{3}He released faster than the extrapolated conversion rate (§2.3) is diffusively lost from the least retentive domains; it corresponds to "Stage 1" helium release of Farley et al. (2024). The kinetic reduction yields simpler, more interpretable Arrhenius plots for the vacuum experiments than the diffusion inversion does. The kinetic inversion collapses the prograde steps at the beginning of the vacuum experiments onto a single line that the diffusion reduction instead shows as an irreversible increase in diffusivity (Fig. 2.8). This initial helium release is most easily read as a diffusion-related loss from a broad distribution of small, poorly retentive domains in the goethite. The Arrhenius plot of the kinetic reduction makes it clear that helium release from the initial steps is in excess of the rate of dehydroxylation, which is not obvious from the Arrhenius plot produced with the diffusion reduction. This shift in the dominance from diffusive loss to dehydroxylation-related loss offers a clear explanation for the change in behavior between "Stage 1" and "Stage 2" in vacuum step heat experiments described by Farley et al. (2024).

Thermogravimetric studies of goethite dehydroxylation report somewhat higher apparent activation energies, 154 ± 15 to 169 ± 8 kJ mol^{-1}, for a reaction controlled by a two-dimensional phase boundary, with a different mechanism at early stages and low temperature (Goss, 1987). This may simply be due to a difference in samples: goethites from different sources have different particle sizes, morphologies, and aluminum contents, all of which affect the kinetics of dehydroxylation (D. Langmuir, 1971; Yapp, 1983; Tardy and Nahon, 1985; Majzlan and Navrotsky, 2003).

Helium Release under Water-Saturated Conditions Is Diffusive

The goal of this work was to attempt to slow the goethite dehydroxylation process below the rate of thermal helium loss in unchanging goethite. The

thermal schedule included several prograde-retrograde heating cycles which together would have completely dehydroxylated goethite in vacuum, using the vacuum kinetics of Eq. (2.8). Even so, only 14.7% of the ^{3}He was released in the water-saturated experiment; we argue below that the helium loss in the water-saturated experiment is controlled by diffusion rather than by conversion.

Fig. 2.8 compares the kinetic and diffusion inversions of the step heat data. These plots show the opposite trend for the vacuum and hydrothermal experiments: the hydrothermal experiment collapses into a simpler, coherent linear array with the diffusivity reduction, whereas the kinetic inversion of the same data records a continuously declining k for any given temperature as the experiment proceeds, forming a zig-zag pattern. This is the expected behavior for a second-order (diffusion-controlled) process for which helium loss slows more strongly than a first-order process as the experiment proceeds. Fig. 2.8 shows that helium loss is first-order for vacuum experiments and second order for the water-saturated experiment. Section 2.4 shows that these observations require the conversion rate at water saturation to be at least $10^{2.5}$ times slower than in vacuum, and that a nucleation control on the dehydroxylation kinetics explains suppression far greater than that.

While it is difficult to prove that decomposition or recrystallization is not playing a role in the results presented in this work, the results presented here suggest that a second-order process (likely volume diffusion) is the dominant mechanism of helium loss for Win-06-01b goethite in water-saturated conditions.

Consistency with MDD Systematics

All existing helium step heat data on goethites are from naturally occurring polycrystalline aggregates with unclear and potentially widely variable grain sizes (H. Monteiro, 2017; Farley et al., 2024), which likely could follow MDD systematics (Farley, 2018). In MDD systems, apparent bulk diffusivities of a homogeneously distributed diffusant at a single temperature will decrease as an experiment proceeds, or for a system where the activation energy of diffusion is known, the temperature dependence can be removed by referring to the data in relation to a temperature-dependent reference line, as in delta or r/r0 plots (where $\Delta = ln\left(\frac{D}{a^2}\right)_{measured} - ln\left(\frac{D}{a^2}\right)_{reference}$) (Lovera et al., 1991; Farley, 2018). For the activation energy of 157.1 kJ/mol retrieved by regression in this work, the homogeneously distributed 3He step heat data follow the expected

MDD systematics and monotonically decrease within a delta plot (Fig. 2.7), which is consistent with volume diffusion in an MDD system.

The strong plateau in the delta plot of the water-saturated goethite experiment and the statistically robust regression over a wide range of fractional cumulative 3He yield (from 7-15% ΣF_{3He} with an R^2 of 0.989) suggest that the frequency factor obtained by the regression may be dominated by a single diffusion domain for $F_c \geq 7\%$ rather than a mix of diffusion domains as is most often observed in the apparent diffusivities for polycrystalline hematite (Farley, 2018).

A multiple diffusion domain inversion of the ^{3}He release (§2.2) tests this claim (Table 2.3; Fig. 2.10). A single domain cannot fit the release (Arrhenius $\chi^2_\nu = 27$), and two domains capture the high early diffusivities but leave a poor Arrhenius fit ($\chi^2_\nu = 3.3$). Three domains reduce the yield misfit by more than half ($\chi^2_\nu = 1.08$). With four, the smallest domain splits in two and the fit improves slightly further ($\chi^2_\nu = 1.00$). Every domain added beyond four coincides in size with the dominant domain and leaves the fit unchanged. We therefore adopt four domains as a complete description of the ^{3}He data:

- 0.57% of the ^{3}He at $r/r_0 = 6.8 \times 10^{-4}$;
- 0.66% at $r/r_0 = 2.9 \times 10^{-3}$;
- 1.5% at $r/r_0 = 0.031$;
- 97.2% at $r/r_0 = 1.12$.

The three small domains, holding 2.8% of the ^{3}He in total, are fully degassed during the experiment and account for the high early diffusivities and the gradual approach of the delta plot to the Stage 1 line (Fig. 2.11). The dominant domain releases 12% of its ^{3}He during the experiment and diffuses at 0.80 times the Stage 1 rate. The Stage 1 line is therefore a slightly fast description of the dominant domain, because the bulk-inverted diffusivities on which it was regressed still carry a small contribution from the fast domains. The four-domain model reproduces the Arrhenius array, including its retrograde segments, and the delta plot, although neither is fitted directly. Fits for $N = 1$–5 are shown in the SI (Fig. 2.24).

Table 2.3: Multiple diffusion domain fits to the ^{3}He release of HT-Win-06-01b (N domains, $k = 2N - 1$ parameters). RMS is the yield misfit ($\times 10^{-4}$, fraction of total ^{3}He) over the 28 steps above the signal threshold. "Fast" is the ^{3}He fraction in domains with $r/r_0 < 0.1$; f_{main} and r_{main}/r_0 are the fraction and size of the largest-fraction domain. For $N \geq 5$, the dominant domain is split among several domains of identical size.

N	RMS	Arrh. χ^2_ν	fast (%)	f_{main} (%)	r_{main}/r_0
1	85.30	27.45	0.0	100.0	0.80
2	20.20	3.30	2.2	97.8	1.05
3	8.45	1.08	2.7	97.3	1.12
4	7.81	1.00	2.8	97.2	1.12
5	7.81	1.11	2.8	95.1	1.12
6	7.81	1.24	2.8	85.6	1.12
7	7.81	1.40	2.8	85.3	1.12
8	7.81	1.62	2.8	76.8	1.12
9	7.81	1.91	2.8	76.5	1.12
10	7.81	2.34	2.8	74.6	1.12

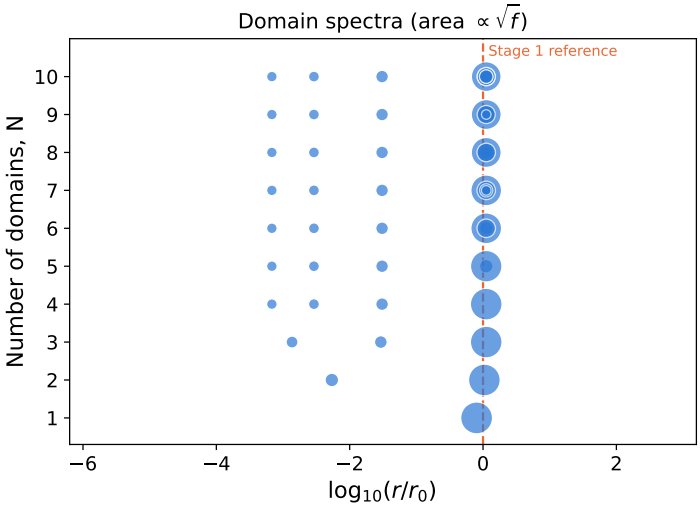


Figure 2.10: Domain spectra of the best-fitting multiple diffusion domain models for $N = 1$–10. Symbol area scales with the square root of the ^{3}He fraction; the dashed line marks the Stage 1 reference domain. The three small domains are unchanged from $N = 4$ onward; domains added beyond $N = 4$ coincide in size with the dominant domain.

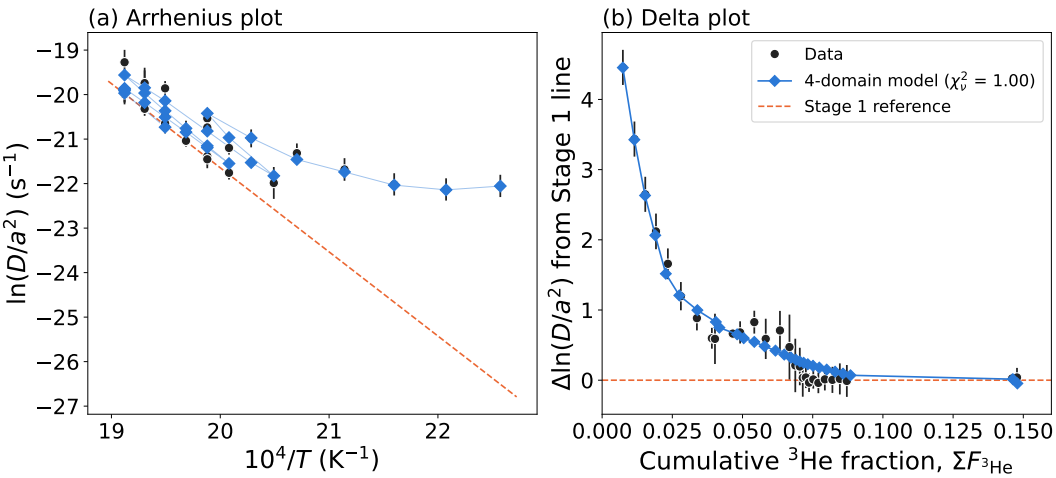


Figure 2.11: (a) Arrhenius plot and (b) delta plot (departure from the Stage 1 line) of the measured ^{3}He diffusivities of HT-Win-06-01b, with the values implied by the four-domain fit to the ^{3}He yields. Neither plot is fitted directly. The six low-signal steps are omitted.

Thermodynamic Model of Goethite Dehydroxylation

The preceding sections establish from helium step heat data alone that the vacuum step heats of Win-06-01b record first order conversion to hematite, whereas the water-saturated step heat records second order diffusion without detectable dehydroxylation. Here we use the thermodynamics of goethite dehydroxylation to explain why water has this effect and to estimate how much solid-state conversion could have occurred in the hydrothermal experiment (§2.4).

Goethite breaks down to hematite by the dehydroxylation reaction

$$2\,\alpha\text{-FeOOH} \rightleftharpoons \alpha\text{-Fe}_2\text{O}_3 + \text{H}_2\text{O}_{(\text{fluid})}, \tag{2.9}$$

whose Gibbs energy change, written per mole of hematite formed, is

$$\Delta G_r(T,P) = G_{\text{hem}}(T,P) + G_{\text{w}}(T,P) - 2\,G_{\text{goe}}(T,P) + \Delta G_{\text{surf}}. \tag{2.10}$$

where ΔG_r is the Gibbs energy change of the reaction and G_{hem}, G_{w}, and G_{goe} are the Gibbs energies of hematite, water, and goethite, respectively. ΔG_{surf} is the surface energy contribution, which must be considered to reproduce observed phase stability (Navrotsky et al., 2008). Goethite is stable where $\Delta G_r > 0$ and hematite plus water where $\Delta G_r < 0$. The values and equations used to evaluate each of these terms are given in the supplement: §2.5s.

This model can be used to calculate the stability field for goethite with respect to dehydroxylation to hematite and water in $\log P_{H_2O}$–T space (Fig. 2.12). This phase diagram communicates several important features of the goethite–hematite–water system that are relevant to the experiments considered here:

1. Goethite is not stable at the conditions of any of the experiments considered here, in water or in vacuum, in spite of the evidence that it did not react in the water-saturated experiment. Thus, we hypothesize that the reaction is kinetically inhibited for goethite under these conditions.

2. The stability of goethite is strongly dependent on the partial pressure of water below and up to water saturation. Above the saturation pressure, water is liquid and contributes far less to the reaction volume than it does as a gas, becoming a secondary control on goethite's stability.

3. The stability of goethite is strongly dependent on grain size, with smaller grains being more stable than larger grains. At water saturation, bulk goethite is stable below 18 °C, 100 nm goethite below 28 °C, and 10 nm goethite below 108 °C.

Stability alone does not determine what happens in a step heat; the rate of conversion controls whether this equilibrium state is reached or if the reaction proceeds at all, and that depends on the chemical potential of the reaction: $\Delta\mu$. Table 2.4 gives the bulk driving forces at the experimental temperatures, evaluated with IAPWS-95 water. At saturation, $\Delta\mu$ is 6–11 kJ, only 1.6–2.4 times RT. Under the vacuum of a step-heating line the water pressure approaches zero, so the chemical potential of water falls without bound and $\Delta\mu$ is unbounded. The chemical potential of water is therefore the key variable separating the two experimental settings.

Water Pressure Control on the Hematite-in-Goethite Nucleation Barrier

The hydrothermal data bound how fast goethite could have converted to hematite in water. Under the vacuum kinetics of Eq. (2.8), the hydrothermal schedule gives $\sum_i k_{vac}(T_i)\,\Delta t_i \approx 60$, enough to convert the sample completely. If water lowers the conversion rate by a factor $R = k_{sat}/k_{vac}$, then even if all of the observed ^{3}He release came from conversion ($\alpha = F/(1-\phi) = 0.147/0.899 = 0.16$), R can be at most $10^{-2.5}$. The release is second order volume diffusion

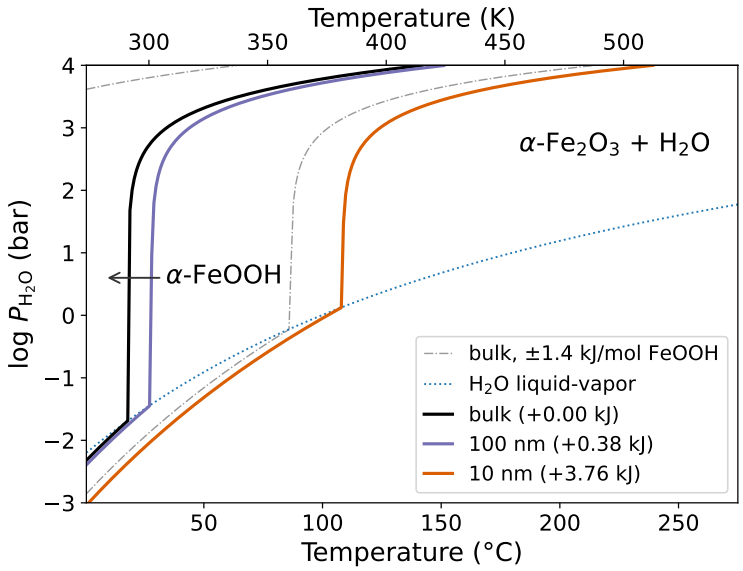


Figure 2.12: Goethite–hematite–water equilibrium in $\log P_{H_2O}$–T space for bulk goethite and for 100 and 10 nm grains with hydrated surfaces, compared at equal surface area per mole of Fe (Eq. (2.22)). Goethite is stable above and to the left of each curve. The dash-dot curves illustrate the absolute uncertainty: it is the bulk stability boundary shifted by the ± 1.4 kJ mol^{-1} FeOOH uncertainty of Majzlan et al. (2003). The blue dotted curve shows the water vapor saturation pressure; liquid water is present at and above this curve. The thermodynamic data used to compute the phase diagram are in Table 2.11.

Table 2.4: Bulk driving force for hematite formation, $\Delta\mu = -\Delta G_r$ (kJ per mole Fe_2O_3), at water saturation. In vacuum ($P_{H_2O} \to 0$) the driving force is unbounded.

T (°C)	P_{sat} (bar)	$\Delta\mu_{sat}$	$\Delta\mu_{sat}/RT$
150	4.76	5.68	1.61
175	8.93	6.88	1.84
200	15.55	8.11	2.06
225	25.50	9.37	2.26
250	39.76	10.66	2.45

(§2.4). Limiting the dehydroxylation fraction to below 1% requires $R \leq 10^{-3.8}$. Here we show that classical nucleation theory predicts suppression of this size for any reasonable value of its one unknown, the goethite–hematite interfacial energy.

Nucleation barrier. In classical nucleation theory (Gibbs, 1878; Volmer and Weber, 1926; Becker and Döring, 1935), reviewed by Christian (2002), Kashchiev (2000) and, for mineral systems, De Yoreo and Vekilov (2003) and Lasaga (1998), a spherical hematite nucleus of radius r has free energy

$$\Delta G(r) = -\frac{4}{3}\pi r^3 \frac{\Delta\mu}{V_{\text{hem}}} + 4\pi r^2 \gamma, \tag{2.11}$$

where $\Delta\mu$ is the driving force per mole of Fe_2O_3 (Table 2.4), V_{hem} the molar volume and γ the energy of the interface surrounding the nucleus. Nuclei larger than the critical radius $r^* = 2\gamma V_{\text{hem}}/\Delta\mu$ grow, and the energetic barrier to forming one is

$$\Delta G^* = \frac{16\pi\,\gamma^3\,V_{\text{hem}}^2}{3\,\Delta\mu^2}. \tag{2.12}$$

In the first-order limit of §2.4, the rate constant is a nucleation rate of the form derived by Turnbull and Fisher (1949),

$$k = A\exp\!\left(-\frac{E_a}{RT}\right)\exp\!\left(-\frac{\Delta G^*}{k_B T}\right),$$
$$\frac{\Delta G^*}{k_B T} = \frac{C\gamma^3}{T\,\Delta\mu^2}, \qquad C \equiv \frac{16\pi V_{\text{hem}}^2}{3k_B} = 1.112\times 10^{15}\ \text{m}^6\,\text{K}\,\text{J}^{-1}\,\text{mol}^{-2}, \tag{2.13}$$

where $A\exp(-E_a/RT)$ describes the atomic mobility and $\exp(-\Delta G^*/k_B T)$ is the equilibrium fraction of sites occupied by critical nuclei. Because the barrier scales as $\gamma^3/\Delta\mu^2$, it vanishes in vacuum, where the water pressure approaches zero and $\Delta\mu$ grows without bound, but is significant at water saturation, where $\Delta\mu$ is only 6–11 kJ (Table 2.4).

Internal nucleation. Goethite dehydroxylates topotactically: hematite nucleates within the parent crystal, inherits its oxygen framework and orientation, and the pseudomorph develops pores and twinned hematite domains (Lima-de-Faria, 1963; Watari et al., 1979; Watari et al., 1983; Cudennec and Lecerf, 2005). We therefore consider only internal nucleation, in which the new interface of the burgeoning nuclei is between hematite and goethite.

Interfacial energy. We know of no measured or calculated energy for the goethite–hematite interface. It is unlikely to be lower than that of a twin

boundary in hematite, 0.24 J m^{-2} for the prismatic twins formed by dehydrating goethite (Saito et al., 2016): a twin boundary separates crystals of the same phase and carries only a structural energy, whereas an interface between two phases adds a chemical term (Porter et al., 2009), here the change from hydroxylated goethite to anhydrous hematite and a 27% reduction in volume.

Threshold for suppression. Rather than assigning a value to γ, we ask how large it must be to explain the data. In the vacuum line the water pressure approaches zero, $\Delta\mu$ is unbounded and the barrier vanishes, so the observed vacuum rate constant is the mobility term $A\exp(-E_a/RT)$ of Eq. (2.13) (Eq. (2.8)). At water saturation the same mobility term is multiplied by $\exp(-\Delta G^*_{\mathrm{sat}}/k_BT)$. For an interfacial energy γ, the conversion predicted over the hydrothermal schedule is

$$\alpha(\gamma) = 1 - \exp\left[-\sum_i k_{\mathrm{vac}}(T_i)\,\Delta t_i\,\exp\left(-\frac{C\,\gamma^3}{T_i\,\Delta\mu_{\mathrm{sat}}(T_i)^2}\right)\right], \tag{2.14}$$

summed over the steps of the experiment, with k_{vac} from Eq. (2.8) and $\Delta\mu_{\mathrm{sat}}$ evaluated at each step temperature (Table 2.4). Because every term decreases with γ, the condition $\alpha(\gamma_{\min}) = 10^{-6}$ has a single root, which we find numerically: $\gamma_{\min} = 0.10$ J m^{-2}, a factor of 2.4 below the twin-boundary value. The sum is dominated by the hottest steps, where $\Delta\mu_{\mathrm{sat}}$ is largest and the barrier smallest; at $\gamma_{\min}$ the barrier at 250 °C is 18 k_BT, close to $\ln(\sum_i k_{\mathrm{vac}}\Delta t_i/10^{-6})$, the barrier that would be needed if it were the same at every step. At this threshold the critical nucleus at 250 °C has a radius of 0.56 nm and contains about 14 Fe_2O_3 formula units, a size at which classical nucleation theory is only approximate. For an interfacial energy near the twin-boundary value, the critical nucleus has a radius of 1.4 nm and contains about 212 formula units (several hematite unit cells), and the barrier is 259 k_BT. Any interfacial energy of this order suppresses solid-state conversion far below detection at water saturation while leaving the vacuum rate unaffected, consistent with the simple Arrhenius behavior of both vacuum experiments. The slowdown observed when the trap was isolated in the KAF experiment (§2.3) has the sign this picture predicts: added water lowers $\Delta\mu$ and raises the barrier. Omitting surface terms is conservative: for pseudomorphic conversion they stabilize goethite (§2.4), lowering $\Delta\mu_{\mathrm{sat}}$ and raising the barrier.

Assumptions. This is a consistent physical explanation of the data, not a proof. It assumes (i) that conversion is limited by internal nucleation of hematite, as

the first-order kinetics of the vacuum experiments imply; (ii) that the mobility term $A\exp(-E_a/RT)$ is the same in vacuum and in water, so that only $\Delta\mu$ changes; and (iii) that conversion is solid-state. Conversion by dissolution of goethite and precipitation of hematite from the fluid is not described by this model (§2.4).

Goethite Dehydration and/or Recrystallization During Helium Step Heat Experiments

The aim of helium step heat experiments is to characterize the mechanisms and rates of helium loss in nature to aid in the interpretation of geochronologic and helium geochemical data. The results of this experiment are encouraging in that they differ remarkably from experiments on the same samples in vacuum that are primarily controlled by dehydration processes (Farley et al., 2024), but we cannot exclude that this experiment may have been affected to some degree by processes aside from volume diffusion that are unrelated to helium loss in goethite in nature. We consider two processes outside of volume diffusion that could have contributed to the helium released in the water-saturated experiment: (i) recrystallization of goethite or (ii) fluid-mediated conversion of goethite to hematite. If either of these processes affected the helium release, then the apparent diffusivities reported here would represent upper bounds for diffusive helium loss in goethite in nature.

The helium step heat data at water-saturated conditions reported here are better fit with a diffusion model than with a conversion model (§2.4). The kinetic analysis of §2.4 and the nucleation analysis of §2.4 indicate that topotactic dehydroxylation has a nucleation barrier that is insurmountable for experimental conditions and timescales; however, this model has no bearing on fluid-mediated dehydroxylation. Equilibrium thermodynamics predict that goethite is unstable with respect to hematite and water at these conditions (Majzlan et al., 2003) (§2.4) and therefore dehydroxylation of goethite in the experiment cannot be outright excluded.

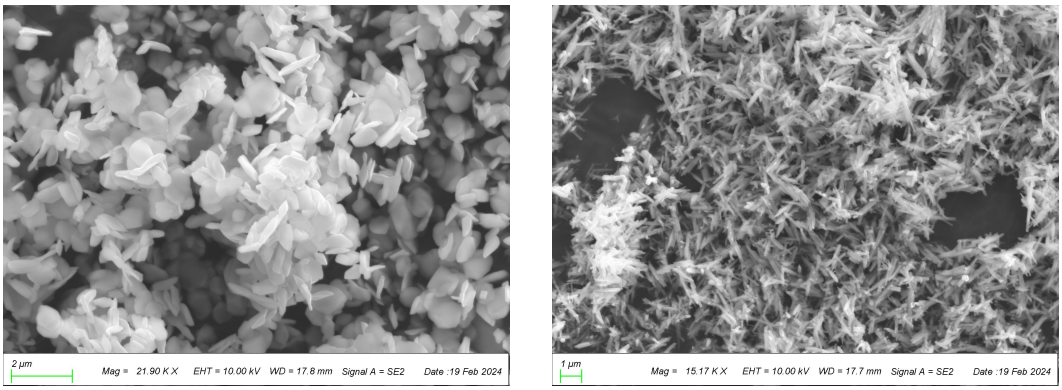
The process of recrystallization is another potential complication in interpreting helium step heats of goethite in hydrothermal conditions. Poorly crystalline synthetic goethite held in hydrothermal conditions is known to recrystallize into more orderly, crystalline goethite aggregates, even at temperatures as low as 125°C in as little as 16 hours (Schwertmann et al., 1985). In this

study, natural goethite was held in contact with liquid water and water vapor at hydrothermal conditions as harsh as ~40 bars and 250°C for many days. These conditions appear to have visibly dissolved and reprecipitated silica from the ampoule's inner walls (Fig. 2.4), qualitatively testifying to the ability of the hydrothermal fluid to dissolve and recrystallize rock-forming oxides. It is unquestionable that recrystallization of some goethite could have occurred under these conditions; however, it is unclear if conditions were favorable to recrystallize this specimen of goethite, Win-06-01b, which is amongst the most helium-retentive and crystalline dated goethite samples (Waltenberg, 2012; Farley et al., 2024). Less crystalline goethites in nature are commonly recrystallized, which appears to result in helium loss, resetting the U-Th/He age (H. S. Monteiro et al., 2018; H. S. Monteiro et al., 2014; Heim et al., 2006). Recrystallization could be the dominant process controlling helium loss in this experiment, but only if it is faster than diffusive loss, in which case the data presented here would represent new upper bounds for apparent diffusivities for helium in Win-06-01b goethite. Recrystallization might also be faster than volume diffusion of helium in goethite in nature over geologic timescales even if it did not occur in this experiment. In that case, recrystallization could be the primary driver of helium loss in nature regardless of the results of this experiment.

Fluid Mediated Conversion of Synthetic Goethite Pigment

We observed the spontaneous conversion of synthetic goethite pigment (§2.2) to hematite when heated in ampoules with excess water at experimental temperatures. Roughly equal proportions of ampoules were completely reacted (red), completely unreacted (yellow), or contained both reacted and unreacted material in isolated patches. In the mixed ampoules, the red and yellow portions were entirely isolated from one another, separated by gas bubbles. Unreacted pigment retains its original acicular habit (Fig. 2.13b). The reacted hematite product is instead composed of equant to platy crystals ~0.5–1 µm across (Fig. 2.13a). The change of shape rules out pseudomorphic dehydroxylation like that observed to occur in goethite heated in vacuum (Figs. 2.2, 2.1, and 2.3), indicating dissolution of goethite and nucleation and growth of hematite from the fluid, a pathway outside the solid-state nucleation model developed in §2.4.

Conversion either went to completion within a fluid-connected volume or



(a) Reacted (red) product (b) Unreacted (yellow) pigment

Figure 2.13: Secondary electron images of synthetic goethite pigment after hydrothermal treatment at 200–250 °C for about a day (§2.2). (a) Material from a completely reacted (red) ampoule: the acicular goethite has been replaced by equant to platy hematite crystals ∼0.5–1 µm across. (b) Material from an unreacted (yellow) ampoule, which retains the unchanged acicular habit of the starting pigment. The change in crystal habit requires dissolution of goethite and precipitation of hematite from the fluid, and the all-or-nothing outcome of the experiments indicates that the reaction is fast once it begins.

did not occur at all, even where reacted and unreacted pigment shared the same ampoule and thermal history. We posit that the fluid-mediated reaction requires some threshold of dissolved Fe(III) in the fluid to nucleate hematite, but because the reaction consumes and produces dissolved Fe(III) in equal proportions, it is not self-limiting and proceeds quickly to completion once that threshold is reached.

We conclude that water-saturated conditions do not by themselves exclude fluid-mediated conversion of goethite, even where topotactic conversion is suppressed (§2.4). Whether that pathway operates likely depends on the solution chemistry. Still, the helium step heat data for Win-06-01b-HT show no evidence for the abrupt and extreme helium loss that would be expected to accompany a fluid-mediated reaction. The step heat data are instead consistent with a smooth, history-dependent release of helium, as expected for diffusive loss (§2.4).

Grain size and bulk thermodynamics cannot explain why the pigment blades reacted when the natural polycrystalline goethite did not. At 1 µm, the surface contribution to ΔG_r is only ∼0.04 kJ, and the bulk driving force at saturation is the same 6–11 kJ for both materials (Table 2.4). The most likely difference

is the fluid itself. Commercial iron-oxide pigments are commonly produced by precipitation from iron-salt solutions. They can carry residual soluble salts such as iron and alkali sulfates or nitrates, and they are often surface-treated with organometallic compounds (Legodi and Waal, 2007; Almeida Silva et al., 2017; M. Xu et al., 2021). Such contaminants would change the pH and ionic strength of the water they are heated in, and with them the solubility of Fe(III). The dissolution–precipitation pathway is limited by how much iron the fluid carries, so a modest increase in Fe(III) solubility could make a nucleation-limited reaction viable. The natural Win-06-01b goethite, by contrast, was heated in distilled water, where near-neutral Fe(III) solubility is very low, potentially offering an explanation for the divergent outcomes.

High Helium Retentivity of Win-06-01b-HT

The apparent diffusivities of the water-saturated experiment are approximately seven natural log units lower than those obtained from the same sample in vacuum by H. Monteiro (2017) and Farley et al. (2024) (Fig. 2.8b, Fig. 2.6, Fig. 2.7). This high helium retentivity is explored through two lenses: closure temperature of cooling samples (Dodson, 1973) and isothermal diffusion (Wolf et al., 1998).

If the slope and intercept on the regression of 3He apparent diffusivities from this experiment are taken to represent a real diffusion domain for goethite, then a closure temperature can be calculated for this domain. Considering uncertainty only in the intercept (D_0/a^2) term and a cooling rate of 10°C/Ma, the closure temperature for that domain would be 98.1± 8.6°C. The multiple diffusion domain inversion (§2.4) indicates that the dominant domain, which holds 97% of the ^{3}He, diffuses at 0.80 times the rate of the Stage 1 line. Its closure temperature is correspondingly about 1.6 °C higher, well within the quoted uncertainty; the Stage 1 parameters are used throughout this section.

Goethite is formed near Earth's surface and thus U-Th/He geochronology is most typically applied to date surface processes (Heim et al., 2006; Yapp and Shuster, 2011; Vasconcelos et al., 2013; Danišík et al., 2013; H. S. Monteiro et al., 2014; Hofmann et al., 2017; Allard et al., 2018). For crystals of goethite grown and collected near the surface, the concept of a closure temperature is poorly applicable, as it necessarily assumes a cooling rate (Dodson, 1973). In such cases, a better measure of retentivity is an analysis of isothermal

retentivity over time (Wolf et al., 1998), which is commonly expressed in terms of t'/t, where t' is the helium age and t is the isothermal holding time (most often the 'real age'). For an isothermal holding time of 100 Ma at 40°C, a diffusive domain with the parameters obtained from the regression of 3He apparent diffusivities of Win-06-01b-HT has a t'/t = 0.968, where t'/t = 1 corresponds to perfect retentivity. Fig. 2.14 shows isothermal holding time against helium age for this same domain for 4 different temperatures. At 25°C, this domain is nearly quantitatively retentive of helium even at billion-year time scales. The results presented here qualitatively support that goethite is highly retentive of helium at surface conditions, even at 100-million-year timescales.

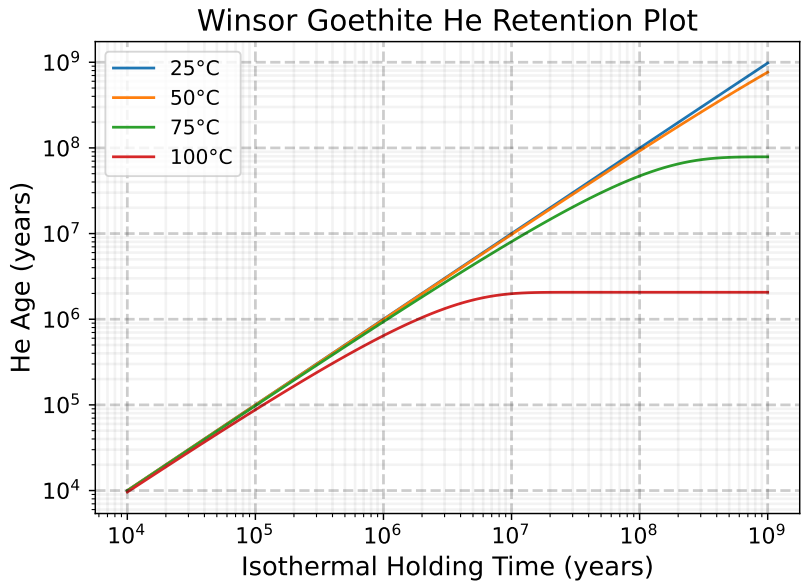


Figure 2.14: Isothermal retention plot (Wolf et al., 1998) for helium in goethite corresponding to the "Stage 1" reference line regression on the hydrothermal experiment. Helium hosted in a diffusion domain with these diffusion parameters would be nearly perfectly retentive at billion-year timescales at 25°C. At 40°C, the equilibrium age for this domain (the helium age at which radiogenic helium production and diffusive loss are equal) is 33.2 billion years, attained after 50.4 billion years.

This study of a single sample has no bearing on potential complexities related to aluminum content, radiation damage, or crystal defects on bulk diffusion (Bassal et al., 2022), nor does it permit estimation of the diffusivity at infinite temperature (D0); this would require experiments on controlled grain sizes.

Profiling Radiogenic Helium Loss with $\frac{^4He}{^3He}$ Ratios of Win-06-01b-HT
Bulk-normalized ^{4}He/^{3}He ratios in step heats of irradiated samples measure the rounding of concentration profiles of radiogenic helium by natural helium loss, if the loss mechanism in nature and in the laboratory share the same geometry (Shuster and Farley, 2004; Farley et al., 2024). Aside from the first experimental step, bulk-normalized ^{4}He/^{3}He ratios for all experiments begin at vanishingly small values and steeply increase at first, with the rate of increase slowing as the experiments proceed (Fig. 2.15). This behavior is consistent with complete then partial loss of radiogenic helium in nature in the least retentive domains and is observed in all step heat experiments on irradiated goethite (Farley et al., 2024). These least retentive domains could represent fine goethite grains, or they could represent interstitial helium trapped between crystallite grain boundaries. Goethite aggregates are known to have interstices that are filled by chemisorbed non-stoichiometric water that represents the earliest water to be lost in vacuum (Naono et al., 1987). Non-stoichiometric hydrogen is known to persist in goethite even after heating to 140°C (Yapp and Poths, 1995). It is therefore reasonable to posit that artificial 3He observed in less-retentive domains may be trapped by non-stoichiometric water or hydroxyl groups within interstices of goethite aggregates. Dehydration of non-stoichiometric interstitial water may be associated with early fractions of degassing 3He from goethite samples during step heats, and could potentially represent the LRD (low retention domain) of Shuster et al. (2005), and the 3He offgassed during "Stage 1" in vacuum step heats (Farley et al., 2024). The high variability of the portion of 3He hosted in early-degassing "Stage 1" fractions within vacuum step heats is sensible considering observations that goethite formed at different conditions is known to have highly variable crystallinity, crystallite ordering, surface area, and non-stoichiometric water contents (Schwertmann et al., 1985).

The hydrothermal experiment records lower bulk-normalized ^{4}He/^{3}He ratios than the vacuum experiments at any given $\Sigma F_{3\mathrm{He}}$ for the duration of the experiment. Similarly, the approach to the ^{4}He/^{3}He plateau varies between the three experiments: a high bulk normalized ^{4}He/^{3}He plateau is most decisively represented in the Farley et al. (2024) experiment, whereas the ^{4}He/^{3}He ratios in the step heats of H. Monteiro (2017) transition more gradually before reaching a plateau later in the experiment, while the increase in ^{4}He/^{3}He ratios is slowest for the hydrothermal experiment (Fig. 2.15). This suggests that the geometry of helium loss differs between the hydrothermal and vacuum experiments,

and could be explained by the relative dominance of dehydration processes or their mode of spatial propagation as a function of water pressure, where the smoothest ^{4}He/^{3}He spectra are generated by higher water pressures. We interpret this experimental control on the ^{4}He/^{3}He spectra to reflect the relative importance of goethite dehydration and helium diffusion through crystalline goethite in governing the geometry of helium loss from the sample.

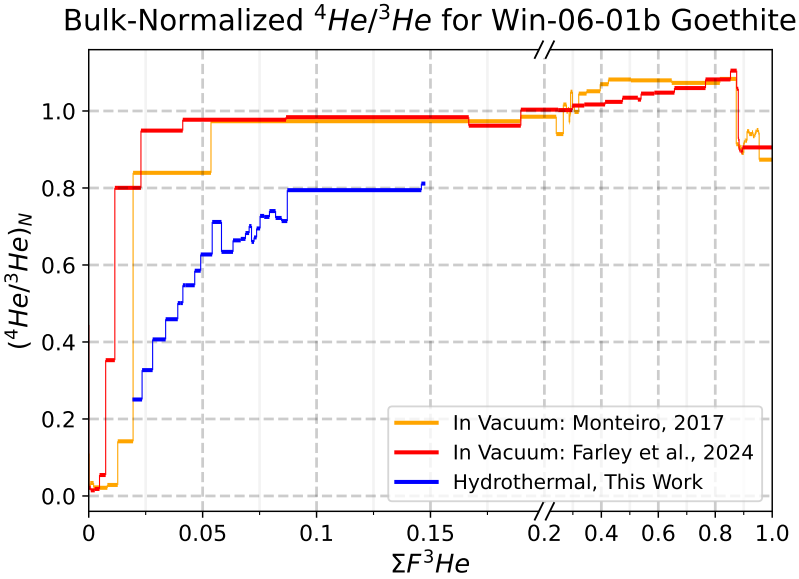


Figure 2.15: Bulk-normalized ^{4}He/^{3}He ratios for Win-06-01b goethite step heats against cumulative fractional ^{3}He yield. The hydrothermal experiment shows lower bulk-normalized ^{4}He/^{3}He ratios at a given $\Sigma F_{3\mathrm{He}}$ for the duration of the experiment, suggesting that ^{4}He/^{3}He spectra are dependent on the partial pressure of water and that the geometry of goethite dehydroxylation differs from the geometry of helium diffusion through goethite. For all experiments, ^{4}He/^{3}He ratios begin at very small values and increase steeply at first, which is consistent with complete then partial loss of radiogenic helium in nature for the least retentive helium domains. Six measurements from the water-saturated experiment are omitted for clarity due to low signals and high errors. For visual continuity, the gap left behind is bridged. The horizontal axis changes scale at $\Sigma F_{3\mathrm{He}} = 0.2$ (double slash) to emphasize the portion of the plot where data exist for the water-saturated experiment.

If the geometry of helium loss due to dehydration and due to diffusion differ, then the ^{4}He/^{3}He spectra of goethite measured in vacuum may not straightforwardly encode the thermal history of the sample in the way that such data faithfully encode this history in phases such as hematite or apatite (Farley and McKeon,

2015; Shuster and Farley, 2004). More ^{4}He/^{3}He spectra of goethite under hydrothermal conditions are needed to investigate this hypothesis further.

These ratios also bear on the natural thermal history of Win-06-01b. Throughout the range of $\Sigma F_{^3\mathrm{He}}$ that defines the Stage 1 reference line (7–15%), step ^{4}He/^{3}He ratios remain at 0.67–0.82 of the bulk value (Table 2.2). Radiogenic ^{4}He is depleted relative to the uniformly distributed ^{3}He in this domain, so even this retentive domain lost helium in nature. A domain with the Stage 1 parameters is nearly quantitatively retentive at Earth-surface temperatures (§2.4), so this loss is difficult to attribute to thermally activated diffusion at surface conditions. Either Win-06-01b experienced temperatures well above surface conditions (perhaps it grew in a hydrothermal setting or experienced burial heating), or non-thermal processes such as fluid-mediated recrystallization caused partial helium loss in nature. Section 2.4 tests whether thermally activated diffusion alone, within the domain structure inferred from the ^{3}He release, can reproduce the measured spectrum.

Thermal History of Win-06-01b

We begin the exercise by limiting our exploration of potential thermal histories to geologically reasonable and simple paths. With the four-domain structure of §2.4 held fixed, we fitted steady-cooling, burial-and-exhumation, and isothermal thermal histories to the ^{4}He/^{3}He spectrum of HT-Win-06-01b (§2.2). Previous studies constrain our exploration of the burial-and-exhumation paths. Gibson and Stüwe (2000) and M. M. Mitchell et al. (1998) interpret regional cooling from roughly 50–70 °C after the mid-Cretaceous from their apatite fission-track studies. M. M. Mitchell et al. (1998) additionally interpret, from limited 1.5 km thick sediment accumulation in Triassic intramontane basins, that parts of the ranges were near the surface in the early Mesozoic. Table 2.5 gives the best fit of each family, Fig. 2.16 compares the best steady-cooling and burial predictions with the spectrum, and Fig. 2.17 shows their χ^2 maps on a common color scale.

Steady cooling. Linear cooling to 25 °C fits the spectrum best. The best age-consistent path starts at 78 °C at 150 Ma ($\chi^2_\nu = 2.00$), and the age-consistent 1σ range is crystallization at 150–175 Ma at 76–80 °C, followed by cooling at $\sim$0.3 °C Myr^{-1}. Such paths pass through $\sim$55 °C at 90 Ma, inside the apatite fission-track window. The spectrum alone prefers somewhat younger, hotter

Table 2.5: Thermal histories fitted to the ^{4}He/^{3}He spectrum of HT-Win-06-01b with the four-domain structure fixed. Each row gives the best path with a predicted He age of 130–180 Ma along with the reduced χ^2 value ($\nu = 22$ for steady cooling and the isothermal hold, $\nu = 21$ for burial). For steady cooling, $T_{\max} = T_0$ is reached at crystallization, so $t_{\text{peak}} = t_{\text{form}}$. For the isothermal hold, $T_{\max} = T_{\text{iso}}$ throughout. Burial crystallization ages are fixed within each row. The last three columns give the ranges of the age-consistent paths within the joint 1σ region about the best fit (see Eq. (2.7)).

Path	t_{form} (Ma)	t_{peak} (Ma)	$T_{\max}$ (°C)	χ^2_ν	t_{He} (Ma)	t_{form} (1σ)	t_{peak} (1σ)	$T_{\max}$ (1σ)
Linear cooling	150	150	78	2.00	132	150–175	150–175	76–80
Burial	180	70	61	5.36	164	fixed	60–82	58–65
Burial	195	80	61	5.04	177	fixed	68–98	58–65
Burial	210	115	71	6.53	180	fixed	105–130	70–74
Isothermal hold	150	–	41	18.5	141	140–190	–	39–43

paths: the unconstrained minimum ($\chi^2_\nu = 1.68$; 89 °C at 80 Ma) predicts a He age of 68 Ma, and the unconstrained 1σ region lies entirely at ages younger than 130 Ma (Fig. 2.17b).

Burial and exhumation. Burial paths fit the spectrum less well ($\chi^2_\nu = 5.0$–6.5), but they agree with the independent regional constraints of Gibson and Stüwe (2000) and M. M. Mitchell et al. (1998). For crystallization at 180 and 195 Ma, the best age-consistent paths peak at 61 °C at 70 and 80 Ma, with 1σ ranges of 58–65 °C between 98 and 60 Ma (Fig. 2.17a). These peaks fall in or just after the apatite fission-track window, within the 50–70 °C range of Gibson and Stüwe (2000) and at the <60 °C limit reported by M. M. Mitchell et al. (1998). For crystallization at 210 Ma, only hotter and earlier peaks (70–74 °C at 105–130 Ma) give an age within the measured range, and they fit worse ($\chi^2_\nu = 6.5$).

Isothermal hold. The spectrum constrains only the product of diffusivity and time, so the χ^2 minimum is a ridge from ∼47 °C at 50 Ma to ∼35 °C at 500 Ma, along which $\chi^2_\nu \approx 18.5$ throughout. No isothermal path reproduces the shape of the spectrum: they underpredict the ratios of the early steps and overpredict those of the late steps. We do not consider this family further.

Where the families differ. The steady-cooling and burial predictions differ mainly in the earliest steps ($\Sigma F_{^3\text{He}} \lesssim 0.035$; Fig. 2.16), which sample the outer rim of the dominant domain. On the burial path the sample sits at 20 °C for

the last 50 Myr, long enough for radiogenic ^{4}He to re-accumulate in the rim, so the early ratios are overpredicted. Steady cooling keeps the rim depleted until geologically recent times, with diffusive loss declining monotonically toward the present. Elsewhere both leave the same residual pattern: ratios underpredicted between $\Sigma F_{3\mathrm{He}} = 0.04$ and 0.06, and overpredicted in the two longest high-temperature steps.

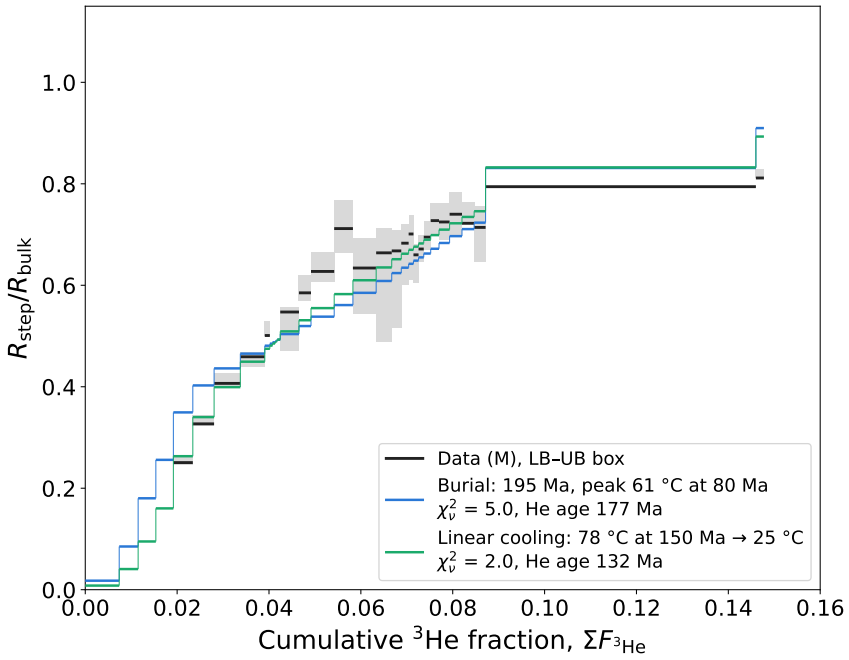


Figure 2.16: Bulk-normalized ^{4}He/^{3}He spectrum of HT-Win-06-01b (black, bounding-method LB–UB boxes) with the best age-consistent burial path for crystallization at 195 Ma (blue) and the best age-consistent steady-cooling path (green), for the four-domain structure.

Consistency with diffusive loss. These two families of thermal paths encode very different geologic origins. We note that any number of arbitrarily complex thermal paths can be evaluated against the data with this workflow, but a thorough search for viable thermal paths is beyond the scope of this study, and is an ill-posed problem for a single sample. The main purpose of this exercise is to demonstrate that the data we present here are consistent with diffusive loss of radiogenic helium in nature. We do not claim to have found the unique thermal history of the sample, but have demonstrated that plausible thermal histories reproduce our results reasonably well.

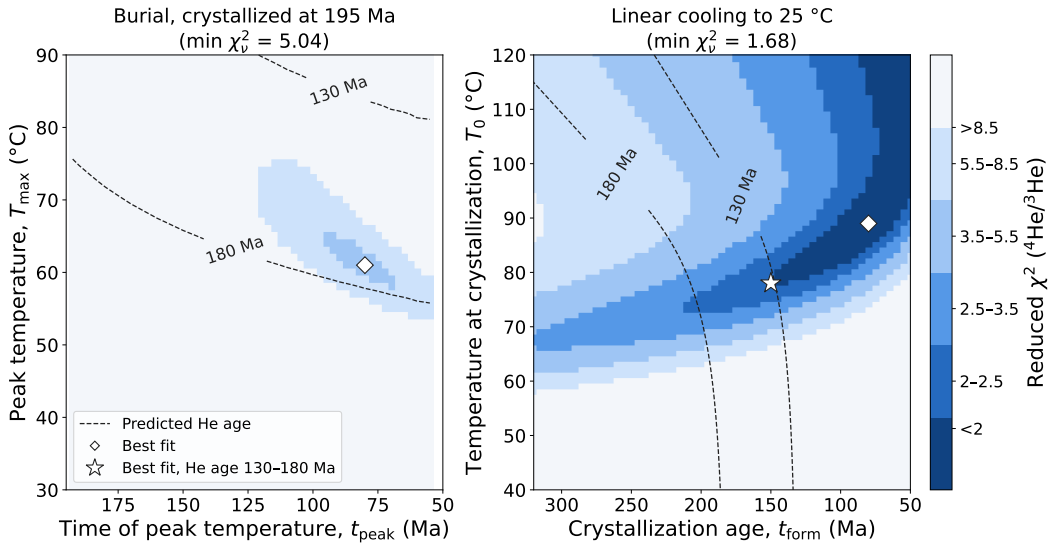


Figure 2.17: Reduced χ^2 of the ^{4}He/^{3}He spectrum for (a) burial paths with crystallization at 195 Ma, as a function of the time and magnitude of peak temperature, and (b) steady cooling to 25 °C, as a function of crystallization age and initial temperature. Both panels share one color scale, anchored to the overall minimum. Dashed lines show the predicted bulk He age at 130 and 180 Ma, diamonds the best fit, and the star the best fit with a He age of 130–180 Ma.

2.5 Conclusions

Predicting the degree of helium retention and loss to guide interpretation of U-Th/He dates of natural goethite based on factors such as grain size, crystallinity, radiation damage, defect density, impurities, and aluminum substitution remains challenging. Previous efforts to understand helium loss in natural goethite with step heating experiments in vacuum have instead queried the kinetics of dehydration (Farley et al., 2024). Because goethite dehydration is suppressed at high water pressure, helium step heats under hydrothermal conditions are more likely to interrogate the processes controlling radiogenic helium loss in natural goethite. The method presented in this study leverages the high helium permeability of silica glass to easily conduct helium step heat experiments under water-saturated conditions, achieving internal water pressures up to 86 bars at 300°C using standard ultra-high vacuum experimental setups (Farley et al., 1999). Natural crystalline goethite does not develop the visible dehydration-induced contraction cracks when heated at water-saturated conditions like vacuum-heated goethite does.

The experiment conducted on irradiated Win-06-01b represents the first helium

step heat experiment for goethite at water-saturated conditions. Vacuum step heats of the same sample follow first-order conversion to hematite, whereas the hydrothermal release shows the history and time dependence of diffusion. The hydrothermal experiment gives apparent diffusivities approximately seven natural log units below those obtained from the same sample in vacuum. The corrected ^{3}He array is linear between 7–15% cumulative yield, with an activation energy of 157.1± 5.3kJ/mol. Extrapolated to surface conditions, a domain with these parameters is essentially quantitatively retentive of helium at billion-year timescales at 25°C. The ^{4}He data require that this domain experienced helium loss in nature. A four-domain diffusion model fitted to the ^{3}He release shows that 97% of the ^{3}He resides in a single domain and can account for this loss by thermally activated diffusion alone. Within that model, steady cooling from crystallization at 76–80 °C at 150–175 Ma to the present surface temperature fits the ^{4}He/^{3}He spectrum best, favoring crystallization from warm, possibly hydrothermal, fluids. Burial of surface-formed goethite to a Late Cretaceous peak of 58–65 °C, consistent with regional apatite fission-track constraints, remains a viable alternative.

The ^{4}He/^{3}He spectrum observed for this sample is distinct from that of the same sample in vacuum, suggesting that the geometry of helium loss induced by goethite dehydration is distinct from that of diffusion. More hydrothermal step heat experiments of goethite are necessary to confirm this hypothesis.

A thermodynamic and kinetic analysis of goethite dehydroxylation explains why water seems to stabilize goethite up to 250°C at experimental timescales. The hydrothermal release itself requires solid-state conversion to be at least $10^{2.5}$ times slower in water than in vacuum. At water saturation, the driving force is only 6–11 kJ, whereas it is unbounded at the near-zero water pressure of a vacuum line; this raises the barrier to hematite nucleation enough to suppress solid-state conversion far below this limit. Fluid-mediated conversion, which transformed synthetic goethite pigment in similar conditions, is rapid and would be accompanied by abrupt and extreme helium loss, which is not observed in the hydrothermal step heat data. This divergent outcome may be explained by different solution chemistry between the two experiments, which would affect the availability of aqueous Fe(III) that supplies the reaction. Recrystallization, however, cannot be excluded. If it contributed, the diffusivities reported here would be upper bounds and the retentivity a lower bound.

Distinguishing these possibilities, and separating the effects of crystallite size, chemical composition, and radiation damage, will require further hydrothermal experiments on irradiated single-crystal goethite of controlled size. The method described here makes such experiments straightforward.

Supplementary Information

for Chapter 2: *Helium Step Heats on Goethite at Water Saturated Conditions Show Diffusive Helium Loss and High Retentivity at Surface Temperatures*

2.1s Permeation Lag Correction and Ampoule Permeability

Permeation Lag Correction

Because HT-Win-06-01b was encapsulated in an imperfectly helium-transmissive silica glass ampoule, permeation lag was an expected complication in these measurements. To address this, experimental steps at temperatures at or above 170°C were bracketed by low temperature (<120°C) steps where the helium in goethite at high pH2O was effectively immobile at experimental timescales, following the permeation step method (Chapter 1, §1.4). For 3He, measured signals at these low temperature permeation steps were entirely controlled by the process of permeation through silica glass, and therefore could be used to estimate the amount of helium remaining within the ampoule between high-temperature experimental steps. However, for 4He, the measured signals at low temperature permeation steps were most often indistinguishable from the blank, so Eq. (1.6) (Chapter 1) could not be applied to 4He and the permeation step method could not be carried through for that isotope.

As a result, permeation lag is instead addressed using the bounding method described in Chapter 1 (§1.4), which constrains the ampoule's residual helium from above rather than measuring it. In this approach, an upper bound constraint for the helium content of the ampoule at the end of each high-temperature step was calculated using Eq. (1.10) (Chapter 1). 3He was reduced this way as well, for consistency with 4He; because 3He can also be reduced by the permeation step method, the two reductions can be compared directly, and that comparison is used below (§2.1s) as an internal test of the bounding method.

Both routes require the ampoule's permeation rate constant λ, which was measured during the experiment itself rather than assumed from the literature (Chapter 1, §1.4). This in-experiment measurement is not an alternative method of handling permeation lag but a prerequisite of the permeation step method and, here, the empirical basis for choosing a conservative λ for the bounding method. It is indispensable in this experiment: the permeability of ampoules used in hydrothermal step heats cannot reasonably be assumed to match literature values for anhydrous silica glass.

Ten series of 3–4 consecutive permeation steps held at a single temperature between 90–110°C for 120-180 minutes each with 26 minutes of cooling time before being inlet into the Helix-MC-Plus^{10K} were identified as suitable for this

exercise. The 3He measurements in these series were fit to Eq. (1.9) using a non-linear least squares optimization (Fig. 2.18). This is accomplished with the Levenberg-Marquardt algorithm implemented in the curve_fit function in the SciPy package of Python (Virtanen et al., 2020), weighted by each measurement's own uncertainty. One additional series that otherwise qualified was not well fit by this model and was not used further for clarity, though its fit value of λ was within uncertainty of the other series' fit values for λ.

An initial guess of $\lambda \approx 6{\times}10^{-6}$ sec^{-1} and $n_0 \approx \frac{m_i}{1-e^{-\lambda t_i}}$ (evaluated at the first measurement of a series) pccSTP was passed to the optimizer for each series. Uncertainties from the model fits were calculated using the covariance matrices generated for each of the model fits. Although the initial helium amount n_0 was a free parameter in the optimizations, the rate constant of permeation (λ) was the only parameter of interest for this exercise. The fitted rate constants span 2.7×10^{-5} to 6.3×10^{-5} s^{-1}, with 2σ uncertainties of 0.08 to 0.46 log units and no trend resolvable against step duration:

All measured apparent 3He permeabilities were equal to or in excess of the literature values for anhydrous silica glass (Fig. 2.19), similar to observations discussed in Chapter 1 (§1.7). 4He permeability itself could not be independently measured in this experiment due to blank-level signals during the low temperature permeation steps and is assumed to similarly exceed the literature values for anhydrous silica glass. To evaluate the uncertainty due to the permeation lag, we use the literature values as a conservative (lower bound) estimate of the true permeability; a λ that underestimates the ampoule's true rate constant yields a conservative (higher) upper bound on the residual helium (Chapter 1, §1.4). λt (where $\lambda = GRTK$) is estimated here using the 4He permeability K reported by Swets et al. (1961), while the 3He permeability is taken to be the product of the 4He permeability and the relative 3He/4He permeability reported by Jones (1953). The geometric constant G is calculated using the measured inner and outer diameter of the ampoule (Fig. 2.5).

The first three experimental steps had values of λt lower than 4, and therefore the upper bound of the helium content of these steps could not be determined. These measurements are omitted from further analysis. Later experimental steps with values of λt lower than 4 could be determined using Eq. (1.57) from the supplement to Chapter 1 because upper bounds on previous steps' residual helium content could be evaluated.

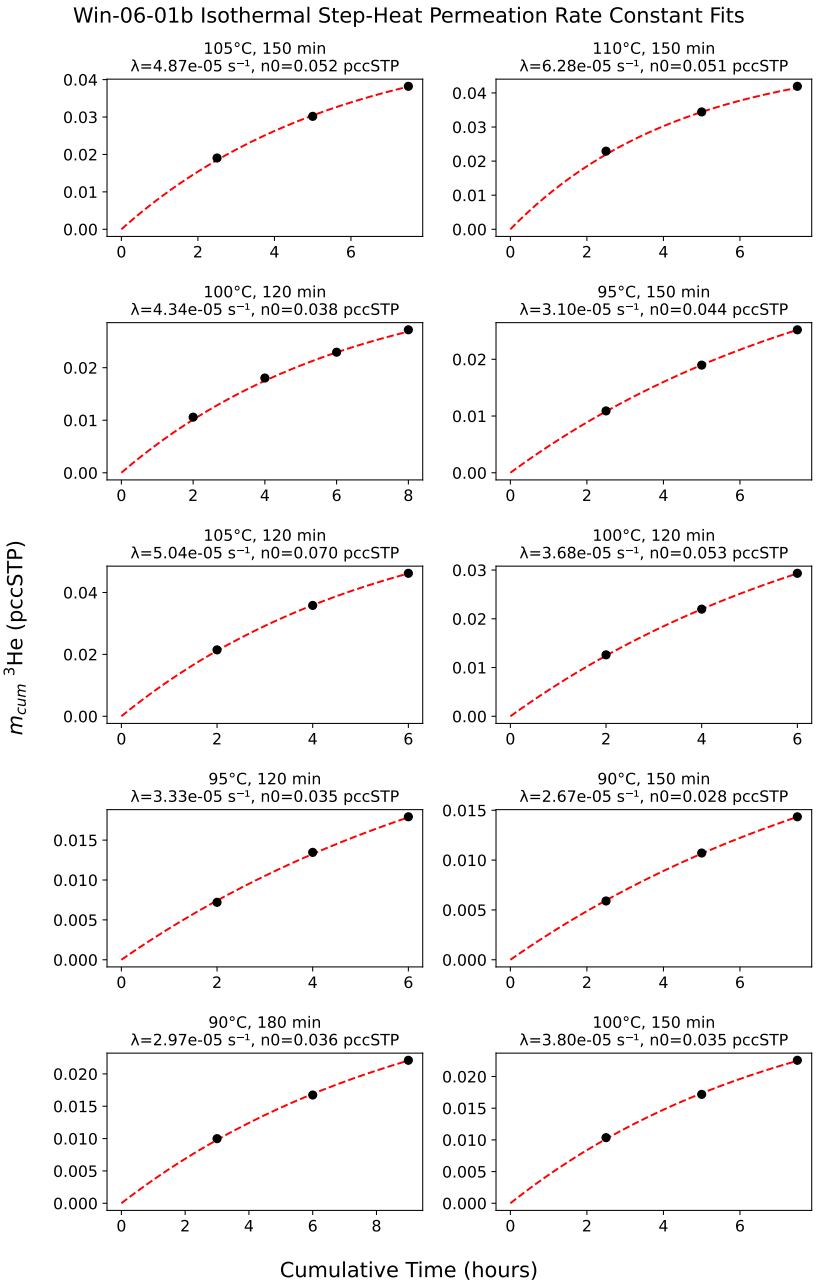


Figure 2.18: Cumulative yield against cumulative time for helium-3 yields for isothermal permeation steps. A non-linear least squares optimization fits λ and n_0 of Eq. (1.9) (Chapter 1) to data from a series of permeation steps like these. The fitted λ is plotted as $\ln(\lambda/T)$ against inverse temperature in Fig. 2.19. The curvature of each series is what conditions the fit: a series in which the permeation rate has not yet slowed appreciably constrains λ and n_0 only weakly.

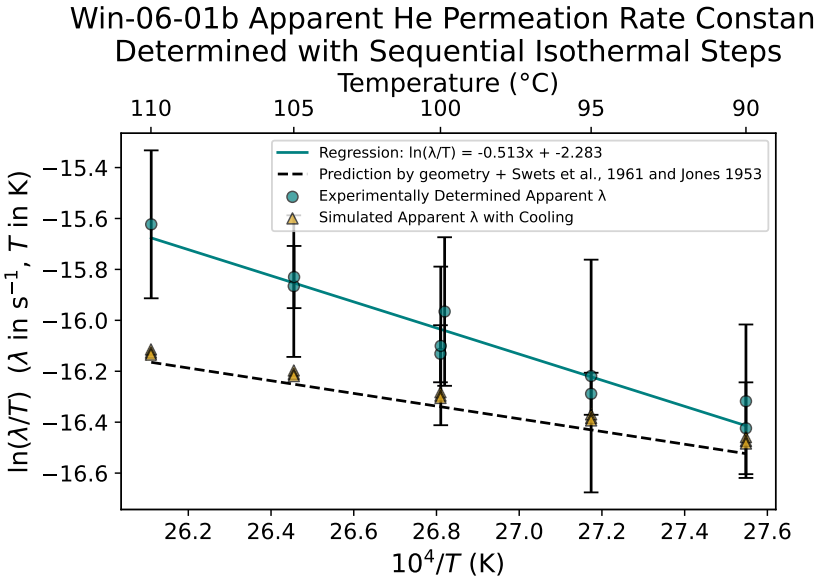


Figure 2.19: Arrhenius plot of the apparent permeation rate constant, $\ln(\lambda/T)$, as a function of inverse temperature for HT-Win-06-01b, using the rate constants fitted in Fig. 2.18. The dashed line is the prediction from the ampoule geometry and the literature permeabilities of Swets et al. (1961) and Jones (1953), converted to a rate constant via $\lambda/T = GRK$. The activation energy is higher than anhydrous silica glass, as expected (Masaryk, 1968; Shelby, 1972). It is unexpected that the glass ampoule of HT-Win-06-01b has a higher apparent 3He rate constant than predicted from the literature values of Swets et al. (1961) and Jones (1953), but this can be explained by an increase in the partial pressure of the gas phase within the tube due to displacement by high He diffusivity, low He solubility water. Yellow points labeled "Diffusion Model with cooling, Anhydrous" were produced to evaluate the expected excess apparent permeability due to the 25 minute cooldown period before measurement. This is accomplished by simulating the measured apparent permeability for a dry silica glass ampoule that continued to permeate helium while cooling for an additional 26 minutes before measurement after being held at 90–110°C for 120–180 minutes, similarly to the measured values. Variability in these points is due to differences in the length of time the ampoule was heated but is very small. They show a near constant offset from the expected values of Swets et al. (1961) and Jones (1953) that is much smaller than the uncertainty in the regression of the fit values determined experimentally in teal.

Measurement and Numerical Simulation of Apparent Permeability of Experimental Ampoule

The apparent permeability values used in the numerical simulation (§2.2s) are taken from the literature (Swets et al., 1961; Jones, 1953) and explicitly apply to an ampoule without water. However, the purpose of these simulations was simply to quantify the excess apparent permeabilities that would be observed for ampoules with a cooling history similar to that of the hydrothermal experiments. Because the predicted offset is universally small and a weak function of temperature or permeability, these simulations demonstrate that the much larger experimentally observed permeability is not an artifact of the cooling history of the hydrothermal experiments, reaffirming the conclusion that the hydrothermal experiments are indeed more permeable than the vacuum experiments.

The difference in log permeability between the actual and apparent permeabilities of the anhydrous simulation was significantly smaller than nearly all the 2σ uncertainties of the apparent permeabilities determined above for the hydrothermal Win-06-01b experiment, which ranged from 0.08 to 0.46 log units. Thus, no trends with step duration could be identified in the effective permeability values.

Therefore, the literature values underestimate the helium permeabilities of the ampoule used in this experiment, which is permissible because these values are used to calculate an upper bound constraint of the helium content of the ampoule, and lower permeabilities produce more conservative (higher) upper bound constraints.

A linear regression of $\ln(K)$ against inverse temperature (Fig. 2.19) fit to these isothermal 3He permeation measurements over 90-110°C yields an R^2 of 0.936 and an activation energy of 42.6±7.9 kJ/mol (2 s.e.), significantly higher than expected for pure silica glass with or without water (Swets et al., 1961; Jones, 1953; Masaryk, 1968; Shelby, 1972), but similar to glasses with a high abundance of network-modifying cations such as Na, K, Pb, or Li (Norton, 1953; Altemose, 1961). Because 4He signals during these isothermal steps were blank-level, only the ampoule's 3He permeability could be measured directly this way, using the fitting procedure described above.

Every 3He permeability measurement for HT-Win-06-01b equals or exceeds the anhydrous literature value, which runs counter to expectations for a hy-

drothermal experiment: water and hydroxyl groups dissolved in silica glass are reported to *decrease* helium permeation at all temperatures (Tsuru et al., 2011; Shelby, 1972; Masaryk, 1968), under the common paradigm that species such as $(OH)^-$ occupy interstitial space within the amorphous glassy network (Norton, 1953; Shackelford, 1999). The offset is nonetheless real: it cannot be accounted for by the ampoule's extended permeation during cooling before measurement (§2.2s), and a similarly elevated permeability was independently observed at 100°C in a separate silica glass ampoule at Caltech partially filled with MilliQ water and equilibrated with ~0.1 torr of tank helium, as discussed in Chapter 1 (§1.7).

The likely explanation is geometric. Helium is only sparingly soluble in water and diffuses through it rapidly, roughly 7×10^{-5} cm^2/s at room temperature (Verhallen et al., 1984) against about 2×10^{-8} cm^2/s in silica glass (Swets et al., 1961). On the timescale of permeation through the glass wall, transport through the water is effectively instantaneous, so the gas phase stays in equilibrium with the glass over the entire inner surface of the ampoule, including where liquid water contacts it. Helium partitions strongly into the gas phase over both water and glass: helium concentrations in the glass and water are about 3% and 1%, respectively, of the gas-phase concentration, using solubilities from Sander (2015) and Swets et al. (1961). Displacing gas-phase volume with water therefore raises the helium partial pressure without changing the surface area available for permeation, and because the rate of permeation is proportional to the gas-phase partial pressure (Norton, 1953; Altemose, 1961; Swets et al., 1961; Masaryk, 1968), the ampoule's rate constant scales with the inverse volume fraction of the gas phase (Chapter 1, Eq. (1.12)), generalized here to any condensed payload: $\lambda_{ampoule} = \lambda_{empty} \left(V_{condensed} + V_{gas}\right) / V_{gas}$. This observed complexity emphasizes the need to measure the ampoule's rate constant in-experiment as described in Chapter 1 (§1.4).

Consistency of the Bounding Method with Direct Permeation-Step Quantification: The Complete Degas Test

All Win-06-01b step heat data presented in this chapter are reduced using the bounding method (Eq. (1.10)), because 4He permeation-step yields were near blank level and could not support the permeation step method directly. As an internal check on this choice, the ampoule's 3He inventory was also reconstructed directly from the permeation steps themselves, using the in-

experiment rate constants of Fig. 2.19 (§2.3s). Reducing 3He by the bounding method gives Stage 1 activation energies between 157.1 and 160.6 kJ/mol and closure temperatures between 97.5 and 101.2°C across its lower-bound, central, and upper-bound reductions, a spread comparable to the regression uncertainty itself. Fig. 2.20 compares the two: n_t, the ampoule's residual helium quantified directly from a permeation step, against $n_{t,SS}$ (equivalent to $n_{t,\mathrm{UB}}$ used elsewhere in this chapter), the upper bound on that same residual calculated independently from the preceding high-temperature step. For every ordinary permeation step in the experiment, $n_t \leq n_{t,SS}$, exactly as expected if the steady-state approximation genuinely bounds the ampoule's residual helium from above. This agreement is the justification for reducing the 4He dataset by the bounding method alone, since no analogous direct check of 4He is possible.

Beyond the ampoule's residual helium content itself, the two methods' derived apparent diffusivities agree just as closely (Fig. 2.22, §2.3s): $\ln(D/a^2)$ from the permeation step method tracks the bounding method's central value along the 1:1 line within uncertainty across the full range of the experiment. Fitting the permeation step method's own diffusivities with the same Stage 1 procedure used for the bounding method in the main text (Fig. 2.23, §2.3s) yields an activation energy of 157.0±4.8 kJ/mol (R^2=0.991, n=12) and a closure temperature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (Fig. 2.6).

The two "complete degas" permeation steps depart from this pattern. For the longer of the two (7200 minutes; predicted $n_t/n_0 \approx 1 \times 10^{-8}$ under simple first-order decay), the directly measured 3He inventory exceeds $n_{t,SS}$ by a factor of 1.5. For the shorter (1580 minutes; predicted $n_t/n_0 \approx 0.02$), the excess is smaller, 25% over $n_{t,SS}$. A measured inventory in excess of $n_{t,SS}$ requires that the source term (Ω: §1.4) increased during the high temperature experimental step, or the goethite sample was not effectively closed to helium loss during the low temperature permeation step (§1.4), or both. The measured diffusivity is constant at a given temperature for these high temperature measurements at the end of the experiment (Figs. 2.6, 2.7), so there is no evidence the source term (Ω) is increasing during the high temperature measurements.

The size of the two excesses scales with the duration of their respective permeation step. The 7200-minute step is associated with the larger (1.5×) excess

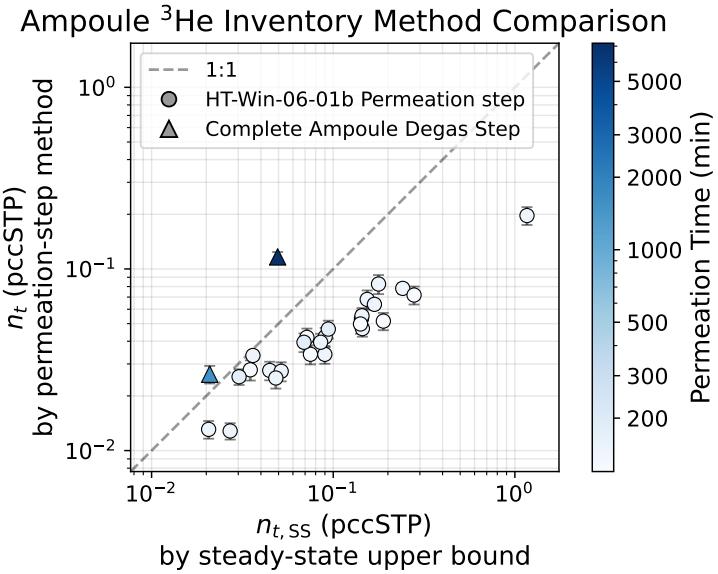


Figure 2.20: Two independent estimates of the ampoule's residual 3He inventory, n_t, following each permeation step of HT-Win-06-01b: quantified directly from the permeation step itself (*permeation step method*, y-axis, §2.3s), against the upper bound calculated independently from the preceding high-temperature step (*bounding method*, x-axis, Eq. (1.10)). Color indicates the duration of the permeation step. Circles are ordinary steps; triangles are the two "attempted complete degas" steps, held at 100°C for far longer than any other permeation step (1580 and 7200 minutes) in an attempt to drive the ampoule's 3He inventory toward zero. The dashed line is 1:1. Excepting the two complete-degas steps, n_t never exceeds $n_{t,SS}$, as required if permeation lag is genuinely bounded above by the steady-state approximation. Both complete-degas steps instead record more helium than their own $n_{t,SS}$ permits.

whereas the 1580-minute step is associated with the smaller (25%) excess. An artifact intrinsic to the first-order process of helium permeation through the ampoule itself, such as an underestimated rate constant, has no reason to strongly scale with the step duration, since both of these long steps would be expected to empty the vast majority of the helium inventory of the ampoule. That the excess does scale this way points to open system behavior in the sample. The preferred explanation is short-circuit or fast pathway diffusion (J. K. W. Lee and Aldama, 1992) of residual helium held within the interstitial space of the goethite crystallite aggregate (§2.4), recharged during the preceding high-temperature step, released slowly and with weak temperature dependence during the following extended low-temperature hold. This requires communication between a lattice or grain-boundary source and an interstitial

reservoir, a departure from the independent, non-communicating domains of a strict MDD interpretation, but it is consistent with the interstitial framework already invoked to explain the early, low-retention fraction of 3He release (§2.4). Unlike dehydration, it requires no change in the intrinsic retentivity of the crystalline lattice recorded by the bulk of the experiment.

2.2s Analytical Methods

Finite Difference Diffusion Model of Cooling Tube

One potential concern about the results of the internally measured permeabilities is that some additional helium was permeated during the cooling period after each permeation step. In order to evaluate the significance of this effect, a finite difference model was built in Python to simulate diffusion, influx, and outflux of helium across a hollow glass cylinder containing an arbitrary amount of helium gas within a helium-free chamber using the procedure described in Chapter 1 (§1.5). The glass cylinder was given the same dimensions of the ampoule used in the Win-06-01b hydrothermal experiment, and was subjected to the same heating and cooling schedule as the hydrothermal experiment's permeation steps. The tube was charged with 100 arbitrary units of helium atoms at the end of each experimental step and a steady state concentration profile was assumed for the new arbitrary charge at T = 24°C. Permeability values for 3He were assumed as the product of the 4He permeability values given by Swets et al. (1961) and the 3He/4He relative permeabilities given by Jones (1953).

To simulate cooling, a cooling rate had to be assumed. To most accurately simulate the cooling of the experimental ampoule, Watlow temperature records of the experimental ampoule during cooling steps were used to calculate the cooling rate dT/dt for discrete time intervals. A quadratic function was fit to these data points to model the cooling rate as a function of temperature with a single root at T = 24°C such that the temperature would stabilize at T = 24°C by design.

$$\frac{dT}{dt} = -\left(9.23081 \times 10^{-6}\right)(T-24)^2 - \left(7.74304 \times 10^{-4}\right)(T-24)$$

(dT/dt in °C/s, T in °C; $R^2 = 0.9633$).

Uncertainty for Diffusivity Inversions

Apparent values of D/a^2 for 3He and 4He were calculated from times and cumulative measured yields using the equations of Fechtig and Kalbitzer (1966). Lower and upper bounds for diffusivity were calculated using the upper bound for the helium contents of the ampoules before and after each step, evaluated using Eq. (1.10). Diffusivity for step i is a function of the cumulative fractional yield for step i $\Sigma^i F$ and the cumulative fractional yield of the previous step $\Sigma^{i-1} F$. The diffusivity is maximized when the difference between the fractional

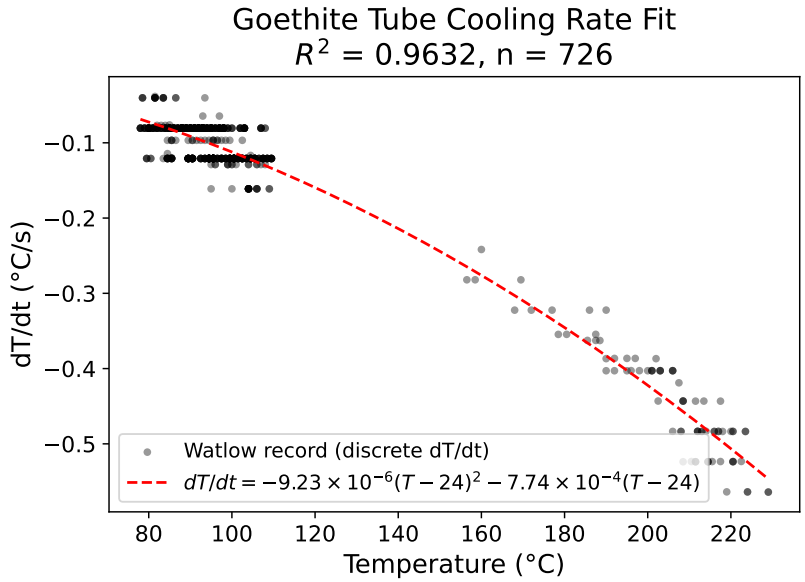


Figure 2.21: Discrete cooling rate (dT/dt) values calculated from the Watlow temperature record for the Winsor Goethite experiment, plotted against temperature, with the quadratic fit used to simulate cooling in the finite difference diffusion model overlain.

yields is largest, and minimized when the difference is smallest. This difference is maximized for the extreme case where the previous step ended with no permeation lag, and the current step ended with the maximum amount of permeation lag (${}^{i}_{SS}n_t$), whereas the opposite is true for the minimum diffusivity.

$$\begin{aligned} {}^{UB}D_\omega &= D\left(\Sigma^{i-1}F_m,\ \Sigma^{i}F_m + {}^{i}_{SS}n_t\right) \\ {}^{LB}D_\omega &= D\left(\Sigma^{i-1}F_m + {}^{i-1}_{SS}n_t,\ \Sigma^{i}F_m\right) \end{aligned} \tag{2.15}$$

For the 3He diffusivities, a linear regression was fit to the steps with cumulative 3He fractional yield ≥7% (12 steps, Fig. 2.6), yielding an R^2 statistic of 0.989. The fit's slope and intercept are used as a reference line to calculate "Delta" values for the diffusivities of 3He and 4He (Fig. 2.7). No uncertainty on the total amounts of 3He and 4He is propagated into the diffusivities because the experiment operated at low cumulative yields where this error affects all diffusivities approximately homogeneously, so it is only relevant for the y-intercept of the regression and the corresponding calculated closure temperature. The uncertainty from the intercept and slope parameters of the linear regression is significantly larger than the uncertainty related to the total yield, so it is

neglected.

Helium Isotope Ratio Uncertainty

The upper and lower bounds of the $\frac{^4He}{^3He}$ isotope ratios for each heating step are taken as the maximum and minimum $\frac{^4He}{^3He}$ isotope ratio calculated out of the 4 hypothetical end-member permeation scenarios, which would have been experienced by both helium isotope systems:

1. $n_t = 0,\ n_0 = 0$
2. $n_t = n_t^{\mathrm{UB}},\ n_0 = 0$
3. $n_t = n_t^{\mathrm{UB}},\ n_0 = n_0^{\mathrm{UB}}$
4. $n_t = 0,\ n_0 = n_0^{\mathrm{UB}}$

Intuitively, higher differential permeability between the two helium isotopes ($\delta = \frac{K_{3He}}{K_{4He}}$ here) will result in greater isotope fractionation between the measured $\left(\frac{m_4}{m_3}\right)$ and true $\left(\frac{\omega_4}{\omega_3}\right)$. Furthermore, lower absolute permeabilities will result in stronger permeation lag, additionally producing higher isotope fractionation in the measured value. The 4He permeability is thus conservatively taken to be identical to Swets et al. (1961), an anhydrous literature value which this experiment's own isothermal 3He permeation measurements show to underestimate the ampoule's true permeability (Fig. 2.19); 4He permeability itself could not be independently measured in this experiment. For the purposes of evaluating the measured isotope ratios, the relative 3He/4He permeability (δ) is taken to be 1.29, the 2σ upper bound of the value measured independently for a water-filled ampoule in Chapter 1 (§1.6): 1.10 ± 0.19 (2σ). This value of 1.29 far exceeds realistic values of the relative 3He/4He permeability, which is expected to be between 1.06 and 1.11 for the temperature range relevant to this experiment (Jones, 1953). These conservative assumptions taken together guarantee that the calculated upper and lower bounds for the isotope ratios accurately constrain the true value. The upper bounds of the ampoule contents are calculated using the steady-state approximation of Eq. (1.10) (Chapter 1): $n_{ss} = \frac{m}{\lambda t - 1}$. For the four extreme cases listed above, the following equations apply:

1. $\dfrac{\omega_4}{\omega_3} = \dfrac{m_4}{m_3}$

2. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1}}$

3. $\frac{\omega_4}{\omega_3} = \frac{m_4 + \frac{m_4}{\lambda t - 1} - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 + \frac{m_3}{\delta\lambda t - 1} - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$

4. $\frac{\omega_4}{\omega_3} = \frac{m_4 - \frac{m_4^{i-1}}{(\lambda t)^{i-1} - 1}}{m_3 - \frac{m_3^{i-1}}{\delta(\lambda t)^{i-1} - 1}}$

Where m_x^{i-1} and $(\lambda t)^{i-1}$ refer to the measured helium content and λt of the previous step, respectively. λ here applies explicitly to 4He, whereas $\delta\lambda$ applies to 3He. The upper and lower bounds of the isotope ratios are then taken to be the maximum and minimum of these four extreme cases.

2.3s Goethite Permeation Rate Constant Dataset

The in-experiment permeation rate constants used to correct for permeation lag are reported below. Table 2.6 gives the raw per-step data for all 10 isothermal sequences, and Table 2.7 gives the lambda/n_0 fit to each sequence (Fig. 2.18) that together define the Arrhenius regression of Fig. 2.19.

Table 2.6: Raw step-heat data ('Fit Lambdas' sheet, cross-checked against 'Raw Data and Corrections') for the 10 isothermal 3He permeation sequences used to determine the ampoule's apparent permeability (Fig. 2.18, Fig. 2.19). Series are separated by horizontal rules; within a series, steps are listed in the order measured.

Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
1	105.00	150	26	0.01905	0.00011
1	105.00	150	26	0.011134	0.000074
1	105.00	150	26	0.008027	0.000053
2	109.99	150	26	0.02292	0.00014
2	110.00	150	26	0.011512	0.000073
2	110.00	150	26	0.007518	0.000052
3	99.86	120	26	0.010584	0.000061
3	100.00	120	26	0.007448	0.000040
3	100.00	120	26	0.004916	0.000025
3	99.99	120	26	0.004263	0.000021
4	95.00	150	26	0.01090	0.00028
4	95.00	150	26	0.00807	0.00020
4	95.00	150	26	0.00622	0.00014
5	104.99	120	26	0.021450	0.000089
5	104.99	120	26	0.014372	0.000060
5	104.99	120	26	0.010381	0.000043
6	100.00	120	26	0.012600	0.000056
6	100.00	120	26	0.009377	0.000035
6	100.00	120	26	0.007355	0.000020
7	95.00	120	26	0.007203	0.000030
7	95.00	120	26	0.006262	0.000026
7	95.00	120	26	0.004461	0.000019
8	90.00	150	26	0.005899	0.000027
8	89.90	150	26	0.004812	0.000024
8	90.00	150	26	0.003635	0.000019
9	90.00	180	26	0.009987	0.000062
9	89.92	180	26	0.006748	0.000051
9	90.00	180	26	0.005370	0.000048

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Series	Temp (°C)	Nominal Time (min)	Cooling Time (min)	^{3}He measured (pccSTP)	$\pm 2\sigma$
10	99.99	150	26	0.010350	0.000043
10	99.98	150	26	0.006832	0.000035
10	100.00	150	26	0.005377	0.000034

Table 2.7: Fitted λ (permeation rate constant) and n_0 (residual helium at the start of the series) for each of the 10 isothermal sequences in Table 2.6, from the weighted non-linear least-squares fit of Eq. (1.9) (Chapter 1) described in §2.1s. n is the number of steps in the sequence. These are the same per-sequence fits plotted in Fig. 2.18 and used to build the Arrhenius regression in Fig. 2.19.

Series	Temp (°C)	Duration (min)	n	λ (s^{-1})	$\pm 2\sigma$	n_0 (pccSTP)	$\pm 2\sigma$
1	105.00	150	3	4.9×10^{-5}	1.4×10^{-5}	0.0521	0.0078
2	109.99	150	3	6.3×10^{-5}	1.8×10^{-5}	0.0508	0.0073
3	99.86	120	4	4.3×10^{-5}	1.3×10^{-5}	0.0377	0.0052
4	95.00	150	3	3.10×10^{-5}	2.6×10^{-6}	0.0444	0.0022
5	104.99	120	3	5.04×10^{-5}	6.2×10^{-6}	0.0696	0.0047
6	100.00	120	3	3.68×10^{-5}	4.1×10^{-6}	0.0534	0.0034
7	95.00	120	3	3.3×10^{-5}	1.5×10^{-5}	0.035	0.011
8	90.00	150	3	2.67×10^{-5}	4.8×10^{-6}	0.0279	0.0035
9	90.00	180	3	2.97×10^{-5}	9.0×10^{-6}	0.0357	0.0070
10	99.99	150	3	3.8×10^{-5}	1.2×10^{-5}	0.0351	0.0070

^{3}He Step Heat Dataset, Corrected with the Permeation Step Method

The complete corrected step-heat dataset produced by the permeation step method, using the in-experiment rate constants above (Fig. 2.19), is reported in Table 2.8, giving the central value and 2σ uncertainty for each quantity, analogous to Table 2.1 in the main text for the bounding method. 4He is not treated by the permeation step method here: 4He permeation-step signals were blank level, so no direct, step-by-step quantification of the ampoule's 4He inventory was possible, and 4He throughout this chapter is instead reduced exclusively by the bounding method of the main text (§2.2s). The permeation step dataset reported here is retained only for 3He, as the internal check on the bounding method discussed in §2.1s.

Table 2.8: ^{3}He step-heat dataset for HT-Win-06-01b under the permeation step method: measured and cumulative ^{3}He, the initial and residual ampoule content (n_0, n_t), the true step production (ω), and the apparent diffusivity $\ln(D/a^2)$, each with its 2σ uncertainty. Six rows have derived values marked with "*": these steps have measured ^{3}He < 0.1 pccSTP and are excluded as low-signal, as in Table 2.1.

Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
170	300	0.9630	0.0000	0	0	0.197	0.022	1.160	0.026	-21.910	0.045
180	463	0.5442	0.9630	0.073	0.013	0.0782	0.0049	0.550	0.016	-22.184	0.089
190	550	0.5292	1.6241	0.0195	0.0025	0.0679	0.0082	0.578	0.013	-21.976	0.085
200	500	0.5190	2.2112	0.0328	0.0054	0.0532	0.0058	0.539	0.012	-21.702	0.088
210	475	0.5770	2.7646	0.0141	0.0021	0.0552	0.0057	0.618	0.012	-21.311	0.082
220	450	0.6333	3.3799	0.0110	0.0018	0.0469	0.0046	0.669	0.012	-20.991	0.080
230	425	0.7899	4.0551	0.0059	0.0019	0.0639	0.0035	0.848	0.012	-20.514	0.066
225	908	0.7513	4.8885	0.0195	0.0024	0.0420	0.0049	0.774	0.014	-21.20	0.12
215	450	0.1415	5.6842	0.0131	0.0018	0.0278	0.0035	0.1563	0.0065	-22.02	0.32
205	520	0.0763	5.8529	0.0072*	0.0012*	0.0125*	0.0016*	0.0816*	0.0081*	-22.79*	0.83*
195	1106	0.0670	5.9491	0.00212*	0.00040*	0.0307*	0.0029*	0.0956*	0.0032*	-23.4*	1.1*
185	740	0.0227	6.0246	0.0125*	0.0016*	0.0090*	0.0011*	0.0192*	0.0020*	-24.6*	2.8*
175	920	0.0180	6.0610	0.00368*	0.00050*	0.0045*	0.0004*	0.0189*	0.0007*	-24.8*	3.8*
200	260	0.0224	6.0814	0.00065*	0.00012*	0.0095*	0.0012*	0.0313*	0.0016*	-23.0*	1.5*
210	520	0.0812	6.1081	0.0075*	0.0010*	0.0145*	0.0013*	0.0882*	0.0038*	-22.7*	1.1*
220	1167	0.5766	6.1914	0.0028	0.0014	0.0276	0.0032	0.601	0.024	-21.50	0.18
230	370	0.3577	6.7796	0.0087	0.0040	0.0422	0.0051	0.391	0.015	-20.71	0.15
240	320	0.6767	7.1563	0.0172	0.0024	0.0826	0.0099	0.742	0.016	-19.807	0.099
245	250	0.5549	7.8582	0.0213	0.0089	0.0516	0.0056	0.585	0.015	-19.71	0.12
250	210	0.6934	8.4743	0.0108	0.0045	0.0718	0.0082	0.754	0.014	-19.20	0.10
245	270	0.4659	9.2086	0.0248	0.0035	0.0498	0.0058	0.491	0.010	-19.81	0.20

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Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	n_0 (pccSTP)	$\pm 2\sigma$	n_t (pccSTP)	$\pm 2\sigma$	ω (pccSTP)	$\pm 2\sigma$	$\ln(D/a^2)$	$\pm 2\sigma$
240	320	0.2857	9.7206	0.0207	0.0028	0.0339	0.0041	0.2989	0.0066	-20.43	0.31
235	370	0.2199	10.0357	0.0166	0.0022	0.0273	0.0033	0.2307	0.0052	-20.81	0.40
230	420	0.1400	10.2736	0.0132	0.0018	0.0255	0.0025	0.1523	0.0040	-21.33	0.60
225	652	0.1494	10.4279	0.0126	0.0016	0.0131	0.0015	0.1499	0.0034	-21.77	0.75
230	500	0.1550	10.5908	0.0007	0.0017	0.0128	0.0013	0.1671	0.0034	-21.38	0.46
235	450	0.1951	10.7582	0.00257	0.00042	0.0334	0.0018	0.2259	0.0033	-20.96	0.36
240	400	0.2425	10.9638	0.0069	0.0014	0.0251	0.0032	0.261	0.012	-20.67	0.32
245	350	0.3150	11.2328	0.0058	0.0010	0.0395	0.0046	0.3487	0.0094	-20.22	0.24
250	300	0.3716	11.5689	0.0165	0.0022	0.0467	0.0052	0.4018	0.0096	-19.89	0.21
250	300	0.3640	11.9627	0.0243	0.0044	0.0339	0.0038	0.3735	0.0080	-19.93	0.24
250	300	0.3452	12.3490	0.0113	0.0016	0.0395	0.0045	0.3733	0.0079	-19.90	0.24
250	10250	8.4829	12.7167	0.0191	0.0025	0.1165	0.0074	8.58	0.13	-19.96	0.19
240	785	0.2261	21.2187	2.9×10^{-9}	6.8×10^{-9}	0.0263	0.0029	0.252	0.022	-20.66	0.58
Total			146.1483								

Permeation Step Approach and Concordance with Bounding Approach

Beyond the ampoule's residual helium content itself (§2.1s), the two methods' derived apparent diffusivities also agree closely. Fig. 2.22 compares $\ln(D/a^2)$ from the permeation-step method against the bounding method's central value for every step in the experiment, with the bounding method's own lower- and upper-bound spread on one axis and the permeation step method's propagated uncertainty on the other. The two diffusivity datasets track the 1:1 line within uncertainty across the full range of the experiment. Fitting the permeation-step method's own diffusivities with the same Stage 1 procedure used for the bounding method in the main text (Fig. 2.23) yields an activation energy of 157.0±4.8 kJ/mol (R^2=0.991, n=12) and a closure temperature of 97.5±7.8°C, indistinguishable from the bounding method's own 157.1±5.3 kJ/mol and 98.1±8.6°C (Fig. 2.6, main text).

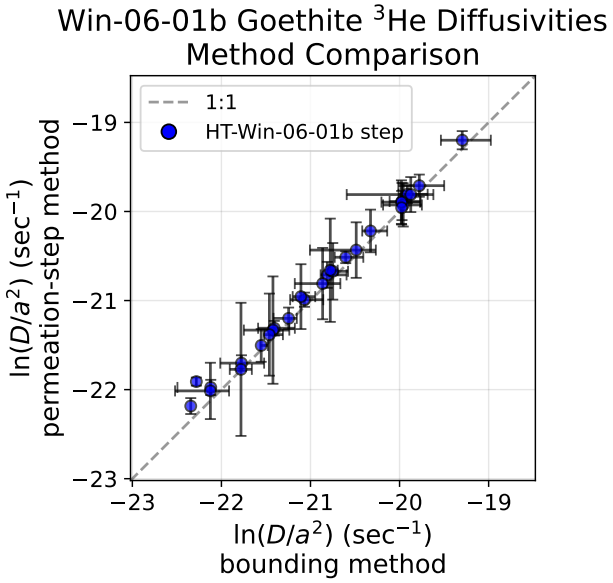


Figure 2.22: Apparent ^{3}He diffusivities for HT-Win-06-01b from the permeation-step method (y-axis) against the bounding method's central value (x-axis, Eq. (1.10)), with the bounding method's lower/upper-bound spread and the permeation step method's own propagated uncertainty plotted on their respective axes. The dashed line is 1:1.

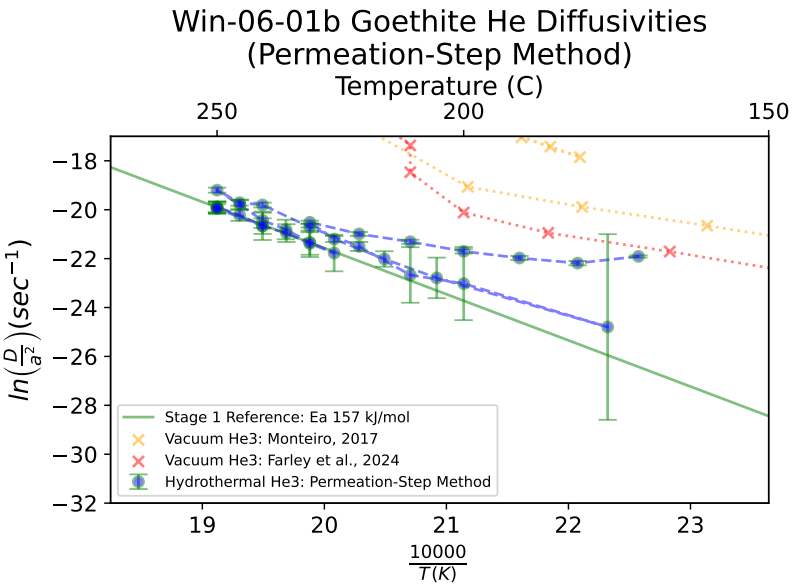


Figure 2.23: Analogue of Fig. 2.6 (main text) using the permeation-step method's own ^{3}He diffusivities and its own Stage 1 linear regression, rather than the bounding method's. The resulting activation energy and closure temperature (157.0±4.8 kJ/mol; 97.5±7.8°C) agree closely with the bounding method's own (157.1±5.3 kJ/mol; 98.1±8.6°C).

2.4s Kinetic mechanisms and the limits of a single heating schedule

Solid-state rate laws separate the temperature dependence of the rate from its dependence on the extent of reaction (Vyazovkin et al., 2011):

$$\frac{d\alpha}{dt} = A \exp\left(-\frac{E_a}{RT}\right) h(\alpha), \tag{2.16}$$

where α is the fraction of goethite converted. The reaction model $h(\alpha)$ expresses the mechanism: it describes how the rate changes as conversion proceeds, which depends on where and how the reaction advances through a grain. For first-order kinetics (F1), grains convert independently once nucleated, so the rate is proportional to the goethite remaining, $h = 1 - \alpha$. For a reaction front moving inward from the grain surface (R2, R3), the rate is proportional to the shrinking area of the unreacted core. For random nucleation followed by growth (A2, A3), the rate first accelerates and then slows. For any heating schedule, each rate law gives α as a function of a single variable,

$$x = A \sum_j \Delta t_j \exp\left(-\frac{E_a}{RT_j}\right), \tag{2.17}$$

the Arrhenius-weighted time delivered by the steps up to the current one; for a single isothermal step, $x = kt$. We tested the mechanisms in Table 2.9, including spherical diffusion on the same footing: for diffusion, x is the Fourier number Dt/a^2 with D/a^2 Arrhenius in T. Under vacuum, the driving-force factor $1 - \exp(-\Delta\mu/RT)$ is unity to better than 10^{-9}, so x depends only on the heating schedule. Diffusion and conversion then differ only in the shape of $\alpha(x)$ and in the fitted E_a.

Table 2.9: Kinetic models: reaction model $h(\alpha)$ and conversion as a function of the Arrhenius-weighted time, $\alpha(x)$.

Model	$h(\alpha)$	$\alpha(x)$
R2 contracting cylinder	$2(1-\alpha)^{1/2}$	$1-(1-x)^2$
R3 contracting sphere	$3(1-\alpha)^{2/3}$	$1-(1-x)^3$
F1 first order	$1-\alpha$	$1-e^{-x}$
A2, A3 Avrami ($n = 2, 3$)	$n(1-\alpha)[-\ln(1-\alpha)]^{1-1/n}$	$1-e^{-x^n}$
D3 sphere diffusion	—	$1-\frac{6}{\pi^2}\sum_{m=1}^{\infty}\frac{e^{-m^2\pi^2 x}}{m^2}$

Fits to the vacuum data

Table 2.10 gives the fit of each model to the cumulative KAF ^{3}He release within the fitting window. It also gives the misfit of the resulting out-of-sample prediction of HDM. F1 gives the best fit and the best prediction. R3 and R2 are close behind, so we describe the mechanism as nucleation- or interface-controlled rather than claim first order specifically. The Avrami fits reach the lower bound of E_a and fail on both data sets. Spherical diffusion with $\phi = 0$ fits KAF 4.4 times worse than F1 and cannot produce the $F \approx 0.89$ plateau. To compare diffusion on the same footing, we also fit D3 with ϕ fixed at the F1 value and with ϕ free, over the same 18 KAF steps. The imposed plateau improves the KAF misfit to 0.026 and 0.019, which still ranks below F1, R3 and R2, and the out-of-sample prediction of HDM does not improve (rms 0.056 and 0.052, against 0.018 for F1). These fits also require E_a of 243–289 kJ mol^{-1}. The rate factor for the trap-isolated steps is fixed at $s = 1$ in all three D3 fits, because a diffusive process should not respond to the water pressure at all.

The fitting window excludes the low-temperature steps, where release is dominated by diffusion from goethite, and the plateau, where it is dominated by diffusion from hematite; the rms misfits are computed over the same steps. Repeating the fits with every step included gives the same result: F1 remains the best model ($E_a = 122.7$ kJ mol^{-1}, $\phi = 0.112$, rms 0.0073 for KAF and

0.026 for the HDM prediction), followed by R3 (99.4 kJ mol^{-1}; 0.0097, 0.030) and R2 (87.9 kJ mol^{-1}; 0.012, 0.033), with D3 at 177.4 kJ mol^{-1} (0.063, 0.062; 0.022 and 0.042 with ϕ fixed at the F1 value).

Table 2.10: Kinetic and Diffusive Model fits to the vacuum experiments (from H. Monteiro (2017) and Farley et al. (2024)). All rms values are over the same 18 KAF and 19 HDM steps. D3 is fitted with ϕ fixed at 0, fixed at the F1 value, and free.

Model	E_a	$\ln A$	ϕ	rms KAF	rms HDM
R2	92.8	10.63	0.149	0.0133	0.0291
R3	103.1	12.73	0.136	0.0114	0.0240
F1	120.8	18.15	0.101	0.0087	0.0176
A2	25.5	−4.73	0.000	0.0212	0.1906
A3	20.0[a]	−5.85	0.203	0.0433	0.2342
D3, $\phi = 0$	195.6	32.43	0.000	0.0385	0.0484
D3, $\phi = \phi_{\mathrm{F1}}$	242.8	43.91	0.101	0.0258	0.0557
D3, ϕ free	288.5	55.04	0.184	0.0186	0.0520

E_a in kJ mol^{-1}, A in s^{-1}. [a]At the lower bound of the fit.

2.5s Equilibrium Thermodynamics of Goethite Dehydroxylation

Gibbs energies of the solids. The Gibbs energy of a phase at (T, P) follows from integrating $dG = -S\,dT + V\,dP$ from the reference state $(T_r, P_r) = (298.15\ \mathrm{K}, 1\ \mathrm{bar})$, first in temperature at P_r and then in pressure at T. At P_r, $H(T) = H(T_r) + \int_{T_r}^{T} C_p\,dT$ and $S(T) = S(T_r) + \int_{T_r}^{T} (C_p/T)\,dT$, so $G(T, P_r) = H(T) - TS(T)$. Adding $\int_{P_r}^{P} V\,dP$, and treating the solids as incompressible with no thermal expansion, gives

$$G_i(T,P) = \Delta_f H_i^0 + \int_{T_r}^{T} C_{p,i}\,dT \;-\; T\left[S_i^0 + \int_{T_r}^{T} \frac{C_{p,i}}{T}\,dT\right] \;+\; V_i\,(P - P_r), \quad (2.18)$$

where $H(T_r) = \Delta_f H_i^0$ and $S(T_r) = S_i^0$. This is the "apparent" Gibbs energy of formation (Berman, 1988); the element-entropy terms that distinguish it from the true Gibbs energy of formation cancel in any balanced reaction. Compressibility and thermal expansion change V_i by $\lesssim 1\%$ over the conditions considered and are neglected.

Goethite heat capacity uses the Berman–Brown form (Berman and T. H. Brown, 1985) refined by Majzlan et al. (2003), $C_p = k_0 + k_1 T^{-1/2} + k_2 T^{-2} + k_3 T^{-3}$. Hematite uses the same polynomial with the coefficients of Berman (1988), which Majzlan et al. (2003) held fixed in their refinement. Both heat capacities are integrated analytically.

Water. Water is evaluated with the IAPWS-95 equation of state (Wagner and Pruß, 2002), implemented in CoolProp (Bell et al., 2014), which is valid for liquid, vapor and supercritical water to 1 GPa. In the same apparent convention,

$$G_{\mathrm{w}}(T,P) = \Delta_f H^0_{\mathrm{w}} + [h(T,P) - h(T_r,P_r)] - T\left[S^0_{\mathrm{w}} + s(T,P) - s(T_r,P_r)\right], \tag{2.19}$$

where h and s are the IAPWS-95 molar enthalpy and entropy and the reference state is liquid water at T_r, P_r. Table 2.11 lists all input data. We use the enthalpies and entropies of Majzlan et al. (2003) refined by mathematical programming against the phase-equilibrium brackets of Voigt and Will (1981).

Table 2.11: Thermodynamic data used for Eq. (2.10).

	Goethite (α-FeOOH)	Hematite (α-Fe_2O_3)
$\Delta_f H^0$ (kJ mol^{-1})	-561.4	-825.8
S^0 (J mol^{-1} K^{-1})	59.5	87.4
V^0 (cm^3 mol^{-1})	20.881	30.274
C_p coefficients	$k_0 = 133.8765$	$k_0 = 146.86$
	$k_1 = -738.5577$	$k_1 = 0$
	$k_2 = -2.177657 \times 10^6$	$k_2 = -5.5768 \times 10^6$
	$k_3 = 2.165124 \times 10^8$	$k_3 = 5.2563 \times 10^8$
C_p(298 K) (J mol^{-1} K^{-1})	74.8	104.0
$\gamma_{\mathrm{hydrated}}$ (J m^{-2})	0.60 ± 0.10	0.75 ± 0.16
$\gamma_{\mathrm{anhydrous}}$ (J m^{-2})	0.91 ± 0.09	1.90 ± 0.30

Enthalpy, entropy, goethite volume and C_p: Majzlan et al. (2003); hematite volume and C_p: Berman (1988); surface energies: Navrotsky et al. (2008). Water: $\Delta_f H^0_{\mathrm{w}} = -285.83$ kJ mol^{-1}, $S^0_{\mathrm{w}} = 69.95$ J mol^{-1} K^{-1} (J. D. Cox et al., 1989).

As a check, at 298.15 K and 1 bar with liquid water, Eqs. (2.10)–(2.19) give $\Delta G_r = -0.26$ kJ: hematite plus liquid water is favored by only 0.13 kJ per mole of FeOOH. This is well inside the ± 1.4 kJ mol^{-1} uncertainty that Majzlan et al. (2003) assign to the goethite–hematite free-energy difference; the two assemblages are essentially balanced at ambient conditions. We carry this uncertainty (± 2.8 kJ per reaction as written) as bounding curves on the phase diagram.

Surface energy. A crystal of finite size carries an excess free energy at its surface in addition to the free energy of its bulk (Gibbs, 1878). For a population of particles of fixed size, this excess is γA, where γ is the surface energy (J m^{-2}) and A the total surface area. Both the bulk and surface terms scale with the amount of material at fixed particle size, so the surface contribution can be written per mole, $G = G_{\mathrm{bulk}} + \gamma A_m$, with A_m the surface area per mole. A_m

grows as $1/d$, so the term is negligible for coarse crystals and significant for nanoparticles (Navrotsky et al., 2008). For the reaction as written (one mole of hematite from two moles of goethite),

$$\Delta G_{\mathrm{surf}} = \gamma_{\mathrm{hem}} A_{m,\mathrm{hem}} - 2\,\gamma_{\mathrm{goe}} A_{m,\mathrm{goe}}, \tag{2.20}$$

where $A_{m,\mathrm{hem}}$ is per mole of Fe_2O_3 and $A_{m,\mathrm{goe}}$ per mole of FeOOH.

A size must be chosen for each phase. We assume pseudomorphic dehydroxylation, in which hematite inherits the external envelope of the parent goethite, so that both phases share the same external surface area per mole of Fe. This is also the basis of the stability diagrams of Navrotsky et al. (2008). For spherical goethite grains of diameter d this shared area is

$$A_{\mathrm{Fe}} = \frac{6\,V_{\mathrm{goe}}}{d} \quad \text{per mole of Fe.} \tag{2.21}$$

Goethite has one Fe per formula unit, so $A_{m,\mathrm{goe}} = A_{\mathrm{Fe}}$. Hematite has two, so $A_{m,\mathrm{hem}} = 2A_{\mathrm{Fe}}$. Substituting into Eq. (2.20),

$$\Delta G_{\mathrm{surf}} = \gamma_{\mathrm{hem}}\,(2A_{\mathrm{Fe}}) - 2\,\gamma_{\mathrm{goe}}\,A_{\mathrm{Fe}} = 2A_{\mathrm{Fe}}\,(\gamma_{\mathrm{hem}} - \gamma_{\mathrm{goe}})\,. \tag{2.22}$$

The factor of two counts the two moles of Fe that pass through the reaction. It multiplies both terms because both phases carry the same area per Fe. For $d = 10$ nm, $A_{\mathrm{Fe}} = 6 \times 20.881 \times 10^{-6}/10^{-8} = 1.253 \times 10^4$ m^2 mol^{-1}, and with hydrated surface energies $\Delta G_{\mathrm{surf}} = 2 \times 1.253 \times 10^4 \times (0.75 - 0.60) = +3.76$ kJ. The corresponding values are +0.38 kJ at 100 nm and +0.04 kJ at 1 µm. A positive ΔG_{surf} stabilizes goethite. If the two phases are instead compared at equal particle diameter, the same hydrated surface energies slightly destabilize fine goethite (−1.41 kJ at 10 nm). The choice affects only the stability discussion; the kinetic analysis uses bulk driving forces.

We use hydrated surface energies. They are referenced to liquid water, so they apply wherever the surfaces are in equilibrium with water at or near saturation, which includes all of the hydrothermal experiments. Anhydrous surface energies apply only to surfaces that have lost their chemisorbed water, i.e. under vacuum.

Phase diagram. The phase diagram (Fig. 2.12) is computed by solving $\Delta G_r(T, P) = 0$ (Eqs. (2.10)–(2.22)) numerically for $P = P_{\mathrm{H_2O}}$ at each temperature, using Brent's method on $\log_{10} P$ between 10^{-12} and 10^4 bar, with the

solids at the fluid pressure. Where no root exists below the melting curve of ice, the curve is truncated.

Driving force and the role of water. The quantity that matters for kinetics is the driving force for hematite formation, $\Delta\mu \equiv -\Delta G_r/N_A$ (per mole of Fe_2O_3). Water controls $\Delta\mu$ through its own chemical potential, because it is a product of reaction (2.9). G_w in Eq. (2.10) is the chemical potential of H_2O at the prevailing water fugacity. It depends on the partial pressure of water, not on the total pressure. An inert gas, or any other mechanical load, acts only through the solid-volume term $\Delta V_s(P - P_r)$, which contributes $-1.15\ \mathrm{J\,bar^{-1}}$ and slightly favors the denser hematite. At 40 bar that is 46 J, three orders of magnitude smaller than the effect of 40 bar of water. The water-buffered step-heating technique therefore stabilizes goethite primarily by reducing the driving force of the reaction. Surface hydration is a second-order correction that matters only for nanometer grains (Eq. (2.22)).

2.6s Multiple Diffusion Domain and Thermal History Modeling

Forward modeling radiogenic ^{4}He in nature and in the laboratory

Each diffusion domain produces ^{4}He uniformly at rate P and loses it by diffusion through a zero-concentration boundary from the time of crystallization, t_{form}. The concentration is expanded in the eigenfunctions of the sphere, $C = \sum_k A_k(t)\sin(k\pi x)/x$, where x is radial position normalized to the domain radius and A_k is the amplitude of the k-th mode (Carslaw and Jaeger, 1959). Each mode amplitude A_k evolves as

$$\frac{\mathrm{d}A_k}{\mathrm{d}t} = -k^2\pi^2\frac{D}{a^2}A_k + c_k P, \qquad c_k = \frac{2(-1)^{k+1}}{k\pi}, \tag{2.23}$$

where the amplitude A_k of each mode k decays by diffusion and grows by production P. For the initial condition, at $t = 0$, all amplitudes are zero. Then, for a discrete timestep Δt at constant temperature, each mode is updated as

$$A_k \leftarrow A_k e^{-\lambda_k} + c_k P\,\Delta t\,\frac{1 - e^{-\lambda_k}}{\lambda_k}, \qquad \lambda_k = k^2\pi^2\frac{D}{a^2}\Delta t. \tag{2.24}$$

Because the eigenfunctions do not depend on D, the present-day amplitudes $A_{k,j}$ carry each domain's non-uniform natural ^{4}He profile into the laboratory, where each mode decays as $\exp(-k^2\pi^2\tau)$. The ^{4}He remaining in domain j when the reference domain has reached laboratory time τ is therefore

$$N_j(\tau) = 3\sum_k A_{k,j}\,\frac{(-1)^{k+1}}{k\pi}\exp\left[-k^2\pi^2\tau\left(\frac{r_j}{r_0}\right)^{-2}\right], \tag{2.25}$$

and the predicted step ratio is

$$\frac{R_{\mathrm{step},i}}{R_{\mathrm{bulk}}} = \frac{\Delta N_{4,i}/N_{4,\mathrm{tot}}}{\Delta F_i^{\mathrm{obs}}}, \qquad \Delta N_{4,i} = \sum_j f_j \left[N_j(\tau_{0,i-1}) - N_j(\tau_{0,i})\right], \tag{2.26}$$

where $N_{4,\mathrm{tot}} = \sum_j f_j N_j(0)$ includes the gas released at fusion. The reference times $\tau_{0,i}$ are solved from the *measured* cumulative ^{3}He by inverting Eq. (2.5), so the prediction does not depend on the heating schedule or on step-to-step scatter in the kinetics, and no uniform ^{4}He distribution is assumed at the start of the experiment. Taking U and Th to be distributed like the ^{3}He, the bulk He age is $t_{\mathrm{He}} = \sum_j f_j N_j(0)/P$.

Multiple diffusion domain fits for $N = 1$–5

Figure 2.24 shows the fitted fractional and cumulative ^{3}He yields, and the Arrhenius and delta plots they imply, for one to five domains. The fits for all N are summarized in Table 2.3 of the main text.

Two-domain alternative

The two-domain structure (2.2% of the ^{3}He at $r/r_0 = 5.4 \times 10^{-3}$, 97.8% at 1.05) fits the ^{4}He/^{3}He spectrum better than the adopted four-domain structure: a burial path with crystallization at 195 Ma and a peak of ∼66 °C at ∼100 Ma gives $\chi^2_\nu \approx 1.3$. However, this structure reproduces the ^{3}He Arrhenius array poorly ($\chi^2_\nu = 3.3$; Fig. 2.24), so we do not adopt it.

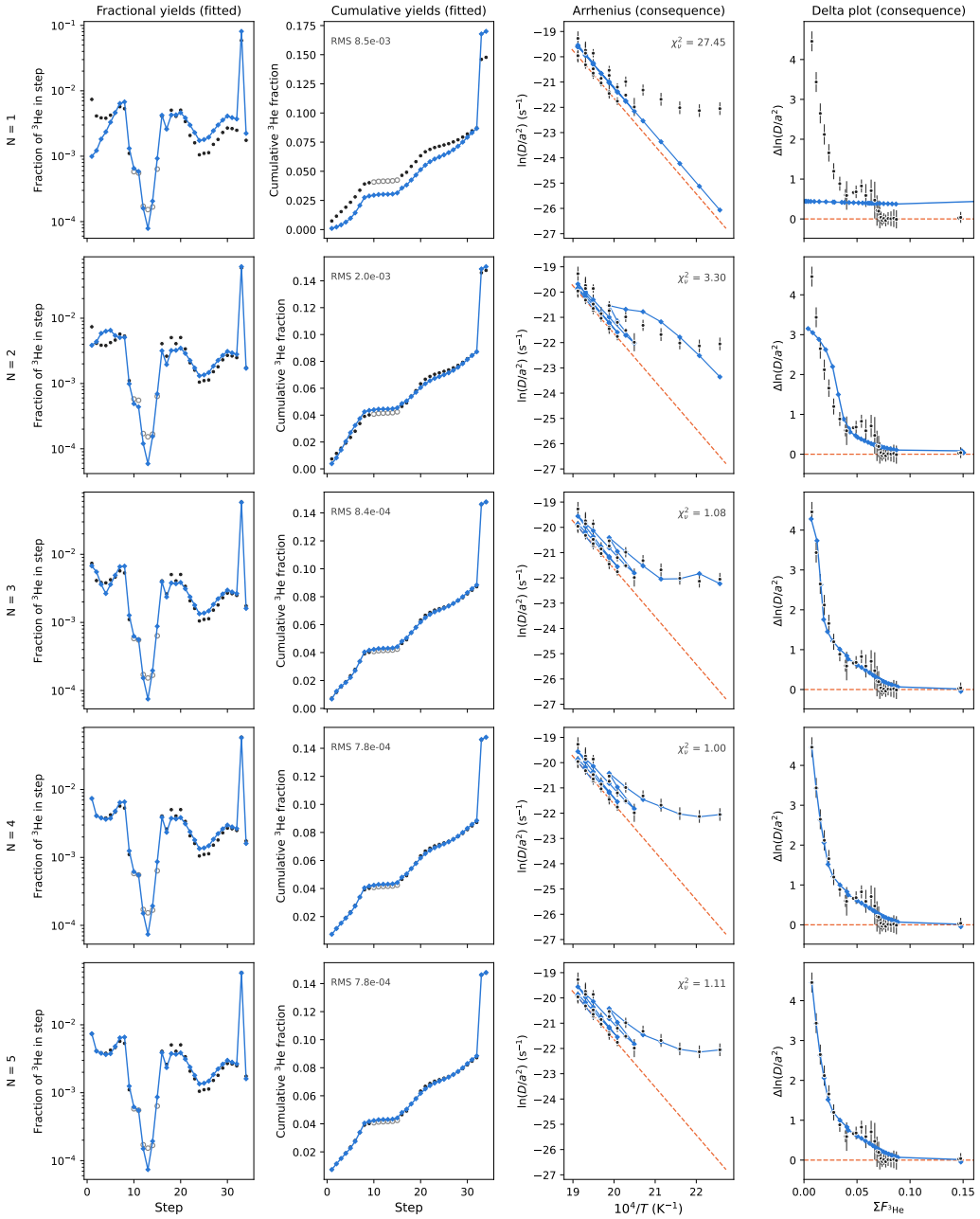


Figure 2.24: Multiple diffusion domain fits to the ^{3}He release of HT-Win-06-01b for $N = 1$–5 domains (rows). Columns: fitted fractional and cumulative yields, and the Arrhenius and delta plots implied by each fit. Hollow grey symbols are the six low-signal steps, which are not fitted and are omitted from the Arrhenius and delta plots.

Chapter 3

(U-TH-SM)/HE SULFIDE GEOCHRONOLOGY

Sulfide mineralization is a near-ubiquitous process in the formation of ore deposits of valuable chalcophile and siderophile elements. Helium isotope studies of sulfides have been applied to infer the origins of mineralizing hydrothermal fluids, and some pioneering studies have applied the U-Th/He method to date sulfides, especially pyrite (O. Yakubovich et al., 2020; O. Yakubovich et al., 2021). Yet, the interpretation of helium isotope data or U-Th/He ages requires an understanding of the conditions under which helium in sulfides behaves as an open or closed system, as well as the mechanisms of helium loss, a process which is poorly understood in sulfides. Furthermore, sulfides systematically reject U-Th and only produce appreciable radiogenic helium when they contain U-Th-bearing inclusions, further complicating the interpretation of U-Th/He data in sulfides. We explore the systematics of U-Th/He in sulfides through two lenses: 1) an example application of the sulfide U-Th/He geochronometer to two test cases, and 2) helium step heat experiments of sulfide phases. We report pyrite/arsenopyrite U-Th/He ages for Au-ores in the Permo-Triassic Gympie formation of Queensland, Australia, and the Archean Kidd Creek VMS deposit of Ontario, Canada. The Gympie arsenopyrite yields a mean age of 258 ± 35 Ma and spans the range between reported Ar/Ar and K/Ar ages of the country rock and the mineralized quartz veins (Holcombe et al., 1997; Little et al., 1995), suggesting the sulfides predate and were variably reset by the primary mineralization event. Pyrite/arsenopyrite from the 2.7 Ga Kidd Creek VMS deposit of Ontario contain large apatite inclusions ($r \leq 30$ μm) and span a range of ages from 465–1600 Ma. Although one aliquot can be explained as a perfectly retentive 2.7 Ga sulfide host of apatite inclusions that record a ∼380 Ma cooling age (in agreement with the apatite ages of Sturrock et al. (2024)), the remaining aliquots are too young to be explained by a closed sulfide host and instead

require that the sulfides lost significant helium over geologic time.

Helium step heat experiments in sulfides are challenging because sulfides readily react/decompose in the high-vacuum, high-temperature conditions of the experiments, producing results that reflect phase decomposition/reaction kinetics that are not geologically relevant. We address this challenge by performing helium step heat experiments at temperatures up to 600°C on natural single-crystal pyrite, arsenopyrite, and galena implanted with uniform 3He through proton bombardment sealed in helium-permeable silica glass tubes under high sulfur (± arsenic) fugacity to stabilize each phase against decomposition during analysis. We were unsuccessful in extending this method to stabilize chalcopyrite, chalcocite, or marcasite during step heat experiments: the helium retentivity of these phases in nature remains unclear. Retrograde-prograde thermal cycles on sulfur-stabilized galena support a Multiple Diffusion Domain (MDD) helium loss mechanism with an activation energy around 174.8 kJ/mol, but with relatively low helium retentivity that makes it unlikely to be useful as a target for helium isotope studies or U-Th/He geochronology. Comparisons of helium step heat data of sulfur-stabilized pristine and deformed pyrite point towards a multipath diffusion model for deformed pyrites, in which dislocations and fine fractures serve as high activation energy (300.6 ± 7.3 kJ/mol) fast pathways, with bulk volume diffusion following a lower activation energy that is constrained by step heats on pristine pyrite samples: 199.1 ± 3.5 kJ/mol. Helium step heat data of arsenopyrite are complex, potentially complicated by exchange of sulfur and arsenic with the vapor phase, but qualitatively demonstrate very high helium retentivity. These experiments demonstrate that pyrite is similarly helium-retentive to monazite-structure phosphates and is more retentive of helium than natural zircon, apatite, hematite, or goethite. Arsenopyrite is less retentive, more similar to the most helium-retentive zircon.

Author contributions. This chapter is being prepared for publication with co-authors J. Treffkorn and K. A. Farley. **B.J.T.:** conceptualization, methodology, investigation, formal analysis, visualization, writing – original draft. **J.T.:** methodology, investiga-

tion. **K.A.F.:** conceptualization, resources, supervision, funding acquisition, writing – review and editing. Generative AI was used as described in the front matter.

3.1 Introduction

Sulfide minerals are nearly ubiquitous in the formation histories of economic ore deposits, and indeed are themselves the primary repositories for many base and precious metals in the crust. The question of when and how they formed is central to understanding ore-forming processes, crustal fluid circulation, and the tectonic contexts in which economic deposits develop. Despite decades of study, the timing of sulfide mineralization remains contested or poorly resolved across a wide range of deposit types, from shallow epithermal systems to deep, high temperature magmatic deposits in mafic intrusions. This uncertainty reflects both the genuine complexity of ore-forming systems (which are frequently multistage, hydrothermal, and overprinted) and the limitations of the geochronological tools available for dating sulfide phases directly.

An Alternative and Complement to Re-Os Dates in Sulfide Geochronology

The rhenium-osmium isotope system is the most widely applied method for direct radiometric dating of sulfide minerals, exploiting the relatively high Re concentrations that pyrite, molybdenite, arsenopyrite, and pyrrhotite can accommodate. Re-Os geochronology has produced geologically meaningful ages across a wide range of sulfide-bearing deposit types, and remains indispensable for directly dating ore-forming events (M. Chen et al., 2015; Hnatyshin et al., 2026; Schoenberg et al., 1999; Ostendorf et al., 2019; Selby et al., 2002). Nevertheless, the Re-Os system is vulnerable to a set of well-documented failure modes, and a growing number of published ages exhibit scatter, implausible geologic interpretations, or internal inconsistencies that motivate the development of independent dating methods (Stein et al., 2003; Suzuki et al., 2001; Y. Li et al., 2025; Lyu et al., 2020).

More practically, the highly specialized and costly n-TIMS method for Re-Os dating restricts its application to a select few labs, limiting the scale of its application (Ackerman, 2025). Furthermore, Re-Os dating of sulfides (other than the rare mineral molybdenite) requires gram-quantities (Qi et al., 2010) and is therefore poorly suited to resolve distinct mineralization episodes in multistage systems.

For multistage systems in particular, the co-application of independent radiochronometers is often useful to distinguish the timings for hydrothermal alteration from uplift, cooling, exhumation, and potentially earlier emplacement of host rocks. For example, Selby et al. (2002) demonstrate that Re-Os molybdenite is more likely to accurately record the age of sulfide mineralization in the Tintina Gold Belt, Alaska, relative to the comparatively thermally-sensitive Ar-Ar mica ages. However, molybdenite is far from a ubiquitous phase in sulfide-bearing ores, unlike pyrite. A pyrite U-Th/He geochronometer would expand such opportunities to co-apply radiochronometers that interrogate processes operating at different times and temperatures to a much more diverse and expansive array of locales that experienced complex histories. Similar geochronometers developed for other sulfide phases such as arsenopyrite, galena, and sphalerite may offer the opportunity to disentangle multiple episodes of sulfide mineralization and alteration within an ore body, allowing geologists to better understand the sequence of events that lead to economic ore formation.

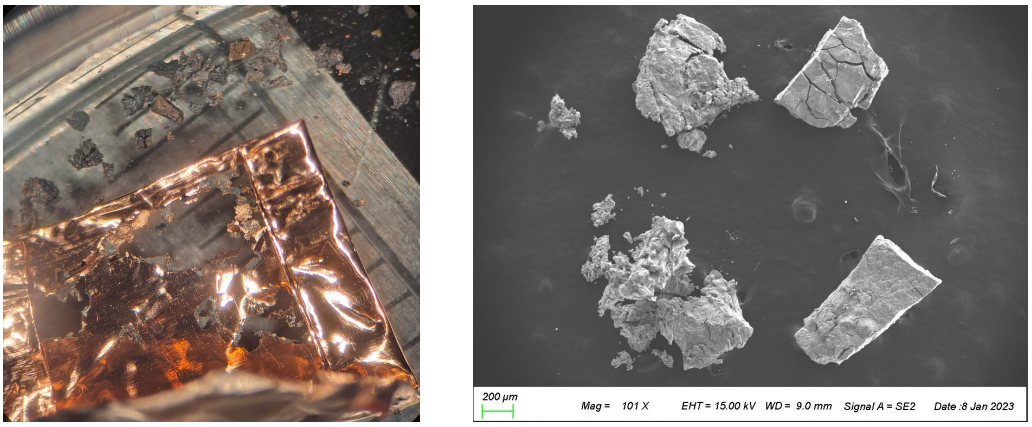
One such example of an opportunity for applying U-Th/He sulfide geochronology is in the world's largest known platinum-group element deposit, the Bushveld Complex of South Africa, which hosts the Merensky Reef and Platreef PGE deposits in the sulfide-bearing upper critical zone, for which geochronometers yield a surprising diversity of ages: zircon U/Pb ages vary from ≈2057-2060 Ma, while rutile ages are significantly younger at ≈2053 Ma (Scoates et al., 2021), yet Re-Os systematics show evidence for multiple sulfide mineralization events within the same horizon (Schoenberg et al., 1999). The Re-Os and Rb-Sr isochron ages they report are unable to resolve these distinct mineralization events from the zircon U/Pb emplacement age. Unlike Re-Os dating, the U-Th/He system produces distinct ages for every dated aliquot, potentially offering the opportunity to resolve the timings of prolonged volatile activity or late hydrothermal Re-Os disturbance.

Similarly, at the Norilsk-Talnakh deposits of the Siberian Traps, the ore-bearing intrusions can be correlated with a specific transition in flood basalt source and eruptive style (Yao and Mungall, 2021), yet only three of many differentiated intrusions in the region are economic, and the precise timing and mechanism of sulfide saturation relative to trap emplacement are not fully understood. In the 1.1 Ga Sonju Lake Intrusion of Minnesota, USA, stable isotopic evidence suggests that meso/epithermal fluids reacted with primary

magmatic sulfides, raising questions about when redistribution occurred and how much metal was moved (Park et al., 2004). These are deposits where independent thermochronometric constraints capable of distinguishing primary crystallization from later hydrothermal reworking are needed.

Stabilizing Sulfides for Helium Step Heat Experiments

Our earliest step heat experiments on sulfides revealed that vacuum experiments produced sudden, profound, irreversible increases in apparent diffusivity at temperatures above 300 °C, similar to the behavior observed of dehydrating goethite (Farley et al., 2024). Such experiments occasionally displayed obvious signs of reaction with the copper packet that housed the pyrite grains within the heating cell (Figure 3.1).



(a) (b)

Figure 3.1: Evidence of pyrite breakdown during early vacuum step heat experiments. (a) A copper packet recovered after a vacuum step heat experiment, corroded and perforated where it reacted with the enclosed, desulfurized pyrite grains; dark, brittle reaction residues are scattered on the surrounding surface. (b) Secondary-electron image of copper packet residues formed from reaction with sulfur emitted from pyrite during a vacuum step heat experiment, showing pervasive cracking and fragmentation. This step heat experiment only went as hot as 315 °C, yet decomposition was pervasive. This experiment was associated with a dramatic spike in helium release after 300 °C. These experiments in vacuum clearly interrogate the decomposition kinetics, motivating another approach to stabilize sulfides for step heat experiments.

Figure 3.2 shows the sulfur fugacities required to keep pyrite stable over the temperature range of our step heat experiments. Encapsulating pyrite in silica glass tubes using the methods developed in Chapters 1 and 2 with sulfur in sufficient excess to maintain a sulfur liquid phase controls the sulfur fugacity,

stabilizing the pyrite against thermal decomposition.

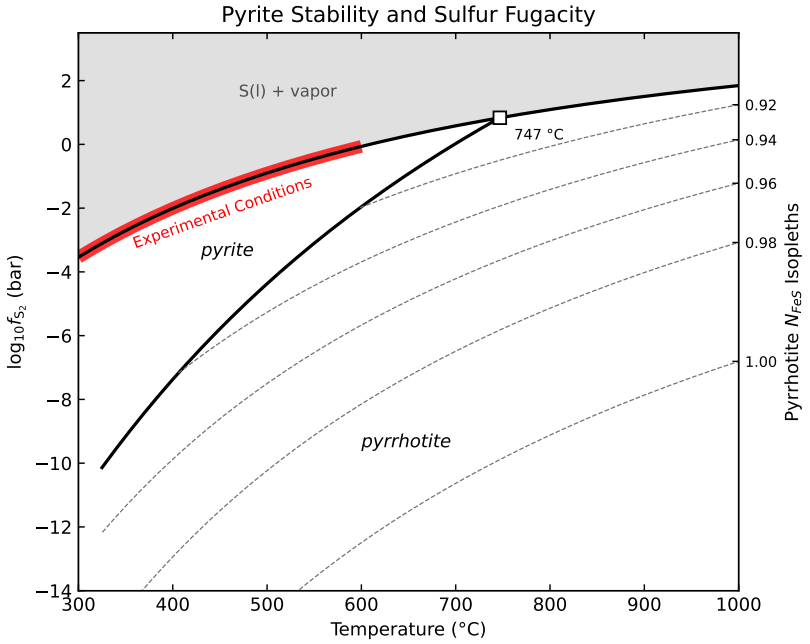


Figure 3.2: Stability of pyrite in $\log_{10} f_{S_2}$–temperature space, figure adapted from Toulmin and Barton (1964). The upper black curve is the sulfur fugacity in equilibrium with pure liquid sulfur, computed from NASA Glenn thermodynamic polynomials (McBride et al., 2002); the grey field above it is inaccessible. The lower black curve is the pyrite–pyrrhotite univariant boundary of Toulmin and Barton (1964), which meets liquid-sulfur saturation at an invariant point near 747 °C. Dashed curves are pyrrhotite composition isopleths from the Toulmin and Barton (1964) pyrrhotite equation, labelled on the right-hand axis by N_{FeS}, the mole fraction of FeS in the FeS–S_2 system. The thick red line marks the 300–600 °C range of our step heat experiments for pyrite along liquid-sulfur saturation, well within pyrite's stability field. Conditions in vacuum experiments (i.e., at very low sulfur fugacity) are well below the pyrite stability field.

SEM images of sulfide grains heat-treated with and without a sulfur (or realgar) buffer (Fig. 3.3; methods in §3.1s) show that encapsulation preserves the surfaces of pyrite and arsenopyrite at step-heat temperatures, but does not prevent the breakdown of marcasite (§3.4).

The Single-Jump Migration Model and its Application to Pyrite

The Russian school of noble gas geochronology spearheaded by Shukolyukov and Yakubovich approaches the problem of helium migration in minerals through a theoretical framework that differs fundamentally from the volume

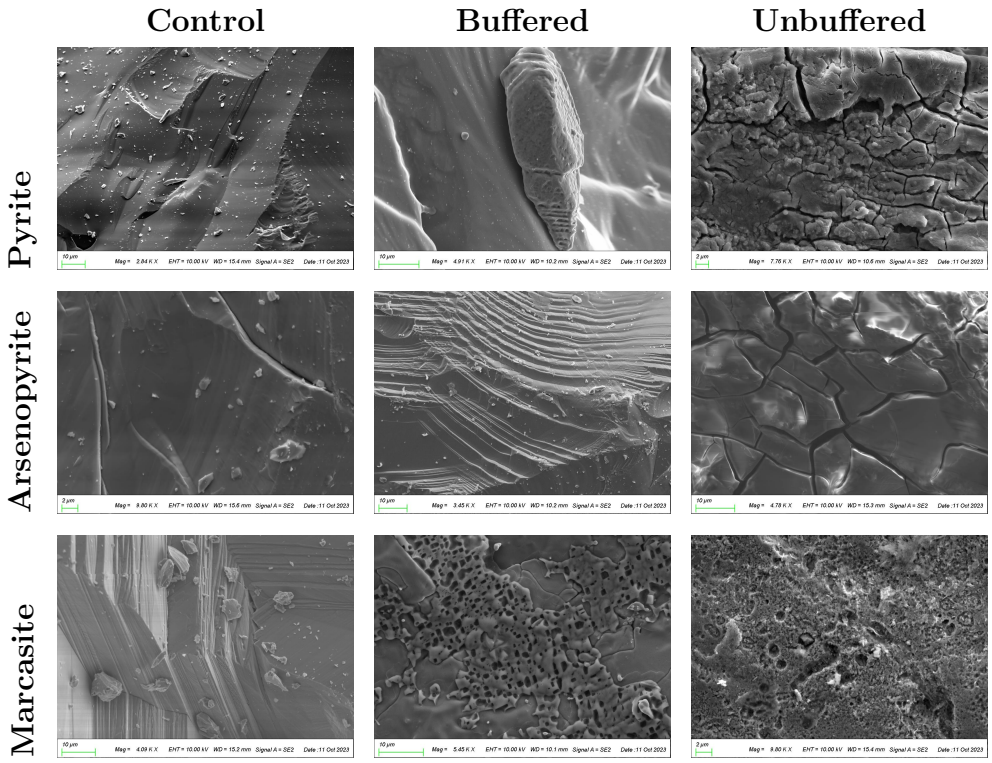


Figure 3.3: Secondary electron SEM images of pyrite (top row), arsenopyrite (middle row), and marcasite (bottom row) surface textures before heat treatment (control), and after heat treatment at 400°C for at least 4 hours, both unbuffered and buffered in a silica glass ampoule with fused sulfur (pyrite and marcasite) or mineral realgar (AsS; arsenopyrite). Unbuffered pyrite was heated in open air, unbuffered arsenopyrite in a sealed silica glass capsule without realgar, and unbuffered marcasite in ultra-high vacuum. The buffered pyrite and arsenopyrite maintain the smooth surfaces and clear cleavage faces that the controls have, showing no evidence of decomposition. The unbuffered samples show clear evidence of decomposition by devolatilization, indicated by contraction cracks and surface roughening (pyrite and arsenopyrite) and recessive ~1 µm rounded craters (marcasite). The buffered marcasite instead develops a raised layer of angular, presumably pyrite, crystals, indicating that encapsulation does not suppress phase breakdown for marcasite as it does for pyrite and arsenopyrite. See §3.1s for details on the experimental methods and sample preparation.

diffusion model used to describe noble gas migration dominant in the Western thermochronological literature (Dodson, 1973; Harrison et al., 1985; Fortier and Giletti, 1989; Lovera et al., 1991; Wolf et al., 1996; Farley, 2000; J. K. W. Lee et al., 1997). In the Shukolyukov "single-jump model," helium atoms occupy discrete energy minima within the crystal structure, and migration proceeds through thermally activated jumps between these sites (Y. A. Shukolyukov et al., 2009; O. V. Yakubovich et al., 2010).

Within this framework, Y. Shukolyukov et al. (2012) developed the ^{190}Pt-^{4}He system as a new geochronometer for native platinum minerals, demonstrating through thermal desorption analysis that radiogenic ^{4}He in isoferroplatinum is stable to temperatures approaching the melting point of platinum, and obtaining ^{190}Pt-^{4}He ages from the Galmoenan and Kondyor massifs consistent with independent Sm-Nd and Rb-Sr estimates. He storage in native platinum was attributed to accumulation in vesicles and clusters at dislocation cores and grain boundaries. Western scientists similarly model helium migration in metals with single-jump mechanisms, although these single jumps are not typically modeled to instantaneously liberate helium; instead, the modeled single jump allows the atom to spontaneously travel a distance of some angstroms with some frequency, introducing a spatial scale from which a diffusivity can be evaluated (Rätzke et al., 1992; Cost and Hickman, 1975; W. D. Wilson and Bisson, 1973).

O. V. Yakubovich et al. (2015) extend this method to sperrylite ($PtAs_2$), a rare mineral found in magmatic PGE deposits that is isostructural with pyrite. ^{190}Pt-^{4}He ages from sperrylite at the Kondyor and Norilsk-Oktyabrsky deposits were geologically consistent, and the high He retentivity demonstrated in this arsenide prophesies similar behavior in pyrite.

Without providing any sulfur buffer, O. V. Yakubovich et al. (2019) interpreted ramped-heating experiments for a variety of silica glass encapsulated sulfides under the single-jump step-heating method. Pyrite, in particular, was found to have anomalously high retentivity of helium. The first application of U-Th/He pyrite geochronometry to a natural ore deposit was reported by O. V. Yakubovich et al. (2019) and O. Yakubovich et al. (2020) for the Uzelga Cu-Zn VMS deposit of the South Urals, yielding a consistent weighted mean age of 382 $\pm$ 12 Ma that is older than regional prehnite-pumpellyite metamorphism, demonstrating that pyrite can survive a metamorphic overprint

without complete isotopic resetting. While these results establish the viability of pyrite as a U-Th/He geochronometer, their analytical approach employs a simple, low-resolution monotonic heating schedule without grain-size scaling experiments, sulfur buffers, or proton-implanted ^{3}He domain mapping, which leaves the diffusion/migration physics of sulfides undercharacterized. Unsurprisingly, given the rapid heating rates used in their experiments, their results for pyrite generally show abrupt helium loss at temperatures corresponding to the upper stability field of pyrite around 700°C (Toulmin and Barton, 1964), and therefore are unlikely to contain substantial information about helium loss in natural pyrite over geologic timescales. Extending these initial observations into a robust, predictive thermochronometric framework requires the more rigorous experimental and theoretical treatment that is one of the aims of this work.

Geologic Setting

Pyrite and arsenopyrite for the geochronologic study were collected from two deposits that strongly differ in the timing and conditions of their formation: the Archean Kidd Creek volcanogenic massive sulfide (VMS) deposit, Ontario, Canada, and the Permo-Triassic orogenic gold system at Gympie, Queensland, Australia. Pairing a ∼2.7 Ga Cu-Zn-Ag VMS sulfide with a ∼235 Ma Au-bearing arsenopyrite provides two independently well-constrained age brackets against which to test He retentivity and closure behavior in pyrite across very different formation temperatures, host lithologies, and post-depositional thermal histories.

Kidd Creek, Ontario, Canada

Kidd Creek is among the largest and most profitable VMS deposits in the world, with past production and reserves exceeding 138.5 Mt at average grades of 2.4 percent Cu, 6.5 percent Zn, 0.23 percent Pb, and 90 g/t Ag (Hannington et al., 1999). The deposit occurs within the Late Archean Kidd-Munro assemblage of the Abitibi subprovince, Superior Province, hosted at the top of a thickened pile of massive rhyolite flows, felsic volcaniclastics, and coarse epiclastic rocks that attain a maximum thickness of ∼300 m beneath the mine (Hannington et al., 1999). The three main stratiform orebodies (North, Central, and South) formed largely by subseafloor infilling and replacement of permeable volcaniclastic rocks within a subsiding, rift-related graben, rather than by high-temperature

venting at the seafloor (Hannington et al., 1999). U-Pb zircon geochronology brackets the ore-forming interval tightly: the oldest footwall rhyolite is dated at 2716.1 ± 0.6 Ma, the onset of bimodal komatiite-rhyolite volcanism at 2717 ± 1.1 Ma, and the youngest, uppermost quartz porphyry unit at 2711.5 ± 1.2 Ma, indicating that mineralization took place during at least one and possibly several million years of intermittent volcanic activity (Hannington et al., 1999). Oxygen isotope studies of the host rhyolites show that, while primary igneous zircon retains magmatic $\delta^{18}O$ values (5.4‰ ± 0.3‰), coexisting quartz phenocrysts are pervasively shifted to much higher and more heterogeneous values (14.2‰ to 16.1‰), recording extensive subsolidus recrystallization during the same long-lived hydrothermal convection that deposited the sulfides (King et al., 1997).

The interior of the Canadian shield has been a stable platform for hundreds of millions of years, as documented by U-Th/He and fission track dating of apatite (Sturrock et al., 2024). Kidd Creek has been of particular interest for its famously ancient groundwaters that sustain a low-biomass but metabolically active community of sulfate-reducing and alkane-oxidizing microorganisms (Lollar et al., 2019). The Kidd Creek deposit is thus an ideal natural laboratory for testing the long-term He retentivity of pyrite and arsenopyrite.

Gympie, Queensland, Australia

The Gympie arsenopyrite comes from a very different setting: an accreted, Early Permian bimodal (basalt-dacite) submarine volcanic sequence, the Gympie Group, interpreted as a fragment of an intra-oceanic island arc that rifted from the Gondwana margin before docking against the New England Orogen of eastern Australia (Sivell and Waterhouse, 1988). Drillcore near the Gympie Eldorado Gold Mine resolves at least nine sequentially emplaced magmatic suites and a >2 km thick volcano-sedimentary succession. This arc terrane is inferred to have accreted to the Gondwana margin by about 220 Ma (Early to Middle Triassic), based on cross-cutting granitoid relationships with adjacent New England Orogen terranes (Sivell and McCulloch, 2001).

Gold-bearing pyrite-quartz mineralization at Gympie occurs in fibrous extensional veins developed within sheared, carbonaceous black slates. Alteration associated with this vein set has yielded a K/Ar age of about 235 Ma (Holcombe et al., 1997). Refractory dolerite dykes and later microdiorite-shoshonite dykes

cross-cut the mineralized structures and bracket the timing of the auriferous quartz veining from below and above, respectively, giving an independent field constraint on the age of mineralization that is broadly consistent with the K/Ar alteration ages (Sivell and McCulloch, 2001).

The nominal "Early Permian" age assigned to the host rocks of the Gympie deposit, the Highbury Volcanics and Rammutt Formation, however, rests mainly on sparse invertebrate and plant fossil biostratigraphy higher in the sequence and on lithologic correlation, rather than on direct radiometric dates on the volcanic pile itself. Holcombe et al. (1997) note that the lower Gympie Block volcanics, like the correlative Berserker beds, "may contain older volcanic units," and fold the entire Gympie succession into a long-lived (∼320-280 Ma, peaking at ∼305 Ma) regional magmatic episode. These volcanics may very well be related to nearby ∼300-310 Ma granodiorite magmatism and rapid uplift of the regional epidote-blueschist-facies basement, which record $^{40}Ar/^{39}Ar$ hornblende cooling ages around 297 Ma (Little et al., 1995; Holcombe et al., 1997).

3.2 Experimental Methods

U-Th/He Geochronology

Sample Preparation

Intact crystals of arsenopyrite (and pyrite from the Kidd Creek deposit) were extracted from Kidd Creek and Gympie samples using a hammer and chisel at Caltech and SELFRAG high-voltage pulse fragmentation at the University of Queensland, respectively. Pyrite is present in the Gympie sample but only in small amounts, and no pyrite dates are reported for it. The majority of the Kidd Creek samples were silver arsenopyrite, but brassy cubes (presumed pyrite) were also analyzed. A subsample of the Gympie arsenopyrites was leached in 1M HCl for 4 hours to remove any surface contamination and then rinsed in MilliQ water and dried. Individual crystals were washed in a sequence of acetone, ethanol, and MilliQ water baths, then promptly dried using clean, lint-free wipes. These samples, along with the hematite MI-43 standard from Gogebic, Michigan (Farley and McKeon, 2015), and the apatite standard from Durango, Mexico (Farley, 2000), were weighed, then encapsulated in silica glass ampoules similarly to the sulfides in §3.2, but without any additional buffer materials. An empty tube was also encapsulated to serve as a procedural blank. Photos and masses of the prepared ampoules were recorded so that the helium

and trace metal analyses could be correctly matched.

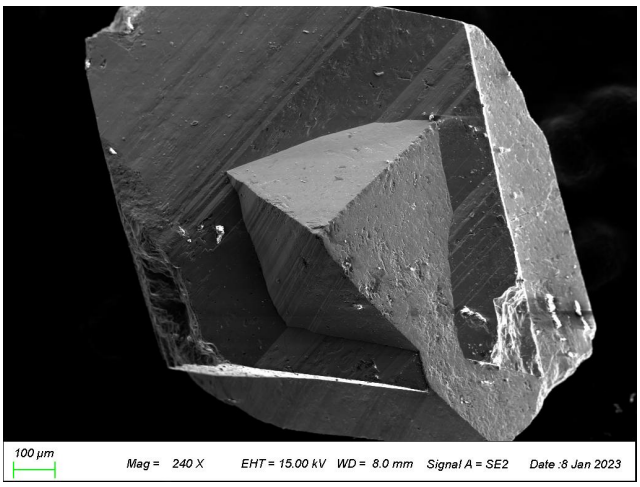


Figure 3.4: Secondary-electron SEM image of crystallographic arsenopyrite twins exceeding 1 cm across, recovered intact from Kidd Creek ore (sample ST05) with a hammer and chisel. Fine-scaled growth striations are preserved on the crystal faces. Euhedral crystals of this size were extracted routinely, showing that simple mechanical excavation preserved crystal faces well.

Helium Analysis

Encapsulated samples were wrapped in Ni-foil and loaded into a custom-built autosampler with the procedural blank loaded first. The autosampler was wrapped in heating tape, which was used to warm the samples to 80-100°C for at least 12 hours in ultra-high vacuum before the first step heat to allow for atmospheric helium to escape. Initial efforts in method development showed that atmospheric helium contamination was significant (Fig. 3.5). This atmospheric helium payload of an ampoule equilibrated with air is on the order of 10s of nccSTP helium per ampoule (Chapter 1, §1.2). It is difficult to mathematically correct for even when 3He can be accurately measured, because helium diffusion and permeation through silica glass fractionates 3He/4He ratios (Shelby, 1971; Jones, 1953), and the ampoules were each geometrically unique and therefore had different permeation rates. Still, we found the best results when we applied a simple correction to the helium data after substantially mitigating this contamination with the warm (80-100°C) degassing step. To correct for the much smaller remaining atmospheric contamination, we subtract 4He proportionally to measured 3He by the atmospheric 3He/4He isotope ratio: 1.39×10^{-6}. The procedural blanks showed that atmospheric helium

contamination was reduced by a factor of 5-10, which was sufficient for these helium-rich samples with 100s of nccSTP helium (Fig. 3.5). Future work may develop a more rigorous approach to experimentally remove atmospheric helium contamination or to mathematically correct for it using the first order framework presented in Chapter 1 (§1.3).

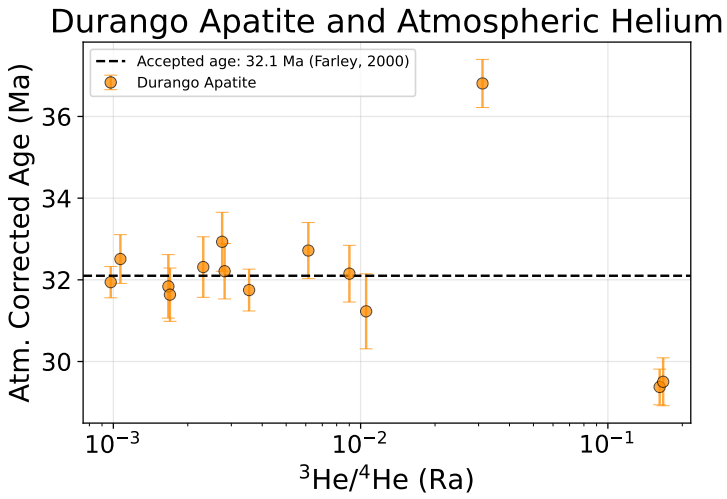


Figure 3.5: The atmosphere-corrected age for encapsulated Durango apatite is plotted against the measured helium isotope ratio ($^3He/^4He$), normalized to atmosphere (Ra), illustrating the atmospheric-contamination problem motivating the experimental correction protocol described in §3.2. Aliquots with low measured Ra (i.e. a small atmospheric 4He contribution relative to radiogenic 4He) require insignificant mathematical correction, preserving the accuracy of the ages and agreeing with the 32.1 Ma accepted age (Farley, 2000). Aliquots measured before implementing the experimental atmosphere-degassing step contain more 3He and require stronger corrections. Because these corrections are unaware of the true isotopic composition of fractionated atmosphere in the ampoule, they yield inaccurate ages when they are large with respect to the sample's total measured helium. The weighted average age of the aliquots with mass >0.5 mg, 32.1 ± 0.2 Ma, agrees well with the accepted age, affirming the fundamental accuracy of the measurements.

After 12 hours, unheated furnace blank measurements were periodically taken until the measured value was similar to such blanks taken for batches of samples not encapsulated in silica glass, then samples were individually dropped into a resistance furnace and baked at 1000-1200°C for 15 minutes to degas their helium contents. Although pyrite and arsenopyrite are unusually retentive of helium at lower temperatures, these temperatures were far above these phases' stability field, and so they were readily degassed. Helium abundances and

isotope ratios were measured on a Thermo Fisher Helix SFT mass spectrometer. Gases evolved during the bake were passed through a charcoal filter held at 77 K, then exposed to an SAES-10 getter and a heated titanium getter before being cryofocused at 14 K, then released at 34 K. Sensitivities were quantified using a calibrated 3He-enriched air standard several times throughout the analysis.

Trace Metal Analysis

The encapsulated, now degassed samples were removed from the furnace and were matched to their pre-bake photos and masses. Around 20% of samples were typically lost during the helium analysis process, with ampoules either broken by the autosampler or by thermal stresses during the bakes. Samples bearing residual nickel-foil that had fused to the surface of the glass were scratched clean with a dental pick, and the remaining glass ampoule was sonicated in a sequence of ethanol then MilliQ water baths. Samples were then placed in an acid-washed Teflon beaker and were broken underwater by an agate pestle, to prevent loss of sample material. The pestle was rinsed with MilliQ water between samples to avoid cross-contamination. Excess water was pipetted away from each sample and disposed of, then the remaining water was evaporated on a hot plate. The vast majority of the samples were then spiked with a ^{235}U, ^{230}Th, and vanadium-enriched spike and digested in concentrated HNO_3 in closed Teflon beakers, leaving the glass intact. The uranium and thorium concentrations of the glass ampoules were measured to be $\approx$ 50 and 30 ppb by mass respectively, but this contaminant payload was only delivered when the glass was dissolved in HF; the uranium and thorium contents of concentrated nitric acid in extended contact with broken glass were below detection limits.

After this point, the Kidd Creek pyrite/arsenopyrite and Durango Apatite solutions were pipetted and diluted to an ICPMS-ready solution. The Gympie samples, however, required further processing to dissolve fine U-Th bearing rutile inclusions. The Gympie and MI-43 hematite samples, now mostly digested in concentrated HNO_3, were dried down and put back up in small volumes of concentrated HCl, then were loaded into Parr vessels and baked at 200°C for 24 hours. The Gympie samples were then dried down again and put back up in concentrated HNO_3 before the concentrated acid was transferred and diluted to an ICPMS-ready solution. U and Th contents were quantified with

isotope dilution using an Agilent 8800 ICP-MS. Other trace elements were measured as ratios with spike vanadium. The U-Th contents of the spikes and the sensitivities of the other quantified trace metals relative to vanadium were continuously monitored by analyzing a standard solution of known composition that was spiked with the same U-Th-V spike as the samples. The same empty tube procedural blank used for the helium analysis was also processed through the same acid digestion and ICPMS analysis, and the measured U, Th, and other trace metal contents were subtracted from the measured values of the samples. The U-Th contents of the procedural blank were below detection limits, and the other trace metals were typically 1-5% of the measured values of the samples.

The Gogebic Iron Range hematite standard (Farley and McKeon, 2015) was carried through this same Parr-vessel digestion step alongside the Gympie arsenopyrite, serving as an in-run check on how completely this digestion protocol recovers a sample's true U-Th-Fe budget. Fig. 3.6 plots each Gogebic aliquot's measured ^{4}He against its eU, normalized by the sample mass calculated from measured Fe content where available and by the pre-analysis balance mass otherwise. Except for one clearly erroneous outlier (with an age in excess of 1.5 Ga), these samples agree with two independently published ages for Gogebic, Michigan hematite: 453 Ma (U-Th/^{21}Ne, aliquot MI-45; Farley and McKeon, 2015) and 579 Ma (U-Th/He; Hofmann et al., 2020). Here the outlier is interpreted to have experienced U-loss, as its helium content is similar to the other aliquots, but it has low eU. Although incomplete digestion would also produce erroneous older ages, the apparent helium concentration would be higher and the eU would match the other samples if that was the case.

Encapsulated Sulfide Step Heat Experiments

Samples

Nine helium step heat experiments of sulfides (pyrite, arsenopyrite, chalcopyrite, galena, chalcocite, and sphalerite) sealed in silica glass ampoules performed at the Noble Gas Laboratory at Caltech are discussed here. Table 3.1 shows the dimensions, geometric factors, and encapsulated materials of this work. All sulfides of this study were sourced from the Caltech Mineral Collection from visually pure, well-formed euhedral crystals with half prism widths greater than 200 microns, often larger than a centimeter. Samples were broken into smaller

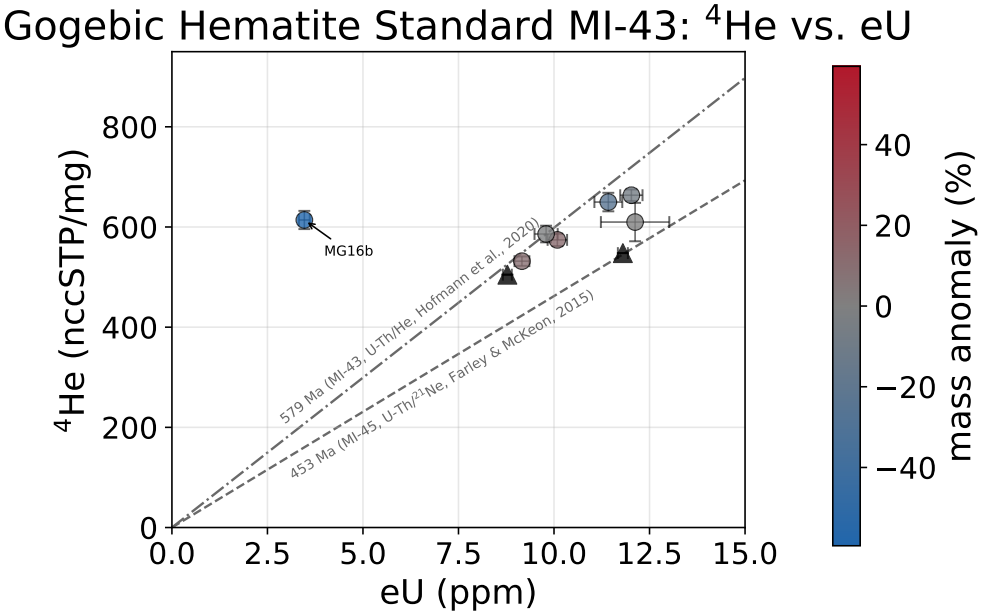


Figure 3.6: Atmosphere corrected ^{4}He (nccSTP/mg) vs. eU (ppm), with 2σ error bars, for the MI-43 Gogebic hematite standard aliquots analyzed alongside Gympie arsenopyrite (Table 3.6), normalized by mass calculated from Fe molar abundances measured on ICPMS. Fe was not measured for two samples: they are plotted as triangles using their balance-measured mass. These two measures of mass are compared to assess internal consistency within samples. Points are colored by "mass anomaly" in percent: $\frac{M_{Fe}-M_B}{M_{Fe}} \times 100$. Anomalies are generally small: with absolute values 11% or smaller for all but 2 samples. The old outlier sample 041323MG16b with low eU, however, has a mass anomaly of 59%, meaning that the balance-measured mass was significantly higher than the mass inferred from the iron abundance. Two isochrons of independently published ages for Gogebic hematite are plotted: 453 Ma (MI-45, U-Th/^{21}Ne; Farley and McKeon, 2015) and 579 Ma (Hofmann et al., 2020). Only the low eU outlier 041323MG16b falls outside of this age range. With anomalously low eU at an otherwise unremarkable ^{4}He content, it is interpreted here as an example of U-loss. Its large mass anomaly may point to the presence of an insoluble phase such as quartz, however, it is difficult to explain why an insoluble phase would affect only the eU concentration and not the helium concentration of the hematite.

pieces under ethanol to avoid irreversible trapping of atmosphere (S. E. Cox et al., 2022). Except for the Rustenburg pyrite, all step heats measured 3He evolution from an initially homogeneous distribution of spallogenic 3He. To produce a homogeneous distribution of 3He, the samples were irradiated with 220 MeV protons for 24 hours at the Francis H. Burr Proton Therapy Center, Massachusetts General Hospital, following standard procedures (Shuster et al., 2004).

Table 3.1: Silica Glass encapsulated sulfides that were the subject of helium step heat experiments in this work. The Rustenburg Pyrite experiment had evolved radiogenic 4He quantified by isotope dilution using a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer, whereas evolved spallogenic 3He was quantified using a Helix-MC-Plus^{10K} multi-collector mass spectrometer for the remainder of the samples, which were proton-irradiated. Experiments Spal230707 and CCT230707 contained no steps for which the uncertainty due to helium permeation through the silica glass capsule could be estimated, and therefore are discussed in a limited, qualitative capacity only in the Supplementary Information (§3.2s)

Experiment Name and Phase	Sample Locality	Tube Inner x Outer Diameters	Geometric factor G	VR	Encapsulated "Buffer" Materials	Mineral Half-prism Width (a)
LVCOPy230630 Pyrite	Leadville, Colorado	*Not Measured*	*3000 cm^{-2}*	-	~0.2 mg Fused Sulfur	~200 µm
BigLVCOPy230707 Pyrite	Leadville, Colorado	*1300 x 1650 µm*	1985 cm^{-2}	1.6	~0.2 mg Fused Sulfur	~400 µm
112123LVCOPyIrr Pyrite	Leadville, Colorado	830 x 1550 µm	1859 cm^{-2}	2.5	~ 0.2 mg Fused Sulfur and >5 mg silica glass shards	~300 µm
Rustenburg Pyrite (4He on Quad)	Rustenburg, South Africa	520 x 810 µm	6675 cm^{-2}	1.4	~0.2 mg Fused Sulfur	106 µm
112123SalAspIrr Arsenopyrite	Mitterberg, Austria	530 x 970 µm	4711 cm^{-2}	2.3	~0.5 mg Crystalline Realgar (AsS)	~175 µm
GalTS230630 Galena	Tri-State District, Missouri	620 x 920 µm	5273 cm^{-2}	1.2	~0.2 mg Fused Sulfur	~200 µm

Experiment Name and Phase	Sample Locality	Tube Inner x Outer Diameters	Geometric factor G	VR	Encapsulated "Buffer" Materials	Mineral Half-prism Width (a)
CCpTS230630 Chalcopyrite	Baxter Springs, Kansas	540 x 1010 µm	4382 cm^{-2}	2.5	~0.2 mg Fused Sulfur	~200 µm
Spal230707* Sphalerite	Chitwood, Missouri	1110 x 1490 µm	2205 cm^{-2}	0.8	~0.2 mg Fused Sulfur	~200 µm
CCT230707* Chalcocite	Messina, South Africa	1100 x 1520 µm	2044 cm^{-2}	0.9	~0.2 mg Fused Sulfur	~200 µm

Encapsulation

Before encapsulating the sulfide samples, glass tubes were cleaned and pulled into pipettes. 10 cm long segments of pure silica glass tubes with an outer diameter of 4 mm and an inner diameter of 3 mm were submerged and sonicated within acetone, ethanol, then MilliQ water baths in sequence, and then were dried overnight in a 100°C oven. The glass tubes were hand-pulled into thinner pipettes using a natural gas and oxygen welding torch and the thin ends were fused closed.

Aliquots of sulfides were weighed and loaded into room temperature glass tubes along with fused sulfur in the presence of an ionizing fan to mitigate static. The arsenopyrite ampoule was loaded with crystalline realgar (arsenic sulfide) instead of fused sulfur, and one pyrite ampoule was loaded with silica glass pieces above and below the pyrite aliquot to separate the pyrite from the sulfur liquid during the experiment. To seal, the top of each pipette was interfaced with a vacuum line and the tube was pumped to a pressure of <60 millitorr, which greatly aided the ease and quality of the final seal. After attaining rough vacuum, a fine-flamed welding torch was used to seal the top of each ampoule. To determine the geometric factors for permeation of the ampoules, the inner and outer diameters of most ampoules were measured using the measurement function of the Amscope stereo microscope software, which was calibrated and tested during early development of this method. LVCOPy230630 was completed without these geometric measurements. For this experiment, a geometric factor of 3000 cm^{-2} was arbitrarily assigned to estimate uncertainties of the measurements due to permeation lag. This assumption is unlikely to

affect the interpretation of the results.

The sealed ampoules were placed in copper foil packets and mounted to a K-type thermocouple within an ultra-high vacuum projector lamp heating cell described in detail by Farley et al. (1999). Before the experiment began, each ampoule was washed with ethanol, air-dried, loaded in the lamp cell, pumped to high vacuum, and was held at 100°C for at least 30 minutes under dynamic vacuum to desorb ethanol, water, and other atmospheric species on the ampoule, copper packet, and lamp cell.

Helium Mass Spectrometry

Heating steps and 3He measurements for all samples were automated using custom LabView routines on a Helix-MC-Plus^{10K} multi-collector mass spectrometer, or a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer for the Rustenburg Pyrite ampoule. For all ampoules, a Watlow temperature controller held the ampoule at the temperature set point for a pre-programmed duration, after which the ampoule was allowed to cool while the evolved gas was prepared for measurement for about 25 minutes.

For the irradiated samples, helium was purified by exposing evolved gases to a charcoal trap held in liquid nitrogen, a titanium sponge getter, then a SAES NP10 getter before freely expanding into the Helix-MC-Plus^{10K}. 3He was measured with a pulse-counting electron multiplier, while 4He was detected on a Faraday detector. The Helix-MC-Plus^{10K} mass spectrometer was operating in peak hopping mode and the majority of the analysis time was spent measuring 3He. Every few hours, the background was mass-scanned to set the 3He and 4He measure points; each was taken to be a constant mass offset from the background HD^+ peak. These mass offsets are precisely measured when tuning the Helix-MC-Plus^{10K} for helium isotope analysis. Months of helium measurements with the Helix-MC-Plus^{10K} at Caltech have shown this approach to be reliable, reproducible, and accurate for helium isotope analysis. Additionally, an electron multiplier monitored atomic mass 3.5 to detect 4He overspray. 4He signals were near-blank for all experiments of irradiated samples, and no attempt is made here to interpret those data. 4He overspray likewise did not affect 3He signals. Measurements of high 3He MM gas standard were analyzed every 8 hours to track instrumental drift of 3He and 4He sensitivities and to quantify internal reproducibility. These standards show 3He was generally internally reproducible to within 1% (2σ) for rolling sets of 5 consecutive standards, except

for the latest measurements on the 112123SalAspIrr Arsenopyrite, during which the ion source began to fail. Those measurements are excluded from further analysis, and any helium evolved during those steps is likely to be a very small proportion of the total helium contents of the sample (<0.2%) and thus would not significantly affect the calculation of the total helium content of the sample.

After the step heat experiments, the LVCOPy230630 and GalTS230630 ampoules were broken and then completely degassed following the step heat experiments by heating to 1100 °C in a resistance furnace, then ^{3}He and ^{4}He yields were quantified using a Thermo Fisher Helix SFT. The total ^{3}He yields for the remainder of the irradiated pyrite samples were inferred assuming the measured ^{3}He concentration of LVCOPy230630. The chalcopyrite experiment was assumed to be entirely degassed in the step heat experiment without the final fusion step. The total ^{4}He content of the Rustenburg pyrite was assumed to be 200 nccSTP, a conservatively low estimate based on the measured ^{4}He content of other Rustenburg pyrite aliquots.

Radiogenic 4He evolved during step heats of the encapsulated Rustenburg Pyrite was quantified using isotope dilution with a high purity 3He spike on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer. A standard gas with a known amount of 4He was measured before and after the experiment, to monitor drift in the relative sensitivities of 3He and 4He and to estimate reproducibility, which was about 6% (2σ). After each step, evolved gas from the ampoule in the lamp heating cell was allowed to mix with the 3He spike and the mixture was frozen on charcoal held at 14 K within a cryostat, then helium was selectively released to the mass spectrometer by raising the temperature to 36 K.

Scanning Electron Microscope Study

The methodology for the SEM study of the sulfides is described in detail in §3.1s.

Analytical Methods For Step Heat Experiments

We designed these helium step heat experiments according to the assumption that the silica glass ampoule was effectively “helium transparent”.Thus, permeation steps (as discussed in Chapters 1 and 2) were rarely used. This assumption was acceptable for most of the pyrite and arsenopyrite step heats, and much of the galena and chalcopyrite step heats. Unfortunately, the as-

sumption was not valid for any of the sphalerite or chalcocite step heats, nor for some of the lower temperature steps of the other sulfides, which can only be discussed here qualitatively. The helium yields for steps at high temperatures are approximately the measured value, and the uncertainty of this estimation due to permeation lag is constrained by upper bounds following the uncertainty-bounding approach presented in Chapter 1 (§1.4). This approach requires no permeation steps or knowledge of the helium content within the ampoule from previous steps, instead calculating the maximum possible helium content remaining within the ampoule following a given high-temperature measurement, provided the step was long and hot enough for the helium inventory of the tube to have reached a steady state concentration and assuming that the rate of helium evolution from the encapsulated mineral is constant or decreasing as is generally expected for first and second order processes. The steady state assumption is taken to be effectively valid once the ampoule loses memory of the helium content remaining from the previous step. According to Eq. (1.57) in Chapter 1, only 1.8% of the difference between the previous step's remaining helium content and the current step's steady state helium content remains when $\lambda t = 4$. In this work, the steady state assumption is only applied for steps long and hot enough for $\lambda t \geq 4$, where $\lambda = KGRT$, K is the temperature-dependent permeation coefficient for 4He or 3He through silica glass, G is the ampoule geometric factor, R is the gas constant, T is the temperature in Kelvin, and t is the time in seconds. The 4He permeation coefficient K is taken from Swets et al. (1961); and these coefficients are multiplied by the permeation coefficient ratio from Jones (1953) to yield the 3He permeation coefficient.

Many heating steps, especially near the beginning of the experiments where lower temperature steps were used, are not interpreted or plotted in this work because $\lambda t < 4$; however, some steps with $\lambda t < 4$ are used when Eq. (1.10) is used to place upper bound constraints on preceding steps, so that Eq. (1.57) can in turn be used to quantify the uncertainty on the yields of steps that have not reached steady state.

This method of calculating uncertainties produces an asymmetric upper and lower uncertainty for each step. These asymmetric uncertainties are propagated through the diffusivity inversion of Fechtig and Kalbitzer (1966) with Eq. (2.15) in Chapter 2. In general, the 2σ analytical uncertainties as calculated using the reproducibility of standards are greatly exceeded by the uncertainties due

to the permeation lag of the ampoule as constrained by the procedure outlined in §1.4.

3.3 Results

Sulfide Step Heat Experiments

The apparent diffusivities calculated from the helium step heat data are reported in Tables 3.2–3.5 (one per pyrite/arsenopyrite experiment; the analogous galena and chalcopyrite tables are given in the Supplementary Information, §3.2s). These experiments qualitatively show remarkably high helium retentivity at high temperatures for pyrite and arsenopyrite. For no pyrite or arsenopyrite helium step heat experiment did the final cumulative fractional helium yield exceed 5%, even after repeated heating cycles at $\geq$ 500 °C.

Table 3.2: ^{3}He step-heat dataset for Pyrite, Leadville, CO, 200 µm half-prism width (sample LVCOPy230630), under the uncertainty bounding method (§1.4)(§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.7 (Arrhenius) and Fig. 3.8 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
31	475.0	100	0.0075	0.4382	0.0022	-24.68	-24.06	-23.81
32	465.0	120	0.0048	0.4430	0.0012	-25.28	-24.68	-24.46
33	455.0	140	0.0047	0.4476	0.0010	-25.15	-24.85	-24.65
34	510.0	60	0.0105	0.4581	0.0046	-23.28	-23.18	-22.81
35	520.0	60	0.0126	0.4708	0.0053	-23.41	-22.97	-22.61
36	530.0	60	0.0177	0.4885	0.0068	-22.94	-22.60	-22.26
37	539.9	60	0.0204	0.5089	0.0075	-22.82	-22.41	-22.10
38	555.0	60	0.0315	0.5404	0.0105	-22.19	-21.93	-21.63
39	545.0	60	0.0181	0.5585	0.0065	-23.30	-22.44	-22.13
40	535.0	80	0.0175	0.5760	0.0049	-23.19	-22.73	-22.48
41	525.0	100	0.0136	0.5895	0.0032	-23.62	-23.18	-22.96
42	515.0	120	0.0114	0.6009	0.0024	-23.85	-23.51	-23.32
43	505.0	140	0.0078	0.6087	0.0015	-24.40	-24.03	-23.85
44	560.0	60	0.0252	0.6339	0.0087	-22.05	-21.98	-21.68
45	570.0	60	0.0371	0.6710	0.0121	-21.81	-21.55	-21.26
46	580.0	60	0.0465	0.7176	0.0145	-21.55	-21.26	-20.98
47	590.0	60	0.0604	0.7780	0.0179	-21.19	-20.92	-20.65

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(continued)

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
48	600.0	60	0.0690	0.8470	0.0195	-21.00	-20.71	-20.45
49	604.7	60	0.0691	0.9161	0.0191	-20.95	-20.63	-20.37
50	595.0	60	0.0381	0.9542	0.0109	-21.85	-21.16	-20.90
51	585.0	60	0.0244	0.9786	0.0073	-22.17	-21.58	-21.31

Table 3.3: ^{3}He step-heat dataset for Pyrite, Leadville, CO, 400 µm half-prism width (sample BigLVCOPy230707), under the uncertainty bounding method (§1.4)(§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.7 (Arrhenius) and Fig. 3.8 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
16	400.0	160	0.0201	0.7190	0.0099	-26.07	-26.03	-25.78
17	420.0	140	0.0386	0.7576	0.0189	-25.36	-25.20	-24.95
18	440.0	120	0.0461	0.8037	0.0234	-25.07	-24.81	-24.55
19	460.0	100	0.0521	0.8558	0.0289	-24.73	-24.45	-24.17
20	480.0	80	0.0490	0.9048	0.0322	-24.62	-24.23	-23.91
21	500.0	60	0.0538	0.9586	0.0483	-24.17	-23.79	-23.40
22	510.0	60	0.0559	1.0145	0.0454	-24.26	-23.70	-23.33
23	490.0	70	0.0237	1.0382	0.0178	–	-24.67	-24.33
24	470.0	90	0.0109	1.0491	0.0065	-27.67	-25.68	-25.40
25	450.0	110	0.0053	1.0544	0.0028	-27.69	-26.59	-26.33
26	430.0	130	0.0033	1.0577	0.0016	-27.87	-27.23	-26.98
27	410.0	150	0.0007	1.0584	0.0004	–	-28.89	-28.64
28	450.0	160	0.0055	1.0640	0.0017	-26.95	-26.91	-26.74
29	470.0	140	0.0100	1.0739	0.0032	-26.30	-26.19	-26.01
30	490.0	120	0.0219	1.0958	0.0073	-25.32	-25.23	-25.05
31	510.0	100	0.0378	1.1336	0.0139	-24.60	-24.48	-24.28

Table 3.4: ^{3}He step-heat dataset for Pyrite, Leadville, CO, sulfur-isolated (glass-shard-separated) (sample 112123LVCOPyIrr), under the uncertainty bounding method (§1.4)(§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.7 (Arrhenius) and Fig. 3.8 (delta). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
46	500.0	120	0.0447	0.1998	0.0138	-22.60	-22.58	-22.27
48	510.1	120	0.0311	0.2327	0.0090	-23.20	-22.74	-22.46
49	520.1	120	0.0259	0.2586	0.0070	-23.20	-22.80	-22.54
51	530.0	120	0.0325	0.2921	0.0082	-22.64	-22.45	-22.21
53	540.0	120	0.0277	0.3203	0.0066	-22.82	-22.51	-22.28
55	550.0	120	0.0326	0.3531	0.0073	-22.46	-22.25	-22.04
57	560.1	384	0.1064	0.4600	0.0061	-22.10	-22.04	-21.98
58	570.1	60	0.0156	0.4755	0.0080	-22.46	-21.97	-21.54
59	580.0	60	0.0263	0.5018	0.0125	-21.76	-21.40	-21.00
60	590.0	60	0.0596	0.5614	0.0263	-20.72	-20.50	-20.11
61	600.0	60	0.0630	0.6244	0.0265	-20.85	-20.33	-19.96
62	595.0	60	0.0468	0.6712	0.0206	-21.36	-20.54	-20.16
63	590.1	60	0.0318	0.7030	0.0146	-21.89	-20.87	-20.48
64	585.1	60	0.0259	0.7290	0.0122	-21.85	-21.03	-20.64
65	580.1	60	0.0165	0.7455	0.0082	-22.79	-21.45	-21.04
66	575.1	60	0.0114	0.7569	0.0058	-23.07	-21.81	-21.39
67	570.1	60	0.0101	0.7669	0.0053	-22.77	-21.91	-21.49
68	565.2	60	0.0075	0.7744	0.0041	-23.43	-22.20	-21.76
69	560.1	60	0.0058	0.7802	0.0033	-23.67	-22.44	-21.99
70	555.2	60	0.0051	0.7853	0.0030	-23.64	-22.57	-22.10
71	600.1	60	0.0337	0.8190	0.0138	-20.75	-20.66	-20.30
72	551.7	79	0.0115	0.8305	0.0047	–	-21.99	-21.64
73	550.2	60	0.0044	0.8349	0.0028	–	-22.66	-22.16
74	550.2	60	0.0040	0.8388	0.0025	-23.98	-22.75	-22.27
75	500.0	420	0.0065	0.8453	0.0005	-24.68	-24.20	-24.13
76	500.1	180	0.0023	0.8476	0.0004	-24.62	-24.39	-24.22

Table 3.5: ^{3}He step-heat dataset for Arsenopyrite, Mitterberg, Austria (sample SalAspIrr), under the uncertainty bounding method (§1.4)(§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.7 (Arrhenius) and Fig. 3.21 (delta, row 1). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
76	360.0	120	0.0052	0.3258	0.0017	–	-24.59	-24.30
78	370.0	120	0.0073	0.3342	0.0022	–	-24.22	-23.96
80	380.0	120	0.0059	0.3417	0.0016	–	-24.41	-24.17
82	390.0	120	0.0074	0.3504	0.0018	–	-24.16	-23.94
84	400.0	120	0.0111	0.3625	0.0025	–	-23.73	-23.52
86	395.0	120	0.0075	0.3704	0.0018	–	-24.09	-23.87
88	390.0	120	0.0049	0.3763	0.0012	–	-24.50	-24.28
90	385.0	130	0.0065	0.4182	0.0015	–	-24.19	-23.98
92	380.0	140	0.0044	0.4226	0.0010	–	-24.65	-24.44
93	410.0	120	0.0154	0.4379	0.0032	-23.28	-23.21	-23.02
95	420.0	120	0.0255	0.4693	0.0049	–	-22.65	-22.47
97	430.0	120	0.0547	0.5262	0.0097	–	-21.79	-21.62
99	440.0	120	0.1123	0.6419	0.0185	–	-20.91	-20.75
101	450.0	120	0.1449	0.7912	0.0222	–	-20.46	-20.30
102	445.0	120	0.0825	0.8737	0.0131	-21.17	-20.87	-20.72
103	440.0	120	0.0510	0.9247	0.0084	-21.57	-21.28	-21.12
104	435.0	130	0.0305	0.9552	0.0048	-22.14	-21.82	-21.68
105	430.0	140	0.0187	0.9739	0.0028	-22.65	-22.36	-22.22
106	425.0	150	0.0149	0.9888	0.0021	-22.85	-22.64	-22.51
107	420.0	160	0.0091	0.9979	0.0012	-23.45	-23.19	-23.06
108	415.0	170	0.0062	1.0040	0.0008	-23.85	-23.62	-23.50
109	410.0	180	0.0048	1.0089	0.0006	-24.11	-23.93	-23.80
110	405.0	190	0.0035	1.0124	0.0004	-24.50	-24.30	-24.18
111	400.0	206	0.0054	1.0178	0.0007	-24.02	-23.94	-23.82
112	460.0	120	0.0588	1.0765	0.0084	-20.99	-20.98	-20.84
113	470.0	100	0.0750	1.1515	0.0123	-20.61	-20.49	-20.34
114	480.0	80	0.0732	1.2247	0.0147	-20.41	-20.23	-20.04
115	500.0	60	0.1425	1.3672	0.0353	-19.29	-19.19	-18.95
116	495.0	60	0.0693	1.4366	0.0178	-20.53	-19.83	-19.60
117	485.0	60	0.0300	1.4666	0.0083	-21.53	-20.63	-20.39
118	475.0	60	0.0100	1.4766	0.0030	-23.47	-21.72	-21.46
119	465.0	60	0.0046	1.4812	0.0015	-23.50	-22.48	-22.20
120	455.0	60	0.0054	1.4866	0.0019	-22.65	-22.33	-22.02
121	445.0	60	0.0022	1.4888	0.0009	-25.23	-23.22	-22.89
122	435.0	60	0.0015	1.4903	0.0006	-24.52	-23.62	-23.26

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Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
123	425.0	80	0.0004	1.4907	0.0001	–	-25.20	-24.93
125	405.0	120	0.0015	1.4922	0.0003	-24.38	-24.30	-24.10
126	395.0	150	0.0018	1.4940	0.0003	-24.51	-24.32	-24.15
127	395.0	150	0.0009	1.4950	0.0002	-25.40	-24.97	-24.81
128	395.0	150	0.0022	1.4972	0.0004	-24.20	-24.12	-23.96
129	395.0	150	0.0017	1.4989	0.0003	-24.64	-24.37	-24.21
130	400.0	150	0.0018	1.5007	0.0003	-24.53	-24.34	-24.18
131	410.0	415	0.0041	1.5048	0.0002	-24.60	-24.52	-24.47
132	420.0	130	0.0010	1.5058	0.0002	-25.02	-24.78	-24.62
133	430.0	120	0.0022	1.5080	0.0004	-24.00	-23.92	-23.75
134	440.0	593	0.0103	1.5183	0.0003	-23.99	-23.95	-23.92

The irradiated Leadville pyrite was the most retentive sample of the sulfide suite. Three separate helium step heat experiments were conducted on this sample, two with varying grain size and a third that attempted to isolate the pyrite from the sulfur liquid by packing glass shards above and below the pyrite grains within the ampoule (Table 3.1). The Arrhenius plots for all three experiments show similar diffusivities (Fig. 3.7). The three experiments shared a similar slope, and a regression was fit to the most linear region of 9 measurements of the LVCOPy230630 experiment with an R^2 of 0.998, yielding an apparent activation energy of 47.59 ± 0.85 kcal/mol (199.1 ± 3.5 kJ/mol; Fig. 3.7), implying a closure temperature of 294.3 ± 6.7°C assuming a cooling rate of 10°C/Ma. The delta plot constructed with this activation energy showed a general agreement between the smaller grain-size experiments, seemingly unaffected by the direct contact with sulfur liquid or lack thereof. The more retentive, larger ampoule yielded lower delta values at the same cumulative fractional yield, revealing that a smaller proportion of its helium resided in these least retentive domains that produce the common diffusivity trend on the Arrhenius plot for the three experiments (Fig. 3.8).

The octahedral Rustenburg pyrite is less retentive of helium than the pyritohedral Leadville pyrite. On the down-temperature retrogrades of heating cycles, the Rustenburg pyrite shows clearly developed reproducible linear arrays with the unusually high implied activation energy of 71.8 ± 1.7 kcal/mol (300.6 ± 7.3 kJ/mol; Fig. 3.9: R^2 of 0.998 on 5 measurements). This feature is completely absent from the three experiments of proton-irradiated Leadville pyrite (Fig. 3.7). The delta plot constructed using this activation energy shows

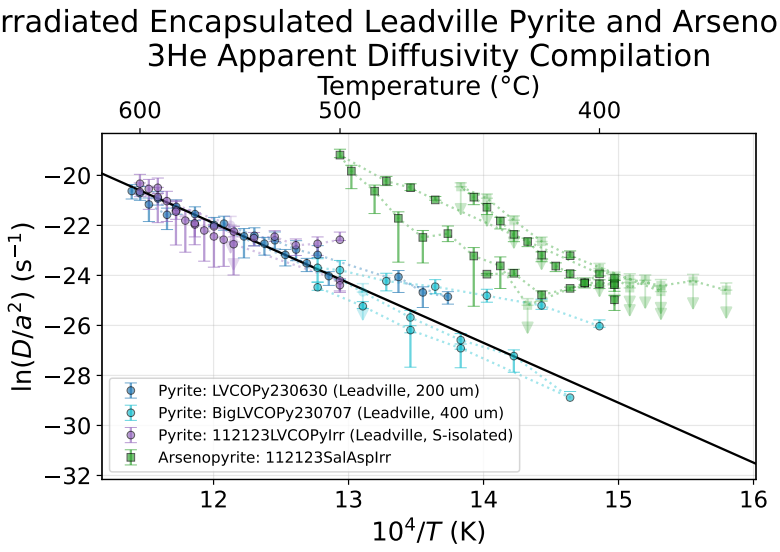

Figure 3.7: Compiled Arrhenius plot of apparent ^{3}He diffusivity for the three irradiated Leadville pyrite step-heat experiments (LVCOPy230630, BigLVCOPy230707, 112123LVCOPyIrr) and the proton-irradiated arsenopyrite experiment (112123SalAspIrr), each with permeation-lag uncertainty bars. Downward-pointing arrows mark steps with an unconstrained lower bound (see §3.2). The solid black line is the regression fit to the most linear region of the LVCOPy230630 experiment (9 points, $R^2 = 0.998$, Ea = 47.59 ± 0.85 kcal/mol), used throughout this section as the shared reference line for all three Leadville pyrite experiments and for arsenopyrite.

a monotonic decrease in delta values throughout the duration of the experiment (Fig. 3.10).

The encapsulated arsenopyrite helium step heat experiment produced an apparent diffusivity Arrhenius plot that showed generally high helium retentivity, but less so than the Leadville pyrites (Fig. 3.7). There were no representative linear regions produced by this experiment that were suitable for linear regression, although the general slope was similar to the Leadville pyrites' and so it was also used as a reference for this experiment. The delta plot using this reference showed complex behavior, where high delta values (far beyond the uncertainty related to the permeation lag of the ampoule itself) were observed for low temperature steps following high temperature steps (Fig. 3.21, row 1, Supplementary Information).

Step heat results for galena (GalTS230630) and chalcopyrite (CCpTS230630) are reported in the Supplementary Information (§3.5). Neither mineral seems to be a promising U-Th/He geochronometer or helium isotope target.

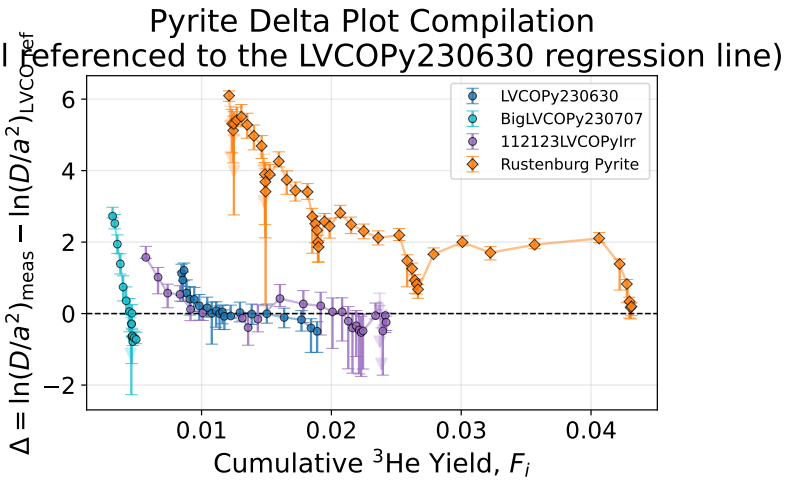

Figure 3.8: Delta plot ($\Delta = \ln(D/a^2)_{\text{meas}} - \ln(D/a^2)_{\text{LVCO ref}}$) vs. cumulative helium yield for all four pyrite step-heat experiments of this study, each referenced to the shared LVCOPy230630 regression line of Fig. 3.7. The three irradiated Leadville pyrite experiments (LVCOPy230630, BigLVCOPy230707, 112123LVCOPyIrr) initially decrease, then plateau near $\Delta = 0$, with the large-grained BigLVCOPy230707 aliquot recording lower diffusivities than the other two at a given cumulative fractional yield. The Rustenburg pyrite (§3.4) departs sharply from all three, reaching Δ values up to +6 at low cumulative yield before decreasing toward $\Delta = 0$ at higher yield.

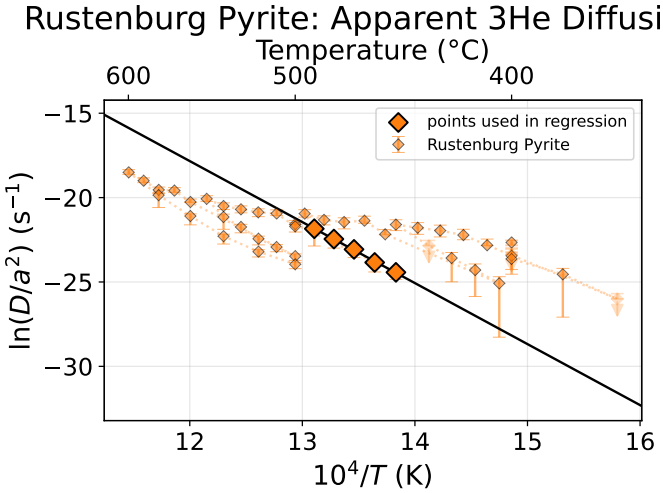

Figure 3.9: Apparent ^{4}He diffusivity Arrhenius plot for the Rustenburg pyrite step-heat experiment (isotope-dilution-quantified on a Pfeiffer Prisma QMS 200 quadrupole mass spectrometer; §3.2). The solid black line is the regression fit to the five down-temperature retrograde steps used to define the Rustenburg activation energy (highlighted points; Ea = 71.8 ± 1.7 kcal/mol, R^2 = 0.998, n = 5), markedly steeper than the shared Leadville/arsenopyrite reference slope of Fig. 3.7.

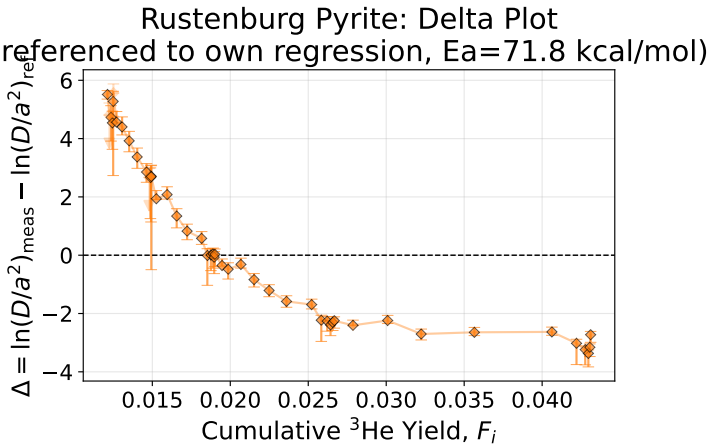

Figure 3.10: Delta plot for the Rustenburg pyrite step-heat experiment, referenced to its own regression (Fig. 3.9; Ea = 71.8 kcal/mol). Delta values decrease monotonically from about +5.5 at low cumulative ^{4}He yield to about −3.5 at high yield, in contrast to the near-flat delta trends of the three Leadville pyrite experiments (Fig. 3.8).

Table 3.6: U-Th/He geochronology results for the four sample suites discussed in the main text: Kidd Creek pyrite, Gympie pyrite, and the Durango apatite and Gogebic (MI-43) hematite standards run alongside them, including U, Th, and Sm (the radiogenic-budget elements quantified by ICPMS, §3.2) plus Zr, also measured as a ratio to spike vanadium; '–' denotes below detection or not measured for that aliquot. The remaining ICPMS trace elements measured the same way (Cu, As, Ag, Sb) are reported separately in the Supplementary Information (Table 3.9) rather than here, to keep this table within the page width. Ages are the atmospheric-He-corrected 'Age (Ma Atm=1Ra Corr.)' value. * marks aliquots analyzed under the early ("old method") protocol that predated the warm atmosphere-degassing step (§3.2, Fig. 3.5), and, for the first four Gympie aliquots, also the Parr-vessel HCl digestion (§3.3); † marks outliers excluded from interpretation (§3.3; §3.2 for Gogebic 041323MG16b). Uncertainties are 2σ and are given to the same decimal place as the value: U, Th, Sm, and eU to 0.01 ppm (0.1 ppm for Durango apatite, and for U and eU in Gogebic hematite), Zr to 0.1 ppm, ^{4}He to 0.1 ncc, and ages to 0.1 Ma, or to 1 Ma where the uncertainty exceeds 10 Ma. The complete compilation, including localities not discussed in the main text, is given in the Supplementary Information (Table 3.10).

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Zr (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)
Kidd Creek Pyrite (Archean VMS, ~2716 Ma host)								
2211KC_Py_ST05_C	4.77	0.92±0.05	1.64±0.04	0.10±0.01	–	1.31	470.8	695±29
2211KC_Py_ST05_A	3.23	0.69±0.01	2.64±0.05	0.13±0.01	–	1.32	305.6	644.1±8.4
2211KC_Py_ST05_B	1.74	0.88±0.05	1.59±0.03	0.20±0.01	–	1.26	159.8	621±26
2211KC_Py_ST05_D	1.98	0.66±0.02	0.91±0.03	0.04±0.00	–	0.88	165.8	811±20
011123KCPyST05C	1.28	0.51±0.02	0.73±0.05	0.38±0.07	9.1±2.6	0.69	158.1	1596±46

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Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Zr (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)
011423KCPyST05A	2.41	1.71±0.02	4.05±0.12	1.18±0.07	28.9±2.2	2.68	389.2	591.0±7.7
011423KCPyST05B	4.04	2.00±0.05	3.38±0.09	2.21±0.13	41.2±3.9	2.81	534.0	465.2±8.8
011423KCPyST05C	3.93	0.83±0.04	1.97±0.06	0.49±0.05	15.5±1.1	1.31	281.6	540±16
011423KCPyST05D	1.46	0.44±0.02	0.85±0.09	0.07±0.03	7.4±2.5	0.64	71.2	731±35
013023KCPyST05As1	1.66	0.90±0.02	7.93±0.21	11.48±0.42	6.8±1.8	2.81	259.4	552±10
013023KCPyST05Cu1	1.05	0.32±0.07	1.00±0.12	0.25±0.09	4.9±2.8	0.56	55.1	860±121
013023KCPyST05Cu2	1.56	0.87±0.02	9.90±0.26	16.54±0.62	8.8±2.2	3.24	268.8	523.4±10.0
Gympie Pyrite (Permo-Triassic orogenic Au)								
013023PVGyPy-1*	3.81	0.06±0.00	0.06±0.01	0.13±0.02	1.2±0.7	0.08	34.7	928±51
013023PVGyPy-2*	1.15	0.27±0.02	0.25±0.04	0.32±0.07	3.6±2.4	0.33	47.7	1008±52
013023PVGyPy-3*	1.81	0.13±0.01	0.16±0.02	0.17±0.05	–	0.17	31.6	763±35
013023PVGyPy-4*	1.82	0.28±0.04	0.36±0.03	0.65±0.05	7.1±3.7	0.37	52.2	638±70
0413PyGy-1	3.54	0.46±0.01	0.59±0.03	0.59±0.07	56.9±3.4	0.60	53.0	233.1±3.4
0413PyGy-2	1.67	0.23±0.01	0.36±0.03	0.37±0.06	40.4±3.2	0.32	15.1	257±13
0413PyGy-3	1.32	0.75±0.02	0.72±0.05	0.77±0.12	72.6±3.3	0.92	30.3	235.2±5.3
0413PyGy-4†	1.85	0.86±0.02	0.79±0.04	0.49±0.07	77.4±5.2	1.05	126.5	623±10
062123GyPyHClb	2.15	0.64±0.02	0.58±0.05	0.50±0.03	89.5±7.2	0.78	32.7	191.1±6.1
062123GyPyHClc	9.43	0.33±0.01	0.39±0.01	0.37±0.02	65.5±2.4	0.43	126.0	306.4±5.0

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Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Zr (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)
230625GyPyHCl	2.38	0.48±0.01	0.46±0.04	0.97±0.05	64.4±3.0	0.59	35.3	227.5±4.5
230625GyPyUnAbr	2.32	0.61±0.01	0.52±0.04	0.78±0.05	61.5±2.8	0.74	50.1	285.0±5.5
230628GyPyHClab	6.17	0.63±0.01	0.72±0.03	0.78±0.05	96.4±3.8	0.80	133.6	264.7±3.6
230628GyPyHClc	4.03	0.58±0.02	0.48±0.03	0.65±0.04	65.0±6.5	0.70	79.0	275.6±9.1
230628GyPyUnAbrA	2.53	0.31±0.01	0.37±0.04	0.40±0.03	64.0±3.1	0.40	31.8	304±12
230628GyPyUnAbrB	1.57	0.47±0.01	0.41±0.07	0.63±0.05	64.4±6.0	0.57	23.9	261.8±8.9
Durango Apatite Standard (accepted age 32.1 Ma; Farley 2000)								
2211DApA*	0.81	10.4±0.1	193.6±3.8	25.0±0.9	–	56.8	141.4	31.7±0.5
2211DApC*	0.82	10.8±0.1	206.2±4.1	24.5±0.7	–	60.3	181.2	36.8±0.6
2211DApD*	0.31	12.6±0.2	272.2±4.7	31.5±1.1	–	77.9	81.5	29.4±0.4
2022StDApA*	0.30	12.6±0.2	230.7±5.5	71.5±2.0	–	68.0	69.6	29.5±0.6
011123DAp1	0.53	10.2±0.2	206.6±7.3	135.6±5.6	10.5±7.2	59.8	96.7	31.2±0.9
011123DAp2	0.58	10.9±0.1	200.4±5.4	127.0±5.0	–	59.0	107.4	32.2±0.7
011123DAp3	0.79	11.2±0.1	226.2±6.0	120.2±4.4	–	65.5	164.7	32.7±0.7
011423DAp1	0.93	9.9±0.1	196.6±5.1	133.1±4.8	–	57.1	165.9	32.2±0.7
011423DAp2	1.51	11.1±0.1	205.0±4.5	129.1±4.8	–	60.3	286.7	32.5±0.6
011423DAp3	0.98	11.5±0.1	223.2±5.9	123.7±4.4	–	65.1	203.5	32.9±0.7
011423DAp4	0.81	11.8±0.1	235.8±7.0	125.7±5.2	8.9±4.0	68.4	170.7	31.8±0.8

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(continued)

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Zr (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)
062123DAp	0.30	13.4±0.2	266.6±7.4	142.9±10.0	5.0±2.8	77.4	72.7	32.3±0.7
062523DAp	0.85	11.1±0.1	211.8±5.3	123.8±8.2	2.7±1.7	61.9	161.3	31.6±0.7
062823DAp	–	–	–	–	–	–	177.6	31.9±0.4
Average for > 0.5 mg	–	–	–	–	–	–	–	32.1±0.2
Gogebic Iron Range BIF Hematite Standard (MI-43; 571 Ma)								
2211HemMSA	1.37	11.8±0.1	0.04±0.01	0.34±0.03	–	11.8	750.6	450.1±4.8
2211HemMSB	2.12	8.8±0.1	0.06±0.01	0.34±0.02	–	8.8	1070.2	554.8±7.0
041323MG16a	0.46	9.2±0.1	0.00±0.00	1.56±0.19	–	9.2	240.7	549.4±7.7
041323MG16b†	0.49	2.2±0.1	0.09±0.08	0.87±0.15	19.4±5.0	2.2	188.7	1540±35
041323MG16d	0.28	11.3±0.2	0.00±0.00	1.55±0.34	–	11.3	174.3	531.8±8.3
041323MG16e	0.42	12.1±0.5	0.44±0.27	1.86±0.46	31.1±24.3	12.2	257.1	490±18
062123MS	0.32	10.1±0.1	0.00±0.26	1.81±0.10	–	10.1	187.1	561.8±5.5
230625MSHm	0.23	11.4±0.1	0.00±0.40	1.81±0.18	–	11.4	148.9	549.4±6.0
230628MSHm	0.42	9.7±0.1	0.00±0.20	1.63±0.17	–	9.7	245.0	579.5±6.4

*Excluded: analyzed under the early (old) protocol (§3.2, §3.3).

†Excluded: outlier (§3.3, Fig. 3.6).

Pyrite/Arsenopyrite U-Th/He Ages from Kidd Creek

Twelve aliquots of Kidd Creek pyrite/arsenopyrite were dated by U-Th/He step-degassing (Table 3.6) and give atmospherically corrected ages ranging from 465 to 860 Ma with the exception of one aliquot (011123KCPyST05C) that gives an older measured age of 1596 Ma (Fig. 3.18). The mean age of the remaining eleven aliquots is 639 ± 36 Ma, 1σ. These dates are decisively older than the independent regional apatite cooling-age of 392 ± 32 Ma (Sturrock et al., 2024), yet far younger than the ∼2716 Ma age of the host rhyolite (Hannington et al., 1999) (§3.4).

Arsenopyrite U-Th/He Ages from Gympie

Sixteen aliquots of Gympie arsenopyrite were dated by the same method (Table 3.6; Fig. 3.11). The first four aliquots analyzed (013023PVGyPy-1 through -4) were prepared under an early iteration of the protocol that predated both the tube-blank atmospheric helium degassing step and the Parr-vessel HCl digestion step (§3.2) later adopted to fully dissolve fine U-Th-Sm-bearing rutile inclusions. The measured ages of these aliquots are unreasonable: ranging from 638 to 1008 Ma (mean 834 Ma, n=4), though it is unclear to what extent the bias is due to atmospheric contamination, incomplete digestion of rutile, or both. The ages of these four early aliquots are therefore considered unreliable and are excluded from further interpretation.

The remaining twelve Gympie aliquots, prepared after both protocol refinements were adopted, define a substantially younger and tighter age population. Eleven of these twelve give atmospherically corrected ages between 191 and 306 Ma (mean 258 ± 35 Ma, 1σ, n=11; these eleven are plotted in Fig. 3.11). The twelfth, 0413PyGy-4, is a clear outlier at 623 Ma and is also anomalous in its measured 4He concentration (68.4 ncc/mg), roughly three times that of the next most helium-rich aliquot in the accepted Gympie suite (230628GyPyHClab, 21.65 ncc/mg: supplementary Fig. 3.22). We interpret this combination as evidence for helium implantation from a U-Th-Sm-rich accessory phase concentrated at the sulfide grain margin and not fully digested during trace-element sample preparation. Aliquot 0413PyGy-4 is therefore also excluded from further interpretation.

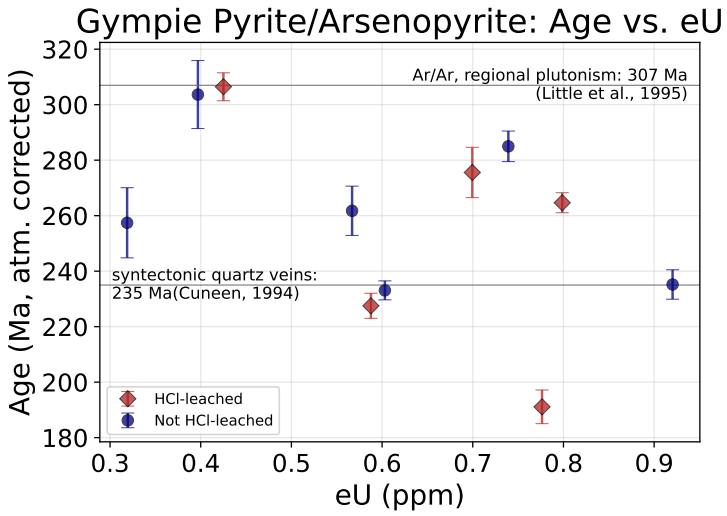

Figure 3.11: Atmospherically corrected age vs. eU for the eleven reliable, newer-protocol Gympie arsenopyrite aliquots (Table 3.6), split by HCl-leaching. Reference lines mark the 235 Ma K/Ar syntectonic quartz-vein age (Cuneen, 1994; Holcombe et al., 1997) and the 307 Ma Ar/Ar regional plutonism age (Little et al., 1995). The four early-protocol aliquots and the 0413PyGy-4 outlier (§3.3) are excluded here; see Fig. 3.22 (SI) for the outlier.

3.4 Discussion

Effectiveness of the Ampoule Method in Stabilizing Sulfides

The methodology for the heating treatments and SEM imagery of pyrite, arsenopyrite, and marcasite surfaces following thermal treatments in different conditions is discussed in §3.1s. The SEM imagery reveals deformed, variably raised, ~10 µm cracked surfaces on pyrite and marcasite that are heated in open air (open-air marcasite not shown), and for arsenopyrite heated in a closed capsule without a realgar buffer (Fig. 3.3). Pyrite and marcasite samples that were heated in ultra-high vacuum developed recessive ~1 µm rounded craters (Fig. 3.3). Contrarily, images of thermally treated, encapsulated pyrite and arsenopyrite with added fused sulfur or realgar show no obvious distinction from the control group: both of which showcase conchoidal fracture (Fig. 3.3). The buffered, encapsulated marcasite sample shows a raised layer of angular (presumably pyrite) crystals growing atop its surface (Fig. 3.3); evidence of phase transformation is present in all three heat-treated marcasite populations, demonstrating that the encapsulation method is not effective in preventing phase breakdown at elevated temperatures for marcasite. These experiments are not conclusive evidence that thermal phase transformation is prevented for pyrite or arsenopyrite (indeed, arsenopyrite's equilibrium stoichiometry

varies with temperature, suggesting that some reaction during step heats would be expected for this phase: Morimoto and Clark (1961) and Barton (1969)). However, the step heats presented here are consistent with the interpretation that reaction is suppressed, encouraging its application for helium step heat experiments.

Remarkable Helium Retentivity of Pyrite and Arsenopyrite

The central finding of this chapter's step heat experiments (§3.3) is that pyrite (and to a lesser extent, arsenopyrite) are extraordinarily retentive of helium against thermal loss: cumulative fractional yields never exceeded 5% for any pyrite or arsenopyrite experiment, even after repeated heating cycles at temperatures between 500-600 °C. These cumulative fractional yields are difficult to interpret without a reference diffusive timescale, so these step heat results are contextualized against the helium retentivity of other phases in Fig. 3.12. This in effect places the step heat experiments of the sulfides of this study on the same diffusive-dose axis as step heat experiments of several other minerals that are often used for U-Th/He dating: single-crystal and polycrystalline hematite, vacuum- and hydrothermally-degassed goethite, zircon, and monazite. To do this, the activation energy of hematite from Farley (2018) is used to convert the time and temperature of each step heat into a cumulative diffusive time, expressed as a Fourier number ($Fo = \frac{D}{a^2}_{\text{ref}} t$). The Fourier number can also be thought of as a dimensionless diffusive time, real time divided by the diffusive timescale of a reference domain, $\tau = \frac{a^2}{D}$. τ here is referenced to a 1 μm single-crystal hematite domain. The aim of such a plot is to flatten complex heating histories into a single value so that phases with complex helium migration behavior can be compared on a common scale. Admittedly, this is not strictly correct because the activation energies of different phases (and indeed diffusive domains within a single phase) are not identical; however, this plot preserves the relative ordering of phases regardless of the choice of reasonable activation energy: we tested 149-171 kJ/mol. Therefore, this plot is a useful tool for comparing the relative retentivity of different phases, even though the absolute diffusive dose is not strictly correct.

On this comparison, pyrite and arsenopyrite sit at or beyond the retentive extreme of the entire compilation. Both plot at diffusive doses orders of magnitude beyond where the reference 1 μm single-crystal hematite domain (Farley, 2018) has gone fully to completion, and beyond essentially the full

range spanned by zircon (Guenthner et al., 2013), comparable to or exceeding that of monazite (Farley, 2007). The polycrystalline hematite and vacuum-degassed goethite aggregates, by contrast, sit orders of magnitude further to the left. The hydrothermally degassed goethite (Chapter 2) plots closer to but less retentive than the sulfides here.

Taken together, this comparison indicates that sulfur-buffered, silica-encapsulated pyrite and arsenopyrite are not merely retentive relative to other sulfides (galena, chalcopyrite, chalcocite, sphalerite; §3.2s), but are among the most helium-retentive phases yet characterized by step heating, on par with or exceeding zircon and monazite and substantially exceeding both single-crystal and polycrystalline hematite and vacuum-degassed goethite. This is the empirical basis for treating pyrite (and, by extension, arsenopyrite) as a credible U-Th/He geochronometer in this chapter's Kidd Creek and Gympie case studies (§3.3, §3.3), notwithstanding complexities introduced by the inclusion-hosted origin of the radiogenic budget (§3.4).

Inferring Bulk Helium Migration Behavior from Low Fractional Yields

Many of the step heats on encapsulated sulfides presented here show very high helium retentivity, and for those experiments only a very small fraction of the total yield evolved (<5% for the arsenopyrite and Rustenburg pyrite; <2% for the Leadville pyrite), similarly to previous work on highly helium retentive phases such as baddeleyite (Baughman and Flowers, 2018). For these sulfides, a mechanism responsible for bulk helium loss besides desulfurization cannot be identified. For some well-studied phases like zircon and titanite, Arrhenius plots of the earliest fractions of evolved helium often reflect processes or conditions that are accessory to the bulk mechanism for helium migration through the mineral, such as radiation damage annealing or surface roughness (Reiners and Farley, 1999; Guenthner et al., 2013), where the primary mechanism for helium migration through the pristine mineral is volume diffusion. These apparent departures from the bulk helium migration behavior can be caused by properties of natural samples, like surface roughness or fine-scale cracks and fractures (Reiners and Farley, 1999) that produce complex grain size distributions even in carefully sieved, (apparently) single grains. Contrastingly, some other complexities in the helium loss behavior at early loss fractions can be ephemeral and altered during experimental heating, such as the annealing of

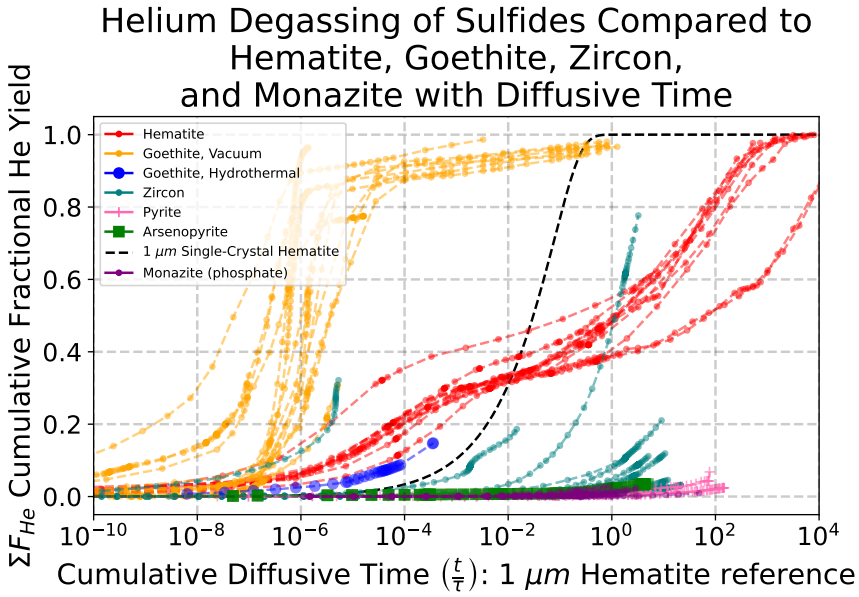

Figure 3.12: Cumulative fractional He yield vs. cumulative diffusive time (Fourier number, $Fo = D_{\text{ref}}t/a^2_{\text{ref}}$) for each heating step of every experiment plotted, referenced to a common 1 μm single-crystal hematite domain diffusivity (Farley, 2018) ($\ln(D_0/a^2) = 0.66$, $E_a = 171$ kJ/mol). Expressing step power this way, rather than as raw time or temperature, puts experiments run under very different heating schedules, temperatures, and durations onto one common, directly comparable dose axis: a phase reaching a given cumulative yield only at a much larger Fo is more helium-retentive than one reaching the same yield at a smaller Fo, regardless of how either was actually heated. Because cumulative fractional yield is dimensionless, isotope-specific normalization is not required to combine step heats of different helium isotopes on one axis: ^{3}He step heats of proton-irradiated pyrite, arsenopyrite, zircon, monazite, and vacuum-degassed hematite/goethite are plotted alongside ^{4}He step heats of the (non-irradiated, isotope-dilution-quantified) Rustenburg pyrite and the hydrothermally degassed goethite. Plotted data: this chapter's own sulfur-buffered pyrite and arsenopyrite step heats; 11 vacuum step heat aliquots of polycrystalline hematite from the Grand Canyon, Arizona (Farley and Flowers, 2012) and the Gogebic Range, Michigan (Farley and McKeon, 2015); 9 vacuum step heat aliquots of polycrystalline goethite (Win-06-01b and Capão, Minas Gerais, Brazil) (H. Monteiro, 2017; Farley et al., 2024); the hydrothermal step heat of Win-06-01b goethite from Chapter 2 of this dissertation; and proton-irradiated zircon (Guenthner et al., 2013) and synthetic monazite (Farley, 2007) step heats. The dashed black reference curve is the forward-diffusion cumulative-yield curve for the reference 1 μm hematite domain itself (i.e. Fo vs. its own resulting cumulative yield, with no data plotted against it), included so every other curve's retentivity can be read directly relative to this single benchmark.

radiation damage (Shuster and Farley, 2009). For radiation damaged samples, understanding the migration behavior of helium before and during annealing is essential to understand the sample's retentivity in nature and therefore the earliest fraction of helium evolved in step heats is the focus of many studies (Shuster and Farley, 2009; Flowers et al., 2009; Flowers et al., 2007; Shuster et al., 2006).

The fraction of total helium evolved before the predominant migration mechanism's behavior can be characterized in helium step heats varies: Guenthner et al. (2013) show that significantly non-zero delta or ln(r/r0) values persist in zircon until about 2% of the total helium is degassed for typical natural samples; Reiners and Farley (1999) observe apparent diffusivities of helium in titanite converging to linearity after 5% of the total helium is evolved. Similar experiments on proton-irradiated synthetic samples seem not to be marred by the same complexities: helium step heat data for synthetic monazite converge nearly immediately to a linear array at very low cumulative helium loss fractions (Farley, 2007).

The step heat experiments on pyrite and arsenopyrite in this study would have benefited from longer steps at high temperature to degas more helium to better understand the bulk behavior of helium, but even at small loss fractions some key observations about helium loss in sulfides can be made.

Comparison of Helium Step Heat Experiments of Natural Pyrites

In this work, four encapsulated, sulfur-buffered aliquots of pyrite taken from two localities are subjected to helium step heat experiments. Step heats for the three irradiated pyrites from Leadville, CO are broadly similar, but contrast markedly with step heats of non-irradiated pyrite from Rustenburg, South Africa; therefore, differences between pyrites of different localities are discussed before the variation within the aliquots from Leadville, CO.

Radiogenic helium evolution during step heats of octahedral pyrite from Rustenburg, South Africa shows that this sample is less retentive than the pyritohedral pyrite from Leadville, Colorado (Fig. 3.8). Apparent diffusivities of these samples also have fundamentally distinct geometries when plotted in an Arrhenius plot. The Rustenburg pyrite shows a striking feature of very steep slopes (corresponding to an activation energy of 300.6 $\pm$ 7.3 kJ/mol) on retrograde-prograde heating cycles at temperatures lower than the previous maximum temperature

step heat, whereas the maximum apparent diffusivity for a given temperature seemed to follow a much shallower slope, similar to the lower activation energy of the Leadville, CO pyrite: 199.1 ± 3.5 kJ/mol. The geometry of the step heats in an Arrhenius plot for the irradiated Leadville, CO pyrite is generally simpler: generally slowly decreasing in a delta plot, or for isothermal steps (Fig. 3.8) as would be expected for helium release governed by MDD systematics (Farley, 2018).

Natural pyrite is known to develop fractures and crystalline dislocations in metamorphic and experimental conditions (Gill, 1969; McClay and Ellis, 1983; Barrie et al., 2011). Abundant fractures of micron-scale widths were clearly visible in the Rustenburg pyrite of this study (Fig. 3.13). Those pyrites were surprisingly easy to crush, presumably breaking easily along fine pre-existing fractures. Although pyrites from Leadville are known to be associated with faulted dolomites, especially along deformation planes which likely served as pathways for mineralizing hydrothermal fluids (Nash, 1979), the samples used here were exceptionally sturdy and visually homogeneous under SEM. Rare veins and coatings of dolomite within and around the pyrites were present but were avoided when preparing the sample for irradiation.

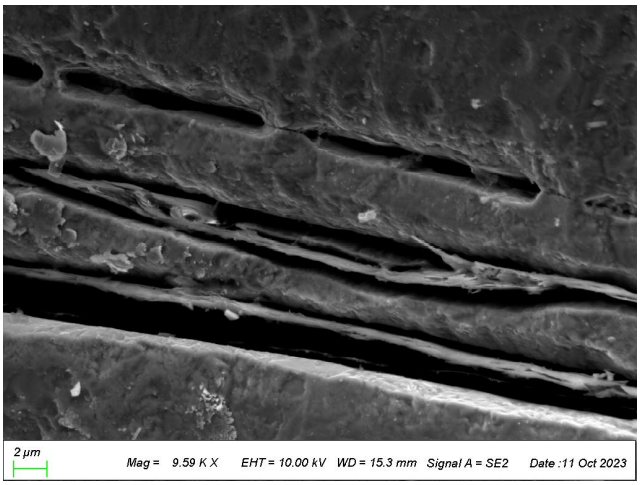

Figure 3.13: Secondary-electron SEM image of Rustenburg (Bushveld Complex) pyrite prior to heat treatment (control), showing the pervasive, micron-scale fractures characteristic of this sample (§3.4). These pre-existing fractures are interpreted as the pathways for the short-circuit helium diffusion mechanism invoked below to explain the Rustenburg pyrite's distinct retrograde Arrhenius behavior (Fig. 3.9).

Within this work, a pipe/short-circuit diffusion mechanism (J. K. W. Lee

and Aldama, 1992) is favored to explain the helium loss behavior of the Rustenburg pyrite, where helium populates the sample's fine fractures at increasing temperatures due to the primary helium migration mechanism, perhaps volume diffusion through crystalline pyrite. This primary mechanism is interpreted to be the same for all pyrite samples regardless of locality. On retrograde heating steps, short-circuit diffusion of helium out of fine fractures in the Rustenburg pyrite is hypothesized to become the rate-limiting process.

Leadville, CO Pyrite: Evidence for a Grain Size Control?

Three helium step heat experiments for the irradiated Leadville, CO pyrite are discussed here. Grain size was varied by a factor of 2 between a half-prism width of 200 microns for LVCOPy230630 and 400 microns for BigLVCOPy230707 to investigate the dependence of helium migration in pyrite on grain size. The aim of the third experiment, 112123LVCOPyIrr, was to test the effect of isolating the pyrite from the liquid sulfur during the experiment to investigate the role of dissolution and (re)crystallization of pyrite by the ampoule's sulfur fluid. Silica glass shards were added to this ampoule below and above the pyrite aliquot, forcing the pyrite grains to remain in the middle of the ampoule, and not at the edges/bottom, where sulfur liquid is presumed to accumulate. All three samples overlap in their Arrhenius plots (Fig. 3.7), and all three samples record declining diffusivities consistent with MDD systematics (Fig. 3.8). The larger grained aliquot recorded lower cumulative fractional yields at these common apparent diffusivities. The smaller-grained aliquots, LVCOPy230630 and 112123LVCOPyIrr, have essentially identical delta plots (Fig. 3.8), providing no evidence for a sulfur-liquid control in driving helium loss for these experiments.

Very little helium was evolved from any of the exceptionally retentive Leadville pyrite samples: none evolved more than 3% of its cumulative fractional yield, in spite of the intense heating schedules (at temperatures up to 600°C) they were subjected to. These samples are only resolved by grain size. Because all Arrhenius plots overlap, it is likely that the differences in the delta plots are due to the relative abundance of the smallest domain size: the smaller aliquots contained a greater proportion of the least retentive domains interrogated by these experiments. One potential explanation is that only helium on extraordinarily small dusty material carried on the surfaces of the larger aliquots is degassed, and thus the primary control on the delta plot is the surface area to volume ratio. This is consistent with the observation that the

smaller aliquots, with half the half-prism width of the larger aliquot, record Δ=0 at twice the cumulative fractional yield of the larger aliquot (Fig. 3.8).

Relevance of Helium Step Heat Data of Encapsulated, Sulfur-Buffered Sulfides in Nature

Given the high helium retention of pyrite and arsenopyrite shown in this work, thermal helium loss in nature may be rare. A more likely mechanism for helium loss in these common sulfides in nature may indeed be recrystallization. Pyrite in metamorphic conditions commonly recrystallizes and may either shrink or grow (Craig et al., 1998; McClay and Ellis, 1983). Until more work has characterized helium loss (or gain) for recrystallizing pyrite or arsenopyrite in metamorphic conditions, helium should cautiously be considered to have behaved as an open or partially open system in sulfides during metamorphic events, although O. Yakubovich et al. (2020) report that helium in the Uzelga pyrite was retained through metamorphism.

Arsenopyrite

Arsenopyrite (FeAsS) adopts a distorted, lower-symmetry variant of the marcasite structure in which arsenic substitutes irregularly for sulfur (Morimoto and Clark, 1961), in contrast to pyrite's simple cubic lattice. Because the degree of distortion scales with arsenic content (Morimoto and Clark, 1961), helium diffusion is likewise expected to vary with composition. Unfortunately, equilibrium arsenopyrite composition is not itself fixed: arsenic content increases systematically with formation temperature (Barton, 1969; Morimoto and Clark, 1961), so the same crystal-chemical property expected to govern helium retentivity should also vary with an arsenopyrite sample's thermal history, complicating any attempt to generalize helium retentivity from a single experimental aliquot to arsenopyrite as a mineral species. Similarly, the composition of the arsenopyrite subjected to step heats herein may itself have changed during the experiment, further distancing these experimental results from helium loss from arsenopyrite in nature.

With that caveat, the step heat results obtained here (§3.3) show the encapsulated arsenopyrite experiment was less helium-retentive than the Leadville pyrite experiments of this study, and it lacked a well-defined linear Arrhenius region. Still, arsenopyrite remained markedly more retentive of helium than any other sulfide examined in this study besides pyrite, in agreement with the

relative helium retentivity ranking of sulfides reported by O. V. Yakubovich et al. (2019).

Qualitatively, arsenopyrite appears to be more retentive of helium than most zircon (Fig. 3.12; Guenthner et al., 2013).

Thermal Helium Migration Systematics for Other Sulfides

Discussion of the thermal helium migration systematics of the four remaining, non-pyrite/arsenopyrite sulfides of this study (chalcopyrite, chalcocite, sphalerite, and galena) is given in the Supplementary Information (§3.5); neither copper sulfide nor galena is considered a promising U-Th/He geochronometer or helium isotope target.

Origin of Radiogenic Helium in Sulfides

The pyrite and arsenopyrite of this study are not, themselves, appreciably radiogenic minerals. Pyrite in particular has very little affinity for U, Th, or Sm, which are large, highly charged cations poorly suited to substitution for Fe^{2+}. Consistent with this, only the aliquots (of any locality or mineral in the compilation, main text or SI) that host measurable U-Th-Sm-bearing accessory phases returned appreciable eU and, correspondingly, appreciable radiogenic 4He. For example, the Leadville pyrite samples are the purest, most visually inclusion-free grains, selected for the sulfur-buffered step heat experiments of §3.2, and they are essentially non-radiogenic. The U-Th/He ages for sulfides may not date growth of the sulfide lattice directly, even if it is perfectly retentive of helium, because some of the helium retentivity of the sample will be determined by the size, mode, and connectivity of inclusions. Sulfide ages are easiest to interpret for the case when the radiogenic helium-source inclusions are smaller than 10 microns across, when only the helium retentivity of the sulfide is significant (Fig. 3.19).

Backscattered-electron imagery of the Kidd Creek arsenopyrite/pyrite directly shows a surprising abundance of apatite inclusions. Superficially, hand-picked crystals appear as optically clean, blocky euhedral crystals set within a finer silicate groundmass (Fig. 3.14), but in cross-section the same grains are pervasively spongy, riddled with abundant tiny inclusions rather than being the pure, homogeneous pyrite their exteriors suggest. These inclusions are dominantly apatite (Figs. 3.15, 3.16), ranging in size from sub-6 micron grains at the limit of SEM resolution up to ∼30 micron half-width. Because apatite preferentially

incorporates U, Th, and especially Sm relative to coexisting sulfide and silicate phases, even a small volume fraction of included apatite can dominate a pyrite aliquot's bulk eU budget and, correspondingly, its radiogenic 4He production.

The Gympie arsenopyrite hosts a texturally distinct, but analogously inclusion-hosted, radiogenic budget. SEM imagery (Fig. 3.17) shows abundant fine, needle-like rutile inclusions. These rutile needles are the most likely U-Th-Sm-He source for the Gympie samples, and variable inheritance of older rutile engulfed by later sulfide may be responsible for its observed age dispersion (§3.3).

This inclusion-host origin of radiogenic helium in sulfides has a direct methodological consequence: sulfide U-Th/He geochronology cannot be treated as a single, generic protocol applied uniformly across localities. The identity, size distribution, and digestion behavior of the accessory phase or phases actually carrying the U-Th-Sm budget are properties of each deposit's own mineral paragenesis, and differ sharply even between the two localities studied here. Reliable sulfide U-Th/He geochronology of a new locality therefore requires SEM characterization of the dated material to identify its radiogenic inclusion population before analysis, and a dissolution protocol tailored to that population's specific mineralogy and grain size.

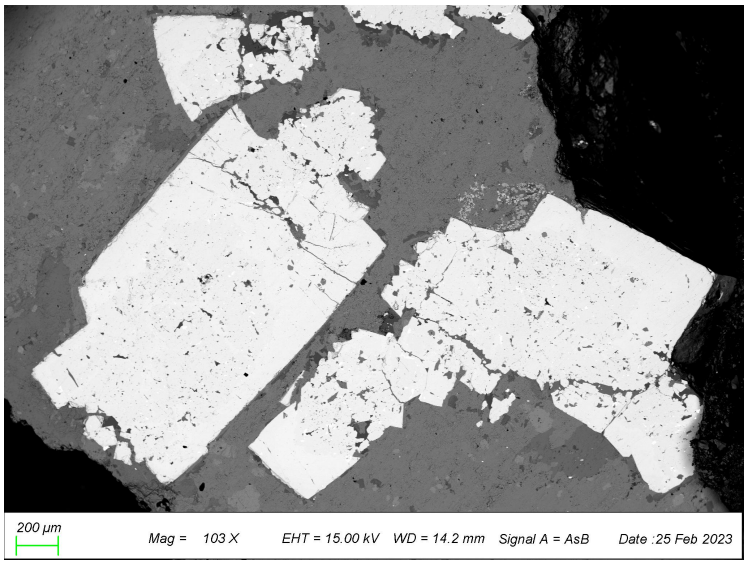

Figure 3.14: Backscattered-electron image of Kidd Creek pyrite (ST05) in cross-section: optically clean, blocky euhedral grains set within a finer-grained groundmass are, internally, spongy and riddled with abundant tiny (predominantly apatite) inclusions rather than homogeneous pyrite.

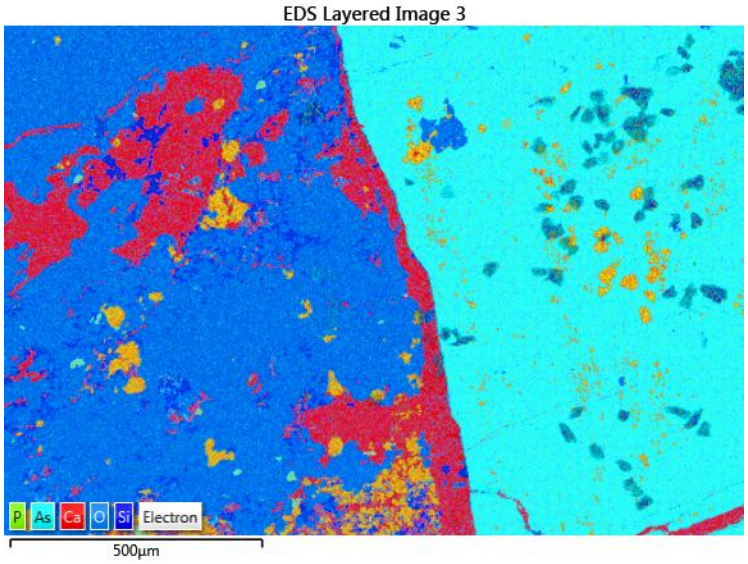

Figure 3.15: Backscattered-electron image of Kidd Creek pyrite showing apatite inclusions, the dominant host of the aliquots' U-Th-Sm budget.

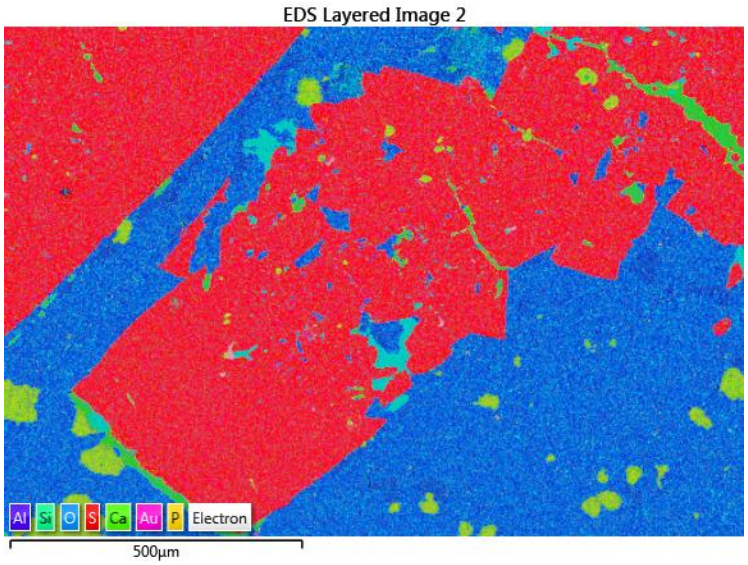

Figure 3.16: Backscattered-electron image of apatite inclusions in Kidd Creek pyrite, ranging from sub-6 micron grains up to ∼30 micron half-width.

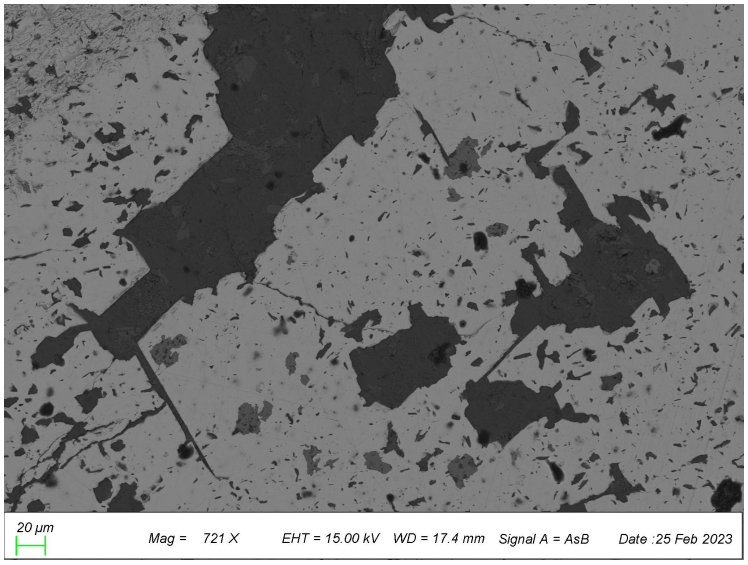

Figure 3.17: Backscattered-electron image of Gympie pyrite showing abundant fine, needle-like rutile inclusions and much larger, blocky feldspar inclusions. Ineffective rutile digestion is implicated in the anomalously old ages discussed in §3.3.

Interpretation of Kidd Creek Pyrite/Arsenopyrite Ages: Mixing Between a Retentive, He-Th-Poor Host and Apatite Inclusions

The Kidd Creek pyrite/arsenopyrite ages are not scattered randomly around some single "true" age. Eleven of the twelve dated aliquots define a coherent array in which the Th- and He-richest aliquots give the youngest ages and the Th- and He-poorest aliquots give the oldest, ranging from 465 to 860 Ma (Fig. 3.18). The youngest of the eleven approach, but are always higher than, the 392 ± 32 Ma regional apatite (U-Th)/He cooling age reported by Sturrock et al. (2024) for sample 01-OE-69, sampled only ∼90 km from Kidd Creek; the oldest approach ∼860 Ma, implying that this population is significantly more helium-retentive than the apatite it includes.

Given the inclusion-hosted origin of radiogenic helium established above (§3.4), the natural reading of this array is a two-component mixture: an old end-member that is essentially He-Th-poor but comparatively retentive, and a population of included apatite with a much younger (∼392 Ma) cooling age (Sturrock et al., 2024). A Kidd Creek aliquot's bulk age then falls between these two end-members depending on what fraction of its total measured Th and He budget was hosted in (and retained by) apatite inclusions since 392 Ma, versus alpha-ejected into the surrounding, retentive sulfide host and retained

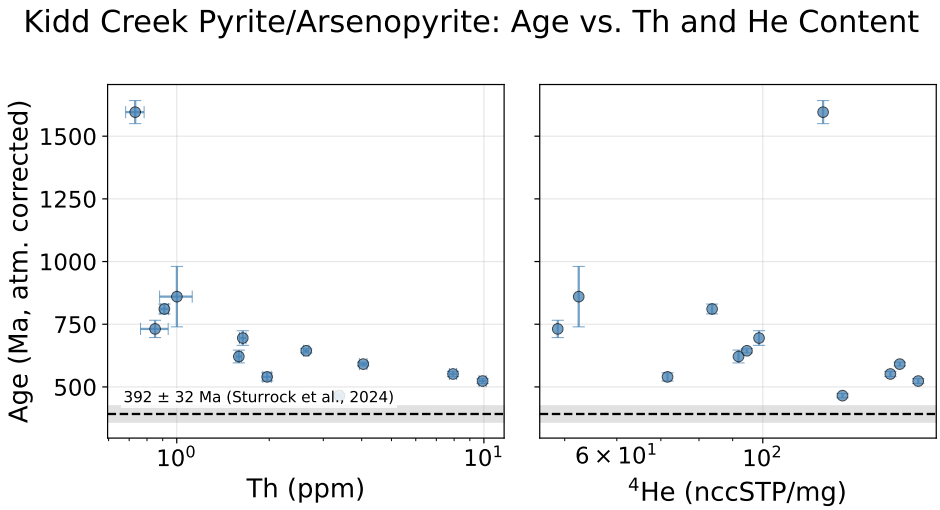

Figure 3.18: Atmospherically corrected Kidd Creek pyrite/arsenopyrite age vs. Th content (left) and vs. measured 4He content (right; nccSTP/mg), sharing a common age axis (§3.3). The shaded band is the independent regional apatite cooling-age estimate of 392 ± 32 Ma for sample 01-OE-69 (Sturrock et al., 2024). Both panels show the same sense of anti-correlation: Th- and He-rich aliquots give younger ages, Th- and He-poor aliquots give older ages.

there since it was crystallized.

A simple model shows that lower age paired with higher Th and He can be explained with a single-age apatite population of variable inclusion size. Fig. 3.19 extends the alpha-ejection retention framework of §3.4 to compute the apparent bulk He age of a hypothetically perfectly helium-retentive sulfide aliquot hosting a single apatite inclusion of half-prism radius R, assuming the sulfide and apatite crystallized 2700 Ma ago, but the apatite itself only began retaining its own self-implanted helium after 380 Ma. At small R, nearly all the produced helium recoils out of the inclusion into the retentive sulfide, and the aliquot's bulk age approaches the old, 2700 Ma crystallization age, but at large R, an increasing fraction of the inclusion's own production is retained within the apatite inclusion itself, so the bulk age falls toward 380 Ma. Because a larger inclusion also simply contains more total U-Th-Sm mass, the same increase in R that drives the age down simultaneously drives up an aliquot's bulk Th and He content, producing exactly the anti-correlation seen in Fig. 3.18.

At the $\leq$30 μm half-width apatite inclusion size actually observed in the Kidd Creek material (§3.4, Fig. 3.16), Fig. 3.19 predicts an apparent age of $\sim$1.6

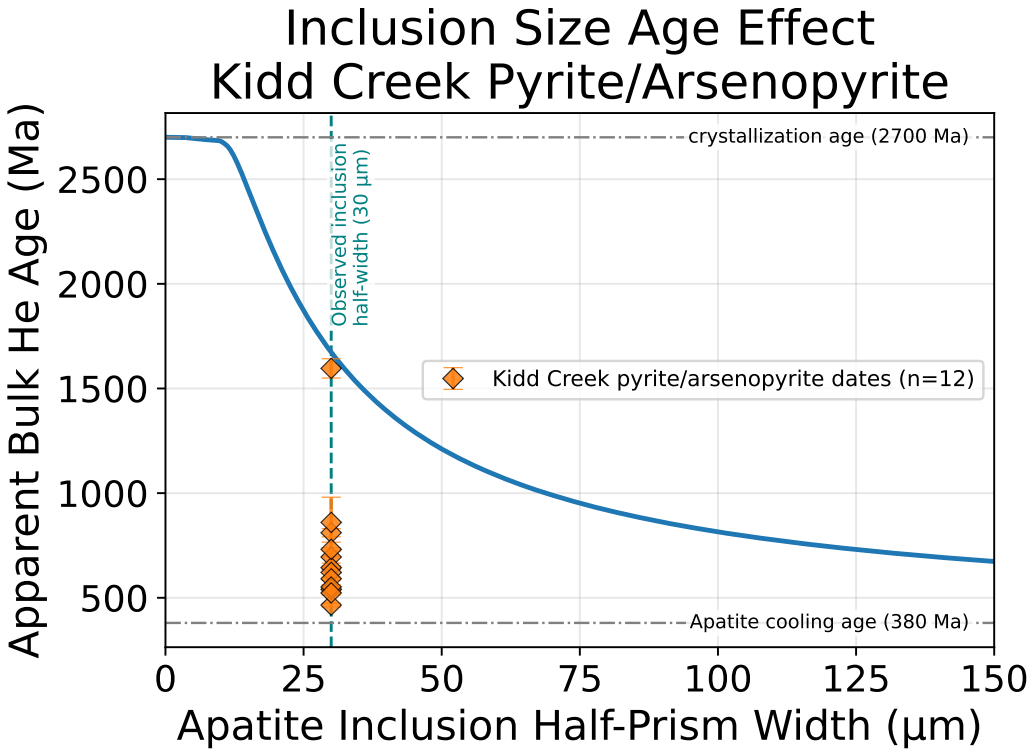

Figure 3.19: Modeled apparent bulk He age of a quantitatively He-retentive, 2.7 Ga Kidd Creek sulfide aliquot hosting a single fluorapatite inclusion serving as its U-Th-Sm source. The apparent age of such a system is a function of the inclusion's half-prism radius R. The alpha-ejection retention fraction $F_t(R)$ (fraction of produced helium self-implanted within the inclusion) is computed via a Monte Carlo simulation of 10^6 decays uniformly distributed by volume within a spherical inclusion with isotropic alpha-ejection directions, using per-isotope (^{238}U, ^{235}U, ^{232}Th, ^{147}Sm) stopping distances for fluorapatite and a fixed U:Th:Sm ratio of 1:2:2. This reproduces Farley et al. (1996)'s closed-form retention factor for $R \geq S$ (the stopping distance) and remains accurate for $R < S$, where the closed form is not valid. Helium is then accumulated separately for the inclusion-retained fraction $F_t(R)$, assuming retention only since 380 Ma, and the sulfide-retained fraction $1 - F_t(R)$, assuming retention since a 2700 Ma inclusion/host source age. The two helium contributions are summed to calculate a single apparent bulk age. Reference lines mark the 2700 Ma source age and 380 Ma cooling age end-members. A vertical line marks the maximum $\sim$30 μm half-width apatite inclusion size actually observed at Kidd Creek (Fig. 3.16). Eleven of the twelve pyrite/arsenopyrite ages fall well below the model's closed-system prediction, but the oldest age (011123KCPyST05C, 1596 ± 46 Ma) agrees with the model prediction (§3.4).

Ga or older for a sulfide host that is perfectly closed to helium loss over its entire 2.7 Gyr history. Eleven of the twelve measured Kidd Creek ages, plotted at this inclusion size in Fig. 3.19, fall far short of this prediction. This model cannot produce ages as young as 465-860 Ma, so most of these sulfides must have lost a substantial fraction of their radiogenic helium since crystallization, rather than behaving as a closed system simply mixing with variably sized apatite inclusions.

The twelfth, oldest aliquot, 011123KCPyST05C, is the one exception: its 1596 ± 46 Ma age sits close to the closed-system prediction above at the observed $\sim$30 μm inclusion size, as though this one aliquot alone retained essentially all of its radiogenic and inclusion-sourced helium since crystallization while the rest of the suite lost helium over time. A single old date is weak evidence for anything on its own: an anomalously old U-Th/He age is also the expected signature of incomplete acid digestion, which would leave some U-Th-Sm-bearing residue undissolved, overestimating the apparent age. We cannot distinguish between these two explanations with the data at hand.

This chapter's own step heat experiments show that sulfur-buffered pyrite is consistently among the most helium-retentive phases measured with step heat experiments, while arsenopyrite, though still strongly retentive, is notably less so (Fig. 3.7, Fig. 3.12; §3.4). If 011123KCPyST05C were confirmed to be closer to pure pyrite than the rest of the Kidd Creek suite, this retentivity contrast would offer a ready explanation for why it alone approaches closed-system behavior. Indeed, the arsenic measurement (Table 3.9) supports this interpretation: 011123KCPyST05C has the lowest measured arsenic concentration of the Kidd Creek aliquots. Unfortunately, this study's arsenic measurements for both the Kidd Creek and Gympie sulfides are not reliable: they are inconsistent with observations of optical habit during sample picking, and they do not agree with estimates of arsenopyrite content implied by mass balance using sample mass and measured Fe.

Whether the rest of the suite's helium loss was thermal or accompanied recrystallization is similarly unclear, but recrystallization seems the less likely explanation, since the most recent regional metamorphic event that could have recrystallized these sulfide porphyroblasts predates 2.6 Ga (P. E. Smith et al., 1993; Hannington et al., 1999), well before the 465-860 Ma ages recorded here.

Arsenopyrite U-Th/He Ages for Gympie Gold Belt

There is no obvious difference in the geochronological sulfide U-Th-Sm/He data between the HCl-leached and non-leached Gympie arsenopyrite populations (Fig. 3.11), thus we interpret leaching to be a harmless but unnecessary step in the Gympie arsenopyrite U-Th/He geochronology protocol. This step was originally intended to remove any potential surface contamination, but these data do not suggest that the scatter in age is related to surface contamination. It is more likely that the scatter in age is related to the variable sizes and interconnectivity of inclusions, which could affect the sulfide's helium retentivity.

Lithostratigraphic age constraints that rely on foraminifera microfossils seem to suggest an Early Permian age for the Gympie Group rocks that host the gold-vein systems (§3.1; Holcombe et al., 1997). However, the oldest ages reported here are indeed resolvably older than the inferred Early Permian upper bound, instead suggesting that the mineralized volcanic pile may be of Carboniferous age. Indeed, these biostratigraphic constraints are sparse and rest mainly on correlation higher in the sequence rather than direct radiometric dates on the volcanic pile itself, and Holcombe et al. (1997) note that the lower Gympie Block volcanics "may contain older volcanic units." The data here support the latter interpretation.

The oldest Gympie ages are interpreted as a record of hydrothermal alteration of the Gympie Block volcanics contemporaneous with ∼300-310 Ma regional granitoid magmatism (§3.1; Little et al., 1995; Holcombe et al., 1997). Because the rutile inclusions identified as the U-Th-Sm source in the Gympie arsenopyrite (§3.4) are small (≤20 µm, Fig. 3.17), they would self-implant little of their own produced helium, and thus the measured dates cannot be explained by rutile inheritance before a hypothetical sulfide crystallization event at ∼235 Ma (Cuneen, 1994; Holcombe et al., 1997). Thus, the arsenopyrites likely crystallized at or before ∼306 Ma, well before the Au-quartz mineralization event at ∼235 Ma. It is possible that the rutile population predates ∼306 Ma, and only started retaining helium after being incorporated into the sulfide hosts, permitting the volcanic pile's emplacement age to be older than the oldest sulfide ages. We prefer the interpretation that the sulfide crystallized contemporaneously with metasomatism related to the ∼300-310 Ma regional granitoid magmatism because it is more feasible that such large sulfides would be metasomatic rather than magmatic in origin. Therefore, the volcanic pile

was likely emplaced sometime before ∼306 Ma. Aside from a single young outlier, the youngest Gympie sulfide ages reported here are contemporaneous with the syntectonic Au-mineralization event at ∼235 Ma (Cuneen, 1994; Holcombe et al., 1997), which we interpret to represent the timing of a recrystallization event for some of the sulfides. At the very least, it is somewhat remarkable that the sulfide U-Th/He ages demonstrate that most samples retained helium through the ∼235 Ma hydrothermal event.

Under this interpretation, the eleven reliable Gympie ages (§3.3) record variable, partial resetting of the ∼300-310 Ma population during the ∼235 Ma hydrothermal mineralizing event, rather than multiple discrete sulfide generations.

The one clear exception is 062123GyPyHClb, at 191 Ma, the youngest age in the suite, and the only one below the ∼235 Ma mineralization age itself. A partial-resetting model bounded between a ∼306 Ma inclusion age and a ∼235 Ma resetting event cannot, on its own, produce an age younger than 235 Ma, so this aliquot requires additional explanation. One speculative possibility is that this particular aliquot's U-Th-Sm-rich inclusions sat near or on the surface of the sulfide grains, and thus some of the radiogenic helium was implanted elsewhere and not measured in the aliquot. Some Gympie samples contain coarser alkali feldspar inclusions identified alongside the rutile (Fig. 3.17). If there were rutile crystals at the interface of the sulfide and the feldspar, then some of the rutiles' radiogenic helium may have been implanted into the feldspar and lost over geologic time, since feldspar is not normally considered a particularly helium-retentive phase (Hurley and Goodman, 1941). In any case, we point out that this single age is only one of eleven, and therefore seems to be the exception rather than the rule.

3.5 Conclusions

Sulfide geochronology has so far relied almost entirely on the Re-Os system, which is compositionally restricted to a handful of phases and analytically restricted to a few specialized laboratories (§3.1). This chapter tested whether U-Th/He dating of pyrite and arsenopyrite can serve as an accessible complement to Re-Os geochronology through two lenses: direct application to two independently age-constrained deposits at Kidd Creek and Gympie (§3.1), and a set of sulfur-buffered helium step heat experiments designed to char-

acterize the underlying diffusion and migration behavior of helium in pyrite, arsenopyrite, galena, and chalcopyrite.

The step heat experiments show that pyrite and arsenopyrite are remarkably helium-retentive against thermal loss: cumulative fractional yields never exceeded 5% for any pyrite or arsenopyrite experiment, and never exceeded 2% for the undeformed Leadville pyrite specifically, even after repeated heating cycles up to 600°C (§3.4). Placed on a common diffusive-dose axis against independently characterized phases, pyrite and arsenopyrite sit at or beyond the retentive extreme of the compilation, on par with or exceeding zircon and monazite and substantially exceeding hematite and goethite (Fig. 3.12). Arsenopyrite is less retentive than pyrite, consistent with its lower-symmetry, arsenic-bearing structure (§3.4), but is similarly retentive to zircon and markedly more retentive than galena or the copper sulfides, which are not considered promising U-Th/He targets (§3.2s).

The step heat data also point to two distinct helium-loss pathways within pyrite itself. Deformed, fractured Rustenburg pyrite shows steep, retrograde-dependent Arrhenius behavior best explained by a short-circuit or pipe-diffusion pathway along pre-existing fractures (J. K. W. Lee and Aldama, 1992), whereas the visually homogeneous, irradiated Leadville pyrite follows a simpler, shallower-sloped trend consistent with multiple-domain volume diffusion (§3.4). Varying grain size and sulfur-liquid contact within the Leadville suite did not resolve a clear control on this behavior beyond a surface-area-driven nugget effect, in which smaller aliquots simply carry a larger proportion of fine pyrite dust (§3.4).

Applied to natural samples, the U-Th/He ages obtained here are not simple crystallization or cooling ages. Eleven of twelve Kidd Creek pyrite/arsenopyrite aliquots give ages of 465-860 Ma (mean 639 ± 36 Ma, 1σ, n=11), decisively younger than the ∼2716 Ma host rhyolite (Hannington et al., 1999) but older than the regional 392 ± 32 Ma apatite cooling age (Sturrock et al., 2024); the twelfth instead gives a much older, 1596 ± 46 Ma age. Gympie arsenopyrite ages range from 191 to 306 Ma (mean 258 ± 35 Ma, 1σ, n=11; §3.3). Except for a single young outlier, these ages are bracketed by the regional ∼300-310 Ma magmatic episode and the ∼235 Ma gold-vein mineralizing event (§3.1), and thus the sulfides are interpreted to have crystallized during metasomatism related to regional plutonism at ∼306 Ma, then variably reset by the final

mineralizing event at ∼235 Ma.

Neither pyrite nor arsenopyrite itself hosts appreciable U or Th, instead relying on U-Th-Sm-bearing accessory inclusions to supply the sulfide hosts' alpha-emitters and radiogenic helium: apatite at Kidd Creek, rutile at Gympie. The helium from small, dispersed inclusions is implanted entirely within the sulfide host (and is thus subject to the sulfide's own helium retentivity), but larger inclusions will partially self-implant, complicating the interpretation of the sulfide U-Th-Sm/He ages (§3.4). At Kidd Creek, a simple nugget-effect model explains the anti-correlation between age and Th/He content among the eleven-aliquot array, but the observed sizes of the apatite inclusions are too small to explain ages this young if this population behaved as a closed system: most of it must have experienced helium loss during its long history. The twelfth, oldest aliquot is instead consistent with the closed-system prediction at the observed inclusion size, though this rests on a single measurement that could equally reflect incomplete sample digestion rather than a genuinely more retentive phase (§3.4).

Together, these results support pyrite and arsenopyrite U-Th/He dating as a viable complement to Re-Os geochronology in multistage sulfide systems, but only once the identity and grain size of the actual radiogenic host phase are established for a given locality (§3.4): no single digestion protocol can be assumed to transfer between deposits. Extending this framework to the other sulfide-hosted geochronometers proposed in §3.1, and to a wider range of deposits than the two tested here, remains the natural next step.

Supplementary Information

for Chapter 3: *(U-Th-Sm)/He Sulfide Geochronology*

3.1s Sample Preparation and SEM Imaging of Buffered and Decomposed Sulfides

Samples of pyrite, arsenopyrite, and marcasite were crushed under ethanol, and grains larger than ~100 µm were dried for each mineral and subdivided into four populations: one control, and three for heat treatment in atmosphere, ultra-high vacuum, or encapsulated within a glass ampoule along with an excess of fused sulfur (or realgar, As_4S_4, for arsenopyrite). The encapsulated populations were loaded into 3 mm inner diameter, 4 mm outer diameter silica glass tubes with fused bottoms and open tops, along with sufficient fused sulfur or crystalline realgar to guarantee vapor saturation at 400°C. These tubes were interfaced with a vacuum line and were pumped to 60 millitorr before the tops of the tubes were sealed.

The open-air populations for each mineral were carefully placed on glass microscope slides within borosilicate petri dishes and the encapsulated populations were submerged in pure quartz sand within capped, deep petri dishes. These petri dishes containing the encapsulated and open-air populations of sulfides were placed on a temperature controlling hot plate under a fume hood and were held at 400°C for 8-11 hours, at which point the hot plate was turned off and the samples were left to cool.

The ultra-high vacuum populations for pyrite and marcasite were loaded into copper packets and mounted to a type K thermocouple within a high vacuum line. A silver wool based sulfur trap was installed between the samples and the turbo pump. The high-vacuum line was closed and pumped to $< 10^{-5}$ torr. A Watlow temperature controller held the samples at 400°C for 4.5 hours with a projector lamp illuminating the sample through a sapphire window. This procedure was repeated for each mineral sample of pyrite and marcasite. The arsenopyrite sample was not heated in vacuum to avoid contaminating the line with arsenic. To simulate thermal decomposition in an anoxic environment, a population of arsenopyrites was thermally treated in a large glass capsule similarly to the encapsulated populations, but without a realgar "buffer".

The control and thermally treated samples were mounted to carbon tape and carbon coated, then were loaded in the Zeiss 1550 Scanning Electron Microscope (SEM) at Caltech. Images of the surface topography of each population were taken using the secondary electron detector and an Electron High Tension (EHT) of 10 kV at a working distance of 10-15 mm.

3.2s Chalcopyrite, Chalcocite, Galena, and Sphalerite Step Heat Results and Discussion

This section reports the step heat results and discussion for the non-pyrite/arsenopyrite sulfides of this study – the copper sulfides chalcopyrite (CCpTS230630) and chalcocite (CCT230707), galena (GalTS230630), and sphalerite (SphTS230630) – moved here from the main text because none of these minerals is a promising U-Th/He geochronometer or helium isotope target, and none is otherwise part of this chapter's main narrative (Table 3.1 in the main text still lists all nine step heat experiments, including these, for completeness).

Results

Chalcocite and Chalcopyrite

Step heat experiments for chalcopyrite (CCpTS230630) and chalcocite (CCT230707) produced macroscopic evidence of phase transformation, including the development of an iridescent blue sulfide or oxide layer on both specimens and dramatic crystal growth on the ampoule walls for the chalcocite experiment. Although permeation lag was not sufficiently constrained to calculate apparent diffusivities for the chalcocite experiment, the data for CCT230707 are consistent with a primary phase-breakdown control of helium migration at temperatures $\leq$300°C. These step heat data therefore do not bear on the helium retentivity of copper sulfides.

A linear regression is fit to 9 measurements defining a linear region of the Arrhenius plot of chalcopyrite sample CCpTS230630 for use as a reference for the delta plot (Fig. 3.21, row 3). This regression has an R^2 value of 0.9811, an implied activation energy of 41.8 $\pm$ 2.2 kcal/mol (174.8 $\pm$ 9.2 kJ/mol), and an implied closure temperature of 128 $\pm$ 13°C assuming a cooling rate of 10 °C/Ma. The delta plot generated using this reference line for chalcopyrite clearly shows a dramatic mid-experiment increase in diffusivity following steps above 400°C, followed by a continuous decrease in delta values (Fig. 3.21, row 3), similarly to that observed for the dehydrating goethite sample Win-06-01b (Farley et al., 2024).

Galena

The majority of the total helium yield was evolved during the step heat experiment of the encapsulated galena, in stark contrast to the comparatively

retentive pyrites and arsenopyrite. The first 70% of helium released in the galena experiment could not be reduced into diffusivities, because $\lambda t \leq 4$ and so steady state permeation out of the ampoule could not be assumed. There were no permeation steps for this experiment. These data are still included in Fig. 3.12, because the cumulative yields are less affected by the permeation lag, which mostly affects the short-term knowledge of helium degassing: earlier degassed helium will have long since permeated out, allowing for generally accurate accounting of evolved helium throughout the experiment. The reference line from the chalcopyrite experiment showed a similar activation energy to the activation energy implied by curved retrogrades of the galena experiment (Fig. 3.21, row 2), so it was used as the reference for the delta plot of galena. This delta plot shows a monotonic decrease in delta values as the experiment proceeds (Fig. 3.21, row 2).

The full ^{3}He step-heat datasets for both experiments, in the same format as the main-text pyrite/arsenopyrite tables (§3.3), are given in Tables 3.7 and 3.8.

Sphalerite

Although we conducted a step heat experiment on translucent sphalerite (SphTS230630), none of the sample's steps reached $\lambda t \geq 4$ and there were insufficient permeation steps to constrain the permeation lag, so no diffusivities could be calculated. Qualitatively, the sphalerite experiment showed that it is somewhat less helium retentive than apatite, and it was consistent with a single diffusion domain, although our data do not permit any confident interpretation. Future work on sphalerite should conduct disciplined permeation steps to track the helium inventory of the ampoule following the procedure outlined in §1.4.

Table 3.7: ^{3}He step-heat dataset for Galena, Tri-State District, Missouri (sample GalTS230630), under the uncertainty bounding method (§1.4)(§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.21 (row 2). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
15	385.0	100	0.2968	11.3209	0.0835	-14.76	-13.97	-13.72
16	375.0	120	0.2004	11.5213	0.0491	-15.02	-14.49	-14.26
17	380.0	110	0.2022	11.7235	0.0528	-14.62	-14.34	-14.11
18	390.0	90	0.1906	11.9141	0.0586	-14.47	-14.15	-13.88
19	400.0	70	0.3558	12.2699	0.1413	-13.38	-13.21	-12.85
20	410.0	60	0.0947	12.3646	0.0473	–	-14.32	-13.90
21	300.0	60	0.0374	12.4020	0.1171	–	-15.23	-13.79
22	350.0	60	0.0241	12.4262	0.0366	–	-15.66	-14.73
23	375.0	60	0.0459	12.4720	0.0327	-16.60	-15.01	-14.46
24	400.0	60	0.1039	12.5760	0.0525	-14.54	-14.17	-13.75
25	415.0	60	0.1722	12.7481	0.0744	-13.98	-13.62	-13.25
26	460.0	60	0.6155	13.3637	0.1765	-12.34	-12.23	-11.94
27	470.0	60	0.6037	13.9673	0.1605	-12.35	-12.03	-11.77
28	480.0	60	0.4856	14.4529	0.1200	-12.39	-12.02	-11.77
29	489.9	60	0.3938	14.8467	0.0908	-12.33	-12.00	-11.79
30	499.7	60	0.3285	15.1752	0.0709	-12.28	-11.99	-11.76
31	504.6	60	0.2357	15.4109	0.0493	-12.44	-12.11	-11.90
32	495.0	60	0.1264	15.5374	0.0282	-13.04	-12.57	-12.35
33	485.0	80	0.0935	15.6309	0.0158	-13.39	-13.05	-12.88
34	475.0	100	0.0667	15.6976	0.0093	-13.78	-13.52	-13.38
35	465.0	120	0.0489	15.7465	0.0059	-14.14	-13.94	-13.82
36	455.0	140	0.0351	15.7816	0.0038	-14.55	-14.37	-14.26
37	510.0	60	0.0648	15.8464	0.0131	-12.89	-12.84	-12.64
38	519.9	60	0.0765	15.9229	0.0145	-12.74	-12.56	-12.38
39	529.9	60	0.0758	15.9987	0.0135	-12.64	-12.44	-12.26
40	539.1	60	0.0709	16.0696	0.0120	-12.56	-12.36	-12.19
41	535.8	60	0.0570	16.1266	0.0098	-12.65	-12.43	-12.26
42	538.6	60	0.0385	16.1651	0.0065	-12.98	-12.70	-12.53
43	532.9	80	0.0330	16.1980	0.0041	-13.24	-13.03	-12.91
44	525.0	100	0.0265	16.2245	0.0027	-13.54	-13.38	-13.28
45	515.0	120	0.0207	16.2452	0.0018	-13.86	-13.72	-13.63
46	503.7	140	0.0139	16.2590	0.0011	-14.35	-14.21	-14.13
49	480.0	60	0.0028	16.2618	0.0007	-14.93	-14.93	-14.71
50	489.9	60	0.0032	16.2651	0.0007	-15.01	-14.77	-14.56
51	500.0	60	0.0050	16.2701	0.0011	-14.47	-14.31	-14.11

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(continued)

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
52	505.0	60	0.0052	16.2753	0.0011	-14.50	-14.26	-14.07

Table 3.8: ^{3}He step-heat dataset for Chalcopyrite, Baxter Springs, Kansas (sample CCpTS230630), under the uncertainty bounding method (§1.4)(§3.2): measured and cumulative ^{3}He, the upper-bound residual ampoule content ($n_{t,\mathrm{UB}}$), and the corresponding lower-bound, central, and upper-bound apparent diffusivities plotted in Fig. 3.21 (row 3). Only steps meeting the $\lambda t \geq 4$ plotability threshold are listed (§3.2); earlier equilibration/permeation steps and any steps manually excluded as unreliable are omitted rather than shown with undetermined bounds. $\ln(D/a^2)_{\mathrm{LB}}$ is left blank where the lower bound is unconstrained (§3.2).

Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
16	374.9	120	0.0782	13.1758	0.0247	-18.32	-17.57	-17.30
17	380.0	110	0.0639	13.2396	0.0220	-18.17	-17.68	-17.38
18	390.0	90	0.0721	13.3118	0.0296	-17.71	-17.35	-17.00
19	400.1	70	0.0943	13.4061	0.0509	-17.20	-16.82	-16.39
20	410.0	60	0.0963	13.5024	0.0610	-17.39	-16.64	-16.14
22	291.8	60	0.0928	13.5961	3.0586	-17.70	-16.66	-12.98
23	345.2	60	0.2696	13.8657	1.0009	–	-15.58	-13.98
24	375.0	60	0.9931	14.8588	1.0408	–	-14.21	-13.44
25	400.0	60	2.0663	16.9251	1.4653	-13.97	-13.32	-12.71
26	415.0	60	2.2228	19.1480	1.3491	-14.06	-13.05	-12.51
27	429.9	60	2.8501	21.9981	1.4512	-13.14	-12.56	-12.08
28	420.0	60	2.4223	24.4204	1.3837	-13.33	-12.48	-11.96
29	410.0	60	1.4757	25.8962	0.9756	-15.50	-12.79	-12.23
30	400.1	60	0.7845	26.6806	0.6038	–	-13.31	-12.71
31	390.0	80	0.5091	27.1898	0.2693	–	-13.96	-13.52
32	380.0	100	0.3706	27.5604	0.1495	-15.74	-14.46	-14.11
33	370.0	120	0.2689	27.8293	0.0892	-15.73	-14.92	-14.63
34	360.0	140	0.1510	27.9803	0.0452	-16.52	-15.63	-15.37
35	355.0	150	0.1416	28.1219	0.0408	-16.13	-15.75	-15.49
36	365.0	130	0.1775	28.2994	0.0557	-15.62	-15.36	-15.09
37	375.0	110	0.2245	28.5239	0.0808	-15.22	-14.94	-14.62
38	385.0	90	0.3152	28.8391	0.1360	-14.66	-14.37	-14.00
39	395.0	80	0.4454	29.2845	0.2065	-14.21	-13.86	-13.46
40	405.0	70	0.4926	29.7771	0.2543	-14.10	-13.56	-13.13
41	415.0	60	0.4935	30.2707	0.2950	-14.05	-13.35	-12.86
42	460.0	60	0.4521	30.7228	0.1759	-14.41	-13.37	-13.03
43	469.7	60	0.1397	30.8624	0.0517	–	-14.50	-14.19
44	465.4	60	0.1654	31.0278	0.0601	-14.68	-14.31	-14.00

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Step	Temp (°C)	Time (min)	^{3}He meas. (pccSTP)	Cum. ^{3}He (pccSTP)	$n_{t,\mathrm{UB}}$ (pccSTP)	$\ln(D/a^2)_{\mathrm{LB}}$	$\ln(D/a^2)_m$	$\ln(D/a^2)_{\mathrm{UB}}$
45	473.5	60	0.1554	31.1832	0.0531	-14.84	-14.35	-14.05
46	480.0	60	0.1691	31.3523	0.0538	-14.62	-14.24	-13.96
47	491.9	60	0.1549	31.5072	0.0452	-14.73	-14.31	-14.05
48	491.0	60	0.1219	31.6291	0.0357	-14.98	-14.53	-14.27

Discussion: Thermal Helium Migration Systematics

Chalcopyrite and Chalcocite

Neither copper-bearing sulfide mineral studied here was successfully stabilized by encapsulation with sulfur, so these results provide little insight into helium migration within copper sulfides in nature, though they suggest high potential for helium retention at surface temperatures, similar to apatite. Still, they are unlikely to be promising targets for U-Th/He geochronology or helium isotope studies as there exist many copper-bearing sulfide phases stable at temperatures between 200-500°C, for which phase stability and Cu/Fe ratios are sensitive functions of temperature, composition, and sulfur fugacity (Zhao et al., 2014; J. H. Chen and Harvey, 1975; Yund and Kullerud, 1966). These reactions are likely to result in helium loss during reheating, burial, diagenesis, or metasomatism after crystallization. Copper sulfides are unlikely to be promising targets for U-Th/He geochronology, but may be suitable for helium isotope studies in young or unaltered samples.

Galena

Because the majority of the cumulative yield was degassed in steps for which $\lambda t < 4$, the bulk helium behavior for galena is somewhat unclear. The data of the remaining 30% of the cumulative yield were consistent with a multiple diffusion domain helium loss mechanism. As expected for an MDD system, the delta plot for galena monotonically decreases as the fractional yield increases (Fig. 3.21, row 2; Farley, 2018). The majority of the 3He cumulative fractional yield was evolved at low temperatures where the uncertainties due to permeation lag could not be constrained, yet these early degassing domains were certainly less retentive than 150 µm half-prism width Durango apatite (Farley, 2000). Additionally, the total 3He concentration for the irradiated galena was less than half that of co-irradiated pyrite, implying poor helium retention even at

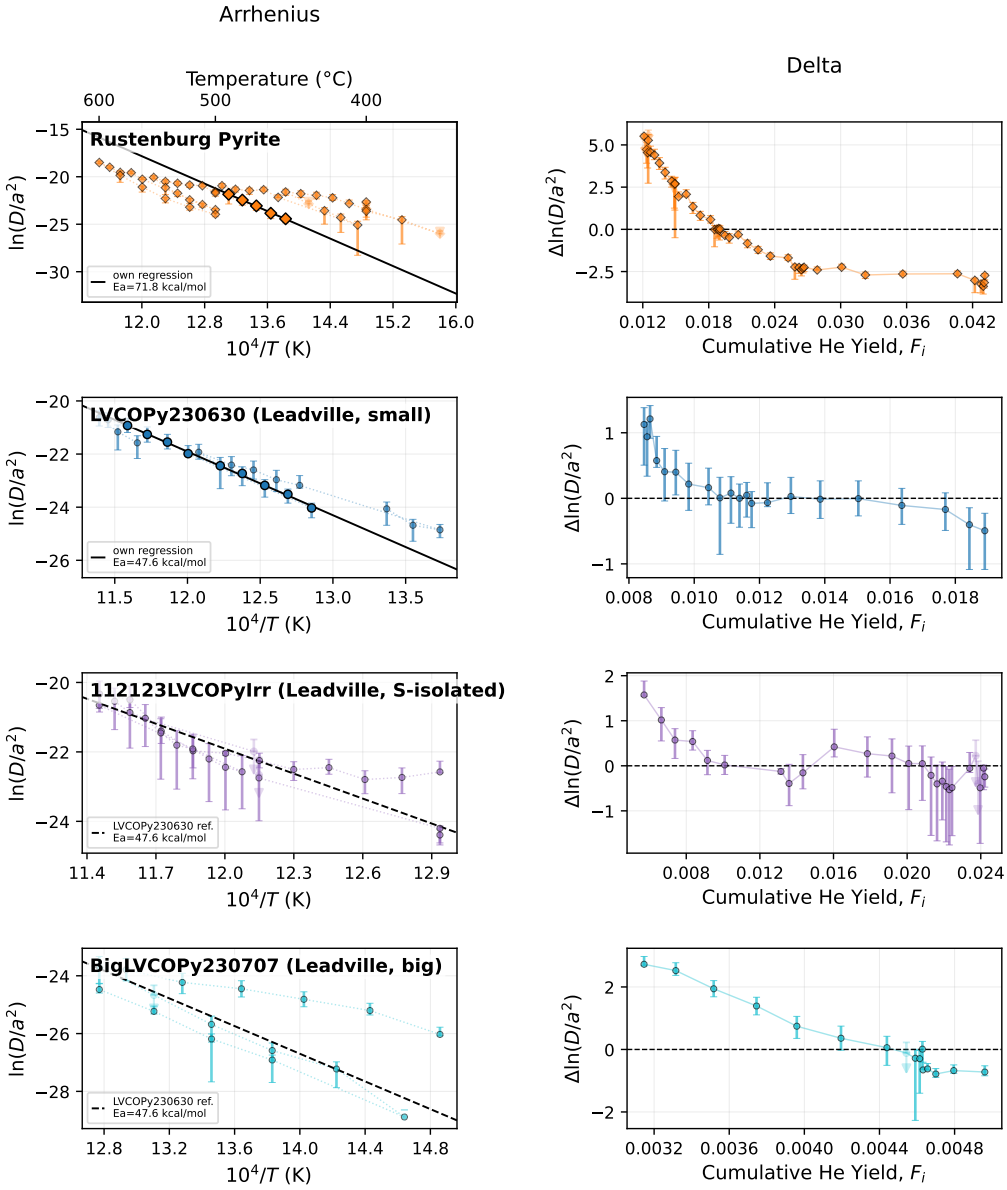

Figure 3.20: Compiled Arrhenius (left) and delta (right) plots for the four plottable Chapter 3 pyrite step-heat experiments: (1) Rustenburg pyrite, (2) LVCOPy230630 (Leadville, small), (3) 112123LVCOPyIrr (Leadville, sulfur-isolated), (4) BigLVCOPy230707 (Leadville, big). Each row's Arrhenius panel plots the reference regression used for that row's delta panel (solid where owned, dashed where borrowed; see §3.3). Each row has its own x-axis scale, labeled on every panel. The three non-pyrite sulfides are compiled separately in Fig. 3.21.

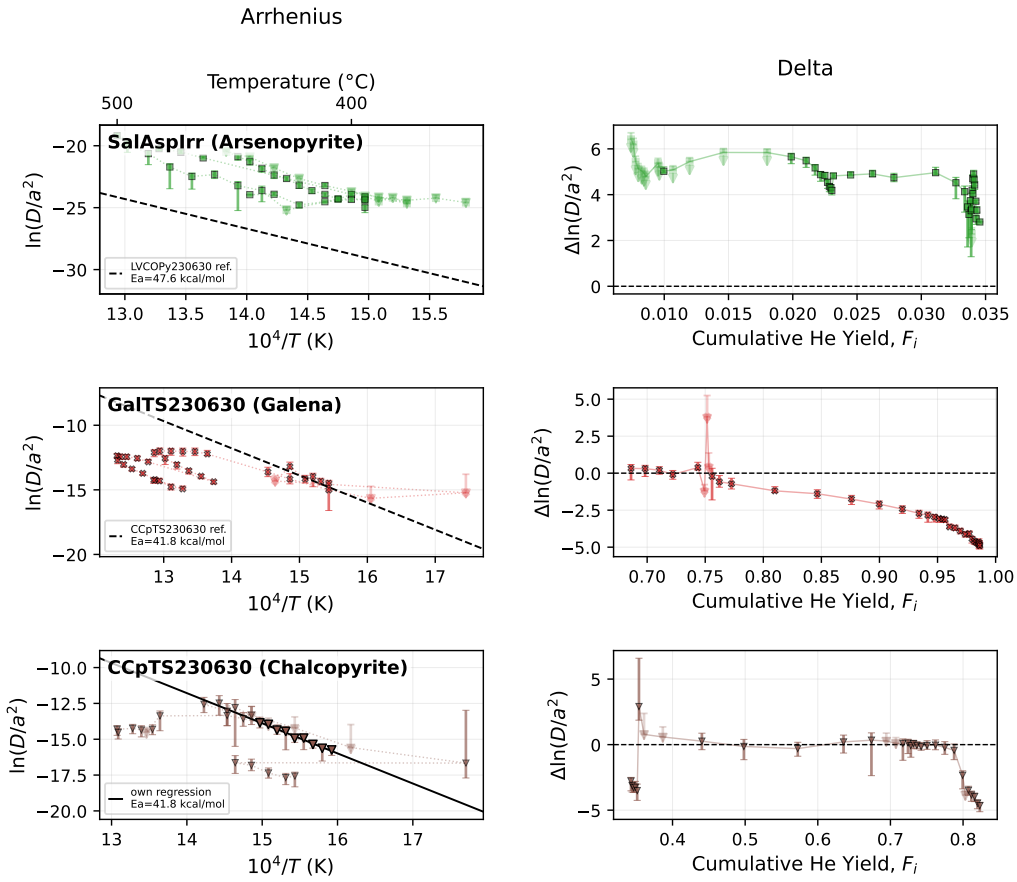

Figure 3.21: Compiled Arrhenius (left) and delta (right) plots for the three plottable Chapter 3 non-pyrite sulfide step-heat experiments: (1) SalAspIrr (arsenopyrite), (2) GalTS230630 (galena), (3) CCpTS230630 (chalcopyrite). Each row's Arrhenius panel plots the reference regression used for that row's delta panel (solid where owned, dashed where borrowed; see §3.3). Each row has its own x-axis scale, labeled on every panel. The four pyrite samples are compiled separately in Fig. 3.20.

ambient temperatures. Galena is unlikely to be a promising target for U-Th/He geochronology or helium isotope studies.

3.3s Complete U-Th/He Geochronology Compilation

Trace Elements for the Main-Text Sample Suites

The main-text geochronology table (Table 3.6) reports U, Th, Sm, and Zr for the Kidd Creek, Gympie, Durango, and Gogebic aliquots. The remaining ICPMS trace elements measured the same way – Cu, As, Ag, and Sb – are reported here instead (Table 3.9), in the same row order, to keep the main-text table within the page width.

Table 3.9: Cu, As, Ag, and Sb (measured as ratios to spike vanadium by the same ICPMS run as U, Th, Sm, and Zr, §3.2) for the same Kidd Creek, Gympie, Durango, and Gogebic aliquots reported in Table 3.6, in the same row order; '–' denotes below detection or not measured for that aliquot. Uncertainties are 2σ, rounded to two significant figures, with each value rounded to the same decimal place. Footnote symbols (*, †) as in Table 3.6.

Sample	Cu (ppm)	As (ppm)	Ag (ppm)	Sb (ppm)
Kidd Creek Pyrite (Archean VMS, ∼2716 Ma host)				
2211KC_Py_ST05_C	–	–	–	–
2211KC_Py_ST05_A	–	–	–	–
2211KC_Py_ST05_B	–	–	–	–
2211KC_Py_ST05_D	–	–	–	–
011123KCPyST05C	248±28	49±13	3.88±0.98	15.91±0.94
011423KCPyST05A	555±39	5920±400	7.8±1.1	35.69±0.95
011423KCPyST05B	1366±58	1600±110	12.6±1.0	22.44±0.41
011423KCPyST05C	781±52	8030±490	11.2±1.6	21.97±0.46
011423KCPyST05D	147±20	11440±800	2.1±1.4	24.09±0.73
013023KCPyST05As1	17.4±7.8	47100±2800	–	38.45±0.81
013023KCPyST05Cu1	314±30	11920±780	3.5±2.1	23.9±1.3
013023KCPyST05Cu2	103±20	410000±23000	–	1235±30
Gympie Pyrite (Permo-Triassic orogenic Au)				

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Sample	Cu (ppm)	As (ppm)	Ag (ppm)	Sb (ppm)
013023PVGyPy-1*	42.5±7.5	202000±12000	0.73±0.41	138.0±3.2
013023PVGyPy-2*	197±15	1045000±64000	3.1±1.3	734±22
013023PVGyPy-3*	15.9±8.9	434000±27000	2.81±0.86	345.2±9.9
013023PVGyPy-4*	39±17	617000±36000	1.67±0.79	333.7±7.8
0413PyGy-1	43.1±8.3	552±45	9.3±1.9	70.2±4.4
0413PyGy-2	33.1±8.9	165±38	8.8±2.1	12.5±3.7
0413PyGy-3	96±28	3450±210	7.9±4.0	61.5±5.7
0413PyGy-4†	30±11	3810±170	7.4±2.9	82.9±4.5
062123GyPyHClb	29.5±5.7	3594±92	4.8±1.6	186.8±5.2
062123GyPyHClc	21.7±3.2	499±18	2.3±1.1	6.36±0.19
230625GyPyHCl	30.6±5.8	13390±340	4.2±1.5	144.7±5.1
230625GyPyUnAbr	11.5±2.5	101400±2300	2.8±1.2	166.2±5.9
230628GyPyHClab	61.0±6.3	297100±8300	72±26	222±13
230628GyPyHClc	23.0±4.2	33600±1100	7.6±3.1	183±12
230628GyPyUnAbrA	30.7±4.0	1616±62	4.6±2.0	96.6±7.0
230628GyPyUnAbrB	12.0±5.3	1499±71	7.3±3.1	45.7±3.4
Durango Apatite Standard (accepted age 32.1 Ma; Farley 2000)				
2211DApA*	–	–	–	–

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Sample	Cu (ppm)	As (ppm)	Ag (ppm)	Sb (ppm)
2211DApC*	–	–	–	–
2211DApD*	–	–	–	–
2022StDApA*	–	–	–	–
011123DAp1	–	925±73	118±11	89.1±4.7
011123DAp2	–	910±100	–	26.3±2.9
011123DAp3	–	804±55	–	49.4±1.0
011423DAp1	–	788±51	1.8±1.4	28.71±0.83
011423DAp2	–	847±58	–	17.66±0.58
011423DAp3	–	820±72	–	23.65±0.54
011423DAp4	–	811±61	–	42.9±2.3
062123DAp	–	670±200	–	–
062523DAp	24.9±9.0	800±160	–	–
062823DAp	–	–	–	–
Average for > 0.5 mg	–	–	–	–
Gogebic Iron Range BIF Hematite Standard (MI-43; 571 Ma)				
2211HemMSA	–	–	–	–
2211HemMSB	–	–	–	–
041323MG16a	–	340±110	–	–

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Sample	Cu (ppm)	As (ppm)	Ag (ppm)	Sb (ppm)
041323MG16b†	77±35	1310±210	10.8±9.0	–
041323MG16d	–	3960±230	20±16	–
041323MG16e	–	8500±1400	–	–
062123MS	–	2720±110	3.6±1.4	18.83±0.96
230625MSHm	–	1850±130	78±30	89.1±4.5
230628MSHm	–	444±20	3.1±1.6	93.2±7.1

*Excluded: analyzed under the early (old) protocol (§3.2, §3.3).

†Excluded: outlier (§3.3, Fig. 3.6).

Additional Sample Suites

This subsection is a data appendix, not a narrative: it reports the complete U-Th/He geochronology compilation (Chapter3/Tables/geochronology_ compilation_BJT.xlsx) for the sample suites and replicate aliquots not discussed in the main text – Kidd Creek replicate/mass-only aliquots, Franklin (NJ) pyrite and fluorite, Rustenburg/Pilanesberg pyrite, Magnet Cove carbonatite pyrite, French Creek magnetite/pyrite, and Salzburg arsenopyrite – so that the full underlying dataset is available even where it is not part of this chapter's argument. Later work extending this chapter's geochronology (e.g. to these additional localities) can build directly on this section.

Table 3.10: U-Th/He geochronology results for all remaining sample suites in the compilation, not discussed in the main text: Kidd Creek replicate/mass-only aliquots, Franklin (NJ) pyrite and fluorite, Rustenburg/Pilanesberg pyrite, Magnet Cove carbonatite pyrite, French Creek magnetite/pyrite, and Salzburg arsenopyrite. Reported here for completeness, with the full trace-element suite (Cu, As, Zr, Ag, Sb) retained since these aliquots are not duplicated in the main text; columns, rounding, and conventions as in Table 3.6 (Cu, As, Ag, and Sb rounded as in Table 3.9).

Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)	Sb (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)
Kidd Creek Pyrite (replicate / mass-only aliquots)												
2211KC_Py_ST05_DU1	2.09	0.78±0.03	1.76±0.04	0.08±0.01	–	–	–	–	–	1.20	–	–
2211KC_Py_ST05_DU2	2.98	0.80±0.01	2.19±0.06	0.02±0.00	–	–	–	–	–	1.33	–	–
2211KC_Py_ST05_DU3	1.77	0.77±0.03	1.93±0.06	0.10±0.01	–	–	–	–	–	1.23	–	–
12122022KCPyST05A	14.35	–	–	–	–	–	–	–	–	–	1422.9	–
12122022KCPyST05B	10.52	–	–	–	–	–	–	–	–	–	960.7	–
12122022KCPyST05C	12.19	–	–	–	–	–	–	–	–	–	1204.0	–
12122022KCPyST05D	9.60	–	–	–	–	–	–	–	–	–	865.7	–
Franklin, NJ Pyrite (marble/volcanics, ∼1.3–1.0 Ga)												
2211GFr_PyA	2.49	0.29±0.01	0.05±0.01	0.02±0.00	–	–	–	–	–	0.31	51.3	529±22
2211GFr_PyB	3.00	1.78±0.02	0.01±0.01	0.01±0.00	–	–	–	–	–	1.78	66.9	106.9±1.3
2211StFr_PyA	1.45	0.64±0.03	0.02±0.01	0.02±0.00	–	–	–	–	–	0.65	42.8	322±15
2211StFr_PyB	2.53	0.12±0.01	0.00±0.00	0.91±0.04	–	–	–	–	–	0.12	57.8	1257±124
2211AtmFr_PyC	1.93	1.00±0.02	0.01±0.01	0.02±0.00	–	–	–	–	–	1.00	55.1	223.2±4.6
Franklin, NJ Fluorite												
011423FrFl2	1.40	3.19±0.03	0.24±0.07	0.21±0.04	32±10	–	1.8±1.1	10.2±1.7	35.81±0.74	3.25	76.8	168.7±1.9
011423FrFl3	1.03	5.35±0.08	0.20±0.06	0.34±0.09	–	–	5.8±1.6	–	30.63±0.71	5.40	99.9	179.3±2.8
Rustenburg/Pilanesberg(?) Pyrite												
013023BVPy H-3026-1	1.03	2.50±0.31	1.04±0.09	0.29±0.07	174±26	3330±230	10.9±5.7	–	152.6±3.8	2.75	226.1	747±79
041323BVPy1	0.95	0.56±0.02	0.17±0.05	0.05±0.05	106±27	392±59	–	–	–	0.60	45.4	725±20

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Sample	Mass (mg)	U (ppm)	Th (ppm)	Sm (ppm)	Cu (ppm)	As (ppm)	Zr (ppm)	Ag (ppm)	Sb (ppm)	eU (ppm)	^{4}He (ncc)	Age (Ma)
041323BVPy2	0.53	2.07±0.28	2.34±0.41	0.77±0.45	480±460	22900±4900	130.3±59.2	–	680±100	2.63	70.8	490±53
062123BVPyHCla	1.22	3.13±0.12	0.81±0.07	0.25±0.01	96±10	582±13	12.2±2.1	2.43±0.97	10.07±0.42	3.33	241.2	572±19
062123BVPyHClb	0.53	1.46±0.05	0.25±0.18	0.16±0.01	103±11	1499±45	7.9±4.0	3.3±1.4	10.48±0.72	1.52	1046.1	5110±58
230628BVPyUnAbrB	0.75	2.15±0.07	0.42±0.13	0.14±0.02	64±12	996±56	–	3.1±1.4	3.54±0.32	2.25	233.7	1244±39
230628BVPyUnAbrC	1.11	0.50±0.13	0.12±0.09	0.09±0.02	979±63	1029±34	–	7.6±3.3	4.60±0.29	0.53	723.2	5051±430
230628BVPyUnAbrC	0.60	29.01±0.40	5.44±0.33	2.01±0.17	138±16	729±54	203.0±10.2	11.8±4.8	10.59±0.76	30.32	881.5	474.1±6.0
Magnet Cove Carbonatite Pyrite, Arkansas (∼100 Ma)												
011123MCARPy1	3.38	0.06±0.00	0.41±0.03	0.31±0.03	15.7±4.7	3.9±1.2	1.6±0.7	1.10±0.48	18.58±0.78	0.16	9.7	177.1±7.5
011123MCARPy2	3.90	0.01±0.00	0.20±0.01	0.02±0.01	57.0±6.5	62.1±8.6	1.6±0.7	–	11.49±0.88	0.06	3.2	52.2±4.7
French Creek, PA Magnetite and Pyrite (∼130 Ma)												
062123FCHCl	4.36	0.01±0.00	0.00±0.02	0.01±0.00	9.0±2.5	29.6±2.1	–	21.2±6.9	7.6±1.1	0.01	1.5	218±52
062123FCMagHCl	3.84	0.06±0.00	0.00±0.02	0.05±0.00	–	81.9±3.2	–	0.71±0.29	15.55±0.44	0.06	2.7	97.7±5.5
230625FCHFa	3.41	0.63±0.05	0.71±0.03	0.34±0.02	210±15	1118±39	7.8±0.9	5.7±2.1	19.75±0.77	0.80	276.5	953±53
230625FCMagHCl	4.07	0.05±0.00	0.00±0.02	0.02±0.00	–	85.6±8.7	–	0.74±0.33	4.22±0.19	0.05	3.6	137.1±7.7
Salzburg, Austria Arsenopyrite												
230625SBAsPy	2.52	0.96±0.01	1.29±0.05	0.11±0.01	79±12	187400±4300	24.7±1.7	3.7±1.4	94.9±3.8	1.27	33.9	106.2±1.3

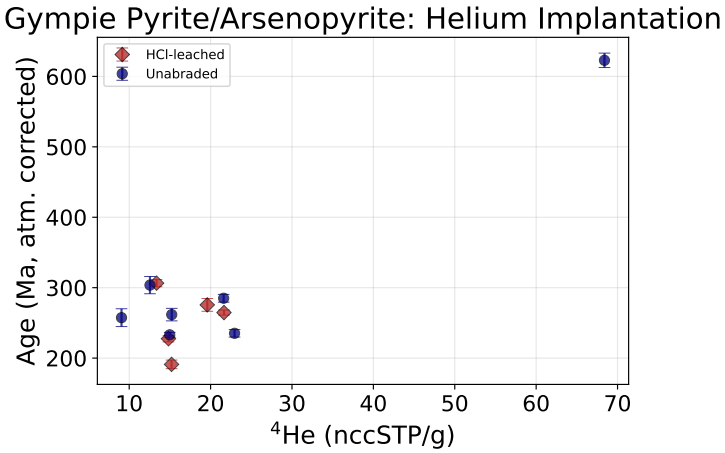


Figure 3.22: Companion to the main-text Gympie age-vs-eU figure (Fig. 3.11), plotting atmospherically corrected age against measured 4He content (ncc-STP/mg) for the newer-protocol Gympie arsenopyrite suite with the 0413PyGy-4 outlier put back in (§3.3, main text). Its 4He content stands well off the trend defined by the other eleven aliquots, consistent with the helium-implantation interpretation offered there.

Chapter 4

NGIBBS: FAST PHASE EQUILIBRIA AND MATERIAL PROPERTY CALCULATIONS FOR GENERAL MULTICOMPONENT SILICATE SYSTEMS VIA DEEP LEARNING

Thermodynamic models that compute equilibrium phase assemblages by free energy minimization, such as MELTS and HeFESTo, are often the computational bottleneck of the workflows that use them. The problems that would benefit most from these models include compositional inversions of seismic data, geodynamic models of a chemically heterogeneous mantle, and physical models of realistic magmatic systems that are heterogeneous in composition and temperature over space and time. These problems require large numbers of equilibria over a composition space that lookup tables cannot cover. Recently, machine-learning surrogate models have been built to address this need, but those that have been built to date each emulate a narrowly defined slice of their parent model. This chapter presents nGibbs*, a set of general deep-learning emulators that reproduce MELTS 1.0.2 and 1.2.0 and the HeFESTo models of Earth's and Mars' mantles across broad ranges of pressure, temperature and bulk composition. Each emulator predicts the full equilibrium assemblage, the abundance of every phase and the endmember composition of every solution phase. Like the parent models* nGibbs *emulates, phase abundances vary continuously across phase boundaries, and mass is conserved. Physical properties are computed from the predicted assemblage with the parent models' own equations of state. On a validation dataset generated from bulk compositions distinct from the training data, the emulators predict the presence of the major silicate and oxide phases with precision and recall of 0.93 or better and conserve every element to a median relative error of about* 5×10^{-6}*%. Along a depleted-mantle isentrope, emulated densities and seismic velocities differ from HeFESTo by 0.02% on average. The HeFESTo emulators run four orders of magnitude faster than*

HeFESTo on an ordinary CPU, and the MELTS emulators two orders of magnitude faster than alphaMELTS. The emulators are differentiable, run on CPUs or GPUs, and are interchangeable with alphaMELTS in existing PetThermoTools workflows. These fast, stable, and compositionally general models make composition a practical free parameter in seismic inversions and coupled geodynamic models.

Author contributions. This chapter is being prepared for publication with co-author P. D. Asimow. **B.J.T.:** conceptualization, methodology, software (nGibbs), investigation (training-data generation with HeFESTo and MELTS), formal analysis, data curation, visualization, writing – original draft. **P.D.A.:** conceptualization, supervision, formal analysis, funding acquisition, writing – review and editing. Generative AI was used as described in the front matter.

4.1 Introduction

Motivation

Free energy minimizers translate a bulk composition and two intensive variables into a stable phase assemblage, the abundance and composition of every phase, and the physical properties that follow from them. The MELTS family of models (Ghiorso and Sack, 1995; Ghiorso et al., 2002; Gualda et al., 2012; Ghiorso and Gualda, 2015) serves this role for silicate liquids and crustal to uppermost-mantle conditions, and HeFESTo (Stixrude and Lithgow-Bertelloni, 2005; Stixrude and Lithgow-Bertelloni, 2011; Stixrude and Lithgow-Bertelloni, 2024) serves it for the solid mantle from the surface to the core–mantle boundary. Perple_X (Connolly, 2005) and MAGEMin (Riel et al., 2022) utilize the database and formulations of Holland and Powell (2011), Holland et al. (2018), and Holland and Powell (1998), which apply from the surface to the core–mantle boundary and support liquids. Each equilibrium requires the solution of a constrained, nonlinear optimization problem, and within a larger workflow that solution is often the slowest step. A single MAGEMin minimization takes on the order of 10^{-1} s depending on the water content of the system (Riel et al., 2022; Candioti et al., 2026), whereas MELTS is an order of magnitude faster (10^{-2} s per assemblage), and HeFESTo is an order of magnitude slower (seconds per assemblage). Efficient coupling to a geodynamic model requires less than 1 ms per point (Kerswell

et al., 2024; Candioti et al., 2026). Minimizers are also largely serial, while geodynamic solvers are built for distributed or shared-memory parallelism, so the thermodynamic calculation dominates the run time of a coupled model (Q. Yuan et al., 2026; Candioti et al., 2026). Robustness is a second cost. A MELTS calculation that fails to converge can halt a batch of calculations or require the library to be reinitialized, which has limited the use of Monte Carlo methods with these models (M. L. M. Gleeson et al., 2025).

The common workaround is to precompute a lookup table of properties over pressure and temperature for one or a few bulk compositions and to interpolate within it during the geodynamic calculation (Connolly, 2005; Zunino et al., 2011; Rummel et al., 2020; Q. Yuan et al., 2026). This works well when composition is fixed but becomes impractical when composition is itself a variable, because tracking compositional evolution requires interpolation in a space of ten or more dimensions (Rummel et al., 2020; Candioti et al., 2026). The problems for which thermodynamic coupling matters most are exactly those in which composition varies: open-ended searches over composition space, inversions that treat composition as an unknown, and forward models in which every parcel of rock carries its own chemical history. These problems need a forward model that is fast, robust, and as general in composition as the thermodynamic model it replaces. Several such problems, described below, motivate this chapter.

Compositional inversion of seismic and geophysical data. Gibbs energy minimization inside a stochastic sampler allows geophysical observations to be inverted directly for mantle composition and thermal state, rather than for seismic velocity alone (Khan and Connolly, 2008). Khan and Connolly (2008) applied this approach to the Love number, tidal dissipation, mean density and moment of inertia of Mars in the system CaO–FeO–MgO–Al_2O_3–SiO_2. Khan et al. (2021) later inverted InSight body-wave travel times for the upper mantle structure of Mars, but assumed a single homogeneous bulk composition derived from the martian meteorites, so that the sampled parameters were primarily thermal. Each forward model in such an inversion is a column of equilibria, and a sampler must evaluate many such columns, so the cost of minimization limits how many compositional dimensions can be explored. On Earth, the seismic character of the mantle transition zone is sensitive to composition at a level that such inversions could exploit. The post-spinel transition in the

Mg–Fe binary spans only ∼0.01 GPa, or about 250 m, at mantle compositions and temperatures, which accounts for the observed sharpness of the 660-km discontinuity without requiring chemically distinct upper and lower mantles (Ishii et al., 2019). The dissolution of aluminous garnet into bridgmanite, in contrast, is expected to produce steep but gradual gradients in velocity and density between about 660 and 750 km (Wood and Rubie, 1996). Surrogate-based sensitivity analysis suggests that the wadsleyite–ringwoodite transition and the akimotoite-out reaction near 660 km both shift with mantle fertility (Hartmeier and Lanari, 2026). Whether lateral variations in discontinuity depth and sharpness reflect temperature, composition, or both is therefore a question that a fast, composition-aware forward model could address directly.

For example, the depth, sharpness and velocity contrast of the transition-zone discontinuities depend on both temperature and composition. The binary post-spinel transition is narrow enough to explain the sharpness of the 660-km discontinuity (Ishii et al., 2019), garnet dissolution into bridgmanite broadens the velocity and density gradients beneath it (Wood and Rubie, 1996), and the wadsleyite–ringwoodite and akimotoite–bridgmanite transitions shift with mantle fertility (Hartmeier and Lanari, 2026). Ishii et al. (2019) also proposed that latent heat can thicken the post-spinel transition by up to ∼7 km where vertical flow is faster than thermal diffusion, so that maps of discontinuity sharpness could locate upwellings and downwellings.

Composition of the lowermost mantle. The large low-velocity provinces (LLVPs) and ultralow-velocity zones (ULVZs) at the base of the mantle are commonly interpreted as compositionally distinct material. Proposed origins vary wildly, including a crystallizing basal magma ocean (Labrosse et al., 2007), subducted banded iron formations (Dobson and Brodholt, 2005), and iron-rich mantle material from the Moon-forming impactor that is 2.0–3.5% denser than the ambient mantle (Q. Yuan et al., 2023). Jackson and Thomas (2021) found that assemblages containing iron-rich (Mg,Fe)O match the seismic velocity reductions and density increases of many ULVZ case studies, and that the range of observations allows multiple distinct but coincident origins. Testing these hypotheses requires searching for compositions, including unusual ones unknown on the Earth's surface, whose equilibrium assemblages reproduce observed velocity reductions at core–mantle boundary conditions. That search is open-ended in composition space.

A chemically heterogeneous mantle. The Sr–Nd–Pb isotopic record of oceanic island basalts is described by four principal mantle end-members (DMM, HIMU, EMI and EMII) with distinct compositions and proposed origins, from recycled sediment to intra-mantle metasomatism (Hart, 1988; Stracke et al., 2005). The major element and lithological consequences of this heterogeneity are also observable. A mechanical mixture of basalt and harzburgite has a different phase assemblage and seismic velocity from pyrolite of the same bulk composition, and such a mixture matches transition-zone seismic observations better than pyrolite (W. Xu et al., 2008; Stixrude and Lithgow-Bertelloni, 2012). Geodynamic models that compute mineral physics self-consistently, however, often represent this heterogeneity with a single variable, the basalt fraction in a basalt–harzburgite mixture (e.g., Nakagawa et al., 2010; Q. Yuan et al., 2026). That parameterization cannot represent the full range of diverse compositions implied by the geochemical record, nor the fate of subducted material that falls off the basalt–harzburgite join, such as sediments, altered oceanic crust, or banded iron formations. A flexible approach to modeling this heterogeneity is needed to capture the full complexity of the mantle's composition and dynamics to be able to explain the observed seismic anomalies and plumes (Keller et al., 2023; Dobson and Brodholt, 2005).

Magma ocean solidification. Fractional crystallization of a magma ocean is expected to produce a gravitationally unstable cumulate stratigraphy that overturns (Elkins-Tanton et al., 2003; Elkins-Tanton et al., 2011). One-dimensional overturn models assume that the cumulates reorganize on every scale into a monotonic density profile, and Elkins-Tanton et al. (2003) note that viscosity is likely to limit that reorganization and leave a laterally heterogeneous mantle. Miyazaki and Korenaga (2022) proposed that Rayleigh–Taylor instabilities with length scales below 100 km leave iron-rich blobs embedded in a high-Mg# pyroxenite matrix, and that a wet mantle of this kind permits plate tectonics and removes atmospheric CO_2 more than ten times faster than a homogeneous pyrolitic mantle. Testing this hypothesis in a forward geodynamic model requires an equation of state for every parcel as its composition and depth change, which is the case that lookup tables handle worst.

Igneous processes at smaller scales. Recent works have made impressive progress developing physical models for the mechanical behavior of open-system

magma mushes (Bergantz et al., 2015). Such models aim to describe thermally and chemically heterogeneous and evolving material whose physical properties are a direct consequence of composition and temperature as described by thermodynamics, but are necessarily simplified for computational efficiency. Similarly, melt migration within a magma mush is a complex process that is well studied, but produces geochemical signatures which are poorly understood (O'Hara and Mathews, 1981; Lesher and Walker, 1988; Maier et al., 2020; Yao and Mungall, 2021). Thermodynamic models of reactive porous flow may elucidate these signatures, but current models treat each melt–mush reaction as a single step, run until the percolating melt and the mush reach thermal equilibrium, whereas natural samples record a protracted, open-system process (Boulanger and France, 2023) that may be better approximated by spatiotemporally-resolved, local-equilibrium thermodynamics that is significantly more computationally expensive.

Previous Work

Machine-learning surrogates have been developed to address the cost of phase equilibrium calculations, and those built to date are each scoped to a particular application. Kerswell et al. (2024) trained models that predict bulk physical properties, such as density and seismic velocity, of peridotite. Candioti et al. (2026) validated the Holland et al. (2018) igneous database in MAGEMin against crystallization experiments on arc magmas and then trained a multi-task neural network on ∼980,000 MAGEMin equilibria. Their training data span 0.1–10 kbar, 650–1300 °C, and 100 synthetic compositions. The network predicts the presence, fraction and density of liquid and fluid, the liquid composition, and the system density. Solid properties are computed afterward by difference from the system, liquid and fluid properties. Individual minerals are not predicted. The authors report that it runs about 3,000 times faster than MAGEMin on 64 CPU threads, framing their results as a proof of concept. They recommend retraining on application-specific data, and argue that a single surrogate spanning porous, mush and suspension flow regimes is impractical (Candioti et al., 2026).

Hartmeier and Lanari (2026) present a particularly impressive example of a machine-learning surrogate. They trained a combined classifier and regressor on two million MAGEMin equilibria computed with the Stixrude and Lithgow-Bertelloni (2022) database. It predicts the fractions of 20 phases

and the compositions of the solid solutions in the system Na_2O–CaO–FeO–MgO–Al_2O_3–SiO_2 over 1–40 GPa and 500–2500 °C. Conservation of mass and compositional closure are softly enforced, entering the loss function as penalties. They achieve relative errors around 1% for phase fractions and 0.5% for solid solution compositions. The model's classifier delivers impressive precision and recall, mostly 95% and up. Hartmeier and Lanari (2026) also showed that the surrogate's analytical gradients identify which inputs control each phase boundary, a use of differentiability that conventional minimizers do not offer.

The IGyTheM model (Integrating Geodynamics with Thermodynamics using Machine Learning; Q. Yuan et al., 2025) couples a HeFESTo-trained surrogate to the Citcom mantle convection code, so that thermodynamically derived density enters the momentum equation. Its training database covers 0–140 GPa and 273–4273 K. It represents the mantle as a mechanical mixture of basalt and harzburgite, so composition enters as a single mixing variable.

These surrogates show that neural networks can replace minimizers at a fraction of the cost. Each, however, emulates a narrow slice of its parent model, defined by a compositional subspace (arc magmas, or the basalt–harzburgite join), a pressure range, a volatile treatment, or a limited choice of outputs. None of them can be used in place of its parent model for an arbitrary task that the parent model could perform.

Properties and Innovations of nGibbs

nGibbs is designed as a general emulator: one that can stand in for its parent thermodynamic model for essentially any calculation that model would be used for, within the parent model's own range of validity. Its released emulators reproduce HeFESTo for Earth (0–140 GPa) and Mars (0–25.5 GPa), and the closed-system isothermal forms of MELTS 1.0.2 and 1.2.0, with and without chromium. Emulators for open-system (buffered oxygen fugacity) MELTS calculations have been trained but are not yet released. Similarly to Hartmeier and Lanari (2026), each emulator predicts the quantities that define the equilibrium assemblage: the phase abundance and chemistry. Building upon Q. Yuan et al. (2026), *nGibbs* computes physical properties which are the thermodynamic consequence of an equilibrium assemblage.

Four properties distinguish *nGibbs* from the surrogates above. First, its training and validation compositions are drawn from whole-rock databases of igneous

and, for MELTS 1.2.0, sedimentary rocks, that are perturbed to cover exotic compositions that are rare or unknown at Earth's surface (§4.2). Second, conservation of mass is enforced at inference rather than only encouraged during training: an iterative, constrained correction adapted from Asimow and Ghiorso (1998) adjusts the predicted assemblage until it reproduces the input bulk composition exactly, without allowing any elemental abundance to become negative (§4.2). Third, phase abundances are continuous across phase boundaries, because the gate and the magnitude of each phase are the same variable (§4.2). Fourth, through PetThermoTools (M. L. M. Gleeson et al., 2025), they are called through the same interface as alphaMELTS, so that existing workflows can seamlessly switch between the emulator and the parent model for benchmarking (§4.4).

4.2 Methods

Network architecture

Motivation: the discontinuity problem

Before the introduction of the continuous-saturation framework described below, the early *nGibbs* architecture predicted phase presence with a binary classifier and phase abundance with a distinct regressor, multiplying the two so that the predicted abundance is the regressed magnitude gated by a thresholded presence probability. This gated model was simple and easy to train, but it has a structural defect. Because the gate and the magnitude were distinct functions, nothing tied the magnitude to zero as the gate closes. Therefore, when the presence probability of a phase crosses its threshold, the predicted abundance jumps discontinuously from a finite value to exactly zero, which is (for most phases) an unexpected behavior for a mixed system.

Natural systems of interest to geologists and planetary scientists are typically complex, chemically-mixed systems. These systems, and the thermodynamic models that describe them, behave continuously. In addition to the continuous quality of phase abundance, thermodynamic models like MELTS and HeFESTo also track each phase's chemical affinity, which is a continuous quality that measures the free energy cost of introducing a metastable phase to the system. Even when the phase is absent, the chemical affinity is a meaningful, continuous quantity that smoothly goes to zero as the phase approaches saturation. Likewise, at a phase-out boundary, the abundance of the departing

phase goes to zero smoothly, linearly in reaction progress. A gated model is therefore structurally wrong precisely at the phase boundaries that constitute the most interesting content of a phase diagram.

The architecture used for all released models removes this defect by making the gate and the magnitude the same variable, analogously to a prediction of chemical affinity. Some emulators predict a scaled chemical affinity, while others simply activate their continuous molar output with a ReLU function. The models trained with a scaled affinity perform slightly better, but require a more complex data processing pipeline that cannot be applied to datasets where the data are not structured in paths that hold a state variable constant. Because the HeFESTo data were generated along mock-isentropic pressure-temperature paths (and not true isentropes), the HeFESTo emulators were trained with the simpler, ReLU-activated architecture, hereafter referred to as the affinity-naive architecture. The MELTS emulators, which were trained on data generated by isobaric cooling, were trained with scaled affinities with an architecture hereafter referred to as affinity-aware.

Variables

Here a table is provided to name the key variables used in the equations below that describe the training and function of nGibbs. These equations rely additionally on general letter codes to describe the dimensionality of the data that are used internally within the nGibbs codebase and are borrowed here for clarity: P phases, VP compositionally variable phases, C components (endmembers of all phases), VC components of the compositionally variable phases, E elements and O oxides. Two further sizes are used here: F, the number of physical non-compositional input features (e.g. pressure and temperature, with or without fO_2), and B, the number of rows in a mini-batch (the number of assemblages processed in a single pass during training). Every quantity in Table 4.1 is per row; a batch stacks B rows. For emulator outputs, the hatted symbol is the prediction and the plain symbol is the ground truth taken directly from the thermodynamic model (e.g. MELTS or HeFESTo).

Inputs

Each emulator takes pressure and one further intensive variable (temperature for the isothermal models, specific entropy for the isentropic models) and a bulk

Prediction	Truth	Dim.	Meaning
Inputs			
	u	F	Physical state (pressure, temperature, ...), min–max normalized to $[0, 1]$
$\hat{b}$	b	E	Bulk composition in element moles, normalized to sum to 1. $\hat{b}$ is the bulk reconstruction $\hat{b} = \hat{m}A$
Emulator outputs			
$\hat{n}$	n	P	Phase abundance (moles); 0 for an absent phase
$\hat{x}$	x	VC	Endmember mole fractions of each variable phase (sum to 1 within a phase: $\sum x_i = s_i\|1 - b\|$)
$\hat{m}$	m	C	Extensive endmember components/species in moles; $\hat{m} = (\hat{x}V + b) \circ (\hat{n}M)$
$\hat{s}$	s	P	Phase presence, 1 if present and 0 if absent; $s = [n > 0]$
Quantities used only in training			
g	–	P	Phase affinity/abundance: the raw output of each phase's mole head; $\hat{n} = \max(g, 0)$
$\tilde{n}$	–	P	Smoothed, positive, non-zero abundance used to evaluate mole loss; $\tilde{n}(g, \Upsilon)$: Eq. (4.41)
–	$\bar{s}$	VC	Phase Presence s projected to phases' endmembers (chemistry mask) $\bar{s} = s\,M\,V^\top \in \{0, 1\}$
Training constants			
Υ_0		P	Reference abundance scale per phase (5th percentile of present abundances)
Υ		P	Annealed smoothing scale; $\Upsilon = a\,\Upsilon_0$
a		1	Annealing factor, decreased linearly over an episode
w		P	Phase weights
ω		VC	Endmember weights; $\omega = w\,M\,V^\top$
$\lambda_s, \lambda_x, \lambda_n, \lambda_b$		1	Weights for the binary phase presence, chemistry, molar, and bulk loss terms
Fixed stoichiometric/projection matrices			
A		C×E	Element moles per component mole
M		P×C	Which components belong to which phase; $M \in \{0, 1\}$
V		VC×C	Indexing matrix that places the VC components into the full component list; $M \in \{0, 1\}$
f		C	Which components belong to fixed-composition (pure) phases; $b \in \{0, 1\}$

Table 4.1: Key to the symbols. "–" marks a quantity with no prediction or no ground-truth counterpart.

composition as inputs. The emulators are trained on and expect the same units for pressure, temperature, and entropy that the real thermodynamic MELTS and HeFESTo use: kelvin and GPa for HeFESTo, and degrees Celsius and bars for MELTS. Both expect specific entropy in units of J/g/K. nGibbs receives as input normalized cation atomic abundances, following $\mathbf{b} \in \mathbb{R}^E$ with $\sum_e b_e = 1$. Compositions are supplied by the user either as oxide weight percents or as element molar abundances for any model; oxide inputs are automatically converted internally to the normalized cation abundances expected by nGibbs.

Entropy is offered as a useful alternative to temperature because isentropic paths are the natural coordinate for the convecting mantle (Davies, 1977; J. M. Brown and Shankland, 1981; Asimow et al., 1997; Asimow, 2001; Ganguly, 2005; Tirone, 2016). Supplying S directly removes an inner iteration for T at every pressure step, which is the dominant cost of constructing an adiabat by conventional means.

The full non-compositional feature vector (i.e. pressure, temperature, entropy, $f\mathrm{O}_2$) is min–max normalized to $[0, 1]$. The minima and maxima used for these normalizations are fit on the training data, and are carried by the emulator and automatically applied during inference. This avoids presenting the network with inputs on a different scale than the one it was trained on, leading to errors that would be difficult to diagnose.

The neural networks driving *nGibbs* expect that the input composition vector sum to a total of one. During inference, this is explicitly enforced by a renormalization so that, similarly to MELTS and HeFESTo, the user can supply a composition that sums to values other than one for convenience. The outputs of *nGibbs*, however, are not rescaled and will always map to a cation sum of one regardless of user input.

Compositional basis

Three compositional bases are used. *Components* (m) are the endmembers of the reference code's solution models (C in total across all phases); *oxides* (O) are accepted as an input because it is conventional to measure and report rock compositions as oxides; and *elements* (E) are the conserved quantities that *nGibbs* conserves. Ferric and ferrous iron are distinguished as oxides but not as elements for models with oxygen buffering, for which $O = E + 1$, but closed system models distinguish ferric and ferrous iron in the element basis as

well so that they are conserved: $O = E$. Conversion between bases is exact stoichiometry, expressed as fixed matrices that relate components to oxides and oxides to elements. The product of the two, A, is the most commonly used stoichiometry matrix because it relates model outputs to the bulk elemental composition it conserves. These matrices are model constants, not fitted parameters. The phases and components of the MELTS and HeFESTo emulators, with the formula each component carries in the component-to-oxide matrix, are listed in Tables 4.2 and 4.3.

Table 4.2: Phases and components of the MELTS emulators, with formulae from the component-to-oxide stoichiometry used by *nGibbs*. Liquid components are oxides.

Phase	Component	Formula
olivine	forsterite	Mg_2SiO_4
	fayalite	Fe_2SiO_4
	monticellite	$CaMgSiO_4$
orthopyroxene, clinopyroxene	diopside	$CaMgSi_2O_6$
	clinoenstatite	$Mg_2Si_2O_6$
	hedenbergite	$CaFeSi_2O_6$
	alumino-buffonite	$CaMg_{0.5}AlTi_{0.5}SiO_6$
	buffonite	$CaMg_{0.5}Fe^{3+}Ti_{0.5}SiO_6$
	essenite	$CaFe^{3+}AlSiO_6$
	jadeite	$NaAlSi_2O_6$
spinel	chromite‡	$FeCr_2O_4$
	hercynite	$FeAl_2O_4$
	magnetite	$Fe^{2+}Fe_2^{3+}O_4$
	spinel	$MgAl_2O_4$
	ulvospinel	Fe_2TiO_4
plagioclase, K-feldspar	albite	$NaAlSi_3O_8$
	anorthite	$CaAl_2Si_2O_8$
	sanidine	$KAlSi_3O_8$
garnet	almandine	$Fe_3Al_2Si_3O_{12}$
	grossular	$Ca_3Al_2Si_3O_{12}$
	pyrope	$Mg_3Al_2Si_3O_{12}$
leucite	leucite	$KAlSi_2O_6$
	analcime	$NaAlSi_2O_6{\cdot}H_2O$
	na-leucite	$NaAlSi_2O_6$
biotite	annite	$KFe_3AlSi_3O_{10}(OH)_2$
	phlogopite	$KMg_3AlSi_3O_{10}(OH)_2$
rhm-oxide	geikielite	$MgTiO_3$
	hematite	$Fe_2^{3+}O_3$
	ilmenite	$FeTiO_3$
	corundum	Al_2O_3
apatite	apatite	$Ca_5P_3O_{12}(OH)$
whitlockite	whitlockite	$Ca_3P_2O_8$
quartz	quartz	SiO_2
tridymite	tridymite	SiO_2
fluid	H_2O	H_2O
	CO_2†	CO_2
liquid	oxides	SiO_2, TiO_2, Al_2O_3, FeO MgO, CaO, Na_2O, K_2O P_2O_5, H_2O, Cr_2O_3‡, Fe_2O_3 CO_2†
alloy-solid	Fe-metal	Fe
muscovite†	muscovite	$KAl_3Si_3O_{10}(OH)_2$
hornblende†	pargasite	$NaCa_2Mg_4Al_3Si_6O_{22}(OH)_2$
	ferropargasite	$NaCa_2Fe_4Al_3Si_6O_{22}(OH)_2$
	magnesiohastingsite	$NaCa_2Mg_4Fe^{3+}Al_2Si_6O_{22}(OH)_2$
graphite†	graphite	C
calcite†	calcite	$CaCO_3$
nepheline†	na-nepheline	$Na_4Al_4Si_4O_{16}$
	k-nepheline	$K_4Al_4Si_4O_{16}$
	vc-nepheline	$Na_3Al_3Si_5O_{16}$
	ca-nepheline	$Na_2CaAl_4Si_4O_{16}$

† MELTS 1.2.0 emulators only. ‡ Cr variants only.

Table 4.3: Phases and components of the HeFESTo emulators, with formulae from the component-to-oxide stoichiometry used by *nGibbs*.

Phase	Component	Formula
plagioclase	anorthite	$CaAl_2Si_2O_8$
	albite	$NaAlSi_3O_8$
spinel	spinel	$MgAl_2O_4$
	hercynite	$FeAl_2O_4$
	magnetite	$Fe^{2+}Fe_2^{3+}O_4$
	picro-chromite	$MgCr_2O_4$
olivine	forsterite	Mg_2SiO_4
	fayalite	Fe_2SiO_4
wadsleyite	mg-wadsleyite	Mg_2SiO_4
	fe-wadsleyite	Fe_2SiO_4
ringwoodite	mg-ringwoodite	Mg_2SiO_4
	fe-ringwoodite	Fe_2SiO_4
orthopyroxene	enstatite	$Mg_2Si_2O_6$
	ferrosilite	$Fe_2Si_2O_6$
	mg-tschermaks	$MgAl_2SiO_6$
	ortho-diopside	$CaMgSi_2O_6$
clinopyroxene	diopside	$CaMgSi_2O_6$
	hedenbergite	$CaFeSi_2O_6$
	clinoenstatite	$Mg_2Si_2O_6$
	ca-tschermaks	$CaAl_2SiO_6$
	jadeite	$NaAlSi_2O_6$
	acmite	$NaFe^{3+}Si_2O_6$
HP clinopyroxene	hp-clinoenstatite	$Mg_2Si_2O_6$
	hp-clinoferrosilite	$Fe_2Si_2O_6$
Ca-perovskite	ca-perovskite	$CaSiO_3$
akimotoite	mg-akimotoite	$MgSiO_3$
	fe-akimotoite	$FeSiO_3$
	corundum	Al_2O_3
	hematite	$Fe_2^{3+}O_3$
	eskolaite	Cr_2O_3
garnet	pyrope	$Mg_3Al_2Si_3O_{12}$
	almandine	$Fe_3Al_2Si_3O_{12}$
	grossular	$Ca_3Al_2Si_3O_{12}$
	mg-majorite	$Mg_4Si_4O_{12}$
	na-majorite	$Na_2MgSi_5O_{12}$
	andradite	$Ca_3Fe_2^{3+}Si_3O_{12}$
	knorringite	$Mg_3Cr_2Si_3O_{12}$
quartz	quartz	SiO_2
coesite	coesite	SiO_2
stishovite	stishovite	SiO_2
bridgmanite	mg-bridgmanite	$MgSiO_3$
	fe-bridgmanite	$FeSiO_3$
	al-bridgmanite	Al_2O_3
	ferric-bridgmanite	$Fe_2^{3+}O_3$
	ferric-bridgmanite-ls	$Fe_2^{3+}O_3$
	ferric-al-bridgmanite	$Fe^{3+}AlO_3$
	cr-bridgmanite	Cr_2O_3
post-perovskite[E]	mg-post-perovskite	$MgSiO_3$
	fe-post-perovskite	$FeSiO_3$
	al-post-perovskite	Al_2O_3
	ferric-post-perovskite	$Fe_2^{3+}O_3$
	cr-post-perovskite	Cr_2O_3
ferropericlase	periclase	Mg_4O_4
	wustite	Fe_4O_4
	wustite-ls	Fe_4O_4
	alpha-naalo2	$Na_2Al_2O_4$
	magnetite	$Fe^{2+}Fe_2^{3+}O_4$
Ca-ferrite	mg-ca-ferrite	$MgAl_2O_4$
	fe-ca-ferrite	$FeAl_2O_4$
	na-ca-ferrite	$NaAlSiO_4$
	high-pressure-magnetite	$Fe^{2+}Fe_2^{3+}O_4$
	cr-ca-ferrite	$MgCr_2O_4$
kyanite	kyanite	Al_2SiO_5
nepheline	nepheline	$NaAlSiO_4$
α-iron	alpha-iron	Fe
γ-iron	gamma-iron	Fe
ε-iron	epsilon-iron	Fe
α-PbO_2-SiO_2[E]	alpha-pbo-type-sio2	SiO_2
NAL phase[M]	mg-nal-phase	$NaMg_2Al_5SiO_{12}$
	fe-nal-phase	$NaFe_2Al_5SiO_{12}$
	na-nal-phase	$Na_3Al_3Si_3O_{12}$

[E] Earth model only. [M] Mars model only.

nGibbs models closed systems that conserve oxygen content, as well as open systems where oxygen fugacity is imposed. Iron is the major redox-active element in *nGibbs*. Every element besides iron is tracked at a single, fixed oxidation state (with a limited exception for carbon) and therefore the oxygen content of the system is fully specified by the ferric-to-total iron ratio. The only other redox-active element is carbon in MELTS 1.2 emulators, wherein carbon exists as an oxide in melt, fluid, and calcite, and as reduced carbon in graphite. Graphite is represented as $C(FeO)_4(Fe_2O_3)_{-2}$ in oxide space in order to conserve oxygen, which introduces strong coupling between Fe^{2+}/Fe^{3+} and graphite for closed-system MELTS 1.2.0 models. This coupling vanishes for open system models where oxygen is not conserved. Similarly, native iron phases are represented as $(FeO)_3(Fe_2O_3)_{-1}$ in oxide space in *nGibbs*.

nGibbs accepts one of two inputs that define the redox state: an explicit, distinct definition of both FeO and Fe_2O_3, which the closed-system models conserve; or an oxygen fugacity (fO_2) in log unit offset from the fayalite-magnetite-quartz buffer (ΔFMQ), which the open-system models impose. Oxygen buffering is available in neither HeFESTo nor its *nGibbs* emulator. In these open models, only cation mass is conserved and the oxygen content of the system is a consequence of the predicted phase assemblage, except for the liquid, for which total iron is speciated following the procedure of Kress and Carmichael (1991).

Forward pass

The network sees the input vector state variables u and bulk composition b, represented as the vector $[u, b]$ and:

1. a shared encoder produces a central working representation of the system's state;
2. one small network per phase gives the signed abundance/affinity g;
3. one softmax-activated network per chemically-variable phase gives that phase's endmember fractions $\hat{x}$.

This workflow is illustrated for a single phase in Fig. 4.1.

During inference, the predicted abundance is the rectified affinity,

$$\hat{n} = \max(g, 0). \tag{4.1}$$

Phases and chemical endmembers that contain an element that is absent from b are forced to be zero.

Component moles and the reconstructed bulk composition follow from stoichiometry:

$$\hat{m} = (\hat{x}V + b) \circ (\hat{n}M), \tag{4.2}$$

$$\hat{b} = \hat{m}A. \tag{4.3}$$

Here $\circ$ is the element-wise product, so $\hat{m}$ is each component's phase abundance multiplied by its fraction in that phase. Conservation of mass (adjusting $\hat{m}$ until $\hat{b} = b$) is enforced only at inference (§4.2).

Thermodynamic justification for softmax activated phase composition prediction

The softmax activation function produces a positive, nonzero output that sums to 1, and is most often applied in machine learning for classification problems in tandem with cross-entropy loss, as its outputs are often interpreted as a probability or likelihood distribution over categorical classes (Holm et al., 2023; Pearce et al., 2021; X. Li et al., 2019; Yang et al., 2018). For class i of ϕ categorical classes, the softmax function is $e^{z_i} / \sum_{j \in \phi} e^{z_j}$ where z_i is the logit (linear dense layer output) for class i.

The softmax function is used here for a bounded regression problem, to predict the mole fractions of chemical endmembers in a phase, which are also positive, nonzero, and sum to 1. Originally, we chose the softmax function because of this quality, and early tests showed it outperformed alternative activations (such as a renormalization of ReLU outputs) for predicting chemical endmember fraction. Although this application of the softmax is unusual, it is justified in that its functional form mirrors that of the thermodynamic problem of predicting chemical endmember fractions in a phase at equilibrium.

For a general solution, the chemical potential of endmember i in phase ϕ is $\mu_i = \mu_i^\circ(P,T) + RT \ln a_i$, with activity $a_i = \gamma_i x_i$ (e.g., Anderson, 2005; Ganguly, 2008). The activity coefficient $\gamma_i \equiv a_i / x_i$ collects the departure from ideal molecular mixing, including excess (interaction) terms and multi-site configurational effects. At equilibrium, μ_i equals the apparent chemical

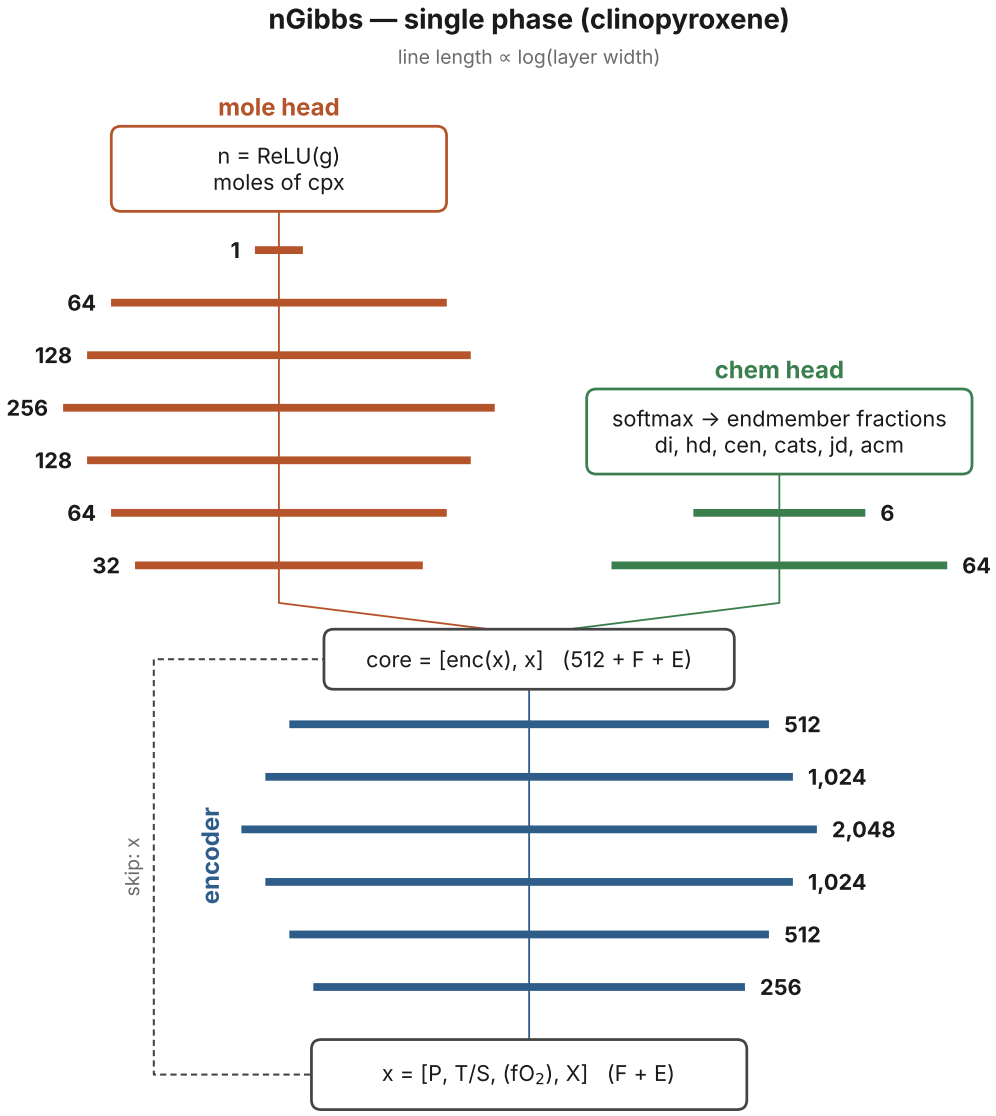


Figure 4.1: Schematic of a forward pass through the neural network geometry, drawn for a single phase (e.g. clinopyroxene). Every phase has its own mole head, which outputs the signed abundance/affinity g, and every phase with variable chemistry also has a chem head, which outputs the softmax-activated endmember fractions $\hat{x}$. The encoder, in contrast, is not specific to any phase but encodes a single system-wide shared state from $[u, b]$ that is passed to all of the heads. These are somewhat large models that carry about 30 MB of weights each. A single trained model thus has a larger storage footprint than all of alphaMELTS or HeFESTo, but remains much faster to run.

potential μ_i^* that the coexisting assemblage imposes on that endmember, so

$$\ln x_i \;=\; z_i \;\equiv\; \frac{\mu_i^* - \mu_i^\circ(P,T) - RT\ln\gamma_i}{RT}. \tag{4.4}$$

The standard state chemical potential $\mu_i^\circ(P,T)$ and the non-ideal contribution ($RT\ln\gamma_i$) may be gathered together in an effective reference chemical potential $\tilde{\mu}_i^\circ(P,T) \equiv \mu_i^\circ(P,T) + RT\ln\gamma_i$.

$$\ln x_i \;=\; z_i \;\equiv\; \frac{\mu_i^* - \tilde{\mu}_i^\circ(P,T)}{RT}. \tag{4.5}$$

Imposing the constraint that the sum of the chemical end members is 1 ($\sum_{i\in\phi} x_i = 1$), it follows that

$$x_i \;=\; \frac{x_i}{\sum_{j\in\phi} x_j} \;=\; \frac{e^{z_i}}{\sum_{j\in\phi} e^{z_j}} \;=\; \mathrm{softmax}(\boldsymbol{z})_i. \tag{4.6}$$

so the softmax is the exact functional form describing the equilibrium composition as a function of the chemical potential with respect to a reference state, normalized by RT. The network does not need to track the non-ideal terms because it predicts the logits $\hat{z}_i$ that satisfy it directly, which absorbs the activity model into the learned mapping from (P, T, bulk composition) to $\boldsymbol{z}$. In the ideal limit ($\gamma_i = 1$), the logit output z represents a unitless chemical potential of each endmember relative to the reference state.

A non-negative, closure-preserving reparameterization of the pyroxene and spinel components of the MELTS models

The softmax function always produces outputs that are strictly positive and sum to one, so the network cannot emit a non-physical composition (e.g. negative atoms) for phases composed of components that are themselves legal compositions, regardless of its unconstrained raw output. For spinels and pyroxenes, however, MELTS predicts legal compositions that are outside the simplex defined by the end members (i.e., some components can be negative). A softmax activation therefore cannot be applied to these components directly, but these compositions can be remapped to another basis that is compatible with a softmax activation: components that are non-negative and sum to one.

Even for spinels and pyroxenes, which can contain negative components, there exists a limit lying outside the simplex that expresses legal compositions,

Table 4.4: Site occupancies of the seven MELTS pyroxene components (Sack and Ghiorso, 1994).

Component	M2	M1	T, T
diopside (x_0)	Ca	Mg	Si, Si
clinoenstatite (x_1)	Mg	Mg	Si, Si
hedenbergite (x_2)	Ca	Fe^{2+}	Si, Si
alumino-buffonite (x_3)	Ca	$\frac{1}{2}$Mg + $\frac{1}{2}$Ti	Al, Si
buffonite (x_4)	Ca	$\frac{1}{2}$Mg + $\frac{1}{2}$Ti	Fe^{3+}, Si
essenite (x_5)	Ca	Fe^{3+}	Al, Si
jadeite (x_6)	Na	Al	Si, Si

implying that there exists some compositional basis that can remap legal compositions to lie within the simplex. We were able to analytically derive a solution for pyroxene, but spinel was instead solved numerically. All other phases' compositional bases in MELTS and HeFESTo are already strictly non-negative.

Pyroxene

We write the mole fractions of the seven MELTS pyroxene components (diopside, clinoenstatite, hedenbergite, alumino-buffonite, buffonite, essenite, jadeite) as $x_0, \ldots, x_6$. Pyroxene has the general formula M2 M1 T_2O_6, with one large octahedral site (M2), one small octahedral site (M1), and two tetrahedral sites (T); the site occupancies of the seven components follow Sack and Ghiorso (1994) and are given in Table 4.4. The atoms-per-formula-unit totals implied by that table are

$$\begin{aligned} \mathrm{Ca} &= x_0 + x_2 + x_3 + x_4 + x_5, & \mathrm{Na} &= x_6, \\ \mathrm{Mg} &= x_0 + 2x_1 + \tfrac{1}{2}(x_3 + x_4), & \mathrm{Al} &= x_3 + x_5 + x_6, \\ \mathrm{Fe}^{3+} &= x_4 + x_5, & \mathrm{Ti} &= \tfrac{1}{2}(x_3 + x_4). \end{aligned} \tag{4.7}$$

Seven components subject to closure leave six independent directions. These separate naturally into three groups. Three elements are carried by exactly one component each and can be read off directly: Fe^{2+} (hedenbergite), Na (jadeite), and Mg on the M2 site specifically (clinoenstatite, the Mg–Ca exchange vector distinct from the M1 magnesium already carried by diopside). Three further

elements—Fe^{3+}, Al and Ti—are each split across two or three components, so no single component fraction stands in for them; the corresponding coordinates are obtained by requiring their pairwise sums to equal the element totals of Eq. (4.7),

$$r_1 + r_3 = \mathrm{Al}, \qquad r_2 + r_3 = \mathrm{Fe}^{3+}, \qquad r_1 + r_2 = 2\,\mathrm{Ti}, \tag{4.8}$$

which is a 3×3 system with the unique solution given below. The remaining two quantities, Ca and Si, are never tracked directly; they are pinned by requiring each crystallographic site to be exactly full—a second, local layer of closure nested inside the overall constraint $\sum_i x_i = 1$. Collecting these,

$$r_0 = x_2, \quad r_1 = x_3 + \tfrac{1}{2}x_6, \quad r_2 = x_4 - \tfrac{1}{2}x_6, \quad r_3 = x_5 + \tfrac{1}{2}x_6, \quad r_4 = x_6, \quad r_5 = x_1. \tag{4.9}$$

Each r_i is bounded for any physically legal pyroxene, but not on $[0, 1]$. Rescaling and recombining gives six quantities that are individually bounded on $[0, 1]$:

$$\begin{aligned} u_0 &= \tfrac{1}{2}r_0, \quad u_1 = \tfrac{1}{2}(r_1 + r_2), \quad u_2 = \tfrac{1}{2}(r_1 + r_3), \\ u_3 &= \tfrac{1}{2}(r_2 + r_3), \quad u_4 = \tfrac{1}{2}r_4, \quad u_5 = \tfrac{1}{2}r_4 + r_5. \end{aligned} \tag{4.10}$$

Unit-sum closure of the full transformed vector is a separate matter and does *not* follow from site positivity. Because $\sum_i x_i = 1$, the constant $u_C \equiv 1 = \sum_i x_i$ is itself a linear function of the mole fractions and can be appended as a seventh coordinate. This makes the map square (and hence invertible), and gives the diopside reference endmember, for which $u_0, \ldots, u_5$ all vanish, a non-zero value in the transformed space. The resulting vector $(u_C, u_0, \ldots, u_5)$ is non-negative for any legal pyroxene but does not sum to one. Closure was imposed by evaluating the linear map of Eqs. (4.9)–(4.10) at each of the seven pure endmembers and dividing each resulting row by its own row sum, yielding the fixed matrix U_{px} of Table 4.5, in which every row is individually non-negative and sums to unity.

Because closure was imposed by endmember normalization rather than derived from the site-occupancy argument, its validity for a general (non-endmember) composition is not guaranteed *a priori*, and was instead established empirically: applying U_{px} to the full set of MELTS-computed pyroxene compositions in the training data yields vectors that are non-negative and sum to unity, and $\mathsf{U}_{\mathrm{px}}^{-1}$ recovers the original mole fractions within numeric precision.

Table 4.5: The pyroxene reparameterization matrix U_{px}. Rows are the seven MELTS pyroxene components; columns are the transformed coordinates. Every row is non-negative and sums to unity. The same matrix is applied to both the orthopyroxene and clinopyroxene solutions.

Component	u_C	u_0	u_1	u_2	u_3	u_4	u_5
diopside	1	0	0	0	0	0	0
clinoenstatite	1/2	0	0	0	0	0	1/2
hedenbergite	2/3	1/3	0	0	0	0	0
alumino-buffonite	1/2	0	1/4	1/4	0	0	0
buffonite	1/2	0	1/4	0	1/4	0	0
essenite	1/2	0	0	1/4	1/4	0	0
jadeite	2/5	0	0	1/5	0	1/5	1/5

Spinel

We were unable to reduce the spinel components (chromite, hercynite, magnetite, spinel *sensu stricto*, ulvöspinel) to a comparably clean set of site-occupancy inequalities. No analogous analytic transform was therefore pursued. Instead a masked, invertible linear transform U_{sp} was fit numerically.

The matrix was parameterized as lower triangular, which guarantees invertibility for any non-zero diagonal. Chromite uniquely expresses Cr contents and is therefore always non-negative. It was mapped onto its own dedicated output coordinate and excluded from the fit so that chromium-free models would be able to use the same transform. The remaining entries were optimized by gradient descent (Adam; PyTorch) against approximately 3×10^6 MELTS spinel compositions drawn from the training data, minimizing a two-term objective: a mean-squared term driving each transformed composition's row sum toward unity, and a leaky-ReLU penalty on negative transformed entries driving them toward zero. The learning rate was annealed by a factor of ten every 2,500 of 10,000 steps; the diagonal was reset to unity whenever it decayed toward zero, preserving invertibility; and coefficients falling below 0.05 in magnitude were periodically hard-thresholded to zero to keep the fitted matrix sparse. The solution satisfyingly converged to one that can be expressed in terms of relatively simple fractions.

The resulting matrix and its analytic inverse were empirically validated in the same way as the pyroxene transform: by histogramming the transformed compositions and their row sums, checking their extrema, and comparing round-

Table 4.6: The spinel reparameterization matrix U_{sp}. Rows are the four (non-chromite) MELTS spinel components; columns are the transformed coordinates. Every row is non-negative and sums to unity.

Component	u_C	u_0	u_1	u_2
Hercynite	1	0	0	0
Magnetite	2/5	3/5	0	0
Spinel	19/20	0	1/20	0
Ulvöspinel	1/5	0	0	4/5

tripped compositions against the originals across the full training set. Original compositions are recovered within numeric precision and the transformed components of all MELTS spinels are confined almost entirely to $[0, 1]$.

Legality constraint for pyroxene and spinel compositions

The transforms derived above in §4.2 guarantee that every legal composition lands in the simplex ($\hat{x} \geq 0$, $\sum \hat{x} = 1$), but not the reverse: the legal compositions fill only a portion of the simplex, and other regions of the simplex are forbidden because they non-physically map to negative atoms. Because the softmax can return any point in the simplex, it is possible for the predicted output to be illegal without the addition of further constraints. To address this, we apply a post-processing step to the softmax outputs of pyroxene and spinel to force them to be legal compositions. This correction is commonly applied to orthopyroxene (6-7% of predicted OPX), for which CaO is low generally and is easily pushed negative. For spinel and clinopyroxene, these corrections are more rare, affecting less than 2% of predicted spinel and 0.1% of predicted clinopyroxene.

Correcting a single negative oxide

A single negative oxide can be safely corrected with trivial arithmetic.

The first step in correcting a single negative oxide is to separate the positive and negative oxide contributions from each component (S_+ and S_- respectively). For oxide i, let q_i be the column of oxide stoichiometry for the transformed components as given in Table 4.7.

$$S^+ = \sum_{q_i>0} q_i\,\hat{x}_i, \qquad S^- = \sum_{q_i<0} |q_i|\,\hat{x}_i, \qquad \text{oxide moles} = S^+ - S^- \tag{4.11}$$

Component	Al_2O_3	CaO	FeO	Fe_2O_3	MgO	Na_2O	SiO_2	TiO_2	Cr_2O_3
Pyroxene									
u_C	0	1	0	0	1	0	2	0	–
u_0	0	1	3	0	−2	0	2	0	–
u_1	0	1	0	0	1	0	0	2	–
u_2	2	1	0	0	−1	0	0	0	–
u_3	0	1	0	2	−1	0	0	0	–
u_4	0.5	−2	0	0	−4	2.5	4	0	–
u_5	0	−1	0	0	3	0	2	0	–
Spinel									
Chr	0	–	1	0	0	–	–	0	1
u_0	1	–	1	0	0	–	–	0	0
u_1	$-\frac{2}{3}$	–	1	$\frac{5}{3}$	0	–	–	0	0
u_2	1	–	−19	0	20	–	–	0	0
u_3	$-\frac{1}{4}$	–	$\frac{9}{4}$	0	0	–	–	$\frac{5}{4}$	0

Table 4.7: Transformed components and their stoichiometries. These are the rows of the inverted transform matrices $\mathsf{U}_{\text{px}}^{-1}$ and $\mathsf{U}_{\text{sp}}^{-1}$. Mapped into an oxide space, these transformed components contain negative atoms and therefore a softmax output in this space is capable of producing illegal negative atoms.

The oxide is negative if $S^- > S^+$. We employ a simple fix that aims to preserve as much of the original prediction as possible. We find two scaling factors to scale the positive and negative contributing components to balance the oxide. We multiply every component fraction on the positive side by one factor a and every fraction on the negative side by one factor b. If we set the oxide to zero and enforce the closure constraint, we formulate a system of two equations that fix the two unknowns a and b:

$$\text{oxide becomes exactly zero:} \quad aS^+ = bS^-, \tag{4.12}$$

$$\text{fractions still sum to one:} \quad aP + bN = 1, \tag{4.13}$$

where P and N are the plain sums of the component fractions on each side.

$$P = \sum_{\hat{x}_i > 0} \hat{x}_i, \qquad N = \sum_{\hat{x}_i < 0} \hat{x}_i, \qquad P + N = 1 \tag{4.14}$$

From Eq. (4.12), $b = aS_+/S_-$; substituting into Eq. (4.13) gives

$$a = \frac{S^-}{S^-P + S^+N}, \qquad b = \frac{S^+}{S^-P + S^+N}. \tag{4.15}$$

Pyroxene: CaO < 0

From Table 4.7,

$$S^+ = \hat{x}_C + \hat{x}_0 + \hat{x}_1 + \hat{x}_2 + \hat{x}_3, \qquad S^- = 2\hat{x}_4 + \hat{x}_5. \tag{4.16}$$

All positive coefficients are 1, so $P = S^+$ and $N = \hat{x}_4 + \hat{x}_5$, and Eq. (4.15) simplifies to:

$$b = \frac{1}{3\hat{x}_4 + 2\hat{x}_5}, \qquad a = \frac{S^-}{S^+\left(3\hat{x}_4 + 2\hat{x}_5\right)}. \tag{4.17}$$

The CaO correction is applied to clinopyroxene only.

Pyroxene: MgO < 0

$$S^+ = \hat{x}_C{+}\hat{x}_1{+}3\hat{x}_5, \quad P = \hat{x}_C{+}\hat{x}_1{+}\hat{x}_5; \qquad S^- = 2\hat{x}_0{+}\hat{x}_2{+}\hat{x}_3{+}4\hat{x}_4, \quad N = \hat{x}_0{+}\hat{x}_2{+}\hat{x}_3{+}\hat{x}_4, \tag{4.18}$$

a, b then can be calculated using Eq. (4.15). The MgO correction is applied to clinopyroxene only.

Spinel: FeO < 0 and $Al_2O_3 < 0$

Spinel has two oxides that can go negative, sometimes simultaneously, substantially complicating the problem. From Table 4.7,

$$\mathrm{FeO} = \hat{x}_{\mathsf{Chr}} + \hat{x}_0 + \hat{x}_1 + \tfrac{9}{4}\hat{x}_3 - 19\,\hat{x}_2, \tag{4.19}$$

$$\mathrm{Al_2O_3} = \hat{x}_0 + \hat{x}_2 - \tfrac{2}{3}\hat{x}_1 - \tfrac{1}{4}\hat{x}_3. \tag{4.20}$$

In models without Cr, $\hat{x}_{\mathsf{Chr}} = 0$.

Three pools. Because the two constraints share components, the components are grouped by the signs they carry in the two equations, and each group gets its own factor:

Pool	Components	FeO	Al_2O_3	Factor
$A = \hat{x}_{\mathsf{Chr}} + \hat{x}_0$	Chr, u_0	+	+	a
$B = \hat{x}_1 + \hat{x}_3$	u_1, u_3	+	−	b
$C = \hat{x}_2$	u_2	−	+	c

In all cases, the closure constraint is enforced:

$$aA + bB + cC = 1, \tag{4.21}$$

If only one oxide is negative, then only the violated oxide is set to zero, and the pool it does not involve held fixed. Then, the second equation for each scenario is:

- *FeO only* ($b = 1$): Eq. (4.21) and $aA + \hat{x}_1 + \frac{9}{4}\hat{x}_3 = 19\,cC$;
- *Al_2O_3 only* ($c = 1$): Eq. (4.21) and $a\hat{x}_0 + C = b\left(\frac{2}{3}\hat{x}_1 + \frac{1}{4}\hat{x}_3\right)$.

Both violated. If both oxides are negative, or if fixing one would push the other below zero, all three factors are solved together with three constraints: closure, $\mathrm{FeO} = 0$, and $\mathrm{Al_2O_3} = 0$,

$$\begin{pmatrix} A & B & C \\ A & \hat{x}_1 + \frac{9}{4}\hat{x}_3 & -19C \\ \hat{x}_0 & -\left(\frac{2}{3}\hat{x}_1 + \frac{1}{4}\hat{x}_3\right) & C \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}. \tag{4.22}$$

Figure 4.2 compares spinel compositions predicted by the MELTS 1.0.2 isothermal emulator with those computed by MELTS, before and after this correction.

Conservation of mass

The assemblage predicted by the emulator is not strictly constrained to reproduce the bulk composition; it is only encouraged to with a loss term during training (§4.2). Therefore, a post-processing step is necessary to ensure that the final assemblage is consistent with the requested bulk composition. *nGibbs* uses a conservation-of-mass correction adapted from Asimow and Ghiorso (1998), who find the smallest relative change to $\hat{m}$ that makes $\hat{b} = b$. The implementation in *nGibbs* differs from their procedure most strongly in two ways: (1) the minimization is solved in parallel for thousands of assemblages at once on a GPU, and (2) the correction is done iteratively, with constraints enforced at each step so that no elemental abundance becomes negative.

The section makes use of variables defined in Table 4.1 as well as those unique to this section in Table 4.8. Index forms are given alongside the matrix forms throughout, with c running over components and e, e' over elements.

The neural network returns the component moles $\hat{m}$ (the endmembers of every phase, in the transformed basis of Section 4.2). The element moles they imply

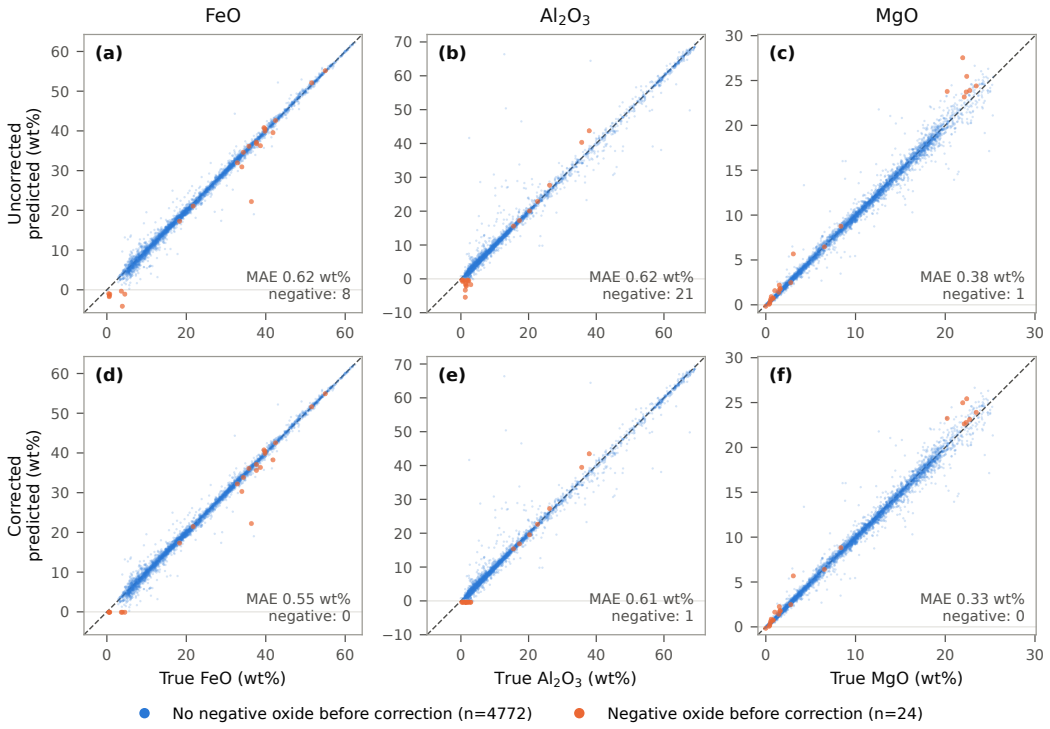


Figure 4.2: Spinel compositions predicted by the closed-system MELTS 1.0.2 isothermal emulator (Cr variant) against MELTS for the 15,000 test assemblage subset. The top row (a–c) is the raw prediction of the emulator, and the bottom row (d–f) is the same data with the legality correction of §4.2. Orange points had at least one negative oxide in spinel before correction and are colored the same in both rows. Dashed lines are 1:1; each panel gives the mean absolute error (MAE) and the number of predicted spinels with that oxide below zero.

Symbol	Dim.	Meaning
δb	E	Bulk residual in element moles, $\delta b = \Gamma\, b - \hat{m}A$
$\delta\hat{m}$	C	Correction to the component moles
ζ	C	Relative correction, $\zeta_c = \delta\hat{m}_c/\hat{m}_c$
D	C×E	Abundance-weighted stoichiometry, $D = \mathrm{diag}(\hat{m})\, A$, i.e. $D_{ce} = \hat{m}_c A_{ce}$
ν	E	Lagrange multipliers, one per element
κ	1	Regularization added to the diagonal of G (10^{-12})

Table 4.8: Symbols added for conservation of mass. Dimensions as in Table 4.1.

are $\hat{m}A$ ($[1 \times \mathrm{C}][\mathrm{C} \times \mathrm{E}] \to [1 \times \mathrm{E}]$). The reconstructed bulk is normalized to sum to one, then its residual is given by

$$\delta b = b - \hat{m}A \quad . \tag{4.23}$$

Conservation of mass seeks a correction $\delta\hat{m}$ to the component moles such that

$$\delta\hat{m}\,A = \delta b, \qquad \text{i.e.} \qquad \sum_c \delta\hat{m}_c\,A_{ce} = \delta b_e \quad \text{for each element } e. \tag{4.24}$$

This is E equations (about ten) in C unknowns (several tens), so the system is underconstrained and has infinitely many solutions. Because the network's prediction is the best available estimate of the assemblage, it is most reasonable to choose the solution that changes it least. We define this as the solution that minimizes the sum of squared relative changes to the component moles. For convenience, we define the relative change ζ_c for each component c as

$$\zeta_c = \frac{\delta\hat{m}_c}{\hat{m}_c}, \tag{4.25}$$

We minimize the sum of squared relative changes,

$$\tfrac{1}{2}\,\zeta\,\zeta^\top = \tfrac{1}{2}\sum_c \left(\frac{\delta\hat{m}_c}{\hat{m}_c}\right)^2. \tag{4.26}$$

The factor $\frac{1}{2}$ is arbitrarily added to elegantly cancel the 2 produced by the differentiation below. We introduce a new variable: D, the component abundance-weighted stoichiometry matrix, to relate ζ to the bulk residual δb.

$$\zeta\,D = \delta b, \qquad D = \mathrm{diag}(\hat{m})\,A \quad (\mathrm{C}\times\mathrm{E}), \tag{4.27}$$

We therefore formulate the problem as a minimization of Eq. (4.26) subject to the constraints of Eq. (4.27). In other words: our solution is the smallest relative change to the component moles necessary to satisfy conservation of mass.

Solving with Lagrange multipliers

Lagrange multipliers reformulate this problem into a set of ordinary equations. One new unknown, the multiplier ν_e, is introduced for each element. Each constraint is written as an expression that must equal zero. The multipliers ν_e relate the derivatives of the objective function to the constraints. Requiring every derivative of the result to vanish then expresses two conditions at once: the constraints hold, and no change that respects them can lower the objective $\frac{1}{2}\,\zeta\,\zeta^\top$. This is the Lagrangian,

$$\mathcal{L}(\zeta,\nu) = \tfrac{1}{2}\,\zeta\,\zeta^\top - \Big(\zeta D - \delta b\Big)\,\nu^\top, \tag{4.28}$$

with ν the row of E multipliers. Now, differentiating with respect to the relative correction ζ gives

$$\frac{\partial \mathcal{L}}{\partial \zeta} = \zeta - \nu\, D^{\top} = 0 \qquad \Longrightarrow \qquad \zeta = \nu\, D^{\top} \quad [1 \times \mathrm{E}][\mathrm{E} \times \mathrm{C}] \to [1 \times \mathrm{C}]. \tag{4.29}$$

This reveals that the C relative corrections are not independent: they are all determined by just E numbers, the Lagrange multipliers. The problem has been reduced from C unknowns to E, and now the system is well-constrained.

Substituting Eq. (4.29) into the $\zeta\, D = \delta b$ constraint gives

$$\nu\, D^{\top} D = \delta b, \tag{4.30}$$

and now the only unknowns are the E multipliers ν, which can be solved with an LU (lower-upper) decomposition as

$$\nu^{\top} = SOLVE(D^{\top} D, \delta b^{\top}). \tag{4.31}$$

With ν known, Eq. (4.29) relates the relative correction ζ to the multipliers.

$$\zeta = \nu A^{\top}\, \mathrm{diag}(\hat{m}) \tag{4.32}$$

The absolute correction $\delta\hat{m}$ is then obtained by multiplying the relative correction by the component moles (Eq. (4.25)).

$$\delta\hat{m} = \zeta\, \mathrm{diag}(\hat{m}) = \nu A^{\top}\, \mathrm{diag}(\hat{m})^2 \quad [1 \times \mathrm{E}][\mathrm{E} \times \mathrm{C}][\mathrm{C} \times \mathrm{C}] \to [1 \times \mathrm{C}], \tag{4.33}$$

Why the Lagrange form is used. Asimow and Ghiorso (1998) obtain the same solution by SVD; however, their problem is distinct from this one in that they correct one assemblage at a time, whereas *nGibbs* applies this operation to tens of thousands of assemblages in parallel. The Lagrange form is used here because batched SVD handles these matrices poorly on a GPU. In an informal test on 15,000 assemblages from the HeFESTo Earth validation dataset (B = 15,000, E = 8, C = 68), `torch.linalg.pinv` took 10.1 s on the GPU (implying that the calculation was run sequentially, perhaps because the matrices were too large) against 0.29 s on the CPU (single precision), whereas the Lagrange solve took 25 ms on the GPU (double precision; 48 ms on the

CPU). The Lagrange form, however, requires double precision floating point arithmetic to recover an accurate solution, whereas the SVD is robust to single precision arithmetic because the $\nu\left(D^\top D\right)$ solve of the Lagrange formulation squares the condition number. This cost is marginal, however, beside the speedup of the Lagrange form on both CPU and GPU, and it elegantly avoids the need to bus data between CPU and GPU when *nGibbs* is run on a GPU.

Regularization

In practice, a small number κ is added to every diagonal entry of $\left(D^\top D\right)$,

$$\nu\left(D^\top D + \kappa I\right) = \delta b, \qquad \kappa = 10^{-12}, \tag{4.34}$$

where I is the E×E identity matrix. This is because $D^\top D$ has all-zero columns and rows for any bulk composition lacking any particular element and is therefore singular. For a row lacking an element, its equation becomes $\kappa\,\nu_e = 0$ for the regularized case and ν is zero because $\delta b_e = 0$ for an absent element. Without regularization, the equation for this row would be $0\nu_e = 0$, for which ν_e is undefined.

On the HeFESTo Earth Model validation dataset, the smallest non-zero diagonal entries of $D^\top D$ are $\sim 3 \times 10^{-7}$ (median over rows; Fig. 4.3), so κ must be far smaller than that. A κ as small as 10^{-12} requires double precision floating point values, but this is already required by the solve of the Lagrange multipliers and therefore poses no additional complication.

Seeding phases for unexplained elements

The conservation of mass correction can only move an element through components that contain it and are present ($\hat{m}_c > 0$). If the predicted assemblage contains no carrier of an element that is in the bulk, that element's residual cannot be removed: the regularized solve leaves it in place (Section 4.2).

Before the mass conservation correction is applied, *nGibbs* first identifies unexplained elements. Element e is unexplained if $b_e > 0$ and no component with $\hat{m}_c > 0$ contains it. For each of these unexplained elements, the phase that is predicted by *nGibbs* to be closest to saturation out of the set of phases that contain the unexplained element is forced to saturation. Its abundance, naturally, is set to the amount that exactly supplies the unexplained element.

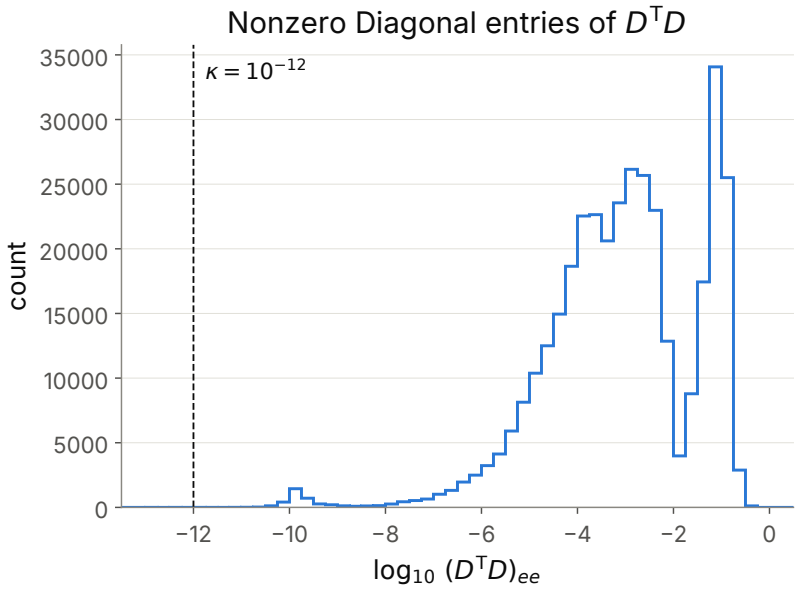


Figure 4.3: Histogram of the non-zero diagonal entries of $D^\top D$ formed by the conservation of mass correction (three iterations) on 15,000 assemblages of the isentropic (NPS) HeFESTo Earth validation dataset. The dashed line marks $\kappa = 10^{-12}$, a regularization term that is small enough to not affect the solution.

We first calculate a_e, the moles of element e in one mole of a given phase p. This is

$$a_e = M_{pc}\, diag(\hat{x}V + f)_{cc}\, A_{ce} \quad [\mathrm{P} \times \mathrm{C}][\mathrm{C} \times \mathrm{C}][\mathrm{C} \times \mathrm{E}] \to [\mathrm{P} \times \mathrm{E}] \tag{4.35}$$

The phase is added with the abundance $\hat{n}_p$ that supplies the missing element in its proportion. For unexplained element $\mathcal{U}$,

$$\hat{n}_p = a_e\, b_e \quad \text{for } e \in \mathcal{U} \tag{4.36}$$

The phase's components get $\hat{m}_c = \hat{n}_p\,(\hat{x}V + f)_c$ for every c in $M_{pc} = 1$ for phase $\mathcal{U}$.

The sequence loops until all elements for each row are explained. $\hat{m}$ is then renormalized so that $\sum_c \hat{m}_c = 1$. Rows that need no seeding are unaffected.

The iterative correction

The mass conservation correction of Section 4.2 places no bound on $\delta\hat{m}_c$, so $\delta\hat{m}_c$ is able to force $\hat{m}$ to become negative. A negative $\hat{m}_c$ is impossible; it means that a phase would have negative atoms, which is non-physical. Generally, this is a

rare occurrence because the high magnitude relative change $d\hat{m}_c/\hat{m}_c < -1$ is highly disfavored by the objective function of §4.2 and therefore is indicative of a poor prediction. This is the "overstepping" failure Asimow and Ghiorso (1998) note. Setting such a component to zero restores a physically sensible assemblage but breaks the conservation of mass again, because the moles it would have supplied are now missing. The step is therefore repeated (three times by default) in order to spread the remaining residual over the components that are still present. For each repetition, $\hat{m}$ is updated as: $\hat{m} \leftarrow \max\left(\hat{m} + \zeta\,\mathrm{diag}(\hat{m}),\ 0\right)$

MELTS emulators in *nGibbs* have another complexity for pyroxenes and spinel: the physically allowed range of components is only a region within the broader accessible range of the transformed component predictions, unlike all other phases. Thus, *nGibbs* additionally pushes these components to lie within their legal simplexes after each iteration following the method described in §4.2.

Figure 4.4 compares the bulk composition reconstructed from the raw network output and from the mass-conserving assemblage with the requested bulk composition for the HeFESTo Earth isentropic emulator.

Data generation

Bulk composition sampling

Bulk compositions are sampled from whole-rock analyses in the GEOROC database in 2022 and the EarthChem PetDB database in April 2025. Only samples with total weight percent within 2% of 100 were kept, then all samples were renormalized to a total = 100%. Evolved compositions are more common in the GEOROC database in particular, so evolved compositions were down-sampled to be symmetric with respect to mafic and ultramafic compositions. The prefiltered dataset included all GEOROC entries, whereas only mafic and ultramafic compositions were pulled from PetDB. A document containing a supersample of citations for the studies that measured these compositions can be found on the nGibbs GitHub page.

For the HeFESTo models, one fifth of the training data are drawn from the population of compositions between 5 and 20 wt% MgO, three tenths of the data are drawn from the population of compositions with greater than 20 wt% MgO, and the remaining half is a mixture of these two mafic and ultramafic populations in random proportions. Roughly 80% of these base compositions were substantially randomly altered so that *nGibbs* would perform well in

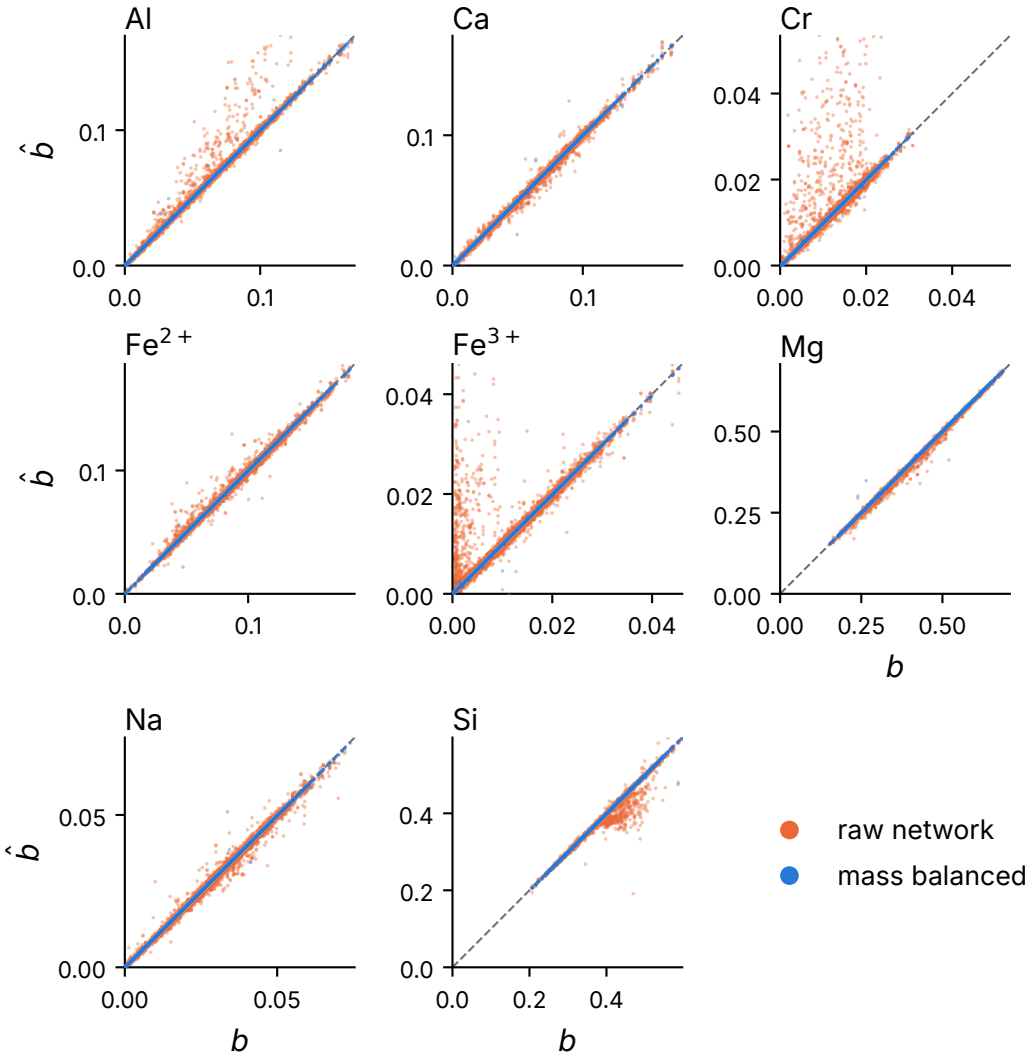


Figure 4.4: Reconstructed ($\hat{b} = \hat{m}A$) against requested (b) bulk composition in element mole fraction for 15,000 assemblages of the isentropic HeFESTo Earth validation dataset. Orange points are the raw network output without the mass conservation correction and the blue points are the results after correction (§4.2). Dashed lines are 1:1. Fe^{2+} and Fe^{3+} are conserved as separate elements by the closed-system models.

a wider variety of parameter spaces, including rock compositions that are uncommon or unknown on Earth's surface. For the Earth model, one quarter of the compositions are not perturbed. For the MELTS emulators, half of the training data is generated from any composition in the compiled database, one fifth of the data are pulled from ultramafic compositions with greater than 25 wt% MgO, and three tenths of the data are pulled from compositions with greater than 5 wt% MgO. These bulk compositions are perturbed by Gaussian noise with a standard deviation of 5%, a substantially more modest perturbation than that applied to the HeFESTo emulator training data. For the MELTS 1.0.2 emulators, half of the isobaric cooling runs are performed with fractional crystallization and half are performed with batch crystallization, whereas only half of the igneous compositions are performed with fractional crystallization for the MELTS 1.2.0 emulators.

For volatile-focused MELTS 1.2.0 emulators in particular, 35% of the training and validation datasets are drawn from batch crystallization of sedimentary compositions within the Gard database (Gard et al., 2019). Of these sedimentary bulk compositions, about one quarter are drawn from carbonate rich compositions, with CO_2 contents greater than 10 wt%, whereas the remaining sedimentary compositions contain less than 10 wt% CO_2. The remaining 65% of the training data were drawn from igneous whole rock data from GEOROC and PetDB, similarly to the other models.

In order to train emulators that perform well in a variety of redox conditions, each sampled composition is randomly assigned an oxygen fugacity from -5 to +5 $\log_{10}\Delta$FMQ for MELTS training data, or -7 to +3 $\log_{10}\Delta$FMQ for HeFESTo training data. Open system models simply carry this oxygen fugacity as a feature, whereas closed models translate this oxygen fugacity into a molar $^{+3}$Fe/$^{+2}$Fe ratio using an empirical parameterization. This is not meant to be explicitly correct; it is meant to generate training data with an appropriate distribution of molar $^{+3}$Fe/$^{+2}$Fe ratios.

$$^{+3}Fe/^{+2}Fe \approx 10^{(0.2\times\log_{10}\Delta\mathrm{FMQ})-1} \tag{4.37}$$

Manganese and nickel are excluded from the compositional basis of all released models. Manganese and nickel are not handled by HeFESTo, and we have observed that MELTS solutions that carry MnO and NiO frequently stabilize

phases with compositions rarely observed in nature, and there is little value in emulating the regions of a model that are not themselves trusted. Excluding these oxides also removes two dimensions from the component basis at small cost to the applications the emulators target.

State-variable sampling

For the MELTS 1.0.2 and 1.2.0 emulator families, each sampled composition is isobarically cooled from 1800°C in 1°C increments at a random pressure between 1 and 12000 bars. These computed assemblages have well-defined entropies and therefore are acceptable for use in training isentropic models.

HeFESTo Earth and Mars models are each trained around a core of (approximately) adiabatic simulations at reasonable potential temperatures (1200–1650°C for Earth; 500–1750°C for Mars). These adiabats are supplemented with cooler conditions due to a conductive lid regime near the surface, which is assumed to be between 0-250 km thick for Earth and 100-700 km thick for Mars. Similarly, warmer conditions are modeled at the approach to the core-mantle boundary (CMB) for the Earth model, which assume a mantle temperature between 3200–4700°C at the CMB and a 100–300 km wide conductive layer.

Although HeFESTo natively calculates isentropes, such a calculation requires an iteration of T at every step to retrieve the imposed entropy, which roughly doubles the computation time. Therefore, adiabatic PT paths were instead approximated to reduce computational expense in the generation of the training dataset. To accomplish this, a second-order linear regression of temperature as a function of specific entropy and pressure was fit to a dataset of dozens of HeFESTo isentropes at representative compositions and potential temperatures. A similar linear regression was fit for specific entropy as a function of temperature at P = 0 GPa on the same dataset to translate potential temperatures to specific entropy. Similarly to the iron speciation parameterization described above, this approach is not intended to be explicitly correct; it is intended to control the distribution of the training dataset in pressure-temperature-entropy space without unnecessary computational expense. These regressions are used to impose a roughly adiabatic pressure-temperature path for training data. These paths are perturbed with small random noise to prevent the emulators from learning relationships between composition and entropy.

Execution

MELTS simulations are orchestrated with Python scripts preparing, running, and parsing MELTS simulations using alphaMELTS versions 2.3.1 and 2.3.2 (P. M. Smith and Asimow, 2005) run with GNU parallel (Tange, 2021). Batch crystallization was run with alphaMELTS 2.3.1 and fractional crystallization was run with alphaMELTS 2.3.2. In future work, we will rebuild the MELTS datasets to consistently use alphaMELTS 2.3.2, but the results are unlikely to significantly differ. Separate datasets are generated for the closed-system emulators, and for the open-system MELTS 1.2.0 emulator. All MELTS calculations were carried out on a 20-core Intel Core i7-14700H CPU @ 3.40GHz with 32 GB of RAM. Additionally, each simulation was run with parameters adjusted to prohibit multiple crystallization of any phase, including clinopyroxenes and spinel, for which different endmembers are well-known to coexist in nature (Lindsley, 1983; Stosch and Seck, 1980). It is important to note in this regard that orthopyroxene and clinopyroxene, as well as alkali-feldspar and plagioclase, are treated as separate phases in MELTS and therefore can coexist in nGibbs as well. Future work will explore the division of clinopyroxenes into distinct low-Ca and high-Ca phases in nGibbs. Additionally, some phases that are modeled by MELTS were deliberately suppressed: sphene, melilite, aegirine, aenigmatite, cummingtonite, orthoamphibole, cristobalite, kalsilite, corundum, rutile, perovskite, ortho-oxide and sillimanite. Both of these deliberate omissions were made for simplicity and to avoid learning MELTS idiosyncrasies for phases that are not well-modeled.

Some HeFESTo simulations were run on the same machine as the MELTS runs, but the majority were run on the Caltech High Performance Computing cluster as SLURM array jobs.

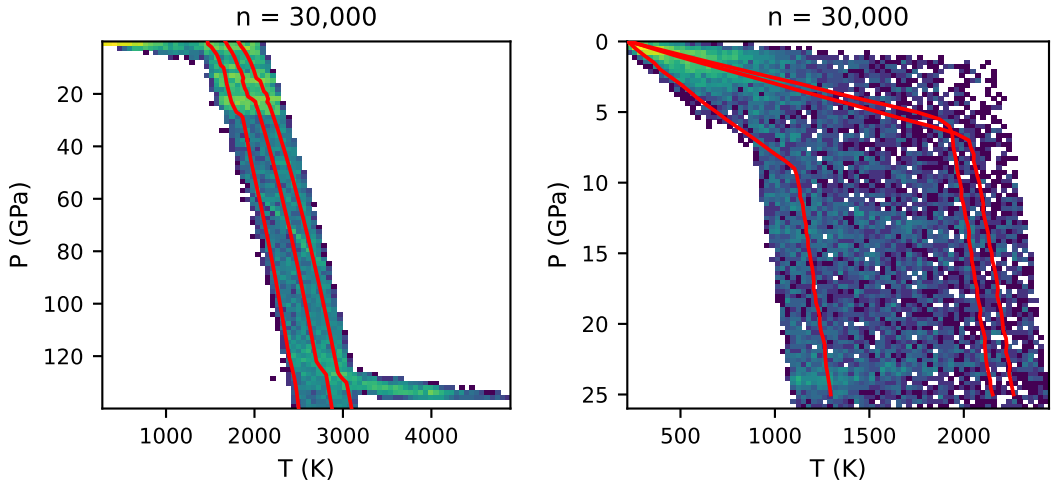
After pressure transects are calculated with HeFESTo, phase changes are identified and iteratively resampled to improve the model resolution around phase boundaries. A random subset of these phase changes is chosen (up to 3x the original simulation count), and the HeFESTo states bounding the phase transition are used to build a P-T grid sampling with higher resolution. The pressure interval is divided into five steps, which is evaluated at at least one isotherm, but more isotherms are added for every full 25 K between the bounding conditions of the phase. Along an isentrope both P and T change between the bracketing states, so this first resampling step produces a two-

dimensional PT search of the region in which the boundary must lie. Each of these isotherms again bounds the phase boundary, but now at a fixed temperature. A second iteration of resampling is performed at 0.01 GPa resolution, between the states that bound the phase boundary. Again, as many as 3x the original number of simulations are called in this step.

A fraction of simulations fail to converge or terminate without usable output. These are detected and discarded at import. Attrition differs sharply between codes and compositional domains: approximately 0.1% for HeFESTo, approximately 15% for MELTS on igneous bulk compositions, and approximately 40% for MELTS 1.2 on sedimentary bulk compositions.

Coverage

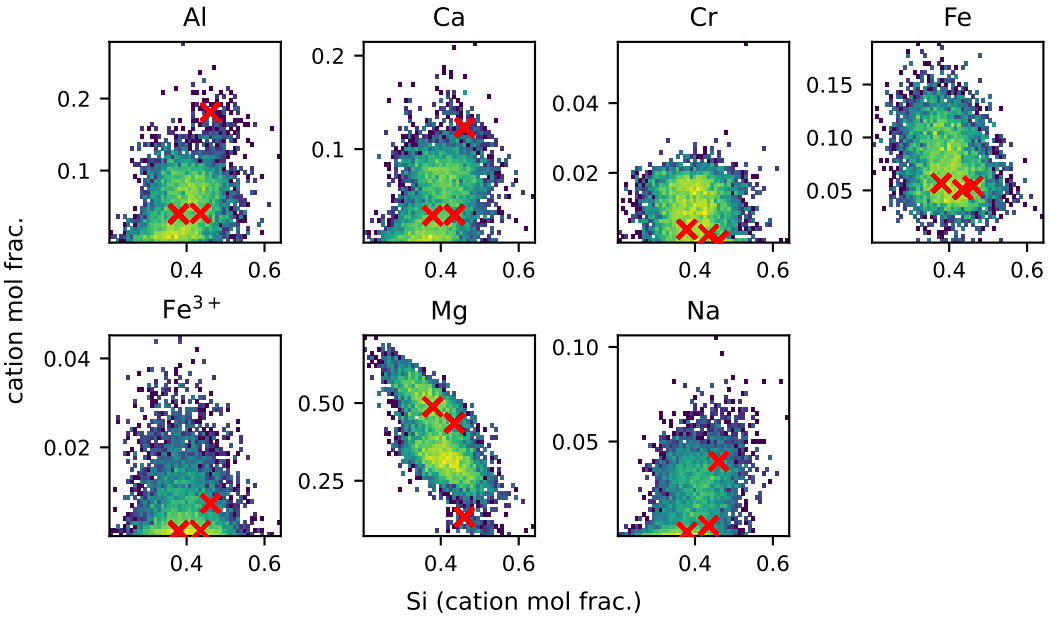
The training domain of each model's pressure and temperature is shown for the HeFESTo models in Fig. 4.5, and in composition space for all models in Figs. 4.6–4.8.



(a) HeFESTo Earth (b) HeFESTo Mars

Figure 4.5: Pressure–temperature distribution of the validation datasets of the HeFESTo Earth (a) and Mars (b) emulators. Red lines mark the deployment-test reference composition paths used in Figs. 4.15–4.21.

(a) HeFESTo Earth



(b) HeFESTo Mars

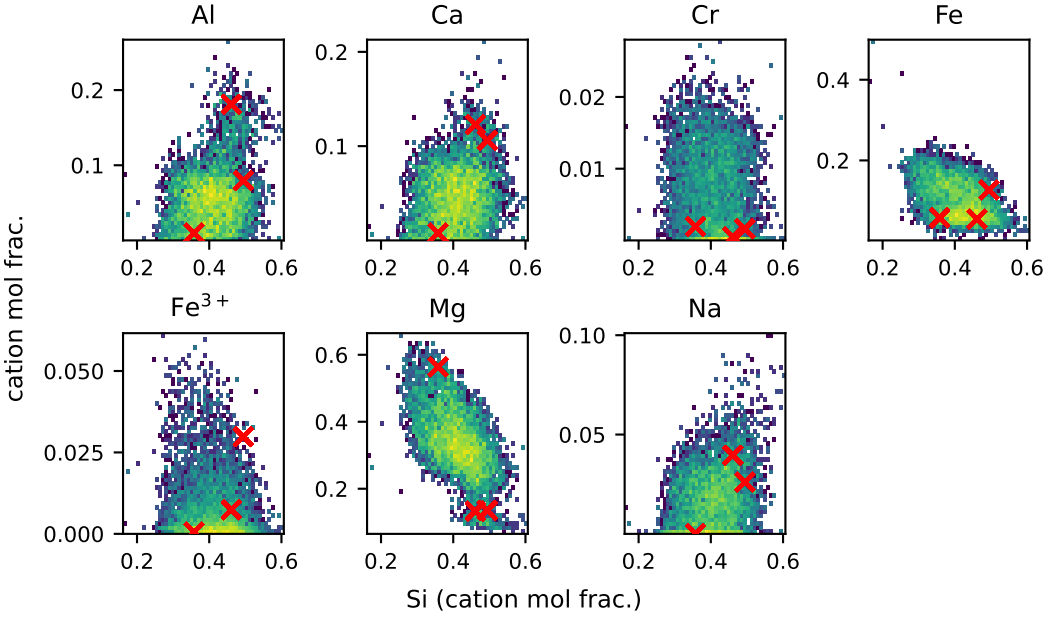


Figure 4.6: Harker diagrams of the validation datasets (as a proxy for the training data) of the HeFESTo Earth (a) and Mars (b) emulators in cation mole fraction, as two-dimensional histograms with a logarithmic color scale. Red crosses mark the deployment-test reference compositions.

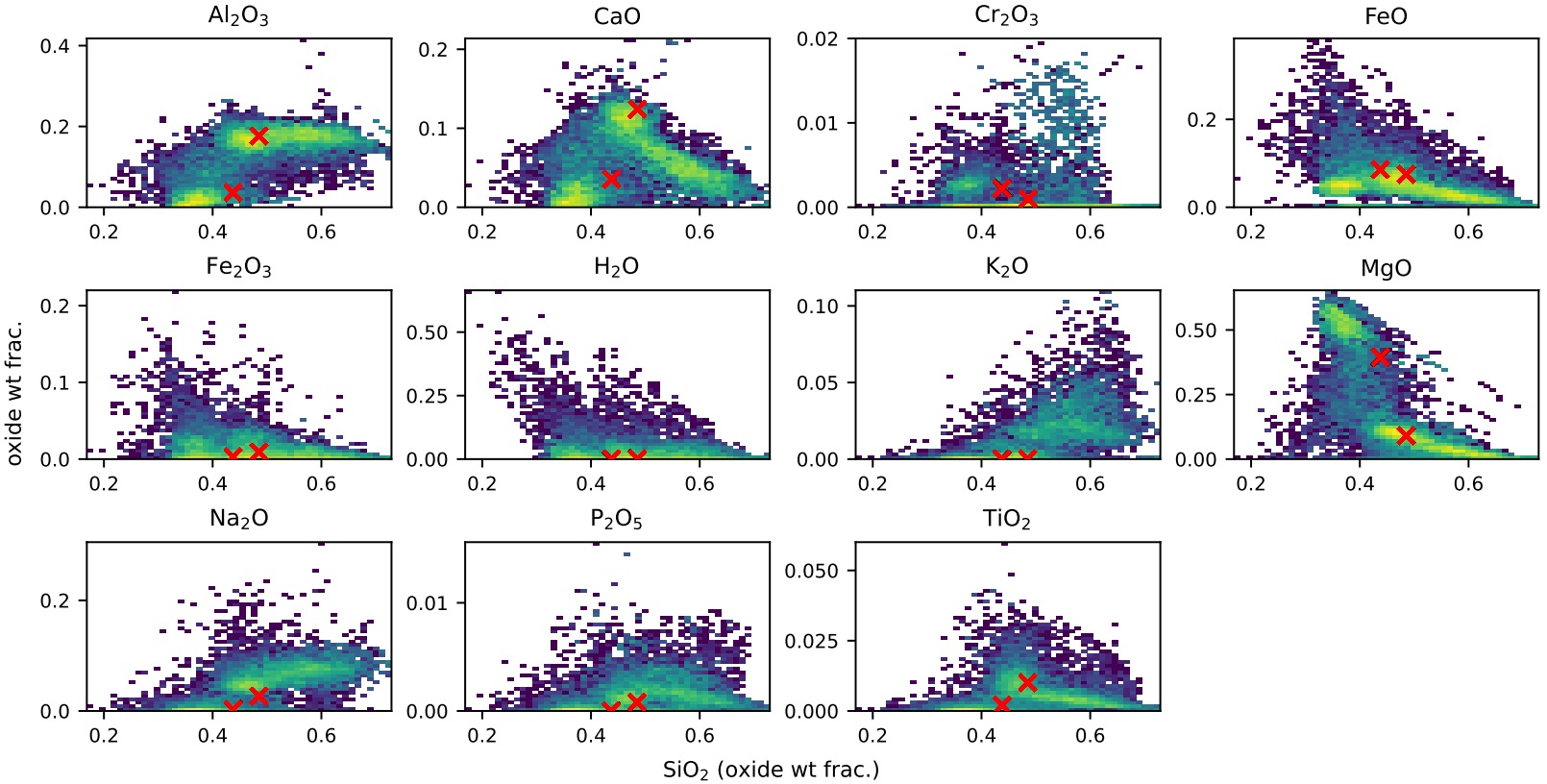


Figure 4.7: Harker diagrams of the validation datasets (as a proxy for the training data) of the closed-system MELTS 1.0.2 emulators (Cr variant) in oxide weight fraction, as two-dimensional histograms with a logarithmic color scale. Red crosses mark the deployment-test reference compositions.

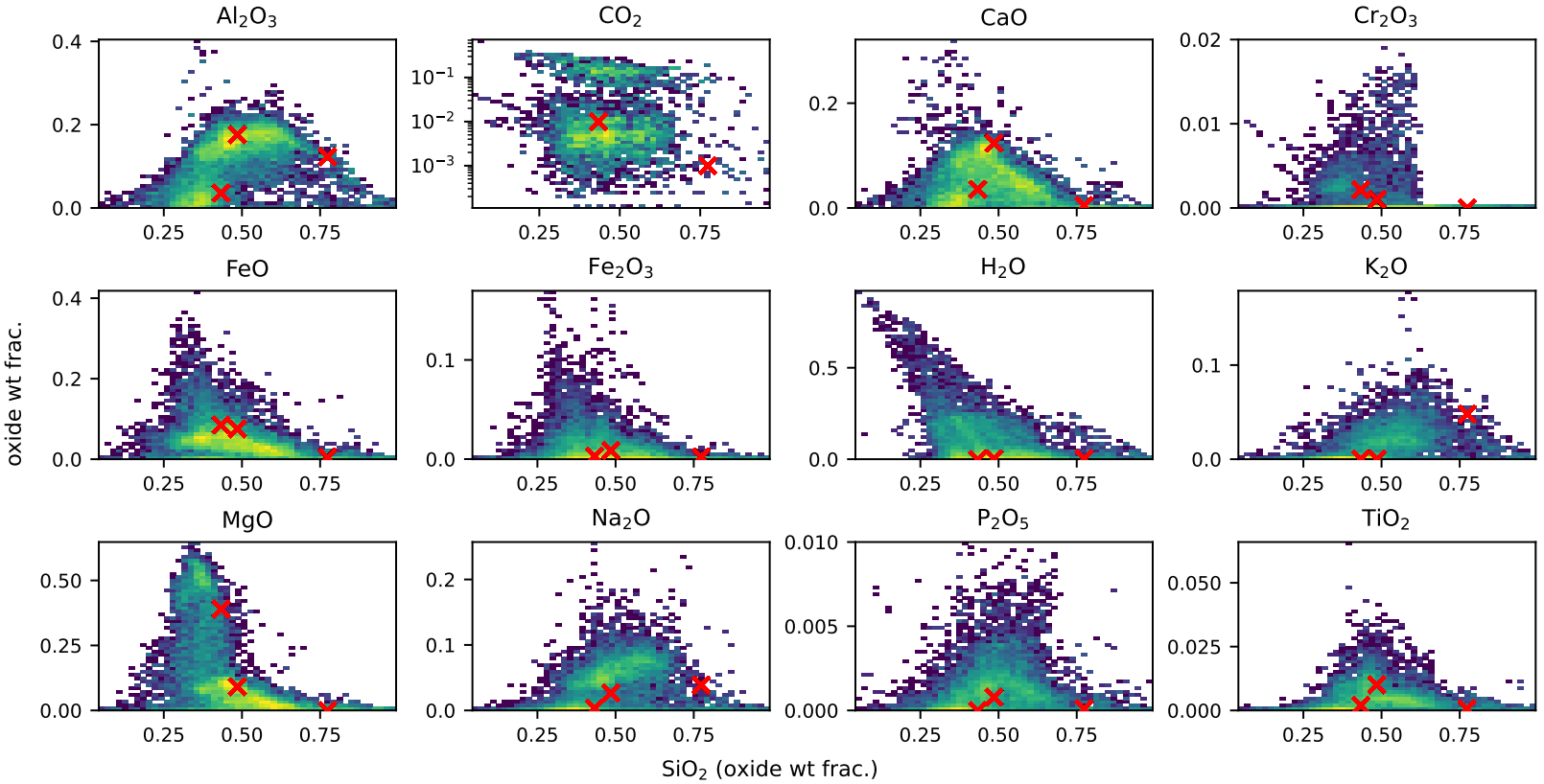


Figure 4.8: As Fig. 4.7, for the closed-system MELTS 1.2.0 emulators (Cr variant), whose training data include sedimentary bulk compositions.

Data preparation

Four operations are applied to prepare MELTS and HeFESTo outputs for training.

Consistency filtering

Rows are removed where the reference code's output is internally inconsistent, e.g. where the composition reconstructed from the reported assemblage disagrees with the stated bulk composition beyond tolerance.

Compositional filtering

A series of filters is applied to remove rows containing phases with unusual compositions generally not seen in nature. These compositions are considered to be MELTS idiosyncrasies that needn't be passed on to nGibbs. These filters include:

1. K-feldspar: at least 6 wt% K_2O and at most 4 wt% CaO.
2. Clinopyroxene: at least 6 wt% CaO, at most 9 wt% TiO_2, and at most 12 wt% Al_2O_3.
3. Orthopyroxene: at most 5 wt% TiO_2, 10 wt% Al_2O_3, and 6 wt% CaO.
4. Silicate melt: at least 30 wt% SiO_2, at most 13 wt% TiO_2, and at most 10 wt% P_2O_5.
5. Plagioclase: the sanidine component may not exceed 0.2 (mole fraction).

Rare phase removal

Phases occurring in less than 1 assemblage per thousand are considered to be too rare in the training set to be effectively learned and are removed from the component basis entirely to simplify the model. This operates on entire rows; assemblages with rare phases are discarded for containing a rare phase at any low modal proportion. It is presumed that this operation is minimally damaging, because very few assemblages are affected by design.

Shuffling

The training and validation datasets are shuffled to remove any correlation between the order of the rows and the order in which they were generated. Conventional dataloaders in PyTorch can automatically shuffle data during training, but this option is not available for *nGibbs*, because its datasets are too large to fit in memory and instead were streamed in static batches of 1,000,000 rows each from disk (SK hynix Platinum P41, 2 TB). The order in which these batches are streamed is shuffled every epoch, but the contents of each bundle are fixed.

Partitioning

To monitor model overfitting during training, a validation dataset distinct from the training set is used. Training and validation datasets are generated with the same pipelines, but are separated *by unique bulk composition*. Bulk compositions were partitioned between training and validation datasets and are treated independently as separate datasets throughout the processing pipeline. This ensures that the validation dataset tracks the same fundamental structure as the training dataset while avoiding data leakage. Successive states along one isobaric or isentropic path are strongly autocorrelated, so a row-wise random split would place very similar data in both the training and validation datasets. The resulting error would measure interpolation between adjacent states of a rock the model had already seen, rather than generalization to a rock it had not, and would be optimistic by a wide and unquantifiable margin. Splitting by composition, before the reference thermodynamic models were applied, makes the validation dataset's error track overfitting to the training dataset more faithfully. This overfitting metric is instrumental to develop an emulator that can be effectively applied to new data.

Compositions were assigned in the ratio 80:20 between training and validation. The validation partition is used every epoch for model selection, to tune adaptive regularization, and for early stopping. Unless otherwise stated, all performance metrics presented here in Section 4.3 are for the validation datasets; the emulator's performance on the training datasets is not reported.

Training

The objective

For each batch the trainer minimizes

$$L = \lambda_s L_s + \lambda_x L_x + \lambda_n L_n + \lambda_b L_b \tag{4.38}$$

covering phase presence (L_s), chemistry (L_x), phase abundance (L_n) and bulk (L_b). There is no phase presence term in the affinity-aware model (§4.2). The weighting factors λ are tuned to balance the contributions of each term to the total loss, preferentially weighting the chemistry and abundance terms. When training some models, the loss from present examples of important rare phases was additionally up-weighted by a per-phase factor to encourage the model to learn phases that otherwise would contribute little to the total loss.

Chemistry and bulk losses use the symmetric relative squared error:

$$\rho(\hat{y}, y) = \frac{(\hat{y} - y)^2}{|\hat{y}| + |y|}, \tag{4.39}$$

with the denominator floored at 10^{-6}. The abundance loss uses the same function in the affinity-naive model, but plain mean squared error (MSE: $\rho(\hat{y}, y) = (\hat{y} - y)^2$) is used in the affinity-aware model (§4.2). The symmetric relative squared error proved effective in training the model to predict both small and large values accurately, for positive targets that span orders of magnitude. MSE is used for the affinity-aware model abundance heads because the network maps to negative labels (affinities) as well as positive labels which behave continuously across zero: a property that MSE preserves but the symmetric relative squared error does not.

Chemistry loss. The chemistry loss L_x is only evaluated for phases that are present in the ground truth: *nGibbs* is not trained to predict metastable phase chemistry. The loss is zeroed for phases without ground truth compositions, but not for phases that are present in the ground truth and not the predicted assemblage: this loss is evaluated everywhere a ground truth composition is available. Because this masking behavior strongly changes the absolute value of the loss as a function of the number of present phases, the loss is normalized by the number of present phases in each batch.

Bulk loss. The reconstructed bulk (Eq. 4.3) is compared with the input bulk over the elements present in it:

$$L_b = \sum_{e:\, b_e>0} \rho(\hat{b}_e, b_e), \tag{4.40}$$

This measures how far the raw predictions are from satisfying conservation of mass, which reduces the correction that has to be applied at inference (§4.2). The backpropagation graph for this loss is complex and covers the entire network, thus it can cause exploding gradients when applied to untrained networks. Here it is only applied late in training to further improve the model's performance after it already produces strong predictions.

Abundance loss. For the abundance loss L_n, every phase is supervised, including absent ones. *nGibbs* emulators can be trained to predict chemical affinities for absent phases (affinity-aware) or it can be trained to predict phase abundances and presence without this information (affinity-naïve). In both cases, they produce an unbounded, unactivated output $\hat{g}$, that corresponds to a phase abundance when present and an effective chemical affinity when negative, tracking how close the phase is to saturation. The affinity-aware models learn this behavior with direct supervision: they are trained with a scaled affinity value, whereas the affinity-naive models are not provided with any negative labels, and instead are trained to treat this negative space as a confidence in the prediction of phase absence, rewarding confidently correct predictions of phase absence, and likewise penalizing confidently incorrect predictions of phase absence.

Affinity-naïve abundance loss. The labels for the affinity-naïve model are the literal phase abundances (n), which are zero for absent phases, never negative. Therefore, the loss is not computed on the predicted abundance $\hat{n} = \max(g, 0)$ itself but on a smoothed version of it ($\tilde{n}$), which is $\hat{g}$ activated with a softplus function (Fig. 4.9).

$$\tilde{n} = \Upsilon \,\mathrm{softplus}(g/\Upsilon) = \Upsilon \ln\left(1 + e^{g/\Upsilon}\right) \tag{4.41}$$

The softplus function is continuous for all real inputs and only approaches zero asymptotically. It is effectively a smoothed version of the ReLU function, where gradients are alive throughout and the transition from positive to zero

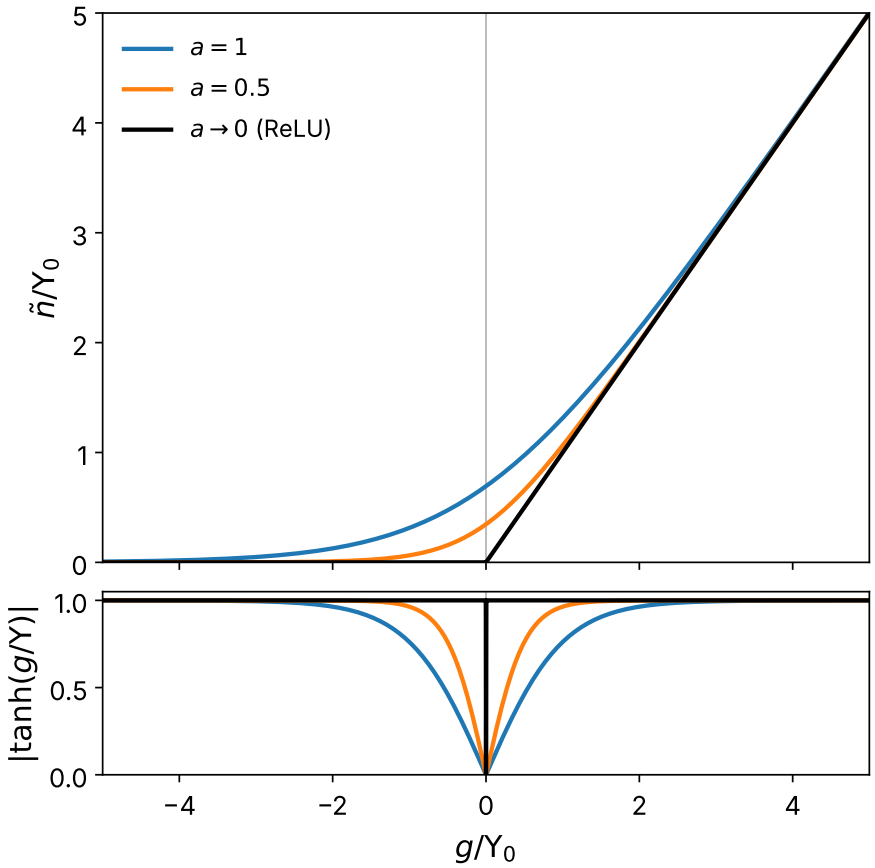


Figure 4.9: The softplus function (Eq. 4.41) generalized as a function of g/Υ_0 for different values of the scheduled annealing parameter a. The softplus function is a smooth approximation of the ReLU function, which is used to activate the predicted phase abundances $\hat{g}$ in the affinity-naïve model. As training proceeds, the softplus function collapses to the ReLU function, while allowing the model to learn a smooth transition from zero to positive values in the earlier stages of training. The smoothing scale Υ controls the strength of the smoothing, with larger values producing a smoother transition from zero to positive values. The lower plot is $\tanh(g/\Upsilon)$, which is the weight of the BCE loss on binary phase presence/absence. This binary loss is similarly scaled by the smoothing scale Υ so that the loss is weak near saturation to avoid teaching the model discontinuous behavior across 0.

values is smooth instead of kinked, following the thermodynamic logic that a phase generally grows from zero abundance at saturation and gradually decays to zero as it becomes undersaturated. The utility of this function applied to this problem is clear given its limits:

$$\lim_{g\to-\infty}\tilde{n}=0, \qquad \tilde{n}\approx g \text{ for } g/\Upsilon>3. \tag{4.42}$$

The cost of the softplus activation in this context is that the smoothed $\hat{n}$ systematically overestimates the true abundance of phases near zero, and therefore the strength of the smoothing needs to be reasonably restricted for the model to learn phase abundances well. The input to the softplus is thus scaled by a smoothing scale $\Upsilon = a\Upsilon_0$, where Υ_0 is the 5th percentile of that phase's abundance where the phase is present and a is a scaling factor within [0,1]. This limits the range of values for which this smoothing error is high to only the smallest abundances that are approaching desaturation. The smoothing is further weakened as training progresses by annealing the a parameter from 1 to ≈ 0, and the softplus function anneals into the ReLU function ($\tilde{n}\approx\max(g,0)$), matching *nGibbs*'s behavior during inference.

Presence loss L_s for the affinity-naïve model. As a means to further incentivize the model to learn the correct assemblages, the affinity g is also used as the logit of a binary cross-entropy against s:

$$L_s=\frac{\sum_j c_j\Big[-s_j\ln\text{sigmoid}\Big(5g_j/\Upsilon_{0,j}\Big)-(1-s_j)\ln\Big(1-\text{sigmoid}\Big(5g_j/\Upsilon_{0,j}\Big)\Big)\Big]}{\sum_j c_j},$$
$$c_j=\Big|\tanh(g_j/\Upsilon)\Big|, \tag{4.43}$$

where the bracketed term is the binary cross-entropy (BCE) of the logit $5g_j/\Upsilon_{0,j}$ against binary label s_j. The logit is scaled by the fixed Υ_0, not the annealed Υ, so the classifier's confidence means the same thing throughout training: it reaches sigmoid(5) $\approx$ 0.99 at $g=\Upsilon_0$. The weight for this binary loss $|\tanh(g/\Upsilon)|$ uses the annealed Υ. It is 0 at the boundary and close to 1 a few Υ away. The gradient is carried by the abundance loss near the boundary, where an almost-saturated phase is thermodynamically similar to a just-saturated phase, a subtlety that the BCE loss cannot capture. The weight c depends on g, a model output. Thus, we normalize the BCE loss by the sum of the weights

$\sum_j^P c_j$ so that the model is not encouraged to make near-zero predictions for g simply to minimize the BCE loss.

Affinity-aware abundance loss

The affinity-aware model is trained to predict an activated abundance y, a transformed version of the signed abundance g that remaps negative values but preserves positive values. Unlike an affinity-naïve model, there exists a ground truth g that is used to supervise the affinity-aware model even for phases that are absent.

Estimating affinities (g) from dn/dT of isobaric transects. Affinities are positive for absent phases and 0 for present phases. They are expressed in units of energy, not molar abundances. This unit mismatch produces an undesirable discontinuity at zero if unaccounted for, reducing the information content of the affinities within the context of the molar abundances that we seek to predict well. If the affinities are scaled well, they smoothly transition into molar abundances at saturation, reinforcing physical dn/dt values even near zero, where other approaches might learn spurious patterns near the boundary in response to the objective function changing character across the boundary. This smoothness constraint is simply that g be continuous at exactly 0, which can be written:

$$\lim_{g\to 0^-}\left(\frac{dg}{dT}\right)_P = \lim_{g\to 0^+}\left(\frac{dg}{dT}\right)_P = \lim_{n\to 0^+}\left(\frac{dn}{dT}\right)_P. \tag{4.44}$$

Affinities are therefore rescaled into a "virtual abundance" unit with a constant unique to each phase and each transect. For an affinity α in an arbitrary unit:

$$\begin{cases} g = n, & n > 0 \\ g = \alpha \frac{lim_{n\to 0^+}(dn/dT)_P}{lim_{\alpha\to 0^+}(d\alpha/dT)_P}, & n = 0. \end{cases} \tag{4.45}$$

$(dn/dT)_P$ and $(d\alpha/dT)_P$ always have opposite signs, so $g \leq 0$ while $n = 0$ because the rescaling factor is always negative. This operation erases the memory of the units of the affinity and instead privileges the slope of the molar abundance (dn/dT) at saturation. Indeed, the affinity is locally linear near zero, as is the molar abundance. The rescaled affinity is therefore well-approximated

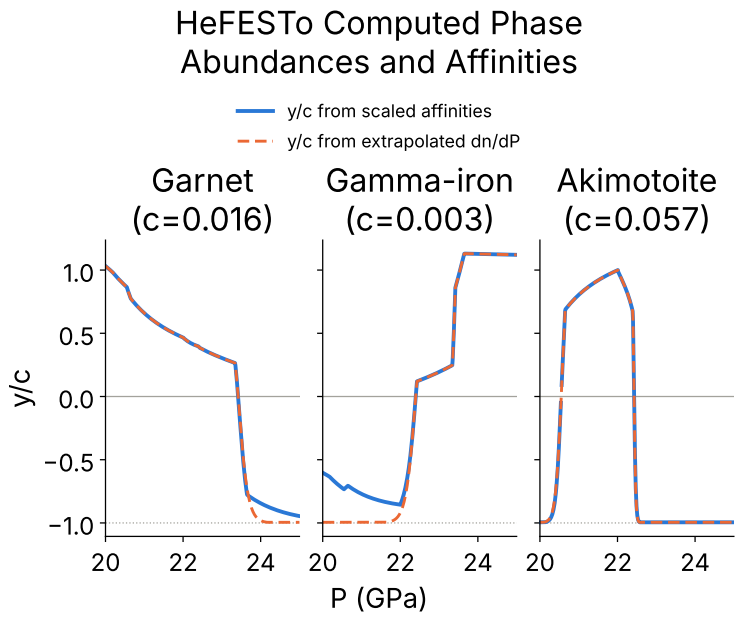


Figure 4.10: Activated and scaled abundance labels y/c for garnet, γ-iron and akimotoite along a HeFESTo isentrope of harzburgite (Stixrude and Lithgow-Bertelloni, 2024, after Workman and Hart, 2005) from 20 to 25 GPa in 10 MPa steps. Blue: scaled abundance and affinity labels built from HeFESTo's own phase affinities, rescaled by the ratio of $|dn/dP|$ to $|dA/dP|$ at saturation (Eq. (4.45), in pressure rather than temperature). Orange dashed: labels built without affinities, by extrapolating dn/dP through zero (Eq. (4.47)). Both are floored at $g = -3c$ and activated with $y = c\tanh(g/c)$ below zero. The scale c, given above each panel, is the median $|dn/dP|$ at saturation times 0.25 GPa, bounded between one fifth of the phase's maximum abundance and that maximum. The extrapolated labels are nearly identical to the affinity-based labels near the saturation boundary, but sometimes diverge further away. The far-field divergences are relatively unimportant because they have little bearing on the predicted assemblage, unlike the near-field behavior.

by a linear extrapolation of the abundance into negative values. Figure 4.10 compares labels built both ways along a HeFESTo transect.

$$\alpha(T)\frac{lim_{n\to 0^+}(dn/dT)_P}{lim_{\alpha\to 0^+}(d\alpha/dT)_P} \approx (lim_{n\to 0^+}(dn/dT)_P) \times (T - T_{sat}) \tag{4.46}$$

With this in mind, there is little benefit to collecting real affinity values from the reference thermodynamic models. We evaluate the ground truth g for each phase as:

$$\begin{cases} g = n, & n > 0 \\ g = (lim_{n\to 0^+}(dn/dT)_P) \times (T - T_{sat}) & n = 0. \end{cases} \tag{4.47}$$

The MELTS training data were generated as isobaric cooling runs, either as batch crystallization (constant bulk composition) or fractional crystallization (solids are removed from the system at every step). Each run is split into temperature transects, and the slope dn/dT is estimated at n=0 with differentials of n and T. Let n_i refer to a phase's molar abundance just after saturation, i.e. the first nonzero value of n going either up or down in T. Then, $n_{i-1} = 0$, and

$$\text{batch:} \quad (lim_{n\to 0^+}(dn/dT)_P) \approx \frac{n_{i+1} - n_i}{T_{i+1} - T_i}, \tag{4.48}$$

$$\text{FxTal:} \quad (lim_{n\to 0^+}(dn/dT)_P) \approx \frac{n_i}{T_i - T_{i-1}}, . \tag{4.49}$$

It is incorrect to use this approach to estimate the affinities for fractional crystallization *sensu stricto*, because the composition (and therefore $(dn/dT)_P$) is not constant as T decreases. However, we found that this approximation is good for small $|T - T_{\text{sat}}|$, and the accuracy of the affinity is less important further from the saturation boundary. Therefore, we accept this rough approximation so that we are able to weight the training datasets towards liquidus assemblages, which are often of interest to the community.

Far from the saturation boundary, the first-order approximation of g is less accurate, and accurate predictions of g for absent phases do little to improve the quality of the predicted assemblage. Therefore, we activate the raw affinity g with a hyperbolic tangent function to compress the negative range of g to limit the influence of strongly undersaturated phases on the abundance loss. Because phases have very different abundances in common rocks (e.g. $n_{\text{clinopyroxene}} \gg n_{\text{apatite}}$), we need to define a scale factor c for each phase which corresponds to some characteristic abundance for a phase. We choose to define c as the typical (median) abundance that a phase accumulates over a temperature interval $\Delta = 25$ K after saturation:

$$c = \text{median}(|\,(lim_{n\to 0^+}(dn/dT)_P) \times \Delta|), \qquad \Delta = 25 \text{ K}. \tag{4.50}$$

Intuitively, the temperature window Δ represents the window of temperature approaching saturation for which we would like to accurately predict affinities. Phases beyond Δ °C from saturation are considered arbitrarily far and so we place them on the plateau of the tanh function. Any confident prediction that such a phase is absent will contribute little to the loss. We use the median rather than the mean slope ($(lim_{n\to 0^+}(dn/dT)_P)$) because it is log-normally distributed.

We define an activated label ($y(g, c)$) that is used during training:

$$y(g) = \begin{cases} g, & g \geq 0, \\ c\tanh(g/c) + \varepsilon g, & g < 0, \end{cases} \qquad \varepsilon = 0.02 \tag{4.51}$$

where epsilon is a small proportionality constant used to invigorate the loss gradient when the predicted $\hat{g}/c$ is lower than -3, for which the tanh term is nearly flat ($\tanh(-3) \approx -1$). This allows the network to learn well at the beginning of training, when predicted $|\hat{g}|$ can be large, inaccurate, and unstable. Additionally, ground truth g is floored at $-3c$ to prevent systematic predictions of very low g values where the gradient is weak (but the network has no such constraint, which would kill the gradient below -3c). Note that g, y, and n are identical for $n \geq 0$.

The activation is shown in Fig. 4.11.

The affinity-aware objective. Within a single isobaric transect, most phases never appear anywhere. These phases in these rows do not permit an estimate of dn/dT at saturation, and therefore g and y are undefined for them. We refer to these phases as "untrusted" ($y \notin \mathbb{R}$), and apply a distinct, weaker loss to them. We also reject low-side outlier estimates of dn/dT at saturation (defined as when dn/dT is a factor of $\sqrt{10}$ smaller than the median dn/dT for that phase across all transects). These values are also untrusted. For present phases, $y \in \mathbb{R}$ and is positive. Unlike the affinity-naïve abundance objective, the affinity-aware objective wraps terms for both regression error and binary phase presence/absence into a single loss function with its own weighting scheme.

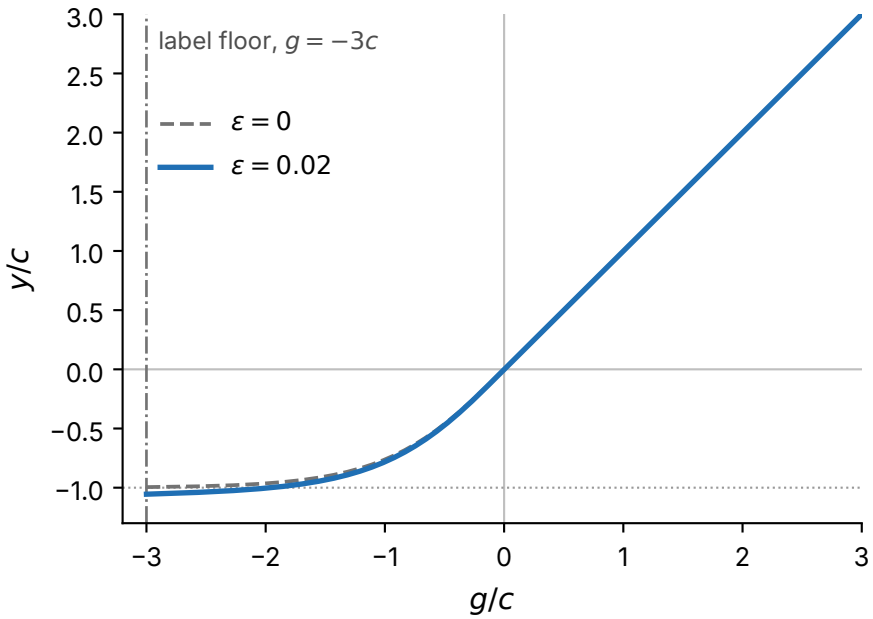


Figure 4.11: The activation of Eq. (4.51) in units of the per-phase scale c, for $\varepsilon = 0.02$ (solid) and $\varepsilon = 0$ (dashed). Above zero $y = g$; below zero the tanh term saturates at $y/c = -1$ (dotted), and the εg term keeps a small slope there. The dash-dotted line marks the floor of $-3c$ applied to the ground-truth g. This saturation behavior ensures that the loss is dominated by saturated or near-saturated phases, which are the cases that matter during inference.

$$\ell = \begin{cases} \frac{(\hat{y}-y)}{c}^2 + s * [w_p * \mathrm{ReLU}\Big(\min(y/c,\ h_p) - \hat{y}/c\Big)^2], & y \in \mathbb{R}, \\ w_n \max(0,\ \hat{y}/c - h_n)^2 + w_\eta\,(\hat{y}/c + 1)^2, & y \notin \mathbb{R}., \end{cases} \tag{4.52}$$

For trusted phases ($y \in \mathbb{R}$), the loss function is a combination of the MSE and a corrective positive hinge loss. The first term is the MSE of the predicted activated abundance $\hat{y}$ against the ground truth y, normalized by the scale factor c. The corrective positive hinge loss penalizes the underprediction of phase abundance for low-abundance phases. This term is a response to the observation that the loss for untrusted phases was leading to underprediction of present phases near saturation, which most obviously manifests in an imbalanced high precision and low recall. A hinge loss uses min or max functions to apply a loss only to one side of a threshold. This hinge loss is applied only to phases which are present in the ground truth ($s = 1$) with abundances less than the threshold $h_p = 0.03$, that are underpredicted by $\hat{y}/c$.

For untrusted phases ($y \notin \mathbb{R}$), the loss function is a combination of a negative

hinge loss and a weak pull towards $-c$. The hinge loss is stronger, but is only applied to predictions that are above the threshold $h_n = -0.2$. The combination of these losses is a compromise that recognizes that untrusted phases are absent and probably would have affinities mapping to decisive undersaturation at $y \approx -c$, however, because this is not known explicitly, only unlikely near-saturation predictions ($\hat{y}$) are substantially penalized.

Table 4.9: Hyperparameters of the affinity-aware loss (Eq. 4.52).

Symbol	Value	Meaning
Trusted phases, $y \in \mathbb{R}$		
h_p	0.03	Predicted abundance threshold for low-abundance, present phases, below which underpredictions are penalized
w_p	3.0	Weight of the underprediction penalty for present phases
Untrusted (absent) phases, $y \notin \mathbb{R}$		
h_n	-0.2	Predicted abundance threshold for untrusted absent phases, above which are penalized
w_n	0.25	Weight of the near-saturation penalty for untrusted absent phases
w_η	0.02	Weight of the weak pull towards $-c$ for untrusted absent phases

Figure 4.12 maps the loss over y/c and $\hat{y}/c$ for the values of Table 4.9.

The HeFESTo emulators within *nGibbs* were trained with the affinity-naïve objective because its transects do not hold entropy constant, so the extrapolation method cannot be applied without rebuilding the training dataset. HeFESTo reports true affinities, and how to scale them into units of "virtual abundance" is left for later work.

The MELTS emulators were trained with the affinity-aware objective because the training dataset was generated as isobaric transects, permitting the estimation of dn/dT at saturation.

Training schedule

All models were trained on an NVIDIA GeForce RTX 5070 Ti GPU with 16 GB of VRAM. Datasets were often too large to fit in memory, and so were streamed from disk in batches of n=10^6.

Quality prediction of equilibrium phase assemblages that conserve mass is a complex task with highly covariate errors. These models are thus trained in

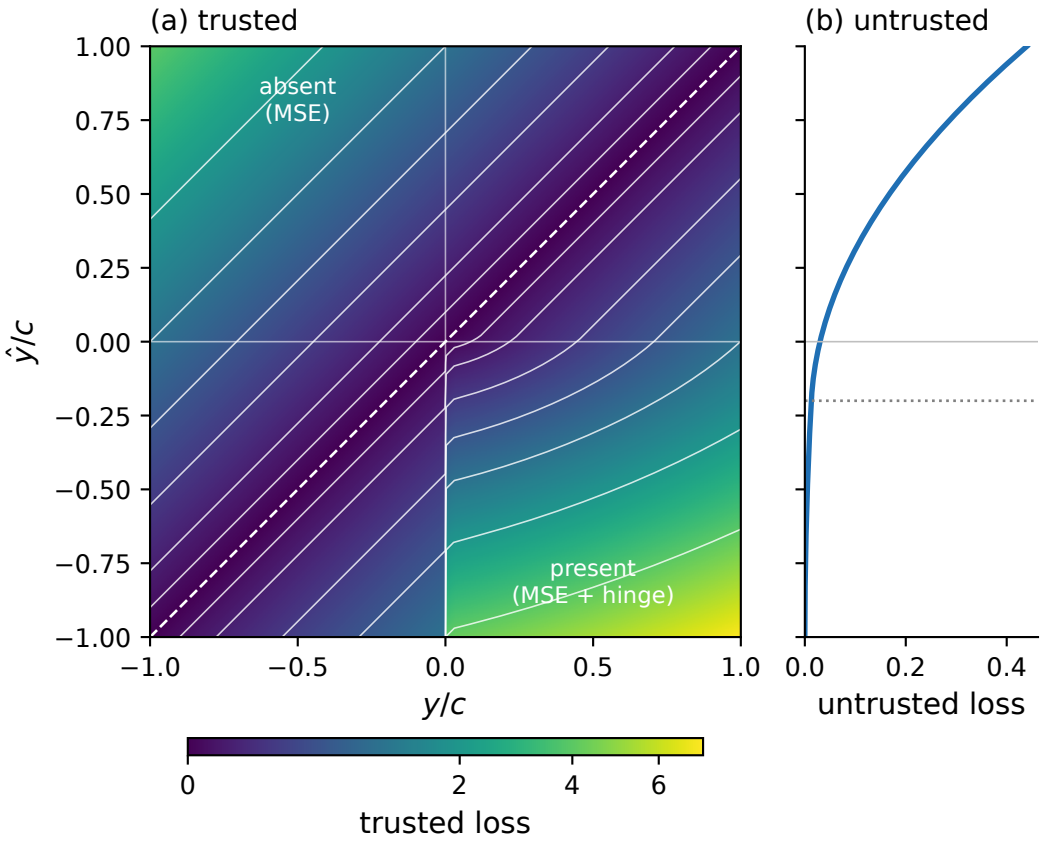


Figure 4.12: The affinity-aware abundance loss (Eq. (4.52)) in units of c, with the values of Table 4.9. (a) Trusted phases, as a function of the label y/c and the prediction $\hat{y}/c$ (square-root color scale; white contours at 0.01, 0.05, 0.2, 0.5, 1, 2 and 4; dashed line $\hat{y} = y$). (b) Untrusted phases, which are absent, have no label, and depend on $\hat{y}/c$ alone. The dotted line marks $\hat{y}/c = h_n = -0.2$, above which the hinge applies. The functional forms follow the *nGibbs* implementation.

episodes that first isolate the distinct abundance and chemistry partitions of the model, training with a high (10^{-3}) maximum learning rate on a single-cycle learning rate scheduler described by L. N. Smith and Topin (2019). Then, the model is trained as an interdependent whole with a more cautious maximum learning rate (10^{-4}) that is periodically annealed to zero then restored following a cosine function. This is effectively a more gentle implementation of the warm restart method described by Loshchilov and Hutter (2017). Initializing the model in this sequential manner ensures that the model does not learn a single task at the expense of the others due to unstable solutions, but optimizes the full unified problem. The model's encoder feeds into both the molar and chemical heads. The encoder must be instantiated in tandem with either the

molar or chemical heads, so that it has an objective to optimize. We choose to instantiate the encoder with the molar heads because they perform the more complex task. Thus, the chemistry heads are frozen and the chemistry loss is zeroed during the first stage of training. During the second stage, the chemistry heads are trained with the chemistry loss, the abundance loss (and phase presence loss for the affinity-naïve model) is zeroed. The weights of both the encoder and molar heads are frozen for this second stage. The encoder feeds into the chemistry heads, but it cannot be optimized for the chemistry loss while the mole heads are frozen, or the abundance predictions will be clobbered as the encoder evolves in isolation from the mole heads that depend on it. It is presumed that an encoded representation of the system state that is useful for predicting phase abundances would also be useful for predicting phase chemistry. Thus, after the second training episode completes, the whole model produces reasonably good abundance and chemistry predictions that both are built upon the same encoder.

Finally, for the third stage, all heads are unfrozen and the model is trained jointly with all losses except the bulk loss. For these two stages, an early stopping patience is implemented that stops training after 10 epochs of no improvement in the model's performance against the validation dataset. For the affinity-naïve model, the smoothing scale $\Upsilon = a\Upsilon_0$ is annealed during this third stage onward, with a linearly decaying from 1 to 0 by the end of training. Bulk loss is the most complex and best captures the systemic error covariation between the abundance and chemistry predictions within the constraint that mass be conserved; it is applied in concert with the other losses in the fourth stage of training, and the adaptive dropout is disabled for stability. Instead, an adaptive input noise is applied as a regularization.

This sequential training schedule is summarized as follows (the losses from an example training history are shown in Fig. 4.13):

1. train the uninitialized encoder and chemistry heads, with the mole heads frozen and $\lambda_n = \lambda_s = 0$;
2. train the uninitialized mole heads, with the encoder and chemistry heads frozen and $\lambda_x = 0$;
3. whole model annealing by training everything jointly with $\lambda_b = 0$.
4. whole model annealing with a mass conservation constraint, by training everything jointly with $\lambda_b > 0$.

The AdamW optimizer is used with weight decay set at 10^{-4} for the encoder and chemistry heads. No weight decay is used on the mole heads.

nGibbs uses an adaptive dropout during training. After each epoch, if the validation loss exceeds 1.05 times the training loss, the dropout rate (outside the mole heads) is raised by a small amount, such as 0.005. If the validation loss is below the training loss, dropout is lowered by an even smaller amount, such as 0.001.

Temperature model for isentropic emulators

The isentropic emulators take pressure and specific entropy as inputs instead of pressure and temperature. Temperature, therefore, is not known *a priori* for an isentropic *nGibbs* emulator, unlike for true free energy minimizers for which temperature emerges as a consequence of the minimization. Because temperature is a required input to calculate material properties (see §4.2), we must retrieve the temperature from the isentropic emulator's output. Thermodynamically, temperature is uniquely defined by the entropy and pressure for an equilibrium assemblage, but recovering it requires an iterative optimization. To avoid this, *nGibbs* implements a separate temperature satellite model for isentropic emulators that predicts the temperature of a state given its pressure, specific entropy and emulated assemblage. This task is divided into two parts: a closed-form reference regression that captures most of the variance, and a fully connected neural network that learns the remaining residual.

The reference regression is an ordinary least-squares fit of temperature to P, P^2, S and S^2, together with a linear and a quadratic term for each element in the bulk composition. Element abundances are expressed as atoms per formula unit, with cations plus oxygen normalized to 24 and total iron counted as a single species. On its own, this smooth polynomial captures the first-order shape of an adiabat and how it shifts with composition. It cannot, however, represent the changes in slope associated with phase transitions. A small neural network corrects for this. It receives the same input as the reference regression, supplemented with the phase abundances and compositions of the equilibrium assemblage predicted by the isentropic emulator. The network then predicts a residual with respect to the reference regression, allowing for more subtle variations in the adiabat's shape to be learned. Because the emulator's predictions are used as inputs during inference, they also serve as the training

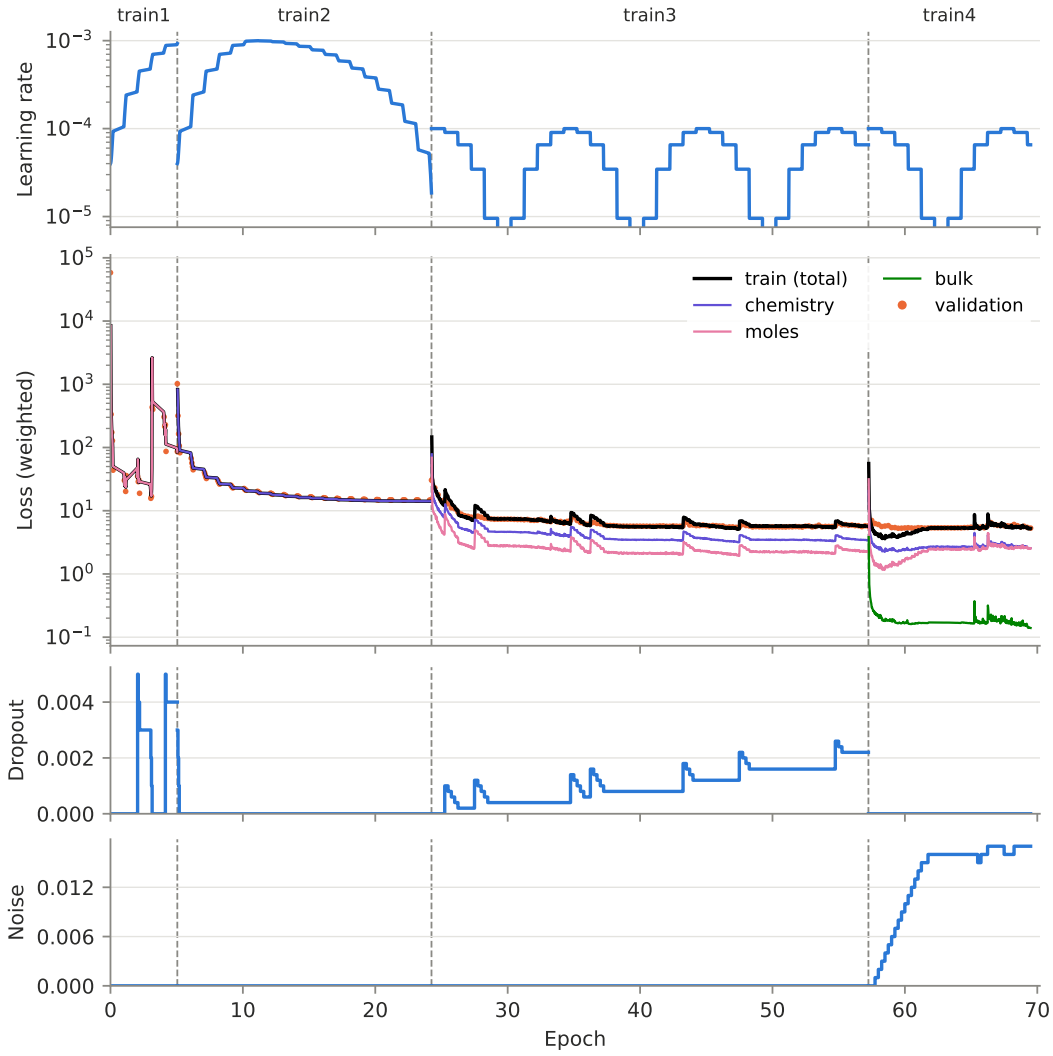


Figure 4.13: Training history of the MELTS 1.0.2 emulator, NoCr variant, across the four stages of the training schedule (dashed lines mark stage boundaries). From top to bottom: learning rate, weighted loss, adaptive dropout rate, and input noise amplitude. The loss panel shows the total training loss and its chemistry, moles, and bulk components, with per-epoch validation loss as points. The bulk loss is zero until the fourth stage, when it is switched on and noise injection begins. The learning rate follows a single-cycle schedule in stages 1 and 2 and cosine warm restarts at a lower maximum in stages 3 and 4. Dropout and noise are adjusted adaptively from the validation-to-training loss ratio.

data. The network has three hidden layers of 256, 128 and 64 units with leaky-ReLU activations. It is trained on a mean-squared-error loss using min–max normalized residuals, with adaptive dropout and early stopping enforced by performance metrics against the validation dataset. The learned residuals span roughly ±200 K. Both the regression and the network are fit only to states with $S \geq 1.75\,\mathrm{J\,g^{-1}\,K^{-1}}$, which encompasses the range spanned by the mantle adiabats the models are designed for, without including temperature conditions at the conductive surface or core-mantle boundary (CMB).

Figure 4.14 compares both stages with HeFESTo for the Mars model, using 10,892 states with $S \geq 1.75\,\mathrm{J\,g^{-1}\,K^{-1}}$ taken from the validation dataset ($T \approx$ 860 K to 2440 K). On its own, the regression reproduces HeFESTo temperatures with an RMSE of 27.5 K ($R^2 = 0.995$), but its errors occasionally exceed 100 K. Adding the network lowers the RMSE to 1.7 K and the mean absolute error to 1.0 K. Ninety-nine percent of states fall within 6.3 K, and the largest error is 47 K.

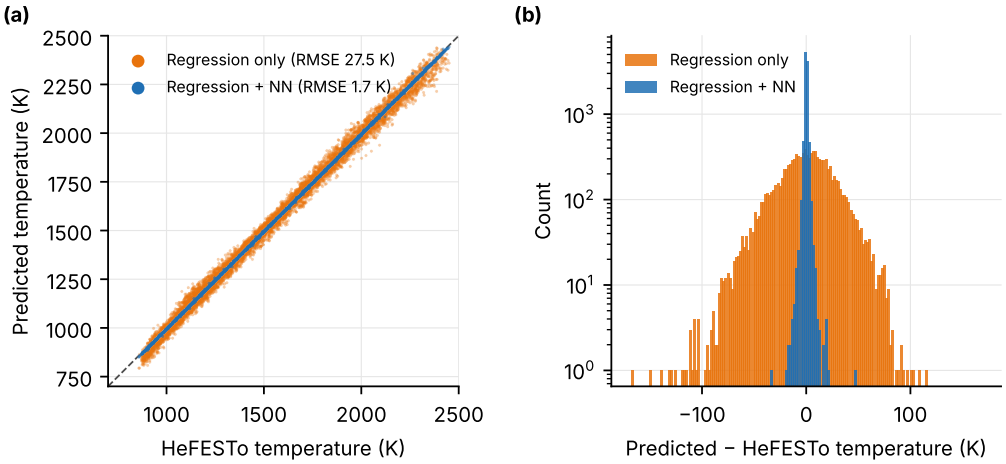


Figure 4.14: Temperature model for the HeFESTo Mars isentropic emulator, evaluated for a subset of the Mars validation dataset. (a) Predicted versus true temperature, for the reference regression alone (orange) and for the regression plus neural-network residual correction (blue); the dashed line is 1:1. (b) Distributions of prediction error (log scale).

Material Properties from Phase Equilibria

To support its use in geodynamic or other physical simulations, *nGibbs* implements the equations of state used by MELTS and HeFESTo to determine the material properties of phases and of the bulk assemblage. These calculations

are cheap relative to the free energy minimizations performed by MELTS and HeFESTo, so there is little benefit to emulating them with a neural network. Furthermore, because these properties are computed from the predicted phase assemblage, they are self-consistent, thermodynamically grounded, and flexible to adapt to changes with the parameters of the reference thermodynamic models. These are implemented as vectorized Python (NumPy/PyTorch) translations of the MELTS models taken from the open source MAGMA C library (commit `705a0fb`), and the HeFESTo models from the open source HeFESTo Fortran repository (commit `9566177`). Parameters for the equations of state are taken from the MAGMA C library for MELTS except for the apatite reference volume, and for HeFESTo from the library (commit `ff99dd0`). The MELTS emulators in *nGibbs* use 15.8228 $\mathrm{J\,bar^{-1}}$ for the apatite reference volume, instead of the 16.4025 $\mathrm{J\,bar^{-1}}$ given in the MAGMA C library so that the volume and density calculations would be consistent with alphaMELTS 2.3.1 output.

Given a phase assemblage, its phase compositions, and a pressure and temperature, *nGibbs* returns density, G, S, V, C_p, for both MELTS and HeFESTo assemblages, H, $\partial V/\partial T$, $\partial V/\partial P$, $\partial C_p/\partial T$ for MELTS only, and V_s, V_p, K_S, K_T, α (referring to the shear wave velocity, compressional wave velocity, isentropic and isothermal bulk moduli, and the thermal expansivity respectively) for HeFESTo only. These properties are computed for each phase and are aggregated to bulk properties. The equations differ from the source codes only in how they are organized and the algorithms used to execute them.

For the HeFESTo properties, the emulator is able to vectorize the isomorphic terms most effectively, whereas the metamorphic terms are more computationally expensive, because they require solving the molar pressure and temperature derivatives $(dn/dp)_T$ and $(dn/dT)_P$. Because the metamorphic terms are only necessary for accurate calculation of the bulk C_p, K_S, K_T, and α (Fig. 4.22), these terms are only evaluated by nGibbs when the user requests properties that need them (Listing 4.1). Thus, it is significantly cheaper to provide S, ρ, V_p, and V_s without the other terms, if they are not necessary.

Listing 4.1: Computing bulk HeFESTo properties for an *nGibbs* assemblage, requesting either the isomorphic terms only or the metamorphic terms as well (pseudocode).

```
from ngibbs.engine.models import HeFESToEmulatorCPU
```

```
input_array = pd.read_csv('bulk_composition.csv')
PROPERTY_VARIANTS = {
    'isomorphic':  ('rho', 'Vp', 'Vs', 'S'),
    'metamorphic': ('rho', 'Vp', 'Vs', 'S', 'cptot', 'KStot'),
    '+fast':       ('rho', 'Vp_fast', 'Vs', 'S', 'cptot', 'KStot'),
}
output = HeFESToEmulatorCPU.ForwardMB(input_array, outputs=['
    component_moles'])
properties = HeFESToEmulatorCPU._compute_bulk_EOS_properties(
    output['component_moles'],
    PT=torch.tensor(input_array[['P(GPa)(System_main)', 'T(K)(System_main)'
    ]]),
    property_names=PROPERTY_VARIANTS['isomorphic'])
```

MELTS and HeFESTo evaluate these equations of state sequentially for one phase at one $(T, P, \mathbf{X})$ state per call, whereas *nGibbs* operates on thousands of states in parallel. The translation, therefore, is a reorganization of the source code to operate on arrays.

These equations of state were empirically validated through comparison against property outputs from MELTS and HeFESTo. Against alphaMELTS 2.3.1 output on the MELTS 1.2.0 reference calculations, bulk S, V, H and C_p computed from the true assemblage agree to within 0.07% (mean relative error). There remains an undiagnosed systematic disagreement in the *nGibbs* implementation for plagioclase H and S. Figs. 4.20–4.23 compare the bulk properties reported by each reference code with those computed by *nGibbs* from the reference code's own assemblage and from the emulator's predicted assemblage.

Performance benchmarking

Both HeFESTo and MELTS are parallelized by running independent simulations in separate processes, so their wall time depends on the number of workers as well as on the number of assemblages N. Each was timed on two machines: at 1, 4 and 8 workers on a 4-core, 8-thread Intel i7-11370H laptop CPU with 24 GB of RAM, and at 1, 4, 8, 16 and 20 workers on the 20-core Intel Core i7-14700H desktop CPU with 32 GB of RAM.

MELTS 1.0.2 was timed through PetThermoTools (`multi_path`) on isobaric cooling paths at 2 kbar from 1200 to 1000 °C in 1 °C steps (201 equilibria per path), each with its own MORB-like bulk composition (every oxide of a

typical MORB multiplied by $\mathcal{N}(1, 0.1)$ noise and renormalized). Every worker count was run with 1 to 5 paths per worker. A PetThermoTools worker process can solve only one MELTS path, so p paths per worker are run as p consecutive calls with one path per worker, and wall time includes process start-up and alphaMELTS library initialization for every path. Equilibria that PetThermoTools failed to return are not counted toward N. Beyond about 10^3 assemblages the wall time is roughly linear in N, so the per-assemblage rate measured for the largest call at each worker count is used to extrapolate to larger N.

HeFESTo was timed on whole-mantle adiabats prepared with the same structure as the training data (§4.2): 141 states each from 0 to 140 GPa. A total of 3 adiabats per worker are measured, run through GNU parallel (Tange, 2021). Preparing the simulation directories was not timed. Both reference codes are timed along paths, where each state is seeded from the last. This is their most efficient mode of use, so the comparison does not favor the emulators. The relative speedup of *nGibbs* is particularly impressive when applied to independent states.

The emulators were timed on a single `ForwardMB` call per batch after an untimed warm-up call. The MELTS 1.0.2 emulator was given the same cooling paths as PetThermoTools, truncated to the same batch sizes as the HeFESTo emulator, $N = 16$–$262{,}144$ rows. The HeFESTo Earth isentropic emulator was given a 0–140 GPa pressure sweep at $S = 2.5\,\mathrm{J\,g^{-1}\,K^{-1}}$, including conservation of mass and temperature prediction. Material properties are then added on top of each emulator. For HeFESTo this is the *nGibbs* equations of state with every term evaluated (§4.2). For MELTS it is the PetThermoTools property calculator (`calc_phase_props_MELTS_parallel`), which calls the alphaMELTS supplemental calculator for the predicted assemblage without minimization, in parallel over PetThermoTools' default worker count. This route is faster than the vectorized translation for large batches (Fig. 4.25). Some rare low-temperature liquids develop negative atoms after the mass conservation correction. For these, the liquid properties cannot be evaluated by alphaMELTS and are removed before timing. MELTS emulator times are the median of three repeats; all other times are single wall-clock measurements.

Table 4.10: Prediction and recall statistics of the closed-system isothermal MELTS 1.0.2 and 1.2.0 emulators, with (Cr) and without (no Cr) chromium, evaluated against their validation datasets. The number of samples (n) in the training dataset is given immediately below the model name. Freq.: fraction of test assemblages in which the phase is present in the ground truth. Prec./Rec.: precision and recall of the predicted phase presence, shaded from white (1.00) to red (0.80 and below). – : phase not in the model.

	1.0.2, Cr $n = 78{,}040{,}752$			1.0.2, no Cr $n = 75{,}621{,}940$			1.2.0, Cr $n = 122{,}908{,}703$			1.2.0, no Cr $n = 122{,}195{,}502$		
Phase	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.
liquid	0.89	1.00	0.99	0.89	1.00	0.99	0.95	1.00	1.00	0.95	1.00	1.00
clinopyroxene	0.42	0.99	0.99	0.47	0.99	0.99	0.26	0.99	0.98	0.27	0.97	0.94
plagioclase	0.41	0.99	0.99	0.46	0.99	0.99	0.26	0.98	0.99	0.28	0.96	0.98
orthopyroxene	0.39	0.97	0.98	0.43	0.97	0.97	0.26	0.97	0.95	0.27	0.93	0.95
olivine	0.37	0.99	0.97	0.40	0.98	0.97	0.26	0.98	0.96	0.24	0.95	0.94
spinel	0.44	0.98	0.97	0.20	0.96	0.96	0.34	0.97	0.95	0.18	0.96	0.80
fluid	0.05	0.98	0.98	0.06	0.97	0.96	0.42	1.00	0.99	0.47	0.99	0.99
rhm-oxide	0.15	0.97	0.97	0.18	0.97	0.96	0.11	0.96	0.96	0.12	0.96	0.95
quartz	0.13	0.98	0.99	0.14	0.98	0.99	0.11	0.98	0.99	0.12	0.97	0.99
apatite	0.13	0.98	0.98	0.14	0.98	0.96	0.10	0.95	0.97	0.11	0.84	0.98
garnet	0.12	0.99	0.98	0.14	0.99	0.97	0.10	0.97	0.97	0.11	0.94	0.96
whitlockite	0.13	0.96	0.99	0.15	0.97	0.98	0.05	0.97	0.97	0.05	0.89	0.93
K-feldspar	0.07	0.98	0.97	0.08	0.98	0.98	0.05	0.97	0.99	0.06	0.95	0.96
tridymite	0.05	0.94	0.97	0.06	0.96	0.96	0.06	0.95	0.96	0.07	0.92	0.95
biotite	0.03	0.94	0.95	0.03	0.95	0.95	0.03	0.95	0.95	0.03	0.93	0.96
graphite	–	–	–	–	–	–	0.02	0.77	0.88	0.02	0.80	0.86
leucite	0.01	0.83	0.93	0.01	0.83	0.85	0.01	0.92	0.88	0.01	0.91	0.88
hornblende	–	–	–	–	–	–	0.01	0.92	0.96	0.01	0.96	0.92
alloy-solid	–	–	–	0.01	0.88	0.68	0.00	0.72	0.83	0.00	0.73	0.88
muscovite	–	–	–	–	–	–	0.00	0.95	0.98	0.00	0.92	0.94
calcite	–	–	–	–	–	–	0.00	0.92	0.92	0.00	0.91	0.84
nepheline	–	–	–	–	–	–	–	–	–	0.00	0.67	0.86

Table 4.11: Prediction and recall statistics of the isothermal (P, T) and isentropic (P, S) HeFESTo Earth and Mars emulators evaluated on their validation datasets. The number of samples (n) in the training dataset is given immediately below the model name. Freq.: fraction of test assemblages in which the phase is present in the ground truth. Prec./Rec.: precision and recall of the predicted phase presence, shaded from white (1.00) to red (0.80 and below). – : phase not in the model.

	Earth (P, T) $n = 26{,}065{,}657$			Earth (P, S) $n = 26{,}065{,}657$			Mars (P, T) $n = 10{,}259{,}324$			Mars (P, S) $n = 10{,}259{,}324$		
Phase	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.	Freq.	Prec.	Rec.
garnet	0.37	0.99	0.97	0.37	0.99	0.98	0.85	0.99	0.99	0.85	0.99	0.99
ferropericlase	0.64	0.99	0.99	0.64	0.99	0.99	0.35	0.97	0.95	0.35	0.96	0.95
clinopyroxene	0.26	0.99	0.98	0.26	0.99	0.98	0.66	0.99	0.99	0.66	0.99	0.99
Ca-perovskite	0.65	1.00	0.99	0.65	1.00	0.99	0.17	0.99	0.95	0.17	0.99	0.95
olivine	0.23	0.99	1.00	0.23	1.00	0.99	0.49	0.99	0.99	0.49	1.00	0.99
bridgmanite	0.52	0.99	0.99	0.52	0.99	0.99	0.09	0.97	0.98	0.09	0.98	0.99
Ca-ferrite	0.40	0.99	0.98	0.40	0.98	0.98	0.14	0.96	0.95	0.14	0.98	0.92
γ-iron	0.28	0.98	0.90	0.28	0.95	0.97	0.24	0.93	0.95	0.24	0.98	0.87
spinel	0.16	0.98	0.98	0.16	0.99	0.98	0.34	0.97	0.99	0.34	0.99	0.97
orthopyroxene	0.13	0.99	0.98	0.13	0.99	0.98	0.28	0.98	0.99	0.28	0.99	0.99
post-perovskite	0.18	0.98	0.99	0.18	0.98	0.99	–	–	–	–	–	–
ringwoodite	0.08	0.97	0.97	0.08	0.97	0.98	0.27	0.98	0.99	0.27	0.99	0.99
stishovite	0.11	0.97	0.97	0.11	0.97	0.97	0.20	0.96	0.98	0.20	0.98	0.98
kyanite	0.04	0.99	0.99	0.04	1.00	0.99	0.13	0.98	0.96	0.13	0.98	0.96
ε-iron	0.16	0.90	0.99	0.16	0.92	0.99	0.01	0.96	0.77	0.01	0.95	0.85
HP clinopyroxene	0.03	0.93	0.97	0.03	0.94	0.97	0.11	0.95	1.00	0.11	0.97	1.00
plagioclase	0.05	0.97	0.97	0.05	0.98	0.96	0.08	0.95	0.96	0.08	0.95	0.95
wadsleyite	0.06	0.96	0.95	0.06	0.97	0.98	0.07	0.94	0.98	0.07	0.97	0.96
akimotoite	0.02	0.86	0.81	0.02	0.86	0.83	0.08	0.77	0.88	0.08	0.79	0.87
quartz	0.02	0.97	0.95	0.02	1.00	0.96	0.07	0.96	0.97	0.07	0.98	0.96
nepheline	0.02	0.96	0.96	0.02	0.98	0.90	0.06	0.98	0.95	0.06	0.98	0.95
α-iron	0.02	0.87	0.86	0.02	0.86	0.90	0.05	0.95	0.82	0.05	0.96	0.77
coesite	0.01	0.96	0.92	0.01	0.97	0.93	0.04	0.96	0.99	0.04	0.98	0.98
α-PbO_2-SiO_2	0.02	0.96	0.97	0.02	1.00	0.99	–	–	–	–	–	–
NAL phase	–	–	–	–	–	–	0.01	0.98	0.88	0.01	0.99	0.83

4.3 Results

The released emulators were evaluated against validation test subsets of 15,000 assemblages per model, and along reference-composition paths computed directly with alphaMELTS and HeFESTo. The training sets contain 10^7–10^8 assemblages per model. We report the results in the order in which a user meets them: which phases are predicted, whether the predicted assemblage conserves mass, how closely phase proportions and physical properties follow the parent model along representative paths, and how long the calculation takes.

Phase presence

Tables 4.10 and 4.11 give the precision and recall of the predicted phase presence. For the MELTS emulators, liquid is predicted with precision and recall of 0.99–1.00, and the common silicate phases (clinopyroxene, orthopyroxene, plagioclase and olivine) with 0.93–0.99. The HeFESTo emulators reach 0.95 or better for the silicate and oxide phases present in more than about 10% of test assemblages, including garnet, the $(Mg,Fe)_2SiO_4$ polymorphs, bridgmanite, post-perovskite, ferropericlase and Ca-perovskite. The exceptions are the Ca-ferrite phase in the Mars isentropic emulator (recall 0.92) and γ-iron, which is present in 24–28% of test assemblages but has a recall as low as 0.87–0.90. The isothermal and isentropic HeFESTo emulators perform similarly.

Classification quality declines for phases that are rare in the training data and stable only over narrow ranges of pressure, temperature or composition. In the MELTS emulators these are graphite (precision 0.77–0.80), Fe–Ni alloy (0.68–0.88 in either metric), leucite (0.83–0.93) and nepheline (0.67 precision in MELTS 1.2.0 without Cr), each present in 2% of test assemblages or fewer. Spinel recall falls to 0.80 in the MELTS 1.2.0 emulator without Cr, where spinel is less frequent than in the Cr-bearing model. In the HeFESTo emulators the weakest phases are akimotoite (0.77–0.88), α-iron (0.77–0.96), ε-iron in the Mars models (recall 0.77–0.85, frequency 0.01) and the NAL phase in the Mars models (recall 0.83–0.88).

Mass conservation

Without correction, the elemental bulk composition reconstructed from the raw network output differs from the requested composition by a median of 0.8–2.3% per element, depending on the emulator, and 45–68% of element

abundances are in error by more than 1% (Fig. 4.4). The largest raw errors are in minor and trace components, such as Cr and Fe^{3+}, where the network assigns small amounts of an element to assemblages whose bulk composition contains little of it. After the correction of §4.2, the median relative error falls to about 5×10^{-6}% for every emulator, the 99th percentile to 0.1% or less, and fewer than 0.4% of element abundances differ from the request by more than 1%.

Phase proportions along reference paths

Figures 4.15 and 4.16 compare phase proportions along isentropes of reference compositions computed with HeFESTo and with the isentropic emulators, given pressure, entropy and bulk composition. For basalt, depleted MORB mantle and Si-enriched pyrolite on Earth, the emulator reproduces the sequence of phase assemblages from the surface to 140 GPa, including the olivine–wadsleyite–ringwoodite transitions, the post-spinel transition near 23–25 GPa and the post-perovskite transition between about 120 and 130 GPa, with phase proportions that are visually indistinguishable over most of each path. The visible differences are confined to thin fields, such as the narrow akimotoite field in basalt near 25–30 GPa, and to small ripples in otherwise smooth phase proportions. The Mars emulator reproduces the basalt, harzburgite and shergottite paths to 25 GPa with the same fidelity, including the stishovite, kyanite and γ-iron fields of the enriched compositions. The largest differences on the Mars paths are small shifts of the clinopyroxene–garnet boundary in the shergottite between about 15 and 18 GPa.

Figures 4.17 and 4.18 show isobaric cooling of the Bishop Tuff, a lherzolite and a MORB with the closed-system isothermal MELTS emulators. Both emulators reproduce the liquidus and the order and temperatures of phase appearance for all three compositions. The lherzolite paths, which span 700–1800 °C in MELTS 1.0.2 and 1330–1800 °C in MELTS 1.2.0, are matched closely throughout, including the clinopyroxene-out reaction near 1380 °C in MELTS 1.0.2. Where alphaMELTS changes abruptly, the emulator smooths the change. The MORB solidus near 1030 °C, at which the alphaMELTS liquid fraction falls from several percent to zero over a few degrees, is spread by the emulator over a few tens of degrees, and a small, sharp step in alkali-feldspar abundance near 780 °C in the MELTS 1.0.2 Bishop Tuff path is absent from the emulator's.

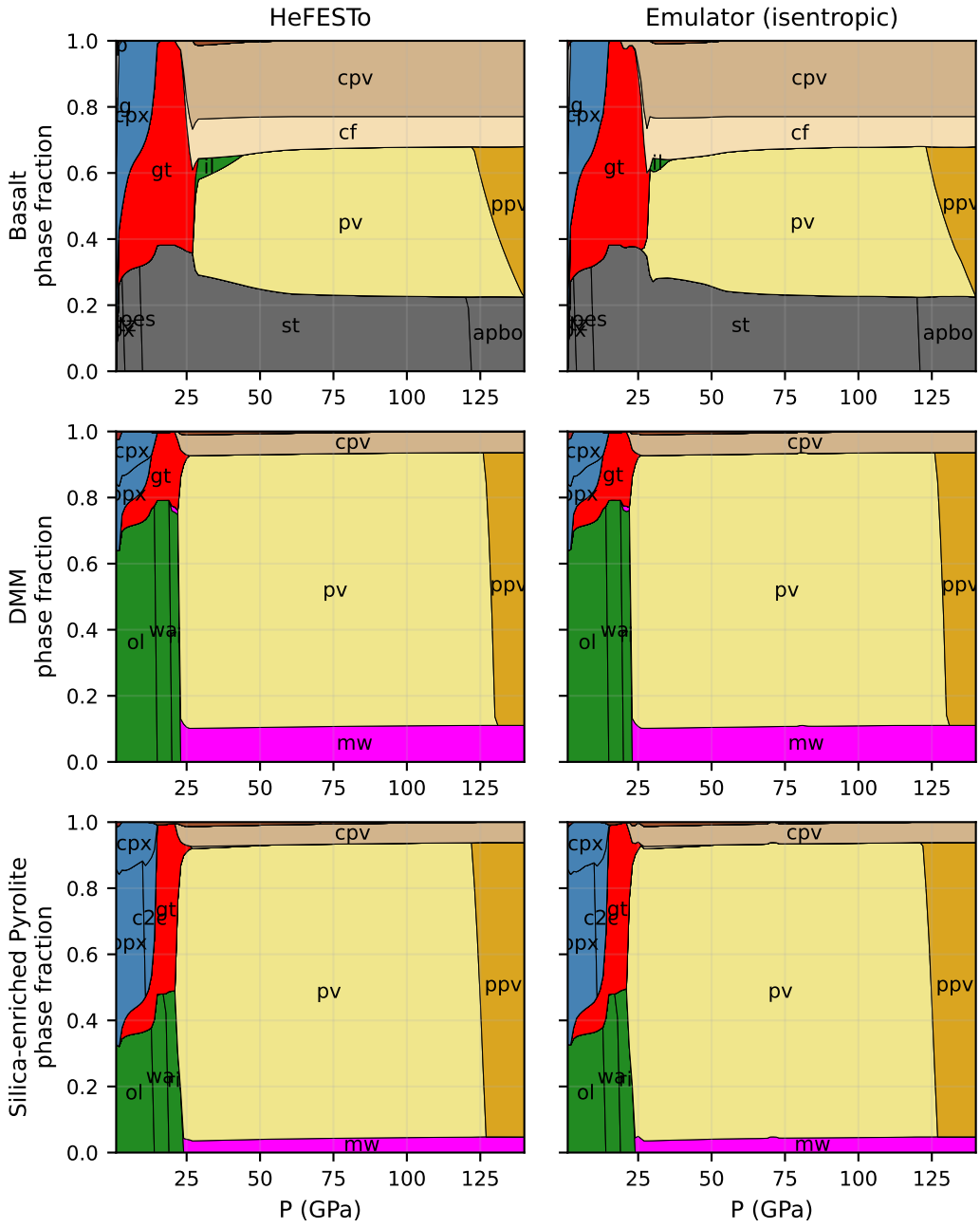


Figure 4.15: Phase abundances for isentropes of reference compositions for the HeFESTo Earth model. Reference compositions are basalt and Depleted MORB Mantle from Stixrude and Lithgow-Bertelloni (2024) after Workman and Hart (2005), and Si-enriched pyrolite (after McDonough and Sun, 1995) reference compositions. Phases: plg, plagioclase; sp, spinel; opx, orthopyroxene; cpx, clinopyroxene; c2c, high-pressure clinopyroxene; gt, garnet; ol, olivine; wa, wadsleyite; ri, ringwoodite; il, akimotoite; pv, bridgmanite; ppv, post-perovskite; cpv, Ca-perovskite; cf, Ca-ferrite-structured phase; nal, NAL phase; mw, ferropericlase; qtz, quartz; coes, coesite; st, stishovite; apbo, α-PbO_2-structured SiO_2; ky, kyanite; feg, γ-iron.

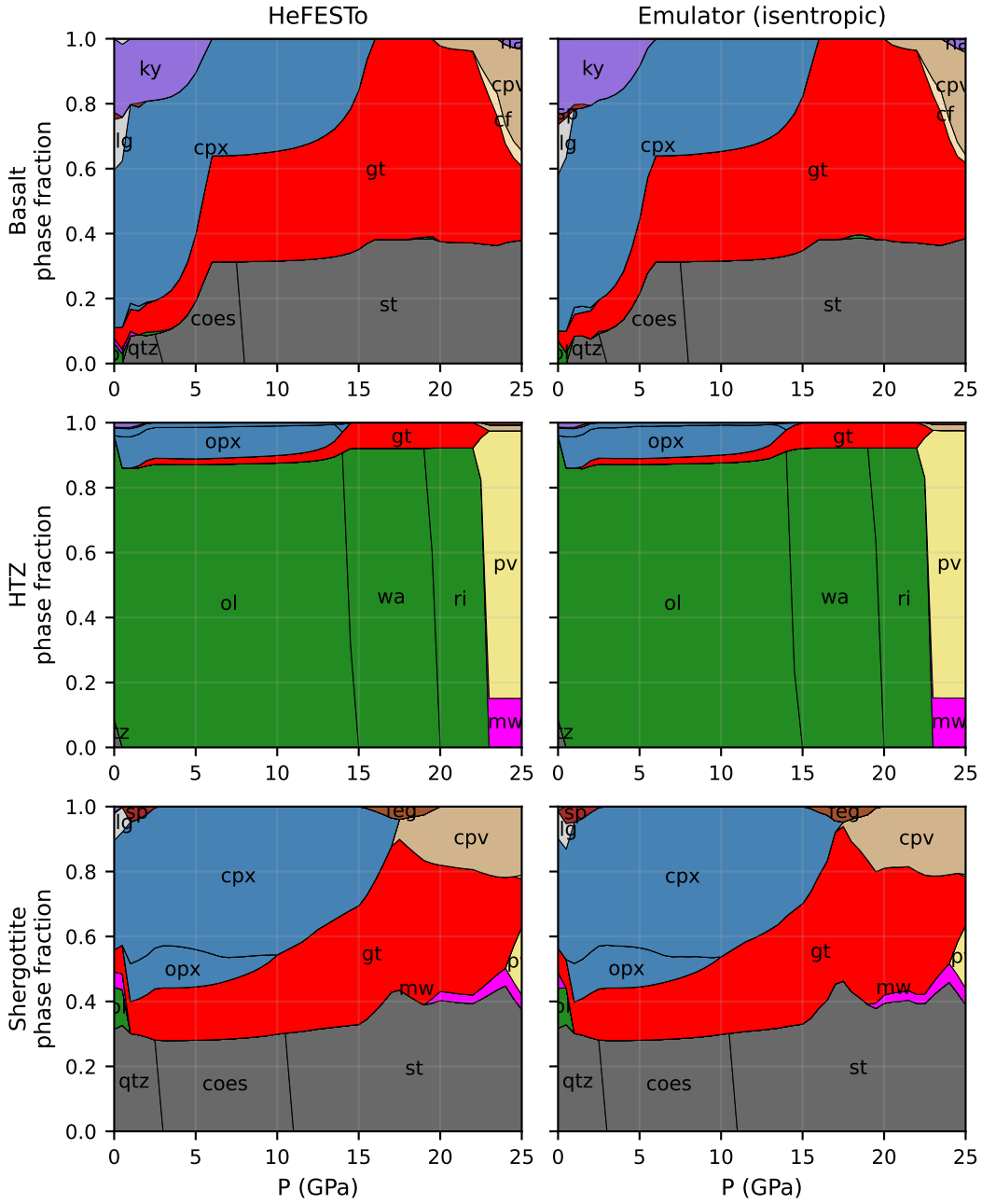


Figure 4.16: As Fig. 4.15, for the HeFESTo Mars model and the basalt, harzburgite (Stixrude and Lithgow-Bertelloni, 2024, after Workman and Hart, 2005), and shergottite (Stolper and McSween, 1979) reference compositions. Phase abbreviations as in Fig. 4.15.

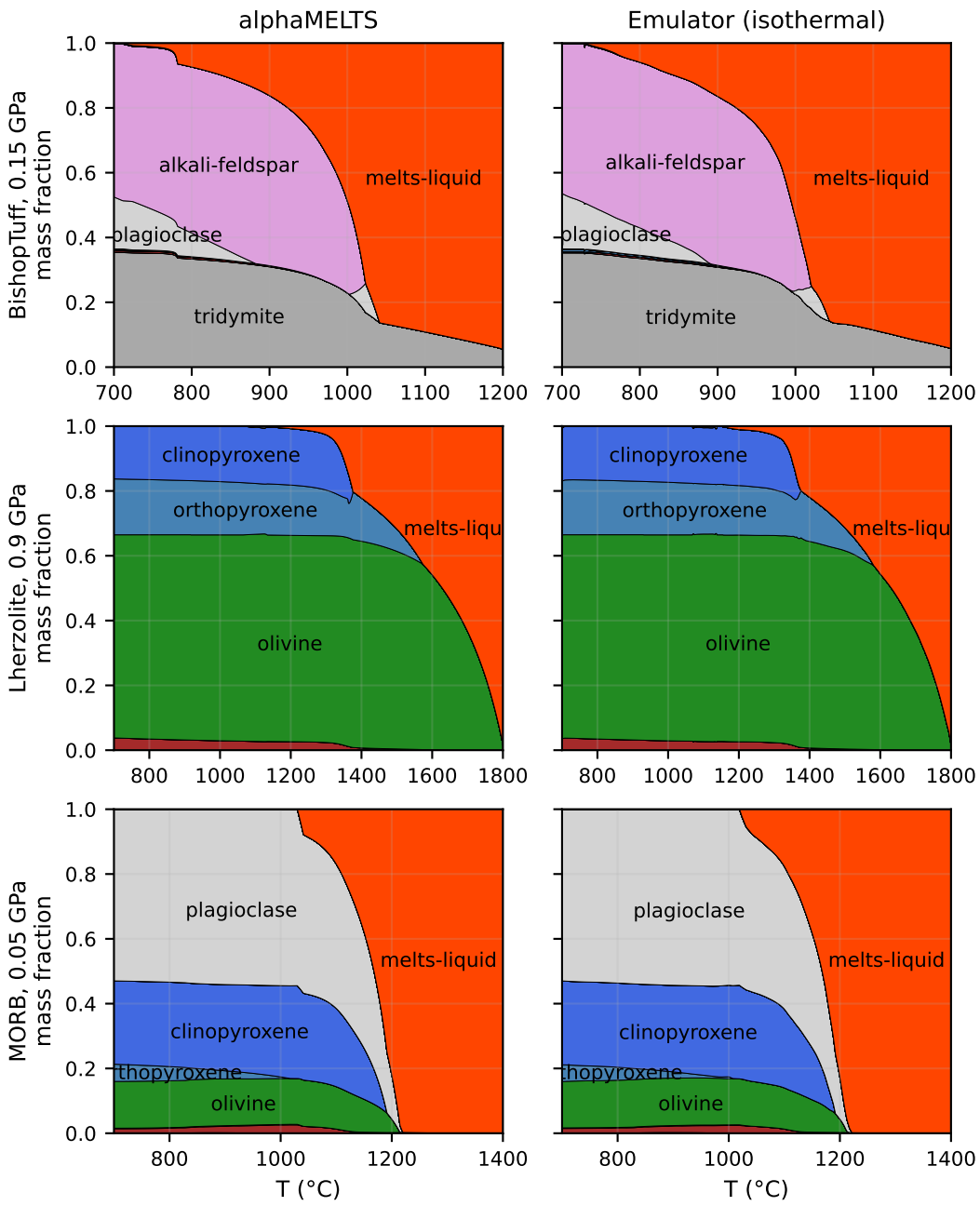


Figure 4.17: Phase mass fractions during isobaric cooling of the MELTS 1.0.2 reference compositions (Cr variant): Bishop Tuff (Hildreth, 1979), lherzolite (Dasgupta and Hirschmann, 2007), and MORB (Allan et al., 1989), each at the pressure given in its panel title. alphaMELTS (left) and the closed-system isothermal emulator, given pressure, temperature and bulk composition (right).

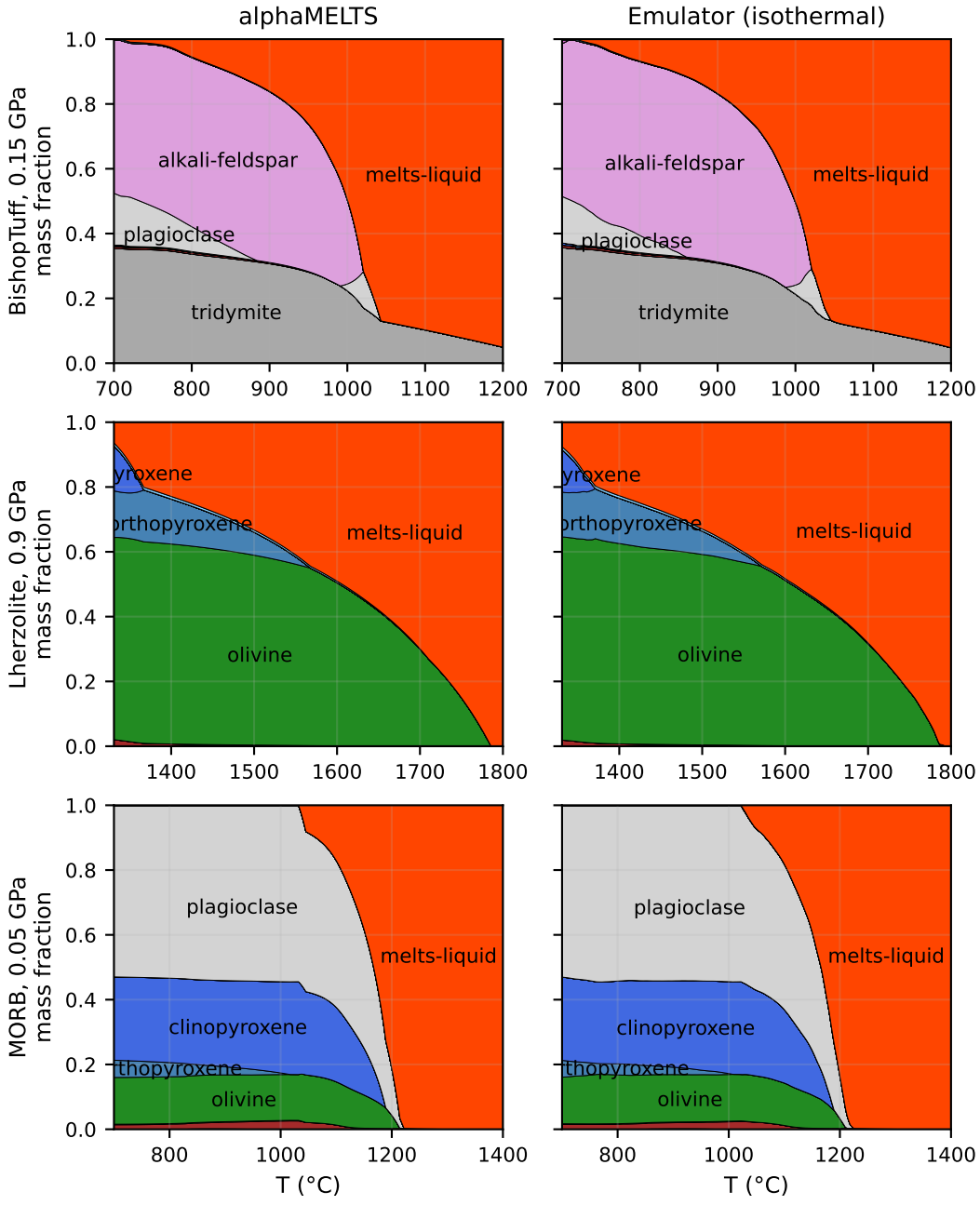


Figure 4.18: As Fig. 4.17, for the MELTS 1.2.0 reference compositions and the closed-system MELTS 1.2.0 isothermal emulator (Cr variant).

Liquid lines of descent

Figure 4.19 compares the liquid compositions along the lherzolite and MORB cooling paths. The emulator follows the alphaMELTS liquid lines of descent closely for MgO, FeO, Fe_2O_3, Al_2O_3 and Cr_2O_3, including the inflections at which phases enter or leave the assemblage. The differences grow as the liquid fraction becomes small. For the most evolved lherzolite liquids ($SiO_2 > 52$ wt%), the emulator underestimates Na_2O and K_2O and overestimates CaO, and its path extends to about 56 wt% SiO_2 where alphaMELTS stops near 54.5 wt%. For the last MORB liquids, H_2O and P_2O_5 diverge from alphaMELTS by up to about 0.4 and 0.15 wt%, respectively. These are the states in which the liquid holds a small fraction of the system's mass, so an error in the liquid composition has little effect on mass balance and is weakly constrained by it.

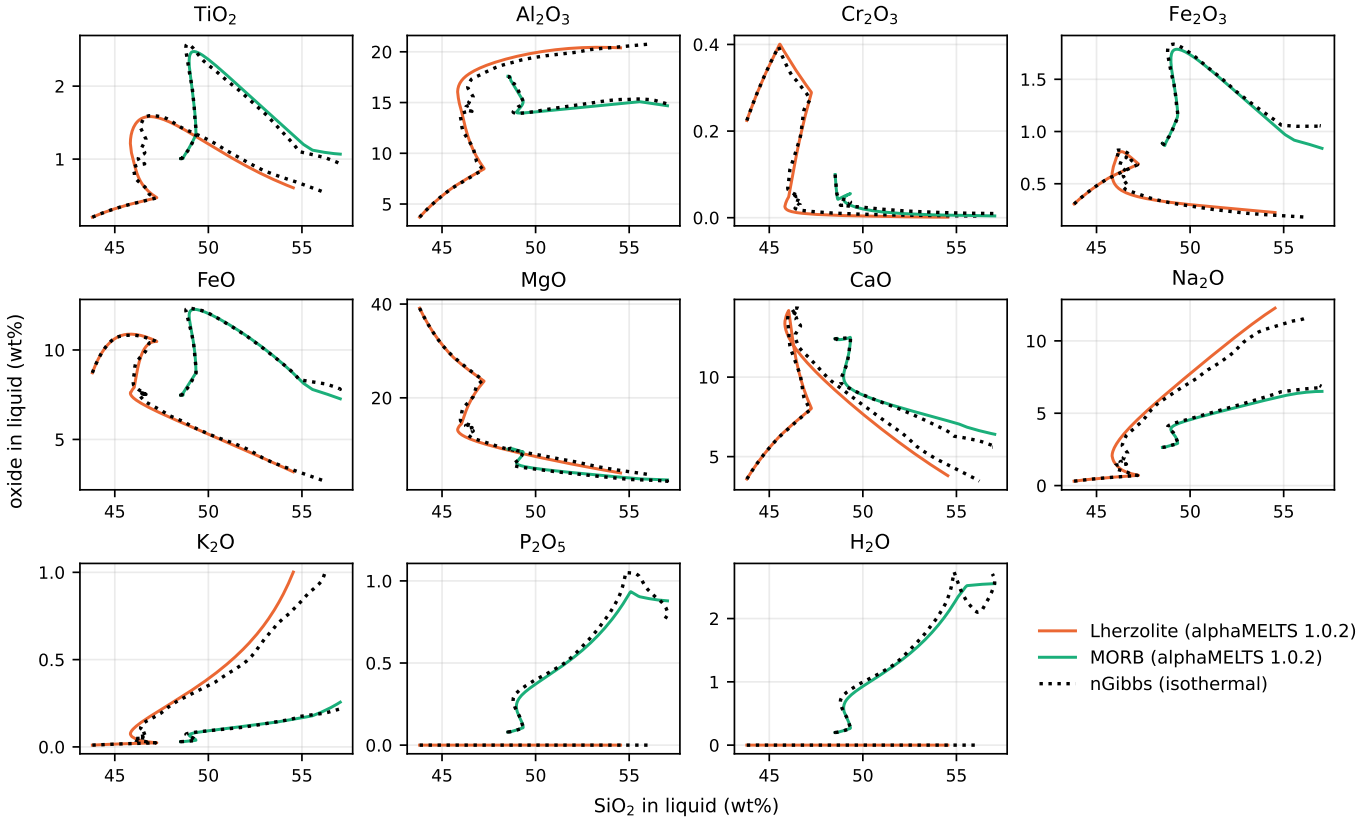


Figure 4.19: Liquid lines of descent during isobaric cooling of the lherzolite and MORB MELTS 1.0.2 reference compositions (Cr variant; Fig. 4.17): Harker diagrams of liquid SiO_2 against each other liquid oxide, in wt%. Solid, colored by rock: alphaMELTS; black dotted: the closed-system MELTS 1.0.2 isothermal emulator, given pressure, temperature and bulk composition. Only states with at least 1% liquid by mass (as predicted by alphaMELTS) are shown.

Material properties and temperature

Along the HeFESTo Earth isentropes (Fig. 4.20), density, V_P, V_S and temperature computed from the emulator's assemblage and predicted temperature cannot be distinguished from those of HeFESTo at the scale of the figure. The emulator also places the peaks in C_p, K_S and α at the phase transitions in the correct positions and with the correct amplitudes. The entropy calculated from the predicted assemblage and temperature stays within about $0.007\,\mathrm{J\,g^{-1}\,K^{-1}}$, or 0.3%, of the entropy imposed on the isentrope, and its mean error on the DMM path is 0.01%. The temperature model that supplies this temperature is evaluated in Fig. 4.14. The HeFESTo Mars isothermal emulator (Fig. 4.21) similarly reproduces the HeFESTo properties along the reference paths to 25 GPa, including the peaks in C_p, K_S and α at the phase transitions for the basalt and harzburgite reference compositions. The shergottite composition is the most exotic to the training domain of the Mars model, and thus the C_p, K_S and α predictions, which are dependent on more temperamental metamorphic terms, are far less accurate. Even for the shergottite composition, the density, V_P, V_S and temperature are indistinguishable from HeFESTo at the scale of the figure.

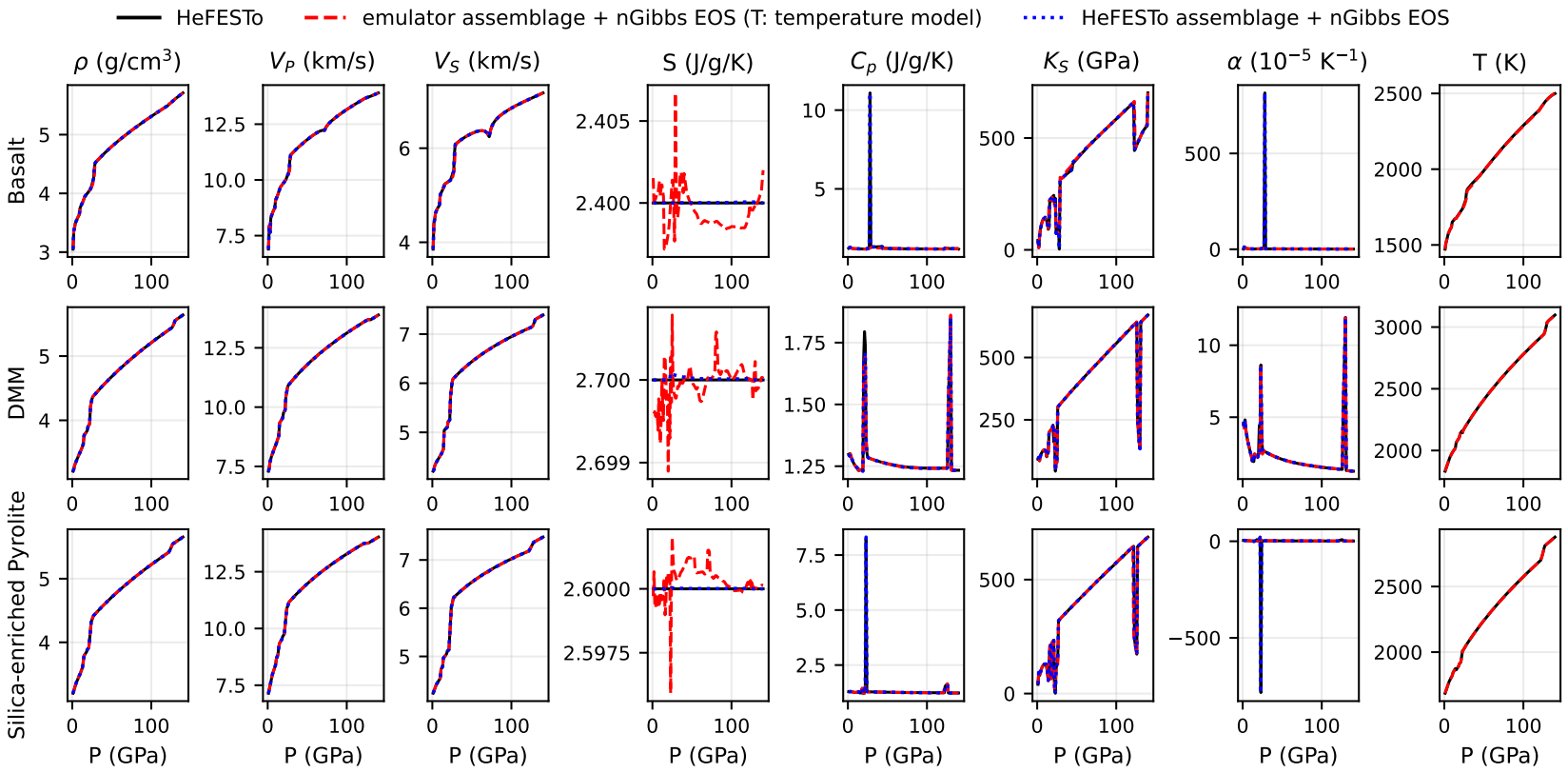


Figure 4.20: Bulk properties along the HeFESTo Earth reference composition paths of Fig. 4.15 (one row per reference composition): density, V_P, V_S, S, C_p, K_S and α against pressure. Black: HeFESTo; red dashed: the isentropic emulator's predicted assemblage, at its predicted temperature, evaluated with the *nGibbs* equations of state (§4.2); blue dotted: the HeFESTo assemblage evaluated with the same equations of state. The right-hand column compares the true temperature with that of the temperature model (Fig. 4.14).

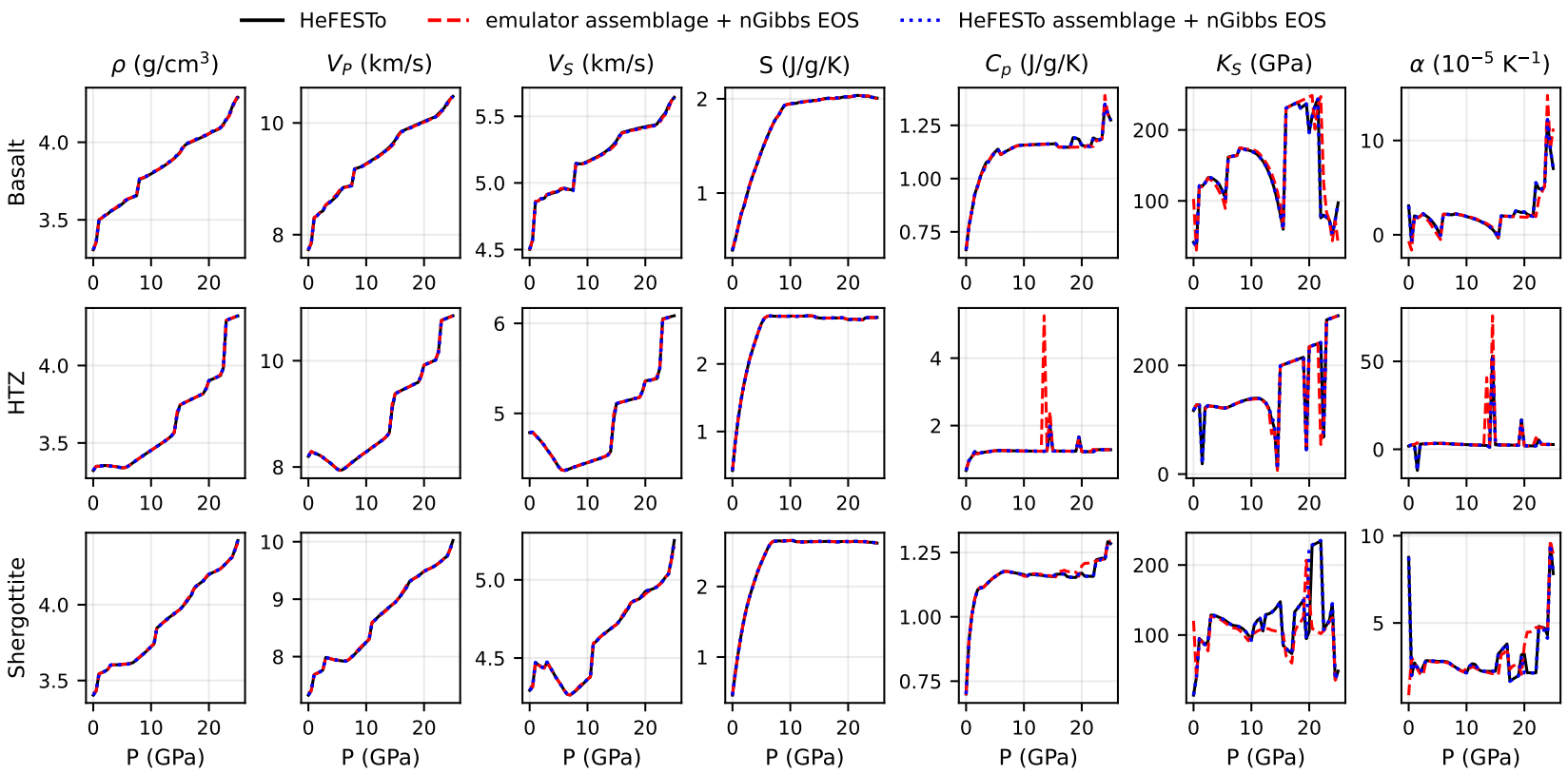


Figure 4.21: As Fig. 4.20, for the HeFESTo Mars model and the reference composition paths of Fig. 4.16, with the isothermal emulator.

Figure 4.22 separates the error of the emulator from that of the equations of state for the DMM isentrope. With every term included, the mean relative errors of the emulator's properties against HeFESTo are 0.02% for ρ, V_P and V_S, 0.01% for S, 0.13% for C_p, 0.63% for K_S and 0.86% for α. The same equations of state applied to the HeFESTo assemblage give 0.11%, 0.53% and 0.69% for C_p, K_S and α, so most of the remaining error comes from the equation-of-state translation rather than from the emulated assemblage. The metamorphic terms are necessary for these three properties: without them, the errors are 2.1% in C_p, 18.8% in K_S and 7.1% in α, because the peaks at the phase transitions are missing. The order–disorder ("fast") term reduces the error in V_P from 0.15% to 0.02% and affects no other property.

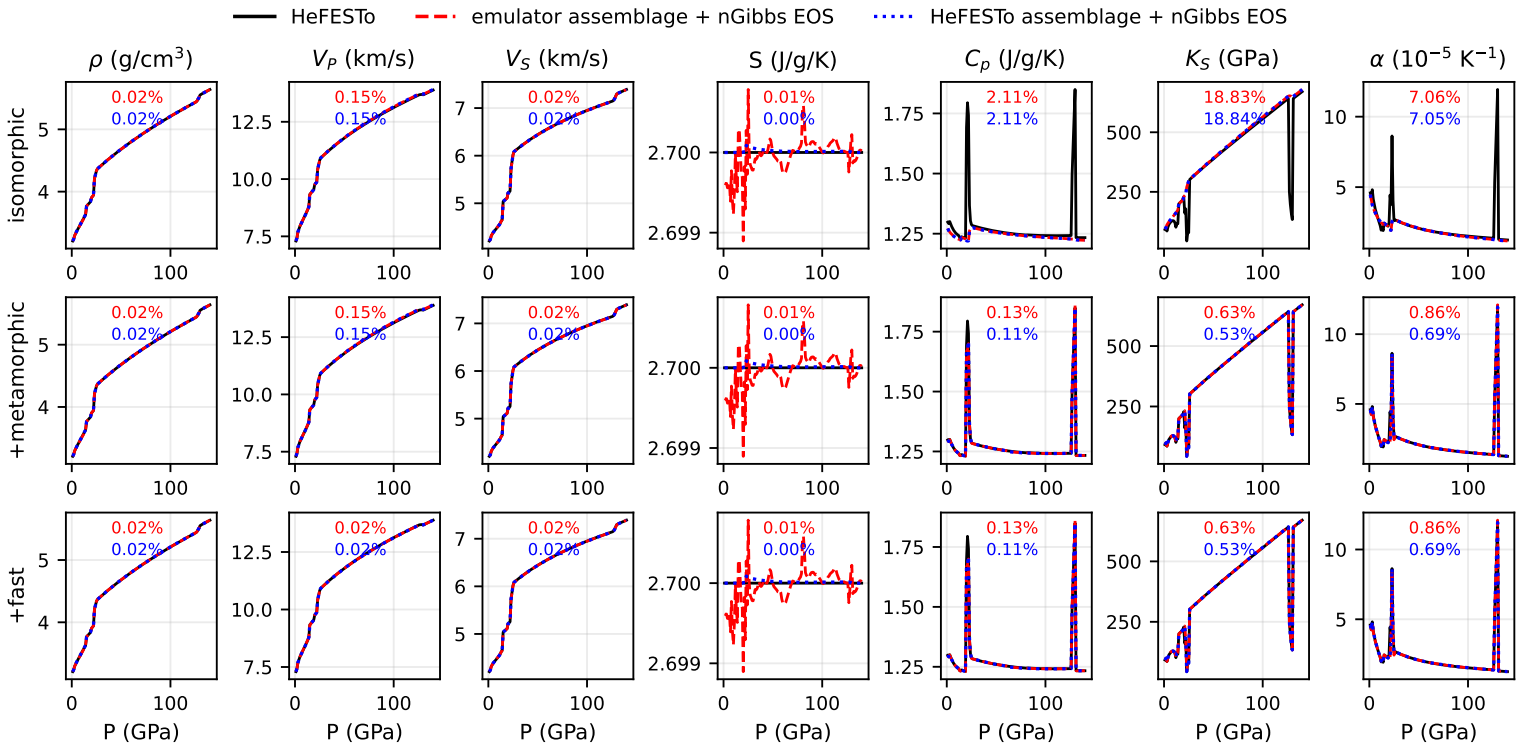


Figure 4.22: Bulk properties along the HeFESTo Earth DMM adiabat of Fig. 4.20, evaluated with successively more terms of the *nGibbs* HeFESTo equations of state (one row per level). Isomorphic: the properties of each phase at fixed composition. +metamorphic: adds the change of the equilibrium assemblage with temperature and pressure to C_p, K_S and α. +fast: also adds the intra-phase order-disorder relaxation to V_P, which completes the set of terms in HeFESTo's reported properties; ρ, V_S and S have no phase-change term and are the same at every level. Line styles as in Fig. 4.20, with the emulator's assemblage at the temperature-model temperature; the numbers in each panel are the mean relative errors against HeFESTo of the emulator assemblage (red) and of the HeFESTo assemblage (blue). Timings for each level are in Table 4.12.

For MELTS 1.0.2 (Fig. 4.23), the properties of the emulated lherzolite and MORB assemblages overlie those of alphaMELTS, including the abrupt changes in density and volume at the MORB solidus. The Bishop Tuff shows the largest differences, in density, volume and enthalpy below about 900 °C, which coincide with the alkali-feldspar step described above.

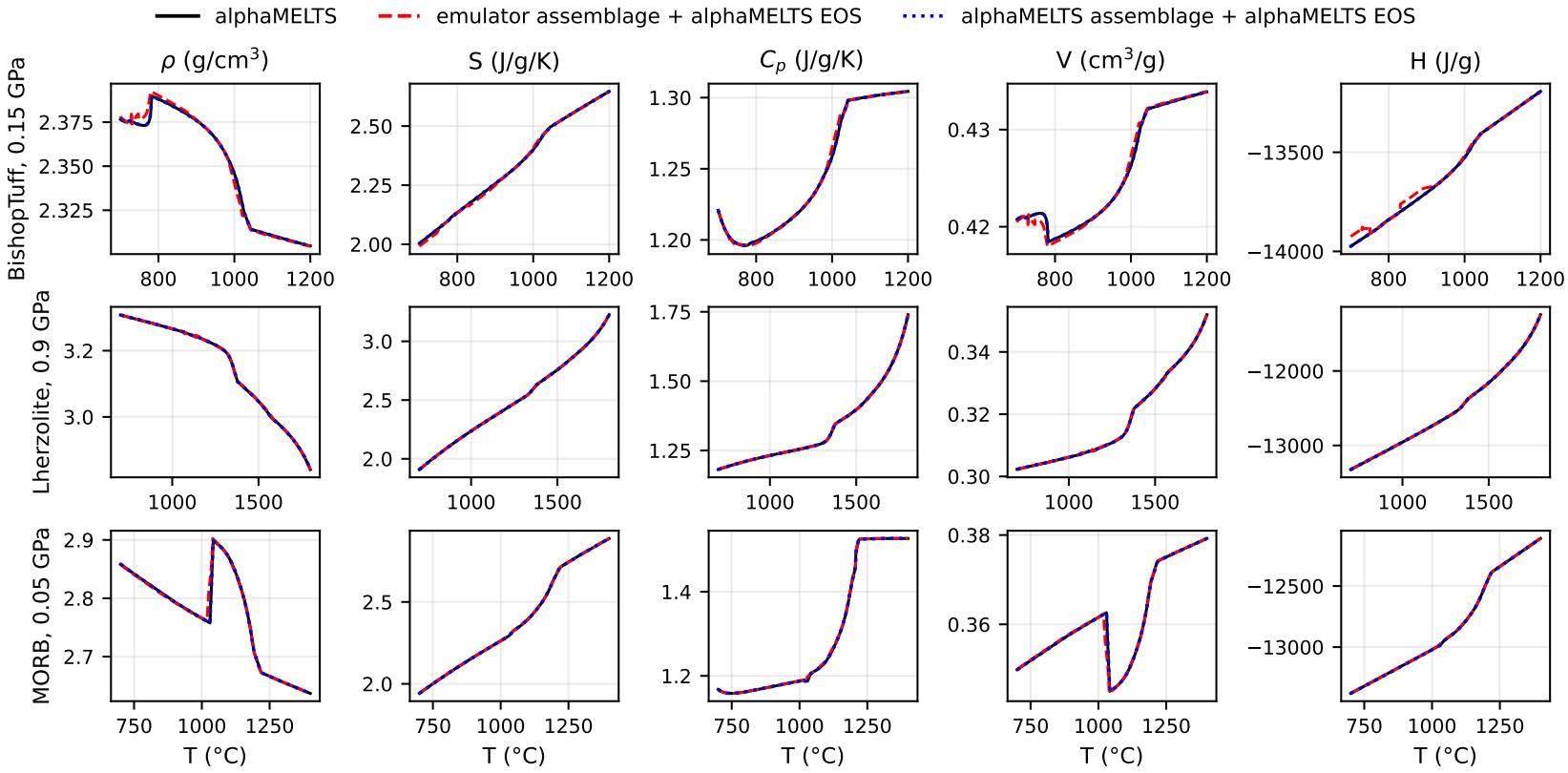


Figure 4.23: Bulk properties during isobaric cooling of the MELTS 1.0.2 reference compositions of Fig. 4.17 (one row per rock): density, S, C_p, V and H against temperature. Black: alphaMELTS; red dashed: the isothermal emulator's predicted assemblage evaluated with the multi-threaded PetThermoTools property solver (§4.2); blue dotted: the alphaMELTS assemblage evaluated similarly.

Table 4.12: Wall time (s) to evaluate $N = 2^{15}$ (32,768) assemblages along the HeFESTo Earth isentrope of §4.2 on the laptop (4-core Intel i7-11370H CPU and NVIDIA GeForce RTX 3060 GPU with 6 GB VRAM) and the desktop (20-core Intel i7-14700H CPU and NVIDIA GeForce RTX 5070 Ti GPU with 16 GB of VRAM). The first column shows the time for the isentropic emulator alone (phase equilibria, mass balance and temperature). Each further column adds the *nGibbs* equations of state with successively more terms (Fig. 4.22): the isomorphic properties only, then the metamorphic (phase-change) terms, then the order-disorder ("fast") relaxation of V_P. The HeFESTo time is extrapolated linearly from the fastest measured run on each machine (8 parallel workers on the laptop, from 3,384 assemblages in 3,479 s; 20 parallel workers on the desktop, from 8,460 assemblages in 1,074 s).

	Emulator	+ isomorphic	+ metamorphic	+ fast	HeFESTo
Laptop CPU	1.52	12.9	22.4	29.6	33,690
Laptop GPU	0.44	2.23	13.7	21.1	–
Desktop CPU	1.38	5.47	8.52	10.6	4,160
Desktop GPU	0.17	0.81	4.03	6.00	–

Computational performance

Table 4.12 and Fig. 4.24 compare the wall time of the emulators and EOS implementations of *nGibbs*, with the source parent models. For 2^{15} (32,768) assemblages on the HeFESTo Earth isentrope, HeFESTo running on all available workers is extrapolated to need about 33,700 s on the 4-core laptop and 4,200 s on the 20-core desktop. The emulator alone, including mass balance and temperature, takes 1.4–1.5 s on either CPU, a speed-up of about 3,000 (desktop) to 22,000 (laptop), and 0.17–0.44 s on a GPU. Adding the full equations of state brings the totals to 10.6 s (desktop CPU) and 29.6 s (laptop CPU), still 390–1,100 times faster than HeFESTo. Most of the property cost lies in the metamorphic and order–disorder terms; with only the isomorphic terms, the GPU totals are 0.81 s (desktop) and 2.2 s (laptop).

The speed-up of *nGibbs* over MELTS is smaller than for HeFESTo because alphaMELTS is itself much faster than HeFESTo, and is timed favorably here along cooling paths on which each equilibrium starts from the previous one. On the laptop, alphaMELTS through PetThermoTools reached at most about 67 equilibria per second (8 workers), against about 8,500 assemblages per second for the emulator on a single CPU call of 262,144 assemblages, a factor of about 130. On the desktop the corresponding rates are about 800

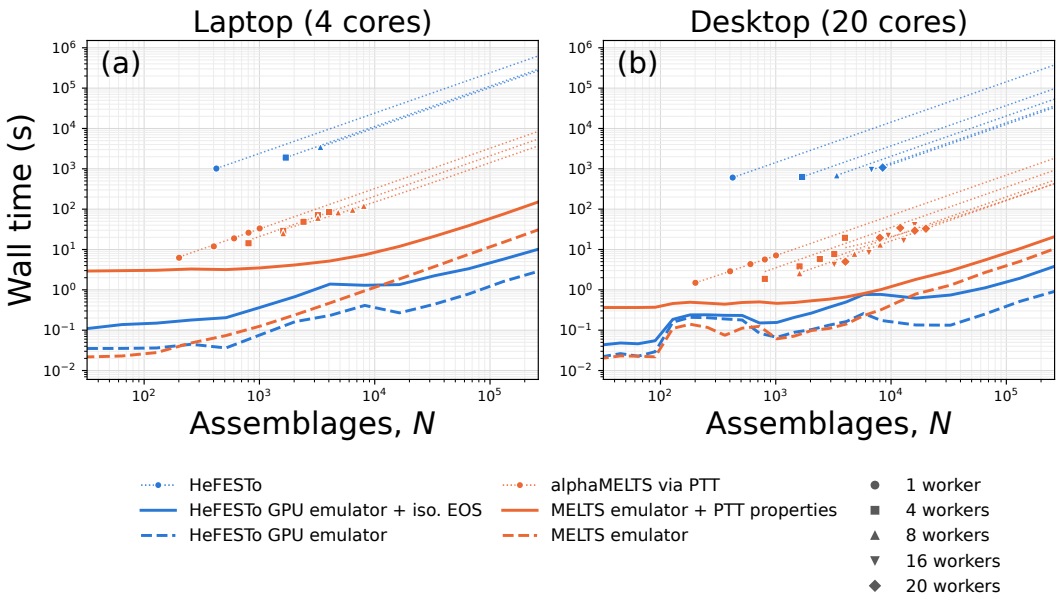


Figure 4.24: Wall time against the number of assemblages N for the reference codes and the emulators (§4.2) on (a) a 4-core Intel i7-11370H laptop and (b) a 20-core Intel i7-14700H desktop, on shared axes. Markers are measured HeFESTo and alphaMELTS (via PetThermoTools) runs, with marker shape giving the worker count; each dotted line is a straight-line fit through the runs of one worker count, or through the origin for a single run, extrapolated to the largest N. The *nGibbs* HeFESTo emulator performance was measured on the CPU for $N < 8192$ and on the GPU for $N \geq 8192$. This cross-over is managed automatically by the *nGibbs* `ForwardMB` function. All other emulator curves are CPU-only timings.

equilibria per second (20 workers) and 24,000 assemblages per second, a factor of about 30. When properties are added through the PetThermoTools property calculator, which dominates the cost for large batches, the factors fall to about 26 (laptop) and 15 (desktop). The emulator's cost is nearly constant below about 10^3 assemblages, where fixed overheads dominate, and grows roughly linearly above about 10^4 (Fig. 4.24). The crossover points between the vectorized and sequential property routes for MELTS are shown in Fig. 4.25.

For the MELTS property calculation benchmark, the sequential approach is more efficient for evaluating fewer states because previous calculations can be reused and previous states become good initial guesses for optimization problems, but a vectorized approach becomes faster when many states are evaluated at once. A multi-threaded approach, however, is able to leverage benefits afforded to both sequential and parallel workflows. For MELTS properties, the

vectorized translation is faster than the single-threaded sequential workflow, but slower than the multi-threaded workflow leveraged by the PetThermoTools `calc_phase_props_MELTS_parallel()` function (M. L. M. Gleeson et al., 2025) for more than about 2000 states (Fig. 4.25). The single-threaded sequential workflow avoids the overhead cost of deploying multiple threads, and is fastest for fewer than about 700 states. The vectorized approach is the fastest between 700 and 2000 states evaluated at once, but because this speedup is marginal and *nGibbs* is meant for applications at much larger batch sizes, we generally recommend that users calculate properties using the multi-threaded implementation in PetThermoTools.

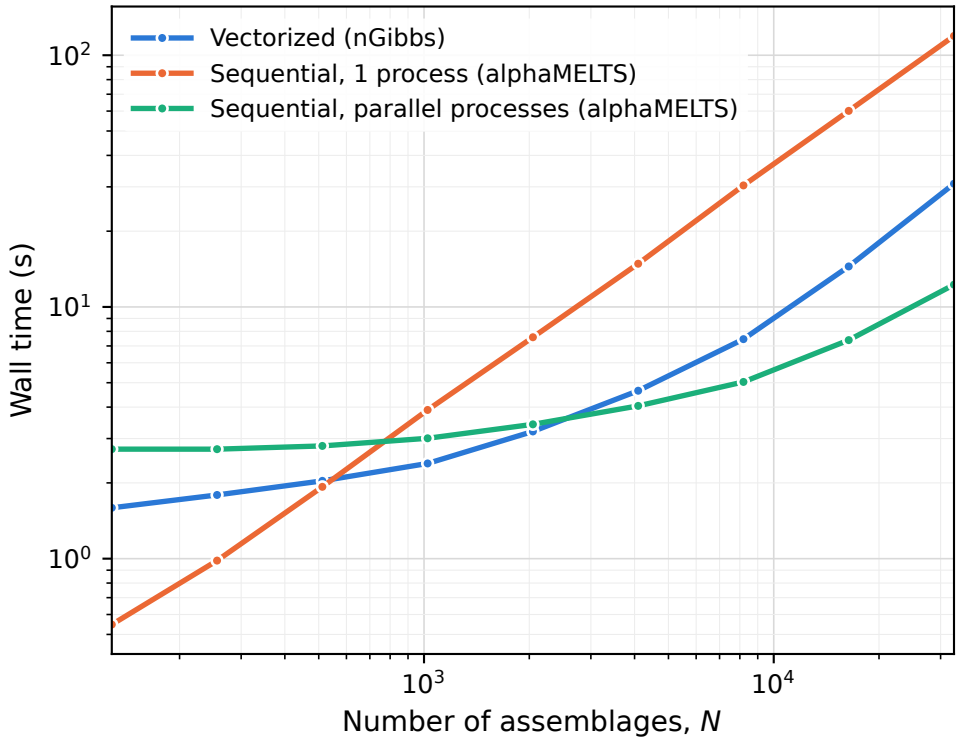


Figure 4.25: Wall time to compute MELTS 1.0.2 phase and bulk properties for an assemblage table predicted by the MELTS 1.0.2 emulator, against the number of assemblages, benchmarked on the 4-core Intel i7-11370H laptop. Vectorized: the *nGibbs* array translation of the MELTS equations of state (§4.2), taken as the difference between an emulator call with and without properties. Sequential: the alphaMELTS supplemental calculator called once per state and phase through PetThermoTools, in one process or split across PetThermoTools' default number of worker processes. No calculation includes free-energy minimization. Inputs are a fixed MORB composition on a P–T grid over 2–10 kbar and 900–1400 °C.

4.4 Application

Availability and installation

nGibbs is open source and available at https://github.com/benjthyer/nGibbs, and the release used in this chapter (v0.1.0b4) is archived on Zenodo (Thyer, 2026). It can be installed from PyPI with

```
pip install ngibbs
```

The MELTS emulators are also available through PetThermoTools (`pip install petthermotools`) with the `"nMELTS"` model keywords described in §4.4. Running the emulators on a GPU requires a CUDA-enabled installation of PyTorch.

PetThermoTools Integration

PetThermoTools (M. L. M. Gleeson et al., 2025) is a Python package that runs alphaMELTS for Python (Antoshechkina and Ghiorso, 2018) and MAGEMin (Riel et al., 2022) through one interface. It serves as a front end interface that allows users to interact with different thermodynamic models through a single, consistent API, which accesses different underlying models through a `Model` keyword argument. Its outputs are pandas DataFrames keyed by phase in the alphaMELTS naming convention, and it runs independent calculations in separate child processes so that a failed MELTS calculation does not halt a batch (M. L. M. Gleeson et al., 2025). The MELTS emulators of *nGibbs* are now also integrated within PetThermoTools, accessible with the `Model` keyword of a MELTS model preceded by the letter `n` (e.g. `"nMELTSv1.0.2"`), and their predicted assemblages are returned in the same structure as an alphaMELTS result. This serves two purposes. First, it makes *nGibbs* accessible to users who already work in PetThermoTools and who need not learn a separate interface. Second, it makes the emulators straightforward to benchmark, because the same function call can be run with the emulator and with the MELTS model it replaces, and the results compared directly. M. L. M. Gleeson et al. (2025) argued that a common interface would encourage systematic comparison of thermodynamic models across P–T–X space, and the same reasoning applies to comparing an emulator with its parent.

By default, the emulator output is passed to the PetThermoTools property calculator, which calls the alphaMELTS supplemental calculator for each predicted phase without performing a minimization, returning material proper-

ties for each phase and for the bulk assemblage. For applications where these properties are not required, this step can be bypassed by appending `NoProp` to the `Model` keyword (e.g. `"nMELTSv1.0.2NoProp"`), resulting in significantly faster performance. At this time, *nGibbs* through PetThermoTools is only supported on the CPU. Users interested in running nGibbs on their CUDA-enabled GPU may do so through the *nGibbs* interface directly.

Phase diagrams

Listing 4.2 computes the same phase diagram for a Mauna Loa primary melt twice, once with rhyolite-MELTS 1.0.2 and once with the *nGibbs* emulator, changing only the `Model` keyword. The emulator needs no initial guess and no path from a neighboring state, so it evaluates the grid as one batch. In the run shown, the emulator returned the diagram in about a tenth of the wall time of rhyolite-MELTS, although the two calls are not identical in workload because PetThermoTools also applies grid refinement to the MELTS calculation (M. L. M. Gleeson et al., 2025). The phase fields produced by the two calculations are compared in Fig. 4.26.

Listing 4.2: Phase diagram of a Mauna Loa primary melt computed with rhyolite-MELTS 1.0.2 and with *nGibbs* through PetThermoTools.

```
import multiprocessing
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from time import time

# Mauna Loa estimated primary melt composition from Wieser et al. 2025
ML_starting = {
    'SiO2_Liq': 48.5858,
    'TiO2_Liq': 1.4381,
    'Al2O3_Liq': 9.73,
    'FeOt_Liq': 10.144,
    'MgO_Liq': 20.0,
    'CaO_Liq': 7.2186,
    'Na2O_Liq': 1.7327,
    'K2O_Liq': 0.3227,
    'H2O_Liq': 0.2,
    'Fe3Fet_Liq': 0.12
}

```

```
# Guard required for multiprocessing on Windows: each PTT worker re-imports this
# file, and without it would rerun phaseDiagram_calc (and reload nGibbs) itself.
if __name__ == "__main__":
    multiprocessing.freeze_support()
    import petthermotools as ptt

    ptt.nGibbs_bridge._ensure_ngibbs_loaded()

    start_time = time()
    PD_ML = ptt.phaseDiagram_calc(Model = "MELTSv1.0.2",
                                  bulk = ML_starting,
                                  T_C = np.linspace(1100,1400,20),
                                  P_bar = np.linspace(250,5000,20),
                                  refine = 2,
                                  i_max = 3)
    print(f"rhyolite-MELTS Phase diagram calculation time: {time() - start_time:.2f} seconds")

    start_time = time()
    nPD_ML = ptt.phaseDiagram_calc(Model = "nMELTSv1.0.2NoProp", # No properties needed for this analysis
                                  bulk = ML_starting,
                                  T_C = np.linspace(1100,1400,20),
                                  P_bar = np.linspace(250,5000,20))

    print(f"nGibbs Phase diagram calculation time: {time() - start_time:.2f} seconds")

    ptt.plot_phaseDiagram_multi({'nGibbs': nPD_ML, 'rhyolite-MELTS v1.0.2': PD_ML})
```

Output:

```
rhyolite-MELTS Phase diagram calculation time: 52.30 seconds
nGibbs Phase diagram calculation time: 1.23 seconds
```

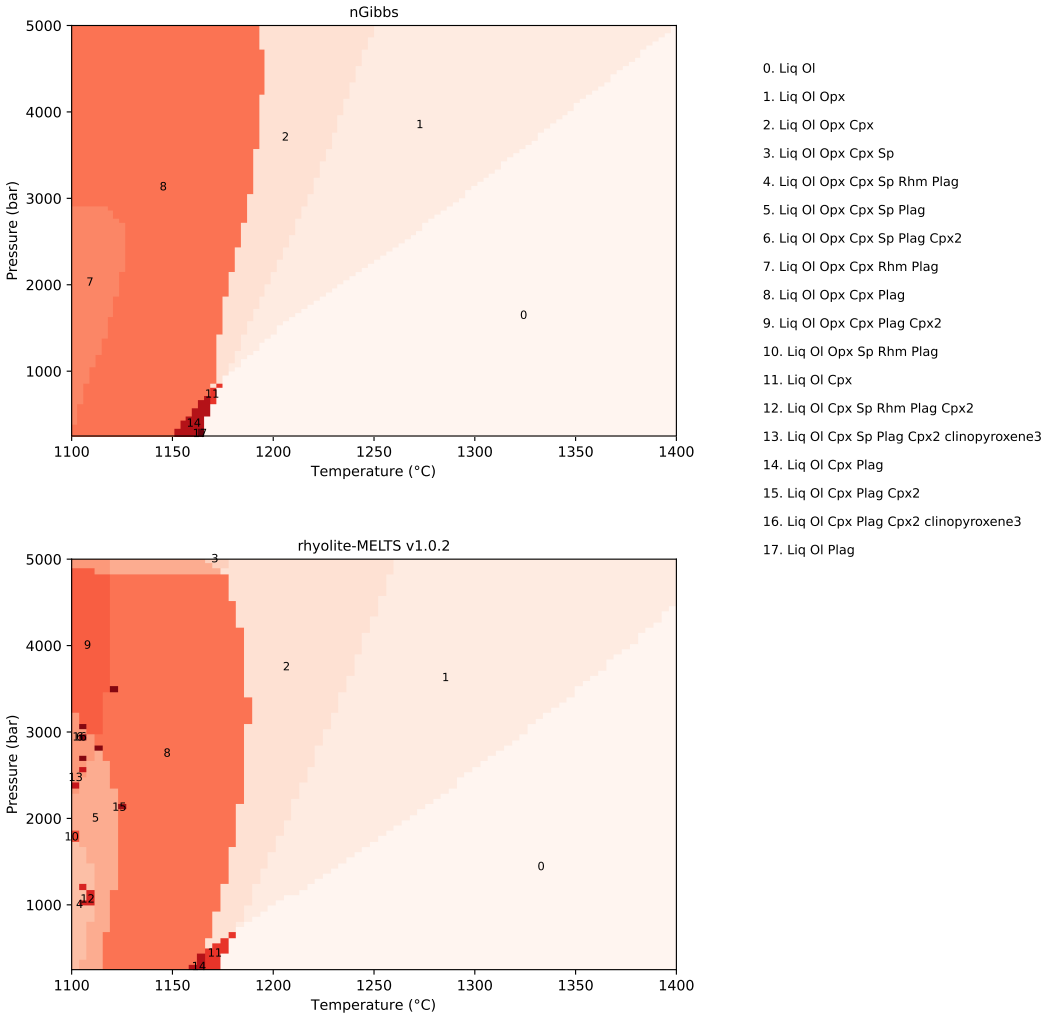


Figure 4.26: Stable phase assemblages for the Mauna Loa primary melt (Wieser, Shi, et al., 2025; Wieser, M. L. Gleeson, et al., 2025) over 1100–1400 °C and 250–5000 bar, from *nGibbs* (top) and RHYOLITE-MELTS 1.0.2 (bottom), produced by Listing 4.2.

Seismic inversions of the thermochemical structure of planetary mantles

Seismic data constrain the thermal and compositional structure of planetary mantles (Stixrude and Lithgow-Bertelloni, 2012; Jackson and Thomas, 2021; Khan et al., 2018; Wood and Rubie, 1996). A thermochemical inversion of seismic data begins with a forward model: bulk composition and thermal profiles are converted to density and seismic velocities that follow from a phase assemblage predicted through free energy minimization. These seismic and density profiles then are compared with seismic and physical observations (Khan and Connolly, 2008; Khan et al., 2021). A Markov chain Monte Carlo

(MCMC) sampler requires many thousands of such likelihood evaluations, which in turn each require many free energy minimizations. Every additional compositional parameter enlarges the volume the sampler must explore. This is why compositional dimensions are often fixed, as in the InSight inversion of Khan et al. (2021), which assumed a single bulk composition derived from the martian meteorites.

The HeFESTo emulators are well-suited to this problem. Listing 4.3 builds 201 whole-mantle profiles which span Earth's interior from 0–135 GPa, as an example workflow for this problem. Each profile is a conductive lid over an isentropic mantle over a conductive thermal boundary layer at the core–mantle boundary; the lid and isentrope follow the geophysical parameterization of Khan et al. (2021). Across the set, adiabat entropy, pressure at the adiabatic-conductive interfaces, temperature of the mantle at the core–mantle boundary, and composition of the mantle are varied. The isentropic segments are evaluated first in parallel as a single batch with the isentropic emulator, which also predicts their temperatures. Both conductive layers are then evaluated as a second batch with the isothermal emulator, and the vectorized HeFESTo equations of state return density, V_P, V_S and bulk modulus using isomorphic terms only for every node (Figs. 4.27 and 4.28). In the run shown, the full set of 35,175 assemblages completes in under 5 seconds on a GPU, whereas the true HeFESTo calculation would take about 10 hours. At that rate an inversion can sample bulk composition alongside the thermal parameters, rather than fixing it in advance, enabling future studies to use seismic data to constrain composition with a level of detail that has never before been possible. The Mars emulators can be used in the same way: they are trained to 25.5 GPa, which spans the pressure at the martian core–mantle boundary of ∼20 GPa estimated by Khan and Connolly (2008). We are already exploring this avenue in ongoing work.

On Earth, where seismic data are abundant, density and velocity profiles are more tightly constrained and carry more chemical information. A joint inversion for temperature and composition beneath plumes and subduction zones could separate the thermal contribution to discontinuity topography and sharpness from the compositional one, and could test whether the transition zone is compositionally stratified relative to the mantle above and below it. The 2024 HeFESTo database, which the Earth emulator reproduces, includes

ferric iron, the high-spin to low-spin transition in ferropericlase, and native iron phases (Stixrude and Lithgow-Bertelloni, 2024). Those additions make the emulator a candidate forward model for the iron-rich compositions proposed for ULVZs and LLVPs (Dobson and Brodholt, 2005; Jackson and Thomas, 2021; Q. Yuan et al., 2023). Some of these candidates, however, such as banded iron formation with up to 85 wt% iron oxide (Dobson and Brodholt, 2005), lie far outside the composition range of the current training data and would require targeted training data before the emulator can be applied to them.

Listing 4.3: Seismic and phase profiles of 201 whole-mantle columns (a conductive lid, an adiabatic mantle and a conductive boundary layer at the core–mantle boundary) of basalt mixed with silica-rich pyrolite, computed in parallel with the HeFESTo Earth isentropic and isothermal emulators and the vectorized HeFESTo equations of state.

```
import numpy as np
import torch
from time import time
from ngibbs.utils.math_utils import mix_compositions, IDX_2D_Split_3Part
from ngibbs.engine.models import HeFESToEmulatorCPU, HeFESToEmulatorGPU

# Each profile is a whole-mantle column: a conductive lid over an
# adiabatic (isentropic) mantle over a conductive thermal boundary layer
# (TBL) at the core-mantle boundary (CMB). All profiles are evaluated in
# one batch, on the GPU when one is available.
HeFESToEmulator = HeFESToEmulatorGPU or HeFESToEmulatorCPU
device = HeFESToEmulator.device
print(f"Using the {str(device).upper()} emulator")

n_pressures = 175  # pressure nodes per profile
n_profiles = 201   # profiles evaluated in parallel
P_CMB = 135        # GPa
T_surface = 300    # K
pressures = np.linspace(0, P_CMB, n_pressures)

# Profile parameters, swept together from the first profile to the last
X_basalt = np.linspace(1, 0, n_profiles)     # basalt fraction
S_ad = np.linspace(2.3, 2.8, n_profiles)     # isentrope entropy (J/g/K)
P_lid = np.linspace(8, 1, n_profiles)        # base of the lid (GPa)
P_TBL = np.linspace(132, 125, n_profiles)    # top of the TBL (GPa)
T_CMB = np.linspace(3500, 5000, n_profiles)  # CMB temperature (K)

```

```
# Snap both transitions to the nearest pressure node. The isentrope
# covers nodes lid_idx to TBL_idx; the lid lies above it, the TBL below.
lid_idx = np.abs(pressures - P_lid[:, None]).argmin(axis=1)
TBL_idx = np.abs(pressures - P_TBL[:, None]).argmin(axis=1)
P_lid, P_TBL = pressures[lid_idx], pressures[TBL_idx]

# (row, column) index pairs into the (profile x pressure) grid for each
# regime. The lid and the TBL are both conductive, so they are evaluated
# together.
TBL_IDX, isentropic_IDX, lid_IDX = IDX_2D_Split_3Part(
    lid_idx, TBL_idx + 1, n_pressures)
conductive_IDX = tuple(np.concatenate(pair)
                       for pair in zip(lid_IDX, TBL_IDX))

# End-member bulk compositions in HeFESTo element moles, from the
# SILICAadiabat and BASALTadiabat standards ('O' sets the ferric iron)
SILICA = {'Si': 4.35196, 'Mg': 4.35196, 'Fe': 0.51993, 'Ca': 0.29376,
          'Al': 0.40504, 'Na': 0.05391, 'Cr': 0.02345, 'O': 14.54576}
BASALT = {'Si': 4.61116, 'Mg': 1.33197, 'Fe': 0.61701, 'Ca': 1.22855,
          'Al': 1.81090, 'Na': 0.39533, 'Cr': 0.00508, 'O': 15.35841}
mixes = [mix_compositions([SILICA, BASALT], [1 - x, x])
         for x in X_basalt]
comp_headers = list(mixes[0])
comp_array = np.array([list(mix.values()) for mix in mixes])

# Results on the (profile x pressure) grid, filled one regime at a time
properties = {prop: np.zeros((n_profiles, n_pressures))
              for prop in ['temperature', 'S', 'rho', 'Vp', 'Vs', 'Kh']}
indexer = HeFESToEmulator.isentropic_emulator.ml_indexer
mass_phasedict = indexer.mass_phasedict  # phase name -> column
phase_fractions = np.zeros((n_profiles, n_pressures,
                            len(mass_phasedict)))

def emulate(IDX, condition, condition_header, outputs):
    """Emulate the stable assemblage at each node in IDX from its
    pressure, its S or T, and its composition."""
    inputs = np.column_stack(
        [pressures[IDX[1]], condition, comp_array[IDX[0]]])
    headers = ['P(GPa)(System_main)', condition_header] + comp_headers
    with torch.no_grad():
        return HeFESToEmulator.ForwardMB(inputs, headers=headers,
                                         outputs=outputs)

```

```
def store(IDX, output, T):
    """Fill the grid at IDX with the phase mass fractions, and with bulk
    properties from the vectorized HeFESTo equations of state evaluated
    on the emulated assemblages."""
    properties['temperature'][IDX] = T
    mass_percent = output['phase_tables'][1].cpu().numpy()
    phase_fractions[IDX] = mass_percent / 100
    PT = torch.tensor(np.column_stack([pressures[IDX[1]], T]),
                      dtype=torch.float64, device=device)
    eos = HeFESToEmulator._compute_bulk_EOS_properties(
        output['component_moles'].to(device), PT,
        property_names=['S', 'rho', 'Vp', 'Vs', 'Kh'])
    for name, values in eos.items():
        properties[name][IDX] = values

# --- Step 1: the adiabatic mantle (isentropic emulator) --------------
# Run first, because both conductive layers start from the isentrope
# temperatures, which the isentropic model predicts at each (P, S).
t0 = time()
output = emulate(isentropic_IDX, S_ad[isentropic_IDX[0]],
                 'S(J/g/K)(System_main)',
                 ['component_moles', 'phase_tables', 'temperature'])
T_isentrope = output['temperature'].cpu().numpy().reshape(-1)
store(isentropic_IDX, output, T_isentrope)
print(f'Adiabatic mantle:  {len(isentropic_IDX[0])} assemblages '
      f'in {time() - t0:.2f} s')

# --- Step 2: the lid and the TBL (isothermal emulator) ---------------
# Temperature is linear in pressure: in the lid from T_surface to the
# isentrope at P_lid, and in the TBL from the isentrope at P_TBL to
# T_CMB. Each linspace drops the node it shares with the isentrope.
T = properties['temperature']
T_lid = [np.linspace(T_surface, T[i, j], j + 1)[:-1]
         for i, j in enumerate(lid_idx)]
T_TBL = [np.linspace(T[i, j], T_CMB[i], n_pressures - j)[1:]
         for i, j in enumerate(TBL_idx)]
T_conductive = np.concatenate(T_lid + T_TBL)

t0 = time()
output = emulate(conductive_IDX, T_conductive, 'T(K)(System_main)',
                 ['component_moles', 'phase_tables'])
store(conductive_IDX, output, T_conductive)
print(f'Conductive layers: {len(conductive_IDX[0])} assemblages '
```

```
114         f'in {time() - t0:.2f} s')
```

Output:

```
Using the CUDA emulator
Adiabatic mantle:  32327 assemblages in 3.33 s
Conductive layers: 2848 assemblages in 1.14 s
```

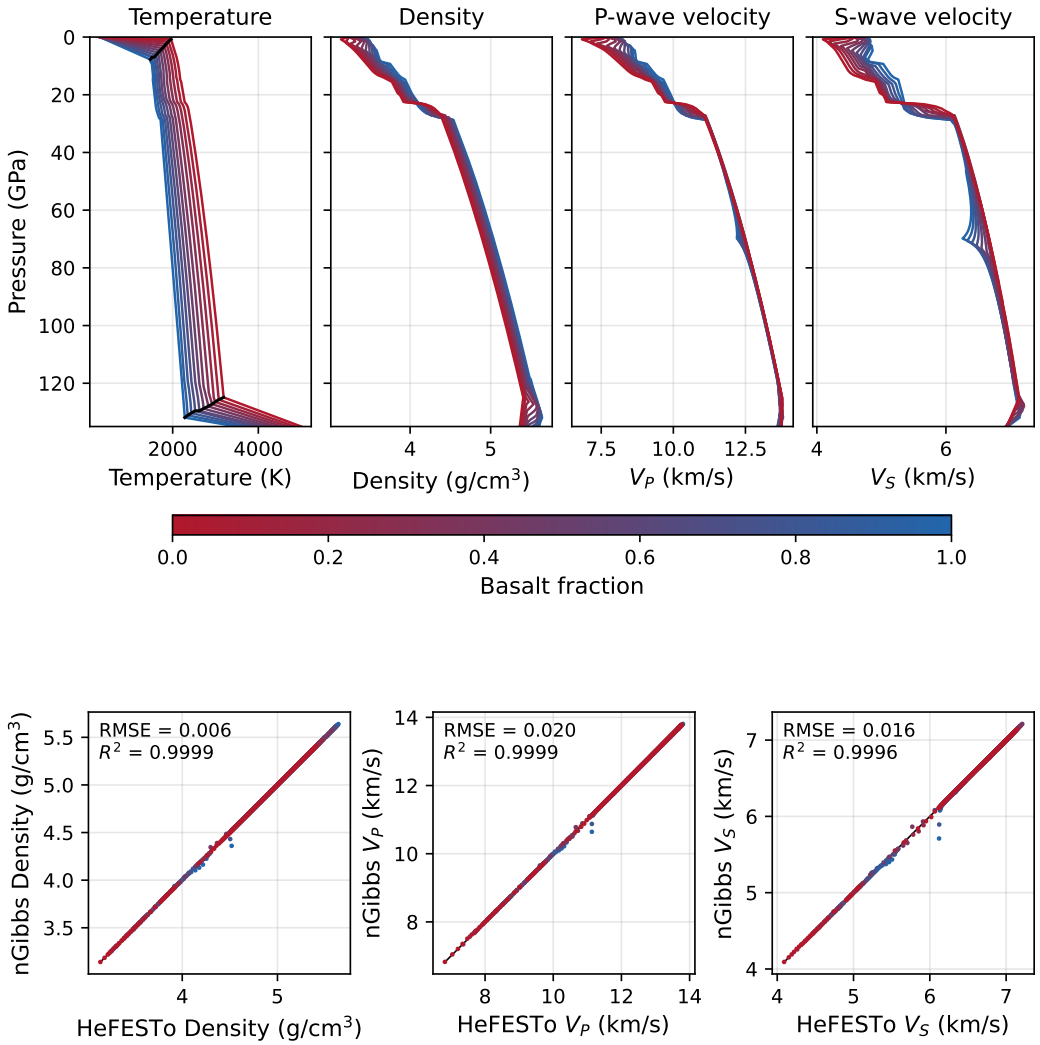


Figure 4.27: Temperature, density, V_P and V_S against pressure for every 20th of the 201 profiles produced by Listing 4.3, colored by basalt fraction. Temperature is linear in pressure in the conductive lid (from 300 K at the surface) and in the core–mantle boundary layer, and follows the isentrope between them; black lines join the transitions of the plotted profiles. The end members are the silica-rich pyrolite (after McDonough and Sun, 1995) and basalt reference compositions from Stixrude and Lithgow-Bertelloni (2024) after Workman and Hart (2005).

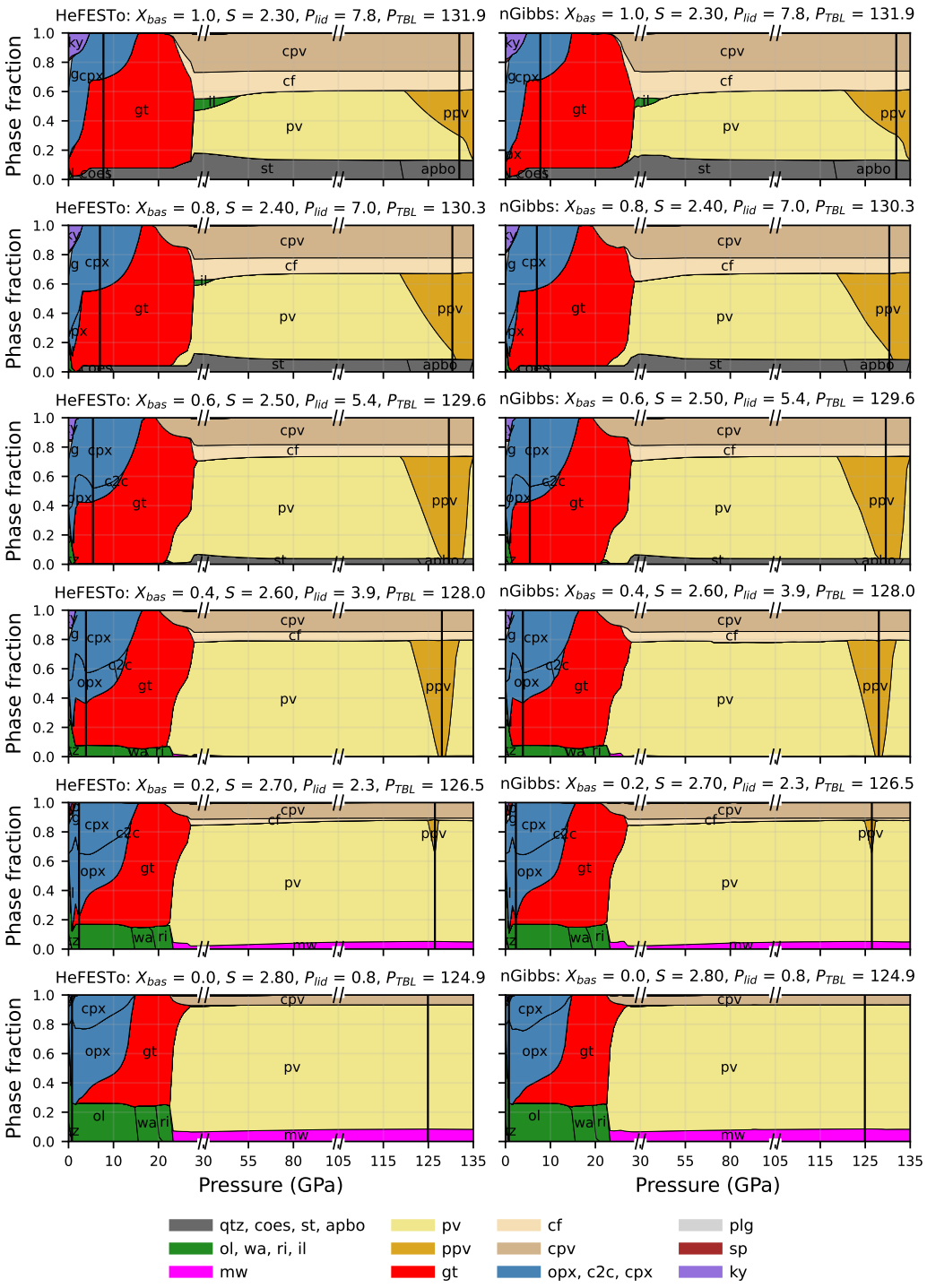


Figure 4.28; caption on the following page.

Figure 4.28: Phase mass fractions along six of the profiles of Listing 4.3, with basalt fractions evenly spaced between 1.0 and 0.0, computed with HeFESTo (left) and with the *nGibbs* emulators (right). Both profiles are computed with the same workflow: the isentrope is evaluated first between the base of the conductive lid and the top of the core–mantle conductive boundary layer, then the conductive layers are evaluated along linear paths between the surface temperature and the CMB temperature and the isentropic temperature at the top and bottom of the isentrope. Each panel gives the basalt fraction X_{bas}, the isentrope entropy S ($\mathrm{J\,g^{-1}\,K^{-1}}$), the pressure at the base of the lid P_{lid} and the pressure at the top of the core–mantle boundary layer P_{TBL} (GPa), which are also marked by solid black lines. The pressure axis condenses the relatively uninteresting middle mantle between 30 and 105 GPa to represent a disproportionately small part of the plot in order to emphasize the portions of the mantle where mineralogy strongly evolves as a function of depth. The 6 profiles evaluated by HeFESTo took 29.8 minutes on the 4-core Intel i7-11370H laptop, while all 201 profiles evaluated by the *nGibbs* emulators are completed in under 5 seconds on the same machine.

Mantle thermochemistry beneath spreading centers

An emulator of pHMELTS is planned. pHMELTS uses the pMELTS calibration for mantle melting and it approximates the storage of water in nominally anhydrous minerals by partitioning H_2O into a hidden reservoir (Asimow et al., 2004; P. M. Smith and Asimow, 2005). This model shows that trace amounts of water can have a significant effect on mantle melting (Asimow and C. H. Langmuir, 2003; Asimow et al., 2004). A fast emulator of polybaric melting followed by crystallization may allow individual mid-ocean ridge basalts to be inverted for the potential temperature, source composition and water content of the mantle that produced them, and repeating the inversion along ridge segments would build a map of mantle thermochemistry beneath spreading centers.

Existing methods for recovering primary magmas show why such an inversion is difficult. PRIMELT2 restores a lava to its primary magma by adding or subtracting olivine alone and infers potential temperature from the primary magma's MgO content (Herzberg and Asimow, 2008). It cannot be applied to lavas that have also fractionated clinopyroxene, which usually happens once the liquid has evolved below 7–10 wt% MgO, and applying olivine addition to such lavas yields primary magmas with too much MgO and source temperatures that are too high (Herzberg and Asimow, 2008). PRIMARSMELT, the martian

member of the same family, has the same restriction: liquids that have fractionated olivine, clinopyroxene and plagioclase cannot be easily inverted by this method without a significant increase in computational cost and uncertainty (Hernández-Montenegro et al., 2024). This restriction effectively excludes every lava that has evolved beyond olivine-only fractionation.

Multiple saturation is also ambiguous evidence of origin. Asimow and Longhi (2004) showed that aggregate liquids from polybaric near-fractional melting can display near-intersections of liquidus surfaces that cannot be distinguished experimentally from true multiple saturation points, so a multiply saturated liquid need not be a single batch that equilibrated with its residue. They did show, however, that such multiple saturation points preserve information about the average pressure at which the constituent increments of liquid were extracted (Asimow and Longhi, 2004). Lavas with complex melting and differentiation history likely encode some of this information in their geochemistry, but attempting to recover it requires a fast and flexible forward model that properly treats melting and multi-phase fractionation together. A pHMELTS emulator would provide such a forward model. Whether the inversion is well posed for a given suite of lavas, given the trade-offs among potential temperature, source fertility and water content, remains to be tested.

4.5 Discussion

Strengths

The central result of this chapter is that a single emulator per parent model can reproduce that model's phase equilibria across a broad range of composition, pressure, and temperature conditions and needn't be restricted to a single compositional path or magma series. The *nGibbs* HeFESTo Earth emulator follows basaltic, peridotitic and Si-enriched compositions from the surface to 140 GPa, and the MELTS emulators follow compositions from lherzolite to rhyolite, with the same network in each case (§4.3). The surrogates reviewed in §4.1 were each trained on a narrower slice: arc magmas with fixed water content (Candioti et al., 2026), or compositions near the basalt–harzburgite join (Hartmeier and Lanari, 2026; Q. Yuan et al., 2025).

The emulators predict the same quantities as their parent models, the abundance of every phase and the composition of every solution phase, so physical properties are computed with the parent model's own equations of state rather

than learned as separate targets. These differences are far smaller than the uncertainty of the thermodynamic databases themselves (Ghiorso and Sack, 1995; Ghiorso et al., 2002; Stixrude and Lithgow-Bertelloni, 2005; Riel et al., 2022; Holland and Powell, 1998). For most applications, therefore, the choice of database matters more than the choice between the emulator and its parent.

As in the parent models, mass balance is enforced exactly rather than encouraged. This is a requirement for reactive transport and for any calculation that removes or adds material step by step, such as fractional crystallization, where a small bias per step would accumulate. Hartmeier and Lanari (2026) penalize mass-balance violations during training and report small residual deviations, while the surrogate of Candioti et al. (2026) closes mass balance only for the bulk solid, by difference.

The emulators are fast without a GPU. On a 4-core laptop CPU, the HeFESTo Earth emulator is four orders of magnitude faster than HeFESTo on the same machine, and a GPU adds a further factor of 3–8 (§4.3). Because *nGibbs* reaches its largest CPU speed-up on a 4-core laptop, it is usable on ordinary workstations. Through PetThermoTools, it is also usable without learning a new interface (§4.4).

Limitations

An emulator can be no more accurate than its parent model. The MELTS emulators inherit the known limitations of MELTS, including its calibration toward low-pressure, nearly anhydrous experiments (Ghiorso and Sack, 1995) and its poor treatment of hydrous minerals such as amphibole (Candioti et al., 2026). The HeFESTo emulators inherit a database that has no silicate liquid. The emulators are also reliable only inside their training coverage (Figs. 4.5–4.8). Compositions far from that coverage, such as banded iron formation or carbonatite, need targeted training data and validation before use.

Three systematic weaknesses appear in the results. First, phases that are rare and stable over narrow ranges are predicted less reliably (§4.3). Akimotoite is the most consequential of these for mantle applications, because its stability near the 660-km discontinuity is one of the composition-sensitive reactions that a transition-zone inversion would aim to resolve (Hartmeier and Lanari, 2026); oversampling its stability field is a priority before that application. Second, the emulator smooths sharp features in the parent model, such as the MORB

solidus in MELTS, over a few tens of degrees. This follows from the continuous architecture (§4.2), which is correct for continuous phase-out reactions but blurs changes that are nearly discontinuous, as at this solidus. Third, properties that depend on derivatives of the phase proportions (C_p, K_S and α) and liquid compositions at small melt fractions carry larger errors than phase proportions do (§4.3 and 4.3) and require higher fidelity predictions. Applications that rely on these quantities, such as latent-heat effects at phase transitions or the composition of the last liquids, should check them against the parent model.

The speed advantage also has limits. Fixed overheads make the emulators no faster than the parent model for very small batches, and the advantage over alphaMELTS on a multicore machine is about one to two orders of magnitude, not four, especially once properties are computed through PetThermoTools (§4.3). The emulators pay off most when many states are evaluated together, as in the inversions and coupled models of §4.4.

Finally, the release is deliberately partial. Only the closed-system isothermal MELTS emulators and the HeFESTo emulators are released; the open-system MELTS emulators await further testing. Emulator error is not yet propagated into downstream calculations, which an inversion would need in its likelihood. Planned work includes a pHMELTS emulator (§4.4), targeted training for rare phases and unusual compositions, and a systematic map of the depth, width and velocity contrast of the transition-zone discontinuities as functions of composition.

4.6 Conclusions

This chapter introduced *nGibbs*, a set of deep-learning emulators of the MELTS 1.0.2 and 1.2.0 and HeFESTo thermodynamic models. Its goal is to stand in for its parent models in essentially any calculation they would be used for, rather than for one composition range or one set of outputs. Four design choices support that goal:

1. Each emulator predicts the full equilibrium assemblage, so thermodynamically consistent physical properties follow from the parent model's own equations of state.

2. Phase abundances are predicted as rectified affinities, which keeps them continuous across phase boundaries like the parent model they emulate.

3. Mass is conserved exactly at inference by a constrained correction that itself is constrained to produce legal (non-negative) atomic abundances.

4. The training compositions are drawn from whole-rock databases of igneous rocks that span the breadth of compositional diversity of igneous rock compositions known on Earth. MELTS 1.2.0 is additionally trained on a database of sedimentary rock compositions, and HeFESTo models are strongly perturbed to cover compositions that are rare or unknown at Earth's surface. The validation data are drawn from distinct bulk compositions to ensure that the emulators generalize to new compositions.

The released emulators reproduce their parent models closely within their training coverage. The major silicate and oxide phases are predicted with precision and recall of 0.93 or better. Element abundances are conserved to a median relative error of about $5 \times 10^{-6}\%$. Phase proportions, densities and seismic velocities along reference paths are nearly indistinguishable from those of alphaMELTS and HeFESTo. Subtle behaviors observed in liquid lines of descent are learned and reproduced by the emulators. Errors are largest for rare phases with narrow stability fields, such as akimotoite and the native iron phases. They are also larger where sharp changes occur in the parent model, for material properties such as heat capacity, bulk modulus and thermal expansivity that depend on molar derivatives; and for liquid compositions at small melt fractions. These differences remain smaller than the uncertainties of the underlying thermodynamic databases for most applications. They do, however, mark where targeted training data are needed before specific applications, such as resolving akimotoite stability near the 660-km discontinuity.

The emulators are fast enough to change what is computationally practical. On a 4-core laptop CPU, the HeFESTo Earth emulator evaluates 2^{15} assemblages about 22,000 times faster than HeFESTo, and about 1,100 times faster with the full equations of state. A set of 201 whole-mantle profiles of phase equilibria, density, and sound velocity, comprising about 35,000 assemblages, takes under 5 s on a GPU, compared with about 10 hours for HeFESTo. The MELTS emulators run two orders of magnitude faster than alphaMELTS and, through PetThermoTools, can replace it in existing workflows by changing a single keyword.

Several applications follow directly from this speed and generality. In seismic

inversions of the Earth's and Mars' mantles, bulk composition can be sampled alongside the thermal parameters rather than fixed in advance. Composition-dependent transition-zone structure and candidate compositions for ultralow-velocity zones can be explored systematically. Geodynamic models can carry a chemically heterogeneous mantle, or the cumulates of a solidifying magma ocean, without reducing composition to a single mixing variable. Planned extensions include an emulator of pHMELTS for inverting mid-ocean ridge basalts for the thermochemical state of the mantle beneath spreading centers, and spatially resolved models of reactive porous flow in crystal mushes.

Code and Data Availability

The *nGibbs* source code is available at `https://github.com/benjthyer/nGibbs` and can be installed with `pip install ngibbs` (§4.4). The release used in this chapter (v0.1.0b4) is archived on Zenodo at `https://doi.org/10.5281/zenodo.21345830` (Thyer, 2026). Subsets (n = 15,000) of the validation datasets are shipped with every PyPI installation of nGibbs. Subsets of the training datasets are available **[TODO: on zenodo]** and the full training datasets are available on request from the author.

Chapter 5

SYNMAGMATIC DEFORMATION AND EXTENSIVE CRUSTAL CONTAMINATION OF THE PLATINIFEROUS UPPER CRITICAL ZONE AT THE MARGIN OF THE BUSHVELD COMPLEX: A FIELD STUDY IN DRAAIKRAAL, MPUMALANGA, SOUTH AFRICA

The Rustenburg Layered Series of the Bushveld Complex in the Mpumalanga, Limpopo, and Gauteng provinces of South Africa is amongst the world's best examples of a large, layered mafic-ultramafic intrusion and has been the focus of decades of research, yet the tempo and the fundamental nature of its formation remain open questions. Field research at key localities in the Bushveld Complex has provided invaluable insights into the emplacement and evolution of these layered intrusions, yet the intrusion is vast; many areas with good exposure remain poorly studied. Draaikraal is an agricultural town located in the eastern limb of the Bushveld Complex in the Mpumalanga province that sits at the platinum-rich intersection of the Critical and Main Zones with the intrusion's margin, presenting a valuable opportunity to study the formation geometry and timing of large mafic intrusions in general, with particular emphasis on its famous Platinum Group Element (PGE) mineralization.

Draaikraal is nested within a conspicuously recessed valley capped by ironstone laterites and hosts several unusual topographic and structural features, including the Kliprivier Synform and the Draaikraal Lineament (named here). The Kliprivier Synform represents a package of modestly rotated, locally pegmatitic plutonic rocks with conspicuously absent modal layering, and country rocks that include fine-grained, melanocratic Marginal Zone sills intercalated with migmatized quartzite and calc-silicate hornfels. The country rocks appear to be displaced over 1 km into the plutonic RLS near the mapped intersection of the Critical and Main Zones. The Draaikraal Lineament is a linear feature of recessive topography stretching from

the country rocks beyond the intrusion's margin to the middle Main Zone. It overlaps with the Kliprivier Synform and separates typical, resistant Main Zone gabbronorites to the north from the recessive Main Zone gabbronorites to the south in the agricultural valley hosting Draaikraal. Using structural measurements and correlation of mottled anorthosite marker beds of the lower Main Zone across the Draaikraal Lineament, I argue that the mottled anorthosite marker beds are not displaced across the Draaikraal Lineament as previous studies have suggested. I show that the Kliprivier Synform was deformed synmagmatically by Bushveld magmatism, and argue that it represents a monocline that formed due to laccolithic deformation of the country rock roof by the intrusion of middle Critical Zone sills below the level of the UG2 chromitite and overlying Merensky Reef and Main Zones, which appear to be undisturbed. The orientation of the Draaikraal Lineament is very similar to the surface-projection of the axial plane of the Kliprivier Synform, which likely continues at depth. The relative recessive topography of the Draaikraal Lineament and the Main Zone to the south is argued to be the result of higher degrees of contamination by hydrothermal fluids than the more resistant Main Zone gabbronorites of the surrounding area. The rocks south of the Draaikraal Lineament are envisaged to be fluxed with upwardly migrating fluids derived from devolatilization of underlying, relatively shallow sediments at the floor of the magma chamber whereas the resistant Main Zone to the north was underlain by thick Critical Zone rocks and thus was less contaminated by crustal fluids.

Author contributions. This chapter is being prepared for publication with co-author J. Roberts (University of Pretoria). **B.J.T.:** conceptualization, investigation (field mapping, sampling, and photography at Draaikraal), formal analysis, visualization, writing – original draft. **J.R.:** conceptualization, resources (hosting at the University of Pretoria and field logistics), investigation (field work), writing – review and editing. Generative AI was used as described in the front matter.

5.1 Introduction

The Bushveld Complex of South Africa in the Mpumalanga, Limpopo, and Gauteng provinces hosts the largest layered mafic intrusion in the world (the Rustenburg Layered Series, or RLS), which in turn bears the world's largest source of platinum group elements (C. A. Lee, 1996) as well as important reserves of chromium, vanadium, titanium, and iron (Cawthorn, 2015). It is exceptionally well-preserved, making it an ideal natural laboratory for studying the geochemistry and petrology of layered mafic intrusions. Even so, the order, tempo, and nature of the formation of the Bushveld Complex remain hotly debated (W. Kruger and Latypov, 2020; Mungall et al., 2016; Scoates et al., 2021). Classically, the RLS was thought to have been emplaced as a cohesive, long-lived magma chamber (F. J. Kruger, 1994; Cawthorn, 2015), but recently many have hypothesized that its formation was more complex, with incremental assembly through sill-like intrusions (A. A. Mitchell et al., 2019; Yao et al., 2021; Scoon and A. A. Mitchell, 2023; Bourdeau et al., 2023). The RLS is divided into the Marginal, Lower, Critical, Main, and Upper zones (von Gruenewaldt et al., 1985), which are often hypothesized to have formed in distinct events. Thus, the intersections of these stratigraphically defined zones at the intrusion's margin are particularly informative locations for detailed field work that seeks to unravel the emplacement history of the RLS. This field study focuses on several unusual but understudied structural and geomorphic features occurring at the economically significant intersection of the Main and Critical Zones with the intrusion's margin in the well-exposed Eastern Limb near the village of Draaikraal, Mpumalanga, South Africa: the Kliprivier Synform, the Draaikraal Lineament, and the unexpected recessive topography of the Main Zone in the agricultural valley surrounding the village of Draaikraal (van der Merwe, 2007; Molyneux, 2008).

5.2 Geologic Setting of the Draaikraal Area

A broad survey of the literature on the Bushveld Complex is presented in Appendix B. Here, the geologic setting in the vicinity of Draaikraal is described.

Previous work near Kliprivier and Draaikraal

Kliprivier was the focus of several early works focusing on the margin of the RLS to better understand the relationship between the RLS and the sedimentary country rock. van Biljon (1949) studied the hornfelsized carbonates

of the area and their relationship to the margin, observing an increase in metamorphic grade to wollastonite skarns near the margin of the RLS and reporting that the transgressive RLS mafic rocks are continuous along strike with the metasedimentary rocks, a finding that I also report here (see §B.5, "The Transformation Hypothesis" for elaboration). He used this and similar pieces of field and geochemical evidence to develop his provocative but unpopular Transformation Hypothesis: claiming that the RLS (and potentially other igneous rocks) were formed from in-situ magmatic-metasomatic reaction of sediment (see critique and discussion within the comparatively well-cited work of Cousins, 1959).

Kliprivier and the surrounding De Berg region were studied in detail by Frick (1967), who distinguished different facies of marginal sills, documented variable assimilation of country rock by marginal sills and the presence of Critical Zone and two chromitite seams in the area, and characterized contact metamorphism within the sedimentary rocks. Frick (1967) emphasized the discordance of the contact between the sediments and the RLS and reported that chromite seams are disaggregated at the contact and not present in the Marginal Zone. Button (1976) reports that the extent of the metamorphic thermal aureole decreases southward in the area as the RLS package thins.

In their influential paper geochemically categorizing the Marginal Zone sills, Harmer and Sharpe (1985) included a broad map of the Marginal Zone along the Bushveld's Eastern limb in addition to a detailed map of the marginal rocks at the boundary of the Main and Critical Zones at the Kliprivier Synform, described more in the "Structural Observations" section. These maps show that the field area at Draaikraal includes two distinct classes of Marginal Zone sills. The large map shows "B2" gabbroic sills to the north abutting the Critical Zone and "B3" gabbroic sills to the south abutting the Main Zone (although later works argue that the upper Critical Zone continues south within and beyond the field area: van der Merwe, 2007; Latypov et al., 2022). Furthermore, fine gabbros of the "B2" and "B3" lineages are also subdivided into "B2N" and "B3N". Because both "B2N" and "B3N" sills are magnetic and visually identical in outcrop they are both described here with the lithologic code "Rz", for fine-grained Marginal Zone. Their coarse-grained "B2" gabbros may (in part) correspond to the Disturbed Gabbronorite (Dcg) lithology mapped in this work.

van der Merwe (2007) mapped and described the geology on Kliprivier and several nearby farms along the eastern lobe of the RLS to assess the area's PGE resource potential. He first described the Kliprivier "Syncline", further detailed in the structural observations, unit descriptions, and discussion in this work. Within, he references several private studies which were contracted by mining companies that are not publicly available. Van der Merwe reported a core log from the area that is helpful to interpret field data.

Latypov et al. (2022) did a systematic but coarse field study of seven separate traverses across ~60 kilometers of the lower Main Zone in order to correlate the mottled anorthosites of the lower Main Zone, arguing that these correlated units, supposedly laterally continuous across a 4 km change in depth of the RLS floor, imply the existence of a kilometer-scale melt column. They argue their correlation requires a voluminous ~10^5 km^3 magma chamber during the emplacement of the Main Zone.

Setera and VanTongeren (2018) conducted a systematic study of the Main Zone in the Draaikraal area, reporting major and trace elements of major cumulus phases. They focused on characterizing magma recharge at the level of the Pyroxenite Marker in the upper Main Zone, but they also used chromium concentrations in pyroxene to identify a less significant magma influx at the level above the mottled anorthosites. This geochemical cross-section was compared with a similar cross-section to the north performed in an earlier work by VanTongeren and Mathez (2013), which notably featured more primitive Main Zone rocks than those observed near Draaikraal. Setera and VanTongeren (2018) argue that this southerly evolving compositional gradient within the Main Zone arose from evolution of migrating melts away from the feeder of the Bushveld, interpreted to be at the Thabazimbi-Murchison Lineament.

Topographic 'Inversion' and the Draaikraal Lineament

The agricultural village of Draaikraal in the Mpumalanga Province of South Africa sits in a local sedimentary basin. Geologically, it is situated within the lower Main Zone of the RLS. The Main Zone is known to be highly resistant to erosion and dominates surface exposure of the RLS in the comparatively poorly exposed North and West Limbs of the RLS. This is thought to be because of the common observation of inversion of small pigeonite grains into larger, continuous OPX grains, serving as a sort of resistant "cement", contributing to

the effectiveness of Main Zone rocks as dimension stone commonly used as a building material (Cawthorn, 2015). Figure 5.1a shows that this considerable change in topography at Draaikraal from the northern mountains to the southern valley occurs despite the continuity of the Main Zone through the lineament.

5.3 Methods

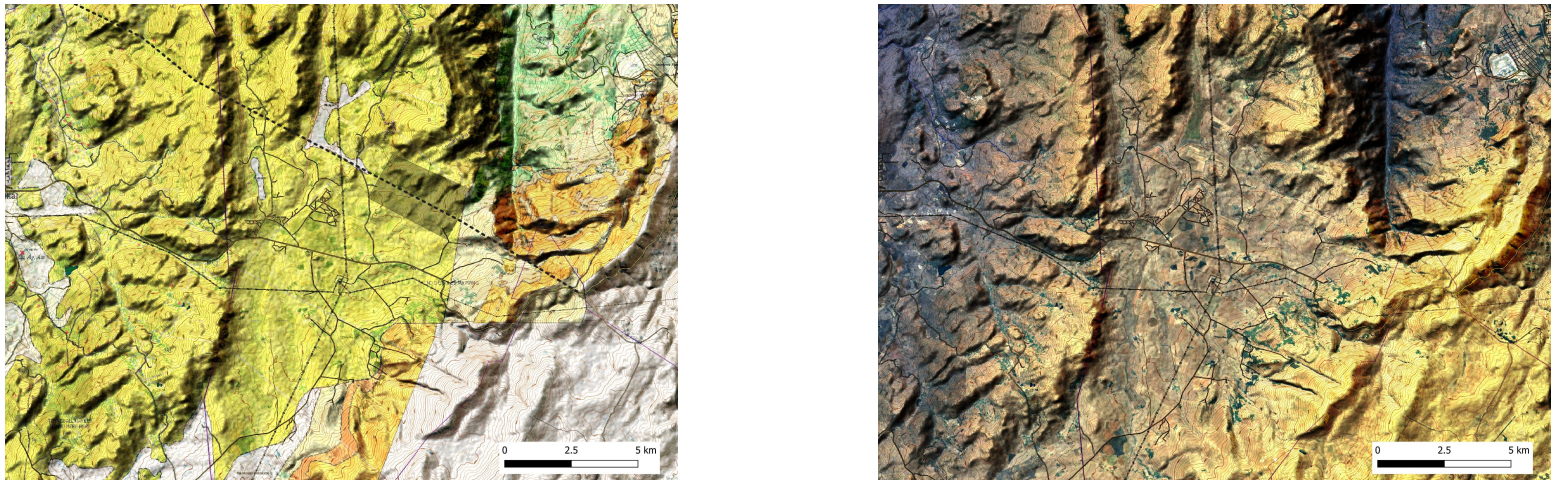
Field Methods

Field observations were made during an 11-day field campaign in July 2023. Field investigation and sampling strategies were adapted as the study proceeded, but initially were focused on correlating distinct rock units across the Draaikraal Lineament and taking structural measurements on each side to characterize it. Geolocated samples were taken across the field area for later geochemical work. The observation that plutonic igneous rocks to the north of the Kliprivier Synform were highly unusual and distinct from those to the south prompted a campaign to more thoroughly characterize its lithologic context and structure. Field observations, samples, and structural measurements were all geolocated and are recorded in Table A.1 (Appendix A). Almost every entry included an interpreted lithological unit to inform the final geologic map of the area (Figure 5.3). Many entries included photographs, which included mostly close-up photos of outcrops, but also some photos of the surrounding landscape to illustrate the area's geomorphology. Field photographs are cited throughout this chapter by number (e.g., photo 95); the photographs that are reproduced in the photo album of Appendix A (§A.2) are linked from their numbers, and the rest are in the supplementary photo album. **[TODO: Link to photo supplement here.]**

Mapping Criteria

The contact between the layered series (Critical and Main Zones) and the Marginal Zone is often gradational, requiring a somewhat arbitrary categorization. Since the Lower Main and Upper Critical Zones are non-magnetic and the finer-grained marginal zone rocks are magnetic, I chose field-detectable magnetism as the simple discriminant to distinguish marginal rocks from the massive Main and Critical Zones of the RLS.

When Layered Gabbronorites (Lg) are observed in the same outcrops as Mottled or Spotted Anorthosites, the locality is assigned to be Mottled or Spotted



(a) (b)

Figure 5.1: (a) The field area around Draaikraal with the geological mapping of Molyneux (2008). Due North is up on this map. The generally resistant, topographically high Steenkampsberg Quartzite and hornfels are mapped as orange units on the eastern margin. The Marginal Zone is mapped as the beige color between the quartzite and the RLS. The Main Zone is featured in yellow, the Critical Zone in the NE quadrant is green and exhibits strikingly recessive topography. A regional SW-NE trending fabric is prominent, which agrees with all structural measurements of rock cleavage taken in the lower Main Zone and upper Critical Zone aside from the rotated units associated with the Kliprivier Synform. The dark box outlines an area where SW-NE rock fabrics appear continuous across the Draaikraal Lineament (dashed line). The Main Zone around Draaikraal is unusually recessive relative to the Main Zone north of the Draaikraal Lineament, or to the west. The compilation of Molyneux (2008) did not consider more recent mapping that revealed the presence of the Upper Critical Zone to the south, terminating beyond the map area (van der Merwe, 2007). (b) The satellite view of the same area as (a), without geology and visual guides for clarity. The Draaikraal Lineament remains clearly visible. Elevations are colormapped such that relative lows are blue and highs are yellow. The Main Zone is generally at higher elevation than the Critical Zone (in the northeast quadrant) and the Upper Zone (to the west), while the quartzitic Steenkampsberg Mountains in the southeast quadrant are locally the highest feature.

Anorthosite.

Structural Measurements

All measurements of plutonic rocks reported here are orientations of near-planar rock cleavage (§5.4). Strike and dip measurements taken from the Kliprivier Synform were plotted on a stereonet and analyzed using Stereonet 11 (Allmendinger et al., 2012; Cardozo and Allmendinger, 2013). Poles to bedding planes of quartzites, hornfels, mafic sills, and plutonic rocks were calculated and plotted together in stereonet projection in Figure 5.7a. In Figure 5.7b, the fold axis is calculated using the poles to the bedding planes. The axial plane also was calculated, assuming it bisected the obtuse angle formed by the limbs. The obtuse fold angle bisector for the axial plane was selected because the rotation within the fold observed is quite gentle. An acute bisector for the axial plane would have been better supported if field evidence showed dramatic changes in dip along the fold, requiring a complete structural overturn that locally annihilated the original horizontality of the deformed rocks in addition to steeply dipping rocks intermediate to the two limbs.

Stratigraphic Column Analytical Methods

The field data of this study north of the Kliprivier Synform are projected onto a stratigraphic column following the assumption that all rocks in the field area north of the Kliprivier Synform follow Steno's Law of Original Horizontality and were only uniformly deformed by tilting due to isostatic flexure towards the center of the RLS after emplacement, so that all regional bedding and structures are currently dipping 15 degrees to the northwest relative to their original orientations. Geolocated field data are projected onto a line to the north of the Kliprivier Synform designed to be orthogonal to the regional strike (trending 301.79 degrees: W-NW) using the 'Snap Geometries to Layer' function in the 'Vector Geometry' Processing Toolbox in QGIS, with a tolerance of 1 degree set to "prefer closest point". Section heights then were calculated for these projected points along the reference line to an arbitrary point along this line, to get relative stratigraphic heights using the "Distance Matrix" function in the "Vector Analysis" Processing Toolbox. A correction was added to these lateral distances to account for elevation, calculated as the product of the elevation differential and the tangent of the regional dip, assumed to be 15°. The column heights are normalized to the height of the Merensky Reef. Because

the Merensky Reef was not observed in the field area, this was instead done by proxy: stratigraphic heights were calculated relative to the UG2 chromitite observed in the field area (§5.4) at a fixed height of -86 meters. This chromitite is mapped by van der Merwe (2007) as the UG2, who profiled drill cores and determined the Merensky Reef to be 86 meters above the UG2 in the field area (van der Merwe, 2007). Note that, for simplicity, it is assumed that there are no local changes in strike and dip, even though the dip of the Main Zone likely shallows and rotates to be roughly due east further from the margin of the intrusion. This approximation is likely to be acceptable locally for the upper Critical and lower Main zones, but becomes less accurate further from the margin. Therefore, the stratigraphic heights presented here for the Homogeneous Gabbronorite lithology far upsection are likely overestimated. The interpreted schematic stratigraphic column for the region is graphically presented in Figure 5.2.

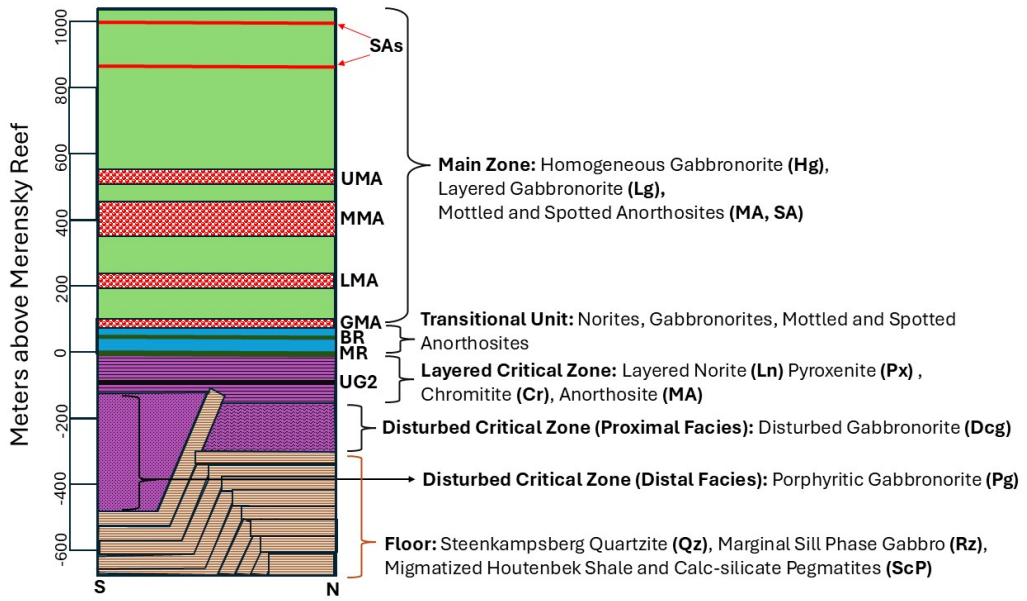


Figure 5.2: The schematic stratigraphic cross section of the RLS in Draaikraal. The Kliprivier Synform is represented by the slab of floor rocks protruding into (and creating) the Disturbed Critical Zone at the halfway point of the column. The Draaikraal Lineament would form from upward percolating fluids trapped and channeled by this protruding slab of floor rocks in a line directly above the slab in this view. The Kliprivier Synform is not interpreted to have deformed the RLS above the UG2.

5.4 Results: Field Observations

Observations are organized into three sections: lithologic descriptions, rock cleavage and modal layering, and structural observations; the first and last each have their own map.

Lithologic Descriptions

Lithologic descriptions are organized by the interpreted map units (see Figure 5.3) and their constituent lithologies. Three mapping units are described in turn, each with a wide variety of lithologies: the marginal rocks (the floor and walls of the intrusion, including sedimentary country rock and concordant pre- and syn-Bushveld magmatic sills); the plutonic rocks of the RLS; and secondary, post-Bushveld rocks that were largely ancillary to the field study.

Margin

Much of this study focused on the eastern margin of the eastern limb of the Rustenburg Layered Series (RLS) east of Draaikraal. The margin displays the most unambiguous bedding planes of the study area, generally dipping 25° to the northwest towards the center of the Bushveld Complex, except for the Kliprivier Synform, which is detailed below in the "Structural Observations" section. The marginal rocks within the field area are composed of gabbroic intrusive igneous sills concordantly intercalated with variably metamorphosed, melted, or otherwise devolatilized Transvaal sediments of the Steenkampsberg and Houtenbek formations. Occasionally, melting and/or reactive porous flow of melts through sedimentary layers remineralizes sedimentary layers with igneous mineralogy and texture (photos 35, 51, and 52). Concordant sills of marginal zone magmas and packages of sediment range in thickness from 10 to 100s of meters, and can be found mutually interbedded at least a kilometer from the margin of the RLS. Here, the marginal rocks are divided into three units: "Rz", the fine-grained Marginal Zone gabbroic sills/magmas; "Qz", relatively undisturbed Houtenbek quartzite; and "ScP", hornfelsed metapelites, marbles, and quartzites with related igneous/hydrothermal pegmatites.

Rz – Fine-grained gabbro

The Marginal Zone gabbroic sills are commonly fine grained (<1 mm), but locally are coarser with some grains up to 2 mm in the groundmass. Unlike

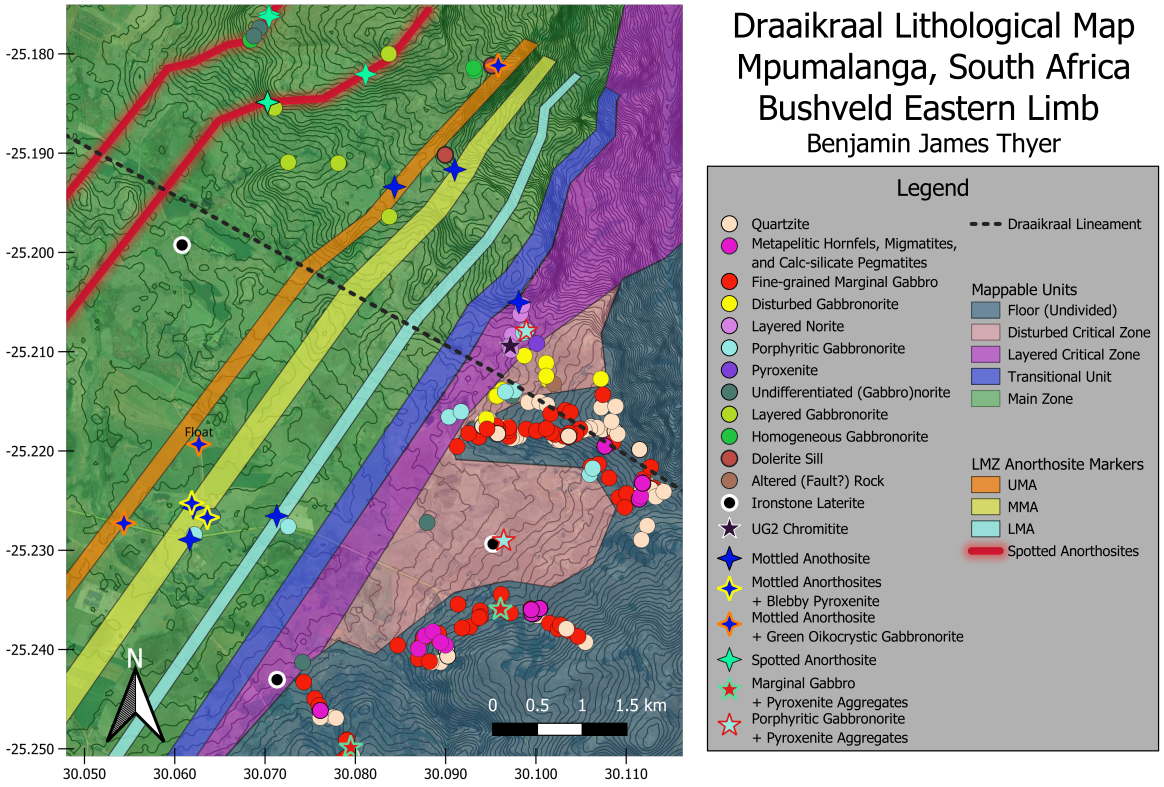


Figure 5.3: Lithological map of the Draaikraal field area, eastern limb of the Bushveld Complex. Field observations of outcrop are colored by lithology over the mappable units (floor, Disturbed Critical Zone, Layered Critical Zone, Transitional Unit, Main Zone). The lower Main Zone anorthosite marker beds (LMA, MMA, UMA, and the spotted anorthosites) are traced in color, and the Draaikraal Lineament is dashed. Field observation data are listed in Table A.1.

much of the plutonic RLS, the Marginal Zone is not visibly modally layered. Marginal Zone magmas within a meter of Steenkampsberg Quartzite often contain conspicuous evidence of silicic contamination: although they retain an igneous texture, the color significantly lightens and the rocks become much harder and less magnetic approaching the contacts with Steenkampsberg Quartzite. Aside from these immediate contacts, the Marginal Zone gabbro is always magnetic. The Marginal Zone contains more pyroxene (distinguishing clinopyroxene and orthopyroxene is difficult with such small crystals) than plagioclase with variable magnetite, sometimes up to 10%, and minor phlogopite. The heavy, fine fraction of mineral separates from one sample was briefly studied under optical microscope to identify trace phases, which include apatite, zircon, sulfides, and euhedral chromite.

The Marginal Zone magmas contain igneous xenoliths and are rich in paleo-vertical, across-strike features that likely record rising, buoyant fluid/magma. Perfect planar resistance banding (photo 95) is the most common structure in the fine grained Marginal Zone and is nearly ubiquitous within the unit. Though the resistance bands have seemingly random strikes, they most often are steeply dipping and may represent fluid conduits along brittle fractures within the Marginal Zone.

In previous works, isolated meta-sedimentary xenoliths (Buick et al., 2000) were emphasized, but in this study, meta-sedimentary rocks were never found as minor xenoliths within magnetic marginal sills, but rather as continuous, concordant strata (photos 42, 70, 71) or as locally abundant disaggregated broken strata cemented within a pink, evolved diatexite (photo 100), both designated as a distinct unit: ScP.

The magnetic sills of the Marginal Zone (Rz) did include enclaves of coarse-grained leucogabbronorite, mottled anorthosite or pyroxenite (photos 2,4,6,96). The anorthosite-rich enclaves are rounded, 6-10 cm along their longest dimension, and are imbricated to be preferentially lying flat paleohorizontally within the Marginal Zone, suggesting they were weathered/eroded by reaction and/or mechanical transport through Marginal Zone magma. Pyroxenitic enclaves are also often spherical (photos 4, 96) and may be related to the formation of the pyroxenite "Tennis Ball Marker" (TBM) of the Main and Critical Zones, described further in the discussion. These plagioclase-rich igneous xenoliths contain large (<3 cm) optically continuous oikocrystic augite that enclose

blocky plagioclase (<5 mm).

More commonly observed within the magnetic Marginal Zone sills east of Draaikraal are paleovertical structures that suggest buoyant upwelling of fluids and melts sourced from the hornfelsized sedimentary rocks. These structures included sheared, viscously deformed lenses of thin pegmatites, including anorthosite (photo 6), magnetite gabbros (photo 60, sheared gabbro ~15 cm; Figure 5.8), and evolved quartz-biotite pegmatites (photo 67: 1×3 cm). In addition to homogeneous pegmatites, the Marginal Zone gabbros host paleovertical invasions of anastomosing anorthosite derived from anatexis of underlying metasedimentary rocks (photos 6, 97). These leucosome intrusions occasionally carry blocks of meta-sedimentary paleosome or brecciate and enclose the Marginal Zone gabbronoritic host rock itself (photo 98).

Near the contact with the migmatite, differential weathering sometimes reveals proud, thin, pinched ovate paleovertical structures (5×1 cm scale) on weathered faces of Marginal Zone norite that are otherwise invisible on a fresh rock face [photo 40].

Fine-grained Marginal Zone gabbros are mapped up to 120 meters below the Merensky Reef stratigraphically.

Qz – Houtenbek Quartzite

The Houtenbek quartzite dips concordantly with marginal zone sills and migmatized sedimentary rocks towards the center of the Bushveld, here dipping about 25° to the northwest. Even within meters of a contact with Marginal Zone sills (Rz), the quartzite often retains a clastic sedimentary texture and possibly could be considered a proper clast-dominant quartzose sandstone with well-sorted and well-rounded coarse quartz grains ~1 mm in diameter with no other mineral visible. While there is little structure visible internally because of the purity of the quartzite, the bedding planes occasionally reveal conspicuous asymmetric ripple structures [photos 2, 68] <1 cm in height. One in-place ripple structure suggests a paleoflow direction towards the south-southwest. Some quartzites are hornfelsized: recrystallized into an interlocking crystalline texture with large, irregular, angular, coarse grains <2 cm across. Many quartzites with preserved clastic textures near the margin of the RLS are stained bright red within rocks' unweathered interior. This staining is hypothesized to be derived from mobilized Fe-rich fluids percolating along grain boundaries, probably

caused by heat and/or fluid circulation related to the emplacement of the RLS. Steenkampsberg Quartzite is mapped up to 140 meters below the Merensky Reef stratigraphically.

ScP – Houtenbek Hornfels, Migmatites, Pegmatites

In addition to quartzite, the Houtenbek formation contains carbonate and pelitic horizons (Button, 1976). At the margin east of Draaikraal, there is a 60 m thick package of melted and metamorphosed sediment containing stratiform pegmatites with variable mineralogy. Hornfelsized sediments come in two main varieties near Draaikraal: pink, diatexite-supported breccias [photo 100] or dark, high-grade schists [photo 29] with thin but conspicuous navy, gray, white, tan, and pink bands of relatively mafic and felsic minerals that have been recrystallized and no longer bear clastic textures except for thin seams of clast-dominated quartzite in a low-fraction glassy biotite-rich matrix [photo 35, large light seam is ~4 cm thick]. This unit is rarely magnetic and most often is visible as continuous, concordant strata [photo 42]. Even when highly brecciated, this unit can be followed along the strike of the quartzites and Marginal Zone, implying that the sediments were not incorporated into Marginal Magmas, but that they rather were melted and devolatilized in-situ due to heating from the emplacement of adjacent marginal sills. Fluids and melts produced in this process are visible as a wide variety of stratiform pegmatites and skarns with undulating erosive contacts within comparatively planar meta-sediments [photo 30: feldspar/wollastonite(?) + amphibole skarn; photo 69: magnesite, quartz pegmatite with minor siderite] and as pockets following fluid flow pipes and void spaces that are accommodated by the host strata through ductile deformation [photo 37, 38: Orthopyroxene and quartz pegmatite]. Hornfels in the field area are mapped up to 190 meters below the Merensky Reef stratigraphically.

The Rustenburg Layered Series (RLS)

The Rustenburg Layered Series is the world's largest individual igneous intrusion (Cawthorn, 2015; Latypov et al., 2022) and is subdivided into the Lower Zone, Critical Zone, Main Zone, and Upper Zone (§B.2). In this study, RLS rocks of the lower Main Zone, Transitional Zone, and upper Critical Zone are mapped. The rocks of the Marginal Zone (Rz) are not layered, nor considered here

to be a part of the RLS on a textural basis. Marginal zone rocks weather distinctly from the plutonic RLS phase with abundant, sharp, ~3 cm-scale potholes and generally are more textured and rough [photo 36] on a weathered surface than RLS rocks, which weather spheroidally and proudly [photos 8, 11: "porphyritic" gabbronorite]. Xenolith blocks of Marginal Zone norites bearing their distinctive resistance banding are found in both the Critical and the Main Zones [Critical Zone: photo 76; Main Zone: 91], though each xenolith sighting within the Main Zone is proximal to a mottled or spotted anorthosite layer. These xenoliths are variably annealed, recrystallized, and even ductily deformed [Critical Zone: photo 77; Main Zone: 84].

The RLS is famous for its distinctive and numerous laterally continuous stratiform marker horizons, sometimes stretching on a 100-km scale (Cawthorn, 2015), which include the Critical Zone chromitite cycles, Merensky and Bastard pegmatoid reefs, three mottled anorthosite packages in the lower Main Zone, the Pyroxenite Marker of the upper Main Zone, and the magnetitite cycles of the Upper Zone. In this study, I observe and describe the three lower Main Zone magnetic mottled anorthosites [photos 49, 53, 86, 93, 105, 106, 108, 111], the magnetic chromitite UG2 [photos 65, 66], and a non-magnetic upper Critical Zone mottled anorthosite that is probably a part of the Merensky Reef footwall [photo 81], and several packages of xenolith-bearing modally layered gabbronorites and spotted anorthosites within the Main Zone (photos 91 and 111). These laterally continuous reefs are mapped and described separately from the bulk cumulate rocks of the RLS.

Cz: Critical Zone

The Critical Zone was previously mapped to pinch out due east of Draaikraal (shown in the compilation of Molyneux, 2008); however, many now recognize that a thin layer of the upper Critical Zone persists much further south towards Tontledoos (van der Merwe, 2007). In this study, diagnostic Critical Zone lithologies were only observed north of the Kliprivier Synform, where typical finely modally layered norites and anorthosites are well-exposed (photos 64 and 80), as well as the UG2 chromitite. The Critical Zone above the Kliprivier Synform dips ~15° to the northwest similarly to the majority of the Main and Marginal Zone rocks in the field area, but gradually rotates to a southwestern dip as it approaches the contact with the rotated block of metasediments and

gabbronorite sills of the Kliprivier Synform. It is essential to note that some rotated Critical Zone rocks are structurally beneath the large, rotated package of intruded metasediments (Rz + Qz + ScP) in the area of the Kliprivier Synform, meaning that this package of country rock is not only serving as the floor for RLS rocks to the west, but is also (at least locally) a roof for Critical Zone rocks below the level of the UG2.

Dcg: Disturbed Gabbronorite This lithology is restricted to the immediate area of the Kliprivier Synform. Disturbed Gabbronorites are coarse-grained, even pegmatoidal (photo 62), with feldspars and pyroxenes up to 3 cm long. This unit was named in response to van der Merwe (2007), who said that the "structural change [of the Kliprivier Synform] is reflected in the Bushveld rocks, which become highly disturbed and fragmented". Disturbed Gabbronorites are most recognizable for their abundance of blobby autoliths of anorthosite (photo 61) and pyroxenite (photos 70 and 72). While it exhibits abundant examples of modal separation, they are never remotely planar, unlike the majority of the RLS.

This lithology is most often non-magnetic, but a few specimens were magnetic. Besides the mottled anorthosites of the Main Zone and the anorthosite-hosted chromitite leader of the UG2, no other rocks of the RLS within the field area are magnetic. Many but not all Disturbed Gabbronorites are rotated within the Kliprivier Synform. Generally, the Disturbed Gabbronorites are CPX-poor and sulfide-rich with abundant, euhedral <1 cm pyroxenes set in an anorthosite matrix that effectively isolates the large, euhedral pyroxenes from one another [photo 62]. This unit generally becomes more feldspathic as one moves away from the intruded metasediments of the Kliprivier Synform into the Critical Zone. This lithologic unit contains recrystallized and annealed xenolith blocks [photos 74, 75, and 77] of fine-grained Marginal Zone gabbros (Rz) as well as coarse-grained pockets containing very large (< 5cm) oikocrystic pyroxene [photo 73] and large clusters of pyroxene often rimmed by thick anorthosite halos [photo 72]. Disturbed Gabbronorite is mapped between 120-145 meters below the Merensky Reef stratigraphically, around the level of the UG1 in core logs (van der Merwe, 2007). van der Merwe (2007) and Latypov et al. (2022) report isolated patches of massive chromite here, but I did not see any.

UG2: Upper Group 2 Chromitite Near the top of the upper Critical Zone in the extreme northern portion of Kliprivier farm, there is a small but high-quality exposure of chromitite (>30 cm thick). The bulk of the exposure is a melanocratic, non-magnetic, anorthositic chromite, which is modally ~50% <1mm euhedral chromite set within a fine matrix of <1 mm plagioclase (photo 66). Structurally above this massive layer are some thin (<1 cm thick) chromitite leaders hosted within bright white mottled anorthosite (photo 65). Both layers contain amoeboid oikocrystic pyroxene-olivine mottles (<2cm). The mafic silicate mottles and anorthosite matrix seem to have crystallized within the interstices of a preexisting framework of a continuous chromite crystal slurry [photos 65, 66], (see Voordouw et al., 2009; Mukherjee et al., 2017, for similar interpretations). The massive melanocratic layer includes pure (relatively chromite-free) white anorthosite blocks (<4×15 cm) that are lying flat and dipping west towards the center of the Bushveld Complex. These pure anorthosite blocks contain thin chromitite stringers that together have been ductilely deformed, and a thin (~1 mm) white seam of pure, apparently undeformed anorthosite forms a plane bounding the top of some of these blocks, also dipping to the west.

The anastomosing chromite leaders within the overlying mottled anorthosite are sometimes highly magnetic and are a much deeper, vitreous black: appearing under hand lens to be >80% modally chromite/magnetite [photo 65].

This chromitite package is interpreted as UG2 for a few reasons. First, although the anastomosing chromite leader resembles the UG1 stringers, the leader is structurally above the massive chromitite, whereas the UG1 stringers are recognized below the massive chromitite (Voordouw et al., 2009; Mukherjee et al., 2017). The UG2 is known to have several small "leaders" a small distance above the main seam (Maier and S.-J. Barnes, 2024). Second, the magnetic chromitite within the leader bears explaining. One heterogeneous magnetic chromitite at the top of the UG2 with discrete magnetic and non-magnetic bands was described by Frankel (1942), 20-30 km to the north of Kliprivier at Tweefontein. Third, a rock at this location was also mapped as the UG2 by Harmer and Sharpe (1985, Fig. 4) and van der Merwe (2007, Fig. 3), who also identified the UG2 and closely overlying mottled anorthosites in drill cores of the area using PGE grades. van der Merwe (2007) also reports that the UG2 at Kliprivier is unusually enriched in iron, which may explain the magnetism

of the leader. The UG2 is 86 meters below the Merensky Reef in the direct vicinity of the Kliprivier Synform (Borehole 3: van der Merwe, 2007).

Ln: Layered Norite: Spectacular outcrops of modally layered norites within the upper Critical Zone are observed within the field area at Kliprivier around and above the stratigraphic level of the UG2 chromitite (photos 64 and 80), at stratigraphic heights from 100 to 50 meters below the Merensky Reef. Modal layering is pronounced, with pyroxenite-rich layers being separated by near-pure centimetric anorthosite bands. This layering is undulose and irregular.

Px: Pyroxenite This refers to rocks predominantly composed of pyroxenes. Pyroxenites are rare in the field area. There was one coarse-grained sulfide-bearing feldspathic pyroxenite horizon in the Critical Zone just below the UG2 (photo 78) as well as a pyroxenite layer just below a mottled anorthosite layer in the UMA in the lower Main Zone (photos 53–54).

Tz: Transitional Zone

There is no perfect division for the Main and Critical Zones. Moving upsection from the Critical Zone into the Main Zone, Cr_2O_3 concentration within OPX decreases, strontium isotopes become more radiogenic, and cumulus clinopyroxene becomes important— but these changes occur at different stratigraphic heights within the transition zone (Cawthorn, 2015) between the Merensky Reef and Giant Mottled Anorthosite (or above). Within the literature, the bottom of the Main Zone is considered to be above the Giant Mottled Anorthosite at the top of the Bastard unit, and the Merensky reef is taken to be the top of the Upper Critical Zone (Cawthorn, 2015; A. A. Mitchell and Manthree, 2002; van der Merwe, 2007). The Bastard and Merensky pyroxenites and Giant Mottled Anorthosite were never observed in the field area, but a ~50 cm thick mottled anorthosite [photo 81] was observed ~50 meters upsection from the UG2 with only finely layered norite [photo 80] and porphyritic gabbronorite outcropping between the UG2 and the mottled anorthosite. Since van der Merwe (2007) reports in core logs that the Merensky Reef is 86 meters above the UG2 on Kliprivier and Latypov et al. (2022) observe unusual pyroxenites just above this mottled anorthosite, this anorthosite layer is interpreted to be a part of

the footwall to the Merensky Reef, and the pyroxenites described in Latypov et al. (2022) are interpreted to be the Merensky Pyroxenite.

Pg: Porphyritic Gabbronorite The "Porphyritic Norite" is a well-recognized marker of the lower Main Zone, with large, rounded pyroxene phenocrysts set in a white, feldspar-rich matrix, used often in mining to locate the Merensky Reef (SACS, 1980; von Gruenewaldt et al., 1985). I observe a general textural similarity in lower Main Zone Gabbronorites with the unlayered upper Critical Zone gabbronorites below the Merensky Reef (photos 10, 79). These gabbronorites are distinct from the dark and homogeneous gabbronorites that compose the majority of the Main Zone (Hg). The porphyritic gabbronorite contains large ($\sim<$ 1cm), equant pyroxenes set in a leuconoritic matrix that is predominantly plagioclase with small spots of interstitial orthopyroxene. I refer here to them collectively as the "porphyritic gabbronorite" because I see cumulus clinopyroxene present throughout, becoming more modally abundant upsection until about half of the cumulus pyroxenes are CPX near the Lower Mottled Anorthosite (LMA). Generally, cumulus CPX is present in the Main Zone and absent from the Critical Zone (Cawthorn, 2015). However, some cumulus CPX has been reported at the top of the Critical Zone (Cameron, 1982; SACS, 1980).

The rocks of this unit have well developed cleavage [photo 63]; however, they are not visibly layered. I located porphyritic gabbronorite south of the Kliprivier Synform at the beginning of the second field day and assumed it was Main Zone given the cumulus CPX I saw, but now I believe that the porphyritic gabbronorite occurs in the upper Critical Zone through the Transition Zone and into the lower Main Zone in the vicinity of Draaikraal. The Porphyritic Gabbronorite in this field area hosts spherical pyroxenite tennisballs (photos 10, 79), in the same locations as mapped by Harmer and Sharpe (1985) and A. A. Mitchell and Scoon (2023). Porphyritic Gabbronorite is mapped in the Critical and Main Zones 480 meters below the Merensky Reef to 380 meters above, stratigraphically.

Mz: Main Zone

The Main Zone is the most volumetrically significant portion of the RLS: as thick as 5 km (Latypov et al., 2022). The Main Zone stratigraphy is (in

general) relatively featureless aside from three prominent mottled anorthosite packages in the lower Main Zone and the Pyroxenite Marker at the upper Main Zone, which was not studied in this work. Mottled anorthosites and associated strongly modally layered sequences are mapped and described here in order to study the effect of the Kliprivier Synform and Draaikraal Lineament on the stratigraphy, as these unique layers can be traced great distances through the Lower Main Zone (Latypov et al., 2022).

Hg: Homogeneous Gabbronorite: The massive Homogeneous Gabbronorite of the Main Zone varies slightly in color due to variations in the modal abundance of pyroxenes and plagioclase, though these variations are subtle and continuous on a scale of 10s of meters. There is no obvious fabric visible in outcrops or hand samples of these massive rocks, although I was shown a polished core of upper Main Zone gabbronorite from a dimension stone mine east of Roossenekal that obviously displayed a fabric where crystals preferentially were aligned with the gravitational plane. Stonemasons favor this rock for building, but are careful to allow only faces cut against this fabric to outwardly show for aesthetics and durability. Homogeneous Gabbronorite (Hg) begins about 450 meters above the Merensky Reef and continues upsection stratigraphically through the Pyroxenite Marker (SACS, 1980; von Gruenewaldt et al., 1985).

(L/M/U)MA: (Lower/Middle/Upper) Mottled Anorthosites Mottled Anorthosites are enigmatic igneous rocks composed of a nearly pure, white anorthosite matrix containing (often quite large: <5 cm) mossy green "mottles" of oikocrystic pyroxene +/- olivine that enclose abundant plagioclase. In the RLS they are found in large stratiform horizons interpreted to be continuous across the majority of the Main Zone in the Bushveld (>100 kilometers: Latypov et al., 2022). The lower Main Zone near Draaikraal contains 3 large mottled anorthosite packages. These mottled anorthosites are magnetic and notably contain rare large (< 2mm) grains of magnetite within the mottles. They are found in association with larger packages of distinct but non-magnetic anomalous lithologies. The associated unusual rocks include: 1) fine green gabbronorites with blebby, anhedral phenocrystic (<2 cm) clinopyroxene (lithologic code Fgog: Fine-grained oikocrystic gabbronorite) associated with the UMA; 2) feature-rich, coarse pyroxenites (lithologic code: Px, [photos 53, 54]) and

3) blobby pyroxenitic gabbronorites (lithologic code Bap: Blobby anorthosite-pyroxenite) that include deformed, irregular, decimeter-scale blobby pyroxene agglomerations within an anorthositic matrix that are interpreted here to be recrystallized blocks [photos 104, 108]. Sometimes these blobby anorthosite-pyroxenites exhibit modal layering of coarse (<2 cm) pyroxene oikocrysts [photo 105] within anorthosite. The mottled anorthosites and their associated pyroxenites and blobby gabbronorites have undulating, nonplanar contacts with one another [photo 49] and with the underlying common gabbronorites [photos 107, 103]. Sometimes these contacts may be drastically discordant from the local strike and dip. In photo 106, the mottled anorthosite - gabbronorite contact is dipping ~70° to the east towards the margin of the chamber: a much steeper dip and opposite strike than observed in the leucogabbronorites of the immediate vicinity. Occasional 3D exposure of the anorthosite-gabbronorite contact allowed me to measure the contact rotating 90+ degrees in this outcrop within meters (photo 107). It is unclear if these undulating contacts are evidence for discordance of the mottled anorthosites, since in other places the contact appears concordant [photo 110], and local apparent discordance could be the result of thermal erosion of the cumulate pile by overlying magmas (Roberts et al., 2007).

In this study, all three lower Main Zone anorthosites described by Latypov et al. (2022) and mapped by van der Merwe (2007) were located within the low-lying valley south of the Draaikraal Lineament at stratigraphic heights between 200 and 600 meters above the level of the Merensky Reef. The Upper and Middle "mottled" anorthosites in the mountains to the north reported by Latypov et al. (2022) were located in this study, but are visually quite distinct and occupy higher stratigraphic heights, argued here to be distinct units that stratigraphically overlie the thick, massive mottled anorthosites in the valley to the south (see the Discussion of this work). von Gruenewaldt et al. (1985) report multiple subordinate anorthositic layers near and at the boundary of the Main Zone subzone A and B much higher above the level of the Main Mottled Anorthosite. Those higher anorthosites are referred to here as spotted anorthosites (SA) to reflect their distinct texture.

SA: Spotted Anorthosites North of the Draaikraal Lineament where exposure is better, two packages of anorthosites were found in the lower Main Zone. These anorthosites are somewhat distinct from the common, unlayered

gabbronorites of the Main Zone, but anorthositic beds are significantly smaller (<1 m thick) and altogether less visually striking than the Mottled Anorthosites (MA) unit described above. The spotted anorthosites are non-magnetic and bear <1 cm spots of equant pyroxenes composing <20% of the rock by volume [photos 86, 87] that sometimes contain recrystallized pyroxenite-gabbronorite blocks [photo 111]. The Spotted Anorthosites are strongly associated with much thicker (<10 m) packages of planar, finely layered gabbronorites that are unusual within the Main Zone and are more reminiscent of the Critical Zone [photos 85, 112]. These modally layered sections also include xenoliths [photos 91, 84], which were only observed within the Main Zone in these modally layered packages. These Spotted Anorthosite packages are argued here to be distinct and stratigraphically above the remarkable and prominent Mottled Anorthosites described above. Two or three packages of Spotted Anorthosites (often including thick layers of Layered Gabbronorite) are interpreted here at stratigraphic heights 700-1050 meters above the Merensky Reef. More packages may exist, although they are unlikely to be exposed to the south of the Draaikraal Lineament.

Lg: Layered Gabbronorite Modal layering is generally less common in the Main Zone than the underlying Critical Zone, and in outcrop it is comparatively understated. Whereas the Ln (Layered Norite) unit of the Critical Zone contains vivid, centimeter-scale, near-monomineralic anorthositic bands, the modal layering in the Main Zone gabbronorites is millimeter-scale, represented by subtle differences in cumulus grain size and modal proportions, and the contacts between layers are planar (photos 85 and 112). The Layered Gabbronorites are associated with Mottled Anorthosites (MA) or Spotted Anorthosites (SA). Layered Gabbronorites are mapped here at stratigraphic heights between 400 and 1050 meters in association with Mottled and Spotted Anorthosites.

Post-Bushveld Rocks

In the valley surrounding Draaikraal, alluvium primarily is composed of well-rounded quartzite cobbles (~10 cm) and ironstone laterites (photo 59). The ironstone laterites are commonly found as both float in alluvium and as in-place outcrops. Bushveld mafic rocks are rare in alluvium, probably due to their relative susceptibility to chemical weathering on Earth's surface. Quartzite was

substantially dominant over RLS mafic rocks in alluvium even 8 km away from the nearest known quartzite source within the RLS, leading Dr. James Roberts and me originally to suspect that there were closer quartzite sources; however, we were unable to identify quartzite sources west of Draaikraal. It is likely instead that the creeks which source their headwaters from the Houtenbek and Steenkampsberg quartzite highlands to the east and flow past Draaikraal to the west transported and rounded these cobbles, which are much more stable on the surface than the RLS rocks, eventually coming to dominate the regional alluvium.

IS: Ironstone Laterites Ironstone laterites are common throughout the relatively low and flat valley surrounding Draaikraal. They are somewhat weakly consolidated and highly porous non-magnetic amalgamations of <1cm iron concretions composed of 1 mm thick deep black vitreous goethite rims surrounding pockets of fine-grained yellow limonite matrix. Interspersed between these concretions are variably rounded medium-coarse Steenkampsberg quartzite gravels. The goethite nodules and quartz gravels are cemented by fine grained, red and yellow hematite-limonite matrix with mm-scale angular, coarse, quartz sand. The existence and abundance of these ironstone laterites implies that the valley surrounding Draaikraal is a longstanding, persistent topographical feature that has existed on geological timescales. They were likely formed by in-place weathering of RLS rocks, explaining the relative dearth of RLS material within the valley alluvium.

Dl: Dolerite Sill: Post-Bushveld Dolerite sills are mapped throughout the field area (van der Merwe, 2007; Molyneux, 2008), and one such discordant sill was observed at the levels of the UMA and MMA north of the Draaikraal Lineament within the field area (photo 89). Despite being younger than the surrounding RLS, the sill is significantly more weathered, likely because of extensive blocky fracturing hypothesized to result from rapid contraction during cooling. The dolerite is moderately magnetic.

Rock Cleavage and Modal Layering

Within many of the phaneritic plutonic rocks, there was measurable, near-planar cleavage visible (e.g. photos 58 and 63). Plutonic rock cleavages without modal layering yielded measurements that were similar to the regional dip

of bedding in the floor metasedimentary rocks to the east, but in areas with prominent modal layering these cleavages had strikes rotated 90 degrees from the regional strike, interestingly dipping similarly to the exotic limb of the Kliprivier Synform. These 90 degree offset structures could be measured very near to one another, sometimes within the same outcrop. Photo 115 shows rock cleavage that is discordant with modal layering in a striking outcrop of the MMA.

It is important to note that one outcrop of mottled anorthosite clearly showed disagreeing cleavage and planar modal layering, with modal layers dipping 50° more eastwardly than the rock cleavage [photo 115, annotated in Figure 5.4].

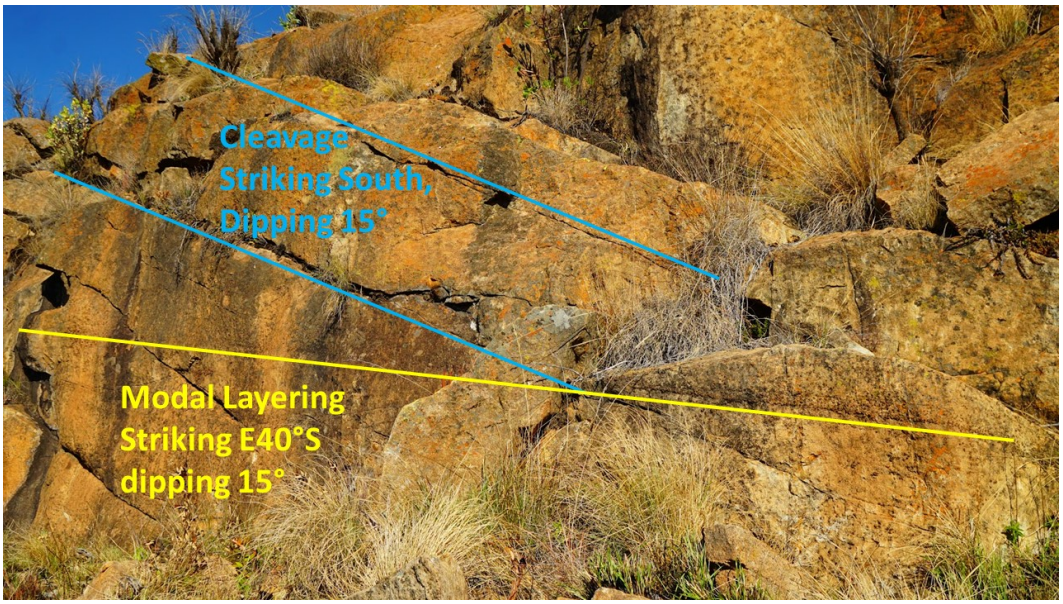


Figure 5.4: Cleavage and modal layering forming distinct planes in a Mottled Anorthosite outcrop (annotated photo 115).

Structural Observations

Structural measurements across the field area, together with the traced lower Main Zone anorthosite marker beds, are mapped in Figure 5.5.

Kliprivier Synform

The most interesting large-scale structure in the mapping area (Figure 5.6) is a rotated package of disturbed, pegmatoidal upper Critical Zone pyroxenites and gabbronorites [photos 61,62, 72–78], well-cleaved leuconorite layers [photo 63], quartzites [photos 28 and 124], migmatized/hornfelsed argillites and calc-silicate

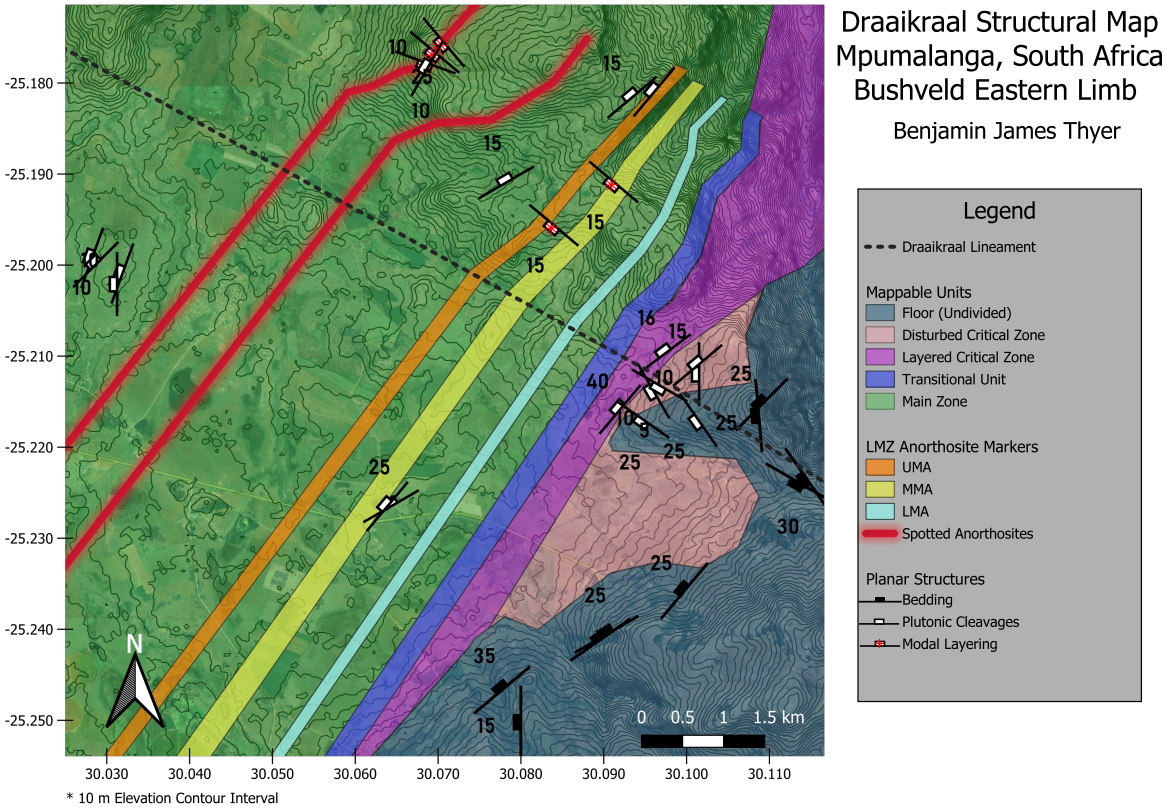


Figure 5.5: Structural map of the Draaikraal field area: strike and dip of bedding, plutonic cleavage, and modal layering, over the mappable units (floor, Disturbed Critical Zone, Layered Critical Zone, Transitional Unit, Main Zone), with the lower Main Zone anorthosite marker beds (LMA, MMA, UMA, and the spotted anorthosites) and the Draaikraal Lineament (dashed). 10 m elevation contour interval. Measurements are listed in Table A.1.

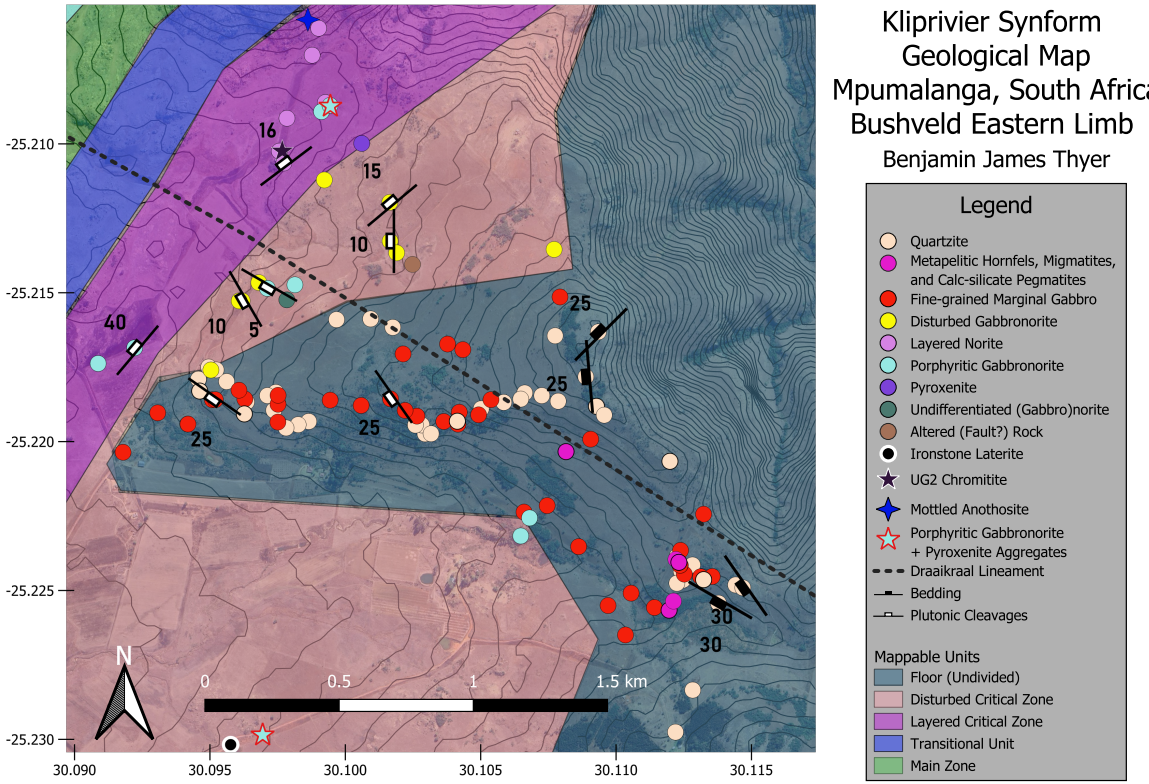


Figure 5.6: Geological map of the Kliprivier Synform. Field observations are colored by lithology over the mappable units (floor, Disturbed Critical Zone, Layered Critical Zone, Transitional Unit, Main Zone), with bedding and plutonic-cleavage measurements and the Draaikraal Lineament (dashed).

layers [photos 29, 30, and 125], and marginal sills [photo 57] in the northern portion of Kliprivier Farm. The structure is remarkable because it apparently displaces a marginal package of metasediments and fine-grained sills over 1 km into the plutonic RLS. This structure is described accurately but qualitatively by van der Merwe (2007), who investigated the area to assess its platinum resource potential. van der Merwe (2007) refers to this feature the "Kliprivier Syncline", though it is not clear if he is naming the structure or it was named elsewhere. A displaced quartzite layer within the structure was also observed during the field mapping of the Main Zone mottled anorthosites by Latypov et al. (2022) and is described in their supplementary information as a "Large Quartzite Xenolith?".

The structure is referred to here instead as the "Kliprivier Synform" to recognize that it is not yet well understood. For instance, no previous work to date has correlated marginal magmas and country rocks from the two limbs of the putative syncline to demonstrate that they are the same and that the structure is indeed a fold and not another structure, such as a laccolithically deformed roof remnant of an RLS subchamber or a package of floor rocks entrained in a rising magma. The map presented here offers additional details that elucidate the nature of this structure. Below, a series of observations of the Kliprivier Synform is presented:

1. To the north of the structure, there is a continuous, 90° gradual change in strike moving from finely-banded noritic Critical Zone, which strikes southwest and dips northwest like the majority of the mapped lithologies east of Draaikraal, to a southeastern strike dipping southwest. This gradual change in structure is accompanied by lithologic changes in the Critical Zone: Near and at the deformed areas, grain-size increases (photo 61) and stratiform modal layering ceases and is replaced by a heterogeneous rock with abundant fine-grained Marginal Zone xenoliths, anorthosite autoliths (photo 62) and irregular clumps of pyroxene surrounded by anorthosite (photos 72–77). I have mapped these lithologies as "Disturbed Gabbronorite" (Lithologic code: Dcg).

2. To the south, there is little exposure and no structural data could be gathered to pinpoint the reversal of the structure back to the common northwestern dip. However, outcrops constrain the contact between the

Marginal Zone magmas and the porphyritic gabbronorite of the upper Critical Zone to occur within 10 meters. The "disturbed" lithologies of the Marginal/Critical zones present on the north side of the synform are not present on the south side. van der Merwe (2007) interprets faulting with downthrows to the south at this contact; however, van der Merwe (2007) does not map any vertical displacement of marker beds such as the UG2 chromitite at this fault.

3. Stratigraphically higher plutonic rocks are undeformed, suggesting that they were intruded and crystallized after the deformation forming the Kliprivier Synform. Interestingly, the western termination of the synform as mapped here occurs at or slightly below the level of the Merensky Reef, where a significant magma recharge event corresponding to the emplacement of the Main Zone is thought to have occurred (Seabrook et al., 2005; Cameron, 1982).

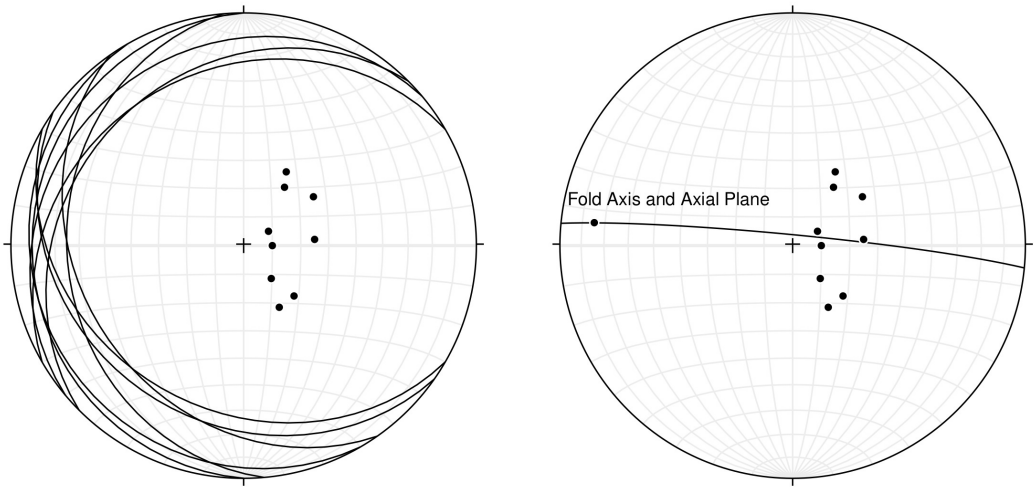
4. Displaced plutonic rocks to the north are stratigraphically below the quartzites and Marginal Zone sills, meaning that the section of sills and sediments serve as a roof to the RLS to the north.

5. Within this study, there is one strike and dip measurement immediately to the west of the structure, that is striking SW like the majority of the field area but is dipping more steeply than other rocks in the region. Latypov et al. (2022) also report an unusually steeply dipping plutonic rock along strike.

6. To the east, the structure is attached to the floor of the RLS. Unfortunately, that area was not mapped in detail and the nature of the rotation from the synform's exotic dip to the common dip of the marginal series was not observed.

The calculated fold axis (Figure 5.7b) trends roughly W5°N, plunging 15°, similarly to the Draaikraal Lineament. The calculated obtuse bisector axial plane strikes W5°N and dips 85°.

The axial plane is similar in orientation to the WNW-ESE trending Draaikraal Lineament that bounds the topographically high and low Main Zone rocks.



(a) (b)

Figure 5.7: Structural analysis of the Kliprivier Synform in stereographic projection (Stereonet 11; Allmendinger et al., 2012; Cardozo and Allmendinger, 2013). (a) Bedding planes and poles to planes of rocks within the Kliprivier Synform. (b) Calculated fold axis and axial plane for the Kliprivier Synform.

5.5 Discussion

Origin of the Kliprivier Synform and Draaikraal Lineament

van der Merwe (2007) described a "synformal structure" on Kliprivier farm and referred to it as the "Kliprivier Syncline", but I identified only a single, exotically-dipping limb within the structure with a gradual southerly change in strike, therefore I interpret it here as a monocline.

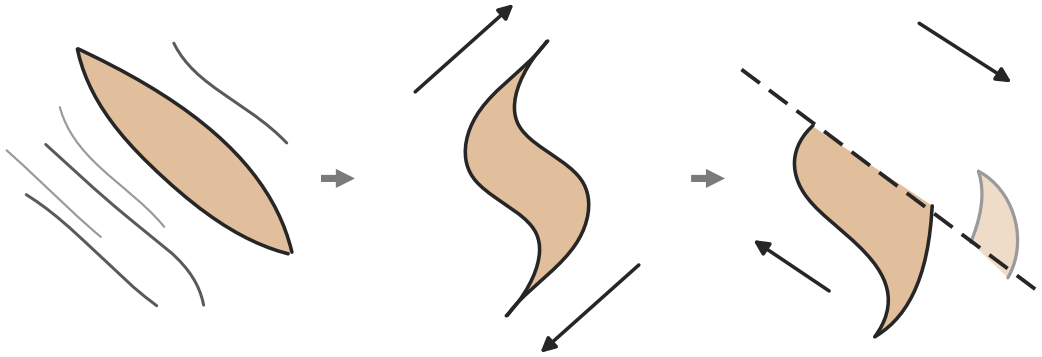
van der Merwe (2007) brings attention to the PGE grade of such structures along the eastern margin of the RLS, which often are rich in sulfides and contain unusually high-grade Pt ores aside from the traditional targets: the UG2 chromitite layer and the Merensky Reef (§B.5, "The Origin of the Merensky Reef and PGE Mineralization"). van der Merwe (2007) believes that the Kliprivier structure and these other sulfide traps predate Bushveld magmatism; however, I identified multiple coarse-grained plutonic rocks of the upper Critical Zone that were rotated within the structure. No Main Zone or Transition Zone rocks exposed on the surface to the west of the Kliprivier are deformed within the Kliprivier Synform, suggesting that deformation preceded the crystallization of the Main Zone. This relative ordering requires that the structure is synmagmatic. This interpretation is also supported by evidence for viscous deformation of gabbroic pegmatoids on the synform (Figure 5.8).

I interpret this unit to be formed by synmagmatic deformation of nascent layered upper Critical Zone mushes during the formation of the Kliprivier Synform, effectively breaking up and redistributing the nascent layering into many blobs.

Structural analysis of the synform is consistent with gentle folding along a fold axis plunging shallowly to the west with a subvertical axial plane trending roughly E(SE)-W(NW) bisecting the limbs' obtuse intersection angle (Figure 5.7b). This deformation is subtle and would not reverse the original ordering of the layered series and the marginal series. A syncline alone cannot explain how the Marginal Zone of the deformed, SW dipping limb came to be structurally higher than some of the deformed plutonic rocks. It is, however, consistent with the idea that much of the stratigraphically lower Critical Zone was intruded as stacked sills from the north and terminated here, laccolithically deforming the overlying country rocks (§B.3). Within this paradigm, the intruded Transvaal sediments "displaced" deeper within the RLS at the Kliprivier Synform are actually roof rocks to the Critical Zone below the level of the UG2. Given the intense melting and devolatilization of the floor rocks in the area, it would not be surprising to find that the thin strand of country rocks dividing the Critical Zone here beneath the level of the UG2 would have been completely reacted to igneous minerals and textures by reactive porous flow of magmas and hydrothermal fluids, as has been argued for some Lower Zone and Marginal Zone rocks (Yudovskaya et al., 2021). Indeed, the ghosts of such transformed strands may be expected at many locations throughout the complex and may have separated sills of distinct composition or served as traps for percolating melts or hydrothermal fluids. They may therefore be expected especially at geochemical discontinuities within the stratigraphic column, such as the Merensky Reef, Giant Mottled Anorthosite, Pyroxenite Marker, Upper Mottled Anorthosite, etc.

Sharpe and Chadwick (1982) also reported that Critical Zone rocks but not Main Zone rocks were deformed in the area, but attributed similar small-scale folding on the margin to isostatic flexure. Isostatic flexure has affected the modern expression of the Bushveld Complex immensely, but a paleomagnetic study shows that isostatic flexure only commenced within the Bushveld after cooling beneath the Curie temperature of magnetite: 575°C (Hattingh, 1986). This means that flexure had to occur after termination of Bushveld magmatism

(a)



(b)

Figure 5.8: (a) Sheared magnetite pegmatoid inclusion within non-magnetic marginal rocks, interpreted as reacted sediments, on the western side of the Kliprivier Synform (photo 60). The pegmatoid is about 15 cm long. (b) Interpreted deformation history: (1) mafic pegmatitic melt accumulates in lenses in migmatized sediment; (2) synmagmatic viscous deformation produces a sigmoid geometry; (3) post-magmatic brittle failure along an annealed fault truncates the lens and displaces its tip (lightly outlined). Black arrows show the sense of shear at each stage. The second stage of deformation may have occurred during isostatic flexure during cooling. The anneal of the fault suggests that it occurred at warm subsolidus conditions.

and cannot explain synmagmatic deformation.

If the southwesterly dipping limb of the Kliprivier Synform is rotated ~20° about a level rotational axis trending ~45° NE to "undo" the local effects of postmagmatic isostatic flexure that affected the RLS and its margin, the exotic limb associated with the Kliprivier Synform would be dipping ~20° due south. This observation is consistent with the hypothesis that an intrusion originating from the Thabazimbi-Murchison Lineament to the north terminated in this area, producing a southerly dipping monocline in its roof rocks.

The axial plane of the Kliprivier Synform is similar in orientation to the Draaikraal Lineament (§5.4). Considering that the Kliprivier Synform also rests upon this lineament, it is likely that the formation of the Kliprivier Synform is related to the Draaikraal Lineament.

Origin of Spheroidal Pyroxenite Aggregates (Tennis balls)

Spheroidal aggregates (SAs) of pyroxenite up to ~5 cm are known to occur as inclusions in the (gabbro)norites near the margin of the RLS, and larger pyroxenite "boulders" within the Merensky Reef are well-documented in the western limb of the RLS (C. A. Lee and Sharpe, 1979; W. D. Smith et al., 2023; Khoza, 2021). The spherical agglomerates (also called "tennis balls" due to their similar size) are richer in pyroxenes, higher in An# and pyroxene Mg#, and richer in quartz and sulfides than their (gabbro)noritic hosts (C. A. Lee and Sharpe, 1979; Khoza, 2021). Originally, dense groupings of these spherical aggregates were thought to represent laterally continuous marker beds, but they cannot be traced along strike of RLS layering for long distances relative to the named reefs or mottled anorthosites– instead they are most commonly found <200m stratigraphically above the margin with the floor rocks regardless of the stratigraphic position within the layered series (C. A. Lee and Sharpe, 1979). This author located isolated pyroxenite tennisballs in porphyritic leuconorite in two places above and below the Kliprivier Synform [photos 8, 79], on regions previously mapped to contain tennisballs (C. A. Lee and Sharpe, 1979). More interestingly, this author located spheroidal aggregates of pyroxenite in the fine-grained Marginal Zone [photos 2, 4, 96]. Only very recently have authors argued the marginal zone hosted spheroidal aggregates at all. A. A. Mitchell and Scoon (2023) used geochemical evidence to claim the coarsely crystalline plutonic gabbronorites hosting the Tennis Ball Marker (TBM), famously known

on the Middelkraal farm, ought to be considered as Marginal Zone and not as Main Zone as previously mapped (C. A. Lee and Sharpe, 1979; Molyneux, 2008; Khoza, 2021). Before this work, no fine-grained marginal zone rocks had been reported to contain pyroxenite tennis balls or spheroidal aggregates, although those observed here in Draaikraal are much more lumpy and irregular than the *spheroidal* aggregates (SA) found in the Middelkraal TBM.

In photo 96, the largest pyroxenite SA (~4 cm) resembles a diapir: It has a long (~15 cm) and thin (~<1 - 2 mm) pyroxenite tail developed behind it, suggesting that it was migrating through a comparatively rigid, stationary crystal-rich mush. The formation of a diapir suggests that the SA was moving relative to the host rock and that the host rock was more rigid than the pyroxenite. This requires explanation, as the spheroidal aggregates contain more primitive pyroxenes and plagioclase than the host matrix and are equilibrated at higher temperatures, meaning that the pyroxenitic SAs would be more rigid than the host rocks if one assumes the two are in thermal equilibrium.

There are three main hypotheses for formation of the spheroidal pyroxenite aggregates in the RLS. The model of C. A. Lee and Sharpe (1979) posits that pyroxenite aggregates are formed when a second, immiscible fluid called the "bridging" liquid intrudes a magma body with suspended pyroxene crystals. The bridging liquid preferentially "wets" the suspended pyroxene crystals, and entrains them as it coagulates with other smaller, individual bubbles of the immiscible fluid entraining the suspended pyroxenes of the host magma.

The second model of Khoza (2021) posits that a magma recharge event introduces magma into a pyroxenite cumulus pile, breaking the pile into pieces that are then subsequently entrained into the secondary magma. This magma does not equilibrate with the pyroxenite; instead, it evenly melts the rough edges of the pyroxenite aggregates, fashioning them into spheres.

The third model posits that a pyroxenitic liquid or mush intrudes a cooler crystal mush and is disaggregated into pyroxenite aggregates, analogously to the formation of mafic enclaves in felsic plutons from the disaggregation of intruded basalts (A. A. Mitchell and Scoon, 2023; W. D. Smith et al., 2023). This is the model most compatible with the observation of diapirism of spherical aggregates in the fine-grained marginal sill phase, since it allows for the SAs to be formed by disaggregation of a comparatively hot ultramafic magma intruded into comparatively cool (and therefore rigid) mafic sills.

The diapiric movement of the pyroxenite spheroid in photo 96 serves as good evidence against the second model of Khoza (2021), as Khoza's model requires the pyroxenite spheroids to be more rigid than the host magma, so that they remain consolidated against the flowing host magma. If the spheroidal aggregate was more rigid than the host magma, then it would not have left a pyroxenitic tail behind it. Finally, if magma recharge introduced into a pyroxenite cumulate pile forms SAs, then one would expect them to be a ubiquitous feature throughout the Bushveld Complex regardless of the distance from the margin, as the conventional and most accepted model for the assembly of the Bushveld Complex evokes many instances of magma recharge atop a preexisting cumulate pile (Holness et al., 2017; Cawthorn, 2015).

The model of C. A. Lee and Sharpe (1979) is interesting because secondary fluid input both can explain the relative softness of the primitive SA migrating through more evolved host rocks, and because it explains the abundant quartz (>10%: C. A. Lee and Sharpe, 1979) in the pyroxenite SA, which would not be expected for an ultramafic magma with high Mg# and An#. Photos 97 and 98 exhibit the migration of a distinct secondary comagmatic silica-rich fluid through the marginal zone magmas. However, A. A. Mitchell and Scoon (2023) point out that multiple distinct SA-rich horizons of the TBM exist but those that they report were over 100 meters from the alleged source of the fluids: the Dullstroom lavas at floor of the magma chamber. Their model simply requires injection of hot pyroxenite lavas into cooler mafic mushes. This author, however, suggests that the secondary bridging liquids of Lee and Sharpe's model could have percolated through homogeneous (gabbro)norites until ultimately meeting a coarse, pyroxenitic horizon susceptible to differential wetting.

The model of C. A. Lee and Sharpe (1979) offers a satisfactory explanation for the SA's ties to the intrusion's margin. The secondary fluids that circulated throughout the Bushveld chamber were predominantly sourced from devolatilized and variably melted country rocks (Schiffries and Rye, 1990; Buick et al., 2000, see also §B.4, "Stable Isotopes of the RLS"), so Bushveld magmas near the margin would be the most likely to encounter poorly mixed, newly-introduced immiscible (or poorly miscible) fluids than magmas further from the margin. The model of A. A. Mitchell and Scoon (2023) also offers a plausible explanation for the SAs' ties to the intrusion's margin, as the margin would cool faster to create a larger temperature (and viscosity) contrast between new

pulses of pyroxenitic magmas and their host rocks.

Correlation of the Mottled Anorthosites in the Lower Main Zone

Although there are three main named mottled anorthosite packages within the lower Main Zone, there are also many other minor anorthosites throughout the Main Zone (SACS, 1980; von Gruenewaldt et al., 1985). There is agreement between van der Merwe (2007), Setera and VanTongeren (2018), and Latypov et al. (2022) about the location of the UMA on the Roossenekal-Lydenburg road. I allow this agreement to calibrate the naming of the area's mottled anorthosites. Additionally, van der Merwe (2007) and Latypov et al. (2022) also mapped the MMA and LMA along the Roossenekal-Lydenburg road at the same location as I have.

van der Merwe (2007) maps the surface exposure of the UMA, MMA, and LMA throughout the southern sector of the eastern limb of the RLS, and later worked with Latypov et al. (2022), who undertook 6 stratigraphic traverses throughout the southeast RLS to prove that these 3 marker beds are laterally extensive throughout the entire Main Zone and do not abut the country rocks, arguing that these continuous mottled anorthosite packages were emplaced at the floor of an entirely molten 5 km thick magma chamber. Three of these traverses overlapped with the field area of this work. Within the field area, the UMA and MMA mapped by van der Merwe (2007) and Latypov et al. (2022) are discordant; they do not follow the regional dip to the northwest. I did locate anorthosites at the coordinates provided in the SI by Latypov et al. (2022); however, I conclude that the UMA and MMA that they have mapped in their traverse I, north of the Draaikraal Lineament, are not the same anorthosites as their mapped UMA and MMA south of the Draaikraal Lineament on the Roossenekal-Lydenburg road in their traverse III. I further argue that what they have mapped as UMA and MMA north of the Draaikraal Lineament are actually spotted anorthosites and some lack mottles entirely.

Two claims should be distinguished here. The first, which I hold with confidence, is that the anorthosites correlated with the UMA and MMA north of the lineament by van der Merwe (2007) and Latypov et al. (2022) are spotted anorthosites: they are thin, non-magnetic, and lack the coarse oikocrystic mottles and associated pyroxenites that define the mottled anorthosites of the lower Main Zone (Table 5.1). The second, which is tentative, is which of the

northern mottled-anorthosite packages corresponds to the UMA and which to the MMA.

Though this particular contested correlation by Latypov et al. (2022) is of minor significance to the overall argument put forward in their work, it is essential evidence to understand the origin and significance of the Draaikraal Lineament as the surface expression of a synmagmatic monocline formed by laccolithic deformation during the intrusion of much of the Critical Zone (see the end of this subsection).

The outcrops behind both correlations are compared in Table 5.1; their photographs are in Appendix A.

The UMA outcrop on the Roossenekal-Lydenburg road (photos 53, 54) contains a distinct package of a striking magnetic mottled anorthosite, ~0.5m thick underlain by a ~0.5m coarse-grained feldspathic pyroxenite with some reacted anorthositic blocks and pure anorthosite bands. This pyroxenite is in turn underlain by a unique rock interpreted to be a fine-grained gabbronorite containing blebby anhedral bottle-green clinopyroxene phenocrysts. Fresh faces of this rock are a uniquely vivid green color. I was able to locate the vivid green fine grained gabbronorite with blebby anhedral pyroxene phenocrysts (Fgog) above the Draaikraal Lineament near a cliff with a view of the Booysendal PGE mine to the east (Fgog not pictured, photos 88 and 90). I correlate this outcrop with the UMA south of the Draaikraal Lineament because it preserves the same sequence of fine-grained green gabbronorite, coarse feldspathic pyroxenite and magnetic mottled anorthosite. It sits ~170 m lower in the stratigraphic column (398 m versus 564 m above the Merensky Reef; Table 5.1). The match is not perfect, but the column assumes a constant strike and dip (§5.3). Over the ~6.5 km separating the two outcrops along strike, small discrepancies between the modeled and true strike and dip could produce significant variation in the stratigraphic heights.

It is possible that this northern package is instead the MMA. That would imply that the fine-grained green gabbronorite is not unique to the UMA but also occurs beneath the MMA. It would not change the argument below: either way, the northern correlatives are magnetic mottled anorthosites with associated pyroxenites, and they lie at heights close to those of the southern packages.

The anorthosite mapped by van der Merwe (2007) and Latypov et al. (2022)

Table 5.1: Lower Main Zone anorthosite outcrops compared across the Draaikraal Lineament. The roadcut of the UMA South of the lineament in particular is well documented and agreed to be a lower Main Zone mottled anorthosite (Setera and VanTongeren, 2018; Latypov et al., 2022). The outcrops north of the lineament as correlated by Latypov et al. (2022) and van der Merwe (2007) are compared with those in this work. Heights are stratigraphic heights above the Merensky Reef (§5.3).

Outcrop	Photos	Height (m)	Thick.	Mag.	Texture and associated lithologies
Upper Mottled Anorthosite (UMA)					
Roadcut UMA, South −25.2274, 30.05494	49, 53, 54	564 m	∼0.5 m	yes	Anorthosite with ∼2cm oikocrystic pyroxene mottles overlying ∼0.5 m coarse feldspathic pyroxenite with anorthositic blocks, over fine-grained green gabbronorite with ∼ 1cm blebby bottle-green CPX; irregular, non-planar contacts
"UMA" of Latypov et al. (2022), North −25.1767, 30.07222	86	1030 m	∼10 cm	no	Anorthosite with <1 cm equant pyroxenes. No other unusual lithologies observed. Planar, concordant contacts
"UMA" of this work, "LMA" of Latypov et al. (2022), North −25.1821, 30.09749	90	398 m	>10 cm	yes	Anorthosite with coarse oikocrystic pyroxene mottles over coarse pyroxenite
Middle Mottled Anorthosite (MMA)					
Valley outcrop MMA, South −25.227, 30.06419	103–109	373 m	> 2 m	yes	Very thick; layers of oikocrystic mottled anorthosite and pyroxenite with both planar and wavy, discordant contacts
"MMA" of Latypov et al. (2022), North −25.1827, 30.08279	87	708 m	not measured	no	Anorthosite with pyroxene spots and foliated (sheared) pyroxenes, no pyroxenites.
"MMA" of this work, North **[TODO: coordinates]**	115	333 m	> 4 m	yes	Thick package of anorthosites with coarse oikocrystic pyroxene mottles and pyroxenites with sharp lower contacts that grade upward to anorthosite in ∼ 20 cm layers; cleavage discordant with modal layering

as the UMA north of the Draaikraal Lineament (photo 86) is very small (~10 cm), non-magnetic, and lacks the associated coarse pyroxenite and fine-green gabbronorite lithologies; it does not resemble the UMA outcrop on the Roossenekal-Lydenburg road. Furthermore, it is significantly stratigraphically higher: ~1030 m above the Merensky Reef. van der Merwe (2007) does not report displacement of the UG2 over the Draaikraal Lineament, so it is interesting that their UMA is given ~470 m of stratigraphic displacement.

Similarly, the discordant MMA correlation they offer does not stand up. The MMA is supposed to be a striking and very large marker in the lower Main Zone (photos 103–109), and yet the outcrop they correlate (photo 87) is a small exposure with subtle, non-magnetic pyroxene spots/mottles, and no associated blebby or coarsely layered pyroxenites. I instead find the concordant correlation of a very large package of magnetic mottled anorthosites with coarsely layered pyroxenites at photo 115 much more plausible.

Neither correlation places the anorthosites at exactly the same height on both sides of the lineament, and neither needs to. The thickness of the anorthosite packages and of the layers between them may vary along strike, shifting their relative positions coherently even without invoking a fault. Nonetheless, I prefer the simplest explanation that explains the data. My correlation follows petrography, and happens to require the smallest lateral change to explain the observations: the mottled anorthosites can be followed across the Draaikraal Lineament without invoking a fault.

Because there are essentially no detailed studies of the lower Main Zone anorthosites (similar to A. A. Mitchell and Manthree, 2002, for the GMA of the transitional unit), it is easy to fall prey to one's confirmation bias: one can find an anorthosite in the lower Main Zone without too much trouble. Detailed and careful petrography of these marker beds is necessary to make correct correlations. The strategy of Latypov et al. (2022) to use these marker beds to understand the structure and petrogenesis of the RLS is excellent; however, I would argue instead that one should first focus on one particularly remarkable anorthosite package (perhaps the MMA or UMA), and should follow it along strike. For this field area, I would want to do a detailed characterization of the Transitional unit, LMA, MMA, and UMA stratigraphy to the north of the Draaikraal Lineament to get complete sections of the remarkable, but incomplete, outcrops exposed along the Roossenekal-Lydenburg road. Then

I would follow the anorthosites up north to examine how their strike might change, as their regionally concordant NW dip would quickly become discordant to the north. I consider the following open questions to remain:

1) Are the LMA, MMA, and UMA generally planar features, irregular contacts aside?

2) For how long are they laterally continuous?

3) Where and why are they discordant to the regional layering, if ever?

4) Do they abut the floor?

5) How do the distances between them vary at different locations within the intrusion?

Origin of the Mottled Anorthosites in the Lower Main Zone

Surprisingly little attention has been given to the spectacular anorthositic sequences of the lower Main Zone. More often, anorthosites from the Critical (or Transitional) Zone are studied (A. A. Mitchell and Manthree, 2002; Maier et al., 2016; A. A. Mitchell et al., 2019; Maier et al., 2020; S. J. Barnes et al., 2021), or at the top of the Main Zone at the Pyroxenite Marker (Bourdeau et al., 2021; Bourdeau et al., 2023). The Upper Mottled Anorthosite (UMA) and Pyroxenite Marker of the Main Zone are known to demarcate significant geochemical and isotopic changes in the RLS stratigraphy (Klemm et al., 1985; F. J. Kruger, 1994; VanTongeren and Mathez, 2013), most often ascribed to magma recharge (§B.4, "Radiogenic Isotope Stratigraphy of the RLS").

In the East and West Limbs, the mottles of the lower Main Zone mottled anorthosites (LMA, MMA, UMA) are surprisingly iron-rich– the pyroxenes have much lower Mg# than the surrounding gabbronorites (Scoon and A. A. Mitchell, 1994; A. A. Mitchell et al., 1998). Though I was surprised to find the mottles of these three units were magnetic at first, it had already been known that they produce subtle but distinct magnetic anomalies observable in aerial surveys, at least within the West Limb (Viljoen and Scoon, 1985; Cole et al., 2013). Viljoen and Scoon (1985) point out that the stratigraphic magnetic anomalies of the mottled anorthosites of the lower Main Zone are not present near structural weaknesses, like faults.

Scoon and A. A. Mitchell (1994) make the provocative argument that the silicate iron-rich ultramafic pegmatites (IRUPs) of the upper Critical and lower

Main Zones are sourced from accumulated evolved fluids interstitial to these mottled anorthosites of the lower Main Zone. These putative residual melts are argued to flow laterally, eventually exploiting structural weaknesses to percolate downward to replace plagioclase-rich, magnesian cumulates with olivine-pyroxene Iron-Rich Ultramafic Pegmatites (IRUP), a lithology where plagioclase is conspicuously absent. This bold argument seems unnecessary. It is easy to explain, as Scoon and A. A. Mitchell (1994) also do, that IRUPs likely originated from downward-percolating immiscible Fe-oxide melts from the Upper Zone. Strontium isotopes within IRUPs are uniform regardless of stratigraphic height and are identical to the Upper Zone (Scoon and A. A. Mitchell, 1994). IRUPs are known to occupy any location along the stratigraphic column between the Upper Zone and their lowest occurrences in the upper Critical Zone, and form a geochemical continuum that records them becoming more magnesian and silica-rich with depth, transitioning from Fe-Ti oxide pegmatites to Fe-rich silicate pegmatites, telling a clear story of evolution/hybridization of downward-percolating, dense Fe-rich liquids.

Still, the strontium isotopes of these mottled anorthosites and some stratigraphically proximal gabbronorites are often discontinuous with the otherwise homogeneous, isotopically evolved Main Zone immediately surrounding them, and instead are, remarkably, identical to the Upper Zone and IRUPs (Karykowski et al., 2017; F. J. Kruger, 1994). It may very well be true that the IRUPs are related to the mottled anorthosites of the lower Main Zone, though I would instead propose that these anorthosites might have served as porous "aquifers" that trapped IRUP melts originating from the Upper Zone, rather than pose the mottled anorthosites as the source of these iron-rich melts.

Additionally, compelling geochemical arguments have been presented on texturally similar mottled anorthosites in the Critical Zone that argue for formation by reactive porous flow of a hydrothermal fluid including the presence of evolved but Ni–rich olivine, hydrous phases and sulfides, and extreme reverse zoning of plagioclase surrounding mottles (Maier et al., 2020, see also §B.5, "Origin of Enigmatic Monomineralic Igneous Rocks"). The presence of water is known to favor olivine crystallization at the expense of pyroxenes, as well as favoring more anorthitic plagioclase over albite (Baker and A. E. Boudreau, 2019).

I envisage mottled, stratiform anorthosites to form where low-permeability structural traps collect upward-percolating reactive hydrothermal fluid that

locally favors plagioclase crystallization and pyroxene dissolution. EPMA work on the mottles of the LMA, MMA, and UMA would bring clarity to the relative importance of evolved interstitial fluids, potential IRUP contaminants, and hydrothermal fluids on the formation of mottled anorthosites in the lower Main Zone and to stratiform anorthosites in general.

Rethinking the Marginal Zone: Arguments for Incremental Assembly The Marginal Zone has been traditionally understood to be the chilled margin of the RLS (Harmer and Sharpe, 1985). However, detailed work reveals that the lowest stratigraphy of the RLS is complex. The Marginal Zone is not always present at the floor of the RLS, especially to the southeast (Molyneux, 2008; van der Merwe, 2007; Latypov et al., 2020). Marginal Zone norites may locally intrude the Lower Zone (A. H. Wilson et al., 2017; Yudovskaya et al., 2021), and the Marginal Zone overlies an additional Bushveld-related ultramafic sequence referred to as the Apiesdooringdrai peridotite body or Basal Ultramafic Sequence (BUS) at Burgersfort that is more likely to contain chilled components of the earliest RLS-related intrusions (A. H. Wilson, 2012; A. H. Wilson et al., 2017). The BUS, Marginal Zone, and Lower Zone all contain reacted sedimentary "rafts" or xenoliths with igneous mineralogy (Yudovskaya et al., 2021). Thinner ultramafic and mafic sills likely related to Bushveld magmatism intrude the floor rocks kilometers below the mapped Marginal Zone (Sharpe and Hulbert, 1985). In my field work, I observed that Marginal Zone sills seemed to concordantly intrude the Transvaal sediments below the RLS for many kilometers.

The lack of a cohesive igneous stratigraphy and the geochemical and isotopic variability of the Marginal and Lower Zones led A. A. Mitchell and Scoon (2023) and Scoon and A. A. Mitchell (2023) to propose that the Marginal and Lower Zones were assembled incrementally by magmas of different compositions (see also §B.5, "Spatial and Temporal Scales of Magma Chambers in the Crust, Pluton Assembly Mechanisms"). I would extend their arguments to the entire RLS. I posit that a series of stacked peridotites, pyroxenites, norites, gabbronorites, and gabbros were emplaced, somewhat self-sorted according to density and the compositional control on stalling depth (more primitive magmas will crystallize at higher temperatures) at the location of the BUS, RLS and Marginal Zones presently. I suggest that postcumulus modification by reactive porous flow of late stage magmas and/or hydrothermal fluids allowed

for dissolution and recrystallization to produce phaneritic textures within the RLS, and that these fluids interacted with the Marginal Zone to a far lesser extent than the greater RLS. These fluids reequilibrated and reacted with the RLS within a depth-controlled thermal gradient which produced the RLS-scale cohesive geochemical trends that are mostly uncorrelated with isotopic and lithological boundaries (see Eales et al., 1986). Subconcordant intrusion of magmas as sills into the country rock results in the inclusion of abundant but comparatively volumetrically small strands of sedimentary rock in the Lower, Marginal, Critical, and Main Zones of the RLS. These strands react to igneous mineralogy by reactive porous flow along grain boundaries and develop planar modal layering due to lateral transit of fluids and evolved magmas sourced from the Thabazimbi-Murchison Lineament (TML) and they serve as traps for hydrothermal fluids that locally supported the reaction, dissolution, and crystallization of mafic lithologies into mottled anorthosites below. The Merensky Reef represents a structural trap for a reduced supercritical carbon-based fluid (Ballhaus and Stumpfl, 1985, see also §B.4, "Unusual Volatile-bearing Phases in the Lower RLS"). The isotopic homogeneity of the Upper Zone (§B.5, "Open or Closed System") allows for a more orthomagmatic interpretation: perhaps it represents a magma chamber where evolved tholeiitic magmas and hydrothermal fluids collected, and magnetite/anorthosite layers were crystallized by cycles of depressurization, cooling, and perhaps magma recharge.

Significance of Rock Cleavage and Modal Layering Orientation in the RLS

The discordance of rock cleavage and modal layering (§5.4; Figure 5.4) suggests that cleavage and modal layering are formed by distinct processes that may have occurred at different times, with modal layering likely predating the formation of rock-cleavage.

There may have been multiple processes generating superimposed fabrics in the plutonic RLS. For instance, rock fabrics may have been developed in two phases, perhaps a primary fabric (modal layering) developed by gravity and crystal settling (§B.1), then a second by lateral flow of hydrothermal fluids, or by differential contractions during cooling. Care must be taken when interpreting planar measurements in plutonic rocks of the RLS. The bimodal planes of the RLS in this field area may be due to an initial development of south-

dipping igneous fabric above the Kliprivier Synform as magmas filled a local basin towards the south before another generation of paleohorizontal fabric developed perpendicular to thermal gradients due to differential contraction during cooling.

5.6 Conclusions

1) The field data of this study support the interpretation that the Kliprivier Synform is a monocline synmagmatic with the Critical Zone below UG2, predating the UCZ at UG2 and up. There is no displacement of marker beds above UG2, including all mottled anorthosites. The country rocks and Marginal Zone sills of the Kliprivier Synform are roof rocks to the Critical Zone below UG2 and are the floor to undeformed Transitional and Main Zone.

2) The Draaikraal Lineament is best explained as the surface projection of the subvertical axial plane of the Kliprivier Synform. It is hypothesized that trapped fluids from the emplacement of the underlying Critical Zone may have exploited the deformed Kliprivier Synform, developing a vertical, planar fluid conduit at the Draaikraal Lineament where upwardly migrating fluids may have escaped, developing recessive lithologies. The regional recessive topography of the Main Zone south of the Draaikraal Lineament is proposed to originate from relatively high degrees of contamination from devolatilized/melted underlying sediments relative to the Main Zone lithologies at the same stratigraphic height to the north of the Draaikraal Lineament.

3) The discordant mottled anorthosite correlation above and below the Draaikraal Lineament by van der Merwe (2007) and Latypov et al. (2022) is locally incorrect. The field work presented here instead suggests that the mottled anorthosite marker beds of the lower Main Zone are concordant and undisplaced by the Draaikraal Lineament. More careful, detailed field work will be necessary to ensure correct correlation of the many understudied anorthosites of the lower Main Zone.

4) Pyroxenite aggregate tennisballs are found in fine-grained Marginal Zone Gabbros. This supports the interpretation that the coarse-grained porphyritic gabbronorite host of the TBM at Middelkraal is a member of the Marginal Zone after A. A. Mitchell and Scoon (2023). It is difficult to

explain the development of a diapiric tail on the nascent pyroxene tennis balls in the fine-grained Marginal Zone with the model of Khoza (2021), although this observation allows the models of A. A. Mitchell and Scoon (2023), W. D. Smith et al. (2023), and C. A. Lee and Sharpe (1979).

5.7 Acknowledgment

Profound thanks are due to Professor James Roberts for his charismatic introduction to the plethora of all-consuming geological problems of "The Beast" (the Bushveld Complex). He is also owed thanks for warmly hosting me as a visitor to the University of Pretoria in 2023, for providing resources to explore the Draaikraal area, for accompanying me for several days of field work in Draaikraal, and for teaching me alongside UP undergraduates on a field course of the wondrous Bushveld.

Appendix A

DRAAIKRAAL FIELD OBSERVATION DATA AND ANNOTATED PHOTOGRAPHS

A.1 Field Observations

Every geotagged field observation of the July 2023 campaign is listed in Table A.1, in order of stratigraphic height (as described in §5.3) relative to the Merensky Reef ($h = 0$ m). The table carries the location, the interpreted map unit and lithology, any structural reading taken at the site, and the IDs of the photographs taken there. Detailed, free-form field descriptions and notes are footnotes to the observations' names, at the foot of the page. These field data are plotted on the lithological and structural maps of Chapter 5 (Figures 5.3 and 5.5).

Table A.1: Field observations of the Draaikraal campaign (July 2023). Lat./Lon.: WGS84, decimal degrees. Elev.: elevation (m). Unit: final interpreted map unit and Lith.: lithology code (§5.4). Structure: bedding (Bed.), cleavage (Clv.), contact (Cont.) or modal layering (M.L.), as strike/dip with strike an azimuth from north. Photos: field photograph numbers; those reproduced in §A.2 or as figures are links, the others are in the supplementary photo album. Field descriptions and notes are footnotes to the observation name, at the foot of the page. '–': not recorded.

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
RZ22	-933	-25.24760	30.11812	2146	Rz	Msp	–	–
QZ27	-724	-25.24040	30.10576	2076	Rz	Qz	–	–
QZ7	-706	-25.23020	30.11221	1958	Rz	Qz	–	–
RZ54	-700	-25.23980	30.10496	2044	Rz	Msp	–	–
CONT MARG E Q W?[1]	-697	-25.22880	30.11288	1960	Rz	Qz	–	–
D3NB	-683	-25.22540	30.11481	1960	Rz	Qz	Bed. 145/30	–
QZ1	-676	-25.22530	30.11455	1956	Rz	Qz	–	–
D3NA[2]	-674	-25.22590	30.11390	1929	Rz	Qz	Bed. 120/30	–
RZ43	-669	-25.23920	30.10392	2015	Rz	Msp	–	–
D7NC[3]	-661	-25.23900	30.10369	2010	Rz	Qz	–	–

continued on next page

[1] Quartzite. Quartzite follows topo line locally here.
[2] Compass trouble, resolved.
[3] Alternating quartzite-marginal zone photo, not very good one.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
RZ8	-654	-25.22500	30.11369	1945	Rz	Msp	–	–
D3NC[4]	-649	-25.22510	30.11337	1940	Rz	QzMsp	–	34
RZ21	-644	-25.22500	30.11326	1947	Rz	Msp	–	–
RZ24	-643	-25.23900	30.10279	1997	Rz	Msp	–	–
RZ36	-642	-25.23860	30.10304	1995	Rz	Msp	–	–
QZ112	-639	-25.23860	30.10289	1995	Rz	Qz	–	–
D3SF[5]	-638	-25.22610	30.11207	1929	Rz	ScP	–	37, 38
SCHIS	-636	-25.22580	30.11224	1937	Rz	Sc	–	–
QZ2	-631	-25.22510	30.11260	1950	Rz	Qz	–	–
QZ3	-631	-25.22460	30.11297	1954	Rz	Qz	–	–
RZ11	-629	-25.22490	30.11266	1952	Rz	Msp	–	–
QZ6	-629	-25.22520	30.11238	1946	Rz	Qz	–	–
QZ5	-626	-25.22480	30.11255	1950	Rz	Qz	–	–
RZ31	-625	-25.22600	30.11153	1926	Rz	Msp	–	–
RZ33	-622	-25.22460	30.11252	1946	Rz	Msp	–	–
D3SE[6]	-621	-25.22460	30.11251	1950	Rz	Msp	–	36, 40
D3SD[7]	-619	-25.22450	30.11247	1946	Rz	ScQz	–	35
RZ71	-617	-25.22690	30.11044	1911	Rz	Msp	–	–
SCHIST	-615	-25.22440	30.11236	1944	Rz	Sc	–	–
QZ4	-615	-25.22430	30.11241	1945	Rz	Qz	–	–
RZ41	-614	-25.22410	30.11254	1948	Rz	Msp	–	–
RZ110	-614	-25.23840	30.10183	1976	Rz	Msp	–	–
RZ51	-614	-25.22290	30.11342	1945	Rz	Msp	–	–
RZ61	-599	-25.22550	30.11068	1907	Rz	Msp	–	–
RZ32	-588	-25.22590	30.10982	1900	Rz	Msp	–	–
QZ26	-575	-25.23760	30.10054	1948	Rz	Qz	–	–
D7SE[8]	-569	-25.23690	30.10081	1946	Rz	ScP	–	70, 71
SHIST	-569	-25.23690	30.10078	1947	Rz	ScP	–	–
RZUP	-565	-25.23740	30.10019	1935	Rz	ScPMsp	–	–
D7SD[9]	-559	-25.23740	30.09991	1936	Rz	ScP	–	69
D2NC[10]	-555	-25.22110	30.11222	1993	Rz	QzMsp	–	14, 23, 28
D7NA[11]	-552	-25.23700	30.09985	1924	Rz	QzMsp	Bed. 220/25	67
SCH1	-551	-25.23700	30.09984	1930	Rz	ScP	–	–
RZMAG	-549	-25.23690	30.09979	1919	Rz	Msp	–	–
RZ	-532	-25.22390	30.10878	1909	Rz	Msp	–	–

continued on next page

[4]No faulting; the quartz and marginal zone is interbedded.

[5]Qz-Px Pegmatite. Hosted in migmatitic quartzite. Schist banding deformed around pegmatite.

[6]Fine grained, light rock. Dacite? Silicified. Pockmarks weathering proud in paleovertical pipes.

[7]Migmatized quartzite. Banded. Light bands of quartz preserve original sedimentary texture. Grains rounded and 2 mm. Some bands pink (feldspar) some green (chlorite, epidote?)

[8]Silicate Pegmatite. within migmatized qzite very close to magnesite skarn. Myrmekitic quartz-K-spar, huge OPX, small vibrant rare green and blue crystals: apatite, CPX, beryl?

[9]Magnesite Skarn. within migmatized qzite. In ground, hard to tell if dike or concordant with schist host.

[10]Contact. On this ridge, there are multiple bands of quartzites with Marginal Zone between them. Is the marginal zone a sill between layers of quartzite, or is there a fault? (Update, I think it's a strike-slip(?) fault striking E-W dipping to the South with a component of left-lateral shear along the ridge)

[11]Fine marginal gabbronorite with shearing pegmatite veins. sheared pegmatite veins in pockets/pipes orthogonal to bedding, paleo-up. This layer is just above migmatized quartzite and just below fine magnetic marginal zone. Represents fluid transport from migmatized sedimentary rocks, origin of resistance banding? Strike and Dip are for migmatized Qz.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
D1ND[12]	-507	-25.23730	30.09753	1926	Rz	Msp	–	6
D1NC[13]	-500	-25.23720	30.09730	1924	Rz	Msp	–	–
RZ3	-485	-25.22250	30.10765	1910	Rz	Msp	–	–
RLS	-481	-25.22350	30.10665	1890	Cz	Pg	–	–
RZ1	-481	-25.22030	30.10931	1973	Rz	Msp	–	–
D1Sb[14]	-479	-25.23690	30.09644	1891	Rz	Msp	–	19
D1A[15]	-479	-25.23690	30.09644	1891	Rz	MspTB	–	4
MZ3	-478	-25.22290	30.10699	1902	Cz	Pg	–	–
D2SB[16]	-477	-25.21950	30.10983	2004	Rz	Qz	–	–
RZ2	-470	-25.22270	30.10679	1901	Rz	Msp	–	–
D2SH[17]	-469	-25.22070	30.10839	1950	Rz	ScP	–	29, 30
RQCONT	-465	-25.21920	30.10955	2014	Rz	QzMsp	–	–
RZ20	-456	-25.23540	30.09654	1903	Rz	Msp	–	–
RZ5	-441	-25.23760	30.09423	1918	Rz	Msp	–	–
D2ND[18]	-440	-25.21820	30.10918	2020	Rz	Qz	Bed. 175/25	–
D11SA[19]	-440	-25.23770	30.09405	1899	Rz	Msp	–	–
QZ	-433	-25.21900	30.10814	1990	Rz	Qz	–	–
FOOLLOW								
TOPO								
RZ4	-431	-25.23700	30.09419	1906	Rz	Msp	–	–
RZ6	-430	-25.23860	30.09298	1923	Rz	Msp	–	–
D2NE[20]	-426	-25.21670	30.10968	2031	Rz	Qz	Bed. 225/25	–
D4NB[21]	-422	-25.24150	30.09048	1932	Rz	QzMsp	Bed. 240/25	52
QZ51	-417	-25.21880	30.10754	1993	Rz	Qz	–	–
D10NK	-415	-25.24210	30.08966	1917	Rz	Qz	Bed. 235/25	–
RZ7	-413	-25.23870	30.09213	1915	Rz	Msp	–	–
D4SA[22]	-412	-25.24090	30.09036	1895	Rz	QzMsp	–	50, 51
SCHI	-403	-25.24040	30.09025	1882	Rz	Sc	–	–

continued on next page

[12]Bladed iridescent white crystals (Plagioclase) in 2 cm thick planar features intersect orthogonally in outcrop. The plagioclase planes are surrounded by 10 cm zone of oikocrystic augite. The southwestern limb pinches in and out and occasionally overlaps itself in parallel, a la boudinage.

[13]Note that all marginal zone rocks are characterized by dark resistance bands, perhaps cataclastic, reflecting warm-brittle deformation with finely recrystallized zones. They stand proud. They frequently strike E-W, secondarily, NE-SW, tertiarily. NW-SE, but can vary and appear pretty random. This feature is diagnostic of the marginal zone.

[14]Common Marginal zone Medium-fine grained gabbronorite.

[15]Fine-grained Magnetite Gabbro with gabbronoritic blocks. Fine-grained magnetite gabbro includes rounded, elongate blocks of oikocrystic gabbronorite preferentially horizontally imbricated, ∼6-10 cm along axis. Also a bunch of rounded gabbronorite inclusions that may represent the spherical aggregates of the TBM.

[16]Sampled because quartzite here all have iron stained interiors. The weathered exteriors are clear. Quartzites served as a conduit for mobilized iron-rich fluids?

[17]Schist with Skarn. Swirled and deformed schist with marginal zone, with thin clear banding. Non-magnetic. Xenolith? Associated with skarn pegmatite veins: 60% Feldspar, 40% amphibole? Big very dark crystals that I think I see 120 degree cleavage on... Likely devolatilizing xenolith.

[18]Measurable bedding planes at top of ridge: Strike: 265, Dip: 25. Also, One bedding plane preserved lovely sedimentary ripples striking E-W.

[19]Coarse grained gabbronorite. Is it critical-margin zone? Are resistance bands chromite bearing?

[20]Different strike, same dip. Strike: 315, dip: 25. Quartzites are rotated around the top of the ridge. NW-SE vertical fault or E-W South dipping fault can neatly explain rotation.

[21]Contact between floor quartzite and marginal zone. Bed plane of quartzite measured.

[22]Contact between marginal zone and migmatized quartzite. Migmatitic quartzite ∼10m thick. At contact is non-magnetic leucocratic gneiss, then fine magnetic marginal zone: 10cm, then coarse grained marginal zone: 30 cm, then common fine marginal zone.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
QZ61	-402	-25.21890	30.10676	1980	Rz	Qz	–	–
QZ41	-402	-25.21870	30.10691	1987	Rz	Qz	–	–
QZ	-395	-25.21680	30.10808	2001	Rz	Qz	–	–
QZ29	-393	-25.24210	30.08861	1907	Rz	Qz	–	–
QZ71	-391	-25.21900	30.10613	1975	Rz	Qz	–	–
RZ28	-388	-25.24200	30.08845	1906	Rz	Msp	–	–
SCH	-387	-25.24010	30.08972	1870	Rz	Sc	–	–
RZ12	-381	-25.23990	30.08955	1866	Rz	Msp	–	–
D2SA[23]	-380	-25.23000	30.09696	1860	Cz	PgTB	–	–
RZ35	-380	-25.21890	30.10566	1965	Rz	Msp	–	–
D2SF[24]	-379	-25.21550	30.10829	1994	Rz	Msp	–	24
RZ42	-379	-25.21940	30.10519	1950	Rz	Msp	–	–
QZ8	-377	-25.21920	30.10529	1954	Rz	Qz	–	–
RZ9	-370	-25.23920	30.08952	1850	Rz	Msp	–	–
RAT	-368	-25.23630	30.09174	1883	Rz	Msp	–	–
RZ511	-368	-25.21970	30.10442	1935	Rz	Msp	–	–
RZLQZR	-366	-25.21960	30.10440	1938	Rz	QzMsp	–	–
RZ53	-362	-25.21930	30.10448	1945	Rz	Msp	–	–
RZ113	-359	-25.24170	30.08722	1876	Rz	Msp	–	–
IS1	-358	-25.23030	30.09577	1868	IS	IS	–	–
RZ62	-355	-25.21960	30.10390	1934	Rz	Msp	–	–
D2SG[25]	-352	-25.21390	30.10811	1956	Cz	Dcg	–	27
QZ9	-352	-25.22000	30.10342	1929	Rz	Qz	–	–
SCI	-350	-25.23900	30.08890	1888	Rz	Sc	–	–
D9NE	-348	-25.25130	30.07965	1929	Rz	Qz	Bed. 180/15	–
QZ10	-347	-25.22000	30.10320	1928	Rz	Qz	–	–
SCH11	-343	-25.24070	30.08725	1881	Rz	Sc	–	–
RZ27	-341	-25.24060	30.08725	1882	Rz	Msp	–	–
D9SD[26]	-340	-25.25150	30.07907	1912	Rz	Msp	–	95
QZ11	-339	-25.21970	30.10308	1938	Rz	Qz	–	–
SCH2	-338	-25.23940	30.08802	1887	Rz	Sc	–	–
RZ82	-338	-25.23960	30.08784	1886	Rz	Msp	–	–
D9NC[27]	-335	-25.25180	30.07858	1908	Rz	Msp	–	–
QZ12	-334	-25.21970	30.10285	1939	Rz	Qz	–	–
D9NF[28]	-333	-25.25050	30.07951	1918	Rz	MspTB	–	96
RZ23	-331	-25.21720	30.10466	1957	Rz	Msp	–	–
RZ72	-330	-25.21940	30.10291	1940	Rz	Msp	–	–
RZ81	-318	-25.21920	30.10248	1940	Rz	Msp	–	–

continued on next page

[23]Common Main Zone Coarse Grained Gabbronorite (Office interp: Leucogabbronorite; porphyritic gabbronorite) On Dirt road to Kliprivier Farms, Boulder piles outcropping of main zone. These rocks are the building materials for much of the structures in the area. Weathering is quite distinct from Marginal zone. Both have exfoliation/onion skin weathering. Main Zone is prouder, forming impressive rounded boulders like a granite. Weathered faces have distinct dalmatian pattern. A large boulder (float) at this outcrop contains a characteristic pyroxene "tennisball", a well-known marker at the base of the Main Zone.

[24]Coarse Grained Magnetite Gabbro. Actually I am now thinking this sample is similar to the magnetite gabbro D1Sa, but coarser grained. That sample too had zones of coarse grains, and this one also had blocks of oikocrystic augite.

[25]Norite. Weathering more closely resembles main zone than nearby marginal zone, but this rock does not look like main zone to me. Non-magnetic.

[26]Fine Grained Marginal Norite. Sampled for prominent resistance band for later analyses.

[27]Chloritized Norite. Plag, OPX, abundant phlogopite, alteration grade lessens up section.

[28]Fine Grained Marginal norite. Droplet forms of coarse pyroxenite xenoliths, pegmatite inclusions?

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
RZ19	-316	-25.21700	30.10410	1948	Rz	Msp	–	–
RZ25	-312	-25.24970	30.07913	1910	Rz	Msp	–	–
D9NG[29]	-312	-25.24980	30.07904	1910	Rz	Msp	–	97, 98
D5NC[30]	-299	-25.21880	30.10195	1946	Rz	Msp	Clv. 145/25	57
RZ44	-290	-25.24030	30.08497	1844	Rz	Msp	–	–
RZ18	-286	-25.21730	30.10245	1940	Rz	Msp	–	–
RZ91	-280	-25.21900	30.10086	1928	Rz	Msp	–	–
QZ31	-264	-25.21640	30.10209	1933	Rz	Qz	–	–
QZ28	-252	-25.24750	30.07796	1893	Rz	Qz	–	–
RZ10	-252	-25.21880	30.09972	1929	Rz	Msp	–	–
QZ13	-248	-25.21950	30.09891	1907	Rz	Qz	–	–
D5SB[31]	-248	-25.21430	30.10287	1919	Cz	A	–	–
D9NH	-244	-25.24740	30.07764	1895	Rz	Qz	Bed. 230/35	–
QZ23	-242	-25.21610	30.10128	1932	Rz	Qz	–	–
QZ14	-241	-25.21960	30.09852	1907	Rz	Qz	–	–
QZ15	-233	-25.21970	30.09807	1907	Rz	Qz	–	–
RPEG1	-229	-25.21390	30.10229	1907	Cz	Dcg	–	–
RZ111	-223	-25.21950	30.09777	1902	Rz	Msp	–	–
D7NF[32]	-218	-25.21350	30.10206	1906	Cz	Dcg	Bed. 180/10	73
QZ110	-215	-25.21610	30.10002	1923	Rz	Qz	–	–
QZ113	-214	-25.24740	30.07621	1874	Rz	Qz	–	–
RZ121	-213	-25.21890	30.09779	1911	Rz	Msp	–	–
QZ16	-213	-25.21910	30.09763	1908	Rz	Qz	–	–
QZ17	-211	-25.21880	30.09778	1915	Rz	Qz	–	–
RZ13	-209	-25.21860	30.09780	1909	Rz	Msp	–	–
QZ18	-205	-25.21850	30.09772	1914	Rz	Qz	–	–
D9NI[33]	-204	-25.24670	30.07624	1865	Rz	ScMsp	–	99, 100, 101, 102
QZ19	-200	-25.21860	30.09741	1912	Rz	Qz	–	–
RZ112	-199	-25.24650	30.07613	1865	Rz	Msp	–	–
D7NG[34]	-197	-25.21220	30.10208	1905	Cz	Dcg	Clv. 230/15	72, 73, 74, 75
SCHI1	-193	-25.24620	30.07609	1863	Rz	Sc	–	–
RZQZLEFT	-193	-25.21920	30.09654	1887	Rz	QzMsp	–	–
QZ20	-187	-25.21880	30.09663	1901	Rz	Qz	–	–
RZ14	-185	-25.21870	30.09657	1900	Rz	Msp	–	–
RZ15	-175	-25.21840	30.09636	1895	Rz	Msp	–	–
RZ26	-173	-25.24550	30.07565	1856	Rz	Msp	–	–
MZ11	-166	-25.22800	30.08851	1837	Cz	Pg	–	–
D5SA[35]	-166	-25.21540	30.09820	1903	Cz	UD	–	–
D6SL[36]	-165	-25.21490	30.09851	1892	Cz	Pg	–	–

continued on next page

[29] Paleovertical Pegmatite Dike Photograph, Marginal zone coarsens upsection towards quartzite.

[30] Cleavage.

[31] Amphibole Gabbro. Weakly magnetic, amphibole bearing? Rock clearly rotten.

[32] Pegmatitic gabbronorite. large (5 cm) oikocrystic pyroxene.

[33] Schist (Pebble marker?) well mixed clasts of schists, zones of pink felsic melt.

[34] Pegmatitic gabbronorite. Here magnetic, slightly upsection (∼2 m) rocks non-magnetic. Large (30 cm) blobs of pyroxenite rimmed by anorthosite. Pyroxene/Anorthosite ranging from 1/4 to 4. Included blocks with preserve resistance banding.

[35] Thought it looked like the "dark facies" main zone in the field,. Compare to D25g (porphyritic gabbronorite), Quartzite in float (alluvial)

[36] Porphyritic gabbronorite. In notes, described as main zone porphyritic gabbronorite. However these rocks are structurally beneath chrome layers. Closer inspection reveals that pyroxenes are

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
RZ16	-162	-25.21870	30.09548	1887	Rz	Msp	–	–
QZ22	-161	-25.21810	30.09591	1898	Rz	Qz	–	–
D5NE	-160	-25.21870	30.09537	1883	Rz	Msp	Clv. 125/25	–
RZ17	-154	-25.21950	30.09445	1875	Rz	Msp	–	–
QZ21	-153	-25.21950	30.09444	1875	Rz	Qz	–	–
QZ25	-147	-25.21770	30.09553	1877	Rz	Qz	–	–
D6SG[37]	-146	-25.21840	30.09491	1872	Rz	QzMsp	–	60
D6NK[38]	-145	-25.21500	30.09749	1888	Cz	Pg	Clv. 120/5	–
D7SH[39]	-144	-25.21020	30.10109	1898	Cz	Px	–	78
D6SH[40]	-144	-25.21770	30.09534	1871	Cz	Dcg	–	62
D6SI[41]	-144	-25.21770	30.09534	1871	Cz	Dcg	–	61
QZ111	-140	-25.21760	30.09526	1869	Rz	Qz	–	–
QZ24	-139	-25.21800	30.09487	1869	Rz	Qz	–	–
RPEQ	-135	-25.21480	30.09718	1883	Cz	Dcg	–	–
PEG	-133	-25.21140	30.09968	1893	Cz	Dcg	–	–
D6NJ[42]	-132	-25.21540	30.09655	1880	Cz	Dcg	Clv. 150/10	63
RPEG	-130	-25.21540	30.09644	1877	Cz	Dcg	–	–
RZ34	-124	-25.21910	30.09334	1858	Rz	Msp	–	–
RZ37	-120	-25.24380	30.07447	1840	Rz	Msp	–	–
RZ52	-118	-25.22040	30.09203	1846	Rz	Msp	–	–
D7NI[43]	-100	-25.20893	30.09996	1876	Cz	PgTB	–	79, 119
D7SJ[44]	-100	-25.20890	30.09995	1876	Cz	Pg	–	–
MZPOR	-96	-25.20910	30.09963	1873	Cz	Pg	–	–
CZ3	-94	-25.20880	30.09978	1877	Cz	Ln	–	–
D6NM[45]	-93	-25.21080	30.09820	1868	Cz	Ln	Clv. 233/16	64
MZ5[46]	-87	-25.24180	30.07439	1829	Cz	UD	–	–
D6SN[47]	-86	-25.21040	30.09815	1863	UG2	UG2	–	65, 66
CZ2	-83	-25.21040	30.09804	1865	Cz	Ln	–	–
CZ1	-72	-25.20930	30.09835	1862	Cz	Ln	–	–
D4ND[48]	-72	-25.21690	30.09254	1847	Cz	Pg	Clv. 220/40	–
CZ	-58	-25.20720	30.09933	1884	Cz	Ln	–	–
IS	-52	-25.24350	30.07152	1840	IS	IS	–	–
MZ2[49]	-51	-25.21740	30.09118	1847	Cz	Pg	–	–
CZ5	-48	-25.20630	30.09959	1898	Cz	Ln	–	–

continued on next page

often oikocrystic and large... Incompletely formed Mottled Anorthosite?

[37] Magnetite pegmatite. Host rock magnetic. Pegmatite weathers proud, cutting the rock paleo-up, "sandy" texture in host rock.

[38] Porphyritic gabbronorite.

[39] Sulfide-pyroxenite.

[40] Coarse-grained Feldspathic Pyroxenite. Non-magnetic, Floor of critical zone? "Flatreef"?

[41] Coarse-grained Feldspathic Pyroxenite. Coarser grained rocks here are more rotten.

[42] Pegmatite gabbronorite.

[43] Pyroxene tennisballs in Gabbronorite. Structurally below chromite?

[44] Gabbronorite. small amount of interstitial CPX. Transitional zone?

[45] Fine Layered Anorthosite-norite.

[46] No field notes. Assigned Cz because it is stratigraphically below my modeled TU.

[47] Magnetic chromite seam w/ oikocrystic pyroxenite and anorthosite. abundant blocks. chromite hosted with pure anorthosite and in massive layer with oikocrystic pyroxene. Top layers are magnetic (magnetite-bearing?)

[48] Coarse Gabbronorite. Indirect measurement, bedding planes in a marsh.

[49] Assigned Mz because it is undeformed / overlying Kliprivier synform.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
D7SK[50]	-33	-25.20600	30.09920	1919	TU	Ma	–	81
D11SB[51]	162	-25.22810	30.07313	1803	Mz	Pg	–	–
D11SC[52]	207	-25.22700	30.07188	1806	LMA	Ma	–	–
D11NH[53]	333	-25.19250	30.09242	1917	MMA	Ma	M.L. 130/15	115
D5NF	362	-25.22690	30.06477	1803	Mz	UD	Clv. 240/25	–
D10SJ[54]	373	-25.22700	30.06419	1808	MMA	MaBap	Clv. 220/15	103–110
MZ1	377	-25.22870	30.06271	1797	Mz	Pg	–	–
MA	378	-25.22920	30.06219	1767	MMA	Ma	–	–
DS	379	-25.19100	30.09140	1903	Dl	Dl	–	–
D8SI[55]	398	-25.18210	30.09749	1981	UMA	MaFgog	Clv. 220/20	90
D8SH[56]	413	-25.18210	30.09678	1984	Dl	Dl	Cont. 185/25	89
D11NF[57]	415	-25.19710	30.08507	1896	Mz	Lg	M.L. 130/15	113
MA1	423	-25.22600	30.06264	1805	MMA	Ma	–	–
D10SI[58]	434	-25.22550	30.06250	1801	MMA	MaBap	–	–
D8NJ	446	-25.18260	30.09486	1979	Mz	Hg	Clv. 230/15	–
D11SG[59]	451	-25.19410	30.08572	1911	UMA	Ma	–	114, 117
RESB1ND	454	-25.18230	30.09475	1983	Mz	Hg	–	–
D5SG[60]	509	-25.21960	30.06343	1789	UMA	MaPxFgog	–	–
D3SG (D4NC)[61]	564	-25.22740	30.05494	1780	UMA	MaPxFgog	–	49, 53, 54
D11NI[62]	622	-25.19160	30.07954	1865	Mz	Lg	Clv. 240/15	–
MODALLAYER	687	-25.18070	30.08540	2054	Mz	Lg	–	–
D8SG[63]	708	-25.18270	30.08279	2022	Mz	Sa	–	87
MZD	739	-25.23480	30.04130	1790	Mz	Hg	–	–
MODAL LAYER	743	-25.19140	30.07401	1828	Mz	Lg	–	–
IS2	865	-25.19950	30.06205	1781	IS	IS	–	–

continued on next page

[50]Mottled Anorthosite. Non-magnetic. Interpreted as proximal Merensky Footwall based distance from UG2.

[51]Porphyritic Gabbronorite. Rotten. Initially confused for mottled anorthosite.

[52]Mottled Anorthosite. Lower Mottled anorthosite; outcrop source of James' favorite boulder.

[53]Large cliffside mottled anorthosite outcrop. Modal Layering.

[54]Mottled anorthosite with blebby/blocky recrystallized pyroxenite. Big Outcrop, undulating contacts of MA, often dipping east. Sometimes appear discordant, though others appear concordant: Can be explained by thermal erosion, deposition of surface with meter-scale hummocks and swales.

[55]Mottled Anorthosite. Interpreted as LMA. Magnetic mottles. Includes fine-grained gabbronorite with large anhedral CPX phenocrysts (<5 cm)

[56]Dolerite Sill. Strongly magnetic, with well-developed fine blocky cleavage from stress field. Inherited phenocrystic blocky 1 cm plagioclase with strong oscillatory zoning. Likely later intrusive, not necessarily concordant.

[57]Layered/porphyritic gabbronorite. Fine modal layering, near "peace" rock.

[58]Mottled anorthosite with blebby/blocky recrystallized pyroxenite. MMA.

[59]Mottled Anorthosite. Very large magnetic mottles (<10 cm) with large (< 0.5 cm) magnetite.

[60]Mottled Anorthosite. Float, outcrop in sight. Didn't want to trespass, Also found green fine grained rock with large anhedral CPX.

[61]Mottled Anorthosite over strongly graded pyroxenite-anorthosite over fine grained Gabbronorite with large (<5 cm) anhedral CPX. Gave a ride to an African man walking home from work at Kliprivier farms who spoke no English. Stopped by an outcrop and sampled in the dark. Vantongeren maps as Upper Mottled anorthosite.

[62]Finely Layered Gabbronorite. Good cleavage, Mottled Anorthosite Nearby?

[63]Sheared Spotted Anorthosite. Strongly foliated, suggesting co-crystallization shearing. Biggest mottles: ∼3cm x 0.5 cm. Interpreted as MMA by RL et al., 2022.

Table A.1 (continued)

Name	Ht.	Lat.	Lon.	Elev.	Unit	Lith.	Structure	Photos
D11SE[64]	867	-25.18590	30.07260	1886	Mz	Lg	–	112
D11ND[65]	893	-25.18530	30.07186	1891	Mz	SaLg	–	111
D8SC[66]	1030	-25.17670	30.07222	1968	Mz	Sa	M.L. 140/25	86
D8NE	1035	-25.17780	30.07114	1955	Mz	UD	M.L. 130/10	–
XENOLITH[67]	1035	-25.17860	30.07051	1946	Mz	Sa	–	91
UMA2	1035	-25.17640	30.07222	1967	Mz	Sa	–	–
D8ND[68]	1036	-25.17900	30.07014	1936	Mz	Lg	M.L. 110/10	85
D8ND[69]	1036	-25.17900	30.07014	1936	Mz	Hg	Clv. 210/10	–
D8NA[70]	1244	-25.17050	30.06651	1814	Mz	Hg	–	84
D6SE[71]	1417	-25.21910	30.02170	1771	Mz	A	–	–
D6SF[72]	1417	-25.21910	30.02170	1771	IS	IS	–	59
D10NC	1468	-25.20220	30.03204	1755	Mz	Hg	Clv. 180/10	–
MZ31	1473	-25.20990	30.02600	1756	Mz	Hg	–	–
D10NB[73]	1475	-25.20090	30.03275	1781	Mz	Hg	Clv. 200/10	–
D6NB[74]	1551	-25.21840	30.01618	1816	Mz	Hg	Clv. -74/6	58
D10ND	1561	-25.19970	30.02966	1772	Mz	Hg	Clv. 225/15	–
MZD1	1562	-25.21950	30.01472	1780	Mz	Hg	–	–
D10NA	1582	-25.19910	30.02914	1767	Mz	Hg	Clv. 200/10	–
MZD2	1623	-25.22540	30.00751	1809	Mz	Hg	–	–
MZ4	1778	-25.19310	30.02448	1733	Mz	Hg	–	–
D10NH1[75]	2050	-25.18090	30.02113	1746	IS	IS	–	–
D10NE	2056	-25.17710	30.02374	1757	Mz	Hg	Clv. 130/10	–
MZ12	2057	-25.17750	30.02334	1740	Mz	Hg	–	–
D10NH[76]	2136	-25.17770	30.01951	1729	IS	IS	–	–
MZ21	2141	-25.17190	30.02370	1746	Mz	Hg	–	–
D10NG[77]	2175	-25.17710	30.01813	1726	Mz	A	–	–
D10NF	2182	-25.17820	30.01698	1730	Mz	Hg	Clv. 225/15	–
D6ND[78]	2682	-25.21330	29.96669	1526	Mz	Hg	Clv. 160/13	–
D6NC[79]	3266	-25.19900	29.95029	1476	Mz	Hg	Clv. 202/12	–

[64] Finely layered gabbronorite. Mottled anorthosite itself is in float here, but graded rocks with blocks are in place.. Mottles are small (<3mm) and non-magnetic.

[65] ∼20 cm Graded spotted anorthosite with irregular contacts and variably recrystallized pyroxenite blocks/blebs. Proper mottled anorthosite in float; non-magnetic. In field interpreted as Upper/main mottled anorthosite (and related package).

[66] Spotted Anorthosite. ∼8cm thick, with prominent igneous layering and including small amounts of pyroxenite below. Interpreted as UMA. I would not call this a mottled anorthosite, the pyroxenes are do not immediately appear oikocrystic.

[67] A second xenolith near photo 91.

[68] Layered Gabbronorite. Centimeter-scale igneous layering with variable color and grain size.

[69] Homogeneous gabbronorite. Different bedding in nearby unlayered rocks...

[70] Porphyritic gabbronorite. Blebby Xenolith noted and photographed.

[71] Green gabbronorite with amphibole "mottles". Amphibole mottles (<5 cm) are strongly imbricated, undulose shine, not perfectly optically continuous.

[72] Ironstone concretion. <2 cm Goethite nodules enclosing limonite cores with Hematite-Limonite matrix. Rounded clastic quarts <2 mm, in float.

[73] Included a range of measured strikes from 250-315, on fault?

[74] Measured another dip (slump?) indirectly from road: 050:30.

[75] Ironstone Concretion in place.

[76] Ironstone Concretion in float.

[77] Amphibole 'mottled' gabbronorite in float. Fault?

[78] Dark and Light Facies.

[79] Dark Facies.

A.2 Annotated Field Photographs

Selected field photographs are reproduced below with their captions, two to a page, in numerical order; photographs taken in portrait orientation follow the others. Photographs keep their field numbers (DrBVn). Photo 60 is shown with an interpretation of its deformation history in Figure 5.8. The remaining field photographs, with their captions, are in the supplementary photo album. Photo numbers in Chapter 5 and in Table A.1 link to the photograph when it is reproduced in this thesis, and are printed without a link when they point to the supplement.

Photo 2. Decimeter scale lenticular inclusions of coarse-grained Anorthositic-Gabbro in magnetic, fine-grained Marginal Sill Phase. Clinopyroxene in inclusions is oikocrystic and up to 3 cm. Found alongside spheroidal pyroxenite aggregates of photo 4.

Photo 4. Pyroxenite Aggregates (colloquially referred to in the literature as "Tennis Balls") 1-5 cm in diameter within fine-grained marginal zone gabbronorite. Pyroxenite aggregates have not been observed before this study outside of coarsely crystalline plutonic rocks, and only recently have authors argued that they could be hosted within the Marginal Sill Phase (A. A. Mitchell and Scoon, 2023). This study offers multiple unambiguous examples of pyroxenite aggregates hosted within fine-grained, magnetic, marginal sill phase.

Photo 49. A roadcut exposure of ∼30 cm's of the UMA, the Upper Mottled Anorthosite, near the village at Draaikraal, south of the Draaikraal Lineament. The mottled anorthosite has an irregular, non-planar lower contact with a coarse-grained feldspathic pyroxenite. The UMA package has a characteristic assemblage of 3 distinct rocks types: Mottled anorthosite, a coarse-grained feldspathic pyroxenite, and a fine-grained green gabbronorite with blebby <2 cm clinopyroxene oikocrysts that weakly define modal layering (not pictured here). These distinct lithologies facilitate correlation with UMA to the north of the Draaikraal Lineament.

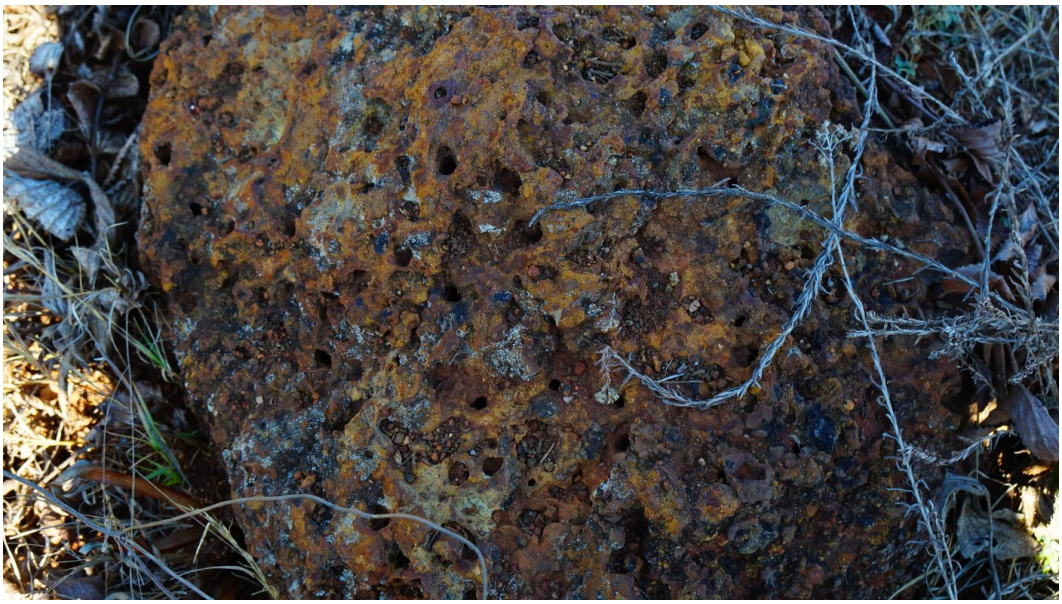


Photo 59. A closeup of a laterite "ironstone" of unit IS. The ironstones are composed of 1-2 cm diameter goethite nodules often surrounding voids or porous-powdery limonite. The rock is cemented with a limonite-hematite matrix, and contains up to 20% clasts of Steenkampsberg quartzite alluvium up to ∼ 1 cm in diameter. This rock type commonly hosted nests of ants.

Photo 61. A specimen containing interesting features commonly mapped as "Disturbed Gabbronorite" here, which generally refers to coarsely crystalline gabbronorites immediately stratigraphically above the deformed floor rocks of the Kliprivier Syncline. Large, Euhedral OPX grains are cemented with plagioclase and clinopyroxene oikocrysts, locally differentiated into bands and regions of pyroxenite or OPX phenocrysts in an anorthositic matrix, somewhat resembling the "porphyritic gabbronorite". Such rocks may record the mechanical separation of pyroxenite and plagioclase to form pyroxenite spheroidal aggregates or "tennis balls". The lens cap is 5.5 centimeters across.

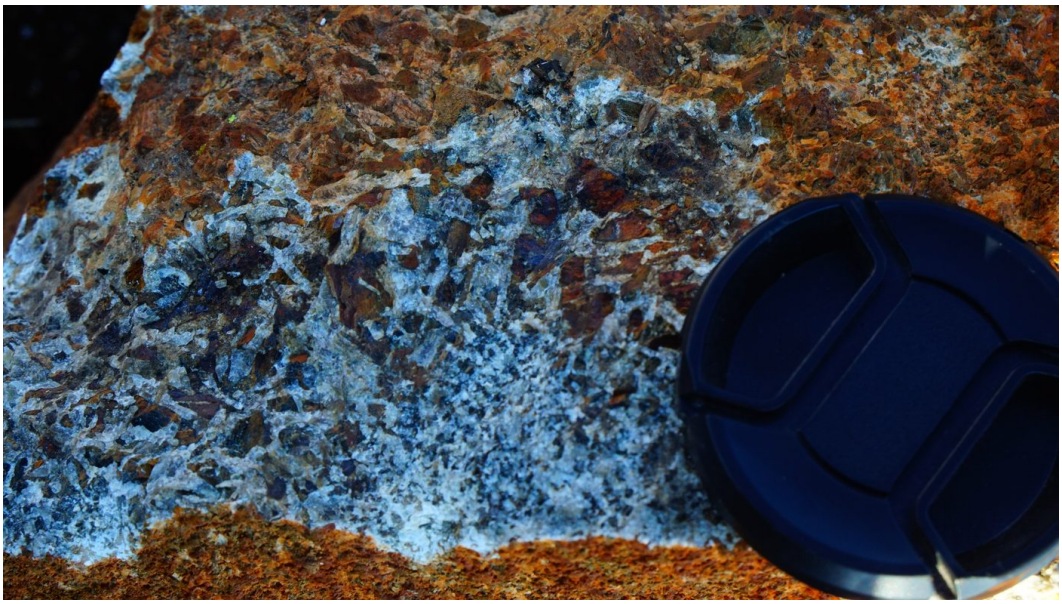


Photo 62. A freshly broken face of coarsely crystalline disturbed gabbronorite. Crystal size varies greatly throughout the hand sample, with some regions dominated by cm-scale crystals and some dominated by mm-scale crystals, suggesting perhaps that these rocks sustained channelized fluid flow in the finely crystalline regions after bulk crystallization. Even in a freshly broken face, the OPX faces appear to have a rusty discoloration. Several elongate feldspar inclusions are visible in the OPX phenocrysts.

Photo 65. Anastomosing cm-scale chromitite bands hosted within mottled anorthosite. The UG-1 layer consistently contains strings of chromitite hosted within mottled anorthosite just below the massive chromitite (Voordouw et al., 2009; Mukherjee et al., 2017). The most famous locality of UG-1 is at Dwars River. Here the oikocrystic pyroxenes are green when fresh, weathering to a rusty red color. A few isolated bands of chromite near the top are quite magnetic, clearly containing significant magnetite. While magnetite spinel is generally absent in the critical zone, including the chromitites, magnetite is common in the Platreef (Kinnaird et al., 2005), the basement-contaminated PGE-rich north limb corollary of the critical zone's Merensky Reef. Because the upper critical zone at Kliprivier Farm also formed at the chamber floor, it may be similar in character and PGE grade to the Platreef (van der Merwe, 2007).

Photo 66. Massive (∼20 cm) non-magnetic chromitite with deformed blocks of anorthosite, preferentially lying along the bedding planes of the area. The lens cap is 5.5 cm. Anorthosite blocks appear to have been ductilely deformed, perhaps by entrainment within a hot chromite slurry. Like the chromitites hosted within the anorthosites, there are oikocrystic pyroxenes which appear to have crystallized late. The groundmass is chromitite with an abundant feldspar matrix, perhaps only 50% cumulus chromite, 30% interstitial plagioclase, 20% oikocrystic pyroxenes. Particularly in the top right corner, one can see a striking increase in modal chromitite approaching the boundary of the entrained anorthosite block. Despite the last photo resembling UG-1 stringers, it is structurally above this massive chromitite (UG-1 stringers are below: Voordouw et al., 2009; Mukherjee et al., 2017). The orange color on the surface is lichen.

Photo 72. Blobby aggregates of pyroxene of the coarsely crystalline Disturbed Gabbronorite (Dcg) of the Kliprivier Monocline. The orange magnet is the size of a typical pen, ∼13 cm long. Could these be precursors to spheroidal aggregates (SAs)? Photo 72 shows a corona of pyroxene depletion surrounding the pyroxene aggregate.

Photo 76. Decimetric xenolith of fine grained marginal zone rock within Disturbed Gabbronorite. The orange magnet is the size of a typical pen, ∼13 cm long. The xenolith exhibits strong resistance banding that is subtle or absent in the coarse grained disturbed gabbronoritic host, and the contacts between the two are sharp. Interestingly, aggregates of pyroxenes have glommed to the rims of the xenolith.

Photo 77. Decimetric xenolith of fine grained marginal zone rock within Disturbed Gabbronorite. The orange magnet is the size of a typical pen, ∼13 cm long. This xenolith has a comparatively diffuse contact with the disturbed gabbronoritic host relative to that of photo 76, perhaps suggesting that it had been incorporated within the magma for longer, or had been reheated more after entrainment.

Photo 86. A thin (<15 cm) band of spotted anorthosite in the mid-lower Main Zone, exposed in weathered and clean faces. The hammer head is about ∼5 cm in width. The spotted anorthosites were much harder to spot in the field than the 3 dramatic packages of mottled anorthosites found in the valley and further downsection, both above and below the Draaikraal Lineament. Latypov et al. (2022) correlated this layer to the Upper Mottled Anorthosite (UMA), and assumed it was laterally continuous with the rocks of photos 49, 53, and 54 south of the Draaikraal Lineament. While I agree that photos 49, 53, and 54 show the UMA, I instead correlate the UMA above the lineament much further to the east (photo 90), where I identified the characteristic fine-grained, bright green gabbronorite with blebby, anhedral phenocrystic (<2 cm) CPX that I had seen nowhere else but as footwall to the UMA. I also interpret photos 114 and 117 as belonging to the UMA.

Photo 87. Similarly to photo 86, this rock was identified as a "mottled anorthosite" by Latypov et al. (2022) and was correlated to the MMA marker bed. The pen used for scale is about 1 cm in diameter. I instead refer to this unit again as a "spotted anorthosite" since it bears none of the striking features of the MMA seen in photos 93 and 103–109 exposed to the south of the Draaikraal Lineament. I argue that the true corollary to the MMA north of the Draaikraal Lineament is much further east, (see the thick mottled anorthosite package in photo 115).

Photo 90. Coarse mottled anorthosite, with a ∼10 cm wallet for scale. Unfortunately, I did not have the foresight to photograph the unique footwall to these anorthosites in place: the fine-grained, bright green gabbronorite with blebby, anhedral phenocrystic (<2 cm) CPX that I found only here and at Draaikraal (at the outcrop in photos 49, 53, and 54) and in the float traced along strike north of that outcrop. I did sample it, however (D3SG).

Photo 95. This a particularly striking example of resistance banding in fine-grained marginal zone sill phase float. The hammer head is ∼10 cm across. This feature was immediately noticeable in the fine-grained magnetic marginal zone rocks, but I didn't photograph it until later, assuming at first that the banding was a secondary, unimportant textural feature. The ubiquitousness of these bands in fine-grained marginal zone rocks and in xenoliths within the plutonic RLS. These bands are most often subparallel and vertical planes. Similar resistance banding is argued to form in sedimentary rocks by localized cataclasis and recrystallization along fault planes during brittle deformation, decreasing the porosity of the affected plane to become more resistant to later chemical weathering (R. E. Hill, 1990). While it is tempting to ascribe the formation of these bands to subsolidus (or near-subsolidus) deformation during isostatic flexure of the cooling RLS, these bands' formation must predate the intrusion of the Main Zone of the RLS, which includes xenoliths of fine-grained Marginal Zone rocks containing these bands (photo 91). An SEM/petrographic investigation of these bands would be helpful to understand their origins and implications.

Photo 96. Two ∼2 cm spherical pyroxenite aggregates found in a leucocratic Marginal Sill Phase outcrop, and a single ∼5 cm pyroxenite aggregate with a long (∼10 cm), thin pyroxenite tail. The hammer's handle is ∼4 cm wide, for scale. This feature is interpreted similarly to a diapir, implying that the pyroxenite aggregate was ductily deformable and migrating through a comparatively rigid/viscous crystal mush. The requirement that the pyroxenite aggregate be ductily deformable is inconsistent with the model of Khoza (2021), in which preexisting and relatively cool pyroxenite layers are brecciated by an intruding (gabbro)noritic magma, but is consistent with the models of C. A. Lee and Sharpe (1979), in which pyroxenite is preferentially wetted by a secondary, immiscible bridging liquid; and A. A. Mitchell and Scoon (2023), who argue that pyroxenite SAs are formed similarly to mafic enclaves in granitoids, by disaggregation of an incoming pyroxenitic magma within a pre-existing, comparatively viscous/rigid (gabbro)noritic host.

Photo 98. Anastomosing, non-planar fine-grained anorthositic (?) veins in the fine-grained Marginal Zone. Similar to veins observed in photo 6. The hammer head is ∼10 cm across. Anorthositic veins include blocks of host fine-grained Marginal Zone rocks with indistinct, irregular contacts. Such veins may represent incompletely formed anorthositic layers formed by reactive porous flow of a magma or hydrothermal fluid sourced from dehydrating/melting country rocks. Note the presence of modal layering within the anorthosite (?) of photo 98.

Photo 100. Highly migmatized Houtenbek shales in weathered and fresh faces. The hammer head is ∼10 cm across. Deformed, fractured, and melted metasedimentary rocks swim in abundant, pink, leucosomal crystalline granite. Partial melt of crustal rocks was generated in-situ within the sedimentary floor rocks of the Bushveld, and it is able to effectively separate from the source rocks. Some amount of melts like these likely migrated upward to contaminate the plutonic RLS and Marginal Zone rocks. This contamination effect would be strongest for magmas near the intrusion floor.

Photo 109. Mottled anorthosite (MMA) south of the Draaikraal Lineament. The hammer head is ∼10 cm across. The interior of a pyroxenite slab separating two layers of mottled anorthosite (as pictured in photo 108) is well exposed by high-angle fracture. The interiors of these pyroxenite slabs are finer grained and more anorthosite rich than the exteriors. In general, I observe coarse grains where mineral phases are well-separated, such as in mottled anorthosites or at anorthosite-pyroxenite interfaces.

Photo 111. ∼ 12 cm bed of Spotted anorthosite north of the Draaikraal Lineament, upsection of the UMA. The hammer head is ∼10 cm across. A blocky, melanocratic xenolith/autolith is included near the middle of the bed, and there is another blobby pyroxene-rich blob mantled by pure anorthosite at the bottom of the bed. The spotty Pyroxenes do not appear oikocrystic. They are larger near the middle of the spotted anorthosite and generally absent near the bottom. The contacts with the host gabbronorite are indistinct and irregular.

Photo 29. A Well-foliated hornfelsized inclusion of Houtenbek shale/dolomite within the Marginal Sill Phase on the Kliprivier Synform. The foliation shows clear plastic deformation, indicative of synmagmatic (hot) deformation. While previous authors have argued that the Kliprivier synform preceded Bushveld magmatism (van der Merwe, 2007), this work best supports the interpretation that deformation the RLS margin on Kliprivier Farm was synmagmatic.

Photo 53. The same UMA outcrop as pictured in 49. The UMA package includes a striking mottled anorthosite above a coarse feldspathic pyroxenite with cumulus textures and decimeter-scale anorthosite inclusions with indistinct contacts with the host pyroxenite. Most of the anorthositic inclusions within the pyroxenite are roughly situated in a horizon ∼15 cm above the base of the feldspathic pyroxenite layer, but one (especially indistinct) inclusion is near the upper contact and is nearly continuous with overlying mottled anorthosite. The two holes in the outcrop were likely from explosives for clearing the outcrop from the road, but may also have been from sampling from previous paleomagnetic or geochemical studies: they are ∼ 4 cm across for scale.

Appendix B

THE BUSHVELD COMPLEX: THE STATE OF KNOWLEDGE AND AREAS OF ONGOING DEBATE

The Paleoproterozoic Rustenburg Layered Series (RLS) of the Bushveld Magmatic Province in northeast South Africa is the largest layered igneous intrusion in the world, and contains the world's largest source of platinum group elements as well as important reserves of chromium, vanadium, titanium, and iron. It is exceptionally well-preserved, and has been the subject of intense study and debate for generations. Studies focusing on the intrusion's margin have been especially fruitful in informing geologists' understanding of igneous petrogenesis and intrusion mechanisms, modes, and tempos of layered mafic intrusions and plutonic rocks more generally.

B.1 Layered Mafic Intrusions

Layered mafic intrusions (LMIs) are compositionally zoned formations of mafic rocks characterized by distinct millimeter- to meter-scale layering of igneous minerals. Traditionally, the layering and chemical zonation of these geologic marvels have been attributed to processes such as fractional crystallization and gravitational settling within a magma chamber (Wager et al., 1960; Charlier et al., 2015). Notable examples of LMIs include the Stillwater (2.71 Ga) in Montana, USA; Great Dyke (2.57 Ga) in central Zimbabwe; the Bushveld's layered series (2.06 Ga) in South Africa's Limpopo, Mpumalanga, and Gauteng provinces; the Muskox intrusion (1.27 Ga) in Nunavut, Canada; the Duluth Complex (1.01 Ga) in Minnesota, USA; the Sept Iles intrusion (565 Ma) in Quebec, Canada; Dufek (189 Ma) in Antarctica; the Rum intrusion (60.5 Ma) in the Inner Hebrides, Scotland; and the Skaergaard intrusion (55.4 Ma) in East Greenland. Despite their significance as supposedly simple natural laboratories for studying the chemical evolution of magmas, numerous aspects of their formation remain subject to debate, even including the presence of significant liquid magma in their formation (W. Kruger and Latypov, 2020; Mungall et al., 2016; Latypov et al., 2022; Yao et al., 2021).

Experimental observations have revealed that plagioclase flotation occurs in mafic magmas at depths corresponding to the Earth's crust (Campbell et

al., 1978). This challenges the notion that gravitational settling alone is responsible for igneous layering. Instead, chemical means of phase separation are often suggested, referencing processes like periodic nucleation during in-situ crystallization (McBirney and Noyes, 1979), in-situ crystallization of a magma with compositional layering developed and sustained through double-diffusive convection (Huppert and Turner, 1981), or preferential dissolution and recrystallization of existing unstructured crystal cumulates or mushes to minimize crystal surface energy perpendicular to temperature gradients (A. Boudreau, 2011) or by preferential attack by chlorine-rich fluids (Sonnenthal, 1992). An alternative mechanical explanation, seismically aided hydrodynamic sorting within a tilting, crystal-rich cumulate pile, has also enjoyed support (Maier et al., 2013).

B.2 The Bushveld Event and RLS

The Rustenburg Layered Series (RLS) is renowned as the world's largest layered mafic intrusion, distinguished by its remarkably continuous layers, sometimes across hundreds of kilometers (Cawthorn, 2015; Latypov et al., 2022; Scoon and Teigler, 1994). Within the RLS, pegmatoid "reefs" and chromitites host the world's largest reserves of platinum group elements or PGEs (C. A. Lee, 1996). In addition to the RLS, the Bushveld event encompasses several satellite intrusions, including the Uitkomst Intrusion, Phalaborwa Carbonatite, the Stavoren Granophyre at the top of the RLS, the massive felsic Rooiberg lava eruption covering 200,000 km^3, and the world's largest A-type granitic pluton at the intrusion's roof: the Lebowa granite suite (M. Hill et al., 1996; Kleemann and Twist, 1989; Scoates et al., 2021; Walraven, 1985).

The mafic plutonic rocks of the RLS are subdivided into the Lower Zone, Critical Zone, Main Zone, and Upper Zone, with unlayered sills and chill sequences at the margin known as the Marginal Zone (von Gruenewaldt et al., 1985). These zones are primarily distinguished by their mineralogy, which transitions from more primitive assemblages in the Lower Zone to more evolved ones in the Upper Zone. The Lower Zone uniquely features dunites and harzburgites, the Critical Zone features chromitites and norites, the Main Zone features gabbronorites, and lastly the Upper Zone features diorites and magnetitites. The RLS exhibits significant compositional zoning, with increasing iron, silica, vanadium, and alkali content as well as chromium and magnesium depletion toward the intrusion's roof (Cawthorn, 2015). While the RLS often maintains remarkable

lateral continuity on the kilometer scale, variations in stratigraphy, PGE mineralization, and thickness sometimes present challenges to stratigraphic correlations (Ashwal et al., 2005; McDonald et al., 2005; Scoon and Teigler, 1994).

B.3 Structure and Relationship to Country Rocks

The layered series outcrops as distinct lobes on the present-day surface, with variations in stratigraphy, PGE mineralization, and mineralogy observed across limbs. These lobes are separated and influenced by geological features like the Thabazimbi-Murchison and Steelpoort lineaments (du Plessis and Walraven, 1990; Ashwal et al., 2005; McDonald et al., 2005; Scoon and Teigler, 1994). Despite these variations, there is remarkable continuity between layers. Chromitite layers with similar thickness and chemical signatures can be correlated across limbs separated by hundreds of kilometers of felsic igneous rocks and sedimentary cover. Additionally, crustal thickness models derived from seismic receiver functions and gravity anomalies support the possibility that the intrusion is connected at depth (Webb et al., 2004; Cawthorn and Webb, 2001). Some researchers have further classified the intrusion into “distal” and “proximal” facies based on the hypothesis that the intrusion's feeders exploited the Thabazimbi-Murchison Lineament, supported by observations such as the best development of ultramafic zones closest to this lineament (Scoon and Teigler, 1994; B. M. Clarke et al., 2000).

The RLS generally dips inward toward its center, with the most significant dips occurring at the margin (Button, 1976; Frick, 1967). While the RLS was historically considered a lopolith (H. D. B. Wilson, 1956), paleomagnetic studies favor an initially horizontal orientation that flexed isostatically after crystallization ceased (Hattingh, 1986). The floor of the RLS exhibits synmagmatic deformation, possibly resulting from interactions during the coalescence of offset interfingered sills (Pollard et al., 1975; B. Clarke et al., 2009). Abundant large sedimentary inliers within the complex can also be explained by the inhomogeneous propagation of finger sills along the complex's margin (B. M. Clarke et al., 2000; B. Clarke et al., 2009).

B.4 Isotopic and Volatile Constraints on the RLS

Radiogenic Isotope Stratigraphy of the RLS

F. J. Kruger (1994) reported a comprehensive strontium isotope stratigraphy of the RLS, revealing radiogenic Sr isotopes indicative of varying degrees of crustal contamination throughout the intrusion. This stratigraphy differentiates the Lower, Critical, and lower Main Zones from the upper Main Zone and Upper Zone. The lower portion of the intrusion records highly variable Sr isotopes, interpreted to arise due to repeated magma replenishments in an open system. F. J. Kruger (1994) refers to this zone as the "Integration Stage." Conversely, the middle Main Zone displays fairly consistent isotopic compositions, transitioning to more primitive values at the Pyroxenite Marker—a well-known bed in the upper Main Zone signifying a shift towards more primitive magmas (Cawthorn et al., 1991). These two stratigraphic regions with internally constant initial strontium isotope ratios are referred to as the "Differentiation Stage" and could possibly have evolved through closed-system fractional crystallization on the basis of strontium isotopes. Studies on neodymium isotopes within the RLS are less detailed than those available for strontium isotopes, but show a broadly similar pattern that requires crustal contamination throughout the entire complex, with the Main and Upper Zones requiring more radiogenic (crustal?) contamination than the Lower and Critical Zones (Maier et al., 2000).

Strontium isotope disequilibrium is well-known within the Merensky and Bastard reefs, where pyroxenes and plagioclase have different and variable initial strontium isotope values (Seabrook et al., 2005; Cawthorn, 2015); however, strontium isotope disequilibrium is not unique to the Merensky and Bastard reefs. Chutas et al. (2012) demonstrated that plagioclase throughout the Lower and Critical Zones was isotopically zoned with primitive cores and evolved rims, and that coexisting pyroxenes and plagioclase were not in isotopic equilibrium. Roelofse and Ashwal (2012) similarly observed significant strontium isotopic disequilibrium in lower Main Zone rocks. This means that crystalline phases from the entire Lower, Critical, and lower Main Zone cumulates interacted with exotic magmas or hydrothermal fluids after substantial crystallization, either by later intrusion of an exotic magma into a preexisting crystal mush at the level of the RLS (Chutas et al., 2012) or by intrusion of xenocryst-bearing crystal-rich magmas derived from a subcompartmentalized chamber at depth (Roelofse and Ashwal, 2012). Any rock in the Critical, Lower, and lower Main Zones

was thus formed by multiple magmatic ± hydrothermal events/environments, and any bulk isotopic or chemical measurement of these zones will inherently integrate over multiple crystallization episodes.

A comprehensive dataset of rhenium-osmium (Re-Os) isotopes provides valuable insights into the history of platinum group elements (PGEs) within the Bushveld Complex. Schoenberg et al. (1999) measured a 2.043 ± 0.011 Ga isochron age for the Bastard Reef, which contained little common osmium. Furthermore, they report a systematic trend in the initial $^{187}Os/^{188}Os$ ratios within chromitites spanning the Critical Zone. These ratios exhibit a distinct progression, starting from values closely resembling chondritic compositions and escalating towards highly radiogenic values as one moves upwards through the stratigraphic sequence reaching their most radiogenic values at the Merensky Reef and upper group (UG) chromitites (Schoenberg et al., 1999; McCandless et al., 1999).

The presence of heterogeneous initial $^{187}Os/^{188}Os$ values within chromitites is commonly invoked to support the argument for an orthomagmatic origin for PGE mineralization in the upper Critical Zone, arguing against a hydrothermal origin (Schoenberg et al., 1999; McCandless et al., 1999). However, the validity of the argument depends upon whether Re-Os isotopes faithfully record magmatic processes. Notably, experimental evidence has demonstrated that osmium diffuses rapidly within sulfides, displaying a closure temperature similar to the closure temperature of argon within biotite (Brenan et al., 2000). Ar-Ar thermochronology of biotite therefore can be used in this case as a proxy to determine the timing of the closure of the Re-Os system. Nomade et al. (2004) reported an average Ar-Ar age of 2.042 ± 0.003 Ga for biotite within the UG2 chromite, which agrees well with the Re-Os isochron age of the Bastard Reef. Furthermore, this congruence extends beyond the Bushveld Complex, as similar Ar-Ar ages are observed on a regional scale within syn-kinematic micas found within low-grade phyllitic metasediments of the Transvaal Supergroup (Alexandre et al., 2006). These Ar-Ar ages effectively date regional compression within the Kaapvaal craton, including deformation into fold and thrust belts (Alexandre et al., 2006). Consequently, it can be inferred that the Re-Os isotope system underwent diffusive reequilibration within the chromitites approximately 10 million years after the cessation of magmatism within the RLS, which ended at 2.055 Ga (age derived from Scoates et al., 2021; Zeh et al., 2015).

The most enigmatic aspect of the Re-Os isotope geochemistry of the RLS

centers around the coexistence of two unusual, isotopically distinct sulfide populations within the Merensky Reef. A substantial population of laurite (RuS_2) exhibits the most radiogenic initial $^{187}Os/^{188}Os$ values of the entire 8-kilometer stratigraphy of the RLS, while the same hand sample contains erlichmanite (OsS_2) with an osmium isotopic ratio mirroring a chondritic value, the least radiogenic $^{187}Os/^{188}Os$ within the entire RLS (Hart and Kinloch, 1989). It is difficult to imagine how sulfides with such different $^{187}Os/^{188}Os$ values were able to coexist within the same rock without isotopically equilibrating.

In a broader geological context, Richardson and Shirey (2008) conducted measurements of sulfide Re-Os isotopes, as well as silicate Rb-Sr and Sm-Nd isotopes, within kimberlitic diamond inclusions in close proximity to the Bushveld Complex. Their aim was to constrain the isotopic reservoirs within the subcontinental lithospheric mantle (SCLM) beneath the Bushveld Complex. Their findings reveal the enrichment of radiogenic isotopes across all three isotopic systems within the SCLM, which they use to argue that the radiogenic isotopic signatures observed within the RLS can be explained as a three-part mixture of mantle reservoirs that requires no contributions from the crust, instead requiring only the harzburgitic SCLM, eclogitic SCLM, and the depleted, convecting mantle.

Stable Isotopes of the RLS

Oxygen isotopic studies show that oxygen within the RLS was equilibrated at magmatic temperatures, remarkably unaffected by interactions with surface-derived fluids (Schiffries and Rye, 1989; Harris et al., 2005). Early studies indicated magmatic temperatures for the Upper Zone equilibrating at 1000°C, with the rest of the intrusion at 1200°C (Schiffries and Rye, 1989). However, later studies produced scattered equilibration temperatures as low as 550°C (Harris et al., 2005). Oxygen isotopes within the RLS exhibit remarkable consistency throughout the intrusion and are slightly heavier (~8‰ $\delta^{18}O$) than mantle values (~6‰ $\delta^{18}O$) suggesting substantial but well-mixed contamination from surface-derived material, possibly from Transvaal Supergroup rocks exposed on the surface or from unexposed, isotopically heavy rocks at depth (Schiffries and Rye, 1989; Harris et al., 2005). Similarly, multiple-sulfur isotopes identified a well-mixed and uniform non-zero mass independent fractionation (MIF) of $\Delta^{33}S$ (Penniston-Dorland et al., 2012) across the RLS, requiring a component of surface-derived sulfur that was very efficiently mixed

throughout the entirety of the RLS. The remarkable homogeneity in volatile element isotopes throughout the RLS contrasts with heterogeneous strontium and neodymium isotopes throughout the intrusion, alluding to a decoupling of magmatic and hydrothermal material sources within the RLS. The volatile element isotopic data require a single source for late-stage hydrothermal fluids across the complex.

Hydrogen isotopes indicate late-stage hydrothermal metasomatism at low water/rock ratios (Schiffries and Rye, 1990). Carbonated fluid within the RLS imparted a carbon isotope signature likely arising from the devolatilization of Transvaal carbonates (Schiffries and Rye, 1990). Sedimentary country rocks display oxygen and carbon isotopic depletions correlating with increasing metamorphic grade and proximity to the RLS margins (Buick et al., 2000; Schiffries and Rye, 1990). The lack of complete isotopic equilibration of metasedimentary country rocks and RLS fluids suggests that fluids flowed from devolatilized metasedimentary rocks into the RLS, and not vice versa (Buick et al., 2000).

Unusual Volatile-bearing Phases in the Lower RLS

Within the Lower and Critical Zones of the RLS, volatile-bearing phases such as apatite, amphibole, and phlogopite are rare but often contain exceptionally high chlorine concentrations (A. E. Boudreau et al., 1986; Willmore et al., 2000). These phases are most concentrated in the lateral Merensky Reef and the vertical metasomatic Driekop dunite pipe, often associated with magmatic graphite and platiniferous sulfides (Ballhaus and Stumpfl, 1985; Schiffries, 1982). Thermodynamic calculations suggest that the initial volatile phase exsolved from RLS magmas was predominantly hydrochloric acid (HCl) and chloromethane (CH_3Cl), with minor amounts of water and sulfur (Mathez et al., 1989). This fluid likely exsolved sulfur during graphite crystallization. Chlorine isotopic studies of the RLS show that these Cl-rich fluids were not derived from the local Transvaal Supergroup, but are consistent with an origin from SCLM charged with volatiles from dehydrating subducting altered oceanic crust (Willmore et al., 2002). The association of these C-Cl rich volatile phases with platiniferous horizons has led many authors to suggest that PGE mineralization is controlled by C-Cl rich fluids, perhaps by chromatographic extraction by reactive porous flow of C-Cl rich fluids within the existing Critical Zone cumulates (A. E. Boudreau and Meurer, 1999).

B.5 Petrological Debates Within the RLS

Open or Closed System

Another quintessential LMI, the Skaergaard intrusion, is thought to have formed primarily through closed-system fractional crystallization from the chamber's outer regions inward (McBirney and Noyes, 1979), albeit with strontium isotopic evidence of country rock assimilation in its early stages (Stewart and DePaolo, 1990). In contrast, the RLS is interpreted as having crystallized mainly in an open system, featuring frequent magmatic recharge. This is supported by numerous chemical reversals in plagioclase and pyroxene compositions, as well as variations in initial strontium isotopes within the Lower, Critical, and lower Main Zones (F. J. Kruger, 1994; Cawthorn, 2015). However, the upper Main and Upper Zones (sometimes lumped together as the UUMZ) of the RLS are often seen as products of closed-system fractional crystallization, as indicated by uniform strontium isotopes (F. J. Kruger, 1994), oxygen isotopes that seem to have evolved through closed-system fractional crystallization (Bucholz et al., 2017), and well-fitting crystallization sequences produced by MELTS simulations of closed-system fractional crystallization of plausible UUMZ parental compositions (VanTongeren et al., 2010). Nevertheless, this view has been challenged by the identification of significant reversals in plagioclase and non-conformable field relationships within the UUMZ (Q. Yuan et al., 2017; Scoon and A. A. Mitchell, 2012).

Spatial and Temporal Scales of Magma Chambers in the Crust, Pluton Assembly Mechanisms

The RLS, along with other LMIs, has become a battleground for debates regarding the temporal and spatial scales of crustal magma chambers and pluton assembly. Some argue that plutonic rocks represent fossilized magma chambers (W. Kruger and Latypov, 2020; Bachmann and Bergantz, 2004; Paterson, 2009), while others propose that these rocks are assembled incrementally through sill-like intrusions. Such arguments are applied to rocks within the RLS by A. A. Mitchell et al. (2019), Yao et al. (2021), Scoon and A. A. Mitchell (2023), and Bourdeau et al. (2023). The latter viewpoint first gained significant support when detailed, high-precision zircon U/Pb geochronology studies revealed wide ranges of crystallization ages within seemingly homogeneous, continuous plutons (Coleman et al., 2004; Glazner et al., 2004). These wide age ranges produced by zircon U/Pb geochronology in plutons greatly exceed the timeframes over

which large magma chambers can be thermally sustained before cooling (Annen, 2009; Cawthorn and Walraven, 1998).

Zircons from the RLS and Stillwater layered mafic intrusions similarly yield some out-of-sequence U/Pb ages that challenge the idea that they represent large, fossilized magma chambers (Mungall et al., 2016; Wall et al., 2018; Scoates et al., 2021). However, resolvably out-of-sequence zircons are somewhat rare and were not detected in a slightly earlier study (Zeh et al., 2015). Additionally, some authors suggest these data are impossible to reconcile with some field relationships, challenging the reliability of U/Pb zircon geochronology for investigating layered mafic intrusions (Latypov and Chistyakova, 2022). Nevertheless, U/Pb zircon geochronology successfully yields narrow age ranges for other igneous rocks when these are geologically expected, such as for erupted ignimbrites (Wotzlaw et al., 2013).

Origin of Enigmatic Monomineralic Igneous Rocks

The RLS contains enigmatic monomineralic igneous rocks that continue to perplex igneous petrologists. For instance, the RLS contains dozens of centimeter to meter scale, near-pure chromitites which are challenging to explain. One common explanation involves the mixing of primitive and evolved magmas inducing chromite saturation (Irvine, 1977). However, an in-situ crystallization hypothesis is problematic because the Lower and Critical Zones contain significantly more chromium than could have been delivered by a plausible amount of chromium-saturated silicate melt (Eales, 2000), and the cumulate rocks above and below some chromitites are petrologically identical (Mondal and Mathez, 2007), seeming to preclude in-situ magma mixing unless over 300,000 km^3 of differentiated mafic lavas were lost from the chamber's top (Cawthorn and Walraven, 1998). The most popular alternative explanation posits that chromite crystallized within staging chambers and was injected into the RLS as chromite-phyric silicate magmas (Mondal and Mathez, 2007; Eales, 2000). Experiments have shown that chromitites can form through magmatic sedimentation from a chromite-bearing magma within realistic timescales (Manoochehri and Schmidt, 2014).

Stratiform anorthosites within the RLS pose another challenge to conventional explanations because plausible RLS magma compositions are rarely, if ever, saturated in plagioclase alone (Latypov et al., 2020; Maier et al., 2020). Various

hypotheses propose that these anorthosites resulted from [1] density-driven mechanical sorting of a magma concomitantly saturated in multiple phases (Cawthorn and Ashwal, 2009); [2] depressurization-forced saturation of plagioclase alone, followed by crystal settling (Q. Yuan et al., 2017) or in-situ growth on the magma chamber floor (Latypov et al., 2020); [3] tectonically-induced mechanical sorting of a preexisting cumulate pile (Maier et al., 2016); [4] reactive porous flow of acidic fluids (Maier et al., 2020; Sonnenthal, 1992); or [5] restites from partial melting at the contact of new intrusions of magma (A. A. Mitchell et al., 2019).

The Origin of the Merensky Reef and PGE Mineralization

The origin of the Merensky Reef and PGE mineralization in the upper Critical Zone remains contentious. One theory suggests that immiscible sulfide fluids scavenged PGEs from a large magma chamber (Campbell et al., 1983). Alternatively, percolating fluids rich in chlorine, carbon, and sulfur may have redistributed PGE elements within an already-solidified Critical Zone (A. E. Boudreau et al., 1986; Ballhaus and Stumpfl, 1985), or magmatic differentiation controlled by chromite may have concentrated PGEs within this zone (Scoon and Teigler, 1994). The platiniferous Merensky Reef, a two-part pyroxenite and pegmatoidal horizon, is debated to be either a composite of multiple synmagmatic or out-of-sequence sills (A. A. Mitchell and Scoon, 2007; A. A. Mitchell et al., 2019) or a mixture of disequilibrated crystals precipitating synchronously from overlying density-stratified Main and Critical Zone magmas (Seabrook et al., 2005). Most authors agree that the Merensky Reef represents a mixture of Main Zone and Critical Zone magmas (F. J. Kruger, 1994; Sharpe, 1985; Cawthorn, 2015).

The Transformation Hypothesis

van Biljon (1949) developed a controversial hypothesis for the origin of the RLS: that it represented an extreme case of in-situ metasomatism of the upper Pretoria Group sedimentary rocks of the Transvaal Supergroup on the basis of field relationships and geochemical arguments. Indeed, terminology and concepts from sedimentology and stratigraphy are often (though arguably erroneously) employed to describe features within the RLS and other LMIs. Van Biljon's Transformation Hypothesis sought to explain the origin of the enormously laterally continuous markers within the RLS, the apparent problem

of an enormous amount of displaced sedimentary rocks, and the transgressive nature of the RLS's contact with the Pretoria Group. van Biljon (1949) argues that chert is transformed to anorthosites, carbonates are transformed to magnetitites/chromitites, variably impure quartzites are transformed to (gabbro)norites, and shales are transformed to pyroxenites. He presents geochemical evidence that seems to show a chemical continuum in different oxide spaces between unmetamorphosed sedimentary rocks, variably hornfelsized sedimentary rocks, and mafic rocks of the RLS.

Button (1976) emphasized Van Biljon's recognition of the "space problem" introduced by the transgressive RLS into relatively undeformed sedimentary rocks, instead arguing that the felsic Bushveld rocks represented the displaced (and transformed) sediments. However, Cawthorn (1998) showed through a simple geometric analysis that the apparently missing sedimentary rocks may have been preserved underneath the RLS given the nature of its transgressive and discordant contact with the Pretoria Group.

One simple counterpoint to Van Biljon's transformation hypothesis for the RLS is that similar LMIs exist that are emplaced entirely or nearly entirely within non-sedimentary granitic, basaltic, and gneissic country rock, such as the chromitite-bearing Great Dyke of Zimbabwe (Worst, 1958) and the finely-layered Skaergaard Intrusion (McBirney, 1996). Furthermore, the RLS itself onlaps granite in the North Limb (van der Merwe, 1976).

Generally, available stable and radiogenic isotopic data require whole-intrusion-scale magmatic and hydrothermal processes to be the dominant processes forming the RLS, and for local contamination with crustal rocks to play a subordinate role (Zirakparvar et al., 2014; Schiffries and Rye, 1989; Schiffries and Rye, 1990; Willmore et al., 2002; Penniston-Dorland et al., 2012; F. J. Kruger, 1994; Richardson and Shirey, 2008). This general statement has notable exceptions:

1. The Marginal Zone contains abundant metapelitic migmatites and calc-silicates (Johnson et al., 2003; van Biljon, 1949; Button, 1976; Frick, 1967, and Chapter 5 of this work).

2. Interstitial phases such as anhydrite, almost certainly derived from sedimentary rocks, are documented within the ultramafic Lower Zone (A. H. Wilson et al., 2017; Yudovskaya et al., 2021).

These observations together suggest that in-situ igneous sediment transformation is not an essential or dominant process in the formation of the RLS or other LMIs, but that it may still occur at small scales. Evidence for this process would become obscured and eventually destroyed by episodic melting and/or reaction with an abundance of incoming magmas and hydrothermal fluids, suggesting that this process could also have been locally important at higher levels within the RLS even when evidence is subtle or completely lost to igneous overprinting.

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